FlyByWire feature user manuals

One manual per `core/*` branch of the fork (31 branches): what each feature does, how to use it, how to test it in the sim, what to expect, and its limitations with the reason

Aircraft	A32NX and A380X
Where to find it	This folder: each manual as HTML (with its `img/NN/` pictures) and as PDF; `FBW-user-manuals.pdf` holds them all, `FBW-user-manuals.docx` is the editable Word version
Branch	The `core/*` branches of the fork `TCdr/aircraft`: what to open the upstream pull requests from
Status	Restructured from the 31 per-PR manuals (their content is merged into the core manuals); each feature carries its own test status.
Pictures	Instrument pictures come from the local browser harness running the installed instrument bundles; the native WASM displays (weather radar, terrain, VD) are in-sim screenshots; audio features use diagrams. Features added after the per-PR manuals are described in text.

1. The manuals
2. Stacks: what must be merged first
3. Companion changes in aircraft-large-files
4. How to read them

1. The manuals

#	Manual	Aircraft	Branch	Based on	Features
1	Weather radar and terrain on the ND and VD (PDF)	A32NX and A380X	`core/nd-weather-terrain`	`master`	8
2	Surveillance: SURV controls, ECAM SURV lines and TCAS (PDF)	A380X (SURV panel, VD messages, ECAM lines) and A32NX + A380X (TCAS)	`core/surveillance`	`master`	8
3	Independent CPT and F/O MCDUs (PDF)	A32NX	`core/a32nx-mcdu`	`master`	3
4	ECAM procedures, EMER CANC and FIRE TEST (PDF)	A380X (ECAM, FWS) and A32NX + A380X (FIRE TEST option)	`core/ecam-fws`	`master`	5
5	Runway awareness: RAAS and OANS (PDF)	A32NX (RAAS) and A380X (OANS)	`core/oans-btv-raas`	`master`	5
6	A380X standby instruments (ISIS) (PDF)	A380X	`core/isis`	`master`	4
7	Lights: strobe and beacon timing, interior lighting presets (PDF)	A32NX and A380X	`core/lights`	`master`	2
8	Fire warnings and the ENG FIRE pushbutton (PDF)	A32NX (fire warnings, LP fuel valve) and A380X (LP fuel valve)	`core/fire`	`master`	3

9	A380X MFD FMS and SURV pages (PDF)	A380X (the shared flight plan code of DIRECT TO, the 50 waypoint limit, the received company winds and the CHECK ALTN WIND check is also in the A32NX FMS; the A32NX keeps its own alternate cruise level and history wind rules; changes the A32NX MCDU too)	`core/a380x-mfd-fms`	`core/surveillance`	18
10	A380X ATC COM and datalink (PDF)	A380X (the ACARS provider connection is shared with the A32NX)	`core/a380x-atccom-datalink`	`core/a380x-mfd-fms`	7
11	A380X fuel jettison and FINAL fuel (PDF)	A380X	`core/fuel`	`core/a380x-mfd-fms`	6
12	GPS receivers, GPS deselection and ECAM STATUS (PDF)	A32NX and A380X (shared systems code); the ECAM STATUS page and STS reminder are A32NX only	`core/gps`	`core/a380x-mfd-fms`	8
13	A380X interactive ND (PDF)	A380X	`core/a380x-interactive-nd`	`core/a380x-mfd-fms`	2
14	flyPad performance calculators (PDF)	A380X and A32NX (one shared flyPad screen per calculator; the company datalink reply time is A380X only)	`core/efb-performance`	`core/a380x-mfd-fms`	11
15	Cockpit printers (PDF)	A380X (printing, buttons) and A32NX (paper door, no torn paper edge)	`core/printer`	`core/a380x-atccom-datalink`	5
16	flyPad ground: taxi route, ATC frequencies and GSX (PDF)	A32NX and A380X (the OANS flags are A380X only)	`core/efb-ground`	`core/efb-performance`	8
17	flyPad Material 3 redesign and independent flyPads (PDF)	A32NX and A380X (the Buffet tab: A32NX only)	`core/efb-redesign`	`core/efb-ground`	19
18	A32NX air conditioning per the FCOM (PDF)	A32NX	`core/a32nx-air-conditioning`	`core/gps`	4
19	A380X cockpit buttons, lights and probes (PDF)	A380X	`core/a380x-cockpit`	`master`	6
20	A32NX exterior: spoilers by panel group and visible RAT (PDF)	A32NX	`core/a32nx-exterior`	`master`	2

21	A380X electrical alerts and AC ESS loads (PDF)	A380X	`core/a380x-elec`	`core/gps`	5
22	A380X flyPad failures (PDF)	A380X	`core/failures-a380x`	`master` + `core/fire` + `core/ecam-fws`	23
23	A32NX flyPad failures (PDF)	A32NX (the fire visual effects, the smoke detection function and the anti-ice systems part also A380X)	`core/failures-a32nx`	`core/gps` + `core/fire` + `core/fcom-audit-fixes` + `core/a32nx-air-conditioning`	48
24	Display unit failures and reconfiguration (PDF)	A32NX and A380X	`core/display-unit-failures`	`core/nd-weather-terrain`	6
25	A380X weather radar GAIN, ELEVN and TILT (PDF)	A380X	`core/a380x-wxr-controls`	`core/nd-weather-terrain` + `core/a380x-mfd-fms`	4
26	A380X VD airports as flat ground (PDF)	A380X (the map view fix: A32NX and A380X)	`core/a380x-vd-runways`	`core/nd-weather-terrain` + `core/oans-btv-raas`	2
27	FCOM audit fixes (PDF)	A32NX and A380X	`core/fcom-audit-fixes`	`master`	15
28	RAT ground stow and A32NX RAT performance (PDF)	A32NX and A380X	`core/rat-ground-stow`	`master`	2
29	A380X cargo doors follow the hydraulics (PDF)	A380X	`core/a380x-cargo-doors`	`master`	1
30	A380X ATC mailbox with the SD on an ND display unit (PDF)	A380X	`core/a380x-mailbox-du-reconf`	`core/display-unit-failures` + `core/a380x-atccom-datalink`	2
31	A380X ELEC, fuel and FCOM gap alerts (PDF)	A380X (A32NX: ADIRU power supplies, engine anti-ice valves and anti-ice key events, through shared code)	`core/a380x-elec-fuel-alerts`	`core/a380x-elec` + `core/failures-a380x` + `core/fcom-audit-fixes`	23

2. Stacks: what must be merged first

Most branches are based directly on `master` (FlyByWire master 76db0aef1 merged in; `core/a380x-cargo-doors` starts from it) and can be merged in any order. The others are stacked: a branch needs every branch under it merged first.

Stack	Merge order
A380X MFD	`core/surveillance` > `core/a380x-mfd-fms` > then any of `core/a380x-atccom-datalink`, `core/fuel`, `core/gps`, `core/a380x-interactive-nd`, `core/efb-performance`
Printer	… `core/a380x-atccom-datalink` > `core/printer`
flyPad	… `core/efb-performance` > `core/efb-ground` > `core/efb-redesign`

GPS	… `core/gps` > `core/a32nx-air-conditioning`, `core/a380x-elec`, `core/failures-a32nx` (which also merges `core/fire`, `core/fcom-audit-fixes` and `core/a32nx-air-conditioning`: its A32NX FWC alerts build on their code)
A380X failures	`core/fire` + `core/ecam-fws` > `core/failures-a380x` (merged in for the engine failures)
A380X ELEC, FUEL and FCOM gap alerts	`core/a380x-elec` + `core/failures-a380x` + `core/fcom-audit-fixes` > `core/a380x-elec-fuel-alerts` (the alerts need the ELEC failures of one and the FWS engine and fuel code of the other; its FCOM alert additions also extend the A380X door, A-ICE and NAV alert files of core/fcom-audit-fixes)
ATC mailbox	`core/display-unit-failures` + `core/a380x-atccom-datalink` > `core/a380x-mailbox-du-reconf` (the mailbox follows the DU reconfiguration of one and the ATC datalink of the other)
ND	`core/nd-weather-terrain` > `core/display-unit-failures`; `core/nd-weather-terrain` + `core/a380x-mfd-fms` > `core/a380x-wxr-controls`; `core/nd-weather-terrain` + `core/oans-btv-raas` > `core/a380x-vd-runways`
Independent (on master)	`core/a32nx-mcdu`, `core/ecam-fws`, `core/oans-btv-raas`, `core/isis`, `core/lights`, `core/fire`, `core/a380x-cockpit`, `core/a32nx-exterior`, `core/fcom-audit-fixes`, `core/rat-ground-stow`, `core/a380x-cargo-doors`
Feature split over two branches	The RAT stow row of `core/efb-redesign` needs `core/rat-ground-stow` (the systems side): merge both for the feature to work; the visible RAT of `core/a32nx-exterior` goes back in only after that ground stow
Feature split over two branches (cargo doors)	The cargo door row of `core/efb-redesign` (Ground > Services) reads the inhibition that only `core/a380x-cargo-doors` writes: merge both for the row to show the new state
Same panel.cfg section	`core/display-unit-failures` (A32NX `DuMessage` gauge) and `core/efb-redesign` (A32NX F/O flyPad) both add `[VCockpit21]`: whichever is merged second renumbers its section to `[VCockpit22]`
Features split over two branches (FCOM alert additions)	A380X smoke detection: the SDF Rust and cargo smoke panel lights are on `core/failures-a32nx`, the SMOKE alerts on `core/a380x-elec-fuel-alerts`. Smart FIVE HUNDRED: the flyPad pin option on `core/efb-redesign`, the callout on `core/a380x-elec-fuel-alerts`. A32NX CHECK APPR SELECTION: the MCDU message on `core/gps`, the PFD FMA line on `core/failures-a32nx`. Merge both in each case. When `core/failures-a32nx` and `core/a380x-elec-fuel-alerts` meet: adirs.rs differs only by the IR 10 min alignment (keep it), keep the a320 navigation.rs of core/failures-a32nx; the anti-ice Rust is identical
Shared engine failure module	`core/failures-a32nx` and `core/failures-a380x` both carry the same fbw-common engine failure files (engine_failure.rs, ata.ts): whichever is merged second keeps one copy; in `failures/mod.rs` and the A380X `a380_systems_wasm` failure list keep the GPS entries of `core/gps` followed by one ATA 72 block

3. Companion changes in aircraft-large-files

Some features change the 3D cockpit models, which live in the separate FlyByWire repository `aircraft-large-files`. Their companion commits are on the fork `TCdr/aircraft-large-files` and must be merged with the core branch.

Companion commit	What it adds	Core branch
`5563e31`	A32NX MCDU screens: both MCDUs on one texture (UV halves)	`core/a32nx-mcdu`
`36b4bd9`	A380X pedestal printer paper	`core/printer`
`27d6123`	A380X printer door, roller unit and paper slit	`core/printer`

`99237fb`	A32NX printer door	`core/printer`
`24f7b02`	A380X F/O flyPad screen (independent flyPads)	`core/efb-redesign`
`d3060ea`	A32NX F/O flyPad screen (independent flyPads)	`core/efb-redesign`
`8c14d21`	A380X printer ON light (green window)	`core/printer`
`fa6c5d1`	A380X fuel jettison plume at the outlets (stacked on 8c14d21)	`core/fuel`
`b8fb739`	A380X MFP (multi-function probe) vane animations	`core/a380x-cockpit`
`b790e8d`	A32NX DU reconfiguration: 17 overlay glasses on the EIS screens, DU_MESSAGE and DU_BLANK materials (stacked on 99237fb and d3060ea)	`core/display-unit-failures`

4. How to read them

- Each manual has one numbered section per **feature** of its branch. The grey line under the feature title gives the aircraft, where to find it and its **status** (sim-tested, or not, with the date).
- **What it is for**: the problem the change solves, in a few sentences.
- **How to use**: numbered steps for normal use.
- **How to test**: a table of steps a tester follows in the sim, each with the expected result. If the sim does something else, it is a bug worth reporting with the branch name.
- **What you should see** tables: the expected behaviour in each situation; **Limitations and why**: what does not work yet, the reason and what to do about it.
- Pictures that do not come from the harness (drawn diagrams, in-sim screenshots taken during development) carry an amber line under their caption: they are not representative of the in-game UI and serve as a reference for the explanation only.

Weather radar and terrain on the ND and VD

A native weather radar and TERR ON ND terrain display on the NDs of both aircraft, and the terrain and weather on the A380X vertical display, drawn from the MSFS radar and terrain data without SimBridge

Aircraft	A32NX and A380X
Where to find it	NDs (ARC and ROSE pages); A32NX weather radar panel and TERR ON ND pushbutton; A380X EFIS WX / TERR overlay, MFD SURV CONTROLS; A380X VD
Branch	`core/nd-weather-terrain`, based on `master` (FlyByWire master 76db0aef1 merged in); independent. `core/display-unit-failures`, `core/a380x-wxr-controls` and `core/a380x-vd-runways` stack on it.
Status	Radar, mode messages and VD tested in the sim over several sessions (September 2026; A380X messages, antenna sweeps, 320 NM limit and ARC bottom by the user); terrain tested by the user: OK; parked views and 20 / 10 Hz redraw user-checked in the sim and measured; OANS mouse fix user-tested; terrain view resend after parking tested in the sim by the user: OK.
Pictures	In-sim screenshots taken during development (the radar and terrain are drawn by a native WASM gauge, which the browser harness cannot run), and two A32NX NDs from the local browser harness for the mode message. Some A380X radar pictures predate later changes: their captions say so.

1. A32NX weather radar on the ND
2. A380X weather radar on the ND and weather on the VD
3. Weather radar mode and messages on the ND
4. Terrain on the ND (TERR ON ND) and A380X VD terrain
5. Radar, terrain and VD after the A380 and A320 manuals
6. Fewer stutters: idle map views paused, 20 Hz / 10 Hz redraw
7. Terrain and radar views get their settings again after a pause
8. A380X: the OANS map takes the mouse through the radar gauges

1. A32NX weather radar on the ND

> **Aircraft:** A32NX | **Where:** Both NDs; pedestal WX SYS and MODE knobs | **Status:** Tested in the sim over several sessions and weather presets (September 2026); antenna sweep tested by the user.

What it is for

The A32NX ND had no working weather radar. A native gauge (`ndwxr`) now draws the MSFS weather radar on the CPT and F/O NDs, aligned with the range rings, in the colours of an airborne radar.

Figure 1. A32NX ND in ARC with weather ahead: green, yellow and red cells.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

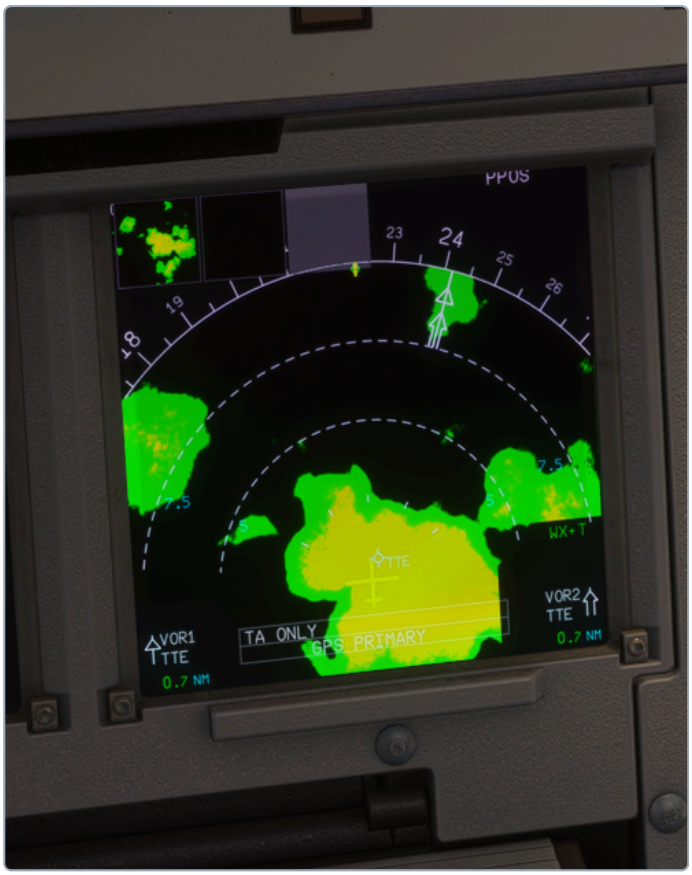


Figure 2. Another cell field, ROSE-like forward view.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

Using it

Control	Position	ND
WX SYS knob	SYS 1 / SYS 2	Radar on (system 1 or 2)
	OFF	No radar
MODE knob	WX	Precipitation
	WX+T	Precipitation and turbulence (magenta)
	TURB	Turbulence only
	MAP	Ground map from the terrain: black water, green ground, amber mountains
EFIS	ARC or any ROSE page (NAV, ILS, VOR)	Radar shown over the antenna sweep (180 degrees, the forward half); nothing in PLAN

The selected mode is written on the ND (see the feature *Weather radar mode and messages on the ND*).

Reading the picture

Colour	Rain rate	Meaning
Green	from 0.7 mm/h	Light precipitation
Yellow	from 4 mm/h	Moderate
Red	from 12 mm/h	Heavy: avoid

Magenta	above 20 mm/h, within 40 NM	Turbulence (WX+T, TURB)

The engine's rain rate is noisy pixel by pixel; each level is smoothed, so the picture shows solid areas with crisp edges.

What to expect

Situation	What you should see
On the ground	Drawn as in flight: the radar scans on the ground once it is switched on with a display mode (A320 FCOM DSC-34-SURV-30-30).
Clear sky	No picture (nothing to return).
WX SYS OFF	No picture; WXR OFF in white on the ND.
CPT and F/O NDs	Independent: each ND uses its own IR position (IR 1 CPT, IR 2 F/O, IR 3 by the ATT HDG knob) and its own range and mode; the two knobs are shared.

Limitations and why

Limitation	Why	What to do
Turbulence is a proxy.	MSFS gives no turbulence data; rain above 20 mm/h within 40 NM is drawn magenta.	-
No tilt, no gain.	The MSFS radar has neither.	-
Drawn only with weather around.	The MSFS radar only returns precipitation.	Test with the MSFS weather presets.

How to use

1. Set the **WX SYS** knob of the weather radar panel to **SYS 1** or **SYS 2**.
2. Set the **MODE** knob: **WX** (precipitation), **WX+T** (precipitation and turbulence), **TURB** (turbulence only) or **MAP** (ground map).
3. On the EFIS control panel select **ARC** or a **ROSE** mode and a range.
4. Read the colours (table above); WX SYS **OFF** takes the radar off.

How to test

#	Do this	Expected result
1	A32NX in flight (or on the ground) with an MSFS weather preset with storms. CPT EFIS: ND **ARC**, **40 NM**. WX SYS **SYS 1**, MODE **WX**.	Green, yellow and red cells ahead of the aircraft, aligned with the range rings; **WX** in green on the right of the ND.
2	MODE **WX+T**.	Heavy rain (above 20 mm/h) within 40 NM turns magenta; message **WX+T**.
3	MODE **TURB**.	Only the magenta areas are left; message **TURB**.
4	MODE **MAP**.	A ground map: black water, green ground, amber mountains; message **MAP**.
5	MODE WX, then ND mode **ROSE ILS**, **ROSE VOR**, **ROSE NAV**, then **PLAN**.	The radar is shown on every ROSE page (forward half), not in PLAN.
6	Change the F/O ND range only.	The F/O ND redraws at its own range; the CPT ND does not change.
7	WX SYS **OFF**.	No weather picture; **WXR OFF** in white on the ND.

8	Select a clear-sky weather preset with WX SYS on.	No picture (nothing to return).

2. A380X weather radar on the ND and weather on the VD

> **Aircraft:** A380X | **Where:** Both NDs and VDs; EFIS WX overlay; MFD SURV CONTROLS | **Status:** Tested in the sim (September 2026); 160 degree sweep and 320 NM limit tested by the user.

What it is for

The A380X ND always showed WXR INOP (both WXR systems were hard-coded as failed) and had no radar picture. It now has the native radar of the A32NX, with the A380 rules, and the VD shows the weather on its vertical cut.

Figure 3. A380X ND with weather cells ahead, and the same cells on the VD below. *Development picture: the WX+T label and the hatched off-path areas seen here were removed later; the A380X ND now reads WX.*

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

Using it

- EFIS control panel: select the **WX** overlay.
- MFD SURV CONTROLS: WXR AUTO or OFF, TURB AUTO or OFF, MODE WX or MAP, WX ON VD. These buttons are wired on `core/surveillance`; with this branch alone the radar runs at their default settings (WXR AUTO, TURB AUTO, MODE WX, WX ON VD AUTO).

- The radar is shown when the WXR & TAWS system selected on the SURV panel is not failed. A system is failed only while its AESU is reset or its power is lost (AC ESS for system 1, AC 4 for system 2).
- The ND range table is -1 (ZOOM, the airport map) and 10 to 640 NM; the radar is hidden at ZOOM and at 640 NM: "The ND can only display the weather, if the selected range is below, or equal to, 320 nm" (A380 FCOM DSC-31-20-50).

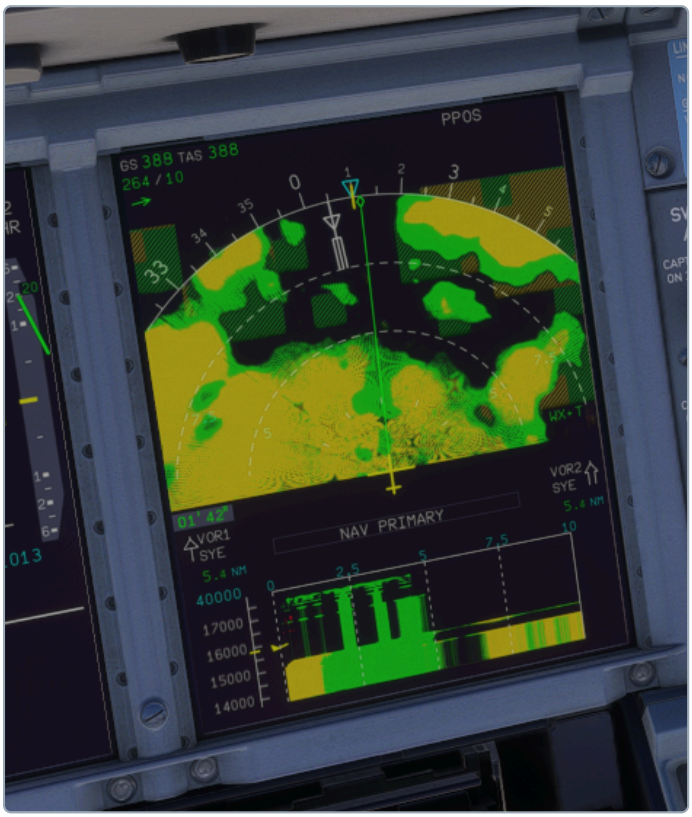


Figure 4. Moderate rain (yellow) over a wide area, with its VD cut. *Development picture: the WX+T label and the hatched off-path areas seen here were removed later; the A380X ND now reads WX.*

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

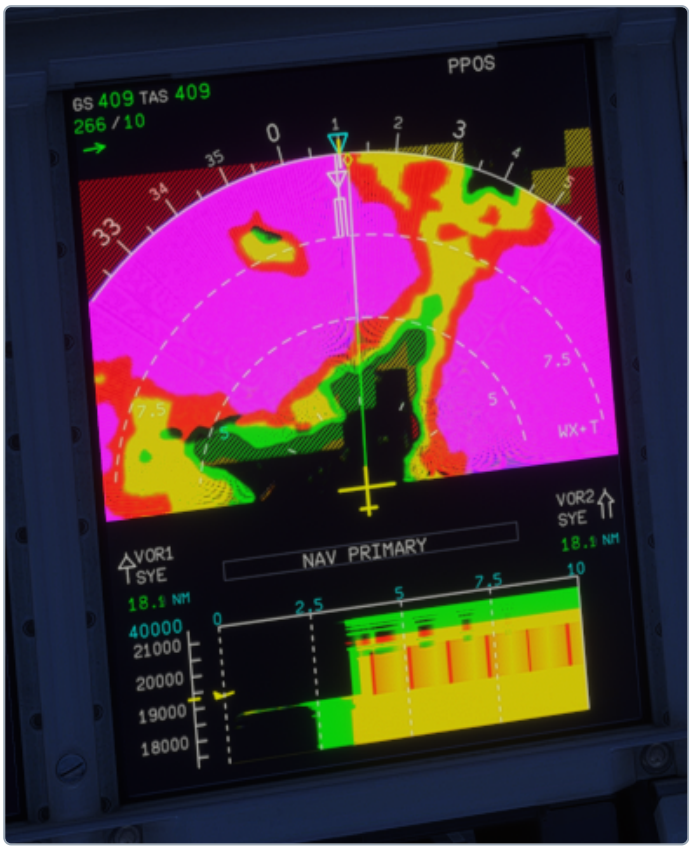


Figure 5. TURB on: heavy rain drawn magenta (turbulence proxy). *Development picture: the WX+T label and the hatched off-path areas seen here were removed later; the A380X ND now reads WX.*

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

Weather on the VD

- WX ON VD (SURV CONTROLS) shows the weather on the VD, along the same cut as the terrain (see *Radar, terrain and VD after the A380 and A320 manuals*).
- Cells are drawn green and yellow with a red core; no magenta: the real VD does not display turbulence (FCOM DSC-31-20-40-10).
- Heights are placed on the ADR baro-corrected altitude, like the VD's aircraft symbol.

Limitations and why

Limitation	Why	What to do
The VD weather is stylised.	The vertical MapView of MSFS is a single thin beam (tilting it crashes the gauge); the VD picture is built from the ND radar views.	-
PRED W/S and GAIN are not drawn.	The MSFS radar has no predictive windshear or gain.	-
No TURB-only mode.	The A380 has none: TURB off gives WX, on gives WX+T.	-

Antenna sweep

The picture covers the real antenna sweep of the A380's Honeywell RDR-4000: 160 degrees (80 degrees each side of the nose; Honeywell IntuVue white paper). The real display fills up to 90 degrees from the radar's memory, which the MSFS radar does not have, so nothing is painted beside and behind the aircraft there.

How to use

1. On the EFIS control panel select the **WX** overlay, the **ARC** or a **ROSE** mode and a range of 10 to 320 NM.
2. With `core/surveillance`: set the WXR block of **MFD SURV > CONTROLS** (WXR, TURB, MODE, WX ON VD). Without it the radar runs at the default settings.
3. Read the weather on the ND and its vertical cut on the VD below.

How to test

#	Do this	Expected result
1	A380X in flight with an MSFS storm preset. CPT EFIS: **WX** overlay, ND **ARC** 40 NM.	Green, yellow and red cells over the 160 degree sweep; **WX** in green on the right of the ND.
2	Look at the VD under the ND.	The cells on the flight path drawn as green and yellow columns with a red core at the aircraft level; no magenta on the VD.
3	ND range **640 NM**, then the smallest range (**ZOOM**).	No weather at 640 NM (the terrain, if selected, is still drawn); none at ZOOM. Back to 320 NM: the weather comes back.
4	With `core/surveillance`: MFD SURV > CONTROLS, **TURB OFF**.	No more magenta on the ND; the message stays **WX**.
5	With `core/surveillance`: **MODE MAP**.	Ground map on the ND; message **MAP**.
6	With `core/surveillance`: **WX ON VD OFF**.	The VD weather goes; the ND keeps it.
7	Deselect the **WX** overlay.	No weather on the ND, no radar message.

3. Weather radar mode and messages on the ND

> **Aircraft:** A32NX and A380X | **Where:** ND, right side above the TERR box | **Status:** Tested in the sim (September 2026); A380X WX / MAP / WXR OFF messages tested by the user.

What it is for

Nothing on the ND told the crew which radar mode was selected. The mode now appears in green on the right of the ARC and ROSE pages while the radar is selected, as on the real ND. The A32NX shows the Honeywell list of the A320 FCOM (DSC-34-SURV-30-30); the A380X shows the A380's own WXR messages (A380 FCOM DSC-34-20-30-20 "WXR MESSAGES"): WX for the WX display function whatever the TURB button, MAP, WXR OFF.

ND message	A32NX (MODE knob)	A380X (SURV CONTROLS)
WX (green)	WX	WX display function, whatever the TURB button
WX+T (green)	WX+T	-- (not an A380 message)
TURB (green)	TURB	-- (no TURB-only mode on the A380)
MAP (green)	MAP	MODE MAP
WXR OFF (white)	WX SYS OFF	WXR button OFF

(none)	-	no WX overlay on the EFIS panel, or the WXR & TAWS system failed

Figure 6. A32NX ARC: WX+T above the TERR box.

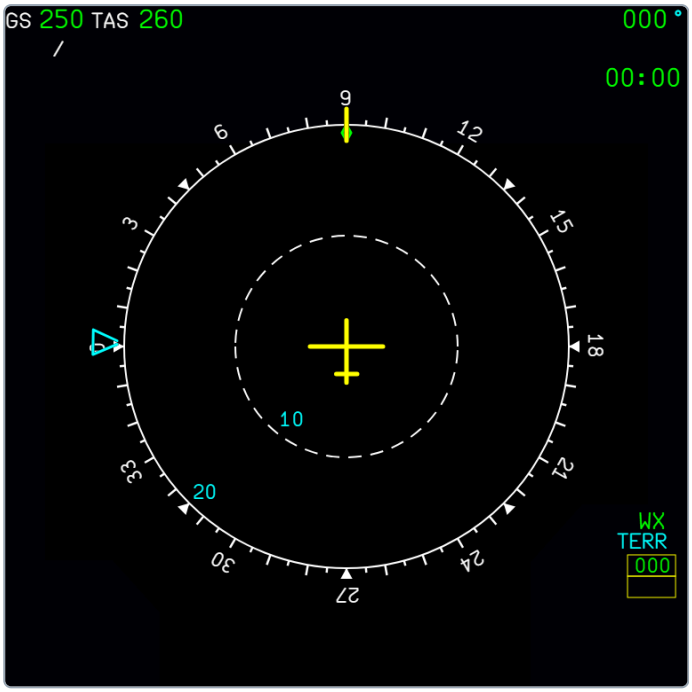


Figure 7. A32NX ROSE NAV: WX.

- Shown on the ARC and ROSE pages (NAV, ILS, VOR), not in PLAN, and not while the terrain replaces the weather (TERR ON ND).
- Shown on the ground too, where the radar also scans.
- The native radar gauge decides and publishes the label (`A32NX_WXR_ND_{L,R}_MODE`), so the ND follows exactly what the gauge draws.

Limitations and why

Limitation	Why	What to do
No label in PLAN.	The radar is not shown in PLAN.	-

How to use

1. Nothing to set: the message follows the radar selection (A32NX MODE knob; A380X WX overlay and SURV CONTROLS).
2. Check the message on the right of the ND, above the TERR box, after each radar change.

How to test

#	Do this	Expected result
1	A32NX, ND ARC: WX SYS SYS 1, turn the MODE knob through WX, WX+T, TURB, MAP.	The message follows in green: WX, WX+T, TURB, MAP.
2	A32NX: WX SYS **OFF**.	**WXR OFF** in white.
3	A32NX: press **TERR ON ND**.	The radar message goes while the terrain replaces the weather.
4	A380X, WX overlay, default SURV settings (TURB AUTO).	**WX** in green (not WX+T).
5	A380X with `core/surveillance`: MODE MAP, then WXR OFF on the SURV CONTROLS page.	**MAP** in green, then **WXR OFF** in white.

6	Either aircraft: ND mode **PLAN**.	No radar message.

4. Terrain on the ND (TERR ON ND) and A380X VD terrain

> **Aircraft:** A32NX and A380X | **Where:** A32NX TERR ON ND pushbutton; A380X EFIS TERR overlay; A380X VD | **Status:** Tested in the sim by the user: OK.

What it is for

TERR ON ND needed SimBridge. The native gauge now draws it from MSFS's own terrain, with the EGPWS colours, on every map page but PLAN, also on the ground.

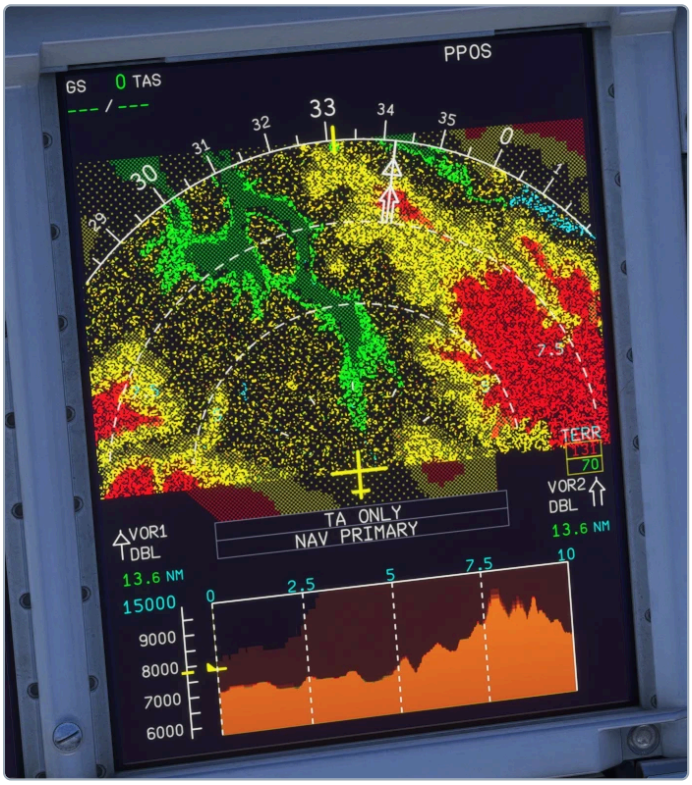


Figure 8. A380X at Aspen, 10 NM: terrain far above the aircraft in red, close in yellow, below in green; the VD terrain profile in brown.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

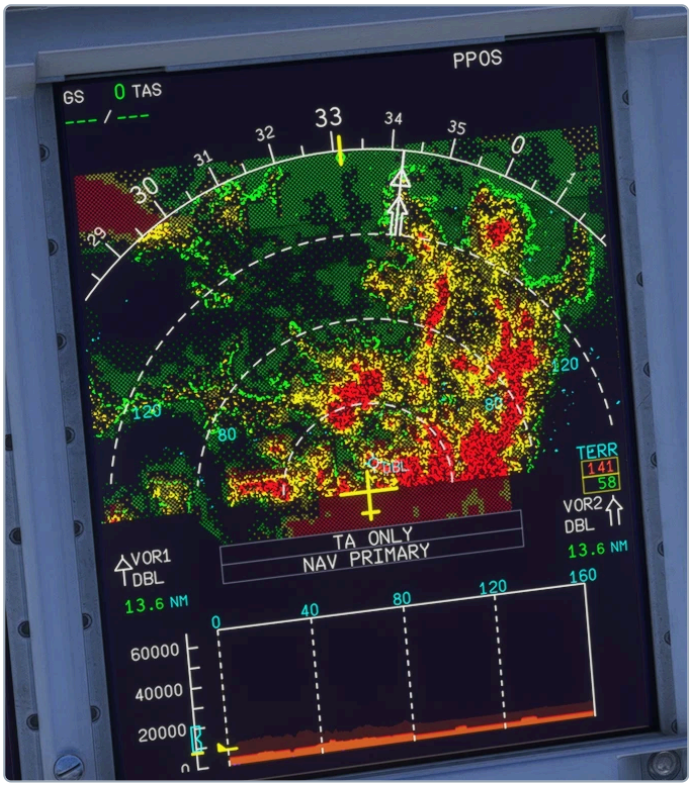


Figure 9. The same place at 160 NM.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

Reading the colours

Terrain relative to the aircraft	Colour
More than 2000 ft above	Dense red
1000 to 2000 ft above	Dense yellow
500 ft below (250 ft gear down) to 1000 ft above	Medium yellow
Down to 1000 ft below (500 ft gear down)	Dense green
1000 to 2000 ft below	Light green
More than 2000 ft below	Nothing
Water	Cyan-blue dots

The dots are a random stipple, as on the real display. The map turns with the ND's heading; ARC shows the forward half, the ROSE modes the whole disk.

A380X vertical display

Figure 10. The A380X VD terrain profile (brown), water in cyan from sea level down.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

- Needs the TERR SYS button of the SURV page not OFF (otherwise NO TERR DATA AVAILABLE).
- The cut follows the flight plan or the track (see *Radar, terrain and VD after the A380 and A320 manuals*).

The peaks box

The TERR box with the highest and lowest elevations of the range needs real elevations, which a gauge cannot read from MSFS: its figures come from **SimBridge** when it runs. Without SimBridge the box stays empty and the terrain is drawn as usual.

Limitations and why

Limitation	Why	What to do
At most 8 map views per module.	MSFS crashes a 9th; the A380X switches its views between weather and terrain.	-
Peaks box only with SimBridge.	No elevation readback from MSFS.	Run SimBridge.
Terrain far above the range top is shown red.	MSFS does not clamp to the first colour entry.	-

Peaks mode (with SimBridge)

When the SimBridge figures say that the highest terrain of the range is more than 2000 ft below the aircraft, the terrain is drawn in green by its elevation within the range: solid for the top 5 %, dense dots down to 65 %, light dots down to 35 %, nothing lower. Without SimBridge the standard colours above apply.

Other details

- When descending faster than 1000 ft/min, the colours use the altitude expected in 30 s (A320 FCOM DSC-31-45).
- The old `terronnd` gauge (SimBridge's own terrain picture) is no longer loaded: SimBridge is only used for the peaks box figures. The A32NX systems host now posts the aircraft status to SimBridge itself.

How to use

1. A32NX: press the **TERR ON ND** pushbutton of the side wanted (ND in ARC or ROSE).
2. A380X: select the **TERR** overlay on the EFIS control panel; keep the TERR SYS of the SURV page not OFF for the VD profile.
3. Optional: run SimBridge for the peaks box figures and the peaks mode.

How to test

#	Do this	Expected result
1	A32NX or A380X on the ground at a mountain airport (for example KASE, Aspen). ND ARC 10 NM. Press TERR ON ND (A380X: TERR overlay).	Terrain far above the aircraft in red, close above in yellow, below in green; lakes as cyan-blue dots; the radar message goes.
2	Range 160 NM.	The same colour logic over the wider area; the map turns with the heading.
3	ND mode **PLAN**.	No terrain.
4	A380X: look at the VD.	Brown terrain profile along the flight plan or the track, water in teal.
5	A380X: set the TERR SYS button of the SURV page to OFF.	VD: NO TERR DATA AVAILABLE.
6	Start SimBridge and wait a few seconds.	The TERR box shows the highest and lowest elevations of the range.
7	Without SimBridge.	The TERR box stays empty; the terrain is drawn as before.

5. Radar, terrain and VD after the A380 and A320 manuals

> **Aircraft:** A380X (VD) and both aircraft (weather picture) | **Where:** A380X VD; both NDs with the radar | **Status:** Tested in the sim (September 2026).

The vertical cut (A380X VD)

- In the managed lateral modes the VD shows the terrain and weather **along the flight plan** (a 200 NM window from just behind the aircraft, with the turns); in the selected modes **along the track** (FCOM: VIEW ALONG ACFT TRK). More than 5 NM off the plan: the track.
- The terrain is the highest across a width of 2 × RNP (1 NM terminal, 2 NM en route above 18 000 ft).
- The grey area starts at the next track change of more than 3° (the VD no longer matches the ND beyond it).

The VD scale and the terrain height

- With a trajectory, the highest of the aircraft and the path sits at three quarters of the window: **in cruise the ground stays on the plot**, as on the real aircraft at FL400.
- The gap under the aircraft symbol is the true height, whatever the altimeter setting (FCOM / FCTM).
- Water in the ND teal; no magenta on the VD; no VD weather when TERR is not available.

Figure 11. A380X VD with the terrain profile and water (in-sim).

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

The weather picture (both aircraft)

- The radar scans and draws on the ground too (A320 FCOM DSC-34-SURV-30-30: "on the ground, the radar is scanning when the flight crew sets one radar to ON and selects a display mode").
- MAP mode is a ground map from the terrain.
- The picture **builds up over 30 s** after the radar starts, sweeping left to right (A380 FCOM).
- The ARC picture keeps the ND bottom corners free and stops just above the TCAS / WXR message box, as on the real display.

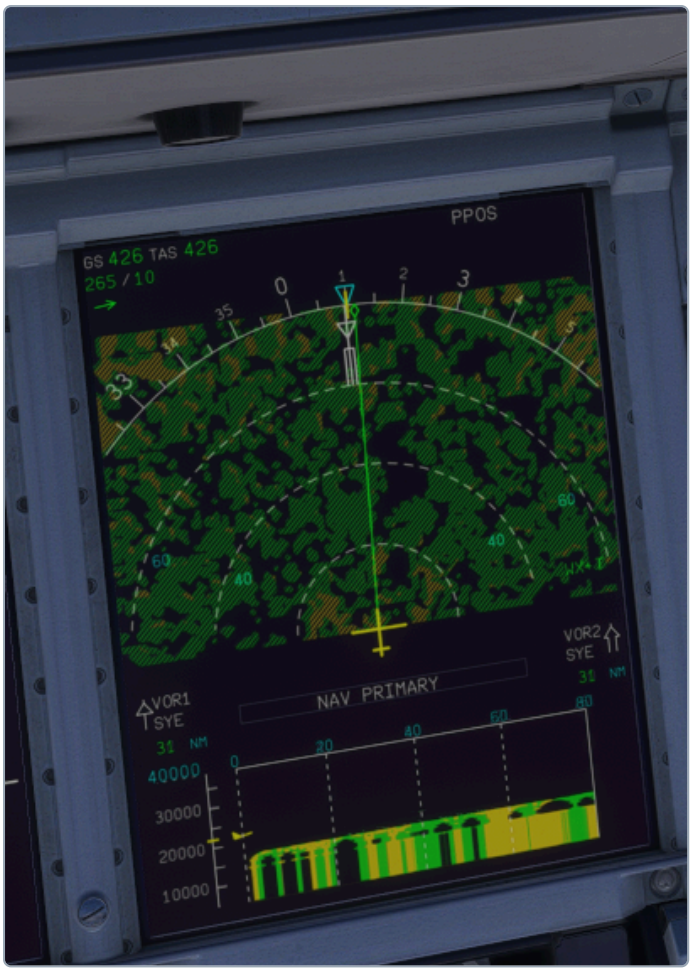


Figure 12. Just after WX is selected: the picture is building up.
In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

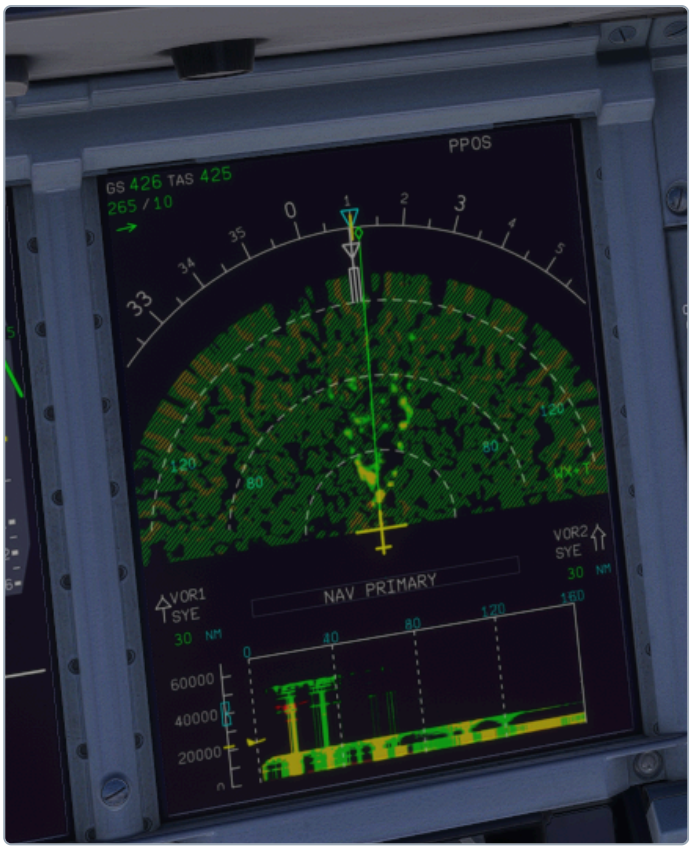


Figure 13. A few seconds later.
In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

Limitations and why

Limitation	Why	What to do
No A380 AUTO off-path display.	MSFS runs one radar per aircraft; a third view changed what the beam views delivered, so it was removed.	-

How to use

1. Nothing to set: the VD cut follows the lateral mode (managed: the flight plan; selected: the track).
2. After switching the radar on, allow 30 s for the picture to build up.

How to test

#	Do this	Expected result
1	A380X in flight in NAV mode with a flight plan that turns ahead, WX and terrain around.	The VD terrain and weather follow the flight plan; the grey area starts at the next track change of more than 3 degrees.
2	Select a heading (HDG mode).	The VD follows the track.
3	Cruise at a high level (for example FL380) over land.	The ground stays on the VD plot (not cut off at the bottom).

4	Either aircraft: switch the radar off, then on (A32NX WX SYS; A380X WX overlay).	The picture builds up over about 30 s, sweeping left to right.
5	ND ARC with weather or terrain.	The picture stops just above the TCAS / WXR message box; both bottom corners stay clear.

6. Fewer stutters: idle map views paused, 20 Hz / 10 Hz redraw

> **Aircraft:** A32NX and A380X | **Where:** NDs and A380X VDs (no control) | **Status:** Tested in the sim by the user: same picture on the NDs and the VD; measured with PresentMon.

What it is for

The radar and terrain gauge uses up to eight MSFS map views per aircraft, and the sim renders every visible view each frame, drawn or not. On the A380X this cost frame rate and caused stutters.

- A view nothing has shown for 120 frames is now paused (made invisible). It is woken up with a 60-frame warm-up before it is drawn again. With WX and TERR off, in PLAN or with the ND unpowered, no view renders at all.
- The ND picture is redrawn at 20 Hz and the VD terrain at 10 Hz instead of every frame, and at once when what the display shows changes (page, range, WX / TERR selection). The first radar sweep is still drawn every frame. At 20 Hz a 3 degree/s turn moves the picture by less than one pixel per step.

What you should see

A380X in cruise, WX on, ARC 40 NM (PresentMon, 2 x 120 s)	Before	Views paused	+ 20/10 Hz
Average FPS	86-90	93-94	95-101
1 % low FPS	18-22	28-30	28-32
Stutters over 50 ms per 120 s	20-30	2-5	-

The A32NX never had the problem (its result is neutral). The picture itself is the same.

Limitations and why

Limitation	Why	What to do
A paused view needs about 1 s to come back.	The 60-frame warm-up lets the sim render the view again before it is drawn (no white flash).	Normal: switching WX / TERR already took about a second on the A380X.

How to use

1. Nothing to set.

How to test

#	Do this	Expected result
1	A380X in cruise, WX overlay on, ND ARC 40 NM with weather around. Fly a turn.	The weather picture turns smoothly with the ND, no stepping.
2	Change the range and the ND mode a few times.	The picture redraws at once at each change.
3	Switch from WX to TERR and back after a few seconds.	The picture comes back after about a second, without a white square or flash.
4	Watch the VD.	Terrain and weather follow as before.

5	Optional: compare the frame rate with an earlier build (WX on, ARC 40 NM).	Fewer stutters and a higher 1 % low frame rate.

7. Terrain and radar views get their settings again after a pause

> **Aircraft:** A380X (seen there) and A32NX (same gauge) | **Where:** NDs and A380X VDs (no control) | **Status:** Tested in the sim by the user: OK: both VDs identical with the same ND mode and range. Native test view_park_test (ndwxr).

What it is for

On the A380X F/O VD showed a terrain profile (flat blocks) that the CPT VD did not, with the same inputs. The gauge pauses the map views nothing shows (previous feature) and wakes them up when needed. It is not known whether the sim keeps a view's settings while it is paused, and a setting the sim ignored was never sent again, since the gauge only sent changes. As a safeguard:

- A view that wakes up gets its whole setup again (terrain, radar, VD view, each by its role).
- The view radius (the range) is sent on a change, after a wake-up, when the sim time goes back, and again every second while the view is in use, not only on a change.

What you should see

Situation	What you should see
CPT and F/O ND with the same mode and range, terrain shown	The two VD terrain profiles are identical.
A view woken up after a pause (WX / TERR switched, page or range changed)	The right terrain zoom and picture, as on the other side.

Limitations and why

Limitation	Why	What to do
The cause is a likely one, not a proven one.	The F/O stale terrain view zoom was cured by the resend (user test); whether the sim drops settings of a paused view is still not known.	-

How to use

1. Nothing to set.

How to test

#	Do this	Expected result
1	A380X powered, CPT and F/O EFIS: TERR overlay, the same ND mode (e.g. ARC) and range on both sides. Compare the two VDs.	Identical terrain profiles; no flat blocks on one side only.
2	On the F/O side switch TERR off and on again, change the range and come back.	The F/O VD and ND come back with the same terrain as the CPT side at the same range.
3	Reload the flight (the sim time goes back) and compare again.	Both sides identical.

8. A380X: the OANS map takes the mouse through the radar gauges

> **Aircraft:** A380X | **Where:** ND, OANS airport map (ZOOM range) | **Status:** Tested in the sim by the user: OK: OANS drag and BTV runway / exit clicks.

What it is for

The radar and terrain gauges sit over the ND and caught every mouse event on the map area: the OANS airport map could not be dragged and its runway and exit labels could not be clicked (BTV selection). The gauges only draw, so they no longer take the mouse.

How to use

1. Select the ND ZOOM range to show the OANS airport map (stock FBW OANS, needs its airport data).
2. Drag the map with the mouse or the KCCU cursor, click runway and exit labels as usual.

How to test

#	Do this	Expected result
1	A380X on the ground at an airport with OANS data, ND ZOOM range. Drag the map.	The map moves.
2	Click a runway label, then an exit label (BTV).	The runway and the exit are selected.
3	With the WX or TERR overlay on.	The same: the gauges do not block the mouse.

Surveillance: SURV controls, ECAM SURV lines and TCAS

The A380X SURV pages and pedestal SURV panel work and stay in step, the ECAM procedures sense the SURV system selection, and the TCAS of both aircraft shows TAs while RAs are inhibited, with the real low-altitude hysteresis, plus the FCOM TCAS messages, THRT display and TRAF / SYS pushbutton fixes

Aircraft	A380X (SURV panel, VD messages, ECAM lines) and A32NX + A380X (TCAS)
Where to find it	A380X MFD **SURV** pages and pedestal SURV panel; A380X VD and E/WD; ND traffic display and TCAS controls of both aircraft
Branch	`core/surveillance`, based on `master` (FlyByWire master 76db0aef1 merged in); independent. `core/a380x-mfd-fms` is stacked on it (it adds the FCOM layout of the SURV pages and the SURV memos).
Status	SURV panel and MFD SURV wiring tested in the sim by the user; VD messages and ECAM SURV lines tested in the sim (September 2026); the three TCAS changes tested in the sim by the user: OK. (TCAS display, ND messages, TRAF pb, SYS pbs; added): not sim-tested, unit tests only.
Pictures	Captain MFD, ND, VD and E/WD in the local browser harness, running the full development build installed in the sim (it also has the FCOM layout of the SURV pages from `core/a380x-mfd-fms`, so the SURV page frames on this branch alone look slightly different); the SURV CONTROLS in-sim screenshot; the TCAS symbol legend is a drawing (the harness has no traffic).

1. A380X SURV controls, STATUS & SWITCHING and the pedestal SURV panel
2. A380X VD: separate NO TERR / NO WX DATA messages
3. A380X ECAM procedures: WXR & TAWS and XPDR & TCAS lines sensed
4. A380X ECAM LINE-UP checklist: CABIN CREW line
5. TCAS: intruders shown and announced as TAs while RAs are inhibited
6. TCAS: low-altitude RA and TA aural inhibitions with the real hysteresis
7. A32NX: TCAS message at the left of the bottom line of the ND
8. TCAS traffic display, ND TCAS messages, TRAF pb, WXR/TAWS SYS pbs

1. A380X SURV controls, STATUS & SWITCHING and the pedestal SURV panel

> **Aircraft:** A380X | **Where:** MFD SURV pages; pedestal SURV panel | **Status:** Tested in the sim by the user: OK (pedestal buttons and MFD pages switch in step; WXR buttons drive the radar).

What it is for

The weather radar block of the MFD SURV CONTROLS page, the STATUS & SWITCHING system selection and the pedestal SURV panel buttons were stubs. They now work and drive the same systems.

What it is for

The A380 has no separate radar, TCAS or TAWS control panels: they are on the MFD SURV pages and the pedestal SURV panel (FCOM DSC-34-20-60-50). In FBW the weather radar block of the page was a stub (every button greyed), STATUS & SWITCHING did not reflect failures, and no button of the pedestal panel worked. Now they do, and the page and the panel stay in step.

SURV CONTROLS

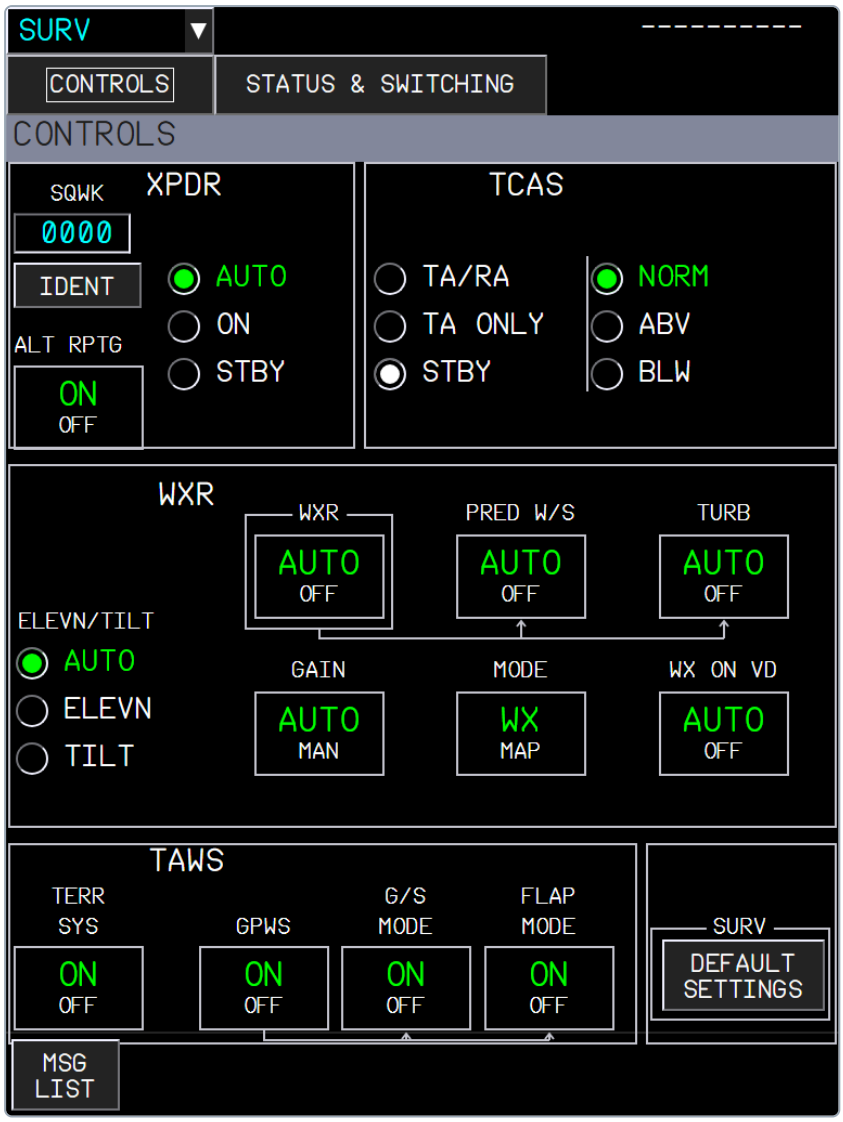


Figure 1. SURV CONTROLS: XPDR (code, IDENT, AUTO / ON / STBY, ALT RPTG), TCAS (TA/RA, TA ONLY, STBY; NORM, ABV, BLW), WXR, TAWS, and DEFAULT SETTINGS.

WXR button	Effect
WXR AUTO / OFF	Weather radar on or off (the ND radar picture of `core/nd-weather-terrain`).
PRED W/S AUTO / OFF	Predictive windshear (written for the systems; MSFS has no predictive windshear).
TURB AUTO / OFF	Turbulence detection (magenta on the ND with the native radar of `core/nd-weather-terrain`).
GAIN AUTO / MAN	Radar gain (written; MSFS has no gain).
MODE WX / MAP	Weather or ground mapping.
WX ON VD AUTO / OFF	Weather on the vertical display.
ELEVN / TILT	AUTO, or a manual elevation or tilt.

The WXR buttons are inhibited only when the selected WXR system has really failed.

STATUS & SWITCHING

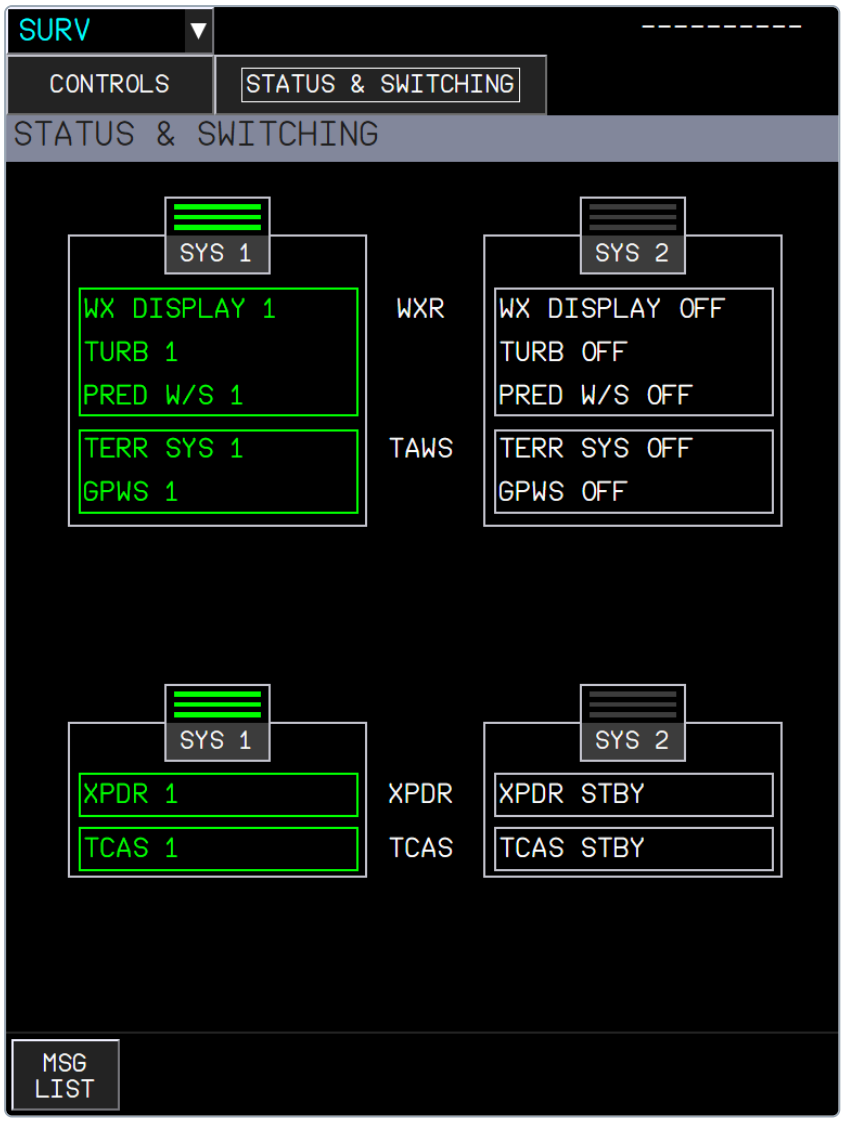


Figure 2. STATUS & SWITCHING: SYS 1 / SYS 2 of the WXR and TAWS (top) and of the XPDR and TCAS (bottom); the selected system is lit, each item shows its real state.

- WXR DISPLAY, TURB, PRED W/S, TERR SYS, GPWS items show the real failure of each system.
- XPDR / TCAS SYS 1 / SYS 2 select the transponder system (the one the TCAS uses) and light from it.

Figure 3. The SURV CONTROLS page in the sim (in-sim screenshot).

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

The pedestal SURV panel

The panel buttons never worked: their light template runs the button code only when the panel is powered, and the pedestal never told it so. They are powered from DC 1 now and linked to the MFD pages:

Panel button	Does	Same as
WXR TAWS SYS 1 / SYS 2	Selects the WXR / TAWS system	STATUS & SWITCHING, top
XPDR TCAS SYS 1 / SYS 2	Selects the transponder system	STATUS & SWITCHING, bottom
G/S MODE	Glide slope mode of the GPWS on / off	SURV CONTROLS, TAWS
TCAS ABV / BLW / TA ONLY	TCAS display and mode	SURV CONTROLS, TCAS

Each button lights from the same value the page uses, so pressing one on the pedestal changes the page, and the reverse.

Limitations and why

Limitation	Why	What to do
PRED W/S and GAIN MAN do nothing visible.	The MSFS radar has no predictive windshear and no gain; the values are written for when it does.	-
One MSFS radar for both systems.	SYS 1 / SYS 2 select the failure state and the lights; the picture comes from the same sim radar.	-

> **NOTE**
> Code clean-up on this branch with no visible effect: the ECAM procedure tables read the selected XPDR & TCAS system through one helper.

How to use

1. Open the SURV pages: MFD system menu **SURV** (or the KCCU SURV key), then **CONTROLS** or **STATUS & SWITCHING**.
2. On CONTROLS set the WXR (WXR, PRED W/S, TURB, GAIN, MODE, WX ON VD, ELEVN/TILT), the TCAS (TA/RA, TA ONLY, STBY; NORM, ABV, BLW) and the TAWS options.
3. On STATUS & SWITCHING, or with the pedestal buttons, select WXR / TAWS SYS 1 or SYS 2 and XPDR / TCAS SYS 1 or SYS 2.
4. Use the pedestal TCAS ABV / BLW / TA ONLY and G/S MODE buttons as shortcuts: the MFD page follows.

How to test

#	Do this	Expected result
1	A380X on the ground, aircraft powered (DC 1 bus powered). MFD > SURV > **STATUS & SWITCHING**.	The selected WXR / TAWS and XPDR / TCAS systems are lit; each item shows the real state of its system.
2	Press **XPDR TCAS SYS 2** on the pedestal SURV panel.	The pedestal button lights; on the MFD page XPDR / TCAS SYS 2 is now the selected one.
3	Select XPDR / TCAS **SYS 1** on the MFD page.	The pedestal SYS 1 button lights, SYS 2 goes off.
4	Press **WXR TAWS SYS 2** on the pedestal, then SYS 1.	The WXR / TAWS selection on the MFD page follows each press.
5	MFD > SURV > **CONTROLS**. Press the pedestal **TA ONLY** button.	The SURV CONTROLS TCAS block shows TA ONLY and the pedestal button is lit; the ND shows TA ONLY.
6	Press the pedestal **ABV**, then **BLW** button.	The TCAS NORM / ABV / BLW selection on the page follows; selecting NORM on the page turns the pedestal light off.
7	Press the pedestal **G/S MODE** button.	The TAWS G/S MODE on SURV CONTROLS changes with it (and back on the second press).
8	On SURV CONTROLS set **WXR** OFF, then AUTO (needs the native radar of `core/nd-weather-terrain` for the ND picture).	The ND radar picture goes away and comes back; the WXR buttons are greyed only if the selected WXR system has failed.
9	Cut the DC 1 bus (or start cold and dark) and press a pedestal SURV button.	Nothing happens and the buttons stay dark.

2. A380X VD: separate NO TERR / NO WX DATA messages

> **Aircraft:** A380X | **Where:** Vertical display (VD) under the ND | **Status:** Tested in the sim (September 2026).

What it is for

TERR SYS OFF removed both the terrain and the weather from the VD message logic, so the VD always said "NO TERR AND WX DATA AVAILABLE". Each source now has its own message.

Messages on the vertical display

The VD message was always "NO TERR AND WX DATA AVAILABLE" when TERR SYS was off. Now:

Situation	VD message
TERR SYS OFF only	NO TERR DATA AVAILABLE
WXR failed or OFF, or WX ON VD OFF	NO WX DATA AVAILABLE
Both	NO TERR AND WX DATA AVAILABLE

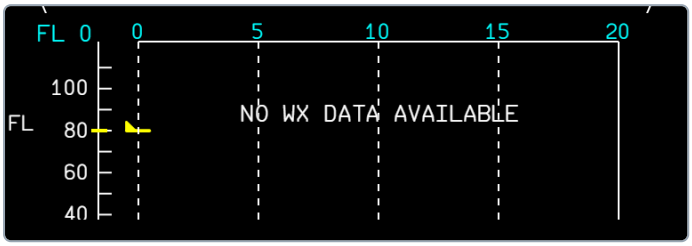


Figure 4. WXR off: NO WX DATA AVAILABLE.

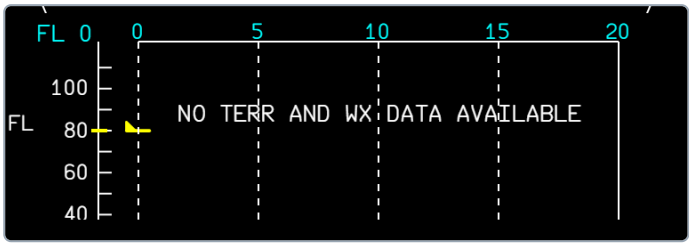


Figure 5. TERR SYS off and WXR off: NO TERR AND WX DATA AVAILABLE.

How to use

1. Nothing to set: the VD message follows the TERR SYS, WXR and WX ON VD buttons of MFD > SURV > CONTROLS.

How to test

#	Do this	Expected result
1	A380X, ND in ARC or ROSE NAV so that the VD shows. MFD > SURV > CONTROLS: TERR SYS, WXR and WX ON VD all on.	No "NO ... DATA" message on the VD.
2	Set **TERR SYS** OFF only.	VD: **NO TERR DATA AVAILABLE**.
3	TERR SYS back ON, set **WX ON VD** OFF.	VD: **NO WX DATA AVAILABLE**.
4	WX ON VD back ON, set **WXR** OFF.	VD: **NO WX DATA AVAILABLE**.
5	With WXR OFF, also set TERR SYS OFF.	VD: **NO TERR AND WX DATA AVAILABLE**.

3. A380X ECAM procedures: WXR & TAWS and XPDR & TCAS lines sensed

> **Aircraft:** A380X | **Where:** E/WD procedures; MFD SURV STATUS & SWITCHING | **Status:** Tested in the sim (September 2026).

What it is for

The dual ADR and IR fault procedures ask the crew to move the WXR & TAWS and the XPDR & TCAS to the system that still has valid data. Those lines were hard-coded as done, so the ECAM ticked them even when the wrong system stayed selected. They now follow the selection made on the MFD SURV STATUS & SWITCHING page, and the TCAS 1 / 2 FAULT procedures show their XPDR & TCAS line, which was hidden.

Where the systems are selected

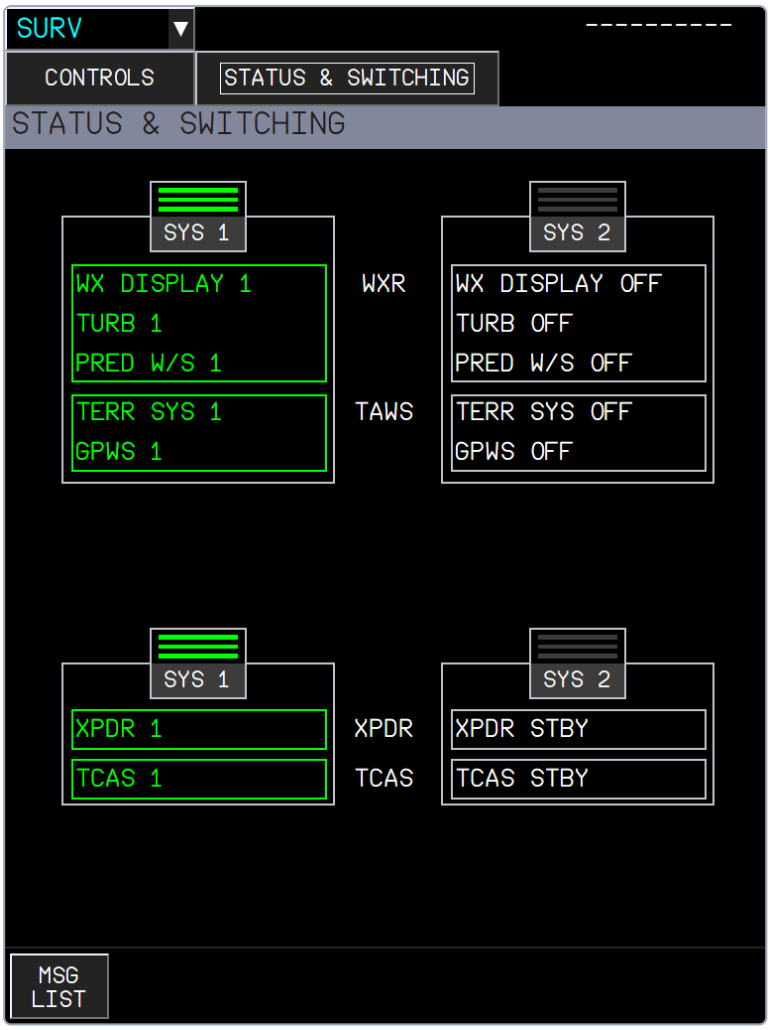


Figure 6. MFD SURV STATUS & SWITCHING: the SYS 1 / SYS 2 buttons of WXR & TAWS (top) and XPDR & TCAS (bottom).

What you see

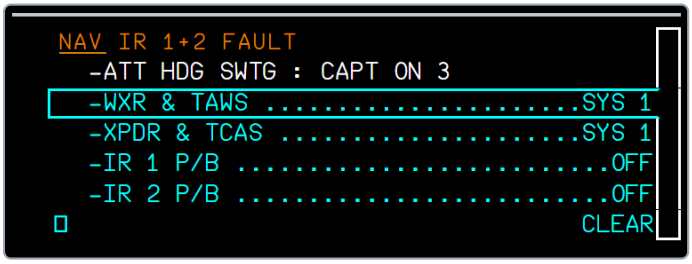


Figure 7. IR 1+2 FAULT with both systems on SYS 2: the two lines wait, in cyan.

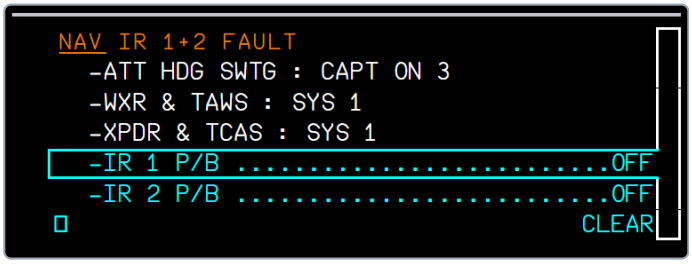


Figure 8. After selecting SYS 1 on the SURV page: the lines are done (white) and the cursor moves on.

Procedure	WXR & TAWS	XPDR & TCAS
ADR 1+2 FAULT, ADR 2+3 FAULT, IR 1+2 FAULT, IR 2+3 FAULT	SYS 1	SYS 1
ADR 1+3 FAULT, IR 1+3 FAULT	SYS 2	SYS 2
TCAS 1 FAULT	-	SYS 2
TCAS 2 FAULT	-	SYS 1

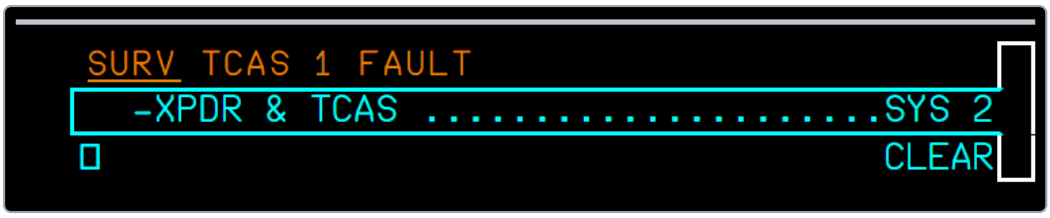


Figure 9. TCAS 1 FAULT: the XPDR & TCAS line is now shown.

Situation	What you should see
The right system is already selected when the fault appears	The line is shown already done.
The crew selects the other system	The line ticks within a second; the cursor moves to the next line.
The crew goes back to the failed side	The line becomes due again (it follows the live selection).

Limitations and why

Limitation	Why	What to do
TCAS 1 FAULT and TCAS 2 FAULT come together.	FBW has one TCAS computer: both procedures read the same TCAS fault (an FBW note in the code).	Select the system the procedure you follow asks for.

How to use

1. When an ADR / IR dual fault or a TCAS fault procedure asks for **WXR & TAWS ... SYS n** or **XPDR & TCAS ... SYS n**, select that system on MFD > SURV > STATUS & SWITCHING (or with the pedestal SYS buttons).
2. The line ticks by itself; the E/WD cursor moves to the next line.

How to test

#	Do this	Expected result
1	A380X on the ground, one engine running (the ADR faults are not raised in flight phases 1 and 12). MFD > SURV > STATUS & SWITCHING: select WXR / TAWS **SYS 2** and XPDR / TCAS **SYS 2**.	Both SYS 2 selected.
2	Overhead ADIRS panel: set the **ADR 1** and **ADR 2** pushbuttons OFF.	E/WD: **ADR 1+2 FAULT**. Its WXR & TAWS ... SYS 1 and XPDR & TCAS ... SYS 1 lines are cyan (to do).
3	Select WXR / TAWS **SYS 1** on the SURV page.	Within a second the WXR & TAWS line turns white (done).
4	Select XPDR / TCAS **SYS 1** (page or pedestal).	The XPDR & TCAS line is done; the cursor moves on.
5	Select XPDR / TCAS SYS 2 again.	The XPDR & TCAS line is due again (it follows the live selection). Set the ADR pushbuttons back ON afterwards.
6	Raise a TCAS fault: flyPad > Failures > ATA 34 **RA SYS A** and **RA SYS B** (the TCAS uses radio altimeter 1, else 2), on the ground.	E/WD: TCAS 1 FAULT and TCAS 2 FAULT, each now with its XPDR & TCAS line (before: hidden). The radio altimeter alerts show too.
7	Select XPDR / TCAS SYS 2, then SYS 1.	With SYS 2 the TCAS 1 FAULT line is done, with SYS 1 the TCAS 2 FAULT line. Clear the failures afterwards.

4. A380X ECAM LINE-UP checklist: CABIN CREW line

> **Aircraft:** A380X | **Where:** E/WD normal checklist LINE-UP | **Status:** Tested in the sim (September 2026).

What it is for

The A380 FCOM BEFORE TAKEOFF checklist has a CABIN CREW item; the ECAM LINE-UP checklist did not.

LINE-UP checklist

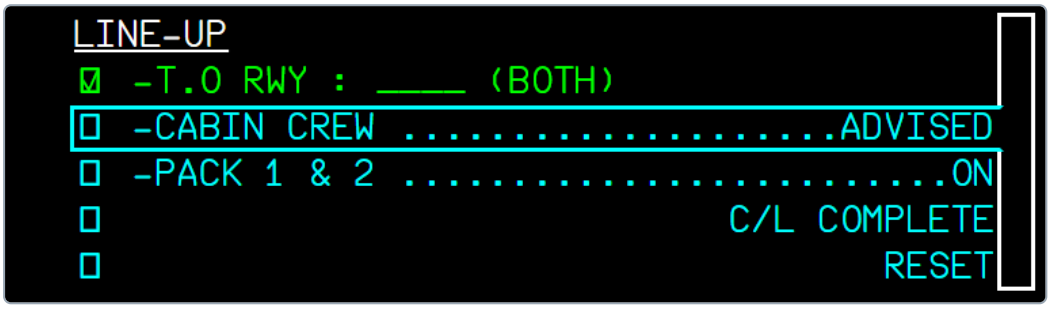


Figure 10. ECAM LINE-UP: the CABIN CREW ... ADVISED line (crew check).

The ECAM LINE-UP checklist gets the CABIN CREW line that the flyPad checklist already had (A380 FCOM BEFORE TAKEOFF checklist: CABIN CREW ... ADVISE). It is ticked by the crew.

Limitations and why

Limitation	Why	What to do
No SURV lines in the normal checklists.	The A380 FCOM normal checklists (PRO-NOR-C-L) have none (SURV, WX & TERR and TCAS are SOP flow items); an earlier version of this change added some and was reverted.	-

How to use

1. At line-up, call the ECAM LINE-UP checklist (C/L key on the ECAM control panel).
2. Advise the cabin crew, then tick the **CABIN CREW ... ADVISED** line.

How to test

#	Do this	Expected result
1	A380X ready for takeoff. Press **C/L** on the ECAM control panel and go to the **LINE-UP** checklist.	The checklist shows a **CABIN CREW ... ADVISED** line, in cyan (to do).
2	Tick the line (ECAM control panel or KCCU).	The line turns white (done); it is not ticked by itself.

5. TCAS: intruders shown and announced as TAs while RAs are inhibited

> **Aircraft:** A32NX and A380X | **Where:** ND traffic display; TCAS mode | **Status:** Tested in the sim by the user: OK.

What it is for

With every RA inhibited (TA ONLY selected, or low: below 900 ft radio altitude in descent, until 1100 ft in climb, see the next feature), the TCAS kept classifying an intruder that met the RA criteria as an RA. The ND drew it as a red RA square, and since it was not a TA, there was no "TRAFFIC, TRAFFIC" and no TA state. A real TCAS in TA ONLY shows it as an amber TA with the TA aural. The classification is now capped at TA while all RAs are inhibited.

Figure 11. The ND traffic symbols: other, proximate, TA, RA, with the relative altitude in hundreds of feet and the vertical trend arrow.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

What you see

Figure 12. A32NX: TA ONLY, centred at the bottom of the ND (A320 FCOM DSC-34-SURV-60-20 figure).

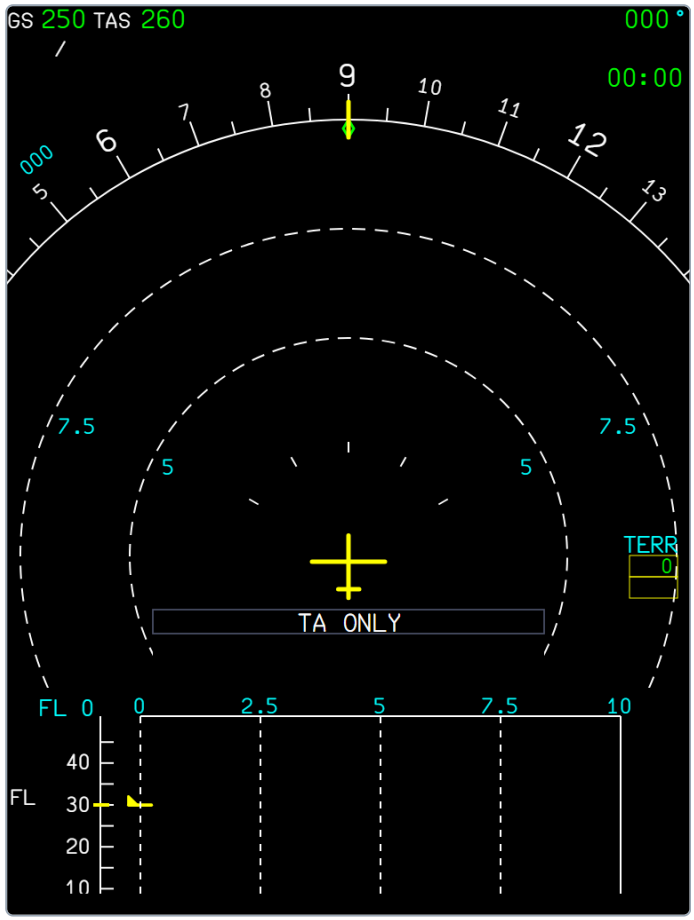


Figure 13. A380X: TA ONLY in the message box.

Situation	What you should see
TA ONLY selected, intruder meeting the RA criteria	Amber TA circle and "TRAFFIC, TRAFFIC" (before: red RA square, no aural).
Below 900 ft radio altitude in descent, until 1100 ft in climb (all RAs inhibited)	Same: TAs only, TA ONLY on the ND.
Partial inhibits (no CLIMB / DESCEND / INCREASE DESCENT RA)	Unchanged: RAs are still issued.
An RA already running when the inhibit starts	It runs to its own end.

Limitations and why

Limitation	Why	What to do

Same change on both aircraft.	The A32NX TcasComputer and the A380X LegacyTcasComputer share the classification code.	-

How to use

1. Select TA ONLY: A32NX ATC/TCAS panel TCAS mode selector to **TA**; A380X MFD > SURV > CONTROLS TCAS **TA ONLY** (or the pedestal TA ONLY button).
2. Traffic that would give an RA is shown as an amber TA with the "TRAFFIC, TRAFFIC" aural.

How to test

#	Do this	Expected result
1	In flight above 1100 ft radio altitude with traffic (AI or multiplayer) converging on you. TCAS in TA/RA.	An intruder meeting the RA criteria is a red square with an RA aural.
2	Select TA ONLY (A32NX: TCAS mode selector TA; A380X: SURV CONTROLS TA ONLY).	TA ONLY on the ND. The same intruder is now an amber TA circle with "TRAFFIC, TRAFFIC" (before: a red RA square and no aural).
3	Back to TA/RA during an RA already running, then TA ONLY again.	An RA already running when the inhibition starts runs to its own end.

6. TCAS: low-altitude RA and TA aural inhibitions with the real hysteresis

> **Aircraft:** A32NX and A380X | **Where:** ND traffic display (TA ONLY message); TCAS aurals | **Status:** Tested in the sim by the user: OK (with the TA ONLY change).

What it is for

The TCAS inhibits all RAs close to the ground, and lower down the TA aural too. Both used a flat threshold (1000 ft and 500 ft radio altitude). The A380 FCTM (Supplementary Information, TCAS) gives a hysteresis, the FCOM "below 1000 ft ±100 ft". Both TCAS computers now use it:

Inhibition	Starts (descending through)	Ends (climbing through)
All RAs (TA ONLY on the ND)	900 ft radio altitude	1100 ft radio altitude
TA aural ("TRAFFIC, TRAFFIC")	400 ft radio altitude	600 ft radio altitude

- Without a valid radio altitude, neither low-altitude inhibition applies.
- The four altitudes are named constants of the TCAS code (no other effect).

How to use

1. Nothing to set: the inhibitions follow the radio altitude.

How to test

#	Do this	Expected result
1	A32NX or A380X on approach, TCAS in TA/RA, descending through 1000 ft radio altitude. Watch the ND.	No TA ONLY message yet at 1000 ft.
2	Continue below 900 ft radio altitude.	**TA ONLY** appears on the ND (all RAs inhibited).
3	Go around and climb.	TA ONLY stays up to 1100 ft radio altitude and disappears above it.

4	With traffic close by on a low approach (AI or multiplayer), descend below 400 ft radio altitude.	The TA symbol stays, the "TRAFFIC, TRAFFIC" aural is not given; it is given again above 600 ft in the climb.

7. A32NX: TCAS message at the left of the bottom line of the ND

> **Aircraft:** A32NX | **Where:** ND, bottom line, left of centre | **Status:** Fix installed, not sim-tested yet. Before it, the centred version showed TA ONLY half outside its box, next to the FM message, in the sim.

What it is for

The TCAS message ("TA ONLY", "TCAS") sits at the left of its line at the bottom of the ND, as the A320 FCOM ND messages figure shows it (DSC-34-SURV-60-20, item 2: TA ONLY at the bottom, left of centre). The bottom centre stays free for the FM messages (GPS PRIMARY...). A first change tried to centre it with a CSS text-anchor plus a CSS transform; the sim renderer applies the text-anchor but not the transform on SVG text, so the message was centred on its left start point and stood half outside its box. That rule is removed. TCAS STBY is not shown on the A32NX ND (hidden by its stylesheet), as before.

How to use

1. Nothing to set: the message shows wherever the TCAS sends one.

How to test

#	Do this	Expected result
1	A32NX, ND in ROSE NAV or ARC. Set the TCAS mode selector on the ATC/TCAS panel to **TA**.	**TA ONLY** in white at the bottom of the ND, left of centre, fully inside its grey box; an FM message such as GPS PRIMARY, when shown, stays readable at the centre.
2	Set the TCAS mode selector back to **TA/RA**.	TA ONLY disappears.

8. TCAS traffic display, ND TCAS messages, TRAF pb, WXR/TAWS SYS pbs

> **Aircraft:** A32NX and A380X | **Where:** ND traffic display and messages; A32NX TRAFFIC selector; A380X FCU TRAF pb, pedestal SURV panel, MFD SURV STATUS & SWITCHING | **Status:** Not sim-tested (added, unit tests only: TcasTrafficSelector, TcasSettingMessage, SurvWxrTawsSysPbs and TcasTrafAutoSelect specs, all failing without the change).

What it is for

Deviations from the FCOM TCAS and SURV descriptions found by the FCOM gap audit (A320 FCOM DSC-34-SURV-40 TCAS, A380 FCOM SURV TCAS and SURV panel descriptions):

- **A32NX TRAFFIC selector THRT:** while a TA or an RA is present, the proximate and the other intruders within ±2700 ft are shown too, not only the threat (A320 FCOM: "THRT: Proximate and other intruders are displayed only if a TA or an RA is present").
- **ND TCAS messages (both aircraft): TCAS: CHANGE MODE** in PLAN mode and **TCAS: REDUCE RANGE** above the 40 NM range while there is a TA or an RA; the message is amber for a TA and red for an RA.
- **A380X only: TCAS: INCREASE RANGE** in ROSE VOR or ROSE LS with the ZOOM range in flight; no such message in ZOOM on the ground.
- **A380X WXR/TAWS SYS 1 and SYS 2 pushbuttons** (pedestal SURV panel and MFD SURV STATUS & SWITCHING): pressing the pushbutton that is already lit keeps that system selected. Before, it selected no system.
- **A380X TRAF pb:** the TCAS now reads the FCU TRAF light (the variable it read before was written by nothing). A TA or an RA selects TRAF on each side once, so the traffic shows; the crew can then deselect it.

Situation	What you should see
TA or RA present, ND in PLAN mode	ND message **TCAS: CHANGE MODE** (amber for a TA, red for an RA).
TA or RA present, ND range above 40 NM	ND message **TCAS: REDUCE RANGE** (amber / red).
A380X, TA or RA present, ROSE VOR or ROSE LS in ZOOM, in flight	ND message **TCAS: INCREASE RANGE**. On the ground in ZOOM: no message.
A380X, TRAF off on the FCU, a TA or an RA starts	TRAF comes on by itself on both sides (once); a press turns it off again.
A32NX, TRAFFIC selector THRT, a TA or an RA	The threat plus the proximate and other intruders within ±2700 ft show on the ND; with no TA/RA, no traffic.

Limitations and why

Limitation	Why	What to do
No TCAS TEST.	Not done yet.	-
No-bearing intruder lines are not shown.	Not done yet.	-

How to use

1. Nothing to set: the messages and the TRAF selection follow the TCAS.
2. A32NX: use the THRT position of the TRAFFIC selector (ATC/TCAS panel) to see traffic only around a TA or an RA.
3. A380X: select the WXR/TAWS system with its SYS pushbutton; pressing the lit one again changes nothing.

How to test

#	Do this	Expected result
1	In flight with traffic (AI or multiplayer) that gives a TA. ND in ARC, range 80 NM.	ND: **TCAS: REDUCE RANGE** in amber (red during an RA).
2	During the TA, select PLAN mode.	ND: **TCAS: CHANGE MODE**.
3	A380X: TRAF off on the FCU before the TA.	At the TA the TRAF light comes on, on both sides; press TRAF: it goes off and stays off for that TA.
4	A380X: ROSE LS, ZOOM range, in flight, during a TA.	ND: **TCAS: INCREASE RANGE**.
5	A380X on the ground: MFD > SURV > STATUS & SWITCHING, WXR/TAWS SYS 1 lit. Press SYS 1 again (MFD page, then the pedestal pushbutton).	SYS 1 stays selected and lit (before: no system).
6	A32NX: TRAFFIC selector THRT, traffic around but no TA.	No traffic on the ND; when a TA comes, the TA and the other intruders within ±2700 ft show.

What PR testers should check

- The ND message position: centred, above the MODE/RANGE CHANGE line.
- A380X: a TA turns TRAF on, and a second press turns it off.
- The TCAS message colours (amber TA, red RA) and the THRT traffic display on the A32NX.
- These changes are not sim-tested: report any message that does not match the FCOM.

Independent CPT and F/O MCDUs

The two A32NX MCDUs stop mirroring each other: each has its own page, scratchpad, messages and keyboard entry, on the one shared FMS

Aircraft	A32NX
Where to find it	The two MCDUs on the pedestal (and the SimBridge remote MCDU)
Branch	`core/a32nx-mcdu`, based on `master` (FlyByWire master 76db0aef1 merged in); independent of the other core branches. The close-up cockpit model needs a companion change in the `aircraft-large-files` repository (commit 5563e31, see the last feature).
Status	Independent MCDUs tested in the sim by the user: OK (pages, scratchpads, shared FMS data, messages, BRT/DIM and power per side). The keyboard entry, SimBridge remote MCDU and ATSU status follow-up is not sim-tested.
Pictures	The MCDU gauge in the local browser harness, running the build installed in the sim; the gauge draws the CPT screen in its left half and the F/O screen in its right half, shown here side by side.

1. Independent CPT and F/O MCDUs

> **Aircraft:** A32NX | **Where:** The two MCDUs on the pedestal | **Status:** Tested in the sim by the user: OK: all 7 test points (pages, scratchpads, shared FMS data, per-screen messages, BRT/DIM, power).

What it is for

The two MCDUs mirrored each other: same page, same scratchpad. On the aircraft each MCDU is its own display, working on the shared FMGC data (A320 FCOM DSC-22_10-10). Now each pilot can work on a different page.

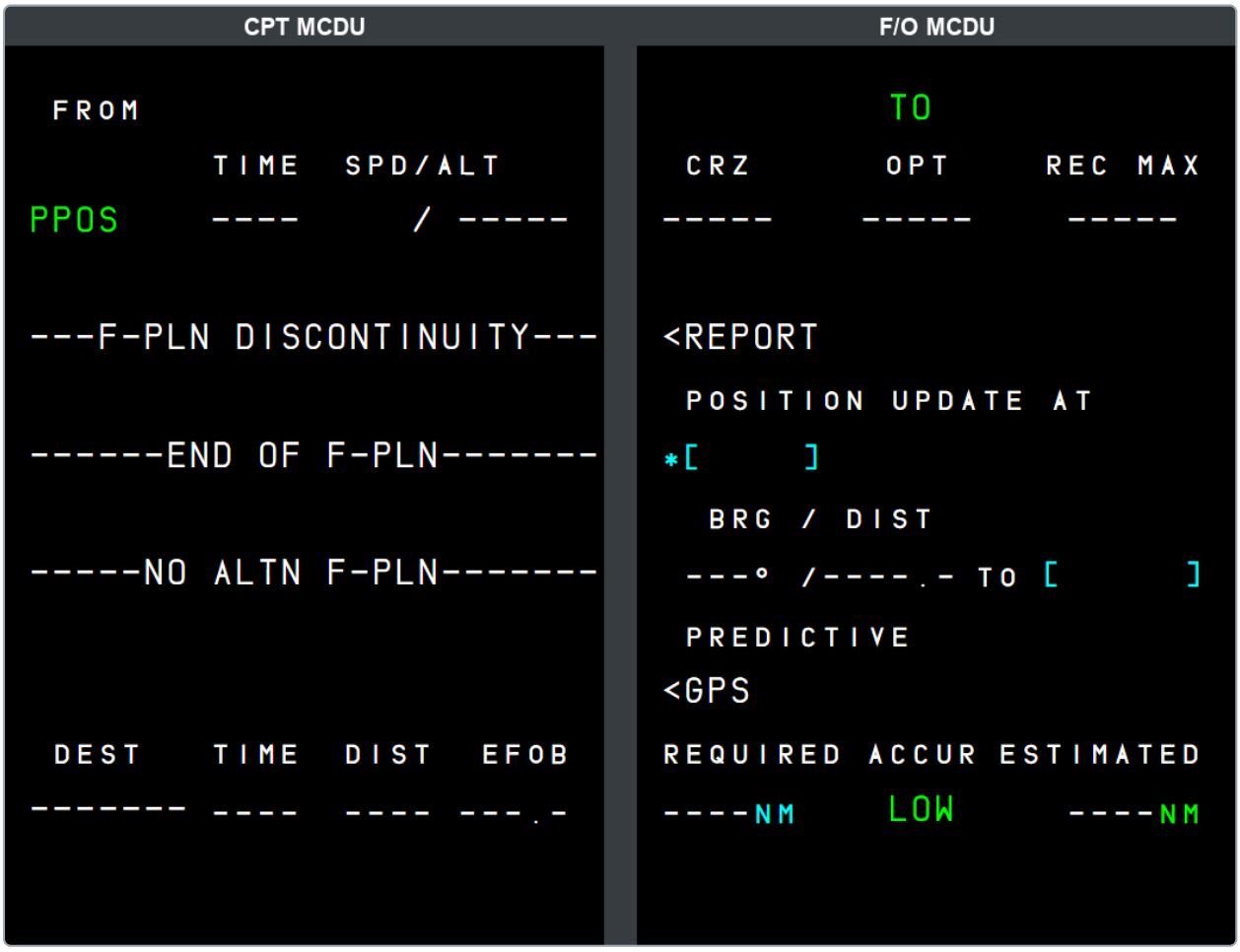


Figure 1. CPT on F-PLN, F/O on PROG at the same time.

What is per MCDU and what is shared

Per MCDU	Shared (one FMS)
The page shown, the line keys	The flight plans, performance, fuel, navigation data
The scratchpad and what is typed in it	The data entered from either MCDU
The scratchpad messages (FORMAT ERROR, NOT ALLOWED ...) of an entry	FMGC messages (queued on both)
The keyboard entry (the MCDU that was clicked)	The EFIS symbols, the guidance

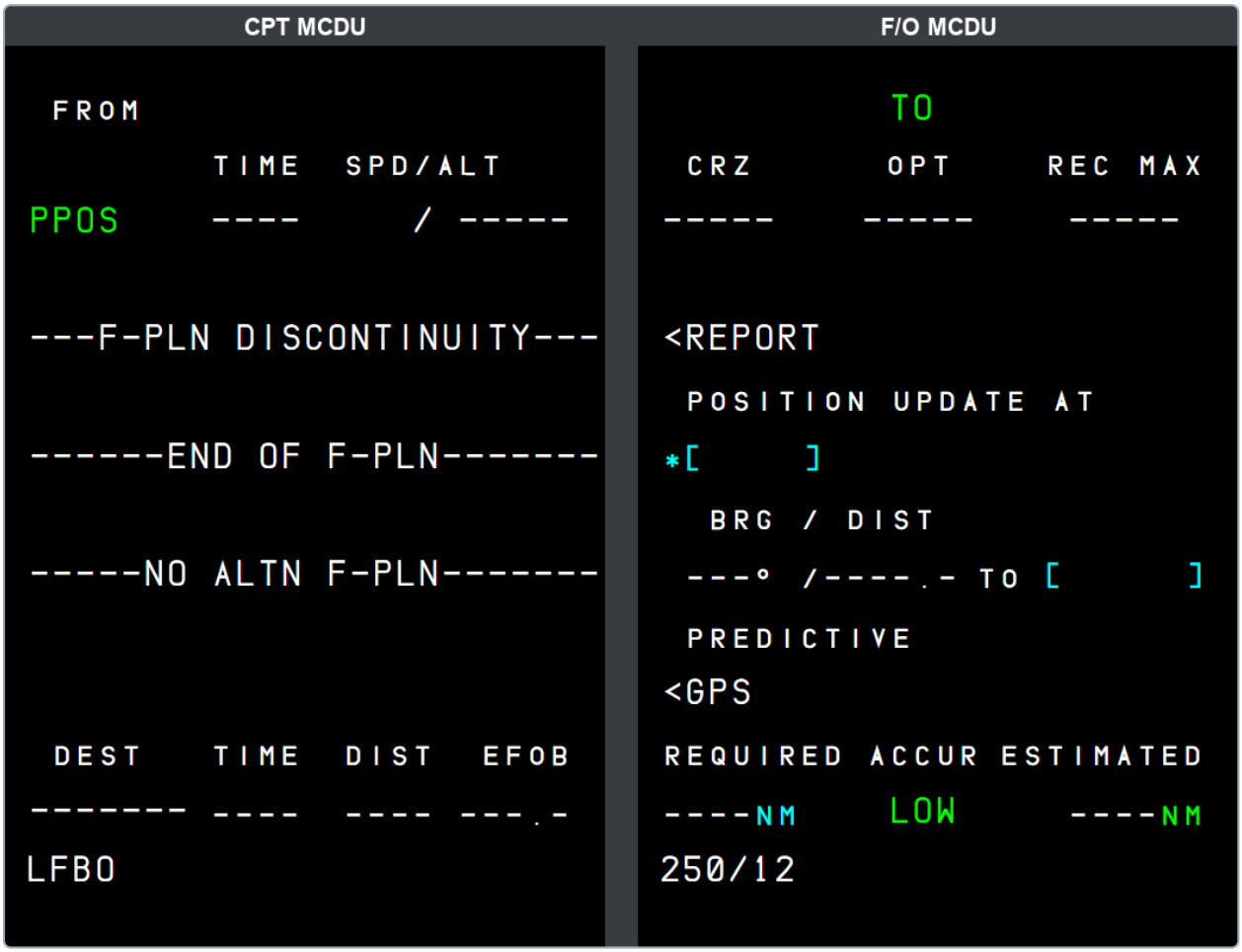


Figure 2. Each scratchpad keeps its own text: LFBO on the CPT, 250/12 on the F/O.

Messages

Figure 3. A wrong entry on the F/O MCDU: FORMAT ERROR there only.

Figure 4. CLR on the F/O MCDU: its message clears and its entry comes back; the CPT scratchpad is untouched.

Situation	What you should see
An entry is rejected	The message shows on the MCDU where the entry was made.
An FMGC message (e.g. CHECK TAKE OFF DATA)	Queued on both MCDUs; CLR on one clears it there only.
The FMS changes pages by itself (flight phase change on PROG / PERF / IDENT, engine-out page, INIT B to FUEL PRED)	Applies to both MCDUs.
ATSU system status (not the answer to an entry)	On the ATSU scratchpad of both MCDUs.
Typing with the PC keyboard (next feature)	Goes to the MCDU that was clicked last; only that MCDU animates its keys.
SimBridge remote MCDU (next feature)	CPT screen on its left side, F/O screen on its right side.

Power

- CPT MCDU: AC ESS SHED bus. F/O MCDU: AC 2 bus.
- An unpowered MCDU ignores its keys; the other one keeps working.

How it works

The FMS cannot be run twice: the MCDU code is the aircraft FMS (flight plan, guidance, EFIS symbols) as well as the display. So the two MCDUs are two screens on that one FMS, the way the A380X draws its two MFD screens: one 2048x1024 MCDU gauge, the CPT MCDU drawn in its left half and the F/O MCDU in its right half, and each screen of the cockpit model samples its own half. The F/O screen has its own copy of what belongs to a screen (page, line key actions, scratchpad, message queue, key handlers); everything else is the shared FMS.

Limitations and why

Limitation	Why	What to do
No return to the MENU page after a power loss.	Same as the CPT MCDU before this change.	-

Close-up cockpit model: both screens show the whole texture until the companion change is merged.	The 4 UV values of the high-detail model (LOD00) live in the aircraft-large-files repository; this branch only changes LOD01 (see the last feature).	Install the aircraft-large-files change (commit 5563e31) and bump the large-files pointer.

How to use

1. Use each MCDU as on the aircraft: each pilot can open any page on his MCDU without changing the other one.
2. Data entered on either MCDU (flight plan, INIT, PERF, fuel) is FMS data: it shows on both MCDUs.
3. A scratchpad message (FORMAT ERROR, NOT ALLOWED...) shows on the MCDU where the entry was made; clear it there with CLR.
4. An FMGC message is queued on both MCDUs: clear it with CLR on each MCDU.

How to test

#	Do this	Expected result
1	A32NX cold and dark at a gate. Set external power or the batteries and the APU so that AC ESS SHED and AC 2 are powered.	Both MCDUs show the MCDU MENU page.
2	CPT MCDU: MCDU MENU > FMGC, then **F-PLN**. F/O MCDU: stay on MCDU MENU, then press **PROG**.	CPT MCDU on F-PLN, F/O MCDU on PROG at the same time (before: both on the same page).
3	Type `LFBO` on the CPT MCDU and `250/12` on the F/O MCDU, without pressing a line key.	Each scratchpad keeps its own text.
4	On the CPT MCDU, open INIT A and enter a city pair (e.g. `LFBO/LFPG`). Open F-PLN on the F/O MCDU.	The F/O F-PLN shows the new flight plan: the FMS data is shared.
5	On the F/O MCDU, enter a wrong value on a page (e.g. `ABC` into a speed field).	FORMAT ERROR on the F/O scratchpad only; the CPT scratchpad is unchanged.
6	Press **CLR** on the F/O MCDU.	The F/O message clears and its entry comes back; the CPT scratchpad is untouched.
7	Press BRT / DIM on one MCDU.	Only that MCDU screen changes brightness.
8	Fail AC bus 2: flyPad > Failures > ATA 24 > **AC 2**. Press keys on both MCDUs.	The F/O MCDU ignores its keys; the CPT MCDU (AC ESS SHED) keeps working.

2. Keyboard entry, SimBridge remote MCDU and ATSU status on both MCDUs

> **Aircraft:** A32NX | **Where:** The MCDUs; flyPad > Settings > Realism; SimBridge remote MCDU | **Status:** Not sim-tested.

What it is for

After the split, three things still knew only the CPT MCDU: the PC keyboard entry, the SimBridge remote MCDU and the ATSU system messages. They now work with both MCDUs.

What you should see

Situation	What you should see
Keyboard entry on, click on an MCDU screen	That MCDU gets the keyboard (cyan band at the top of its screen, highlighted scratchpad); the other MCDU loses it. Only one MCDU has the keyboard at a time.

Typing on the PC keyboard	The text goes into the scratchpad of the MCDU that has the keyboard; only the keys of that MCDU animate (before: both).
That MCDU is not powered	Typing does nothing (CPT MCDU: AC ESS SHED; F/O MCDU: AC 2).
A click outside both screens	The keyboard goes to the CPT MCDU, as before.
SimBridge remote MCDU	Left side = CPT MCDU, right side = F/O MCDU (before: the CPT MCDU on both sides).
An ATSU system status that is not the answer to an entry (ATIS auto-update failure, a failed send from the mailbox, mailbox full)	Shown on the ATSU scratchpad of both MCDUs (before: CPT only).
An ATSU status that answers an entry made on a page	Shown on the MCDU of that page only.

Limitations and why

Limitation	Why	What to do
An MCDU does not go back to the MENU page after a power loss.	Neither MCDU did before (only the model blanks the screen); a reset could also cause nuisance resets on short bus transfers.	-
The SimBridge remote MCDU needs SimBridge.	It is an external program.	Install and start SimBridge.

How to use

1. Turn on **flyPad > Settings > Realism > MCDU Keyboard Input** (optionally set the MCDU Focus Timeout).
2. Click the screen of the MCDU you want to type into, then type. Click it again (or press Alt) to give the keyboard back.
3. For the remote MCDU, enable **flyPad > Settings > Sim Options > Enable MCDU Server Connection** and open the SimBridge remote MCDU.

How to test

#	Do this	Expected result
1	A32NX powered, both MCDUs on the MCDU MENU. Turn on flyPad > Settings > Realism > MCDU Keyboard Input.	The setting is on; the focus timeout field appears.
2	Click the F/O MCDU screen, then type `ABC` on the PC keyboard.	The F/O MCDU top band turns cyan, ABC appears in the F/O scratchpad only, and only the F/O MCDU keys animate.
3	Click the CPT MCDU screen and type `123`.	The keyboard moves to the CPT MCDU: 123 in the CPT scratchpad; the F/O MCDU loses its cyan band.
4	Activate the flyPad failure ATA 24 > **AC 2**, click the F/O MCDU and type.	Nothing is typed on the F/O MCDU.
5	With SimBridge running: enable flyPad > Settings > Sim Options > Enable MCDU Server Connection, open the SimBridge remote MCDU, put the two MCDUs on different pages.	The remote MCDU shows the CPT page on its left side and the F/O page on its right side.
6	Optional (hard to trigger on purpose): make the ATSU report a system status, e.g. an ATIS auto-update that fails.	The ATSU message appears on both MCDUs.

3. Close-up cockpit model: the MCDU screens in aircraft-large-files

> **Aircraft:** A32NX | **Where:** Cockpit model, high detail (LOD00) | **Status:** The same 4 UV values were tested in the sim with the code above; the large-files commit is byte-identical to the tested model. Pushed to the fork TCdr/aircraft-large-files.

What it is for

The high-detail interior model `A320_NEO_INTERIOR_LOD00`, the one shown in the cockpit, is not in the aircraft repository: it lives in `flybywiresim/aircraft-large-files`. Its two MCDU screens (`SCREEN_MCDUL`, `SCREEN_MCDUR`) both sampled the whole MCDU texture. The companion commit 5563e31 (branch `feat/a32nx-mcdu-screen-uv-split`) maps each screen to one half:

Mesh	Texture U before	Texture U after
SCREEN_MCDUL (CPT)	0 .. 1	0 .. 0.5 (left half)
SCREEN_MCDUR (F/O)	0 .. 1	0.5 .. 1 (right half)

Four float16 values change in `A320_NEO_INTERIOR_LOD00.bin.part01`, nothing else. This branch makes the same change to `A320_NEO_INTERIOR_LOD01` (in the aircraft repository); the lower LODs have no MCDU screens.

Limitations and why

Limitation	Why	What to do
Without the companion change, each MCDU screen shows both MCDU pictures side by side, squeezed.	This branch draws the 2048x1024 two-screen texture; the large-files pointer of the branch still points to the unchanged model (5f07408).	Merge the large-files change first, then bump the `large-files` submodule pointer on this branch.
The companion alone does nothing useful.	With the stock 1024x1024 MCDU gauge, each screen would show half of the MCDU picture.	Always install it with this branch.

How to use

1. Merge the aircraft-large-files commit 5563e31, bump the `large-files` submodule pointer, then build the A32NX with this branch.
2. Nothing to set in the cockpit.

How to test

#	Do this	Expected result
1	Build and install the A32NX with this branch and the large-files change. Load a flight, power the aircraft and look at the MCDUs from the pilot seat (high-detail model).	Each MCDU shows one full MCDU page, not stretched or squeezed.
2	CPT MCDU: MCDU MENU > FMGC. F/O MCDU: stay on MCDU MENU.	Two different pages.
3	Type on one MCDU keypad.	Only that MCDU scratchpad changes.

ECAM procedures, EMER CANC and FIRE TEST

The A380X ECAM gets the FCOM procedures and their STATUS, a working EMER CANC pushbutton and MORE key, and both aircraft can keep the FIRE TEST running a few seconds after the button is released

Aircraft	A380X (ECAM, FWS) and A32NX + A380X (FIRE TEST option)
Where to find it	A380X ECP (ABN PROC, EMER CANC, RCL, MORE, STS); EWD and SD STATUS page; overhead FIRE TEST; flyPad > Settings > Realism
Branch	`core/ecam-fws`, based on `master` (FlyByWire master 76db0aef1 merged in); independent.
Status	EMER CANC tested in the sim by the user: OK. ABN PROC STATUS tested by the user: OK (CKPT WINDOW CRACKED and FUEL LEAK). FIRE TEST option tested by the user: A32NX OK, A380X OK. ECAM procedures: tested in the sim (September 2026). ECP MORE key and light: not sim-tested (13 unit tests).
Pictures	EWD pictures (ECAM procedures) and the flyPad Realism page (FIRE TEST) from the local browser harness, running the build installed in the sim. No pictures for EMER CANC, the STATUS lines and the MORE key: the flight warning system does not run in the harness.

1. A380X ECAM procedures from the FCOM
2. A380X STATUS of the ABN PROC procedures
3. A380X EMER CANC pushbutton
4. A380X ECP MORE key and light on the STATUS page
5. FIRE TEST extended after the button release

1. A380X ECAM procedures from the FCOM

> **Aircraft:** A380X | **Where:** ECP ABN PROC; EWD | **Status:** Tested in the sim (September 2026). CKPT WINDOW CRACKED and FUEL LEAK were also run by the user with their STATUS.

What it is for

Many ABN PROC procedures were marked WIP: they had no lines and no FWS entry, so selecting one showed nothing. They now carry the FCOM text (PRO-ABN-ECAM-10 and -20). Some sensed alerts also got their FCOM lines, with the lines the FWS can check ticking by themselves.

Opening a not-sensed procedure

1. Press **ABN PROC** on the ECP. The EWD lists the red emergency procedures, then the categories (ENG, F/CTL, L/G, NAV, FUEL, MISCELLANEOUS).
2. Move the cursor with the ECP **UP / DOWN** keys and press **CHECK**: the category opens, then the procedure.
3. The procedure shows in grey with **ACTIVATE**: a preview. CHECK on ACTIVATE runs it: the lines turn cyan and the cursor walks through them.

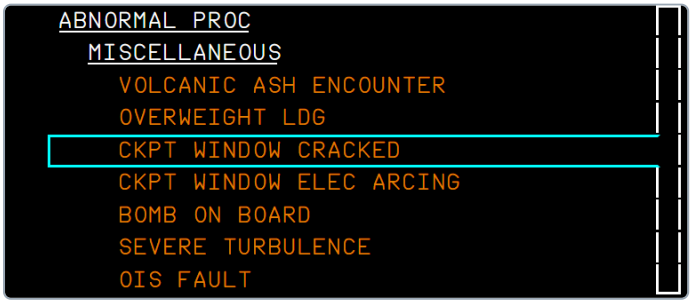


Figure 1. ABN PROC > MISCELLANEOUS.

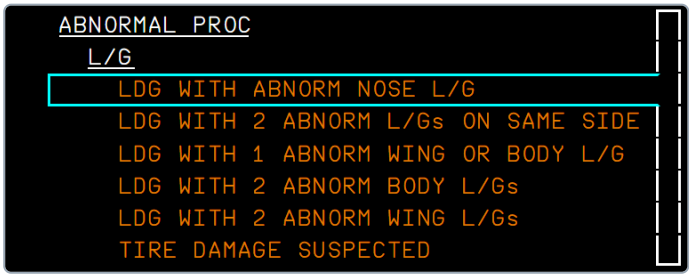


Figure 2. ABN PROC > L/G: the landing gear procedures.

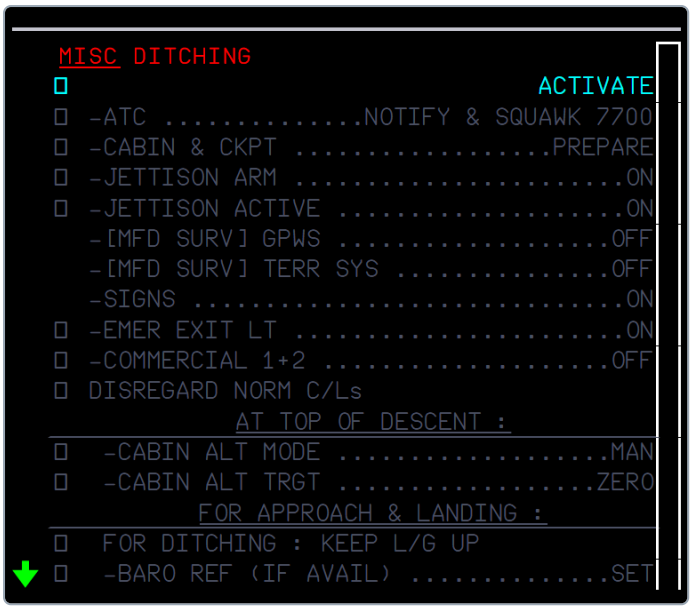


Figure 3. DITCHING before ACTIVATE (grey preview).

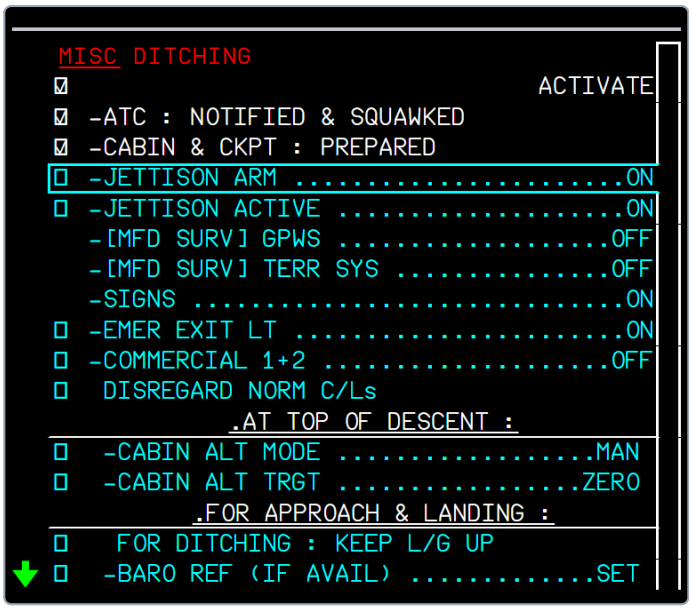


Figure 4. DITCHING activated: ATC and CABIN & CKPT checked by the crew; the cursor on JETTISON ARM.

The procedures added

Menu	Procedures
Emergency list (red)	DITCHING, FORCED LANDING
MISCELLANEOUS	CKPT WINDOW CRACKED, CKPT WINDOW ELEC ARCING, OIS FAULT
FUEL	FUEL LEAK (the FUEL LEAK DETECTED procedure, with LAND ANSA), MAN BALANCING PROCEDURE
L/G	LDG WITH ABNORM NOSE L/G, WITH 2 ABNORM L/Gs ON SAME SIDE, WITH 1 ABNORM WING OR BODY L/G, WITH 2 ABNORM BODY L/Gs, WITH 2 ABNORM WING L/Gs (they call the WHEEL page)
NAV	IR ALIGNMENT IN ATT MODE
ENG	HI VIBRATIONS (was missing from the list)

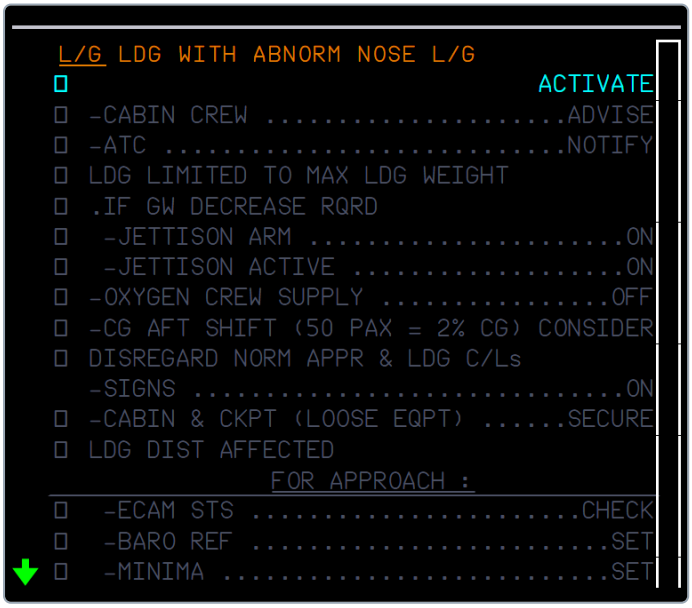


Figure 5. LDG WITH ABNORM NOSE L/G (preview).

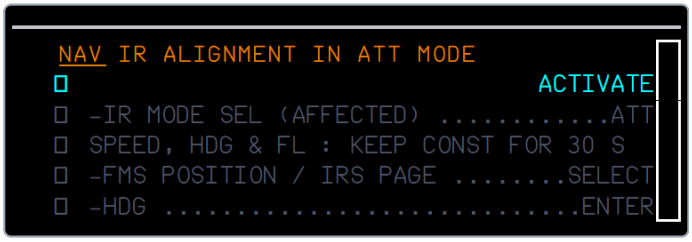


Figure 6. IR ALIGNMENT IN ATT MODE (preview).

Sensed alerts with FCOM lines

These appear by themselves when the failure is detected.

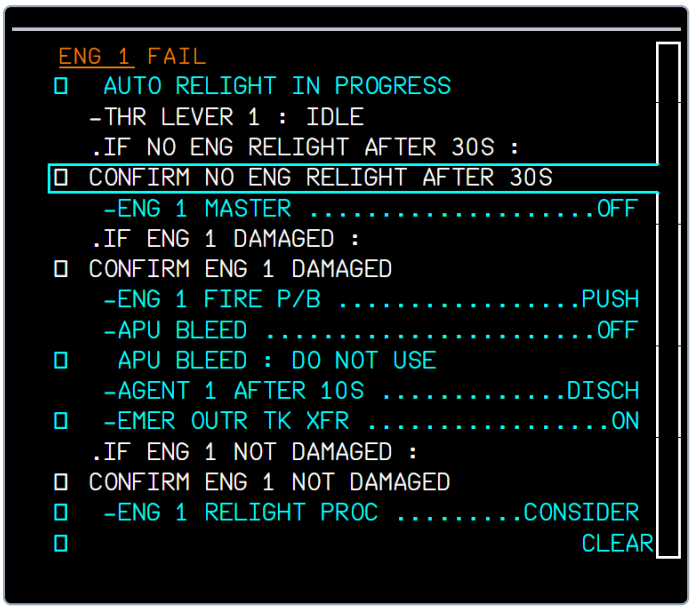


Figure 7. ENG 1 FAIL in flight: AUTO RELIGHT IN PROGRESS, THR LEVER 1 IDLE (ticked, the lever is at idle), then the relight / damaged branches.

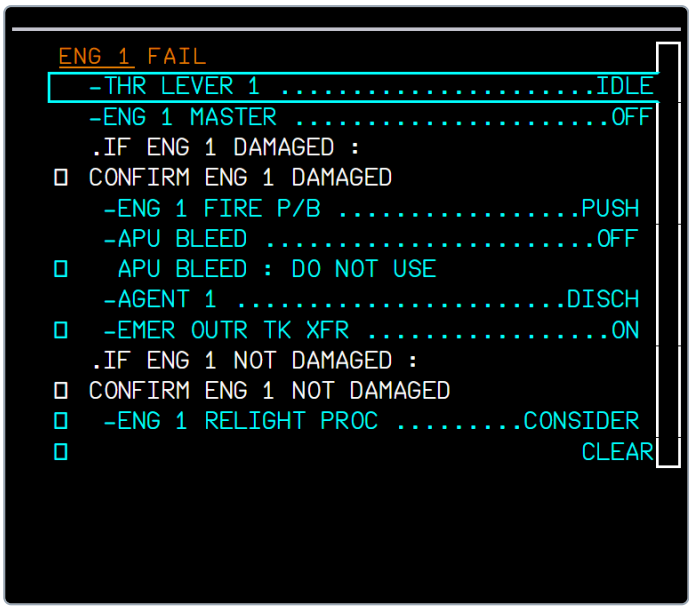


Figure 8. ENG 1 FAIL on the ground: the ground lines only.

Situation	What you should see
ENG x FAIL just after lift-off	The first line is L/G ... UP, until the gear lever is up.
ENG x FAIL in flight	AUTO RELIGHT IN PROGRESS; THR LEVER x IDLE, ENG x MASTER OFF, FIRE P/B, APU BLEED, AGENT: these tick from the cockpit controls.
ENG x FAIL on the ground	Only the ground part of the procedure.
All four engines fail in flight	ALL ENGINES FAILURE (red, LAND ASAP): the FCOM ALL ENG FLAME OUT procedure, fuel remaining / no fuel, relight, then forced landing or ditching; it scrolls with the cursor.
COND HOT AIR 1 (2) FAULT (hot air valve fault)	CARGO TEMP MONITOR; HOT AIR 1 (2) ON shows only while the HOT AIR pushbutton is OFF.
COND TEMP CTL FAULT	Now with its FCOM lines.

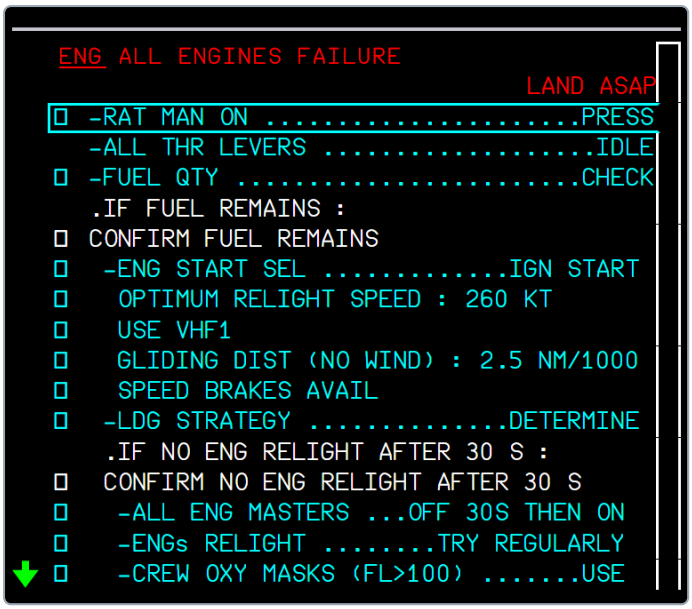


Figure 9. ALL ENGINES FAILURE, first page.

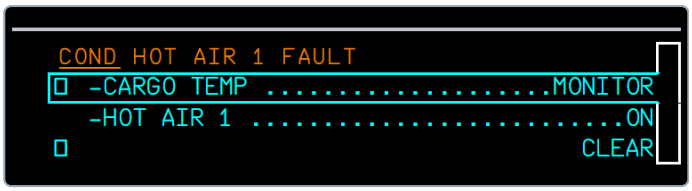


Figure 10. COND HOT AIR 1 FAULT.

Reading the lines

- Upper case **.IF ...** lines are ECAM conditions: the crew confirms them (CONFIRM ...), then the lines below become active.
- Headers such as **AT TOP OF DESCENT :** are centred sub-headings.
- Sensed lines tick by themselves once the action is done; the others are ticked by the crew with the ECP CHECK key.
- A line fits the 41 characters of the EWD; nesting is capped at three levels.

Limitations and why

Limitation	Why	What to do
About 85 sensed alerts show their title only.	The FCOM gives no lines for them ("Crew awareness").	-
Limitations, INOP SYS, INFO and deferred procedures of the new procedures.	They are on the STATUS page: see the feature A380X STATUS of the ABN PROC procedures below.	-
FLUCTUATING VERTICAL SPEED stays WIP; HOT AIR 1(2) OFF / VLV OPEN, T.O V1/VR/V2 DISAGREE, OVERSPEED LOAD ANALYSIS REQUIRED have no lines.	They are not in the 2011 FCOM used as the source.	-
ALL ENGINES FAILURE is long.	It is the full FCOM procedure; the EWD scrolls with the cursor.	-

How to use

1. For a failure the ECAM does not detect: press **ABN PROC** on the ECP.
2. Choose the category, then the procedure, with the ECP UP / DOWN keys and **CHECK**.
3. Read the grey preview; put the cursor on **ACTIVATE** and press CHECK to run it.
4. Tick each line with CHECK once it is done; the sensed lines tick by themselves.
5. Sensed alerts (ENG FAIL, COND...) show by themselves: read and do their lines the same way.

How to test

#	Do this	Expected result
1	A380X on the ground, powered. Press **ABN PROC** on the ECP.	The EWD lists DITCHING and FORCED LANDING in red at the top, then the ENG, F/CTL, FUEL, L/G, NAV and MISCELLANEOUS categories.
2	Open **MISCELLANEOUS**, then **DITCHING**.	The procedure in grey with ACTIVATE at the bottom (preview).
3	Put the cursor on ACTIVATE and press **CHECK**.	The lines turn cyan; the cursor is on the first line (ATC ... NOTIFY & SQUAWK 7700).
4	Press CHECK on each line in turn.	Each line is ticked and the cursor moves to the next one.
5	ABN PROC > ENG.	HI VIBRATIONS is in the list.
6	ABN PROC > L/G > LDG WITH ABNORM NOSE L/G, then ACTIVATE.	The procedure runs; the SD shows the WHEEL page.
7	ABN PROC > NAV > IR ALIGNMENT IN ATT MODE; ABN PROC > FUEL > FUEL LEAK and MAN BALANCING PROCEDURE; MISCELLANEOUS > CKPT WINDOW CRACKED, CKPT WINDOW ELEC ARCING, OIS FAULT.	Each one opens with its FCOM lines (none is empty any more).
8	In flight, make engine 1 run down with its ENG MASTER ON and its FIRE pushbutton in (no flyPad engine failure on this branch: use the sim engine failure or a SimConnect tool).	ENG 1 FAIL with AUTO RELIGHT IN PROGRESS; THR LEVER 1 IDLE ticks by itself when the lever is at idle.
9	Same on the ground.	ENG 1 FAIL with the ground lines only.

2. A380X STATUS of the ABN PROC procedures

> **Aircraft:** A380X | **Where:** SD STATUS page; PFD limitations | **Status:** Tested in the sim by the user: OK with CKPT WINDOW CRACKED (both layers) and FUEL LEAK. The L/G, ELEC ARCING and MAN BALANCING lines: no record of a sim test.

What it is for

The not-sensed procedures of the feature above had no STATUS: after them, the STATUS page stayed NORMAL. The FCOM gives each one its limitations, inoperative systems, information and deferred procedure (A380 FCOM PRO-ABN-ECAM-10-28, -32 and -80). The FWS now writes them from the lines the crew ticks: the STATUS follows the branch of the procedure that was flown.

What you see

Procedure	Line ticked	STATUS
L/G LDG WITH ABNORM ... (the five)	LDG LIMITED TO MAX LDG WEIGHT	LIMITATIONS (approach and landing): **MAX LDG WEIGHT**, **LDG DIST AFFECTED**; MAX LDG WEIGHT on the PFD. INOP SYS (approach and landing): **BTV**, always.
MISC CKPT WINDOW CRACKED	EXTERNAL LAYER CRACKED	No limitation. Deferred procedure **AT TOP OF DESCENT: CKPT WINDOW CRACKED**.
	INTERNAL LAYER CRACKED (or no diagnosis)	**MAX FL : 230/MEA** (STATUS and PFD); INFO: the cabin altitude table (ACFT FL / CABIN ALT TRGT: 100 = LDG ELEVN, 150 = 3000, 200 = 6000, 230 = 8000); deferred procedure at top of descent: CABIN ALT MODE AUTO.

	AICU 1 PULL / AICU 2 PULL	INOP SYS: L (R) WINDSHIELD HEATG, SLIDG WINDOW HEATG, FIXED WINDOW HEATG; with both AICU pulled, WING A-ICE too.
MISC CKPT WINDOW ELEC ARCING	AICU 1 PULL / AICU 2 PULL	The same window heating INOP SYS.
FUEL LEAK	-	**LAND ANSA** (STATUS and PFD), and FUEL PARTLY UNUSABLE unless LEAK STOPS is ticked.
	LEAK CONTINUES	INFO: MIN FUEL USABLE : OTHER FEED TKs, (ALL OTHER FUEL MAYBE LOST).
FUEL MAN BALANCING PROCEDURE	-	**DELAY T.O** while on the ground.
	FEED TKs 1 & 4 (or 2 & 3) NOT BALANCED	INFO: FUEL CONSUMPT INCRSD : FMS PRED DISREGARD.

Limitations and why

Limitation	Why	What to do
A deferred procedure that depends on a crew choice carries that choice as its first line (for example INTERNAL LAYER CRACKED OR NO DIAGNOSIS).	The FWS creates the deferred procedures when the abnormal procedure is first shown, before the crew has ticked anything.	Read the condition line before doing the deferred line.

How to use

1. Run the ABN PROC procedure and tick the lines that match your case (for example INTERNAL LAYER CRACKED).
2. Press **STS** on the ECP (or wait for the STATUS page at the end of the procedures).
3. Read the LIMITATIONS, INOP SYS and INFO; the deferred procedure comes up at top of descent.

How to test

#	Do this	Expected result
1	A380X in flight (or on the ground). ABN PROC > MISCELLANEOUS > CKPT WINDOW CRACKED, ACTIVATE. Tick EXTERNAL LAYER CRACKED and its lines.	STATUS page: no limitation; the deferred procedure AT TOP OF DESCENT CKPT WINDOW CRACKED is listed.
2	Run it again (or reload) and tick INTERNAL LAYER CRACKED instead, then AICU 1 PULL.	STATUS: MAX FL : 230/MEA (also on the PFD), the cabin altitude table in INFO, INOP SYS L WINDSHIELD HEATG, L SLIDG WINDOW HEATG, L FIXED WINDOW HEATG.
3	Also tick AICU 2 PULL.	INOP SYS adds the R window heating lines and WING A-ICE.
4	ABN PROC > FUEL > FUEL LEAK, ACTIVATE.	STATUS: LAND ANSA (also on the PFD) and FUEL PARTLY UNUSABLE.
5	Tick LEAK STOPS.	FUEL PARTLY UNUSABLE goes; LAND ANSA stays.
6	On the ground: ABN PROC > FUEL > MAN BALANCING PROCEDURE, ACTIVATE.	STATUS: DELAY T.O.
7	ABN PROC > L/G > LDG WITH ABNORM NOSE L/G, ACTIVATE, tick LDG LIMITED TO MAX LDG WEIGHT.	STATUS (approach and landing part): MAX LDG WEIGHT, LDG DIST AFFECTED, INOP SYS BTV; MAX LDG WEIGHT on the PFD.

3. A380X EMER CANC pushbutton

> **Aircraft:** A380X | **Where:** ECP, right of RCL | **Status:** Tested in the sim by the user: OK (again: caution cancelled for the flight, RCL > 3 s recall, warning aural and MASTER WARN silenced).

What it is for

EMER CANC stops an alert that is wrong or no longer useful: a spurious warning whose aural alert would sound for the rest of the approach, or a caution the crew has dealt with and does not want to see again. Until now the pushbutton did nothing on the FBW A380.

What it does

It acts on the alert at the top of the EWD (A380 FCOM DSC-31-40-20):

Top alert	EMER CANC
A **warning** (red)	Its aural alert and the MASTER WARN light stop. The procedure stays on the EWD. If the warning goes away and comes back, the aural alert and MASTER WARN come back.
A **caution** (amber)	The caution, its procedure and the MASTER CAUT light go away **for the rest of the flight**, even if the failure comes back. It is listed in white under **CANCELLED CAUTION** on the STATUS page (MORE).
In every case	All the aural alerts stop (the one sounding and those waiting), and the procedures you called from ABN PROC are removed.
Nothing to cancel	The memo **EMERGENCY CANCEL ON** shows for 3 seconds.

Getting the cancelled cautions back

1. Press and **hold RCL** for more than 3 seconds.
2. The cancelled cautions that are still active come back on the EWD with their procedures.

The cancelled cautions are also forgotten when the warning system loses power, or 50 minutes after the engines are shut down, as the other ECAM memories.

What to expect

- A cancelled caution never sounds again in the flight, even if the failure comes back: check the STATUS page.
- A silenced warning comes back only if it goes away and triggers again.
- EMERGENCY CANCEL ON lasts 3 seconds: the FCOM gives no time; 3 s as the RCL "NORMAL" message.

How to use

1. Lift the guard and press **EMER CANC** on the ECP: it acts on the alert at the top of the EWD.
2. A warning: its aural and MASTER WARN stop, the procedure stays. A caution: it goes away for the rest of the flight.
3. Check the cancelled cautions on the STATUS page (**MORE**, CANCELLED CAUTION).
4. To get them back, press and hold **RCL** more than 3 seconds.

How to test

#	Do this	Expected result
1	A380X on the ground, engine 1 running. Set the **ENG 1 BLEED** pushbutton (overhead AIR panel) OFF.	Caution AIR ENG 1 BLEED OFF in amber, MASTER CAUT on, single chime.
2	Press **EMER CANC**.	The caution and MASTER CAUT go away.

3	Set ENG 1 BLEED back to AUTO, then OFF again.	The caution does not come back (cancelled for the flight).
4	Press **STS** on the ECP.	The STATUS page shows MORE at the bottom; the MORE page lists AIR ENG 1 BLEED OFF in white under CANCELLED CAUTION.
5	Press and hold **RCL** more than 3 seconds (ENG 1 BLEED still OFF).	AIR ENG 1 BLEED OFF comes back on the EWD.
6	Press EMER CANC with no alert on the EWD.	The memo EMERGENCY CANCEL ON for 3 seconds.
7	On the runway, parking brake ON, thrust levers to FLX or TOGA (then back to idle).	Warning CONFIG PARK BRK ON in red, MASTER WARN, continuous repetitive chime.
8	Press EMER CANC.	The chime and MASTER WARN stop; CONFIG PARK BRK ON stays on the EWD.
9	Release the parking brake, set it again and advance the thrust levers again.	The warning triggers again with its chime and MASTER WARN.

4. A380X ECP MORE key and light on the STATUS page

> **Aircraft:** A380X | **Where:** ECP MORE key; SD STATUS page | **Status:** Not sim-tested (13 unit tests).

What it is for

A380 FCOM DSC-31-40-10 (STATUS MORE PAGE): the crew displays the STATUS MORE page "if MORE appears at the bottom of the STATUS page, by pressing MORE pb on the ECP"; pressing the pb again clears it (DSC-31-40-20). In FBW the ECP MORE key did nothing: the SD paused and resumed its STATUS page in an order that lost the value telling the FWS that MORE was available, and the MORE pb light read a variable nobody wrote. The STATUS MORE page holds the INOP SYS REDUND items and the CANCELLED CAUTION list (see EMER CANC above).

What you should see

Situation	What you should see
STATUS page with REDUND LOSS items or a cancelled caution	MORE in a box at the bottom of the page.
MORE key pressed	The STATUS MORE page (INOP SYS REDUND, CANCELLED CAUTION); the light of the MORE key comes on.
MORE key pressed again	Back to the STATUS page; the light goes off.
Another SD page selected, or the STATUS page has nothing MORE any more	The MORE page and the light go away.
STATUS page without MORE, or another SD page	The MORE key does nothing (only STATUS has a MORE page in FBW).

Limitations and why

Limitation	Why	What to do
Only the STATUS page has a MORE page.	The other SD MORE pages are not built in FBW.	-
A full MSFS restart is needed after an update for the light.	The MORE key light is in the cockpit behaviour file (ecam-cp.xml).	Restart the sim after installing.

How to use

1. Press **STS** on the ECP.
2. If MORE shows at the bottom of the STATUS page, press **MORE** to see the STATUS MORE page.
3. Press MORE again (or another page key) to leave it.

How to test

#	Do this	Expected result
1	A380X on the ground, engine 1 running. Make a cancelled caution: ENG 1 BLEED pushbutton OFF, then **EMER CANC**.	The caution AIR ENG 1 BLEED OFF goes away.
2	Press **STS**.	STATUS page with MORE boxed at the bottom.
3	Press **MORE** on the ECP.	The STATUS MORE page with CANCELLED CAUTION: AIR ENG 1 BLEED OFF. The MORE key light is on.
4	Press MORE again.	Back to the STATUS page; the light is off.
5	Press MORE, then a page key (for example ENG).	The ENG page; the MORE light goes off. Back on STATUS, the normal page (not MORE) shows.
6	On a STATUS page without MORE (no REDUND LOSS item, no cancelled caution), press MORE.	Nothing happens; no light.

5. FIRE TEST extended after the button release

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Settings > Realism; overhead FIRE TEST | **Status:** Tested in the sim by the user: A32NX OK (incl. the AGENT / SQUIB DISCH lights), A380X OK.

What it is for

On the aircraft, the FIRE TEST lasts as long as the button is held. In the simulator, holding the button with the mouse while looking at the ECAM, the master warning and every FIRE and AGENT light is awkward. This option keeps the test running for 7 seconds after you release the button. It is marked **unrealistic** and is on by default.

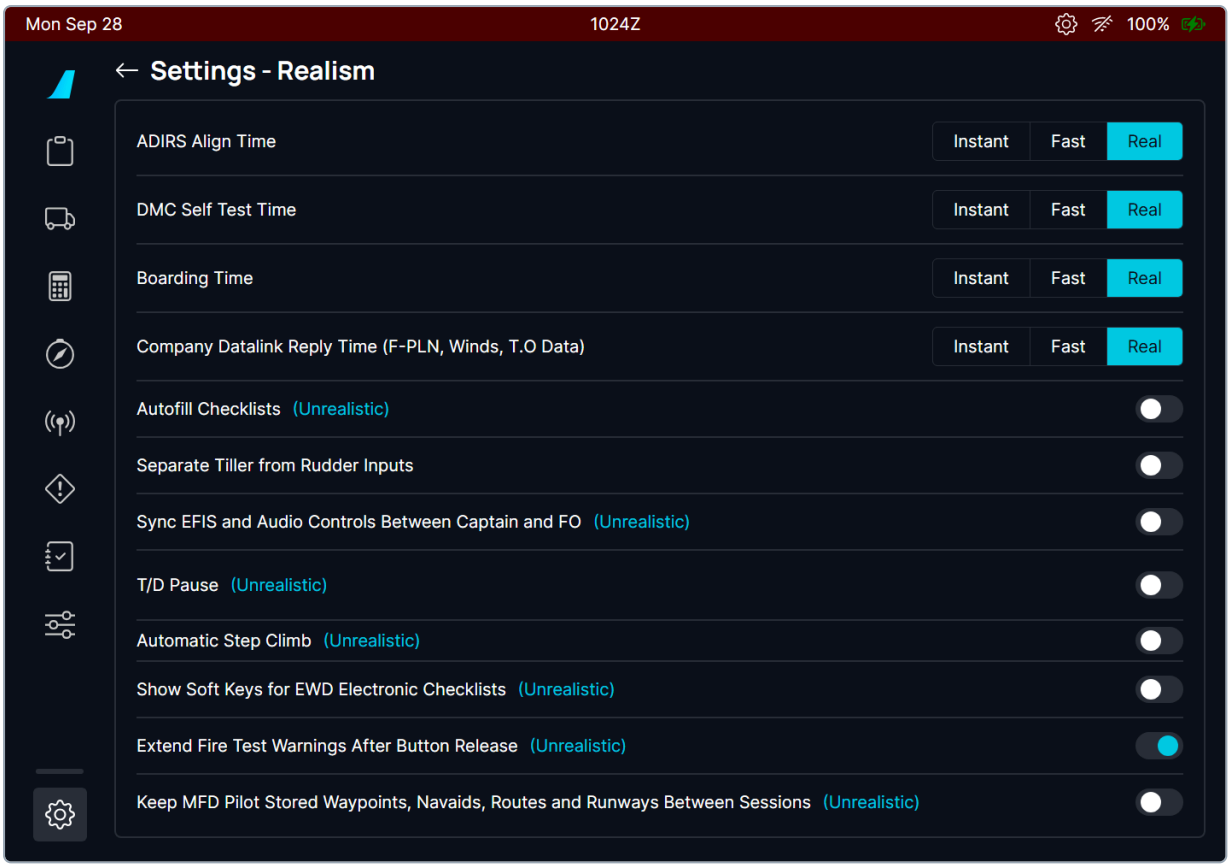


Figure 11. Settings > Realism on the A380X: "Extend Fire Test Warnings After Button Release".

What it does

Aircraft	Test	Kept 7 s after the release
A32NX	APU, ENG 1 and ENG 2 FIRE TEST pushbuttons	master warning and CRC, ECAM, FIRE pushbutton lights, AGENT / SQUIB DISCH lights, together
A380X	FIRE TEST pushbutton (overhead)	the fire detection test itself: FIRE lights, squib lights and the FWS warnings, together

Option off: the test stops the instant you release the button, as on the aircraft.

What to expect

- The test starts half a second after you press (A380X), as before.
- Pressing again during the 7 seconds starts a new 7 seconds after the next release.
- The A380X systems host and fire protection system changed: a full MSFS restart after an update.

How to use

1. flyPad > **Settings** > **Realism**: set "Extend Fire Test Warnings After Button Release" ON (the default) or OFF.
2. Press and release the FIRE TEST pushbutton: with the option ON, check the warnings and lights during the 7 seconds that follow.

How to test

#	Do this	Expected result

1	A32NX, powered, option ON. Press and release the **ENG 1 FIRE TEST** pushbutton (overhead FIRE panel).	For 7 seconds: ENG 1 FIRE test warning on the E/WD, MASTER WARN, continuous repetitive chime, ENG 1 FIRE pushbutton light and the AGENT 1 / AGENT 2 SQUIB DISCH lights. Then everything goes off together.
2	Same with the **APU FIRE TEST** and the **ENG 2 FIRE TEST**.	Same 7 seconds, with their own warning and lights (one AGENT for the APU).
3	Press again during the 7 seconds and release.	A new 7 seconds starts at the new release.
4	Set the option OFF and repeat.	Everything stops at the instant the button is released.
5	A380X, powered, option ON. Press and release the overhead **FIRE TEST** pushbutton.	The test starts half a second after the press; the FIRE and squib lights and the FWS fire test warnings stay together 7 seconds after the release.
6	A380X, option OFF, repeat.	The test stops when the button is released.

Runway awareness: RAAS and OANS

The A32NX gets the Honeywell RAAS runway callouts, and the A380X OANS gives RWY AHEAD without a Navigraph map, loads the FCOM default airport by itself and shows one airport per side; the runway search keeps only the airports with a runway

Aircraft	A32NX (RAAS) and A380X (OANS)
Where to find it	A32NX: flyPad > Settings > Aircraft Options / Pin Programs > RAAS. A380X: ND at the ZOOM range or in PLAN (OANS), ARPT SEL panel; RWY AHEAD on the PFD and ND
Branch	`core/oans-btv-raas`, based on `master` (FlyByWire master 76db0aef1 merged in); independent. `core/a380x-vd-runways` merges it (the VD uses its runway search).
Status	RAAS and RWY AHEAD without a map tested in the sim by the user: OK. OANS default airport tested by the user: OK. One airport per side: not sim-tested. Runway search with airports with a runway only: OK in the sim through the A380X VD, not sim-tested with the RAAS or the OANS.
Pictures	flyPad setting and the Captain ND from the local browser harness, running the build installed in the sim (RAAS callouts drawn as diagrams from the code; the RWY AHEAD advisory and the test airport XTST set up by the harness). No pictures for the airport per side.

1. A32NX RAAS runway awareness callouts
2. A380X RWY AHEAD without an airport map
3. A380X OANS default airport
4. A380X one OANS airport per side
5. Runways around the aircraft: airports with a runway only

1. A32NX RAAS runway awareness callouts

> **Aircraft:** A32NX | **Where:** flyPad > Settings > Aircraft Options / Pin Programs > RAAS | **Status:** Tested in the sim by the user: first callout heard; "RAAS is working well" (approaching on the ground and in the air, extended holding, distance remaining, volume).

What it is for

RAAS is an option of the A320's Honeywell EGPWS (A320 FCOM GEN aircraft configuration, used as a line-up aid in PRO-NOR-SOP-11 TAKEOFF RUNWAY CONFIRM). It tells the crew which runway they are approaching or are on, warns of a taxiway takeoff or landing, and calls the distance remaining. It needs no airport map: the runways come from the sim's airport database.

Turning it on

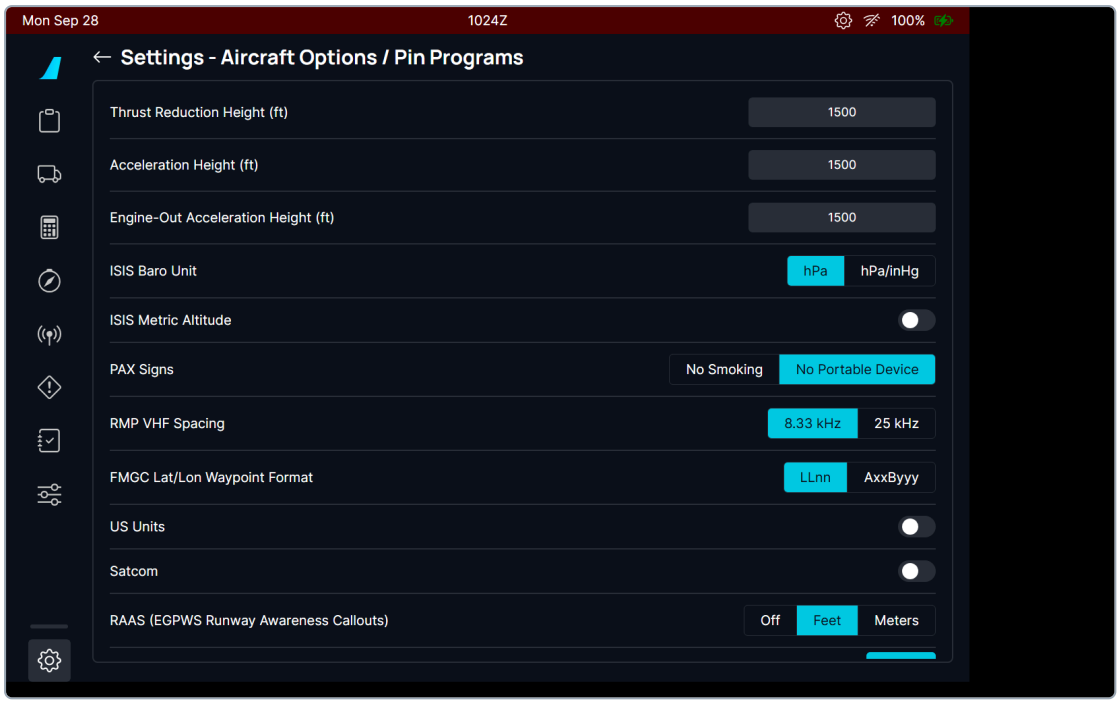


Figure 1. flyPad Settings > Aircraft Options / Pin Programs.

Figure 2. RAAS: Off, Feet (default) or Meters. The unit is the one of the distance callouts.

- The callouts run while the EGPWC is powered (AC BUS 1) and the **GPWS SYS** pushbutton is not OFF.
- The EGPWS aural alerts (e.g. "TOO LOW TERRAIN", "GLIDE SLOPE") have priority: RAAS waits for them.
- The voice is played word by word.

On the ground

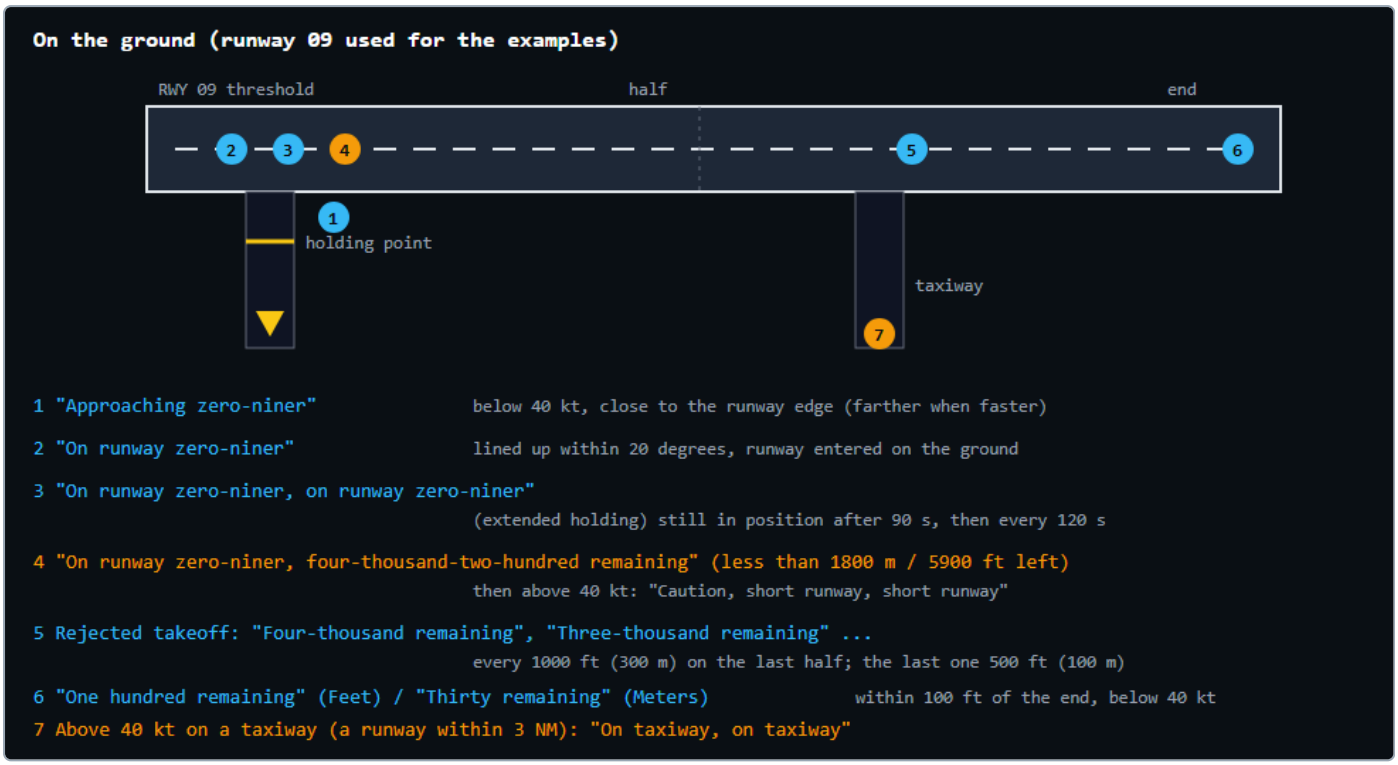


Figure 3. The ground callouts, in blue the advisories, in amber the cautions.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

Situation	What you should see
Taxiing towards a runway (below 40 kt)	"Approaching" and the runway end nearest to where you will enter; "Approaching runways" when two at once.
Lining up	"On runway" and the runway, once, when the runway was entered on the ground.
Lining up with less than 1800 m (5900 ft) ahead	The length remaining is added; at the start of the roll (above 40 kt), "Caution, short runway".
Holding on the runway for 90 s	"On runway ..." twice, then every 120 s while you stay.
Starting a takeoff roll on a taxiway (above 40 kt)	"On taxiway, on taxiway".
Rejected takeoff (ground speed down 7 kt from its maximum)	The distance remaining on the last half of the runway.
Slow near the end (within 100 ft, below 40 kt)	"One hundred remaining" (Feet) or "Thirty remaining" (Meters).

Approach and landing

Figure 4. The approach callouts and the suppress window.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

Situation	What you should see
Between 750 and 300 ft, lined up, within 3 NM	"Approaching" and the runway, once. Not between 450 and 350 ft (the Airbus suppress window, where the crew is busy with the auto callouts).
The runway is shorter than 1500 m (4900 ft)	The available length is added; below the window: "Caution, short runway".
150 to 250 ft, not lined up with any runway	"Caution taxiway, caution taxiway".
Landing roll	The distance remaining on the last half: every 1000 ft (Feet) or 300 m (Meters), the last one at 500 ft / 100 m; then the runway end.

Limitations and why

Limitation	Why	What to do
Only runways known to the sim.	The runways come from the MSFS airport database, not from an airport map: no taxiway geometry, so "on taxiway" is judged as "not aligned with a runway".	-
One synthetic voice.	The clips are generated with the Piper text-to-speech engine (voice trained on the public-domain LJ Speech dataset); its licence allows the clips in the aircraft.	-
No RAAS on the A380X.	The A380X uses the OANS runway alerts instead (see A380X RWY AHEAD without an airport map below).	-

> **NOTE**
> The A380X OIT flyPad settings pages carry the same RAAS option entry, only so that the A380X builds: the A380X has no RAAS.

How to use

1. flyPad > **Settings** > **Aircraft Options / Pin Programs** > **RAAS**: **Feet** (the default) or **Meters** for the distance callouts, **Off** to remove them.
2. Keep the **GPWS SYS** pushbutton on (not OFF): RAAS is part of the EGPWS.
3. Taxi, take off and land as usual: listen to the callouts.

How to test

#	Do this	Expected result
1	A32NX on the ground at an airport, AC BUS 1 powered, GPWS SYS not OFF, RAAS set to **Feet**. Taxi towards a runway below 40 kt.	"Approaching" and the runway name (for example "Approaching zero six right") before the holding point.
2	Line up on the runway.	"On runway" and the runway name, once.
3	Stay lined up for 90 seconds.	"On runway ..." twice, then again every 120 s while you stay.
4	On a runway shorter than 1800 m (5900 ft): line up.	The length remaining is added; above 40 kt, "Caution, short runway".
5	Start a takeoff roll on a taxiway (above 40 kt).	"On taxiway, on taxiway".
6	Reject a takeoff (slow down by 7 kt or more from the maximum ground speed).	The distance remaining on the last half of the runway.
7	In flight, final approach lined up with the runway, within 3 NM, between 750 and 300 ft.	"Approaching" and the runway, once; never between 450 and 350 ft.
8	Landing roll.	Distance remaining every 1000 ft on the last half (last one at 500 ft), then the runway end.
9	Set RAAS to **Meters** and land again.	Distance remaining every 300 m, the last one at 100 m.
10	Set the **GPWS SYS** pushbutton OFF, or RAAS Off.	No more RAAS callouts.

2. A380X RWY AHEAD without an airport map

> **Aircraft:** A380X | **Where:** PFD and ND, automatic while taxiing | **Status:** Tested in the sim by the user: OK (without a Navigraph map). 4 unit tests.

What it is for

While taxiing, the A380 OANS warns the crew before the aircraft enters a runway: **RWY AHEAD** with the runway name, on the PFD and the ND. In FBW this advisory came from the Navigraph airport map (AMDB): without a Navigraph subscription, or at an airport without a map, it never came. Now, while no airport map is loaded, the OANS uses the runways of the sim's own airport database with the same rule.

The rule

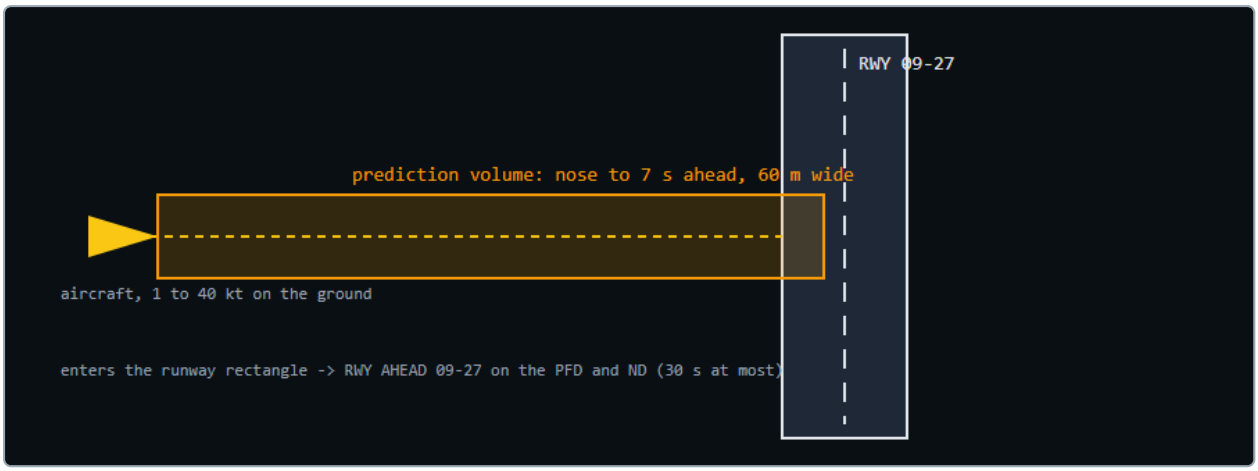


Figure 5. The prediction volume: from the nose to 7 seconds ahead at the current ground speed, 60 m wide. When it reaches a runway, RWY AHEAD appears for that runway.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

- On the ground, from 1 to 40 kt.
- The advisory stays at most 30 s, and goes when the aircraft is on the runway.
- With an airport map loaded, nothing changes: the map is used as before.

What it looks like

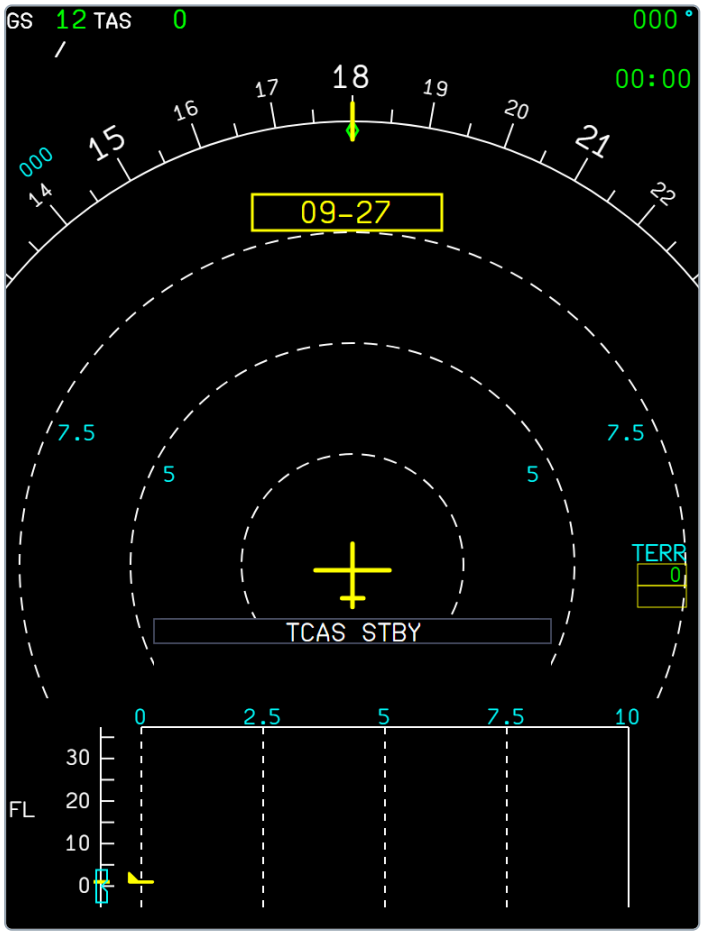


Figure 6. The ND advisory: the runway name in a yellow box at the top (here 09-27, when the aircraft taxis south towards runway 09/27). The PFD shows RWY AHEAD too.

What to expect

Situation	What you should see
Taxiing towards a runway, no Navigraph map	RWY AHEAD with the runway name, up to 30 s.
Taxiing towards a runway, with the Navigraph map	RWY AHEAD from the map (as before).
Stopped, or above 40 kt	No advisory.
On the runway	The advisory goes.

Limitations and why

Limitation	Why	What to do
Runway rectangles only.	The sim database gives each runway as a rectangle (threshold, length, width); displaced thresholds and runway shapes of the airport map are not known.	-
Shares its runway search with the A32NX RAAS.	One runway module (sim airport database) serves both; both are on this branch.	-

How to use

1. Nothing to set: taxi as usual. Before entering a runway, RWY AHEAD shows on the PFD and the ND with the runway name.
2. With the Navigraph airport map loaded the advisory comes from the map; without it, from the sim runways.

How to test

#	Do this	Expected result
1	A380X on the ground, without a Navigraph airport map (flyPad not linked to Navigraph, or an airport not in the airport database). Taxi towards a runway between 1 and 40 kt.	About 7 seconds before the nose reaches the runway: RWY AHEAD on the PFD, and the runway name in a yellow box at the top of the ND.
2	Stop before the runway.	The advisory goes after 30 s at most; no new one while stopped.
3	Taxi onto the runway.	The advisory goes once the aircraft is on the runway.
4	Repeat with the Navigraph map loaded.	RWY AHEAD as before, from the map.

3. A380X OANS default airport

> **Aircraft:** A380X | **Where:** ND at ZOOM range or PLAN; OANS ARPT SEL | **Status:** Tested in the sim by the user: OK. 4 unit tests.

What it is for

The OANS draws the airport around the aircraft on the ND. The FCOM (DSC-34-10-70-20, MOVING AIRPORT MAP) says no manual selection is needed: the OANS shows a **default airport**. FBW loaded the nearest airport on ground, but in flight only the destination (within 50 NM, at any height), and a manual selection stopped the automatic choice for 10 minutes. Now the FCOM rules apply.

The rules

ND mode	Default airport

ARC, ROSE-NAV (on ground)	The airport where the aircraft is.
ARC, ROSE-NAV (in flight)	The **origin, destination or alternate** of the FMS while the aircraft is inside its virtual cylinder: **20 NM** radius, **5000 ft** above the airport. The OANS keeps it while the aircraft stays in the cylinder.
PLAN (in flight)	The origin if it is closer than 50 NM, otherwise the destination; the destination when origin and destination are less than 300 NM apart.
Manual (ARPT SEL)	Only in PLAN: another airport can be displayed. It stays until the ND leaves PLAN, or the aircraft lands or takes off. In ARC / ROSE-NAV the OANS says SET PLAN MODE.

Each ND has its own OANS and its own airport (see A380X one OANS airport per side below): the rules apply to each ND by itself.

What it looks like

Figure 7. In flight, 12 NM from the destination XTST at 3000 ft: the destination loads by itself (name top right; the arrow points to the airport, still outside the zoom range).

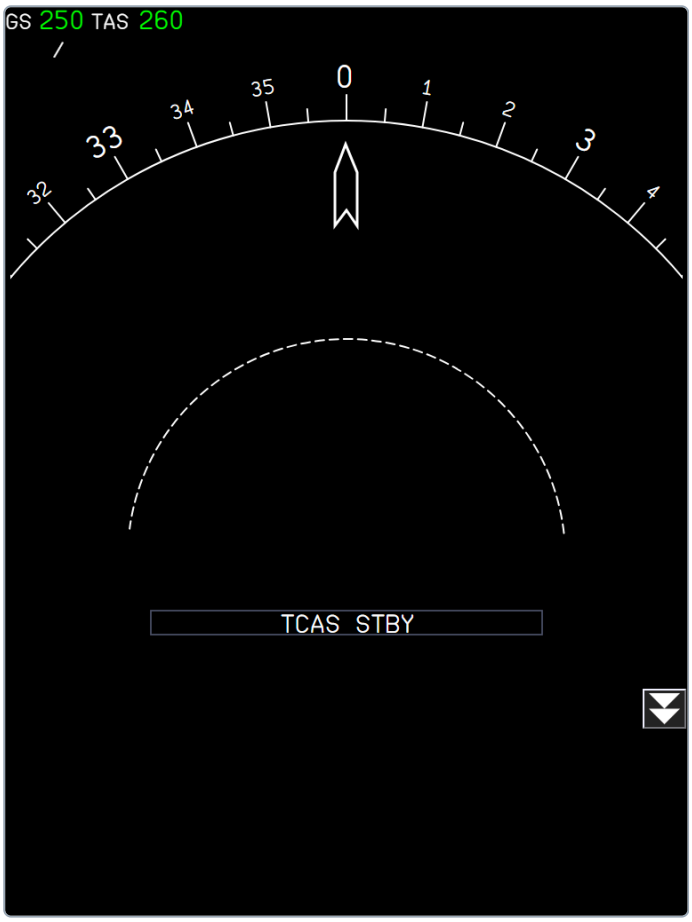


Figure 8. 30 NM away at 9000 ft: outside the cylinder, no airport is loaded.

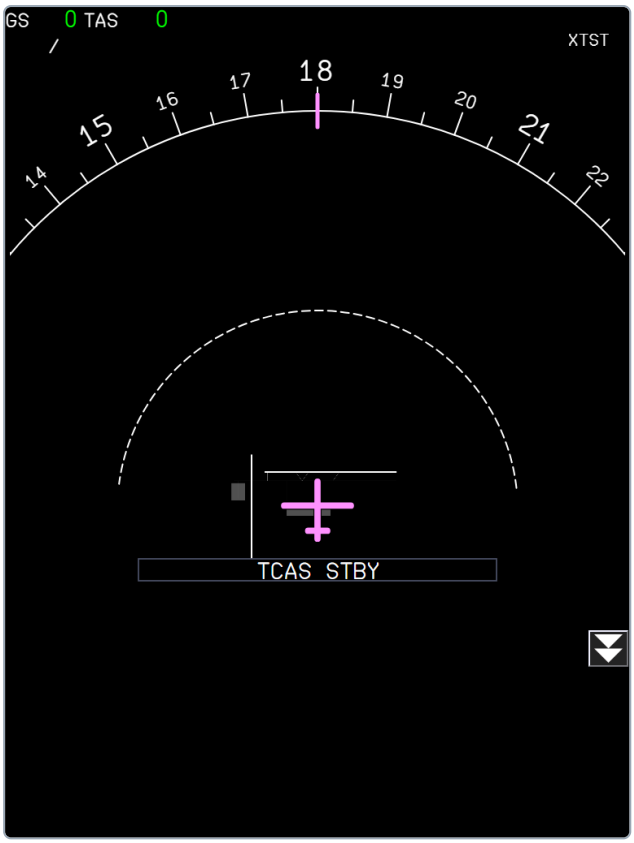


Figure 9. On ground at XTST: the current airport, with the aircraft at its gate.

What to expect

Situation	What you should see
On ground	The airport where you are, whatever the flight plan.
Climbing out of the origin	The origin until you leave its 20 NM / 5000 ft cylinder, then nothing (or the alternate / destination when inside theirs).
Approach to the destination	The destination from 20 NM and 5000 ft above it.
Diversion to the alternate	The alternate from 20 NM and 5000 ft above it.
ARPT SEL in PLAN	Your airport until you leave PLAN, land or take off.
No present position	No automatic airport.

Limitations and why

Limitation	Why	What to do
Needs the Navigraph airport database.	The OANS map is the Navigraph AMDB.	Link Navigraph in the flyPad.
The Captain and F/O NDs can show different airports.	Each side has its own OANS (FCOM DSC-34-10-70-10); see the airport per side below.	-
The destination arrow (5 NM range, map out of range) is FBW's.	Not part of this change.	-

How to use

1. Set the ND to ARC or ROSE-NAV and turn the range knob to **ZOOM**: the OANS shows the default airport by itself.

2. To look at another airport: ND in **PLAN**, open the OANS **ARPT SEL** panel, choose the airport. It stays until the ND leaves PLAN, or the aircraft lands or takes off.

How to test

#	Do this	Expected result
1	A380X on the ground at an airport with a Navigraph airport map; ND ARC, range ZOOM.	The OANS shows the airport where you are, whatever the flight plan.
2	Take off and climb out (ND ARC ZOOM).	The origin stays until you leave its 20 NM / 5000 ft cylinder, then no airport.
3	Approach the destination of the FMS flight plan.	The destination loads by itself from 20 NM and 5000 ft above it.
4	In flight, ND in PLAN.	The origin if it is closer than 50 NM, else the destination (the destination when origin and destination are less than 300 NM apart).
5	In PLAN, choose another airport with ARPT SEL, then set the ND back to ARC.	Your airport while in PLAN; back in ARC ZOOM, the default airport again.
6	In ARC or ROSE-NAV, try ARPT SEL.	The OANS asks SET PLAN MODE.

4. A380X one OANS airport per side

> **Aircraft:** A380X | **Where:** Captain and F/O NDs (OANS) | **Status:** Not sim-tested.

What it is for

A380 FCOM DSC-34-10-70-10: there is a Captain's OANS (ADIRS 1, MMR 1, FMS 1) and a First Officer's OANS (side 2), each shown on its own ND. FBW loaded one airport for both NDs, so an airport chosen on one side replaced the airport of the other. Now each side displays its own airport.

What you should see

Situation	What you should see
Both NDs at ZOOM on the ground	Each OANS shows the current airport (each finds it by itself).
F/O ND in PLAN, another airport chosen with ARPT SEL	The F/O ND shows that airport; the Captain ND keeps its own airport.
PLAN default airport	Follows the mode of that ND only (before, it also looked at the other ND).
RWY AHEAD	Still one advisory for the aircraft, computed by the Captain side OANS: from its airport map when the aircraft is within that airport, else from the sim runways.
An airport unloaded on one side	The BTV runway and exit selection is cleared only if it was made on that side.

Limitations and why

Limitation	Why	What to do
The runway crossing and flag events are not split by side.	An item of the other side's airport is not found on this side's map.	-
RWY AHEAD comes from the Captain side only.	It is one advisory for the aircraft (one PFD / ND message word).	-

How to use

1. Use each ND's OANS as usual: ZOOM range for the default airport, PLAN + ARPT SEL for another one.
2. Each pilot can look at a different airport without changing the other's map.

How to test

#	Do this	Expected result
1	A380X on the ground at an airport with a Navigraph airport map. Both NDs ARC, range ZOOM.	Both NDs show the current airport.
2	Set the F/O ND to PLAN and choose another airport with the F/O OANS ARPT SEL.	The F/O ND shows the chosen airport; the Captain ND still shows the current airport.
3	Set the Captain ND to PLAN and choose a third airport.	Each ND shows its own choice.
4	Set both NDs back to ARC ZOOM.	Both go back to the current airport.
5	Select a BTV runway and exit on the Captain OANS, then unload the airport on the F/O side.	The Captain BTV selection stays.

5. Runways around the aircraft: airports with a runway only

> **Aircraft:** A32NX and A380X | **Where:** Used by the A32NX RAAS and the A380X OANS (RWY AHEAD without an airport map) | **Status:** Tested in the sim by the user through the A380X VD airport list: OK (group 2 : 10 airports on the ground, 11 to 12 in the climb). Not sim-tested with the RAAS or the OANS. Unit test: NearbyRunwayProvider.spec.

What it is for

The RAAS callouts and the A380X RWY AHEAD without an airport map take their runways from one search of the sim airports around the aircraft (the closest 20 airports within 8 NM). That search took every airport class, so near a city the heliports filled the list: in a sim test at RJTT (25 NM / 24 airports for the VD) only RJTT itself had runways, though three other airports are within 25 NM, and RJTT dropped out of the list a few miles after take-off.

The search now keeps only the airports with a land runway (hard, soft or private) whose longest runway is at least 100 m long: no heliports, no water-only seaplane bases. The same search feeds the A380X VD airports on `core/a380x-vd-runways`.

Limitations and why

Limitation	Why	What to do
Airports with only runways shorter than 100 m are left out.	Design choice: it removes the private heliports the class filter keeps; no runway that matters to RAAS, RWY AHEAD or the VD is that short.	-

How to use

1. Nothing to set: the RAAS callouts and RWY AHEAD use the new list by themselves.

How to test

#	Do this	Expected result
1	A32NX at a large airport in a city with heliports around (for example RJTT), RAAS set to Feet, GPWS SYS not OFF. Taxi towards a runway below 40 kt.	"Approaching" and the runway name before the holding point (the airport is found even with many heliports around).

2	A380X at the same airport without a Navigraph airport map. Taxi towards a runway between 1 and 40 kt.	RWY AHEAD on the PFD and the runway name at the top of the ND before the runway.
3	With `core/a380x-vd-runways` merged: take off and climb out; watch the VD.	The departure airport stays in the VD airport list after take-off (see that manual).

A380X standby instruments (ISIS)

The lower ISIS becomes the Standby Navigation Display (SND) with its power-up test, and the MODE pb swaps the SFD and the SND between the two ISIS, back to normal at each power-up

Aircraft	A380X
Where to find it	Centre instrument panel: the two ISIS (upper: SFD, lower: SND) and their MODE, MENU, LS/DIR TO, +/- and SET/SEL controls
Branch	`core/isis`, based on `master` (FlyByWire master 76db0aef1 merged in); independent.
Status	SND tested in the sim by the user: OK. SFD/SND reconfiguration tested by the user: OK. SND power-up test and the normal configuration at power-up: not sim-tested (unit tests).
Pictures	The SND instrument in the local browser harness, running the build of the branch, with ADIRU data typed in (the harness has no aircraft). No pictures for the power-up test and the reconfiguration.

1. Standby Navigation Display (SND)
2. SND power-up self-test
3. SFD/SND reconfiguration with the MODE pb
4. Normal ISIS configuration at each power-up

1. Standby Navigation Display (SND)

Aircraft: A380X | **Where:** Lower ISIS | **Status:** Tested in the sim by the user: OK. 10 unit tests.

What it is for

The A380 has two ISIS (integrated standby instrument systems): the upper one is the Standby Flight Display (SFD: attitude, speed, altitude), the lower one the **Standby Navigation Display** (SND): heading, position and a small navigation of its own, independent of the FMS (A380 FCOM DSC-34-10-20). On the FBW A380 the lower screen was black; it is now the SND.

The screen

Figure 1. The SND: ground speed, heading source IR3, heading rose with the yellow heading index and the green track diamond, aircraft symbol, position from ADIRU 3.

Item	Meaning
GS 370	ground speed, kt
IR3	the heading comes from ADIRU 3 (IR1 when ADIRU 3 has none); **TRU** in cyan in true reference
Heading rose	240° shown, a mark every 5°, numbers every 30°; white markers at 45° and 90°
Green diamond	the aircraft track
GPIR3 N 48 15.5 / E 003 32.3	the aircraft position (degrees and minutes) and its source
HDG, PPOS (amber)	heading or position not available

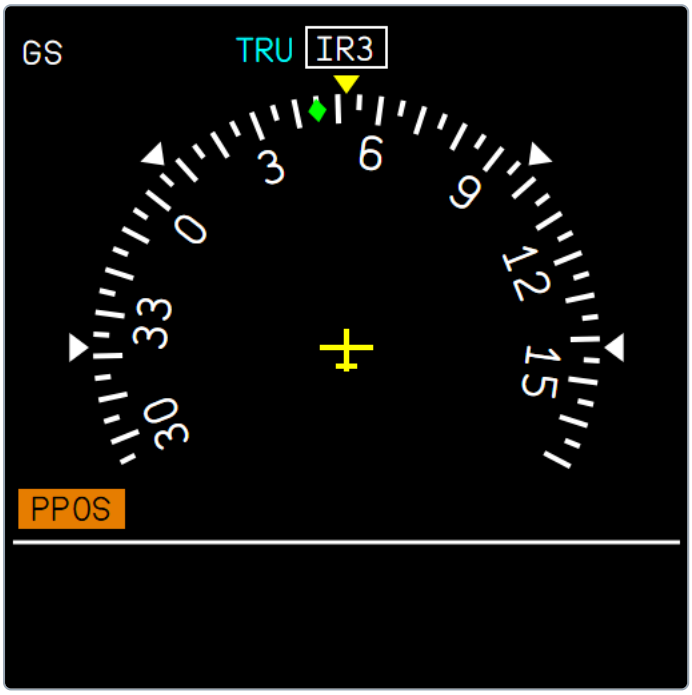


Figure 2. True reference (TRU), no position (PPOS), no ground speed.

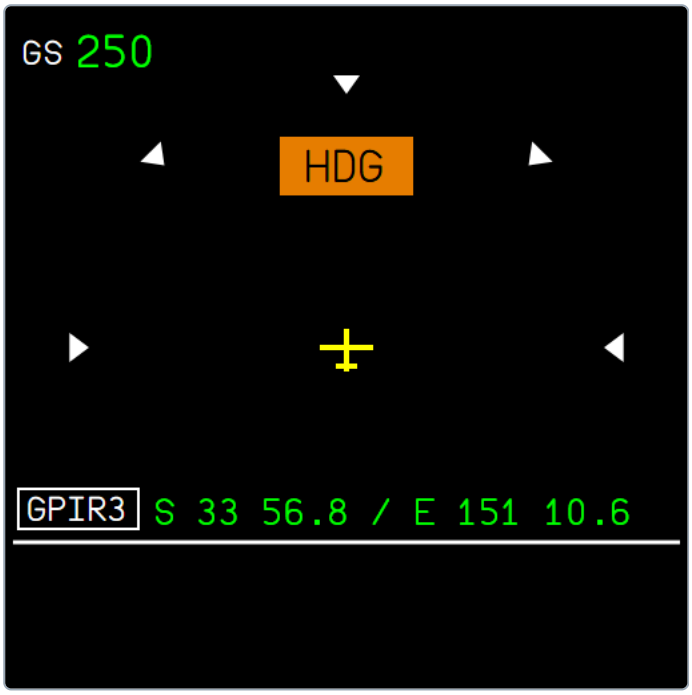


Figure 3. No heading from either ADIRU: HDG replaces the rose.

Buttons

Button	Action
MODE	SFD/SND reconfiguration: see the feature below
+ / -	brightness
MENU	opens the menu under the white line; press again to leave it
SET/SEL knob	turn: next item or value; push: select, or go to the next field
LS/DIR TO	DIR TO (see below); press again to leave it

The waypoint list

The SND has its own list of waypoints, entered by coordinates (it has no database). Press MENU: the menu shows INSERT WPT, INSERT FIX, EDIT WPT / FIX and CLEAR WPT / FIX in cyan; turn the knob to choose, push to select.

1. MENU, then push on **INSERT WPT**: the coordinates show, starting from the last waypoint (or your position).
2. Turn the knob to change the underlined field, push to go to the next one: N/S, degrees, minutes, tenths, then E/W, degrees, minutes, tenths.
3. Push on the last field: the waypoint is added at the end of the list.

Figure 4. The menu: INSERT WPT.

Figure 5. Entering the coordinates: the degrees of latitude being set.

EDIT and CLEAR WPT / FIX: turn to choose the waypoint ("2/3 N5015/E00232", then the FIX last), push to edit or to delete it.

DIR TO and the navigation

1. Press **LS/DIR TO**. With a list, turn to choose the waypoint and push. Without a list, enter the coordinates of the point as above.
2. The navigation starts from your position: FROM, TO (white) and NEXT under the line, the bearing and distance to the TO waypoint top right, and the course on the rose.
3. Passing abeam the TO waypoint, the SND goes on to the next one.

Figure 6. Two waypoints, navigation not active: the true bearing and distance between them.

Figure 7. After DIR TO: FROM / TO / NEXT, TO WPT 342° 126 NM, the course arrow.

Figure 8. About 5 NM right of the track: the deviation bar one large mark left of the aircraft (large marks every 5 NM, small every 2.5 NM).

The FIX

One point you want to watch (a beacon, an airport): MENU, **INSERT FIX**, enter its coordinates. The SND shows its distance top left, a magenta pointer towards it on the rose, and its coordinates in magenta at the bottom of the list. A new FIX replaces the old one; EDIT / CLEAR WPT / FIX offer it after the waypoints.

Figure 9. A FIX alone: 291 NM, bearing 060°.

Figure 10. The FIX together with a DIR TO: FROM / TO / NEXT / FIX.

Limitations and why

Limitation	Why
No automatic SFD/SND reconfiguration	FBW has no ISIS display failure; the MODE pb reconfigures by hand (see SFD/SND reconfiguration below)
The list is lost when the flight is reloaded	the ISIS keeps it while it runs

Position source always GPIR	the FBW ADIRS has no separate GPS / IRS positions yet
Coordinate entry, 10 waypoints at most, sequencing abeam	the FCOM figures of these procedures stop at the first frame: this is our reading

How to use

1. Read heading, track, ground speed and position on the lower ISIS.
2. Waypoints: **MENU**, **INSERT WPT**, set the coordinates with the SET/SEL knob (turn to change, push for the next field).
3. **LS/DIR TO**, choose a waypoint and push: the SND guides you there (TO WPT bearing and distance, course, deviation bar).
4. One point to watch: MENU, **INSERT FIX**. EDIT / CLEAR WPT / FIX to change the list.
5. **+ / -** for the brightness.

How to test

#	Do this	Expected result
1	A380X powered, ADIRS aligned (NAV). Look at the lower ISIS.	GS, IR3 (or IR1), the heading rose with the yellow heading index and the green track diamond, GPIR3 and the position at the bottom.
2	Press the **TRUE** north reference pushbutton (MAG / TRUE).	TRU in cyan; heading and track in true. Press again: back to magnetic.
3	Press **MENU**.	INSERT WPT, INSERT FIX, EDIT WPT / FIX, CLEAR WPT / FIX in cyan.
4	Push the SET/SEL knob on INSERT WPT; turn and push through the 8 fields.	The waypoint is added to the list under the white line.
5	Add a second waypoint the same way.	The true bearing and distance between the two show in the list.
6	Press **LS/DIR TO**, choose the first waypoint, push.	FROM / TO / NEXT, TO WPT bearing and distance top right, the course arrow on the rose.
7	Fly off the course by about 5 NM.	The deviation bar moves one large mark (5 NM) from the aircraft.
8	MENU, INSERT FIX, enter a point.	FIX distance top left, a magenta pointer towards it, its coordinates in magenta at the bottom.
9	Press **+** and **-**.	Brighter, dimmer.

2. SND power-up self-test

> **Aircraft:** A380X | **Where:** Lower ISIS | **Status:** Not sim-tested (2 unit tests). It was installed with the SFD/SND reconfiguration that the user tested, with no separate record.

What it is for

The two ISIS of the A380 are the same unit, SFD or SND (A380 FCOM DSC-34-10-20-10). The SFD (upper ISIS) already ran its power-up tests: a yellow INIT box and a countdown for 90 seconds. The SND came on at once. It now behaves as the SFD. The A380 FCOM gives no time: the 90 s and the 10 s below are the ones of the FBW SFD.

What you should see

Situation	What you should see
Cold and dark spawn, first power-up	INIT and the seconds left in a yellow box for 90 s; HDG and PPOS flags meanwhile; then the normal SND.
Spawn with the aircraft powered (runway, cruise...)	The SND is on at once, no test.
Power loss	The display stays on for 10 s; power back within 10 s: no new test. Longer: off, and the tests run again at the next power-up.
Power supply	As the SFD: DC ESS bus, or DC HOT 1 above 50 kt.

How to use

1. Nothing to do: after a cold and dark start, wait for the end of INIT (90 s) before using the SND.

How to test

#	Do this	Expected result
1	A380X cold and dark. Switch the batteries on (DC ESS powered).	The upper (SFD) and lower (SND) ISIS both show INIT with a countdown from 90.
2	Wait 90 seconds.	The SND shows its rose; HDG / PPOS flags until the ADIRS give heading and position.
3	Remove the power for less than 10 seconds, then restore it.	The SND stays on; no new INIT.
4	Remove the power for more than 10 seconds, then restore it.	The SND goes off, then runs INIT again for 90 s.
5	Spawn on a runway (powered).	The SND is on at once, without INIT.

3. SFD/SND reconfiguration with the MODE pb

> **Aircraft:** A380X | **Where:** MODE pb of either ISIS | **Status:** Tested in the sim by the user: OK. 2 unit tests.

What it is for

A380 FCOM DSC-34-10-20-20-10: the MODE pb "enables to switch from SFD to SND display or to switch off the corresponding standby display". Before, MODE only switched the SND off and on. Now each press goes to the next configuration, and one SFD is always displayed. Both ISIS screens carry both displays; the knob and the pushbuttons of an ISIS act on the display it shows.

What you see

MODE pressed	Upper ISIS (1)	Lower ISIS (2)
Normal	SFD	SND
1st press	SND	SFD
2nd press	SFD	off
3rd press	SFD	SND (back to normal)

- The SND waypoint list, DIR TO and FIX are the same whichever ISIS shows the SND.
- On the SFD, the **MENU** pb opens the BUGS page (the A380 FCOM puts the bugs in the SFD menu); the A320 ATT RST on that pb is gone (not in the A380 FCOM).

Limitations and why

Limitation	Why	What to do
No automatic reconfiguration.	The FCOM reconfigures automatically on an SFD display failure; FBW has no ISIS display failure.	Use the MODE pb.
A full MSFS restart is needed after an update.	The cockpit behaviour files and panel.cfg change.	Restart the sim after installing.

How to use

1. Press **MODE** on either ISIS to go to the next configuration (swapped, SND off, normal).
2. Use the knob and the pushbuttons of the ISIS that shows the display you want to work on.

How to test

#	Do this	Expected result
1	A380X powered, both ISIS after INIT. Upper ISIS: SFD, lower ISIS: SND.	Normal configuration.
2	Press **MODE** on the lower ISIS.	The upper ISIS shows the SND, the lower one the SFD.
3	Turn the knob of the upper ISIS (now SND), press its MENU.	The SND menu opens on the upper ISIS; the waypoint list is the one entered before.
4	Press MENU on the lower ISIS (now SFD).	The SFD BUGS page.
5	Press MODE again.	Upper ISIS SFD, lower ISIS off.
6	Press MODE again.	Back to normal: upper SFD, lower SND.

4. Normal ISIS configuration at each power-up

> **Aircraft:** A380X | **Where:** Both ISIS | **Status:** Not sim-tested (2 unit tests).

What it is for

The configuration chosen with the MODE pb survived a power cycle: an ISIS could come back swapped or off. The ISIS now start in the normal configuration (upper ISIS SFD, lower ISIS SND) at each power-up. The FCOM (DSC-34-10-20-20-10) says nothing about it: this is a design choice.

What you should see

Situation	What you should see
ISIS off, then powered (they run INIT)	Normal configuration, whatever MODE showed before.
Power loss shorter than the 10 s standby	The configuration is kept (not a power-up).
Spawn with the aircraft powered	The configuration is not changed by the spawn.

How to use

1. Nothing to do: after a power-up the ISIS are back to normal; use MODE again if you need another configuration.

How to test

#	Do this	Expected result

1	A380X powered, both ISIS on. Press MODE once.	Upper ISIS SND, lower ISIS SFD.
2	Remove the power for more than 10 seconds (batteries and external power off), then restore it.	Both ISIS run INIT; after it, upper SFD and lower SND (normal).
3	Press MODE once, then remove the power for less than 10 seconds and restore it.	The swapped configuration is kept.

Lights: strobe and beacon timing, interior lighting presets

The wing strobes flash the double flash of the real Airbus LED strobes, the A380X beacons flash halfway between the strobes, and the flyPad interior lighting presets save and load again

Aircraft	A32NX and A380X
Where to find it	Exterior lights: overhead EXT LT panel, STROBE and BEACON switches, seen from the outside view; interior lighting presets: flyPad > Presets > Interior Lighting
Branch	`core/lights`, based on `master` (FlyByWire master 76db0aef1 merged in); independent.
Status	Strobe and beacon timing checked in the sim by the user at night on both aircraft: OK. Lighting presets tested in the sim on the A32NX (save, load of a saved row, load of an unsaved row at 50 %): OK; the A380X uses the same fixed code but was not sim-tested.
Pictures	No pictures: a flash timing does not show on a still picture, and the presets page is the stock FBW flyPad page.

1. Strobe double flash and beacon phase

> **Aircraft:** A32NX and A380X | **Where:** EXT LT panel: STROBE ON, BEACON ON; outside view | **Status:** Checked in the sim by the user at night on both aircraft ("both look right").

What it is for

The FCOM gives no flash rates (A320 DSC-33-20-20, A380 DSC-33-30-20); the A380 FCOM only says that the strobes and the beacons alternate. The timing of the real lights comes from the Collins LED light data: a wing strobe double flash of 50 ms on, 60 ms off, 50 ms on, about 60 times a minute, the beacons flashing between the strobes. The FBW effects had a 50 ms gap (A32NX) and a 150 ms gap (A380X) in the double flash, and the A380X beacons flashed too close to the strobes.

What changed

Light	Before	Now
A32NX wing strobes	double flash 50-50-50 ms	double flash **50-60-50 ms**, 60 per minute (unchanged rate)
A380X wing strobes	double flash 50-150-50 ms	double flash **50-60-50 ms**, 45 per minute (unchanged rate)
A380X beacons	at +0.5 s in a 1.33 s cycle	at **+0.67 s**: exactly halfway between two strobe flashes
A32NX beacon, tail strobe	-	unchanged (beacon already halfway, at +0.5 s in a 1 s cycle)

Limitations and why

Limitation	Why	What to do
The A380X keeps its 45 flashes per minute.	There is no A380 source for the rate; 45 per minute is inside the Collins LED range (45 +/- 2).	-
A full MSFS restart is needed after an update.	The effect files (.fx) are loaded at sim start.	Restart the sim after installing.

How to use

1. Nothing to set: switch the **STROBE** and **BEACON** lights ON on the overhead EXT LT panel as usual.

How to test

#	Do this	Expected result
1	A32NX at night (or dusk), on the ground. Switch **STROBE ON** and **BEACON ON**. Go to an outside view that shows a wing tip and the fuselage.	Each wing strobe gives a crisp double flash (two short flashes very close together) about once a second.
2	Watch the beacon and the strobes together.	The beacon flashes halfway between two strobe double flashes: strobe, beacon, strobe, beacon, evenly spaced.
3	Repeat with the A380X.	Same crisp double flash of the wing strobes, a little slower (45 per minute); the beacons (top and belly) flash exactly halfway between the strobes, not right after them.

2. Interior lighting presets save and load again

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Presets > Interior Lighting | **Status:** Tested in the sim by the user on the A32NX: save, load of a saved row, load of an unsaved row at 50 %: OK. Not sim-tested: the load with no presets file yet (fresh install), and the A380X (same code). Native unit tests: extra-backend test/run_tests.sh.

What it is for

The flyPad interior lighting presets did nothing or the wrong thing on both aircraft, because of three bugs in the C++ extra-backend:

- **No preset was ever saved.** The shared ini library refused the existing work folder (it read the mkdir result the wrong way), so every save failed silently and no presets file was written. It now accepts an existing folder. The FADEC fuel configuration save uses the same library and benefits too.
- **An unsaved row faded every light to off.** Loading a row that was never saved left the target values unset, so every light, the PFD, ND, ECAM and MCDU brightness included, faded to 0. Such a row now loads the default preset, **all lights at 50 %**, as the flyPad page says. The A32NX default also had its DCDU and MCDU values in the wrong scale (0.5 instead of 50).
- **A load before the first save failed.** With no presets file yet (fresh install), a load now gives the default 50 % preset instead of leaving the lights untouched.

What you should see

Situation	What you should see
Save Preset on a row (aircraft powered)	The current lighting levels are stored; the row loads them later, also after a sim restart.
Load Preset on a saved row	The cockpit lights and display brightness fade to the saved levels.
Load Preset on a row never saved	Every light and display brightness fades to 50 %; nothing goes dark.
Automatic loading (Day / Dawn/Dusk / Night set to a row)	The selected row loads as above: its saved levels, or 50 % if it was never saved.

Limitations and why

Limitation	Why	What to do
Presets saved before the fix do not exist.	No save ever reached the file before.	Save the rows again once.

The aircraft must be powered to save or load.	Stock flyPad rule.	Power the aircraft first.
A full MSFS restart is needed after an update.	The extra-backend WASM module changes.	Restart the sim after installing.

How to use

1. Power the aircraft and set the cockpit lights and display brightness as you like.
2. Open flyPad > Presets > Interior Lighting, optionally rename a row, and press **Save Preset** on it.
3. Later, press **Load Preset** on that row to get the same lighting back.

How to test

#	Do this	Expected result
1	A32NX (or A380X) powered, on the ground. Set the panel and flood lights and the display brightness to clearly different levels. flyPad > Presets > Interior Lighting: press **Save Preset** on row 1 and confirm.	The save completes.
2	Change the lights, then press **Load Preset** on row 1.	The lights and displays fade back to the saved levels.
3	Press **Load Preset** on a row that was never saved.	Every light and display brightness fades to 50 %: the PFD, ND, ECAM and MCDU stay readable.
4	Restart the sim, power the aircraft and load row 1 again.	The saved levels come back (the presets file was written).

Fire warnings and the ENG FIRE pushbutton

The A32NX fire warnings come only from a detected fire or the FIRE TEST, and the ENG FIRE pushbutton closes the engine LP fuel valve on both aircraft, so the engine runs down as in the FCOM

Aircraft	A32NX (fire warnings, LP fuel valve) and A380X (LP fuel valve)
Where to find it	Overhead FIRE panel (ENG and APU FIRE pushbuttons, AGENT pushbuttons, TEST); E/WD; SD FUEL page
Branch	`core/fire`, based on `master` (FlyByWire master 76db0aef1 merged in); independent of the other core branches. `core/failures-a32nx` merges it in (its fire failures use these warnings).
Status	All three features tested in the sim by the user on their feature branches: OK; the A32NX LP valve starvation again (engine failures, verified live). The core branch itself was not sim-tested (its fire warning commit is adapted to the FWC of master).
Pictures	No pictures: the warnings and the SD FUEL page are the stock FBW displays.

1. A32NX fire warnings only from a detected fire or the FIRE TEST

> **Aircraft:** A32NX | **Where:** E/WD, master warning, overhead FIRE panel | **Status:** Tested in the sim by the user on the feature branch: OK, with an engine fire set through SimConnect. Not sim-tested on this branch (adapted to the FWC of master).

What it is for

The ENG 1 FIRE, ENG 2 FIRE and APU FIRE warnings were raised by the FIRE TEST *or by the FIRE pushbutton*. Releasing (pushing) a FIRE pushbutton without any fire showed the fire warning: a regression of the FBW E/WD port. On the aircraft the fire warning comes from the fire detection or the FIRE TEST only (A320 FCOM DSC-26-20-10). The warnings now use the sim fire signals that the red FIRE pushbutton lights already read (`ENG ON FIRE:1/2`, `APU ON FIRE DETECTED`).

- The fire chime is a normal red warning chime: **MASTER WARN** cancels it (FCOM DSC-31-10), and pushing the FIRE pushbutton silences it (FCOM DSC-26-20-20).
- The forced master warning and chime of the FIRE TEST pushbuttons are unchanged.
- The **APU AGENT AFTER 10 S** countdown now starts with the **APU FIRE** pushbutton (FCOM PRO-ABN-APUF). Before, it started from the ENG 2 FIRE pushbutton and the ENG 1 AGENT 1 pushbutton (a copy-paste error).

What you should see

Situation	What you should see
ENG or APU FIRE pushbutton pushed, no fire	No fire warning, no chime (before: ENG / APU FIRE warning).
Engine fire detected	ENG 1(2) FIRE warning, master warning, continuous chime, red FIRE pushbutton light.
MASTER WARN pressed during the fire	The chime stops; the warning stays on the E/WD.
FIRE pushbutton pushed during the fire	The chime stops (the pushbutton silences the aural).
APU fire, APU FIRE pushbutton pushed	The APU AGENT AFTER 10 S line counts down from the APU FIRE pushbutton.
FIRE TEST pushbutton held	Same as before: test warnings, master warning, chime, lights.

Limitations and why

Limitation	Why	What to do

A fire set in the sim stays on while the engine runs.	MSFS keeps `ENG ON FIRE` at 1 while the engine turns and ignores the extinguish event; FBW AGENT pushbuttons do not send it.	Stop the engine (ENG MASTER OFF, or the LP valve below) to end the fire.
There is no flyPad fire failure on this branch.	The fire detection model and the fire failures are on `core/failures-a32nx`.	Use that branch, or set the sim variable with a SimConnect tool.

How to use

1. Nothing to set: the warnings follow the fire detection.
2. In a fire, follow the ECAM: MASTER WARN to silence, then the ENG FIRE procedure (ENG MASTER OFF, FIRE pushbutton PUSH, AGENT 1 after 10 s...).

How to test

#	Do this	Expected result
1	A32NX on the ground, engines running, no fire. Push (release) the **ENG 1 FIRE** pushbutton on the overhead.	No ENG 1 FIRE warning on the E/WD, no master warning, no chime. Reset the pushbutton afterwards.
2	Push the **APU FIRE** pushbutton with the APU running and no fire.	No APU FIRE warning, no chime.
3	Hold the **ENG 1 FIRE TEST** pushbutton.	ENG 1 FIRE test warning, master warning, chime and the FIRE pushbutton light, as before.
4	Start an engine 1 fire: set the sim variable `ENG ON FIRE:1` to 1 with a SimConnect tool (or, with `core/failures-a32nx`, the flyPad engine 1 fire failure).	ENG 1 FIRE warning in red, master warning flashing, continuous chime, red light in the ENG 1 FIRE pushbutton.
5	Press **MASTER WARN**.	The chime stops and the master warning goes off; the ENG 1 FIRE warning and its procedure stay on the E/WD.
6	Start the fire again, then push the ENG 1 FIRE pushbutton without pressing MASTER WARN.	The chime stops when the pushbutton is pushed.
7	With an APU fire, push the APU FIRE pushbutton.	The APU AGENT AFTER 10 S line counts down from that moment (not from the ENG 2 FIRE pushbutton).

2. A32NX: the ENG FIRE pushbutton closes the LP fuel valve

> **Aircraft:** A32NX | **Where:** Overhead ENG FIRE pushbuttons; SD FUEL page | **Status:** Tested in the sim by the user: OK: the engine starved, then restarted with IGN/START; again (FIRE pb pushed: LP valve 0 %, residual fuel burned, engine down after about 2 min, verified live).

What it is for

A32NX FCOM DSC-28-10-30: "The LP fuel valve is closed by either the engine master switch or the ENG FIRE PUSH pushbutton". Nothing linked the FIRE pushbutton to a fuel valve, so an engine kept running for ever with its FIRE pushbutton pushed. Each engine now has its LP valve: closed by the ENG MASTER switch OFF or the ENG FIRE pushbutton, with the 1.7 s travel of the engine valves.

Once the valve is closed, the fuel left between the valve and the engine is burned at the engine fuel flow. The FCOM ENG FIRE procedure says the engine stops "UP TO 2 MIN 30 S" after the LP valve closes, when that fuel is burned (PRO-ABN-ENG). The amount is sized on the lowest ground idle fuel flow (14 100 ft ISA), so the engine stops after about 99 s at a sea-level idle, sooner at higher thrust. The fuel is refilled when the valve opens again.

What you should see

Situation	What you should see
ENG FIRE pushbutton pushed, ENG MASTER still ON	SD FUEL: the LP valve symbol closes (crossline). The engine keeps running on the remaining fuel, then shuts down (about 1 min 40 s at sea-level idle, never more than 2 min 30 s).
ENG MASTER OFF	The LP valve closes too (the engine also shuts down through its HP valve, as before).
Engine starved	The FADEC handles it as a closed fuel valve: shutdown, no relight while starved.
FIRE pushbutton reset and ENG MASTER ON again	The LP valve opens, the line refills, the engine can be restarted with the normal start (IGN/START).

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules and flight_model.cfg / systems.cfg change (new MSFS fuel valves 13 and 14).	Restart the sim after installing.

How to use

1. In an engine fire, follow the ECAM procedure: **ENG MASTER OFF**, then **ENG FIRE pushbutton PUSH**. Both close the LP valve.
2. Watch the LP valve on the **SD FUEL** page: green inline = open, amber crossline = closed.

How to test

#	Do this	Expected result
1	A32NX on the ground, engines running at idle. Open the SD **FUEL** page.	Both LP valve symbols green, inline (open).
2	Push the **ENG 1 FIRE** pushbutton. Leave the ENG 1 MASTER switch ON. Start a timer.	The ENG 1 LP valve symbol turns to an amber crossline within about 2 s (amber while in transit). No fire warning (no fire). Engine 1 keeps running for now.
3	Wait, watching N1 / N2 and the fuel flow of engine 1.	Engine 1 runs down by itself after about 1 min 40 s (sea level idle), never later than 2 min 30 s.
4	Reset the ENG 1 FIRE pushbutton, set ENG MASTER 1 OFF, then restart the engine with IGN/START and ENG MASTER 1 ON.	The LP valve opens again (green inline) and the engine starts normally.
5	Set ENG MASTER 2 OFF on a running engine.	The ENG 2 LP valve closes on the SD and the engine shuts down as before.

3. A380X: the ENG FIRE pushbutton closes the LP fuel valve

> **Aircraft:** A380X | **Where:** Overhead ENG FIRE pushbuttons; SD FUEL page | **Status:** Tested in the sim by the user: OK.

What it is for

A380 FCOM DSC-28-10 ENGINE LP VALVES: "The ENG MASTER sw control the LP valves. The LP valves automatically close, when their associated ENG FIRE pb is pushed." Pushing an ENG FIRE pushbutton closed no valve, so the engine

kept running. The four engines now have the same logic as the A32NX, with a 3 s valve travel.

The A380 FCOM gives no shutdown delay. The fuel left after the valve is the 1 gallon tank the FBW fuel model already has between each LP valve and its engine: the engine stops about **17 s** after the pushbutton at a sea-level idle (about 24 s at 14 100 ft).

What you should see

Situation	What you should see
ENG FIRE pushbutton pushed	The engine LP valve closes; the engine shuts down after about 17 s at idle; no start or relight while it is starved.
SD FUEL page	LP valve per FCOM DSC-28-20: green inline = open, amber inline = abnormally open, amber crossline = closed.
ENG MASTER switches	Unchanged: they still open and close their engine valves as before.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules, flight_model.cfg and systems.cfg change (new feed valves 60 to 63).	Restart the sim after installing.

How to use

1. In an engine fire, follow the ECAM: **ENG MASTER OFF**, **ENG FIRE pushbutton PUSH**.
2. Check the LP valve of that engine on the **SD FUEL** page.

How to test

#	Do this	Expected result
1	A380X on the ground, four engines at idle. Open the SD **FUEL** page.	Four LP valves green, inline.
2	Push the **ENG 1 FIRE** pushbutton, ENG MASTER 1 left ON. Start a timer.	The ENG 1 LP valve turns to an amber crossline after about 3 s.
3	Watch engine 1.	It runs down by itself about 17 s after the pushbutton (sea-level idle).
4	Try to restart engine 1 with the FIRE pushbutton still pushed.	No start: the engine is starved.
5	Reset the FIRE pushbutton, ENG MASTER 1 OFF then ON with the start selector.	The valve opens (green) and the engine starts.

A380X MFD FMS and SURV pages

Every A380X MFD FMS and SURV page on its FCOM figure with the FCOM entry formats, the missing pages and functions (FUEL PLANNING, computed ALTN fuel, printing, DIRECT TO options, pilot-stored elements), the FBW WIND page with the FCOM additions (history winds, SAT, ground wind, company wind request), the KCCU keyboard rules and the SURV memos; and the FCOM PLACE/DIST waypoints, HOLD at PPOS, EOSID and PRED TO of both FMSs

Aircraft	A380X (the shared flight plan code of DIRECT TO, the 50 waypoint limit, the received company winds and the CHECK ALTN WIND check is also in the A32NX FMS; the A32NX keeps its own alternate cruise level and history wind rules; changes the A32NX MCDU too)
Where to find it	MFD FMS pages (ACTIVE, POSITION, SEC INDEX, DATA menus, WIND) and SURV pages; KCCU; E/WD memos; flyPad Settings > Realism and Dispatch > Printouts
Branch	`core/a380x-mfd-fms`, based on `core/surveillance` (stacked: needs core/surveillance merged first). `core/a380x-atccom-datalink`, `core/fuel`, `core/gps`, `core/a380x-interactive-nd` and `core/efb-performance` are stacked on it, and `core/a380x-wxr-controls` merges it in. It contains FlyByWire master 76db0aef1 (the FBW A380X WIND page, #10929), which replaced this branch's own WIND page and company wind request; the fork fixes listed in the WIND page feature were put back on top of it.
Status	FMS pages used in the sim since September 2026; tested in the sim by the user: FUEL PLANNING, DEST EFOB fix, printing, SEC page fix, clickable checkboxes. WIND page (FBW page with the fork fixes): sim-tested by the user, PASS (report "WIND page OK"). Computed ALTN fuel seen in the user's sim screenshots. Not sim-tested: EXTRA time, company fuel figures, pilot-stored elements kept across sessions, 50 waypoint limit, DIRECT TO options, KCCU entry and switch rules (checked in the browser harness only), SURV memos, SURV page layout as a whole, received company winds and FIX INFO fixes, and the WIND page items that the recorded test does not cover (history winds, 4 minute NO COMPANY REPLY, SEND F-PLN REQUEST block, FMC step cap). Unit tests: 34 (DIRECT TO 10, pilot-stored elements 17, ALTN fuel 5, waypoint limit 2). (added): not sim-tested, 17 unit tests.
Pictures	Captain MFD and flyPad in the local browser harness, running the full development build installed in the sim (pages that other core branches extend may show their extra items). The harness has no navigation database and no flight plan, so most values are dashes or amber boxes; the DIRECT TO geometry is a drawing. No pictures for the features without an old manual.

1. MFD FMS pages to the FCOM
2. WIND page: FBW wind page, draft winds, and the fork additions
3. FUEL PLANNING: minimum BLOCK computed and confirmed
4. DEST EFOB before takeoff without the TAXI fuel, ALTN EFOB units
5. Computed ALTN fuel and time
6. EXTRA time and the FINAL holding fuel flow
7. Fuel figures of the company flight plan
8. FMS print functions and the flyPad printouts
9. Pilot-stored elements kept across sessions
10. 50 pilot-stored waypoints
11. DIRECT TO options: ABEAM, CRS IN, CRS OUT
12. KCCU keyboard entries: ENT validates, ESC cancels
13. KCCU KBD and CCD switches: FCOM rules, ON by default
14. SURV pages laid out and wired per the FCOM
15. SURV memos, and TCAS STBY with ALT RPTG OFF
16. FMS function fixes: SEC pages, received company winds, FIX INFO, STORE WPT, RTE SEL
17. MFD display fixes: sizes, labels, menus and clickable controls
18. PLACE/DIST waypoints, HOLD at PPOS, DIR TO radials, OPT FL, EOSID, PRED TO

1. MFD FMS pages to the FCOM

> **Aircraft:** A380X | **Where:** MFD FMS pages, both sides | **Status:** Built in September 2026 and used in the sim since; individual pages were fixed after the in-sim checks (see the two fix features at the end).

What it is for

Every FMS page of the MFD is laid out on its figure of the A380 FCOM (DSC-22-FMS-20-30, PDF pages 1009 to 1409), the fields accept the FCOM entry formats, and the pages and functions the FCOM describes and FBW did not have are added.

What changed, in short

- **Layout:** each page is placed on its figure of the A380 FCOM (DSC-22-FMS-20-30, the 2011 KAL FCOM): field positions, 40 px entry fields, 22.5 px labels, 38 px tab bar, the title bar. Where FBW followed a newer FMS standard than this FCOM, the FCOM figure was followed (the WIND page is the exception: it is FlyByWire's own page, see the WIND page feature).
- **Entry formats:** every field accepts the FCOM entry formats (DSC-22-FMS-20-100) and answers with the FCOM messages (FORMAT ERROR, ENTRY OUT OF RANGE, NOT IN DATABASE...).
- **New pages:** ALTERNATE, CLOSEST AIRPORTS, OFFSET, EQUI-TIME POINT, LL XING / TIME MKR, WIND, the POSITION pages, DATA / WAYPOINT, NAVAID, ROUTE, PRINTER, company requests (wind, flight plan, report, takeoff data), TRANSFER TO MAILBOX, REJECTED ATC INFO.
- **New functions:** FUEL PLANNING with CONFIRM BLOCK, LRC and MAX TURB speeds, VERT REV RTA, FMS P/N, pilot-stored elements, automatic airway connection, printing.

The menus at the top of the FMS pages give access to everything: **ACTIVE** (the active flight plan and its pages), **POSITION**, **SEC INDEX** (the secondary flight plans) and **DATA**.

ACTIVE menu

INIT

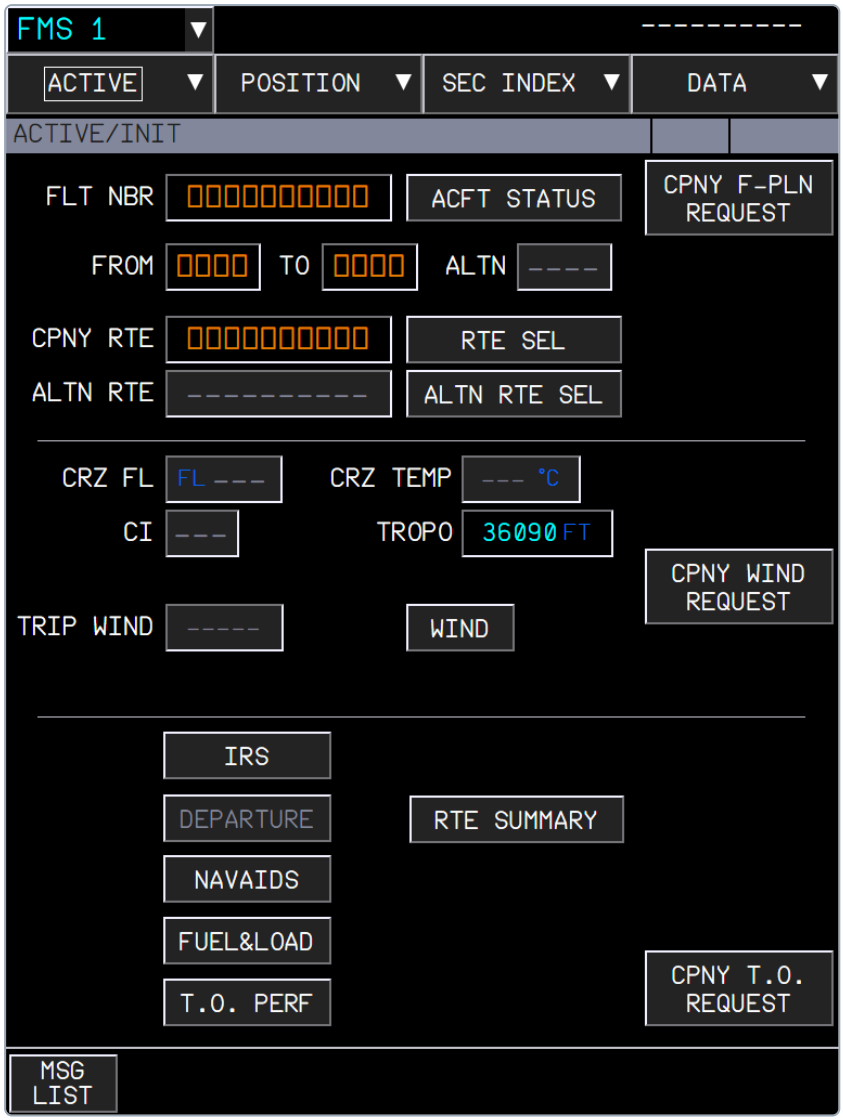


Figure 1. INIT: every button is live. CPNY F-PLN REQUEST (flight plan from the airline, i.e. SimBrief), RTE SEL / ALTN RTE SEL (route selection from the stored and company routes), CPNY WIND REQUEST, CPNY T.O. REQUEST, WIND, and the shortcuts IRS, DEPARTURE, NAVAIDS, FUEL&LOAD, T.O. PERF and RTE SUMMARY.

- Amber boxes are mandatory entries; blue dashes are optional ones.
- **CPNY RTE**: type a stored or company route of the city pair: it is inserted; otherwise NOT IN DATABASE.
- **RTE SUMMARY** stores the active route as a pilot-stored route (DATA / ROUTE).

FUEL&LOAD and FUEL PLANNING

Figure 2. FUEL&LOAD. FUEL PLANNING (next to BLOCK) is available when the flight plan and the weights allow the predictions.

FUEL PLANNING computes the minimum block fuel from the FMS predictions (trip, reserves, alternate, final, taxi) and offers **CONFIRM BLOCK**, which inserts it (FCOM P 179; see the FUEL PLANNING feature below). The EXTRA line shows what is left above the minimum.

PERF

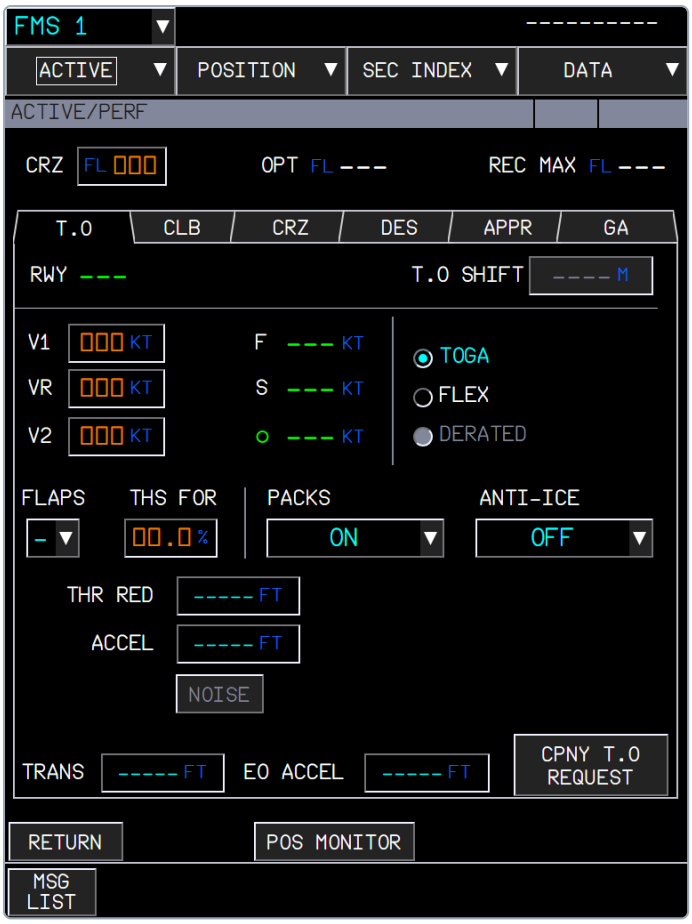


Figure 3. PERF T.O with the company takeoff data button (the flyPad takeoff calculator that answers it is on `core/efb-performance`).

Figure 4. PERF CRZ: MANAGED, PRESEL, and the **LRC** and **MAX TURB** speeds.

- **LRC** (long range cruise): the Mach giving 99 % of the maximum specific range (FCOM PER-IFT).
- **MAX TURB**: the recommended speed for flying in turbulence.

F-PLN and its revisions

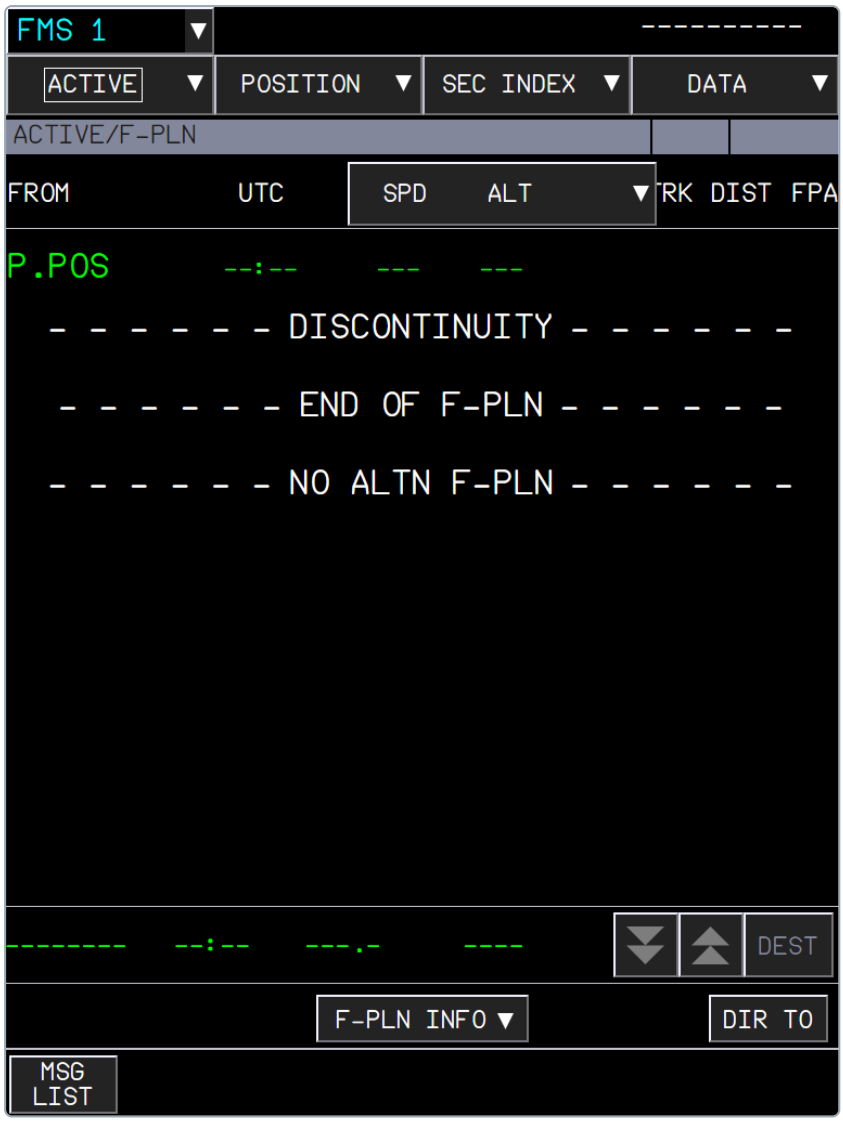


Figure 5. F-PLN without a flight plan: P.POS, DISCONTINUITY, END OF F-PLN, NO ALTN F-PLN.

Click a waypoint name to open the lateral revision menu, a speed/altitude to open the vertical revision. The new pages below open from these menus or from F-PLN INFO.

Figure 6. ALTERNATE: the selected alternate, the database alternates with their company route, track, distance and EXTRA fuel, and two OTHER ALTN fields.

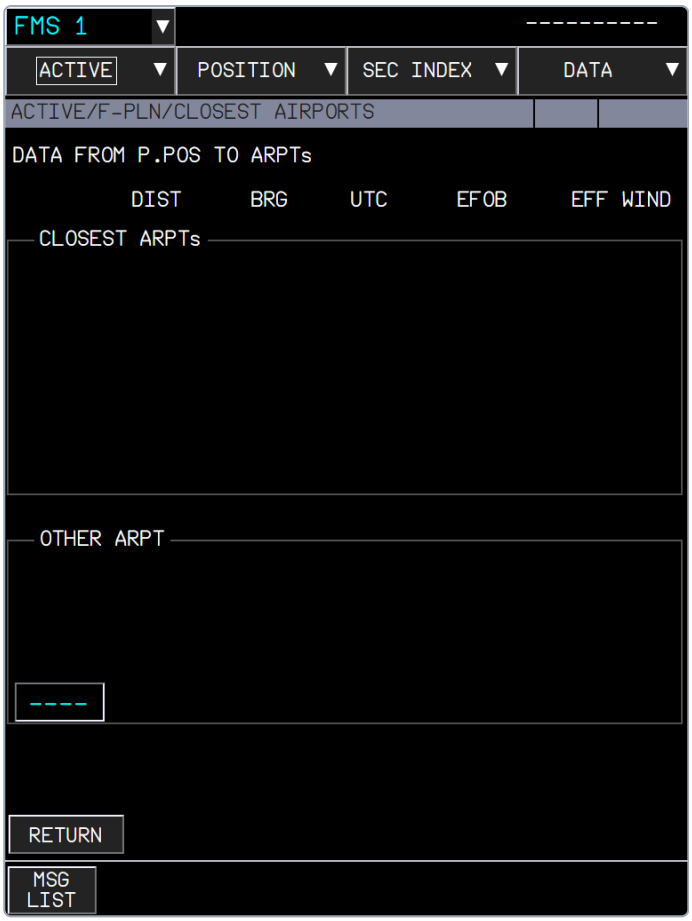


Figure 7. CLOSEST AIRPORTS: the 4 closest airports from P.POS with distance, bearing, UTC, EFOB and effective wind, and up to three OTHER ARPTs.

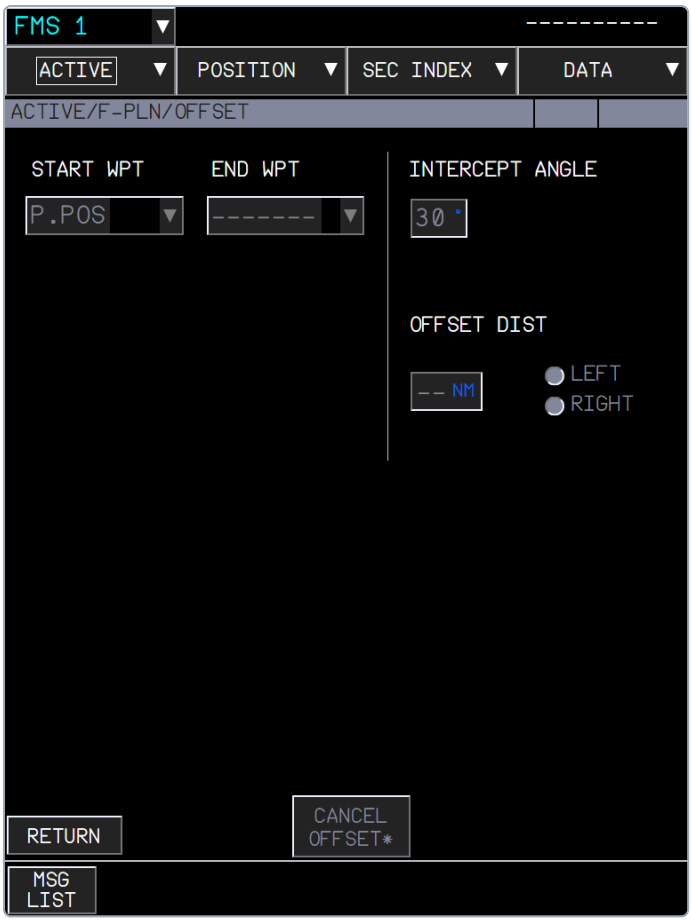


Figure 8. OFFSET: start and end waypoints, distance LEFT / RIGHT, intercept angle (30° by default), CANCEL OFFSET.

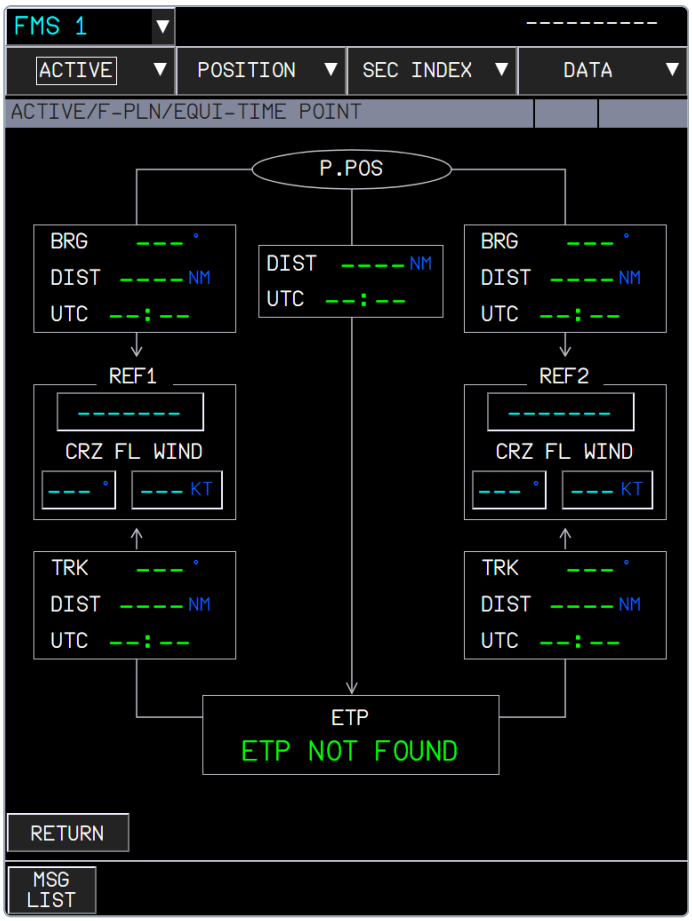


Figure 9. EQUI-TIME POINT: two reference airports, their cruise wind, and the ETP with its bearing, distance and time.

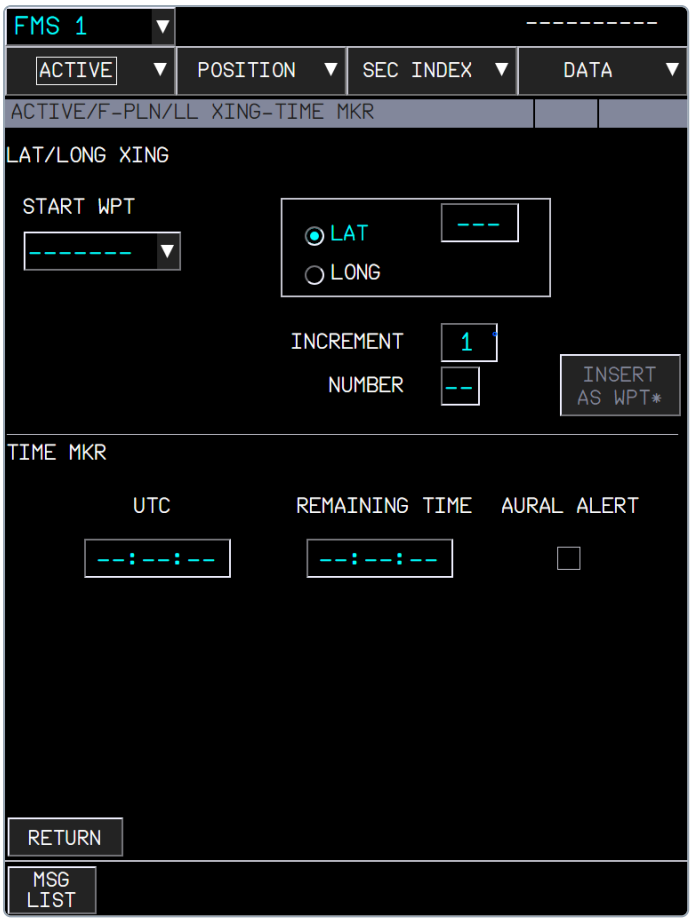


Figure 10. LL XING / TIME MKR: a waypoint at every LAT or LONG crossing (start, increment, number, INSERT AS WPT); a time marker with remaining time and an aural alert.

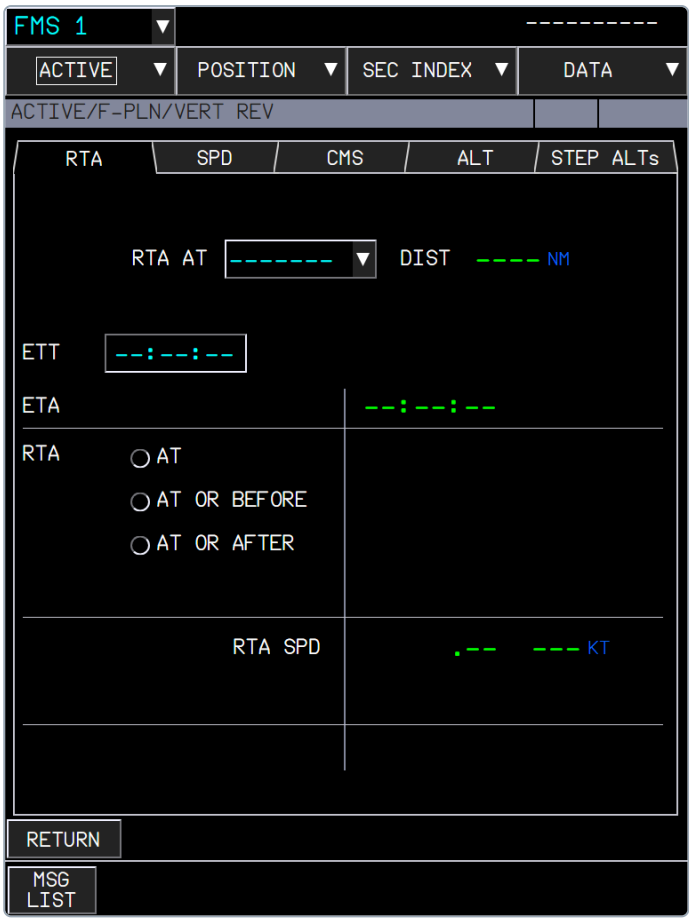


Figure 11. VERT REV, RTA tab: time at a waypoint (AT, AT OR BEFORE, AT OR AFTER), ETA and the speed to meet it; the time error thresholds of the FCOM apply.

Figure 12. AIRWAYS: up to 31 segments; typing the next airway after a waypoint connects the two airways automatically at their intersection.

Company requests (flight plan, report)

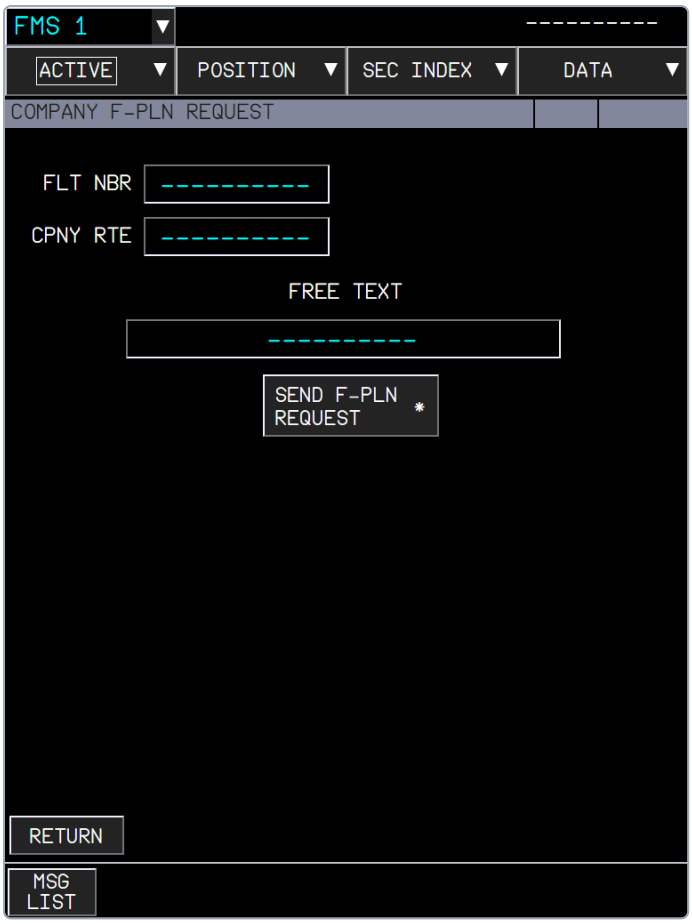


Figure 13. COMPANY F-PLN REQUEST: flight number, company route, free text.

Figure 14. COMPANY F-PLN REPORT: free text, SEND REPORT TO CPNY.

> **NOTE**
> The company functions use SimBrief (flight plan) and the flyPad (takeoff data). A request without a company able to answer ends after a timeout with the FCOM message.

POSITION menu

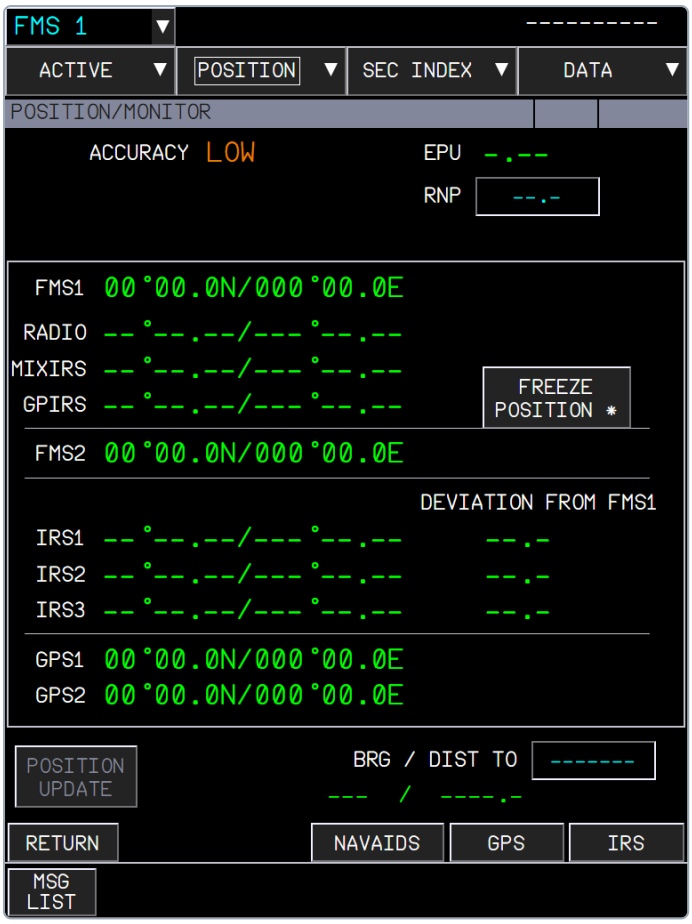


Figure 15. MONITOR: FMS1/2, RADIO, MIXIRS, GPIRS, IRS and GPS positions, accuracy, EPU and RNP, FREEZE POSITION, BRG / DIST TO.

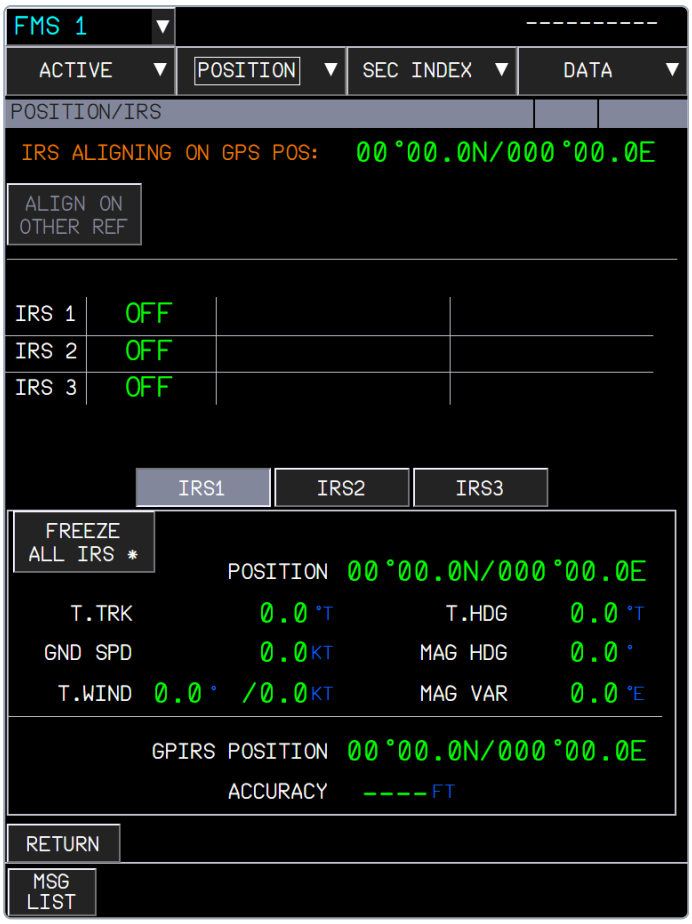


Figure 16. IRS: alignment, the three IRS with their data, FREEZE ALL IRS.

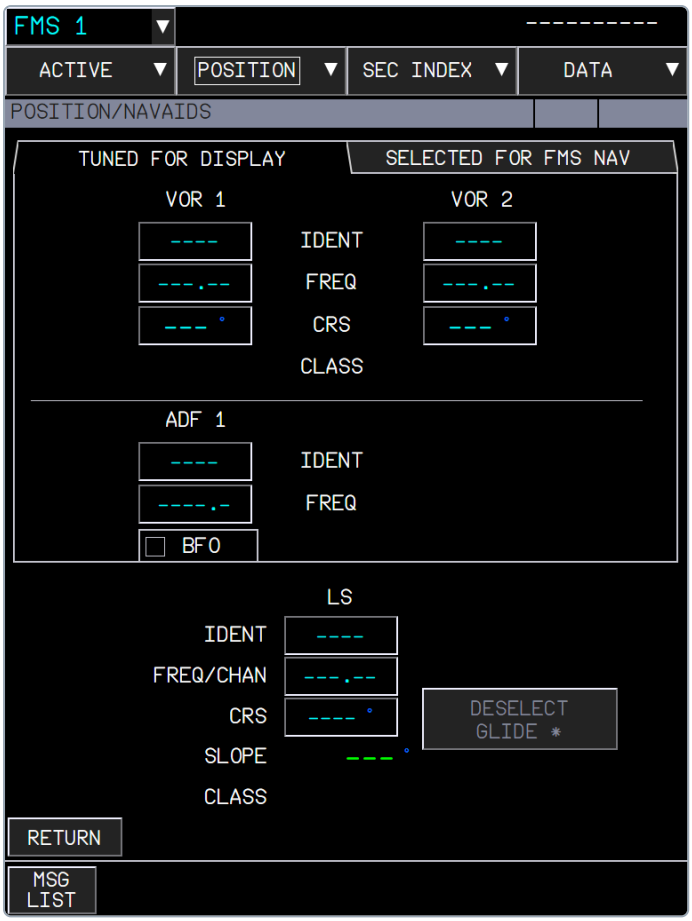


Figure 17. NAVAIDS: tuned for display (VOR, ADF, LS) and selected for FMS NAV, DESELECT GLIDE.

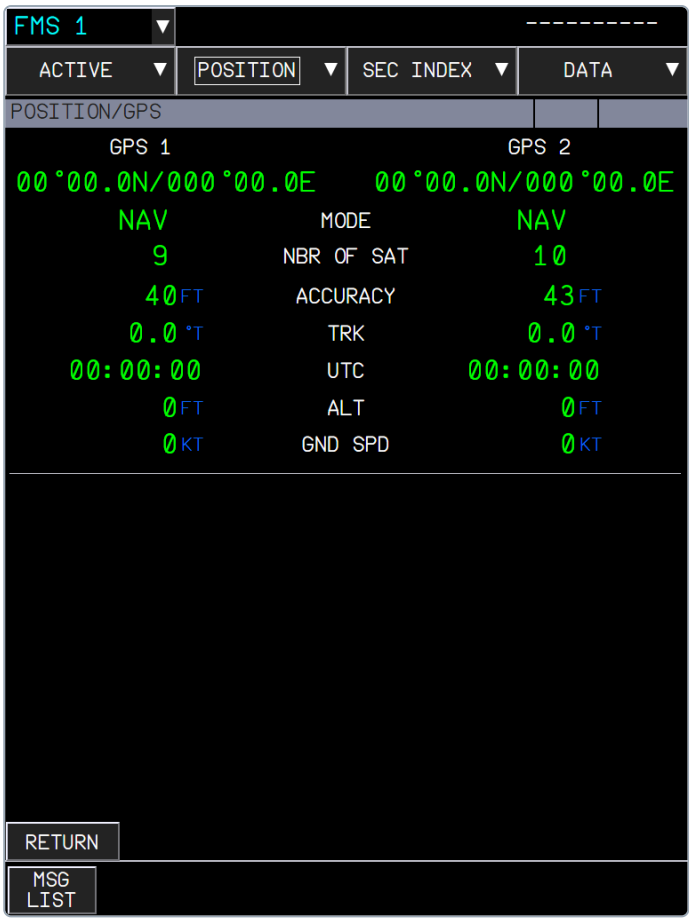


Figure 18. GPS: both receivers: position, mode, satellites, accuracy, track, UTC, altitude, ground speed.

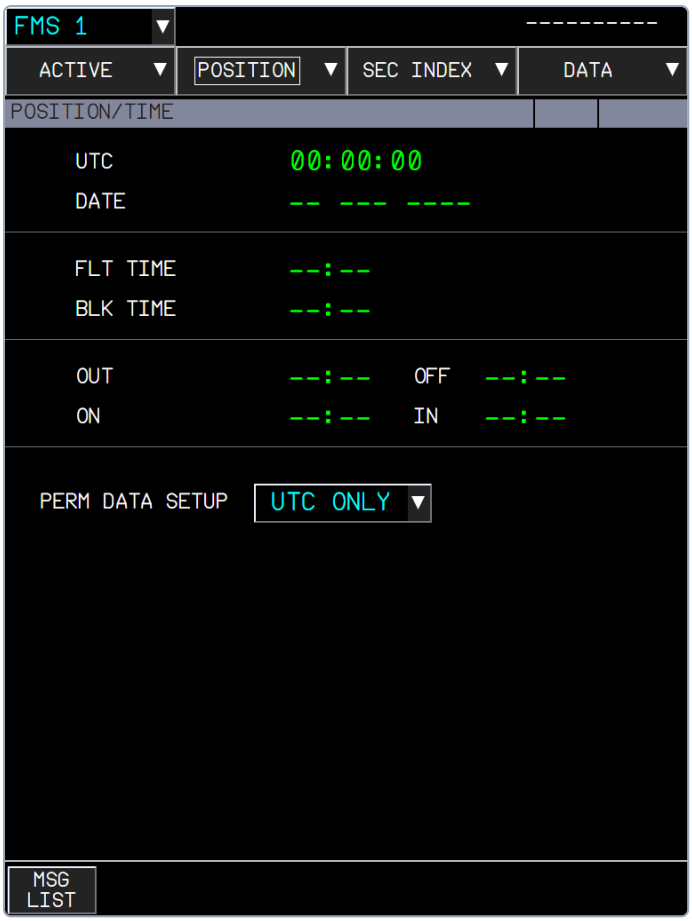


Figure 19. TIME: UTC, date, flight and block time, OUT / OFF / ON / IN, and PERM DATA SETUP (what the permanent data shows).

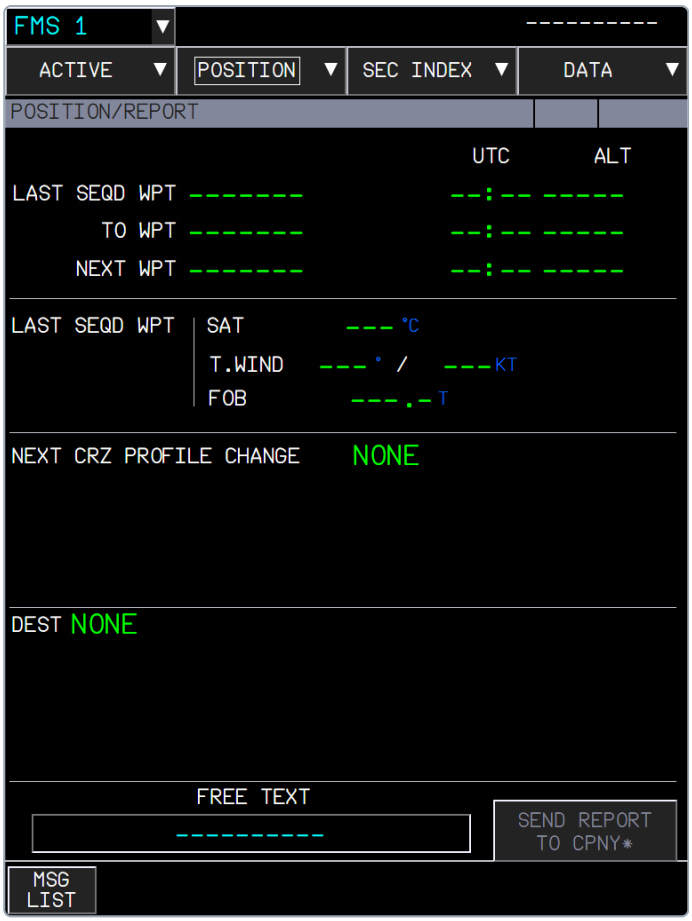


Figure 20. REPORT: last, to and next waypoints with time and altitude, the last sequenced waypoint data, the next cruise profile change, destination.

> **NOTE**
> The TIME page is not in the 2011 FCOM; it follows a later FMS standard at the user's request.

SEC INDEX

Figure 21. SEC INDEX: SEC 1 / 2 / 3, IMPORT, CPNY F-PLN REQUEST into a SEC, the pages of the SEC, DELETE, XFER TO MAILBOX and PRINT.

Figure 22. TRANSFER TO MAILBOX: sends the SEC flight plan to the ATC mailbox with a free text.

REJECTED ATC INFO (FCOM P 321-323) lists the elements of an ATC flight plan inserted in SEC 3 that the FMS rejected, 4 per page. MSFS cannot send an ATC flight plan, so the list stays empty.

DATA menu

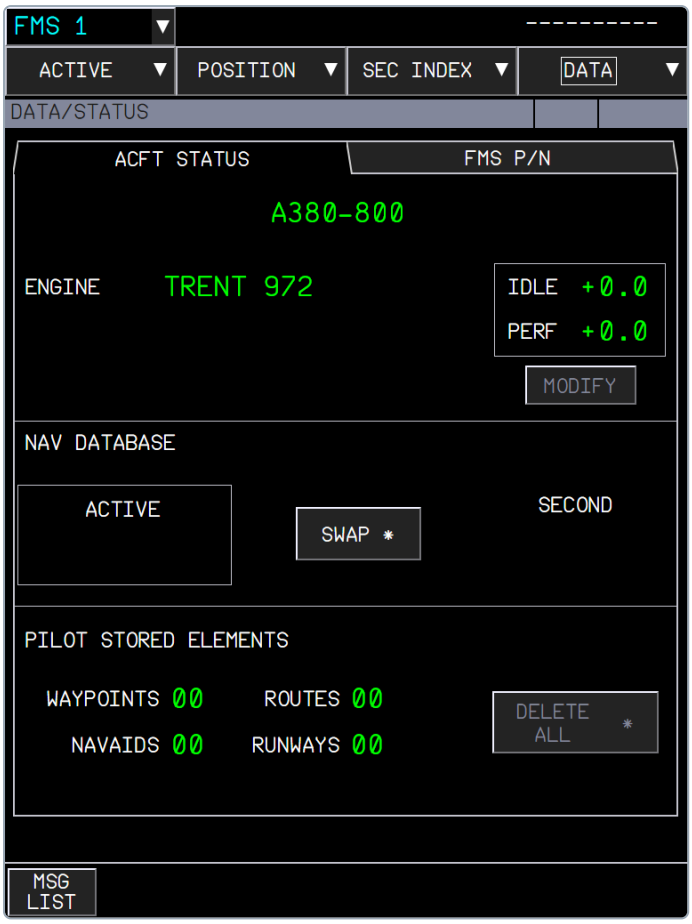


Figure 23. STATUS: aircraft and engine, IDLE / PERF factors (FMS P/N tab), navigation database SWAP, and the count of the pilot-stored elements with DELETE ALL.

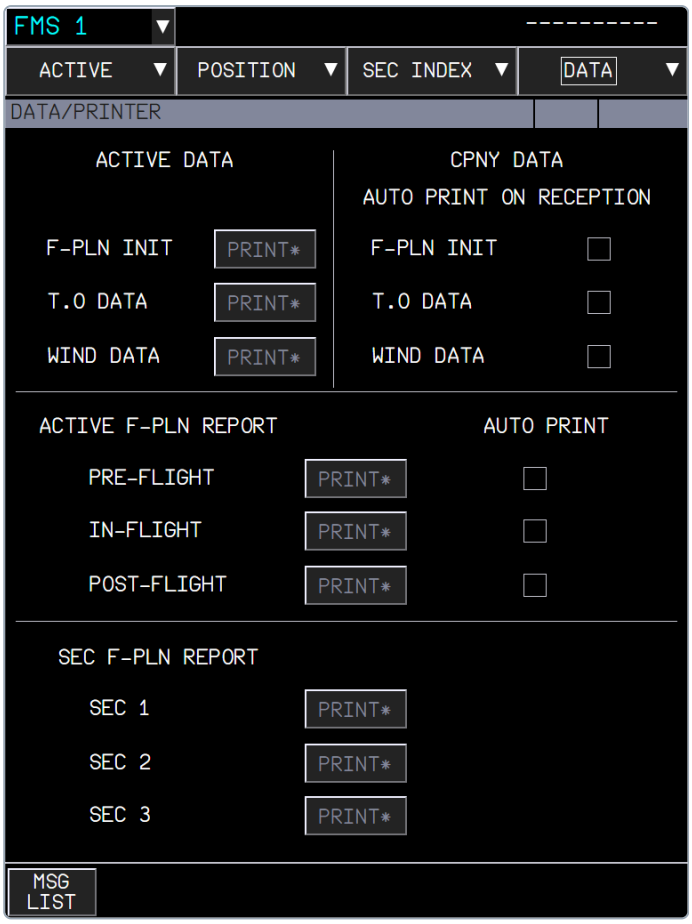


Figure 24. PRINTER: print the F-PLN INIT, T.O and wind data, the active and SEC flight plan reports, and the AUTO PRINT options.

Figure 25. WAYPOINT: database waypoints by ident; pilot-stored waypoints (lat/long, place/bearing/distance, place-bearing/place-bearing), NEW WPT.

Figure 26. NAVAID: database navaids (class, position, frequency, ILS course and slope...); pilot-stored navaids, NEW NAVAID.

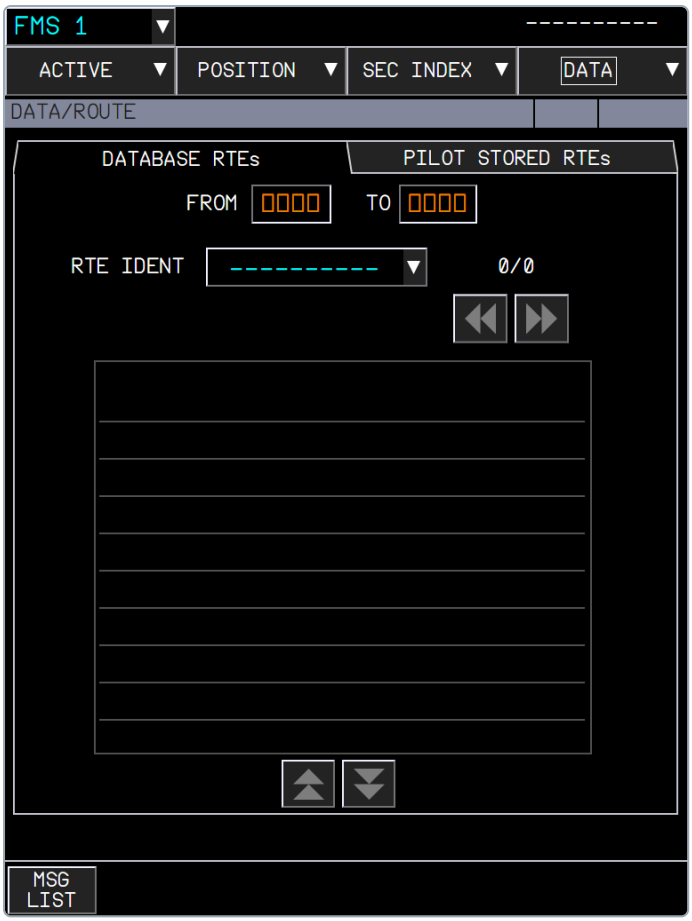


Figure 27. ROUTE: database (company) routes of a city pair, pilot-stored routes with their summary, NEW RTE, DELETE.

Figure 28. AIRPORT: database airports, and the PILOT STORED RWYs tab.

The printouts appear in the flyPad printouts list (see the print feature below).

What to expect

Situation	What you should see
An entry in a wrong format	FORMAT ERROR in the message area; the field keeps its value.
A value out of the FCOM range	ENTRY OUT OF RANGE.
A waypoint or navaid not found	NOT IN DATABASE (or the NEW WPT / NEW NAVAID page when the FCOM says so).
A company request with no company answer	The FCOM timeout message after a few minutes.
A page not available in the flight phase	Its button is greyed.

Limitations and why

Limitation	Why	What to do
The layout follows the 2011 A380 FCOM.	It is the reference available; later FMS standards differ in places.	Report a difference with its FCOM page.
Some values are FBW model stand-ins.	For example GPS satellites and accuracy, like on the A32NX MCDU GPS page; MSFS does not simulate them.	-
Company functions answer only through SimBrief and the flyPad.	There is no airline datalink in MSFS.	-

Pilot-stored navaids cannot be tuned.	The radio tuning uses the MSFS navaids only.	-

How to use

1. Open the FMS pages with the MFD menus: **ACTIVE** (F-PLN, PERF, FUEL&LOAD, WIND, INIT), **POSITION**, **SEC INDEX**, **DATA**.
2. Fill the amber boxes (mandatory) and the blue dashes you need (optional), in the FCOM formats.
3. Open the flight plan revisions by clicking a waypoint (lateral) or a speed / altitude (vertical) on F-PLN.
4. Read a wrong entry's message in the message area (FORMAT ERROR gives the expected format), clear it with CLR INFO.

How to test

#	Do this	Expected result
1	A380X powered on the ground, MFD on. MFD > ACTIVE > **INIT**.	The INIT page as its picture above: CPNY F-PLN REQUEST, RTE SEL, ALTN RTE SEL, CPNY WIND REQUEST, CPNY T.O. REQUEST, the shortcuts at the bottom.
2	Type **ABC** in the TROPO field of INIT (or any altitude field).	FORMAT ERROR with the expected format in the message area; the field keeps its value.
3	Type **99999** in TROPO.	ENTRY OUT OF RANGE.
4	With a flight plan: F-PLN, click a waypoint, open the revision menu.	The menu items open the new pages: ALTERNATE, CLOSEST AIRPORTS, OFFSET, EQUI-TIME POINT, LL XING / TIME MKR, AIRWAYS, FIX INFO, HOLD.
5	AIRWAYS: type an airway, then the next airway right after it.	The two airways are connected at their intersection automatically.
6	PERF > CRZ.	The LRC and MAX TURB speeds are shown.
7	Open POSITION > MONITOR, IRS, NAVAIDS, GPS, TIME, REPORT.	Each page as its figure in the POSITION menu section.
8	Open DATA > STATUS, PRINTER, WAYPOINT, NAVAID, ROUTE, AIRPORT.	Each page as its figure; DATA / STATUS counts the pilot-stored elements.
9	ACTIVE > WIND: CPNY WIND REQUEST, SEND WIND REQUEST (SimBrief account set in the flyPad).	The company winds come back from SimBrief for insertion; without an answer, NO COMPANY REPLY after 4 minutes (see the WIND page feature below).

2. WIND page: FBW wind page, draft winds, and the fork additions

> **Aircraft:** A380X (shared wind code: A32NX too) | **Where:** MFD ACTIVE > WIND (SEC INDEX > SEC n > WIND for a secondary plan), F-PLN wind column, INIT CPNY WIND REQUEST | **Status:** FBW page merged from FlyByWire master 76db0aef1 and tested in the sim by the user: PASS (draft winds and the F-PLN wind click; ALTN CRZ FL, SAT, ground wind, request page, ALTN refusal, typed VERT REV waypoint; the moved pages still refresh; "WIND page OK"). Not run separately in the sim: history winds, NO COMPANY REPLY after 4 minutes, the SEND F-PLN REQUEST block, the FMC step cap. Specs for the fork parts: CpnyWindButtonRule, CpnyWindButtonStateTracker, historyWindRows, HistoryWind, FlightPlanService.historyWinds, NXSystemMessages.

What it is for

The WIND page enters, shows and inserts the climb, cruise and descent winds and the alternate trip wind that the FMS uses for its predictions, and loads the winds of the previous flight or of the company. This branch used to carry its own WIND page. FlyByWire has since added a WIND page of its own to the A380X (FlyByWire master 76db0aef1, "wind page", #10929: the page, a new wind model in the flight plan, draft winds, and a company wind request through the ATSU). That page now replaces this branch's own WIND page and its company wind request; the wind model of the flight plan is shared with the A32NX FMS. What this branch adds on top is listed below.

> **NOTE**
> The page is FlyByWire's layout, not the FCOM figure: the other FMS pages of this manual are laid out on the FCOM figures. The FCOM rules (DSC-22-FMS-20-30 WIND pages) are what the additions below follow.

What you see

- **Four tabs:** HISTORY, CLB, CRZ, DES. A secondary flight plan has no HISTORY tab (history winds are only for the active plan). On the active plan the page opens on the tab of the flight phase: CLB in preflight, takeoff and climb, CRZ in cruise, DES afterwards.
- **CLB and DES:** up to 5 rows of altitude (feet below the transition altitude, flight level above) and wind direction (true) and speed (kt). **CRZ:** choose the waypoint in the CRZ WIND AT list (or with the two arrow buttons); up to 4 levels per waypoint; an entry copied down the route from an earlier waypoint cannot have its level changed or its direction or speed cleared (NOT ALLOWED). **DES** also has the ALTN TRIP WIND row, with the ALTN CRZ FL.
- **Draft winds:** adding, changing or deleting a wind creates draft winds: a DRAFT label on the CLB, CRZ and DES tabs and the buttons **INSERT WIND*** and **CANCEL WIND*** (amber on the active plan). Draft winds are not used for predictions until inserted (A380 FCOM). Inserting shows DRAFT WIND INSERTED.
- **HISTORY:** the winds of the previous flight, with **INSERT HISTORY WIND***, which inserts them directly as the climb winds (no draft) and shows the CLB tab. It is not offered while draft winds or a temporary flight plan exist (FCOM).
- **Phase rules:** the climb winds can be changed only in preflight, the descent winds before the descent phase, and nothing while a temporary flight plan exists; the page then shows the entries as inactive. The cruise winds go inactive after the cruise phase.
- **Bottom:** RETURN (shown only when the page was opened from another page, as the FCOM says), the **CPNY WIND REQUEST** button, and the draft buttons.
- **F-PLN:** clicking the wind of a waypoint in the F-PLN wind column (or the wind item of the waypoint revision menu) opens the WIND page on the CRZ tab with that waypoint selected.

What this branch adds on top of the FBW page

Addition	Before (FBW page as merged)	Now
ALTN CRZ FL (A380 FCOM)	The shared flight plan used the A320 rule (FL100 / FL220 / FL310) for the alternate cruise level.	FL220 when the alternate flight plan distance is under 200 NM, FL310 from 200 NM; it cannot be changed. The A32NX keeps its 100 / 220 / 310. The ALTN fuel computation uses the same level.
SAT at the cruise flight level (FCOM)	The SAT row of the CRZ tab was a pair of disabled dummies.	The row shows the cruise flight level of the loaded plan (not editable) and a SAT entry for it (the pilot temperature entry of the plan). It is disabled without a cruise level, while a temporary plan exists, and after the cruise phase of the active plan (INIT then shows the CRZ TEMP as dashes).
Ground wind (FCOM)	Only GND, or an altitude below the airport elevation, gave the ground wind.	GND, or any altitude below 400 ft, is the ground wind (shown as GND).

History winds (FCOM)	Five rows (the A320 list).	The CRZ FL of the previous flight and FL 250, FL 150 and FL 050, in this order; the row in the CRZ slot carries the CRZ label and a previous cruise at a fixed level (e.g. FL 250) is listed once with the measured cruise wind. The inserted check compares only the levels that INSERT inserts, so INSERT HISTORY WIND is not offered again after an insertion (a level without wind is listed with dashes and not inserted). The A32NX keeps its list.
History winds kept across the update	The history winds moved to a new storage key, so the winds of the last flight would have been lost.	Winds stored under the old keys are converted and moved once, with directions in 0 to 360 degrees and the measured cruise level in 100 ft steps; the old key is removed.
COMPANY WIND DATA REQUEST page (FCOM)	CPNY WIND REQUEST sent the request directly and was disabled while its own request was pending.	The button opens the COMPANY WIND DATA REQUEST page (free text field, SEND WIND REQUEST*, RETURN), also while the request is pending (then SEND WIND REQUEST reads REQUEST PENDING...). Only a company flight plan uplink in progress disables it. The free text stays on the page: SimBrief, the ground station, has no use for it. The INIT page button works the same.
Company wind messages (FCOM)	COMPANY WIND RECIEVED, RECEIVED COMPANY WIND NOT VALID, DRAFT WINDS INSERTED.	COMPANY WIND DATA RECEIVED WAITING FOR INSERTION IN ACTIVE / SEC 1 (2, 3), RECEIVED COMPANY WIND DATA NOT VALID, DRAFT WIND INSERTED.
Reply time, one request per plan, 4 minutes	The answer came at once; REQUEST PENDING... could stay up for ever after a reload of the systems.	Described in the flyPad manual (`core/efb-performance`, company datalink reply time): the flyPad reply time applies to the winds again, NO COMPANY REPLY appears after 4 minutes, and SEND F-PLN REQUEST is blocked while a wind request is pending.
INSERT with four drafts	INSERT did nothing when the CLB, CRZ, DES and ALTN drafts all existed.	All four are inserted.
No false CHECK ALTN WIND	Every company wind INSERT raised CHECK ALTN WIND: the uplinked alternate wind level (feet) was compared with the default level (a flight level), so FL 310 never matched FL 310.	The levels are compared as flight levels (A380 FCOM, A320 FCOM), for both aircraft.

Other fixes of the same update

Fix	Before	Now
Typed waypoints in VERT REV SPD/ALT CSTR AT and STEP ALTs	A waypoint typed with the keyboard reached the page as "no waypoint": SPD/ALT selected nothing, STEP deleted the step.	The typed waypoint is selected in the list; a text that is not in the list shows NOT IN LIST (as the RTA list does).
INIT ALTN refused ident	The typed ident stayed displayed after NOT IN DATABASE, and CLR only blanked the field.	An unknown ident is refused with the alternate unchanged; NONE or CLR deletes the alternate.
FMC update step cap	The FMC step was capped at 1 s of sim time: at 8x or 16x sim rate a normal step is 1.6 s or 3.2 s, so guidance and VNAV ran slow.	The cap is 5 s, a jump protection only (pause, sim time jump); the first step stays 0.
Pages on the new base class	Thirteen pages of this branch sat on the older page base class.	They use FBW's new page base class and still refresh (checked in the sim for ALTERNATE, CLOSEST AIRPORTS, DATA / ROUTE and POSITION / REPORT).

Limitations and why

Limitation	Why	What to do
The SAT is entered for the cruise flight level only.	The FCOM propagates the temperature to the down path cruise waypoints like the wind and allows other altitudes; only the cruise level entry is modelled.	Enter the cruise level SAT.
The WIND page is not on the FCOM figure.	It is FlyByWire's page, kept as it is to follow its development upstream.	-
A company wind request has no real operations centre behind it.	The answer comes from SimBrief (the OFP of the flyPad account); the free text of the request page is not sent.	Set the SimBrief account in the flyPad.

How to use

1. MFD > FMS > ACTIVE > WIND (or click a wind in the F-PLN wind column).
2. On CLB, CRZ and DES: type the altitude (or GND) and the wind (direction / speed) on a row; the DRAFT label appears. For cruise winds, choose the waypoint first.
3. Press **INSERT WIND*** to use the winds in the predictions, or **CANCEL WIND*** to drop them.
4. In preflight, HISTORY > **INSERT HISTORY WIND*** loads the winds of the previous flight as the climb winds.
5. **CPNY WIND REQUEST** asks the company for the winds (SimBrief account set in the flyPad); when the button reads RECEIVED CPNY WIND, click it and choose INSERT* or CLEAR*.

How to test

Flight reload only, no restart: no WASM changed.

#	Do this	Expected result
1	A380X cold and dark at a gate (e.g. EDDF), powered (external power or APU), MFD on the FMS pages. ACTIVE > INIT: FROM/TO **EDDF/EGLL**, CRZ FL **350**, a departure, an approach and a route so that F-PLN shows at least 3 waypoints. WIND, CLB tab: first row, altitude **100**, wind **270/020**.	The row shows FL100 270 / 020 in the draft look, a DRAFT label, and INSERT WIND* and CANCEL WIND*.
2	Press **INSERT WIND***.	DRAFT and the two buttons disappear; the row stays.
3	ACTIVE > F-PLN: click the wind (right-hand column) of the 2nd waypoint.	The WIND page opens on the CRZ tab with that waypoint selected in CRZ WIND AT.
4	INIT: ALTN **EGKK** (about 25 NM from EGLL). In the same field type **XXXX** and press ENTER.	NOT IN DATABASE and ALTN still shows EGKK.
5	WIND, DES tab: read ALTN CRZ FL.	FL220 (the FBW page alone showed FL100).
6	CRZ tab: SAT field, enter **-55**.	SAT -55 in large font (pilot entry); the level next to it shows FL350 and cannot be edited.
7	DES tab, an empty row: altitude **00300** (5 digits = feet), wind **180/010**.	The row shows GND, not 300 ft.

8	Click CPNY WIND REQUEST.	The COMPANY WIND DATA REQUEST page opens (free text, SEND WIND REQUEST*, RETURN). Optional, with a SimBrief flight loaded: SEND WIND REQUEST: REQUEST PENDING... for the flyPad reply time, then RECEIVED CPNY WIND and COMPANY WIND DATA RECEIVED WAITING FOR INSERTION IN ACTIVE; INSERT*: no CHECK ALTN WIND message when the SimBrief alternate level is the default one.
9	F-PLN > the destination > ALTERNATE; F-PLN > CLOSEST AIRPORTS (wait 10 s); DATA > ROUTE > PILOT STORED RTEs; POSITION > REPORT.	Each page is filled and refreshes (alternates list, airport distances and times within a few seconds, stored routes or NONE, TO / NEXT waypoints that follow the plan).
10	F-PLN > a waypoint > VERT REV, ALT CSTR AT: type the ident of the next waypoint with the keyboard, ENTER.	The typed waypoint is selected and the page stays usable.
11	Not covered by the card. Preflight with a previous flight recorded: WIND > HISTORY.	The CRZ FL row and FL 250 / 150 / 050; INSERT HISTORY WIND* loads them as climb winds and is not offered again.

3. FUEL PLANNING: minimum BLOCK computed and confirmed

> **Aircraft:** A380X | **Where:** MFD > ACTIVE > FUEL&LOAD | **Status:** Tested in the sim by the user: OK; the click and convergence fixes user-confirmed in the sim.

What it is for

The FUEL PLANNING button of FUEL&LOAD (A380 FCOM DSC-22-FMS-20-30 P 179) could not be activated and did nothing. It now computes the minimum block fuel and offers CONFIRM BLOCK to enter it.

- **Available** before engine start, with a flight plan and a cruise flight level, the ZFW and ZFWCG entered and no BLOCK entered by the crew.
- **BLOCK = TAXI + TRIP + RTE RSV + MIN FUEL AT DEST.** The TRIP comes from the flight plan predictions, which need a fuel on board: they run with a trial BLOCK, replaced by the result until two computations in a row agree within 50 kg (no result after 8 iterations). A prediction is used only when it was computed with the current trial fuel (before, the first prediction still at the VNAV starting estimate of 4000 lb made the BLOCK grow by about 18 t per iteration: CYUL-CYVR 197.9 t instead of about 63 t).
- While it runs (about 20 to 30 s, several prediction cycles) the button reads **COMPUTING...**, greyed. Not in the FCOM: the real FMS computes the BLOCK at once.
- The minimum BLOCK shows in yellow next to the BLOCK field and the button becomes **CONFIRM BLOCK**, which enters it. A BLOCK entry or an engine start ends the fuel planning.
- FUEL PLANNING did not react at first: the hidden CONFIRM BLOCK at the same place took the click. Hidden FCOM page elements no longer take clicks (checked on 16 FMS and SURV pages).

Limitations and why

Limitation	Why	What to do
A few seconds of COMPUTING...	The FBW predictions need a fuel on board, so the BLOCK is found by iteration.	Wait for the yellow value.
The TRIP comes from the FBW VNAV model.	It can differ from a SimBrief OFP (the user measured the model lower than SimBrief).	Compare with your OFP.

How to use

1. Before engine start, with the flight plan, cruise FL, ZFW and ZFWCG entered, open ACTIVE > FUEL&LOAD.
2. Press **FUEL PLANNING** and wait for the yellow minimum BLOCK.
3. Press **CONFIRM BLOCK** to enter it (or type your own BLOCK).

How to test

#	Do this	Expected result
1	A380X at the gate, engines off. Flight plan with a cruise FL, ZFW and ZFWCG entered, BLOCK empty. ACTIVE > FUEL&LOAD.	**FUEL PLANNING *** next to BLOCK is active.
2	Press FUEL PLANNING.	The button reads COMPUTING..., greyed, for about 20 to 30 s.
3	Wait.	A yellow BLOCK value next to the BLOCK field; the button reads CONFIRM BLOCK. The value is about TAXI + TRIP + RTE RSV + MIN FUEL AT DEST of the page (not tens of tonnes more).
4	Press CONFIRM BLOCK.	The BLOCK is entered (big cyan); EXTRA is about 0.
5	Clear the BLOCK, start an engine, look at the button.	FUEL PLANNING is no longer available after engine start.

4. DEST EFOB before takeoff without the TAXI fuel, ALTN EFOB units

> **Aircraft:** A380X | **Where:** MFD FUEL&LOAD, PERF, F-PLN, ALTERNATE | **Status:** Tested in the sim by the user: OK.

What it is for

- Before takeoff the predictions start from the BLOCK, which still holds the TAXI fuel. The DEST EFOB is now BLOCK - TAXI - TRIP, as the FCOM FUEL&LOAD formulas (EXTRA, LW) and the A32NX FMS do. This also corrects the ALTN EFOB, the DEST EFOB BELOW MIN check, the PERF page DEST EFOB and the ALTERNATE page EXTRA.
- FUEL&LOAD: the ALTN EFOB in tonnes was set as kilograms (136.3 t displayed 0.1 t).

How to use

1. Nothing to set: read DEST EFOB and ALTN EFOB on FUEL&LOAD as usual.

How to test

#	Do this	Expected result
1	A380X before takeoff, flight plan with alternate, ZFW and BLOCK entered, TAXI e.g. 1.5 t. ACTIVE > FUEL&LOAD.	DEST EFOB = BLOCK - TAXI - TRIP (1.5 t less than before this fix).
2	Read the ALTN EFOB.	A realistic tonnage (e.g. a few tonnes below DEST EFOB), never 0.1 t.
3	Open PERF and the ALTERNATE page.	The same DEST EFOB on PERF; the ALTERNATE EXTRA uses it.

5. Computed ALTN fuel and time

> **Aircraft:** A380X | **Where:** MFD FUEL&LOAD (ALTN line), F-PLN > ALTERNATE | **Status:** No dedicated sim test; the computed ALTN fuel shows in the user's sim screenshots of the fuel audit (RJTT-RKSI: ALTN 4.1 t / 0:11 for a 67 NM alternate). Unit tests: 5.

What it is for

The block fuel includes the fuel to divert to the alternate. FBW used a fixed 6.5 t, whatever the alternate. The A380 FCOM (DSC-22-FMS-20-30, FUEL&LOAD page) defines the ALTN fuel as the fuel (and time) for the trip from the primary destination to the landing at the alternate, computed with **CI 0**, at **FL220** when the alternate route is shorter than 200 NM and **FL310** otherwise. This is what the FMS computes now.

Where you see it

Figure 29. FUEL&LOAD: the ALTN line gets its fuel and its time; the arrival at the alternate (destination + ALTN time) is under ALTN in the UTC column.

Figure 30. ALTERNATE page: the fuel to each alternate (EXTRA column) uses the same computation.

- The distance is the alternate flight plan distance, without its missed approach; without an alternate route, the direct distance.
- The weight is the weight at the destination (ZFW + MIN FUEL AT DEST), with the ALTN wind of the WIND page.
- No ZFW: no default ALTN fuel. The value is recomputed only when its inputs change.

How much to expect

Results of the predictor for an alternate close to sea level, fuel on board 12.5 t at the destination (fuel in tonnes / time in minutes / cruise level):

Alternate distance	ZFW 300 t	ZFW 344 t	ZFW 344 t, 30 kt headwind	ZFW 344 t, 30 kt tailwind
60 NM	2.3 t / 13 min / FL140	2.3 / 13 / FL120	2.5 / 14	2.1 / 12

100 NM	3.7 / 19 / FL220	3.7 / 19 / FL200	4.0 / 21	3.4 / 18
150 NM	4.9 / 26 / FL220	5.1 / 27 / FL220	5.5 / 29	4.8 / 25
199 NM	6.1 / 34 / FL220	6.3 / 34 / FL220	6.8 / 37	5.9 / 32
200 NM	6.3 / 32 / FL310	6.5 / 33 / FL310	6.9 / 36	6.2 / 31
300 NM	8.5 / 46 / FL310	8.8 / 46 / FL310	9.4 / 50	8.3 / 44
400 NM	10.8 / 59 / FL310	11.1 / 59 / FL310	11.8 / 64	10.4 / 56
500 NM	13.0 / 72 / FL310	13.3 / 73 / FL310	14.2 / 78	12.5 / 68

- At 200 NM the level jumps from FL220 to FL310 (the FCOM rule): the time drops a little, the fuel rises a little.
- Short alternates (60-100 NM) fly lower than FL220: the cruise level is lowered until the climb and the descent fit in the distance.
- The old fixed value, 6.5 t, is about right only for an alternate 200 NM away.

The computation

- Climb at climb thrust, cruise, idle descent, with the aircraft's VNAV prediction models (the same as the FMS predictions).
- CI 0 speeds of the A380 FMS: 290 kt climb and cruise, 288 kt descent, M0.84, 250 kt below FL100.

What to expect

Situation	What you should see
No ZFW entered	ALTN fuel shown as dashes (no default).
Alternate route entered	ALTN fuel and time from its distance; the arrival time at the alternate in UTC.
No alternate route, alternate airport set	Computed on the direct distance.
ALTN wind entered	More fuel with a headwind, less with a tailwind.

> **NOTE**
> The FL220 / FL310 rule is the A380 FCOM default alternate cruise level (DSC-22-FMS-20-30). After the FBW WIND page merge the shared flight plan applied the A320 rule (FL100 / FL220 / FL310) to the A380X too: the rule is now a value of the aircraft FMS configuration (A380X FL220 below 200 NM and FL310 from 200 NM; the A32NX keeps its 100 / 220 / 310). The same level is shown as ALTN CRZ FL on the WIND page.

Limitations and why

Limitation	Why	What to do
One mean wind for the alternate trip.	The FCOM computation uses the ALTN wind of the WIND page.	-
No holding or approach fuel in ALTN.	The FCOM definition ends at the landing at the alternate; FINAL covers the reserve.	-

How to use

1. Enter the ZFW and an alternate (INIT ALTN, or an alternate route), and the ALTN wind on the WIND page if known.
2. Read the ALTN fuel and time on FUEL&LOAD; the arrival time at the alternate is under ALTN in the UTC column.

How to test

#	Do this	Expected result
1	A380X, flight plan with destination, no ZFW. ACTIVE > FUEL&LOAD.	ALTN fuel as dashes (no default).
2	Enter the ZFW (e.g. 300 t) and an alternate about 100 NM from the destination.	ALTN fuel and time computed, in the range of the table above (about 3.7 t / 19 min at 100 NM); the UTC column shows the arrival at the alternate.
3	Choose an alternate about 300 NM away.	About 8.5 t / 46 min (FL310 from 200 NM on).
4	Enter a 30 kt ALTN headwind on the WIND page (DES tab, ALTN TRIP WIND).	The ALTN fuel rises (and falls with a tailwind).
5	Open F-PLN > ALTERNATE.	The EXTRA column of each alternate uses the same computation.

6. EXTRA time and the FINAL holding fuel flow

> **Aircraft:** A380X | **Where:** MFD FUEL&LOAD (EXTRA line) | **Status:** Not sim-tested (checked in the browser harness: 7.4 t shows 00:37, 12.0 t shows 01:00).

EXTRA time

The EXTRA time next to the EXTRA fuel (FCOM: the fuel, and time, that remains after landing at the destination, minus the MIN FUEL AT DEST) was never computed and always showed --:--. It is now the time the EXTRA fuel lasts at the holding fuel flow of the FINAL fuel default, **0.2 t/min** (30 min = 6 t): 7.4 t of EXTRA shows 00:37, 12.0 t shows 01:00.

Figure 31. FUEL&LOAD with 7.4 t of EXTRA fuel: EXTRA time 00:37. The harness has no flight plan, so the EXTRA fuel is set by hand for this picture; the other values stay as dashes.

- The FCOM leaves the FINAL default to the airline fuel policy (AMI file) and gives no rule to turn the EXTRA fuel into a time: the same holding flow keeps FINAL and EXTRA consistent on the page.
- No EXTRA fuel (no flight plan predictions, or a SEC flight plan): --:--, as the fuel.

The 0.2 t/min holding flow is now one named constant used by both the FINAL fuel default (30 min = 6 t) and the EXTRA time.

> **NOTE**
> On `core/fuel` (stacked on this branch) the FINAL default and the EXTRA time use the A380 holding fuel flow at the ZFW instead of the fixed 0.2 t/min.

What to expect

Situation	What you should see
EXTRA fuel shown	EXTRA time = EXTRA fuel at 0.2 t/min (6 t = 30 min).

Limitations and why

Limitation	Why	What to do
EXTRA time at a fixed holding flow.	The FCOM gives no fuel-to-time rule for EXTRA; the flow is the FINAL default one, not a computed holding at the actual weight.	-

How to use

1. Read the EXTRA time next to the EXTRA fuel on FUEL&LOAD.

How to test

#	Do this	Expected result
1	A380X, flight plan, ZFW and BLOCK entered so that EXTRA fuel shows. ACTIVE > FUEL&LOAD.	EXTRA time = EXTRA fuel / 0.2 t per min (6.0 t shows 00:30, 7.4 t shows 00:37).
2	Leave the FINAL fuel empty.	FINAL default 6.0 t / 00:30.
3	Open a SEC flight plan FUEL&LOAD, or remove the flight plan.	EXTRA time --:--, as the EXTRA fuel.

7. Fuel figures of the company flight plan

> **Aircraft:** A380X | **Where:** MFD INIT (CPNY F-PLN REQUEST), FUEL&LOAD | **Status:** Not sim-tested (checked in the browser harness).

Fuel figures of the company flight plan

Before engine start, performance data come with the company flight plan (FCOM COMPANY F-PLN REQUEST). When the flight plan received from SimBrief is inserted (INIT CPNY F-PLN REQUEST, then INSERT), FUEL&LOAD takes from the OFP, as if typed by the crew: **TAXI**, **RTE RSV** (the OFP contingency, the % is cleared), **ALTN** (only with an alternate) and **FINAL** (the OFP final reserve).

- The **BLOCK is not taken**: press FUEL PLANNING, which computes it from these figures and the FMS TRIP.
- After engine start the flight plan comes in without its fuel figures.
- The FINAL time is computed from the FINAL fuel at 0.2 t/min (4.8 t shows 00:24).
- Any value can be overwritten as usual.

What to expect

Situation	What you should see
Company flight plan inserted before engine start	TAXI, RTE RSV, ALTN and FINAL from the OFP; BLOCK empty until FUEL PLANNING or an entry.

How to use

1. Set your SimBrief account in the flyPad and generate the OFP.
2. Before engine start: INIT > **CPNY F-PLN REQUEST**, then **INSERT** the received flight plan.
3. Check TAXI, RTE RSV, ALTN and FINAL on FUEL&LOAD, then press FUEL PLANNING for the BLOCK.

How to test

#	Do this	Expected result
1	A380X at the gate, engines off, SimBrief OFP with an alternate. INIT > CPNY F-PLN REQUEST, then INSERT.	FUEL&LOAD: TAXI, RTE RSV (the OFP contingency, no %), ALTN (the OFP alternate fuel) and FINAL (the OFP final reserve) as crew entries (big cyan).
2	Read the FINAL time.	FINAL fuel / 0.2 t per min (4.8 t shows 00:24).
3	Look at the BLOCK.	Empty: press FUEL PLANNING to compute it.
4	Request and insert the company flight plan again after engine start.	No fuel figure is taken.

8. FMS print functions and the flyPad printouts

> **Aircraft:** A380X | **Where:** MFD DATA > PRINTER, SEC INDEX PRINT; flyPad Dispatch > Printouts | **Status:** Tested in the sim by the user: printouts reach the flyPad, AUTO PRINT checkboxes, MFD PRINT* OK (on the pedestal printer of `core/printer`).

What it is for

The DATA / PRINTER page and the SEC INDEX PRINT button work (A380 FCOM DSC-22-FMS-10-70, DSC-22-FMS-20-30 P 74-77). The FMS prints the active flight plan initialization, takeoff and wind data, and the pre-flight, in-flight, post-flight and secondary flight plan reports, laid out as the FCOM printout examples, with dashes for data the FMS does not have.

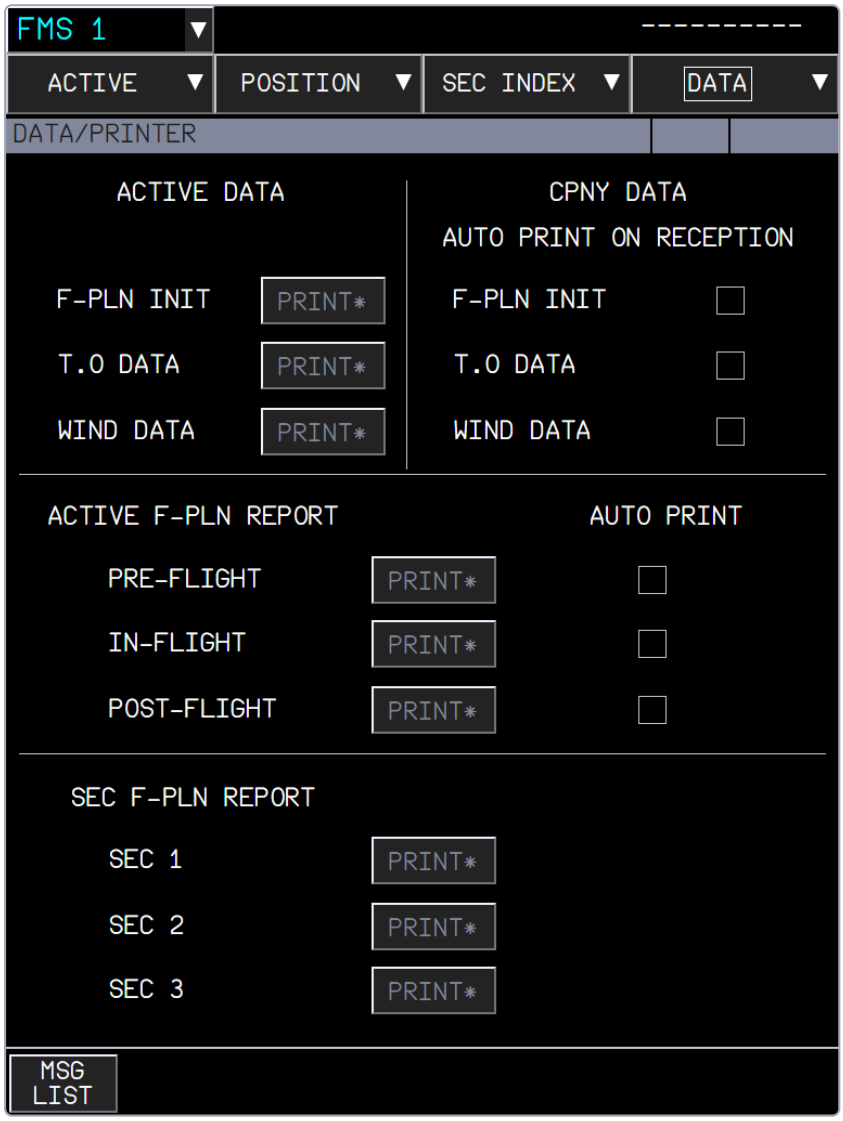


Figure 32. DATA / PRINTER: ACTIVE DATA and CPNY DATA (F-PLN INIT, T.O DATA, WIND DATA), AUTO PRINT ON RECEPTION, the ACTIVE F-PLN REPORT (PRE-FLIGHT, IN-FLIGHT, POST-FLIGHT) with AUTO PRINT, and the SEC F-PLN REPORT.

Print	Available (FCOM)	Automatic print (AUTO PRINT ticked)
PRE-FLIGHT report	In PREFLIGHT, with the flight plan data entered	At engine start
IN-FLIGHT report	From TAKEOFF to engine shutdown	At the transition to TAKEOFF
POST-FLIGHT report	After engine shutdown	At engine shutdown
SEC F-PLN report	With the secondary flight plan	-
CPNY F-PLN INIT / WIND DATA	With company data received	At reception (AUTO PRINT ON RECEPTION)

- On this branch the printed pages go to the flyPad: **Dispatch > Printouts** (A380X only), which keeps the last 30 pages. A page appears there without any notification. With `core/printer` they also come out of the pedestal printer.
- The pages travel on the event bus, synced between the instruments (a view listener trigger did not cross instruments, so the first version never reached the flyPad).
- The AUTO PRINT checkboxes are clickable (they did not react before).

How to use

1. MFD > DATA > **PRINTER**: press the **PRINT*** button of the data or report you want.
2. Tick the AUTO PRINT boxes for the automatic reports.
3. Read the pages on the flyPad: **Dispatch > Printouts**.

How to test

#	Do this	Expected result
1	A380X at the gate, flight plan entered (PREFLIGHT). MFD > DATA > PRINTER.	PRE-FLIGHT PRINT* active; IN-FLIGHT and POST-FLIGHT greyed.
2	Press ACTIVE DATA F-PLN INIT **PRINT***.	flyPad > Dispatch > Printouts: a new page FM ACTIVE FLIGHT PLAN INITIALIZATION DATA, laid out as the FCOM example.
3	Press PRE-FLIGHT PRINT*.	A pre-flight report page in the flyPad list.
4	Tick and untick each AUTO PRINT box.	Each box ticks and unticks.
5	Tick AUTO PRINT IN-FLIGHT, take off.	An in-flight report is printed at the transition to TAKEOFF.
6	SEC INDEX with a SEC flight plan: **PRINT**.	A secondary flight plan report page.

9. Pilot-stored elements kept across sessions

> **Aircraft:** A380X | **Where:** flyPad Settings > Realism; MFD DATA pages | **Status:** Not sim-tested (unit tests: 17; checked in the browser harness over two sessions and with the setting off).

What it is for

On the A380 the crew can store its own waypoints, navaids, routes and runways in the FMS (DATA pages). The FCOM says they are all deleted when all FMCs are shut down (DSC-22-FMS), and that is what happens by default: a new flight starts with an empty list.

In the simulator you may want to keep them from one session to the next (your own departure procedures, a favourite route). The setting below does that. It is marked **Unrealistic** because the real aircraft does not keep them.

The setting

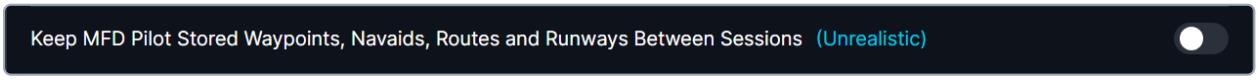


Figure 33. flyPad > Settings > Realism: the last line. Off by default.

- **Off** (default, as the FCOM): the stored elements are deleted at the end of the session.
- **On**: they are saved in the simulator storage and restored at the next session, before the FMS loads them.
- A reload of the flight (same session) never restores them twice.
- Turning the setting off drops the kept copy: the next session starts empty.

- The A380 OIT flyPad has the same option.

Storing elements in the MFD

The DATA menu pages have a PILOT STORED tab each. The DATA / STATUS page counts them and DELETE ALL clears everything.

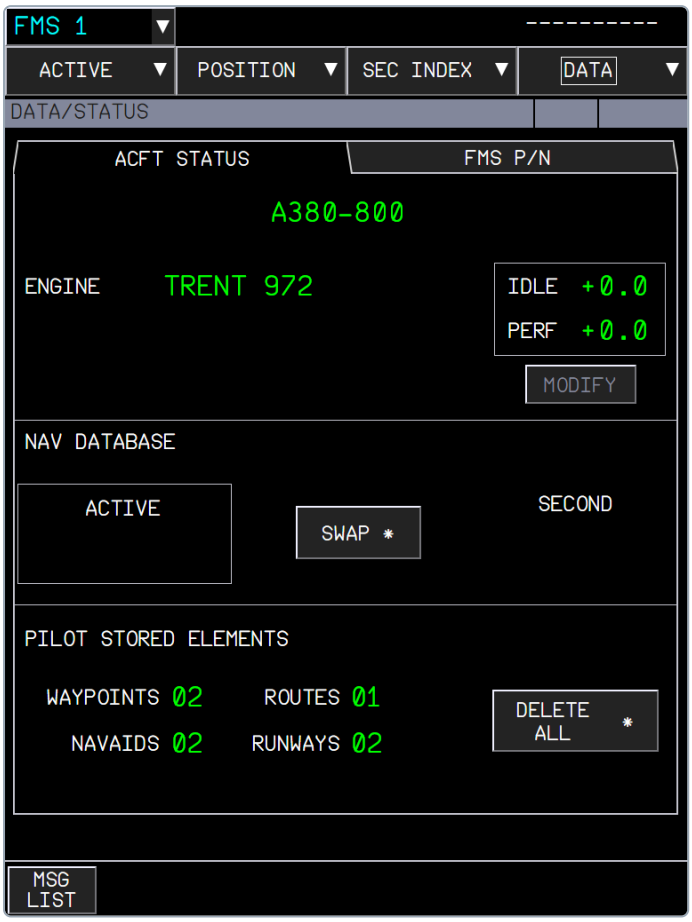


Figure 34. DATA / STATUS: PILOT STORED ELEMENTS: 2 waypoints, 2 navaids, 1 route, 2 runways.

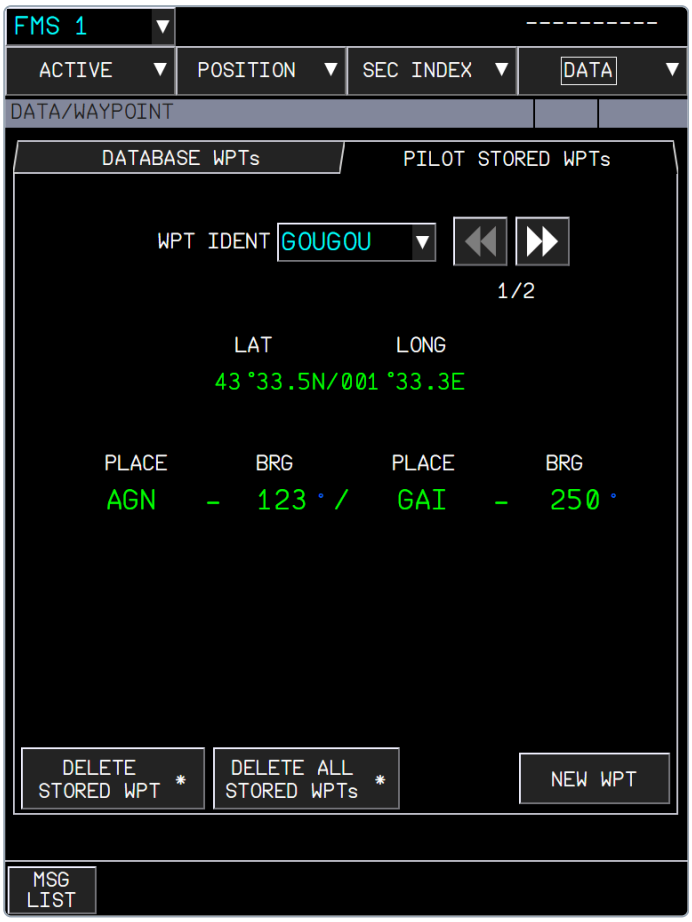


Figure 35. PILOT STORED WPTs: GOUGOU, defined by place/bearing - place/bearing (AGN 123° / GAI 250°), 1/2.

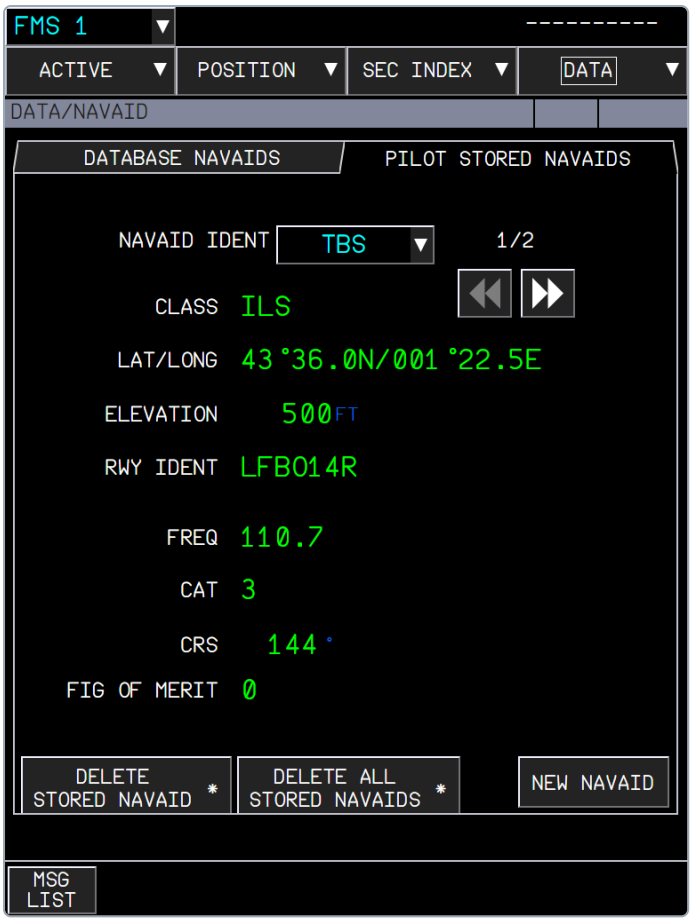


Figure 36. PILOT STORED NAVAIDS: an ILS with its class, position, elevation, runway, frequency, category, course and figure of merit.

Figure 37. PILOT STORED RTEs: NEWROUTE LFBO to WSSS with its summary (runway, SID, airways and waypoints).

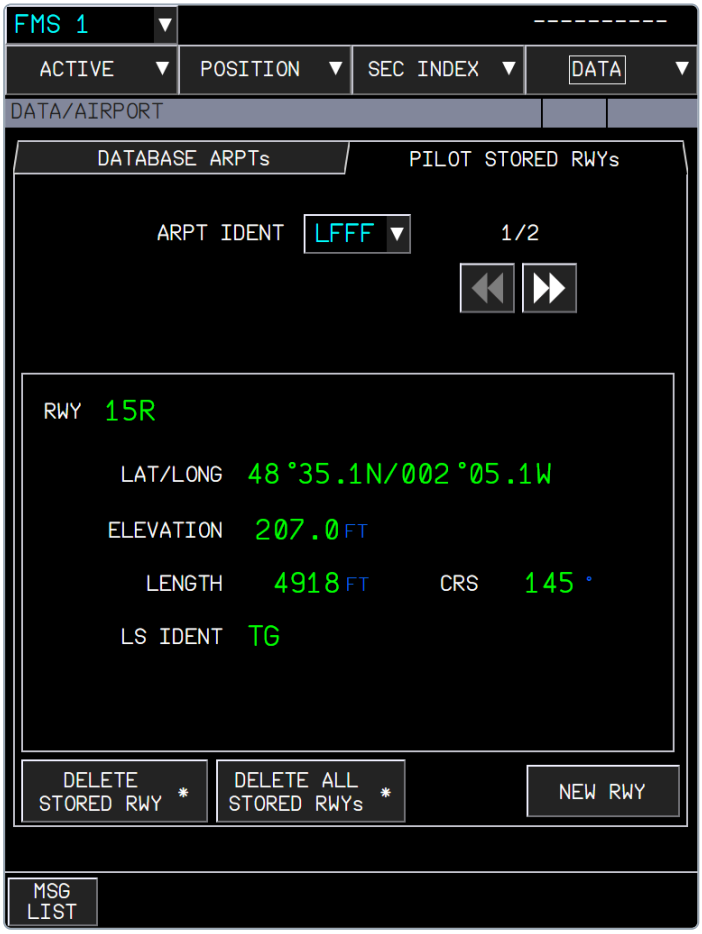


Figure 38. PILOT STORED RWYs (DATA / AIRPORT): runway 15R of LFFF with its position, elevation, length, course and LS ident.

Limits of the database (FCOM)

Element	Limit	When full
Navaids	20	The first one created is deleted.
Runways	10	The first one created is deleted.
Routes	5	PILOT RTEs LIST FULL.
Waypoints	50	The oldest one not used in a flight plan is replaced; if all are used: 50 WPTS MAX : ALL IN USE (see the next feature).

Storing an element with the name of an existing one replaces it (no duplicates).

What to expect

Situation	What you should see
Setting off, new session	DATA / STATUS shows 00 everywhere.
Setting on, new session	The elements of the last session are back.
Setting on, flight reloaded	The elements are there once (no duplicates).
Setting turned off	The kept copy is deleted; the current elements stay until the end of the session.
Stored route of another city pair	Not offered in INIT / RTE SEL (routes are per city pair).

Limitations and why

Limitation	Why	What to do
Pilot-stored navaids cannot be tuned.	The radio tuning uses the MSFS navaids only.	-
Keeping the elements is not realistic.	The real FMS deletes them at shutdown.	Leave the setting off for realism.

How to use

1. flyPad > **Settings** > **Realism**: turn on *Keep MFD Pilot Stored Waypoints, Navaids, Routes and Runways Between Sessions* (off by default, as the FCOM).
2. Store your elements on the MFD DATA pages (WAYPOINT, NAVAID, ROUTE, AIRPORT > PILOT STORED RWYs).
3. At the next session they are back; DATA / STATUS counts them, DELETE ALL clears them.

How to test

#	Do this	Expected result
1	A380X, setting off. Store a waypoint (DATA > WAYPOINT > NEW WPT) and a route (DATA > ROUTE or INIT RTE SUMMARY).	DATA / STATUS counts 1 waypoint and 1 route.
2	Close the session and start a new flight.	DATA / STATUS shows 00 everywhere.
3	Turn the setting on, store the elements again, start a new session.	The elements are back, once each.
4	Reload the flight (same session).	No duplicates.
5	Turn the setting off, start a new session.	Empty again (the kept copy is deleted).

6	Store 21 navaids.	Still 20: the first one created is deleted (FCOM limit).

10. 50 pilot-stored waypoints

> **Aircraft:** A380X | **Where:** MFD DATA > WAYPOINT; DATA / STATUS | **Status:** Not sim-tested (unit tests: 2).

What it is for

A380 FCOM DSC-22-FMS-20-30 (P 117) and DSC-22-FMS-10-40-10: the FMS stores 50 pilot-defined waypoints. FBW shared the A32NX list of 99 and its LIST OF 99 IN USE message.

- When the 50 are stored, a new one replaces the **oldest waypoint not used in a flight plan** (active, temporary or secondary).
- When all 50 are used in flight plans, the new entry is rejected with **50 WPTS MAX : ALL IN USE** (FCOM DSC-22-FMS-20-110).
- The A32NX keeps its 99 waypoints (same flight plan code, its own limit).

How to use

1. Store waypoints as usual (DATA > WAYPOINT > NEW WPT, or the places you create in the flight plan).
2. Watch the count on DATA / STATUS; the oldest unused one is replaced from the 51st on.

How to test

#	Do this	Expected result
1	A380X. Store 50 waypoints (DATA > WAYPOINT > NEW WPT, lat/long), the first one named e.g. WP.	DATA / STATUS: 50 waypoints.
2	Store a 51st waypoint.	Still 50; WP (the oldest, not in a flight plan) is gone from PILOT STORED WPTs.
3	Insert the now oldest stored waypoint into the F-PLN, then store another one.	The oldest waypoint that is not in a flight plan is replaced; the one in the F-PLN stays.
4	(Long; also covered by the unit tests) with all 50 used in the active, temporary or SEC flight plans, store one more.	50 WPTS MAX : ALL IN USE in the message area; nothing is replaced.

11. DIRECT TO options: ABEAM, CRS IN, CRS OUT

> **Aircraft:** A380X | **Where:** MFD F-PLN > DIR TO (or the KCCU DIR key) | **Status:** Not sim-tested (unit tests: 10 on the flight plan geometry).

What it is for

The DIRECT TO page of the A380 (FCOM DSC-22-FMS-20-30, DIRECT TO) has four options. FBW showed them but only the first one worked. All four work now:

Option	What the FMS builds
DIRECT	A direct leg from the aircraft to the target waypoint (as before).
DIRECT WITH ABEAM	The direct leg, plus an **abeam point** for each waypoint it bypasses: the point of the direct leg abeam that waypoint, named **AB** + the first 5 letters of the waypoint (BERGI gives ABBERGI).
CRS IN	An **inbound course** to the target: the aircraft continues on its current track to the intercept point **INTCPT** on that course, then flies the course to the target.

CRS OUT	An **outbound course** from the target: the aircraft continues on its current track to the intercept point INTCPT on the outbound course, then follows it (without overflying the target).

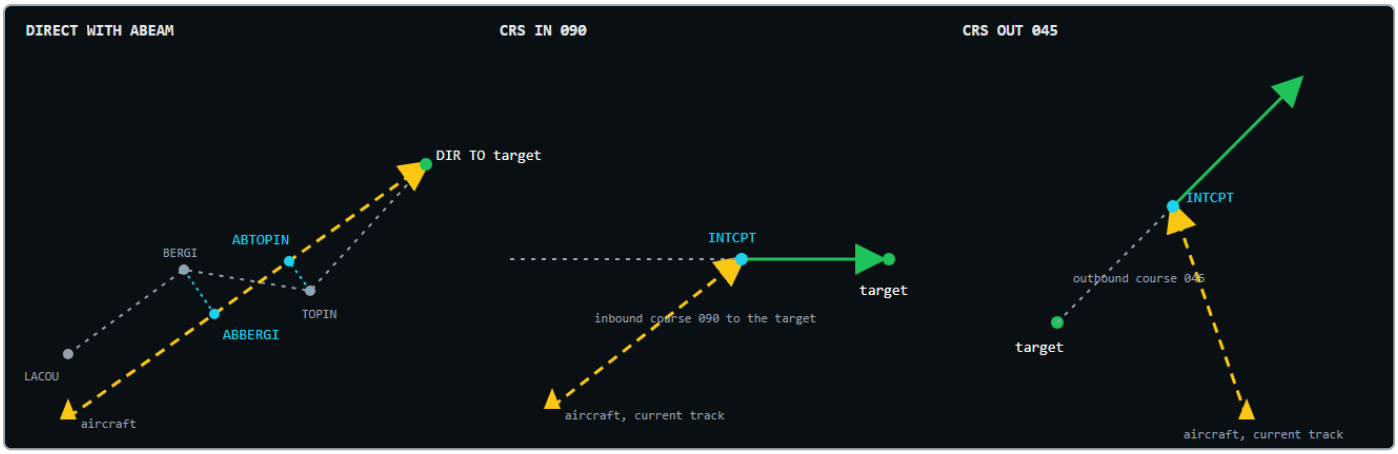


Figure 39. The three new options. Yellow: the aircraft and its path to the first point; green: the course flown after it; blue: the points the FMS creates.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

Using the page

1. Choose the target in **DIRECT TO** (a flight plan waypoint from the list, or type any fix).
2. Choose the option. For CRS IN / CRS OUT a course field appears next to the option.
3. For a flight plan waypoint the course is already filled: CRS IN = the course from the previous waypoint to the target, CRS OUT = the course from the target to the next waypoint (FCOM P 131). Type another one if ATC asks (**NNN** magnetic, **NNNT** true).
4. Check the temporary flight plan (TMPY, yellow) and press **INSERT DIR TO**.

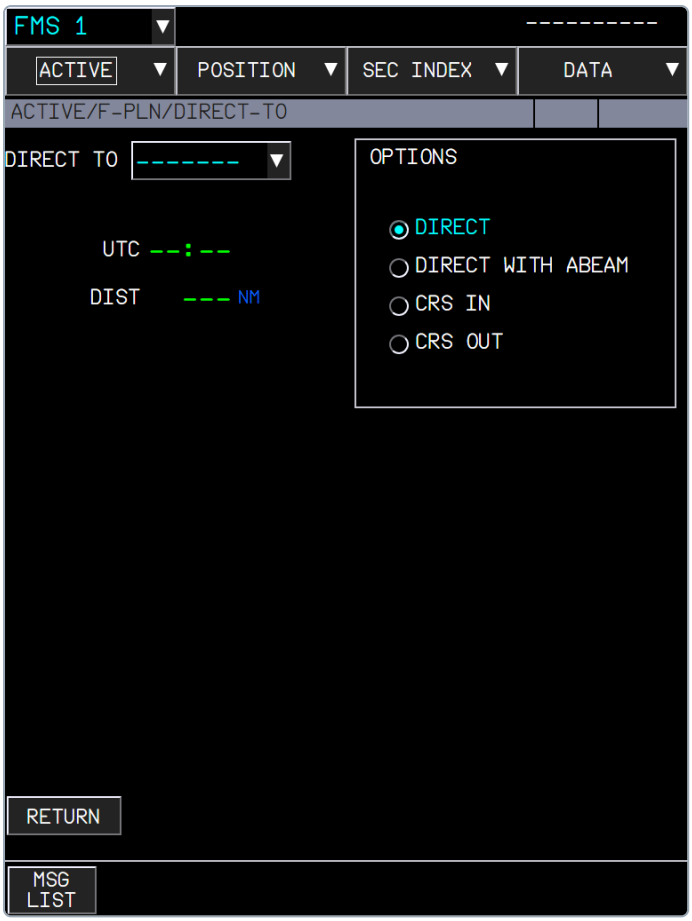


Figure 40. The DIRECT TO page: target, UTC and distance, and the four OPTIONS.

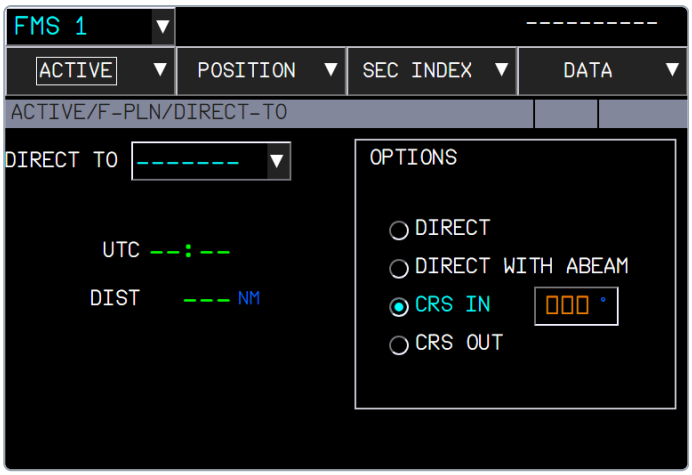


Figure 41. CRS IN selected: the course field (amber boxes until a course is known).

With DIRECT WITH ABEAM, the temporary F-PLN shows **(ABEAM PTS)** in place of the abeam points until the direct is inserted (FCOM P 162).

What to expect

Situation	What you should see
DIRECT WITH ABEAM, waypoints bypassed	One ABxxxxx point per bypassed waypoint, on the direct leg.
DIRECT WITH ABEAM, a waypoint whose abeam is not between the aircraft and the target	No abeam point for it.
CRS IN / OUT with a course that can be intercepted ahead	INTCPT on the course, then the course.
CRS IN more than 160° from the current track, parallel, or behind	No intercept point: a course to the target.
CRS OUT parallel to the track	No intercept point: a course from the target.
CRS IN / OUT without a course	No DIR TO until a course is entered.

Limitations and why

Limitation	Why	What to do
Abeam point names can repeat letters of the waypoint.	The FCOM rule is AB + 5 characters; two waypoints starting with the same 5 letters give the same name.	-

No intercept point when the geometry does not allow one.	Following the FCOM limits (160°, parallel, behind).	Use DIRECT, or another course.

> **NOTE**
> The intercept limits are named constants of the flight plan code (no other effect).

How to use

1. F-PLN > **DIR TO**: choose the target waypoint.
2. Choose DIRECT, DIRECT WITH ABEAM, CRS IN or CRS OUT; for CRS IN / OUT check or type the course (NNN or NNNT).
3. Check the yellow TMPY flight plan and press **INSERT DIR TO**.

How to test

#	Do this	Expected result
1	A380X in flight on a flight plan, NAV engaged. F-PLN > DIR TO, target three waypoints ahead, DIRECT WITH ABEAM.	TMPY F-PLN: the direct leg with (ABEAM PTS) in place of the abeam points.
2	INSERT DIR TO.	One ABxxxxx point (AB + the first 5 letters) per bypassed waypoint, on the direct leg.
3	DIR TO a waypoint ahead with **CRS IN**.	The course field is filled with the course from the previous waypoint; TMPY shows INTCPT on that course, then the target.
4	Type a course more than 160 degrees from the current track.	No INTCPT point: a course to the target.
5	DIR TO with **CRS OUT**.	INTCPT on the outbound course; the aircraft does not overfly the target.
6	CRS IN with the course field empty (fix not in the flight plan).	No DIR TO until a course is entered.

12. KCCU keyboard entries: ENT validates, ESC cancels

> **Aircraft:** A380X | **Where:** KCCU; every MFD, OIT and OANS entry field | **Status:** Not sim-tested as a test of its own (checked in the browser harness); installed, so the MFD entries of the later sim tests use it.

What it is for

On the A380 the crew types on the KCCU keyboard. Before this change a value typed on the 3D KCCU keys was often lost: the field was validated only when the display lost the focus, and clicking a 3D key ends the focus before the key reaches the field. Now the field follows the FCOM (DSC-31-30-20, KCCU):

- **ENT** validates the entry directly.
- **ESC** "*cancels the current field edition. The field returns to the last valid value*".
- Once a field is selected with the cursor, the edition continues whatever happens to the display focus and whatever the **KBD** switch: only ENT, ESC or the selection of another field end it (the previous field is then validated). One field is in edition per side.
- A click on a field selects it even when it already had the focus.
- KBD switch OFF: the KCCU keyboard keys are not active (see the KCCU switches feature below); the PC keyboard still types in the selected field.

Entering a value

1. Set the **KBD** switch of your KCCU ON.
2. Click the field with the cursor (trackball or mouse).
3. Type the value on the KCCU keys (or the PC keyboard).
4. Press **ENT**. The value is checked with the FCOM entry format of the field.

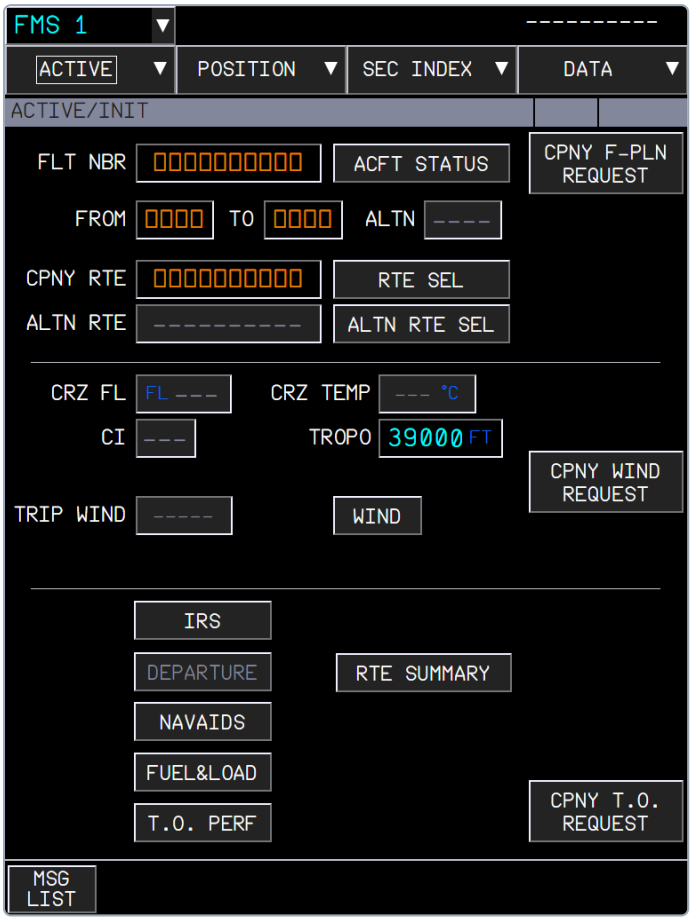


Figure 42. TROPO selected, 39000 typed on the KCCU keys.

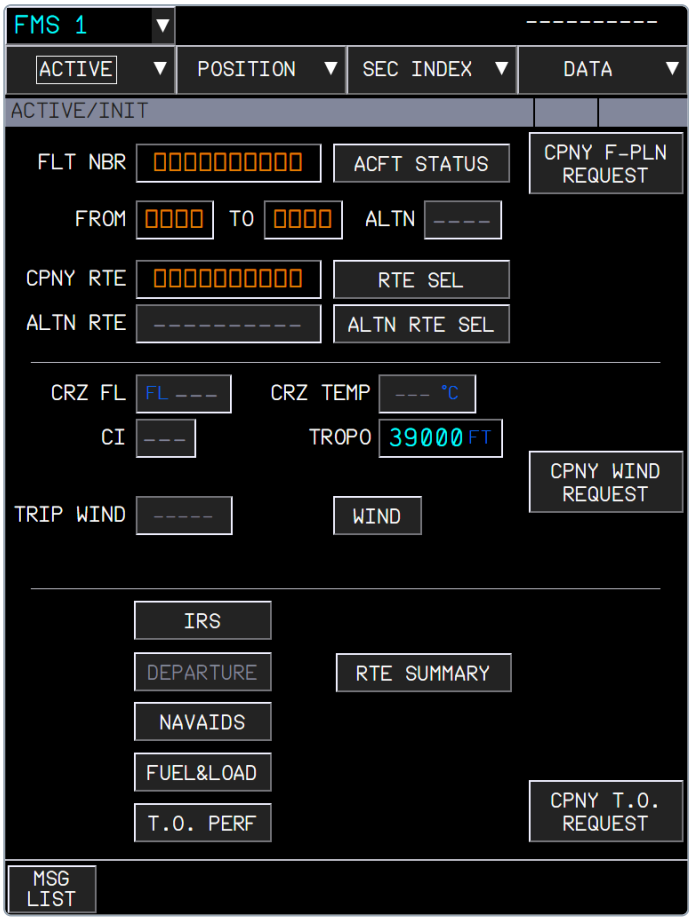


Figure 43. After ENT: TROPO 39000 FT entered (big cyan: a pilot entry).

Wrong entries

ENT on a wrong value leaves the field unchanged and shows the FCOM message in the message area; the message gives the expected format. CLR INFO (KCCU key) clears it.

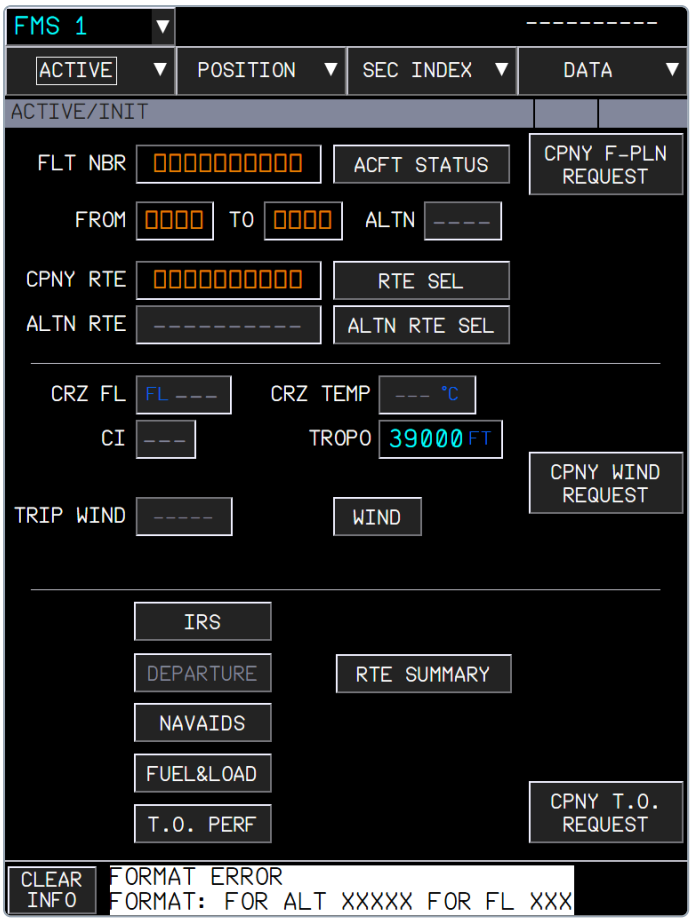


Figure 44. "ABC" in TROPO: FORMAT ERROR, FORMAT: FOR ALT XXXXX FOR FL XXX.

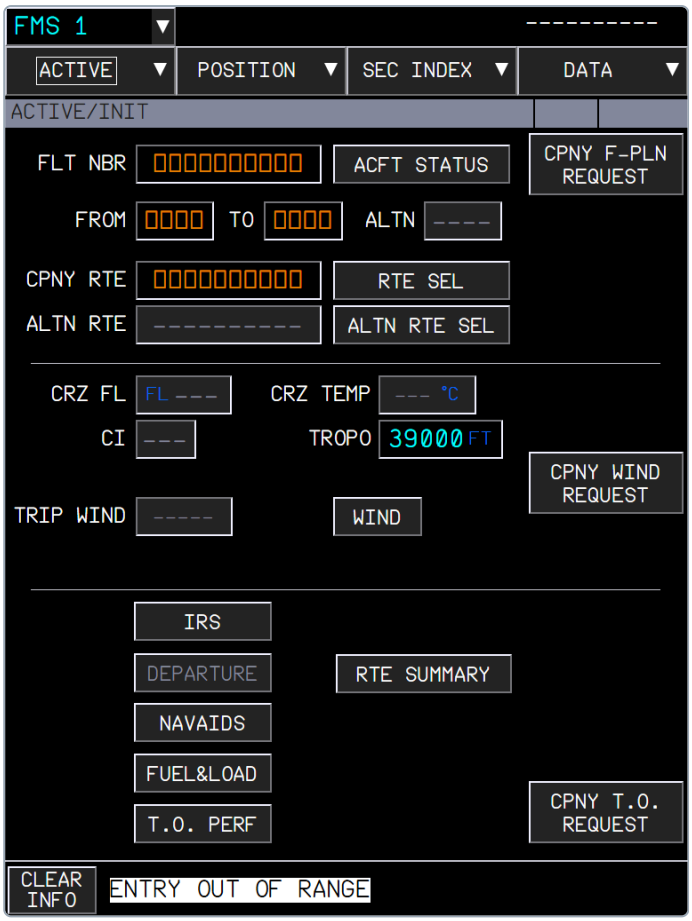


Figure 45. 99999 in TROPO: ENTRY OUT OF RANGE.

Cancelling with ESC

Figure 46. 41000 typed in TROPO, not validated...

Figure 47. ... ESC: the field returns to its last valid value, 39000.

What to expect

Situation	What you should see
Value typed, ENT	The value is entered (or an FCOM error message).
Value typed, ESC	The field returns to its last valid value.
Value typed, another field clicked	The first value is validated, the new field is in edition.
Value typed, click on a 3D KCCU key or elsewhere in the cockpit	The field stays in edition, the value is not lost.
KBD switch OFF	The KCCU keyboard keys do nothing; the PC keyboard still types in the selected field, which stays in edition until ENT, ESC or another field.

Limitations and why

Limitation	Why	What to do
One field in edition per side.	As on the aircraft: each KCCU drives its own side.	-

How to use

1. Click the field with the cursor (trackball or mouse).
2. Type on the KCCU keys (KBD switch ON) or the PC keyboard.
3. Press **ENT** to enter, or **ESC** to go back to the last valid value.

How to test

#	Do this	Expected result
1	A380X, KCCU KBD switch ON. MFD > ACTIVE > INIT, click TROPO.	TROPO in edition.
2	Type 39000 on the 3D KCCU keys, press **ENT**.	TROPO 39000 FT entered (big cyan).
3	Click TROPO, type 41000, click somewhere else in the cockpit (a 3D key, the panel).	The field stays in edition; 41000 is not lost.
4	Press **ESC**.	TROPO back to 39000.
5	Type ABC in TROPO, ENT.	FORMAT ERROR, FORMAT: FOR ALT XXXXX FOR FL XXX; the field is unchanged.
6	Type a value in one field, click another field.	The first value is validated; the second field is in edition.

13. KCCU KBD and CCD switches: FCOM rules, ON by default

> **Aircraft:** A380X | **Where:** KCCU KBD and CCD switches (both KCCUs) | **Status:** Not sim-tested (part of the KCCU change checked in the browser harness, installed).

What it is for

A380 FCOM DSC-31-30-20: the keyboard and the cursor control device of each KCCU have their own ON/OFF switch ("The KCCU keyboard is not active" when OFF), independent of each other.

- The keyboard keys work only with the **KBD** switch ON; the cursor control device keys (ESC2, KBD, REWIND, FORWARD) only with the **CCD** switch ON. Both the MFD and the ND follow this.
- An entry field clicked with the mouse keeps the keyboard input until ENT or ESC whatever the KBD switch, so typing cannot land in another field; a field that disappears ends its edition. The 20 s unfocus timer and the end of edition on a display blur are removed.
- Both switches are **ON** in every flight start (hangar, apron, taxi, runway, climb, cruise, approach, final) and the cold and dark preset no longer turns them off. The FCOM sets them ON during the preliminary cockpit preparation (PRO-NOR-SOP): ON by default is a convenience.

How to use

1. Leave the KBD and CCD switches ON (default). Set KBD OFF to stop the KCCU keyboard, CCD OFF to stop the cursor device keys.

How to test

#	Do this	Expected result
1	Start an A380X flight cold and dark (or at the gate).	Both KCCU KBD and CCD switches are ON.
2	Set the captain KCCU **KBD** switch OFF, click an MFD field, press letter keys on the 3D KCCU.	Nothing is typed.

3	Type on the PC keyboard in that field.	The PC keyboard still types; ENT / Escape end the edition.
4	KBD back ON, set **CCD** OFF, press the KCCU ESC2 key during an edition.	Nothing happens; with CCD ON ESC2 works again.

14. SURV pages laid out and wired per the FCOM

> **Aircraft:** A380X | **Where:** MFD SURV > CONTROLS and STATUS & SWITCHING | **Status:** No full in-sim check recorded; the GAIN and ELEVN/TILT entry fields were used in the sim (ND message test of `core/a380x-wxr-controls`).

What it is for

The SURV CONTROLS and STATUS & SWITCHING pages are laid out on their figures (A380 FCOM DSC-34-20-60-50 P 4 and P 6): area frames, 120 x 78 SURV buttons, the WXR and GPWS link lines, the SURV DEFAULT SETTINGS group, and SYS boxes that no longer run off the right of the screen. The options follow the FCOM rules of the XPDR, TCAS, WXR and TAWS sections (DSC-34-20-40, -50, -30-20, -10).

Item	Rule
XPDR mode	Option list AUTO / ON / STBY; ON keeps the transponder replying as in flight on the ground.
ALT RPTG OFF or XPDR STBY	The TCAS option lists are not available.
ALT RPTG ON	Selects TA/RA and NORM.
WXR button	Needs a confirmation. AUTO resets PRED W/S, TURB, ELEVN/TILT, GAIN, MODE and WX ON VD; OFF sets PRED W/S, TURB and WX ON VD to OFF, inhibits them and hides GAIN and MODE.
ELEVN/TILT	Option list (not available in MAP) with an ELEVN field (FL with the STD reference, FT with QNH: FL 0 to 600, 0 to 60 000 ft) and a TILT field (±NN.N, M for minus, -15.0 to +15.0 degrees).
GAIN	Entry field with GAIN MAN: 0 to 100 %.
DEFAULT SETTINGS	Needs a confirmation.
STATUS & SWITCHING	The FCOM indication texts: WX DISPLAY, and XPDR / TCAS STBY when the system is not in use.

Limitations and why

Limitation	Why	What to do
The ELEVN, TILT and GAIN values do not change the radar picture.	They are stored in new documented LVars (A380X_WXR_ELEVN_TILT_MODE, A380X_WXR_ELEVN, A380X_WXR_TILT, A380X_WXR_GAIN); the MSFS radar has no gain and the native radar does not use them.	The ND messages for them are on `core/a380x-wxr-controls`.

How to use

1. MFD > SURV > CONTROLS: choose the options in their lists; confirm the WXR and DEFAULT SETTINGS buttons.
2. For a manual radar setting, choose ELEVN or TILT and type the value, or GAIN MAN and the gain.

How to test

#	Do this	Expected result

1	A380X powered. MFD > SURV > CONTROLS.	The page as the FCOM figure: frames, link lines, DEFAULT SETTINGS group.
2	Open the XPDR mode list.	AUTO, ON, STBY.
3	Set ALT RPTG OFF.	The TCAS option lists are greyed (TCAS STBY on the ND).
4	Set ALT RPTG ON.	TCAS TA/RA and NORM are selected.
5	Press the WXR button to OFF.	A confirmation is asked; after it PRED W/S, TURB and WX ON VD are OFF and inhibited, GAIN and MODE hidden.
6	WXR back to AUTO (confirm).	PRED W/S, TURB, ELEVN/TILT, GAIN, MODE and WX ON VD back to their AUTO values.
7	ELEVN/TILT: choose TILT, type **M3.5**.	TILT -3.5 degrees; typing 20 gives ENTRY OUT OF RANGE.
8	Choose MODE MAP.	ELEVN/TILT is not available.
9	Press DEFAULT SETTINGS.	A confirmation is asked before the defaults are set.
10	Open STATUS & SWITCHING.	The SYS boxes fit on the screen; the unused XPDR / TCAS system reads STBY.

15. SURV memos, and TCAS STBY with ALT RPTG OFF

> **Aircraft:** A380X | **Where:** E/WD memo area; ND; MFD SURV CONTROLS | **Status:** Not sim-tested.

What it is for

The ECAM follows the SURV CONTROLS page as in the A380 FCOM (DSC-34-20-60-140, flight phase inhibitions from its figures).

Memo	When
TERR SYS OFF (green)	TERR SYS OFF on SURV CONTROLS.
TERR STBY (green)	The TAWS of the selected system is operative, but the TERR function cannot rely on the aircraft position, which the TAWS takes from ADIRU 1 (DSC-34-20-20-10).
WXR OFF (amber)	WXR button OFF. It also sets PRED W/S and TURB to OFF: only WXR OFF shows then.
PRED W/S OFF	PRED W/S OFF by the crew: green in flight phases 2 and 8, amber in the takeoff and landing phases (the colours of the A320 PRED W/S OFF memo).
WXR TURB OFF (green)	TURB OFF by the crew.
WXR ON (green)	On the ground, one engine running, WX selected on an EFIS CP and the WXR not OFF or failed: the WX pb turns the WXR on (DSC-34-20-30-20 P 1).
TCAS STBY (green)	ALT RPTG OFF puts the TCAS in standby (DSC-34-20-50-20 P 7): also TCAS STBY on the ND and, in cruise, the **SURV TCAS STBY** caution (PRO-ABN-ECAM-10-34-20 P 18).

How to use

1. Nothing to set: the memos follow the SURV CONTROLS settings. Restore the normal settings (WXR AUTO, TERR SYS ON, ALT RPTG ON) to clear them.

How to test

#	Do this	Expected result
1	A380X on the ground, one engine running. MFD > SURV > CONTROLS: set **TERR SYS** OFF.	E/WD memo **TERR SYS OFF** in green.
2	TERR SYS ON; set **PRED W/S** OFF.	Memo **PRED W/S OFF**, green (taxi).
3	Set **TURB** OFF.	Memo **WXR TURB OFF**.
4	Set the **WXR** button OFF (confirm).	Memo **WXR OFF** in amber; the PRED W/S OFF and WXR TURB OFF memos go away.
5	WXR AUTO again; press **WX** on the captain EFIS CP.	Memo **WXR ON**.
6	Set **ALT RPTG** OFF.	Memo **TCAS STBY**, TCAS STBY on the ND.
7	In cruise, set ALT RPTG OFF.	Caution **SURV TCAS STBY**.
8	Takeoff with PRED W/S OFF.	The PRED W/S OFF memo is amber in the takeoff phases.

16. FMS function fixes: SEC pages, received company winds, FIX INFO, STORE WPT, RTE SEL

> **Aircraft:** A380X | **Where:** MFD SEC INDEX, INIT, WIND, F-PLN > FIX INFO, DATA > WAYPOINT | **Status:** SEC page fix tested in the sim by the user: OK. SEND F-PLN REQUEST asterisk checked in the browser harness. The other fixes: not sim-tested.

What it is for

Fix	Before	Now
SEC page kept when another flight plan is created	A page opened for an empty SEC (e.g. its COMPANY F-PLN REQUEST) jumped to the ACTIVE copy as soon as any flight plan was created; RETURN had to be clicked twice and the page was bound to the active flight plan.	The page moves to ACTIVE only when its loaded SEC flight plan is deleted.
One asterisk on SEND F-PLN REQUEST	Two asterisks.	One, on the right, as the FCOM figure (DSC-22-FMS-20-30 P 28).
Received company winds kept until INSERT	WIND INSERT did nothing after a flight plan revision: a flight plan copy (revision, temporary plan inserted) lost the received winds.	The received winds stay pending across copies until the crew inserts or clears them (A32NX and A380X). The request and the answer path are now FlyByWire's (see the WIND page feature).
FIX INFO reference fix	An unknown ident opened NEW WAYPOINT.	Looked up only (database and pilot-stored elements): an unknown ident gives NOT IN DATABASE; FIX INFO never creates or stores a waypoint (A380 FCOM DSC-22-FMS FIX INFO).
DATA / WAYPOINT STORE WPT	Stayed inactive (the waypoint was worked out before the last value was stored).	Active as soon as the entries define the waypoint.

INIT (ALTN) RTE SEL	Did not always open ROUTE SELECTION.	Always opens the ROUTE SELECTION page; only the route entries are limited (FCOM P 199-200).

How to use

1. Nothing new to learn: use the pages as described in the FCOM; the fixes remove the wrong behaviours above.

How to test

#	Do this	Expected result
1	A380X, SimBrief account set in the flyPad, active flight plan, SEC 1 empty. SEC INDEX > SEC 1 > CPNY F-PLN REQUEST.	The COMPANY F-PLN REQUEST page; one asterisk, right of SEND F-PLN REQUEST.
2	Press SEND F-PLN REQUEST; wait for the flight plan; press RETURN once.	Back on SEC INDEX with one RETURN; the page stayed on SEC 1.
3	Delete SEC 1 while one of its pages is open.	The page moves to its ACTIVE copy (unchanged behaviour).
4	WIND: CPNY WIND REQUEST, SEND WIND REQUEST; before the answer, revise the flight plan (insert a waypoint); then, when the button reads RECEIVED CPNY WIND, press it and INSERT*.	The company winds are inserted.
5	F-PLN, a waypoint > FIX INFO: type an unknown ident (e.g. ZZZZZ) as the reference fix.	NOT IN DATABASE; no NEW WAYPOINT page. A database ident is accepted; DATA / STATUS counts no new waypoint.
6	DATA > WAYPOINT > NEW WPT, enter a LAT/LONG.	STORE WPT becomes active after the last entry.
7	INIT: RTE SEL and ALTN RTE SEL.	Both open the ROUTE SELECTION page.

17. MFD display fixes: sizes, labels, menus and clickable controls

> **Aircraft:** A380X | **Where:** MFD FMS and SURV pages | **Status:** Found and fixed in the sim (September 2026); printer, BFO and LL XING AURAL checkboxes tested in the sim by the user: OK. The other items: not individually sim-tested.

What it is for

- **FCOM sizes in the sim:** the cockpit renderer counts padding and frame in a box size, so the buttons were 28 px narrower and 18 px lower than the FCOM figures and labels ran out. Buttons now have the full FCOM size as a minimum and grow around their label; labels keep the lines of the FCOM figure, 26 px apart (a one-line button is 44 px high); entry fields and selection lists are 44 px high, without a background painted over their frame. The AIRWAYS list and the RECEIVED COMPANY T.O DATA INSERT / CLEAR buttons moved to make room.
- **Button labels that change** (REQUEST PENDING..., RECEIVED CPNY WIND, FREEZE / UNFREEZE...) showed a raw "<br />": they now use line breaks.
- **Selection menus** that would leave the screen at the bottom (NEW RTE on DATA / ROUTE lost SEC 3) open upwards.
- **F-PLN EFOB / T.WIND:** the wind digits no longer run over the slash.
- **Clicks:** a hidden FCOM page element no longer takes the clicks of the visible one; the DATA / PRINTER AUTO PRINT checkboxes, the POSITION / NAVAIDS ADF BFO toggle and the LL XING / TIME MKR AURAL checkboxes react (an onClick on a plain element never fired).
- **DATA / NAVAID search:** written without an await used as an operand, which the sim's JavaScript engine can reject and blank the whole MFD. Same behaviour.

- Code clean-up with no visible effect: shared FREE TEXT controls of the company pages, one tonnes helper on FUEL&LOAD, a test set-up fix for Node 22 and later, typing fixes.

How to use

1. Nothing to set: the pages look and react as the FCOM figures.

How to test

#	Do this	Expected result
1	A380X powered, open several FMS pages (INIT, FUEL&LOAD, PERF, POSITION, DATA).	No label runs out of its button; fields are one height; no " " text.
2	DATA > ROUTE: open the NEW RTE menu.	It opens upwards; SEC 3 is visible.
3	F-PLN with winds: read the EFOB / T.WIND column.	Wind direction and speed fit their cells.
4	DATA > PRINTER: tick and untick each AUTO PRINT box.	Each box ticks and unticks.
5	POSITION > NAVAIDS: click the ADF BFO toggle.	It toggles.
6	F-PLN revision > LL XING / TIME MKR: click the AURAL checkboxes.	They tick and untick.
7	DATA > NAVAID: search a navaid ident.	The navaid data shows; the MFD stays up.

18. PLACE/DIST waypoints, HOLD at PPOS, DIR TO radials, OPT FL, EOSID, PRED TO

> **Aircraft:** A32NX and A380X | **Where:** A32NX MCDU F-PLN, LAT REV, HOLD AT, DIR TO and PROG pages; A380X MFD F-PLN revisions, DEPARTURE and PERF pages | **Status:** Not sim-tested (added; unit tests: 6 new spec files, 17 tests, all failing before the change).

What it is for

Flight planning functions that the FCOM describes and the FMS did not have (or had wrong), found by the FCOM gap audit. The flight plan code is shared, so both aircraft get the common parts:

- **PLACE/DIST waypoints (both):** a waypoint at a distance along the flight plan path after a flight plan waypoint, or before it with a negative distance (for example `LMG/-15`). It is created as a pilot waypoint **PDnn** on the path, without a discontinuity. A32NX: on the MCDU F-PLN page and as LAT REV NEXT WPT (A320 FCOM DSC-22_20 F-PLN and LAT REV pages). A380X: in INSERT NEXT WPT of the revised waypoint; a place that is not the revised waypoint gives **PLACE / WPT DISAGREE**, a distance beyond the neighbouring waypoint **ENTRY OUT OF RANGE** (A380 FCOM, F-PLN revisions, INSERT NEXT WPT).
- **HOLD at the present position (both):** the HOLD revision of the FROM waypoint (PPOS) creates a T-P and a computed hold at the present position, on the aircraft track. Before, the A32NX put the hold at the waypoint behind the aircraft, and on the A380X the item was greyed out.
- **A32NX DIR TO page:** ABEAM PTS, RADIAL IN (the default inbound radial of a flight plan waypoint is proposed) and RADIAL OUT build the DIR TO temporary flight plan instead of NOT YET IMPLEMENTED.
- **A32NX PROG OPT FL:** the best specific range level for the gross weight, from the FCOM long range cruise ISA tables (PER-CRZ-CRT-30), never above REC MAX; dashes with an engine out. Before, it was REC MAX rounded down.
- **A380X PERF CLB and DES tabs:** the time and distance to the PRED TO altitude instead of dashes.
- **A380X EOSID:** the DEPARTURE page shows the engine-out SID of the runway; an engine failure in the takeoff phase below the thrust reduction altitude creates the EOSID temporary flight plan (A380 FCOM, engine failure at

takeoff, EOSID).

- Code fix with no visible effect today: the remote flight plan client erases the EOSID (it activated it) and sends the manual hold default in the right argument. No production code uses that client yet (specs only).

Situation	What you should see
A32NX: type `LMG/-15` on the F-PLN page at a waypoint (or as LAT REV NEXT WPT)	A waypoint **PD01** (then PD02...) on the path 15 NM before LMG, no discontinuity.
A380X: INSERT NEXT WPT of waypoint X, entry `X/10`	A PDnn waypoint 10 NM after X on the path.
A380X: INSERT NEXT WPT of X, entry `Y/10` (another place)	**PLACE / WPT DISAGREE**.
A380X: distance beyond the next (or previous) waypoint	**ENTRY OUT OF RANGE**.
HOLD revision of the FROM waypoint, in flight	T-P and a hold at the present position on the aircraft track (TMPY first).
A380X engine failure in the takeoff phase below the thrust reduction altitude, EOSID on the DEPARTURE page	The EOSID temporary flight plan is created; the crew inserts it or not.
A32NX with an engine out	PROG OPT FL shows dashes.

Limitations and why

Limitation	Why	What to do
A380X OPT FL and EO MAX FL are not computed.	No source data for the A380 (no table found).	-
The A32NX OPT FL table is for the PW engines.	The closest data of the FCOM long range cruise tables.	-
PLACE/DIST is not accepted on DIR TO or for the STEP ALTs.	Left for later.	Create the PDnn waypoint on the F-PLN first.
No A32NX DCDU LOAD of route clearances; the A32NX DIR TO page has the older layout (not the 2012 one).	Left for later.	-

How to use

1. PLACE/DIST: A32NX, type `PLACE/DIST` (for example `LMG/-15`) in the scratchpad and press the left key of the waypoint on the F-PLN page, or use LAT REV > NEXT WPT; A380X, F-PLN > waypoint revision > INSERT NEXT WPT, type `<that waypoint>/DIST`.
2. HOLD at PPOS: LAT REV (A32NX) or the revision menu (A380X) of the FROM waypoint > HOLD; check the hold and insert the temporary flight plan.
3. A32NX DIR TO: choose the waypoint, then ABEAM PTS, RADIAL IN or RADIAL OUT (type the radial if needed), then INSERT.
4. A380X EOSID: check the EOSID on the DEPARTURE page before takeoff; after an engine failure below the thrust reduction altitude, review the temporary flight plan and insert it.
5. A380X PRED TO: PERF > CLB or DES tab, type the altitude in PRED TO; read the time and distance.

How to test

#	Do this	Expected result
1	A32NX in flight on a flight plan with waypoint LMG (or any waypoint ahead). Type `LMG/-15` and press the LMG left key on the F-PLN page.	PD01 appears before LMG on the path, without a discontinuity; the ND draws it 15 NM before LMG.
2	A380X: F-PLN > revision of a waypoint ahead > INSERT NEXT WPT, type `<that waypoint>/10`; then type another waypoint ident with /10.	First: a PDnn waypoint 10 NM after it. Second: PLACE / WPT DISAGREE.
3	Both aircraft in flight: HOLD revision of the FROM waypoint (A32NX LAT REV at PPOS).	TMPY: T-P and a hold at the present position on the aircraft track; after INSERT the aircraft enters it there.
4	A32NX: DIR TO a flight plan waypoint ahead with RADIAL IN (default radial), then with RADIAL OUT and with ABEAM PTS.	Each gives a TMPY DIR TO flight plan (no NOT YET IMPLEMENTED).
5	A32NX cruise: PROG page.	OPT FL below or at REC MAX (not just REC MAX rounded down); dashes with one engine shut down.
6	A380X: PERF > CLB tab in the climb, PRED TO an altitude above.	Time and distance to that altitude (not dashes); same on the DES tab.
7	A380X: DEPARTURE page with a runway that has an EOSID; take off; at about 500 ft, below the thrust reduction altitude, flyPad > Failures: an engine flameout (or ENG MASTER OFF).	The EOSID shows on the DEPARTURE page; after the failure a TMPY EOSID flight plan appears.

What PR testers should check

- HOLD at PPOS on both aircraft (where the hold is drawn and flown).
- PLACE/DIST entries (positive and negative distances, the PDnn names, the A380X error messages).
- The A32NX DIR TO radial and abeam temporary flight plans.
- The A380X EOSID after an engine failure below the thrust reduction altitude.
- The A380X PERF CLB/DES predictions to the PRED TO altitude.
- None of this is sim-tested yet.

A380X ATC COM and datalink

The MFD ATC COM pages and the SD mailbox of the A380 FCOM on the FBW datalink, with a datalink that connects reliably, ATC route clearances and constraints loaded into SEC 3, the OIT COMPANY COM inbox and the MFD System Message Area

Aircraft	A380X (the ACARS provider connection is shared with the A32NX)
Where to find it	MFD system menu **ATC COM** (or the KCCU ATC COM key); the SD **MAILBOX**; MFD FMS **INIT** (FLT NBR); flyPad > Settings > ATSU / AOC; OIT COMPANY COM; MFD header
Branch	`core/a380x-atccom-datalink`, based on `core/a380x-mfd-fms` (stacked: needs `core/surveillance` and `core/a380x-mfd-fms` merged first). `core/printer` builds on it.
Status	ATC COM pages and mailbox: tested in the sim (ATIS, frames, fixes), logon and MAYDAY, PAN PAN PAN. Flight number connection and SEND FAILED: OK. ACARS provider: seen working with BeyondATC. Mailbox FILE FULL fix: OK; @ markers: OK. LOAD-SEC3: route clearance OK; crossing constraints not sim-tested. COMPANY COM inbox and MFD System Message Area: not sim-tested, unit tests only.
Pictures	Captain MFD and SD in the local browser harness, running the build installed in the sim; the messages come from the harness (no ATC network). flyPad page in the same harness. No pictures for the mailbox fixes and LOAD-SEC3.

1. ATC COM pages and the SD mailbox
2. Datalink connection with the flight number, and SEND FAILED
3. ACARS provider connection
4. SD mailbox fixes: FILE FULL after CANCEL, uplink @ markers
5. LOAD-SEC3: ATC route clearances and crossing constraints into SEC 3
6. OIT COMPANY COM inbox with the real AOC messages
7. MFD System Message Area: CHECK FMS MESSAGE, NEW ATIS RECEIVED, ATIS SYS MSG

1. ATC COM pages and the SD mailbox

> **Aircraft:** A380X | **Where:** MFD ATC COM pages; SD MAILBOX | **Status:** Tested in the sim by the user: ATIS, REQUEST / OTHER REPORTS / EMERGENCY frames and the fixes of that test (frame delete, label layout, font sizes, ATIS time): OK. Logon to BeyondATC (RJJJ), MAYDAY sent, PAN PAN PAN sent. Answering uplinks of a live controller (WILCO, UNABLE...) not recorded. Unit tests: 33.

What it is for

The A380 talks to ATC by datalink: the crew logs on to a center, sends requests and reports, receives clearances and answers them, and gets the airport ATIS as text. The FCOM describes the MFD **ATC COM** pages (DSC-46-10-20-30) and the SD **MAILBOX** (DSC-46-10-10-70). They are all built on their FCOM figures and connected to the FBW ATSU, the same datalink system as the A32NX DCDU.

> **NOTE**
>
> The network is chosen in the flyPad: Settings > ATSU / AOC (Hoppie, BeyondATC or SayIntentions). The datalink needs VHF3 in DATA mode on the RMP.

Page (header)	Use
CONNECT	Logon (NOTIFICATION), CONNECTION STATUS with ADS, MAX UPLINK DELAY.
REQUEST	Build a request from frames (climb, direct, offset, weather deviation, speed, clearances...) and send it to the mailbox.
REPORT & MODIFY	Position report, other reports, modify a prepared answer.

MSG RECORD	The record of the closed dialogues.
ATIS	Request and read the ATIS of three airports.
EMER	MAYDAY / PANPAN and the emergency reports.

Logging on: CONNECT

1. CONNECT > NOTIFICATION: type the **ATC CENTER** (4 letters) and press **NOTIFY**. The page shows NOTIFYING, then the center in NOTIFIED TO CENTERS (with the time), or NOTIFICATION FAILED.
2. When ATC accepts, CONNECTION STATUS shows the ACTIVE ATC (and the NEXT ATC after a transfer). DISCONNECT ends the connection.

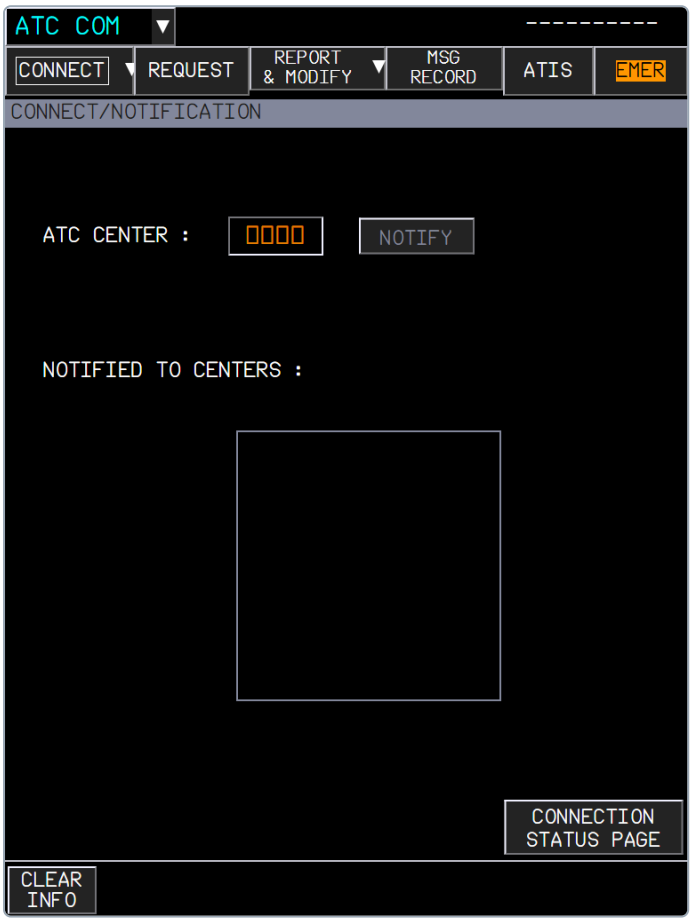


Figure 1. CONNECT / NOTIFICATION.

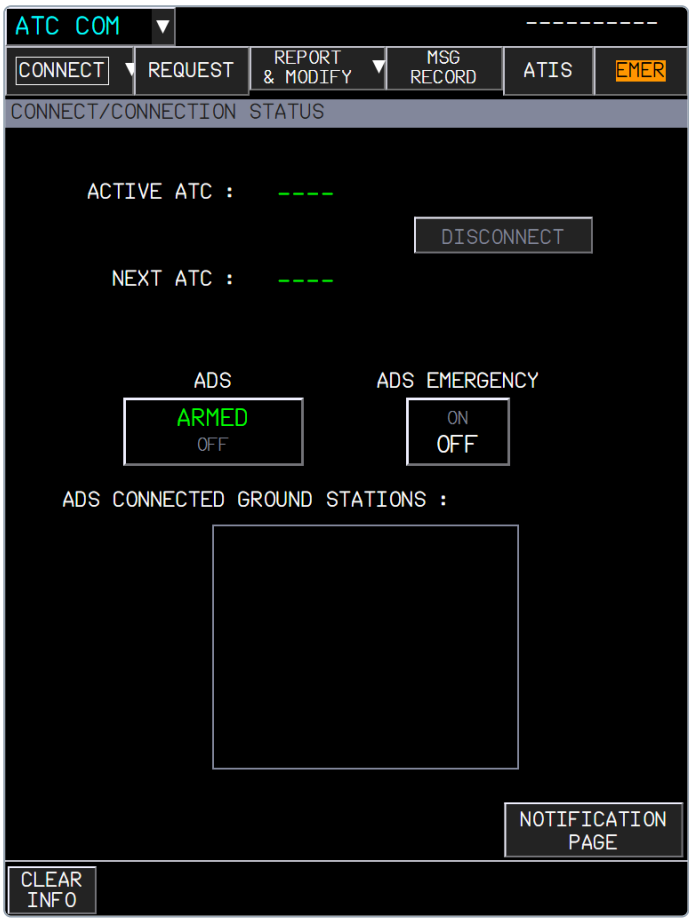


Figure 2. CONNECTION STATUS: active and next ATC, ADS ARMED / OFF, ADS EMERGENCY.

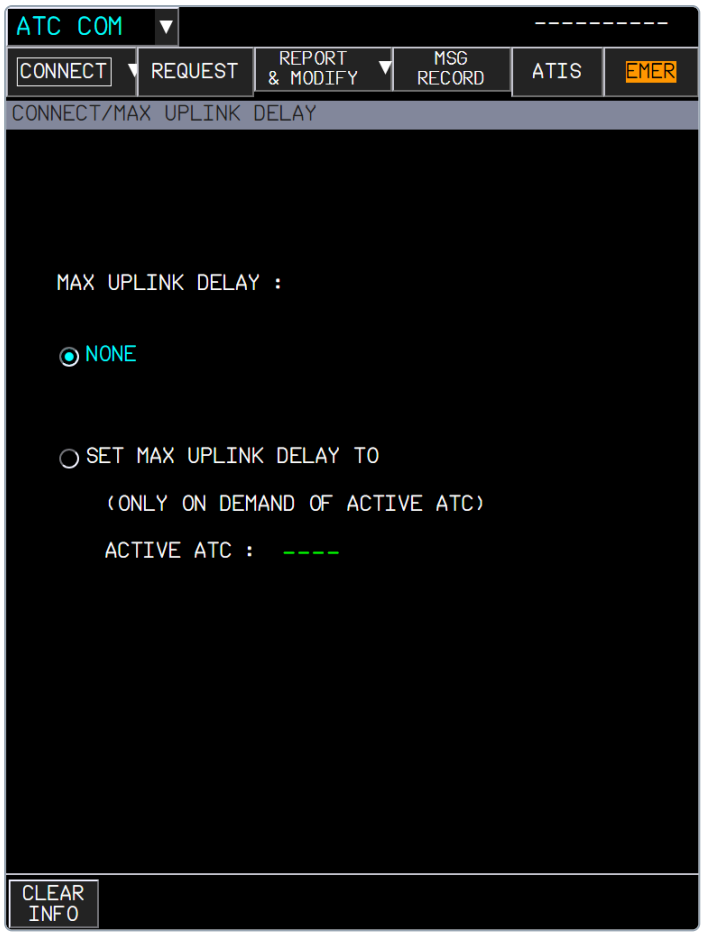


Figure 3. MAX UPLINK DELAY: NONE, or a delay set on the demand of the active ATC.

Asking ATC: REQUEST

1. Open the menus on the right (VERTICAL, LATERAL, SPEED, CLEARANCE, WHEN CAN WE EXPECT, OTHER) and pick the frames you need; up to 5 frames, each with its fields.
2. Fill the fields (FCOM entry formats; amber boxes are mandatory). ADD TEXT adds free text. The bin deletes a frame.
3. Press **XFR TO MAILBOX**: the request goes to the SD mailbox, where you **SEND** it.

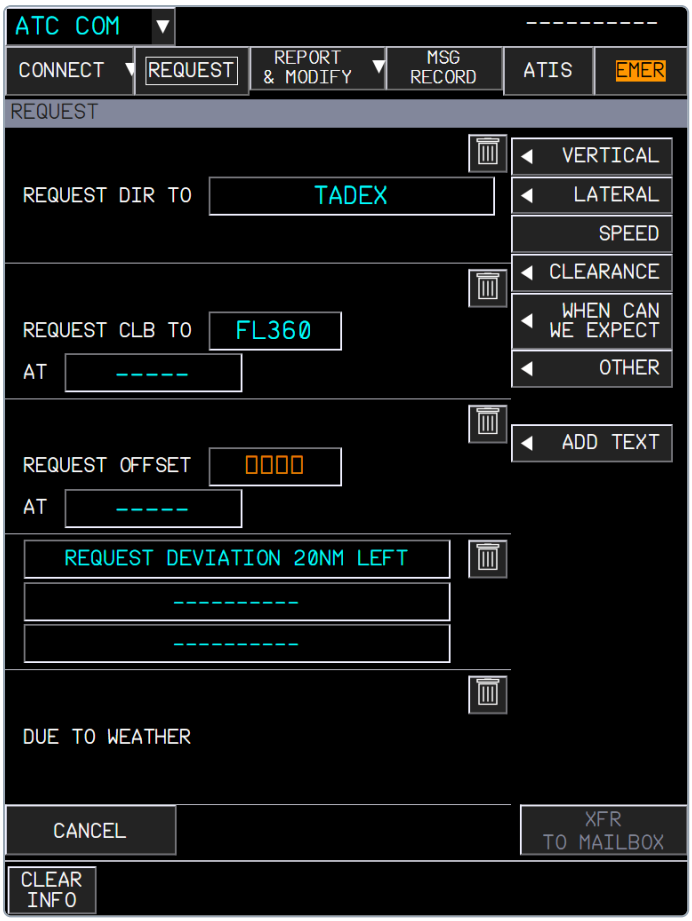


Figure 4. A request of 5 frames: DIR TO TADEX, CLB TO FL360, OFFSET (amber: to fill), free text, DUE TO WEATHER.

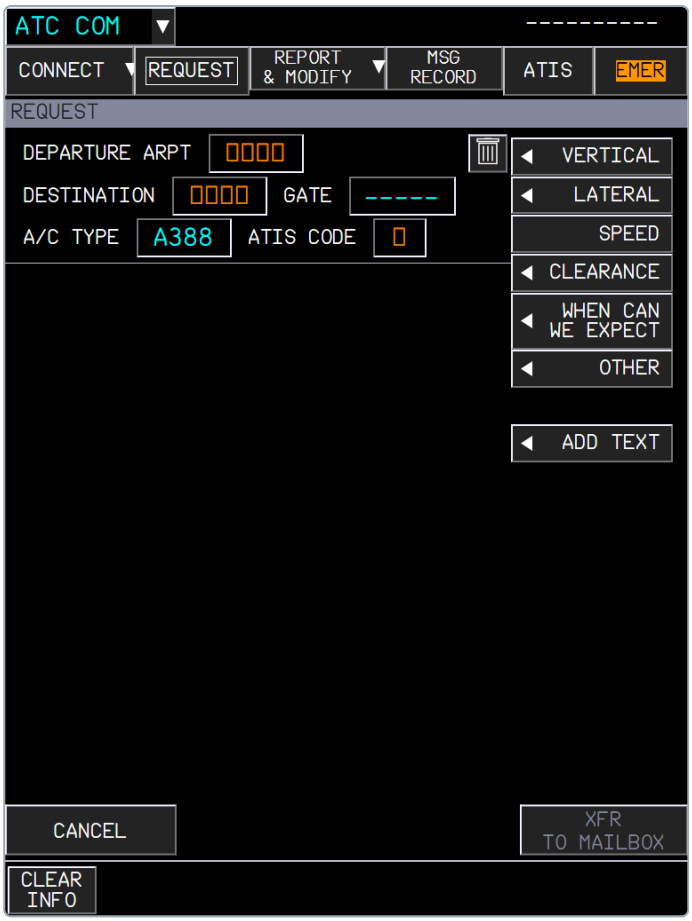


Figure 5. CLEARANCE > DEPARTURE: the departure clearance request (DCL): departure airport, destination, gate, aircraft type, ATIS code.

Reports: REPORT & MODIFY

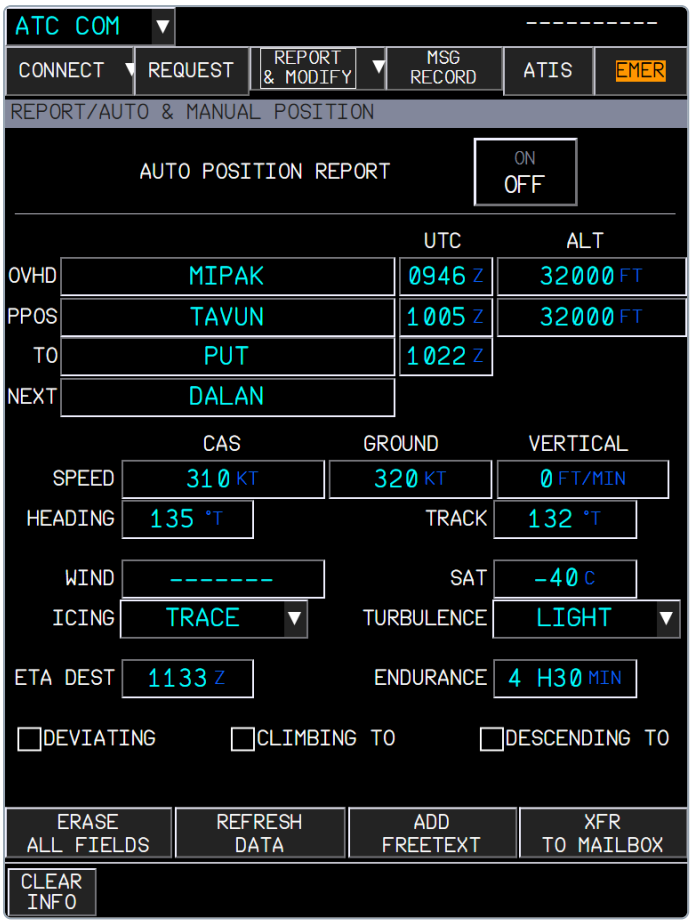


Figure 6. AUTO & MANUAL POSITION: the position report filled by the FMS (overhead, present position, to, next, times, levels, speeds, wind, SAT, ETA, endurance); AUTO POSITION REPORT ON sends it at each reporting point.

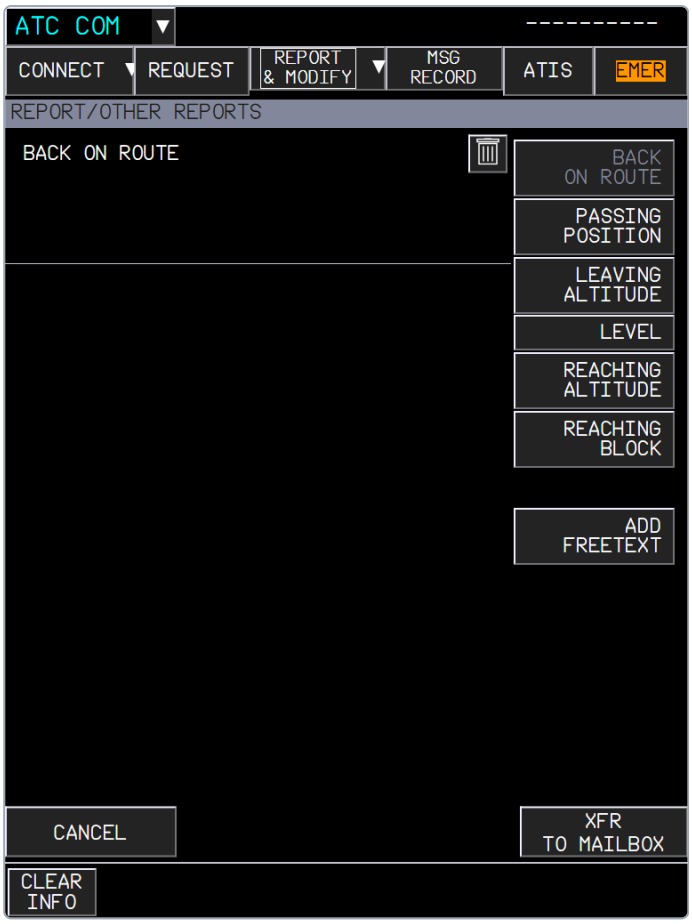


Figure 7. OTHER REPORTS: back on route, passing position, leaving / reaching altitude, level, reaching block, free text.

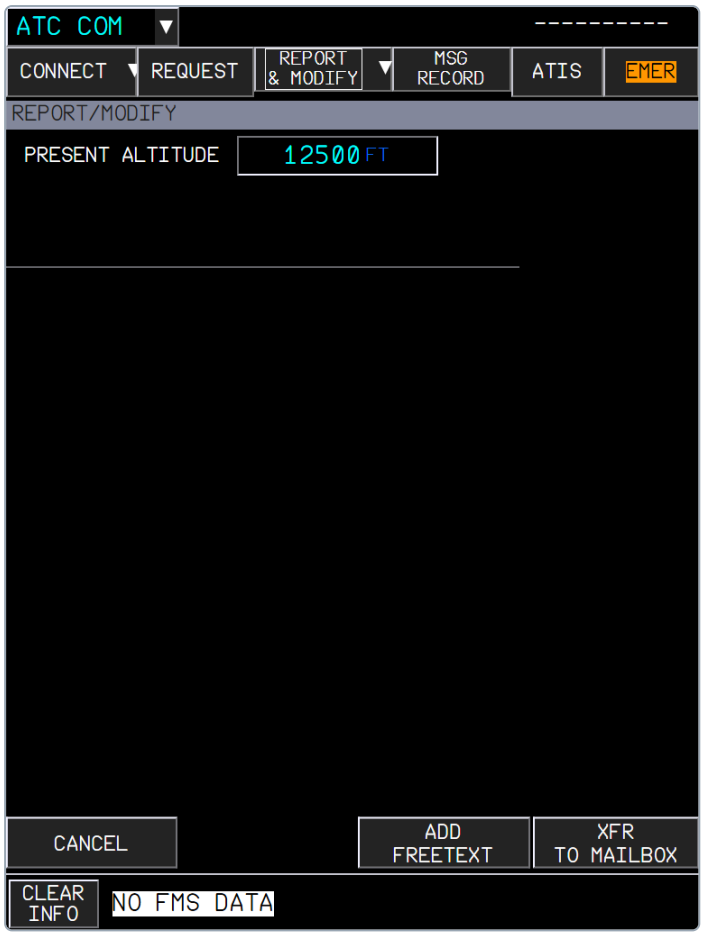


Figure 8. MODIFY: a prepared answer (here PRESENT ALTITUDE 12500 FT) that the crew can change before sending.

The SD MAILBOX

Every message to and from ATC goes through the mailbox at the bottom of the SD. It shows the active ATC, the ADS status, the message and the answer buttons the message expects.

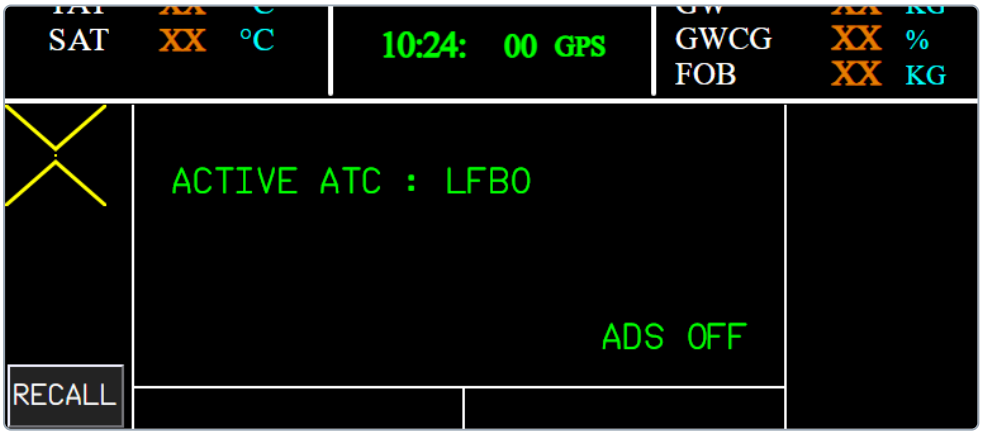


Figure 9. Mailbox without a message: ACTIVE ATC LFBO, ADS OFF.

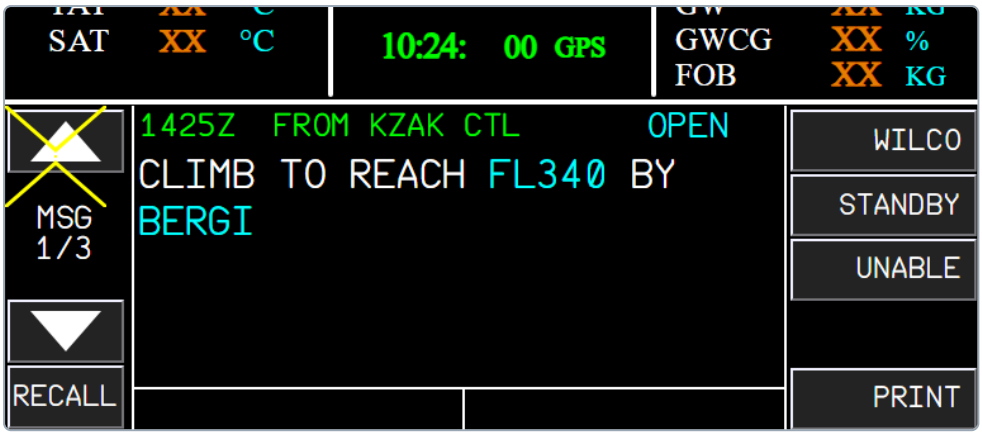


Figure 10. An uplink: CLIMB TO REACH FL340 BY BERGI, OPEN, with WILCO / STANDBY / UNABLE. MSG 1/3: three messages are waiting.

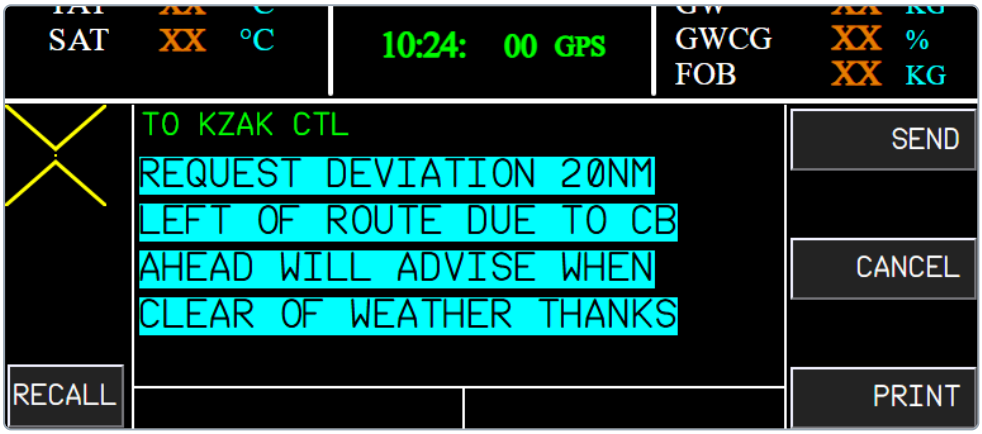


Figure 11. A request ready to go: SEND or CANCEL.

The message expects	Buttons
WILCO / UNABLE (a clearance)	WILCO, STANDBY, UNABLE
AFFIRM / NEGATIVE (a question)	AFFIRM, STANDBY, NEGATIVE
ROGER (information)	ROGER
A departure or oceanic clearance	ACK / REFUSE

UNABLE and NEGATIVE can add a reason (DUE TO WEATHER, A/C PERFORMANCE, TURBULENCE) and free text; MODIFY opens the answer on the MFD. The message colours follow the FCOM status legend.

MSG RECORD and ATIS

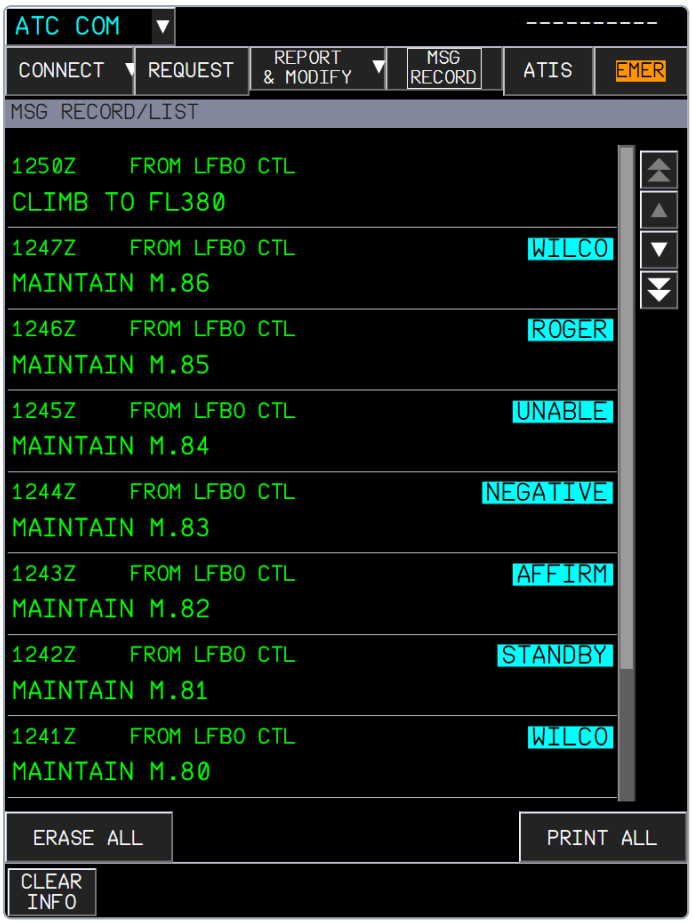


Figure 12. MSG RECORD: the closed dialogues with their time and the answer given; ZOOM shows one in full; ERASE ALL, PRINT ALL.

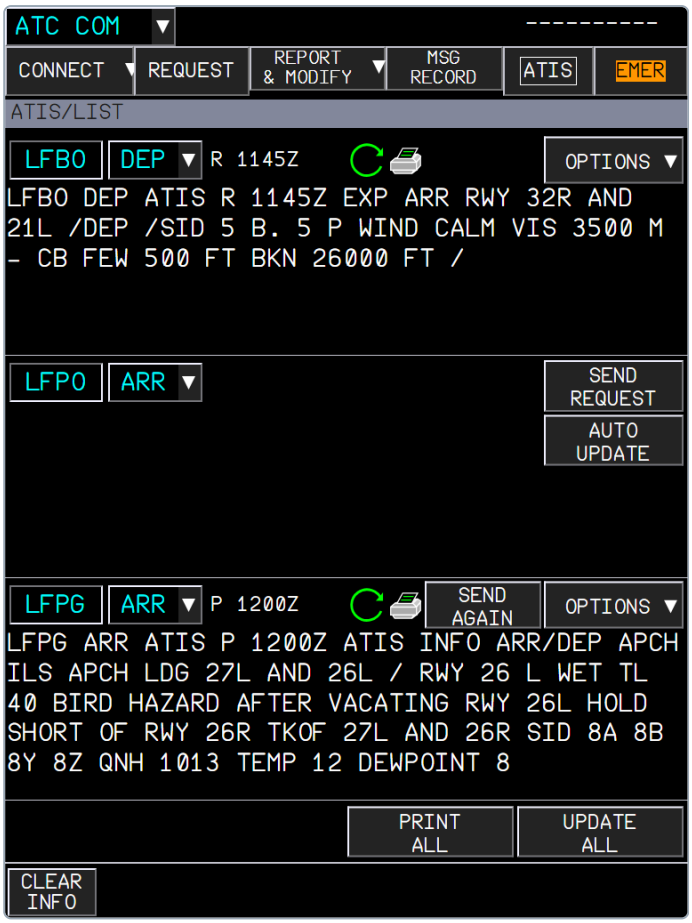


Figure 13. ATIS: three request areas (departure, arrival, alternate, following the FMS airports), each with its ATIS text, AUTO UPDATE, and SEND REQUEST / SEND AGAIN.

Figure 14. ATIS RECEIVED: the full text of an ATIS, RETURN TO LIST.

Emergency: EMER

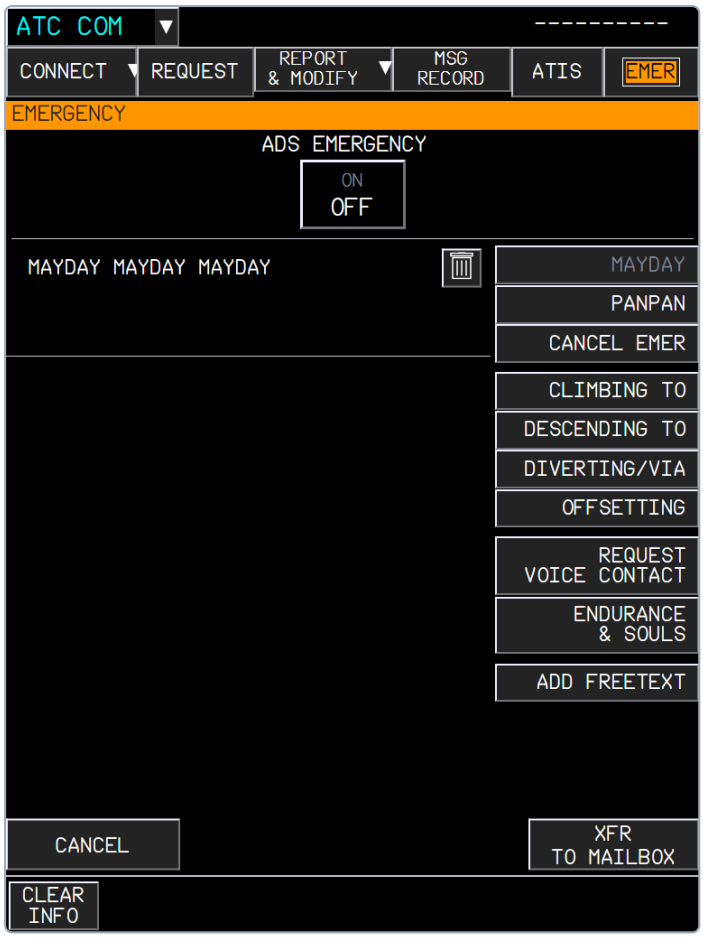


Figure 15. EMERGENCY (orange title): MAYDAY MAYDAY MAYDAY frame, PANPAN, CANCEL EMER and the emergency reports (climbing / descending to, diverting, offsetting, voice contact, endurance and souls). MAYDAY also switches the ADS to EMERGENCY.

What to expect

Situation	What you should see
NOTIFY with no network set	NOTIFICATION FAILED.
Request with an empty mandatory field	XFR TO MAILBOX stays greyed.
Uplink received	Mailbox message with its answer buttons.
Answer sent	The dialogue closes and goes to MSG RECORD.
ATIS requested	The text appears in its area; a newer version replaces it (AUTO UPDATE).
MAYDAY sent	ADS EMERGENCY ON.

Limitations and why

Limitation	Why	What to do
Needs an online datalink network.	ATC messages come from Hoppie, BeyondATC or SayIntentions through the flyPad ATSU settings.	Set the network in the flyPad.
No real ADS contracts.	The ATSU networks do not handle ADS-C; ADS ARMED / EMERGENCY is shown locally.	-
On this branch alone, every ATC COM PRINT says PRINTER NOT AVAIL.	Printing (ATIS, MSG RECORD, mailbox) is on `core/printer`, which builds on this branch.	-

Answering the uplinks of a live controller is not recorded as tested.	Tested in the sim: logon, MAYDAY and PAN PAN downlinks, the ATIS, a route clearance uplink (LOAD-SEC3). The answer buttons were checked with harness messages.	Report any wrong answer element.

> **NOTE**
> The A380X README (list of what is not implemented, ATC / datalink line) now says that the MFD ATC COM pages and the SD mailbox work through the FBW ATSU and its ACARS providers, and that the datalink functions outside those pages are not implemented.

How to use

1. Set the network in the flyPad (Settings > ATSU / AOC), VHF3 in DATA mode on the RMP, and your network callsign as FLT NBR on the MFD FMS INIT page.
2. MFD system menu > **ATC COM** > CONNECT > NOTIFICATION: type the ATC CENTER and press **NOTIFY**.
3. To ask ATC: REQUEST, pick the frames, fill them, **XFR TO MAILBOX**, then **SEND** in the SD mailbox.
4. To answer ATC: use the buttons of the message in the SD mailbox (WILCO / UNABLE, AFFIRM / NEGATIVE, ROGER...).
5. For the ATIS: ATIS page, airport of the request area, SEND REQUEST; tick AUTO UPDATE to get the new versions.
6. In an emergency: EMER, MAYDAY or PANPAN frame, XFR TO MAILBOX, SEND.

How to test

#	Do this	Expected result
1	A380X, any phase. flyPad > Settings > ATSU / AOC: ACARS provider set (Hoppie with its logon code, or BeyondATC running); RMP VHF3 in DATA; MFD FMS INIT: FLT NBR = your network callsign. Open MFD system menu > **ATC COM**.	Header CONNECT / REQUEST / REPORT & MODIFY / MSG RECORD / ATIS / EMER; the REQUEST page opens.
2	CONNECT > NOTIFICATION: type a center (with BeyondATC: the one in its Frequencies tab, COM tuned to it), press **NOTIFY**.	NOTIFYING, then the center in NOTIFIED TO CENTERS with the time (or NOTIFICATION FAILED). Once ATC accepts, CONNECTION STATUS shows the ACTIVE ATC and the SD mailbox shows it too.
3	ATIS page: in the first request area type an airport, press **SEND REQUEST**.	The ATIS text in the area; its time is the time written in the ATIS (e.g. 1151Z), ---- when it has none. Opening it shows ATIS RECEIVED with the full text.
4	REQUEST: VERTICAL menu > a climb frame, type FL360; add a second frame and delete it with the bin symbol.	The labels and fields of a line follow each other without overlap; the bin removes the frame.
5	Press **XFR TO MAILBOX**.	The request in the SD mailbox with SEND and CANCEL.
6	Press **SEND** in the mailbox (logged on).	The request is sent; when the dialogue closes it goes to MSG RECORD with its time.
7	EMER page: MAYDAY MAYDAY MAYDAY frame, XFR TO MAILBOX, SEND.	The MAYDAY is sent and CONNECT shows ADS EMERGENCY. BeyondATC answers NO MESSAGE HANDLING AVAILABLE FOR THIS REQUEST (not supported on its side).
8	Receive an uplink (from a live controller, or self-sent, see LOAD-SEC3 below) and answer it.	The message in the mailbox with the buttons it expects; after the answer the dialogue closes and goes to MSG RECORD.

2. Datalink connection with the flight number, and SEND FAILED

> **Aircraft:** A380X | **Where:** MFD FMS INIT (FLT NBR); FMS message area; SD MAILBOX | **Status:** Tested in the sim by the user: logon to BeyondATC after the FLT NBR fix OK; flight number connection from the FMS and SEND FAILED without ATC OK. FMS DATALINK NOT AVAIL not recorded as seen in the sim. Unit tests: 5.

What it is for

The INIT page FLT NBR field wrote the flight plan only: the field did not refresh and the entry never connected the datalink networks (FBW TELEX and the ACARS provider), so ATC logons and downlinks failed. And a message sent from the mailbox with no ATC center was dropped without a word. Now the FMS connects the networks whenever the flight number of the active flight plan changes, and the mailbox says SEND FAILED (A380 FCOM DSC-46-10-20-60).

What you see

The ATC COM pages send and receive through the aircraft datalink, which connects with the flight number of the active flight plan (FMS INIT, FLT NBR). It must be your callsign on the network (Hoppie, BeyondATC or SayIntentions: the one set in the flyPad, Settings > ATSU / AOC).

- The flight number connects the datalink however it gets there: typed on the INIT page, from the company flight plan (SimBrief), from an activated secondary flight plan, or already there when you reload the flight.
- If the connection is refused, or the datalink does not answer, the FMS message area shows **FMS DATALINK NOT AVAIL** (white). It goes away by itself once connected; typing the flight number again tries again.
- A message sent from the mailbox with no ATC center connected (no logon done, or the center logged you off) shows **SEND FAILED** instead of nothing.

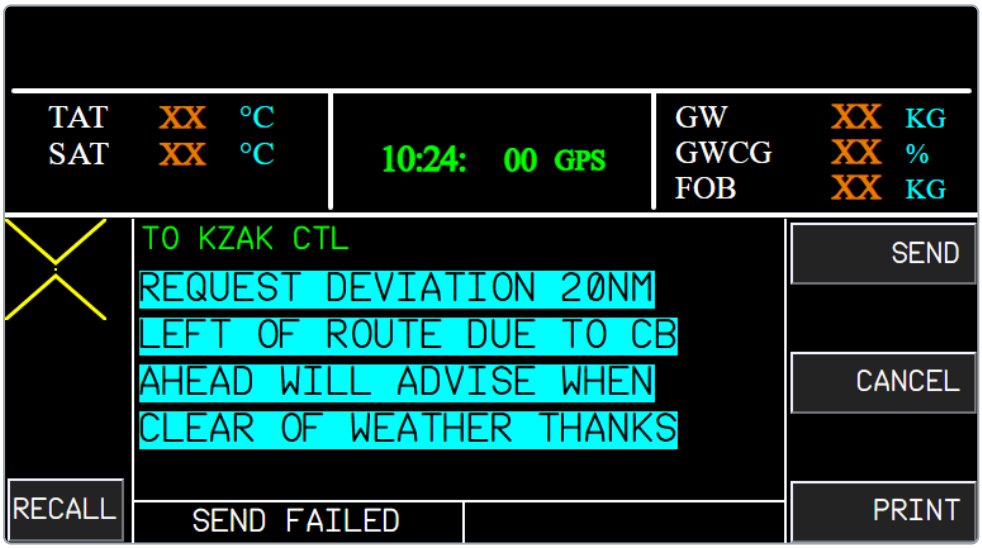


Figure 16. SEND FAILED under a message the mailbox could not send (no active ATC center).

> **NOTE**
> With BeyondATC: log on to the center shown in its Frequencies tab (e.g. RJJJ), above FL145. BeyondATC does not handle MAYDAY or PAN PAN by datalink: it answers NO MESSAGE HANDLING AVAILABLE FOR THIS REQUEST. Declare an emergency by voice with it.

How to use

1. Enter the flight number on the MFD FMS **INIT** page (FLT NBR), or get it with the company flight plan. It must be your callsign on the network.
2. If **FMS DATALINK NOT AVAIL** shows, check the ACARS provider, then type the flight number again to retry.
3. Log on (ATC COM > CONNECT) before sending: without an active ATC the mailbox says SEND FAILED.

How to test

#	Do this	Expected result
1	A380X on the ground, ACARS provider running and set in flyPad > Settings > ATSU / AOC. MFD FMS **INIT**: type the flight number in FLT NBR.	The field shows the new flight number at once. flyPad ATSU / AOC: the provider shows Connected (green).
2	ATC COM > CONNECT > NOTIFICATION: NOTIFY a center.	The logon goes out (NOTIFYING, then NOTIFIED TO CENTERS): the networks are connected.
3	Load a flight with the flight number already in the active flight plan (or activate a SEC plan with a flight number).	The datalink connects without typing the number again.
4	Without logon (no ACTIVE ATC), prepare a request (REQUEST, XFR TO MAILBOX) and press **SEND** in the SD mailbox.	**SEND FAILED** under the message (before: nothing happened).
5	If the connection is refused (for example FBW TELEX refuses the flight number), look at the FMS message area.	**FMS DATALINK NOT AVAIL** in white; it goes away once connected, and typing the flight number again retries.

3. ACARS provider connection

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Settings > ATSU / AOC | **Status:** Late start and close / restart of BeyondATC seen working in the user's flights; not re-checked after the final build. Unit tests: 4.

What it is for

The ATC and company datalink of the aircraft (CPDLC logon, requests, messages, ATIS and METAR) goes through an ACARS provider you choose on the flyPad: Hoppie, BeyondATC or SayIntentions. The aircraft must find it, then connect your flight number to it.

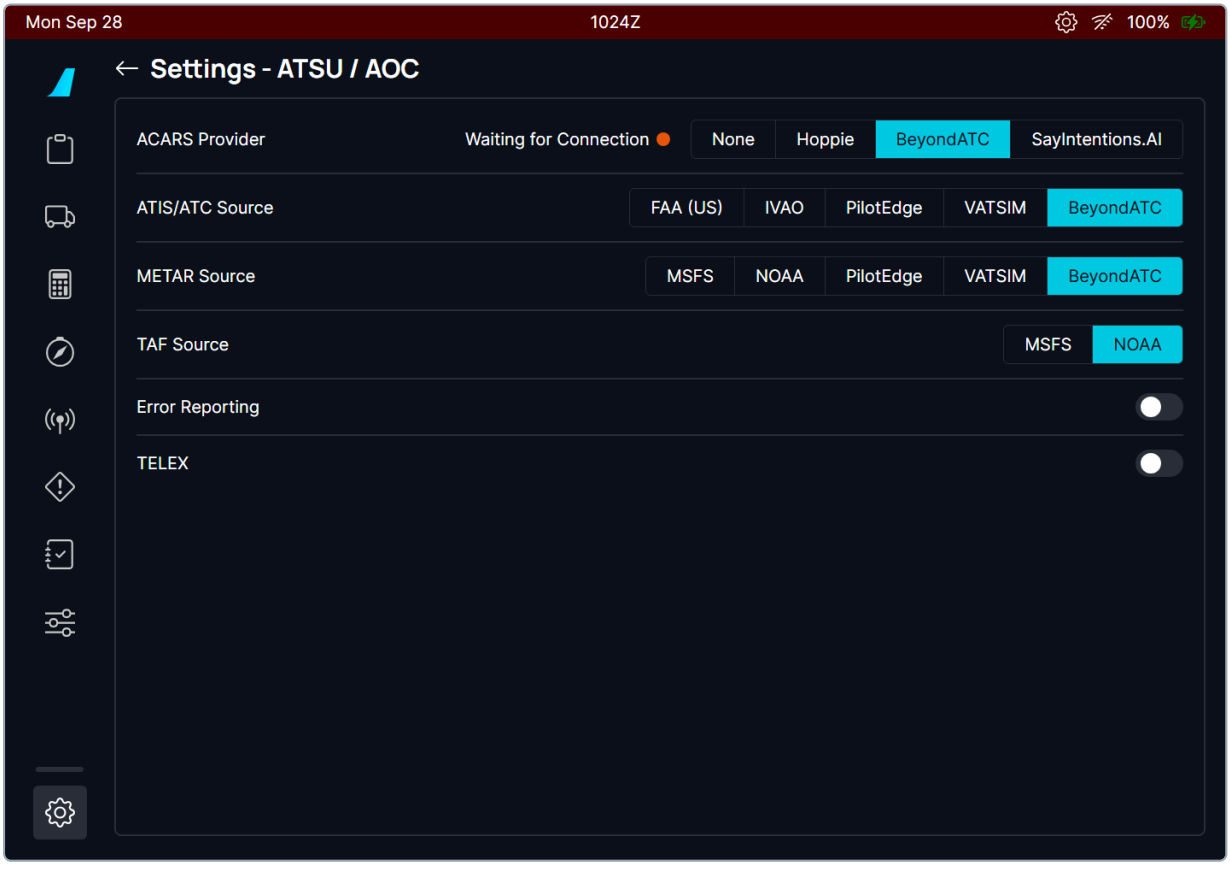


Figure 17. ACARS Provider: BeyondATC, "Waiting for Connection" until BeyondATC answers (then "Connected", green).

What happens now

- **Start the provider when you like.** The aircraft looks for it every 5 s for 5 minutes, then once a minute, without end. Your provider, ATIS and METAR settings are never changed.
- **Flight number.** Entered before the provider answers, it is connected as soon as it answers.
- **Provider closed.** Within about a minute the status goes back to "Waiting for Connection"; start it again and the aircraft connects your flight number again by itself. With BeyondATC, log on to the ATC center again (it forgets the logon when it closes).
- **TELEX on.** If FBW's TELEX service refuses your flight number, the ACARS provider is connected anyway (on the A380X, the FMS shows FMS DATALINK NOT AVAIL for the TELEX failure).

What to expect

- No extra traffic with Hoppie: once your flight number is connected, the regular polls (every 45 to 75 s) check the provider.
- Before this change, a provider started more than 5 minutes after the aircraft set the flyPad to NONE: select your provider again once if it shows NONE.

How to use

1. flyPad > Settings > **ATSU / AOC**: choose the ACARS provider (Hoppie, BeyondATC or SayIntentions).
2. Start the provider program whenever you like, before or after loading the aircraft.
3. Enter the flight number (MFD INIT on the A380X, MCDU INIT A on the A32NX): it is connected as soon as the provider answers.

How to test

#	Do this	Expected result
1	Load the aircraft with BeyondATC NOT running. flyPad > Settings > ATSU / AOC: ACARS provider BeyondATC.	"Waiting for Connection".
2	Wait more than 5 minutes, then start BeyondATC.	Within about a minute the status turns to "Connected" (green). The provider, ATIS and METAR settings are unchanged (not reset to NONE).
3	Enter the flight number before BeyondATC answers (restart the test with the flight number entered first).	The flight number is connected as soon as BeyondATC answers.
4	Close BeyondATC.	Within about a minute the status goes back to "Waiting for Connection".
5	Start BeyondATC again.	The aircraft connects the flight number again by itself; log on to the ATC center again (BeyondATC forgets the logon when it closes).

4. SD mailbox fixes: FILE FULL after CANCEL, uplink @ markers

> **Aircraft:** A380X | **Where:** SD MAILBOX; MFD ATC COM MSG RECORD | **Status:** Tested in the sim by the user: FILE FULL no longer sticks after cancels, PAN PAN PAN sent to BeyondATC; @ markers shown as cyan values.

What it is for

Two mailbox problems found in the sim with BeyondATC:

- **FILE FULL for good.** The mailbox keeps at most five open messages per direction and holds the next ones back with FILE FULL. A cancelled downlink (CANCEL of a prepared message) left the ATC function but kept its place in

the mailbox file, so after five cancels the file stayed full: every new downlink, a PAN PAN PAN included, waited behind FILE FULL. A cancel now frees the place, the held messages move in, and FILE FULL goes when the file has room again (A380 FCOM DSC-46: "File no more full").

- **Triangles around values.** Hoppie and BeyondATC mark the variable fields of a free text uplink with @ (CLRD TO @CYVR@ RWY @06R@), and the A380 display font draws @ as a triangle. The mailbox now shows the marked fields as main parameters, in blue (FCOM DSC-46-10-10-70), without the markers, as the A32NX DCDU does; an unclosed marker runs to the end of the text, and a word glued to a field ("@4611@.") stays glued. MSG RECORD drops the markers too.

What you should see

Situation	What you should see
Five downlinks prepared and cancelled, then a new one	The new downlink is shown with SEND / CANCEL; no FILE FULL.
More than five open downlinks	FILE FULL for the extra ones, until a place is freed (sent, closed or cancelled).
BeyondATC departure clearance "CLRD TO @CYVR@ RWY @06R@ ..."	CLRD TO CYVR RWY 06R ..., the values in blue, no triangles.

How to use

1. Nothing to set: CANCEL a prepared downlink as usual, read the uplinks as usual.

How to test

#	Do this	Expected result
1	A380X, logged on (ATC COM > CONNECT). Prepare a request (REQUEST, XFR TO MAILBOX) and press **CANCEL** in the SD mailbox. Do it six times.	Each new request shows SEND / CANCEL; FILE FULL never stays.
2	Then prepare an emergency (EMER, PANPAN frame, XFR TO MAILBOX) and SEND.	The PAN PAN PAN is sent (no FILE FULL).
3	With BeyondATC, ask for a departure clearance (REQUEST > CLEARANCE > DEPARTURE, XFR TO MAILBOX, SEND).	The clearance uplink in the mailbox: values (destination, runway, squawk...) in blue, no triangle symbols.
4	Close the dialogue and open ATC COM > **MSG RECORD**.	The recorded text has no @ characters.

5. LOAD-SEC3: ATC route clearances and crossing constraints into SEC 3

> **Aircraft:** A380X | **Where:** SD MAILBOX **LOAD-SEC3**; MFD FMS SEC INDEX | **Status:** Route clearance tested in the sim by the user (self-sent uplink): LOAD OK, ATC F-PLN INSERTED IN SEC 3. Crossing constraints and the REJECTED ATC INFO page not sim-tested. Unit tests: route clearance, constraints, mailbox logic.

What it is for

A route clearance or a crossing constraint received by CPDLC can be loaded into the FMS instead of being typed: the FCOM puts it in SEC 3, as a copy of the active flight plan with the clearance in it, for the crew to check and activate (A380 FCOM DSC-22-FMS-10-40, DSC-46-10-40). The mailbox has the **LOAD-SEC3** key in the 4th button position, on the uplinks that can be loaded only; it stays after WILCO, with CLOSE.

Uplink (CPDLC element)	What is loaded into SEC 3

Route clearance: CLEARED TO ... VIA ... (UM79), CLEARED ... (UM80), AT ... CLEARED ... (UM83)	SID (not once airborne), airways, waypoints, lat/long points, STAR and a new destination. The rest of the en route part is replaced; a UM79 route ends at its clearance limit.
Crossing constraints (UM46 to 53, 55, 56, 58 to 61): CROSS ... AT / AT OR ABOVE / AT OR BELOW / BETWEEN ..., AT ... speed, AT ... time	An altitude constraint, a speed constraint (90 kt to VMO 340 kt, in knots) or an RTA on that waypoint of the flight plan. The RTA is kept per secondary flight plan and follows it on a swap or an activation; the VERT REV RTA panel now works on a SEC plan.

What you should see

Situation	What you should see
LOAD-SEC3 pressed	Mailbox information area: LOADING, then LOAD OK, LOAD PARTIAL or LOAD FAILED.
Everything loaded	FMS message ATC F-PLN INSERTED IN SEC 3. SEC 3 holds the active flight plan with the clearance.
Some elements rejected	ATC F-PLN INSERTED IN SEC 3 / REJECTED INFO SEE SEC / INDEX. SEC INDEX, SEC 3: **REJECTED ATC INFO** button, listing each element (description, value, AT fix) and why: NOT IN DATABASE, AWY/WPT MISMATCH, DATA NOT ALLOWED IN PHASE, NOT IN F-PLN, FORMAT ERROR, ENTRY OUT OF RANGE.
Nothing usable	RECEIVED ATC MSG NOT VALID.
No FMS answer within 30 s, or a second LOAD-SEC3 while loading	LOAD NOT AVAIL.
A speed as a Mach number (e.g. M.82)	Rejected: FORMAT ERROR (the FCOM speed format is in knots).

> **NOTE**
> The constraint upload awaits in a plain statement, not as an operand: the sim's JavaScript engine can reject an await used as an operand and then blanks the whole MFD. No visible change.

Limitations and why

Limitation	Why	What to do
A duplicate waypoint ident is taken as the one nearest to the previous point.	Design choice: no NON UNIQUE rejection.	Check the route in SEC 3 before activating it.
A constraint on an en route waypoint with no climb or descent leg type is set as a descent constraint in cruise, else from the second half of the en route part.	The FMS needs CLB or DES for a constraint; this is a guess.	Check the constraint in SEC 3.
Needs an uplink of one of these CPDLC elements.	Live controllers rarely send them; BeyondATC route clearances were not seen.	Send yourself the uplink (How to test).

How to use

1. Receive the clearance in the SD mailbox and press **LOAD-SEC3**.
2. Wait for LOAD OK (or LOAD PARTIAL), then read the FMS message.
3. MFD FMS: SEC INDEX, SEC 3: check the flight plan (and REJECTED ATC INFO if shown).
4. Answer the uplink (WILCO / UNABLE) and, if you accept the clearance, activate SEC 3 (or swap it).

How to test

#	Do this	Expected result
1	A380X with an active flight plan whose en route part has waypoints ahead, datalink connected (flight number entered). Send yourself a route clearance. With BeyondATC (tested way, ask the owner of the setup first): open its local relay `http://localhost:57698/acars/system/connect.html` with `from=CZUL&to=<callsign>&type=cpdlc&packet=/data2/1//WU/CLEARED TO @<DEST>@ VIA @<FIX1> DCT <FIX2>@` (URL-encoded). With Hoppie, send the same CPDLC packet to your callsign through its connect.html.	Within a minute (next poll) the uplink is in the SD mailbox, the values in blue, with **LOAD-SEC3** in the 4th button position.
2	Press **LOAD-SEC3**.	LOADING, then LOAD OK in the mailbox; FMS message ATC F-PLN INSERTED IN SEC 3.
3	MFD FMS > SEC INDEX > SEC 3 F-PLN.	The active route with the clearance: the cleared fixes joined, the waypoints after them removed, the route ending at the clearance limit.
4	Send a clearance with a fix that does not exist (e.g. VIA @ZZZZZ@).	LOAD PARTIAL (or LOAD FAILED); SEC INDEX shows REJECTED ATC INFO with ZZZZZ and NOT IN DATABASE.
5	Send a crossing constraint on a waypoint ahead (CPDLC UM46, e.g. CROSS <FIX> AT FL240) and press LOAD-SEC3.	SEC 3: an altitude constraint FL240 on that waypoint (not sim-tested yet).
6	Press LOAD-SEC3 on a second uplink while the first one is still LOADING.	LOAD NOT AVAIL on the second one.

6. OIT COMPANY COM inbox with the real AOC messages

> **Aircraft:** A380X | **Where:** OIT (side console) AVNCS > COMPANY COM inbox; E/WD COMPANY MSG memo | **Status:** Not sim-tested (added; unit tests: CompanyComMailbox.spec).

What it is for

The OIT COMPANY COM inbox showed two hard-coded test messages. It now lists the company (AOC) messages received by the datalink, newest first, with an unread mark, the sender and the UTC time (A380 FCOM DSC-46-40-20, COMPANY COM application). Displaying a message marks it read; the COMPANY MSG memo goes once every message is read (FCOM: COMPANY MSG, "The flight crew received a company message")..

Situation	What you should see
A company message (for example a Hoppie free text) arrives	A new line at the top of the inbox with the unread mark; the E/WD shows COMPANY MSG.
The crew opens the message	The text shows; the line loses its unread mark.
Every message read	The COMPANY MSG memo goes.

Limitations and why

Limitation	Why	What to do
The other OIS FLT OPS applications are not done.	They are separate work.	-

How to use

1. Connect the datalink (flight number on MFD FMS INIT, AOC provider in flyPad > Settings > ATSU / AOC).
2. When COMPANY MSG shows, open the OIT COMPANY COM inbox and click the message to read it.

How to test

#	Do this	Expected result
1	A380X, datalink connected with Hoppie. Send yourself a free text telex from the Hoppie website.	The OIT COMPANY COM inbox lists it at the top, unread, with the sender and the UTC time; COMPANY MSG on the E/WD.
2	Click the message in the inbox.	The message shows; it is marked read and COMPANY MSG goes (if it was the only unread one).

What PR testers should check

- A Hoppie free text reaching the inbox, and COMPANY MSG cleared after reading it.
- Not sim-tested yet.

7. MFD System Message Area: CHECK FMS MESSAGE, NEW ATIS RECEIVED, ATIS SYS MSG

> **Aircraft:** A380X | **Where:** MFD header, all applications | **Status:** Not sim-tested (added; unit tests: MfdSystemMessageArea.spec, failing first).

What it is for

The A380 FCOM MFD header has a System Message Area that tells the crew about a message of another application while it is not shown (A380 FCOM MFD description, System Message Area). It now shows:

- **CHECK FMS MESSAGE**: an amber FMS message is pending while another application (ATC COM, SURV, FCU BKUP) is shown; removed when the crew opens an FMS page.
- **NEW ATIS RECEIVED**: a new ATIS report of the ATC COM datalink arrived while another application is shown; removed when the crew opens an ATC COM page.
- **ATIS SYS MSG**: an ATIS system message of the ATC centre (USE VOICE, SEND AGAIN, NO AUTO UPDATE, END OF UPDATE) while another application is shown; removed when the crew opens an ATC COM page.

Design choices: the area shows one message, the most recent one (the FCOM gives no priority); a message comes only from a new event, so going back to another application does not bring back a message already seen in its application.

Limitations and why

Limitation	Why	What to do
The KCCU MAILBOX key and the ATC MSG flashing are on `core/a380x-mailbox-du-reconf`.	They belong with the SD mailbox.	-

How to use

1. When a message shows in the MFD header, open the application it names (FMS, or ATC COM for the ATIS) to read it.

How to test

#	Do this	Expected result
1	A380X, MFD on ATC COM (or SURV). Cause an amber FMS message (any amber line of the FMS message list).	The MFD header shows **CHECK FMS MESSAGE**.
2	Datalink connected, MFD on the FMS. Request an ATIS on ATC COM, then go back to the FMS before it arrives.	When the ATIS arrives the header shows **NEW ATIS RECEIVED**.
3	Open the named application.	The header message goes.

What PR testers should check

- The layout of the message area in the header (fit, colour, position).
- Not sim-tested yet.

A380X fuel jettison and FINAL fuel

Fuel jettison down to the JTSN GW with its ECAM memo and procedures, the jettison valve failures with their alerts, a fuel mist plume at the jettison outlets and a fuel drip there on the ground, the FINAL fuel and time from the A380 holding fuel flow, and TRIP and cruise fuel predictions that match the fuel the engines burn

Aircraft	A380X
Where to find it	Overhead FUEL panel: JETTISON ARM and ACTIVE; MFD FMS ACTIVE/FUEL&LOAD: JTSN GW; E/WD and SD FUEL; flyPad > Failures > ATA 28; outside view of the wings; MFD FMS FUEL&LOAD: FINAL and EXTRA time, TRIP fuel; F-PLN EFOB predictions
Branch	`core/fuel`, based on `core/a380x-mfd-fms` (stacked: needs `core/surveillance` and `core/a380x-mfd-fms` merged first). The plume needs the aircraft-large-files companion change fa6c5d1 (jettison outlet nodes, fork TCdr/aircraft-large-files); so does the ground drip.
Status	Jettison: tested in the sim (September 2026), rate checked live (2.5 t/min, feed tanks untouched). Valve failures with FUEL JETTISON FAULT, VLV NOT CLOSED and INOP SYS JETTISON: OK. Plume: look and logic OK in flight, position at the outlets OK. Ground drip: drops seen with the plume (the later change that unties it from the flyPad Situation Visual Effects switch was not re-flown). SD FUEL jettison lines white during a normal jettison: sim-tested. FINAL fuel and time from the holding fuel flow: OK at the gate (user word); its EXTRA time: not sim-tested. TRIP and cruise fuel predictions: tested in the sim by the user on a combined test build before the commit, climb and 20 min cruise at M.85 (tests/): OK. The core branch itself was not sim-tested.
Pictures	MFD and EWD in the local browser harness, running the builds installed in the sim; the EWD states are the ones the FWS sends in flight. No pictures for the failures and the plume.

1. Fuel jettison with the JTSN GW, ECAM memo and procedures
2. Jettison valve failures: FUEL JETTISON FAULT, VLV NOT CLOSED, INOP SYS JETTISON
3. Fuel jettison plume at the outlets
4. Fuel jettison on the ground: fuel drip at the outlets
5. FINAL fuel and time from the A380 holding fuel flow
6. TRIP and cruise fuel predictions match the engines

1. Fuel jettison with the JTSN GW, ECAM memo and procedures

> **Aircraft:** A380X | **Where:** Overhead JETTISON ARM / ACTIVE; MFD FUEL&LOAD JTSN GW; ECP ABN PROC | **Status:** Tested in the sim by the user (September 2026); jettison rate measured live: 2.5 t per minute from the outer, mid and trim tanks, feed tanks untouched; SD FUEL jettison lines white during a normal jettison: sim-tested. Unit tests: FuelJettison.spec.ts (FCOM DSC-28-40), JettisonValveIndication.spec.ts (SD colours, DSC-28-20).

What it is for

FBW had the overhead JETTISON ARM and ACTIVE pushbuttons, but they did nothing. The fuel management (FQMS) now jettisons the fuel of the transfer tanks down to the gross weight entered on the MFD, and the ECAM shows the memo and the procedures of the FCOM.

1. Set the jettison gross weight

1. MFD FMS **ACTIVE/FUEL&LOAD** page. **JTSN GW** is shown when the active flight plan has its mandatory data.
2. Enter the gross weight at which the jettison must stop, in tonnes (or 1000 lb with the LBS unit).
3. Allowed range: from **ZFW + 11 000 lb** (5.0 t) to the maximum gross weight (FCOM DSC-22-FMS-20-100). Outside it: ENTRY OUT OF RANGE with the range.

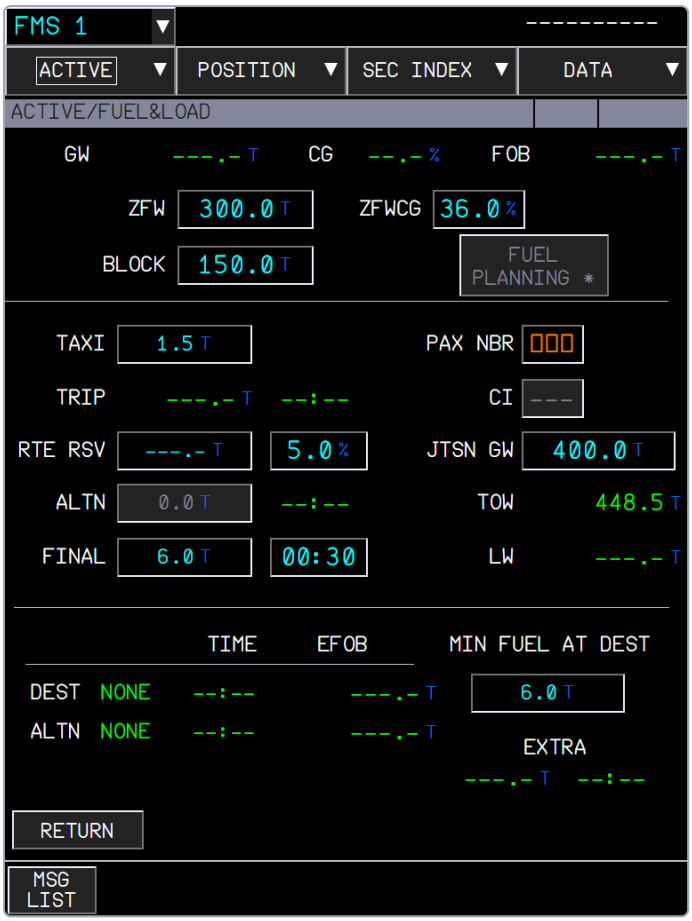


Figure 1. JTSN GW 400.0 t entered (ZFW 300.0 t).

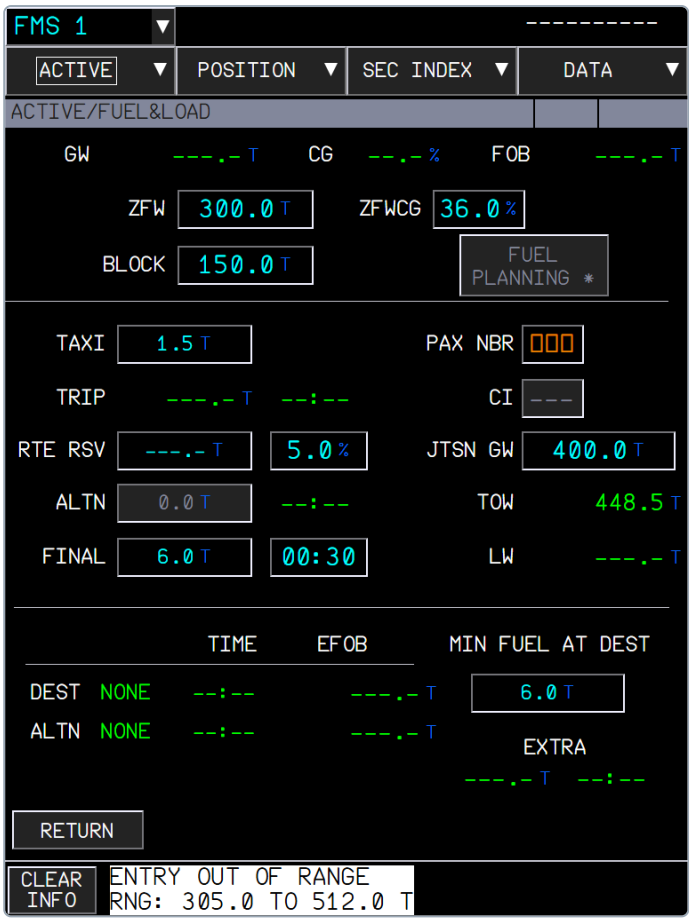


Figure 2. 290 t is below ZFW + 5.0 t: the range is shown in the message area.

> **CAUTION**
>
> Without a JTSN GW the FQMS has no target weight: the jettison then runs until the transfer tanks are empty (or a pushbutton is set OFF).

2. Run the procedure

The ABN PROC procedure takes the crew through the steps. ECP **ABN PROC**, then FUEL, then JETTISON:

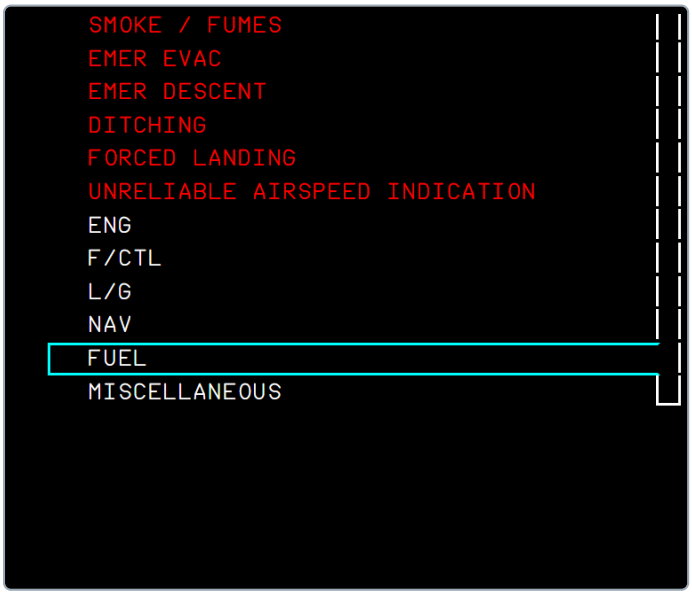


Figure 3. ABN PROC menu: FUEL selected.

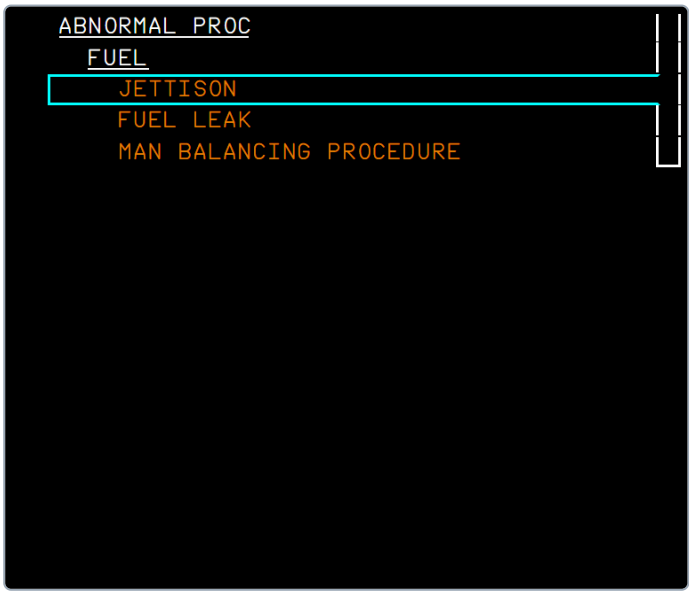


Figure 4. FUEL sub-menu: JETTISON.

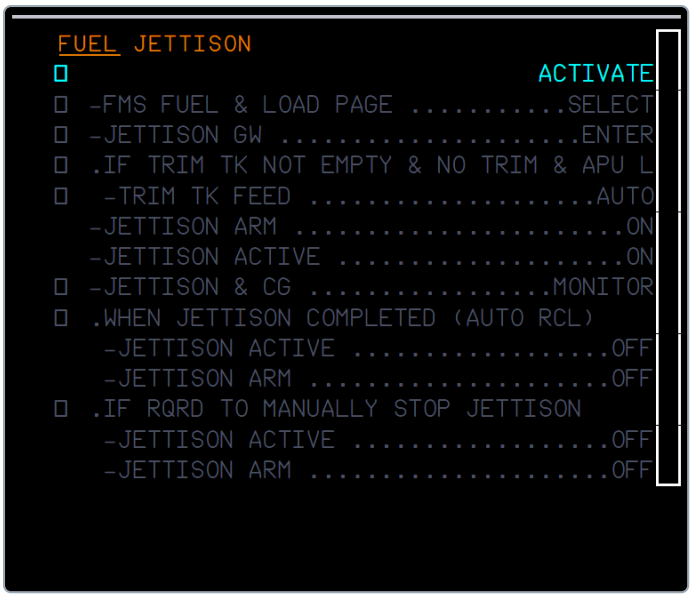


Figure 5. The procedure before ACTIVATE: all lines in grey.

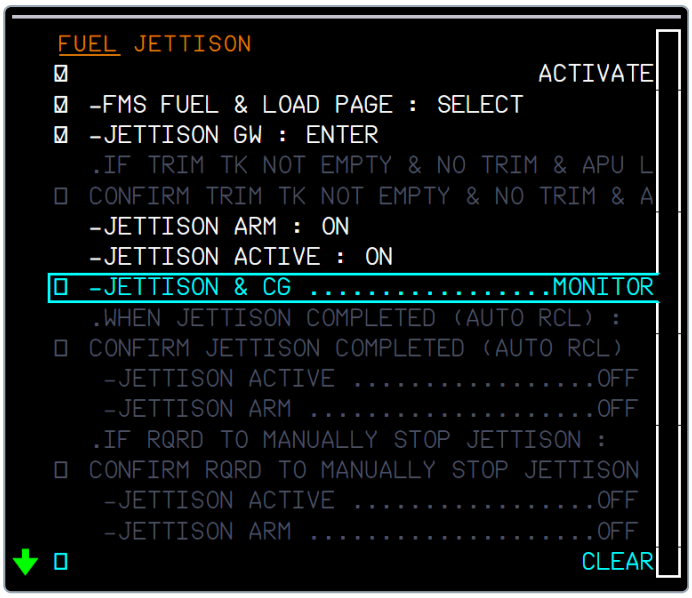


Figure 6. Activated: the FMS lines checked by the crew, JETTISON ARM and ACTIVE ticked by the FWS as the pushbuttons are set (sensed lines).

- FMS FUEL & LOAD PAGE ... SELECT, JETTISON GW ... ENTER: crew check lines.
- IF TRIM TK NOT EMPTY & NO TRIM & APU LINE FAULT: TRIM TK FEED AUTO (condition line, confirmed by the crew).
- **JETTISON ARM ON**, **JETTISON ACTIVE ON**: sensed, they tick themselves.
- JETTISON & CG ... MONITOR: watch the SD FUEL page and the CG.
- WHEN JETTISON COMPLETED (AUTO RCL) / IF RQRD TO MANUALLY STOP JETTISON: ACTIVE OFF then ARM OFF, sensed.

3. During the jettison

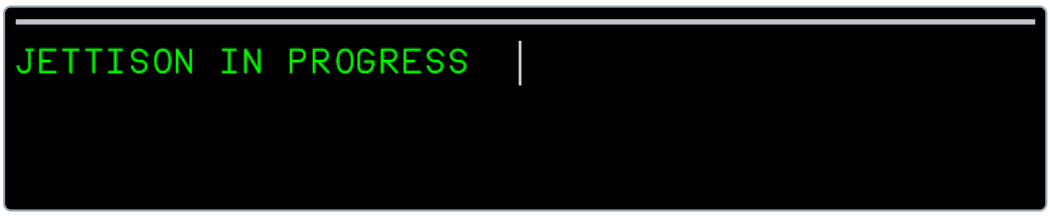


Figure 7. The EWD memo while the fuel is jettisoned.

- Memo **JETTISON IN PROGRESS** (green).
- Fuel leaves the **transfer tanks** (outer, mid, inner and trim) at the FCOM rate of **330 693 lb/h** (about 2.5 t per minute), in proportion to their content, so they empty together. The **feed tanks are never jettisoned**.
- The trim tank is held when the CG reaches the forward T.O/LDG limit, and resumes when the CG is 0.5 % aft of it.
- The other automatic transfers pause during the jettison.
- The SD FUEL page shows the two jettison valves open, with the jettison lines and JETTISON legends in white (normal, FCOM DSC-28-20), and the quantities decreasing. Before the fix on this branch they were amber during every jettison (the page did not know a jettison was running).

4. When it stops

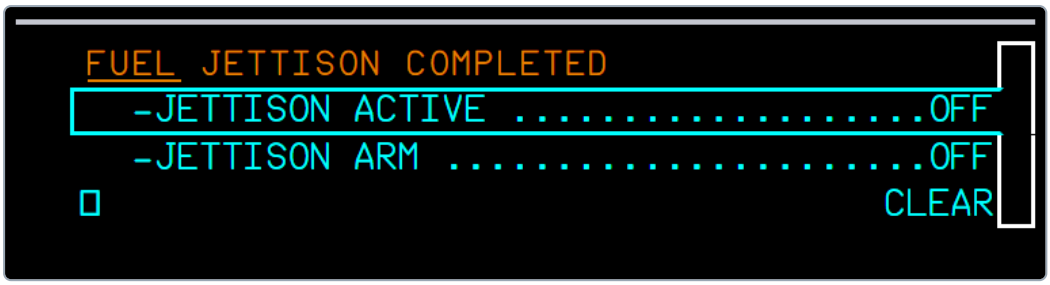


Figure 8. FUEL JETTISON COMPLETED: set ACTIVE then ARM OFF; the lines tick as the pushbuttons are released.

Situation	What you should see
Gross weight reaches the JTSN GW	The jettison stops, FUEL JETTISON COMPLETED shows, the ACTIVE ON light goes off.
All transfer tanks empty (or no JTSN GW entered)	Same: stops and COMPLETED.
ACTIVE or ARM set OFF by the crew	Stops at once (manual stop). No COMPLETED alert.
Both pushbuttons OFF after COMPLETED	The alert and the COMPLETED state clear; a new jettison can be started.
ACTIVE ON without ARM	Nothing: ARM is needed.
AC 1 bus not powered	No jettison (FQMS channel unpowered).

Limitations and why

Limitation	Why	What to do
JETTISON FAULT, VLV NOT CLOSED and INOP SYS JETTISON come only from the flyPad valve failures.	No other cause of a jettison valve fault is simulated; the four valve failures are described in the next feature.	-
The fuel does not really flow out through the outlets.	The MSFS fuel system has no overboard dump: the FQMS sets the tank quantities directly and opens valves 57 / 58 for the SD picture. A fuel mist plume is drawn at the outlets (last feature).	-
After an update, the first flight needs a full MSFS restart.	The change touches the behaviour XML of the overhead, which MSFS loads at start.	Restart MSFS once after installing.

How to use

1. MFD FMS **ACTIVE/FUEL&LOAD**: enter **JTSN GW** (the gross weight at which the jettison stops).
2. ECP **ABN PROC** > FUEL > JETTISON, ACTIVATE, and follow the lines.
3. Overhead FUEL panel: **JETTISON ARM** ON, then **ACTIVE** ON. Monitor the SD FUEL page and the CG.
4. At FUEL JETTISON COMPLETED (or to stop earlier): ACTIVE OFF, then ARM OFF.

How to test

#	Do this	Expected result
1	A380X with fuel in the transfer tanks (e.g. 150 t block), AC power, active flight plan with its mandatory data. MFD FMS ACTIVE/FUEL&LOAD: type a JTSN GW below ZFW + 5.0 t.	ENTRY OUT OF RANGE with the allowed range in the message area.

2	Type a JTSN GW about 20 t below the current gross weight.	JTSN GW shows the value.
3	ECP ABN PROC > FUEL > JETTISON, ACTIVATE.	The ABN PROC FUEL JETTISON procedure; JETTISON ARM and ACTIVE lines not ticked.
4	Set **JETTISON ACTIVE** ON without ARM.	Nothing happens (ARM is needed).
5	Set **JETTISON ARM** ON (ACTIVE ON).	Memo JETTISON IN PROGRESS (green); the ARM and ACTIVE lines tick; SD FUEL: both jettison valves open, jettison lines and JETTISON legends white (not amber), the outer, mid, inner and trim quantities fall together (about 2.5 t per minute); feed tanks unchanged.
6	Wait until the gross weight reaches the JTSN GW.	The jettison stops; FUEL JETTISON COMPLETED with ACTIVE OFF / ARM OFF; the ACTIVE ON light goes off.
7	Set ACTIVE OFF, then ARM OFF.	The lines tick and the COMPLETED alert clears.
8	Start a new jettison and set ACTIVE OFF halfway.	It stops at once, without COMPLETED.

2. Jettison valve failures: FUEL JETTISON FAULT, VLV NOT CLOSED, INOP SYS JETTISON

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28; E/WD; SD STATUS; overhead JETTISON ACTIVE OPEN light | **Status:** Tested in the sim by the user: one valve stuck closed, both stuck closed, stuck open, all OK (after the FAULT / INOP SYS fix); the cases again with the plume. Unit tests: FuelJettison.spec.ts, FuelJettisonFlags.spec.ts.

What it is for

The A380 FCOM has two jettison alerts, FUEL JETTISON FAULT and FUEL JETTISON VLV NOT CLOSED (PRO-ABN-ECAM-10-28 P 61-63), and INOP SYS JETTISON on the STATUS page. FBW had no jettison valve failure to trigger them. There are now four failures, one per valve and position. A first version never showed the FAULT caution and the INOP SYS: the FWS read the FQMS flags as numbers (1) and combined them with a function that only counts `true`; they are now read as real booleans.

The failures

flyPad > Failures > ATA 28 (Fuel)	Effect
Jettison valve L stuck closed / **Jettison valve R stuck closed**	That valve does not open: the fuel goes through the other one at **half the rate**. The FQMS records the valve fault when the jettison is selected (ARM and ACTIVE ON), until the failure is removed.
Both stuck closed	The jettison is **not available**: no fuel leaves.
Jettison valve L stuck open / **Jettison valve R stuck open**	That valve stays open: the **OPEN** light of the JETTISON ACTIVE pushbutton stays on and the SD shows the valve open, also with no jettison running. No fuel is lost by it (the jettison itself still needs ARM and ACTIVE).

What you should see

Situation	What you should see
One valve stuck closed, jettison selected	Amber caution **FUEL JETTISON FAULT**; on the ground with the line JETTISON L (R) VLV FAULT. Jettison at half rate. STATUS: INOP SYS **JETTISON** (the FWS lists it whenever FUEL JETTISON FAULT is active).

Both valves stuck closed, jettison selected	**FUEL JETTISON FAULT** with JETTISON NOT AVAIL, then JETTISON ACTIVE ... OFF and JETTISON ARM ... OFF (they tick as the pushbuttons are set OFF). STATUS: INOP SYS **JETTISON**. No fuel jettisoned.
A valve stuck open, no jettison running	Amber caution **FUEL JETTISON VLV NOT CLOSED** with JETTISON ACTIVE ... OFF and JETTISON ARM ... OFF; OPEN light on.
Takeoff and landing	The alerts are inhibited in the takeoff and landing flight phases (FCOM inhibitions).

How to use

1. flyPad > **Failures** > **ATA 28**: activate one of the four jettison valve failures.
2. Run a jettison (or not, for a valve stuck open) and follow the ECAM.
3. Deactivate the failure in the flyPad to repair the valve.

How to test

#	Do this	Expected result
1	A380X on the ground (or in cruise), fuel in the transfer tanks, AC power, JTSN GW entered. flyPad > Failures > ATA 28: activate **Jettison valve L stuck closed**. Set JETTISON ARM and ACTIVE ON.	Amber **FUEL JETTISON FAULT** (on the ground with JETTISON L VLV FAULT); SD FUEL: only the right jettison valve open; the transfer tanks empty at half the normal rate (about 1.25 t per minute). STATUS: INOP SYS JETTISON.
2	Also activate **Jettison valve R stuck closed**.	FUEL JETTISON FAULT with JETTISON NOT AVAIL, JETTISON ACTIVE OFF, JETTISON ARM OFF; no fuel leaves; STATUS: INOP SYS JETTISON.
3	Set ACTIVE OFF, then ARM OFF.	The ACTIVE and ARM lines tick.
4	Deactivate both failures, activate **Jettison valve L stuck open** (pushbuttons OFF).	Amber **FUEL JETTISON VLV NOT CLOSED**; OPEN light of JETTISON ACTIVE on; SD FUEL: left jettison valve open; the fuel quantities do not change.
5	Deactivate the failure.	The valve closes, the OPEN light goes off and the alert clears.

3. Fuel jettison plume at the outlets

> **Aircraft:** A380X | **Where:** Outside view: wing trailing edge, at the second-outermost flap track fairing | **Status:** Tested in the sim by the user in flight: look, both / left only / right only and the valve failure cases OK; position at the outlets OK.

What it is for

The jettisoned fuel left no trace outside. A fuel mist plume now trails from each wing's jettison outlet while fuel flows out of that side. The FCOM says there is "one discharge point on the trailing edge of each wing" (DSC-28-40); the outlet is the tube at the aft end of the second-outermost flap track fairing (MFTF 5, forum source), where the companion model adds the effect points.

What you should see

Situation	What you should see
Jettison in progress, both valves open	A mist plume behind each wing outlet; it gets longer with the true airspeed (its particles stay where they were released).
One valve stuck closed	The plume on the other side only.

Valve stuck open, no jettison running	No plume (no fuel flows).
Below 40 kt true airspeed	The low-speed version of the mist (the stock MSFS water-ballast effect switches at 40 kt).

Limitations and why

Limitation	Why	What to do
Needs the aircraft-large-files companion.	The effect hangs on the FX_JETTISON_LH / RH nodes added at the outlets in the A380_WINGS models (fa6c5d1); without them no plume shows.	Install it; a full MSFS restart is needed after the update.
The look is the stock water-ballast mist.	Closest stock MSFS effect: design choice.	-
No fuel puddle on the ground.	Jettisoned fuel evaporates; nothing is drawn on the ground. On the ground, fuel drops fall from the outlets (next feature).	-

How to use

1. Run a fuel jettison (first feature) and look at the wings from the outside view.

How to test

#	Do this	Expected result
1	A380X in flight (companion models installed, MSFS restarted), JTSN GW entered. Set JETTISON ARM and ACTIVE ON. Outside view of a wing.	A mist plume trails from the trailing edge at the second-outermost flap track fairing, on both wings.
2	Speed up or slow down.	The plume gets longer at a higher true airspeed.
3	flyPad > Failures > ATA 28: **Jettison valve L stuck closed**.	The left plume stops; the right one goes on.
4	Set ACTIVE OFF.	Both plumes stop.
5	Remove the failure, activate **Jettison valve R stuck open** without a jettison.	No plume.

4. Fuel jettison on the ground: fuel drip at the outlets

> **Aircraft:** A380X | **Where:** Outside view: wing trailing edge, at the jettison outlets | **Status:** Tested in the sim by the user: drops seen with the plume during a ground jettison. The drip was then made independent of the flyPad Situation Visual Effects switch (user decision); that change was not re-flown. Unit tests: JettisonDripContrailEffects.spec.ts (8). The core branch itself was not sim-tested.

What it is for

The A380 FCOM allows a jettison on the ground: "It is possible to jettison in any aircraft configuration, and at any speed" (PRO-ABN-ECAM-10-28 FUEL JETTISON), and the FQMS does not stop it on the ground. The fuel then falls on the ground. With the aircraft on the ground, fuel drops now fall from each jettison outlet whose valve is open, together with the plume of the previous feature.

What you should see

Situation	What you should see

On the ground, jettison running, both valves open	Fuel drops fall from both outlets, with the mist plume.
On the ground, one valve open	Drops and plume at that outlet only.
Below 40 kt true airspeed	The slow drip; from 40 kt the fast drip (the stock MSFS fuel leak effects, switched at 40 kt as Asobo does).
In flight	No drip: the plume alone, unchanged.
Valve stuck open, no jettison running	No drip (no fuel flows).
flyPad Situation Visual Effects switch OFF (the switch comes with the fire effects of `core/failures-a32nx`)	Drip and plume still show: the jettison effect is not a situation effect.

Limitations and why

Limitation	Why	What to do
Needs the aircraft-large-files companion.	The drip hangs on the same FX_JETTISON_LH / RH outlet nodes as the plume (fa6c5d1).	Install it; a full MSFS restart is needed after the update.
The look is the stock MSFS fuel leak drip.	Closest stock MSFS effect: design choice. No puddle or wet patch is drawn on the ground.	-

How to use

1. On the ground, enter a JTSN GW on the MFD FMS ACTIVE/FUEL&LOAD page and set JETTISON ARM and ACTIVE ON on the overhead FUEL panel.
2. Look at the wing trailing edges from the outside view; set ACTIVE OFF to stop.

How to test

#	Do this	Expected result
1	A380X parked on the ground (companion models installed, MSFS restarted), engines running or GPU on, JTSN GW entered below the gross weight. Set JETTISON ARM and ACTIVE ON. Outside view of a wing.	Fuel drops fall from the jettison outlet of each wing, with the mist plume.
2	flyPad > Failures > ATA 28: **Jettison valve L stuck closed**.	The left drops and plume stop; the right ones go on.
3	Set ACTIVE OFF.	Drops and plume stop on both sides.

5. FINAL fuel and time from the A380 holding fuel flow

> **Aircraft:** A380X | **Where:** MFD FMS ACTIVE/FUEL&LOAD: FINAL fuel and time, EXTRA time | **Status:** FINAL fuel and time: tested in the sim by the user at the gate: OK (optional A380X card: ZFW 342.6 t, FINAL entry, ZFW 300.0 t; user word, no simvar to record). The EXTRA time: not sim-tested. Unit tests: FinalHoldingPredictor.spec.ts. The core branch itself was not sim-tested.

What it is for

The FINAL fuel default of the FUEL&LOAD page was a fixed 6.0 t per 30 min (0.2 t per minute, the same flow for the EXTRA time), whatever the weight. The A380 FCOM defines the holding fuel as 30 min at 1 500 ft above the alternate, at holding speed (green dot + 25 kt, CONF CLEAN), ISA, with the landing weight at the alternate (PER-IFT-FPL-MRF HOLDING FUEL). The FINAL fuel now comes from that holding.

- The fuel flow comes from the one A380 holding computation in the FCOM (PER-IFT-HLD-HMI, IN-FLT PERF example: 940.0 klb, 30 min, 13 100 lb), brought back from ISA+10 to ISA, and taken proportional to the weight.

- FINAL fuel: a holding of the FINAL time (30 min by default) that ends at the ZFW, rounded up to the 0.1 t of the entry field, like the ALTN fuel. Check: RJTT-RKSI at ZFW 342.6 t gives 4.70 t, shown 4.8 (the airline OFP gives 4.8 t).
- FINAL time from a FINAL fuel entry: the time that fuel lasts in the same holding.
- EXTRA time: the time the EXTRA fuel lasts in the same holding, ending with the ZFW plus the MIN FUEL AT DEST on board.
- Without a ZFW: the FCOM conservative quantity, 10 600 lb per 30 min.
- This replaces the fixed 0.2 t per minute flow (30 min = 6 t) of `core/a380x-mfd-fms`.

What you should see

Situation	What you should see
RJTT/RKSI, ALTN RKSS, ZFW 342.6 t, no FINAL entry	FINAL 4.8, time 00:30.
ZFW 300.0 t	FINAL 4.2, time 00:30.
FINAL fuel 6.0 entered (ZFW 342.6 t)	FINAL time 00:38.
FINAL fuel cleared (CLR)	Back to FINAL 4.8, 00:30.
5.0 t of EXTRA fuel at ZFW 342.6 t	EXTRA time about 00:30 or 00:31 (about 6.3 min per tonne; before: 5 min per tonne).

Limitations and why

Limitation	Why	What to do
The FINAL time stays 00:30 with a calculated fuel.	Design choice: the time shown is the one the fuel was computed from, not the slightly longer time of the rounded-up fuel.	-
The holding altitude and the temperature are not used.	The A380 FCOM has no holding table; the A320neo table varies by less than 5 % from FL15 to FL140, and the FCOM figure is brought back to ISA (design choice).	-
The fuel flow is taken proportional to the weight.	Maximum endurance: thrust follows the weight; the A320neo holding table confirms the near linear law (design choice).	-

How to use

1. MFD FMS **INIT**: FROM/TO and ALTN. **FUEL&LOAD**: enter the ZFW and ZFWCG.
2. Read the FINAL fuel and time. Overwrite the FINAL fuel or the FINAL time if the company policy asks for another value; CLR returns to the calculated value.
3. Read the EXTRA time next to the EXTRA fuel once the BLOCK is entered.

How to test

#	Do this	Expected result
1	A380X at the gate, aircraft powered, no flight plan. MFD (FMS) INIT: FROM/TO **RJTT/RKSI**, ALTN **RKSS**. FUEL&LOAD: ZFW **342.6**, ZFWCG **30.0**.	FINAL **4.8**, time **00:30** (before: 6.0).
2	Change the ZFW to **300.0**.	FINAL **4.2**, time 00:30. Set the ZFW back to 342.6.
3	Enter **6.0** in the FINAL fuel field.	FINAL time **00:38** (before: 00:30).

4	CLR the FINAL fuel.	FINAL back to **4.8**, time 00:30.
5	Run FUEL PLANNING, then add 5 t to the BLOCK (not sim-tested yet).	EXTRA time about **00:30** or **00:31** for 5.0 t (before: 00:25).

6. TRIP and cruise fuel predictions match the engines

> **Aircraft:** A380X (the shared VNAV engine model also changes the A32NX predictions) | **Where:** MFD FMS FUEL&LOAD TRIP; F-PLN EFOB and DEST EFOB | **Status:** Tested in the sim by the user on a combined test build before the commit: climb FL100-FL300 and 20 min cruise at M.85 : OK. Unit tests: EngineModel.spec.ts, A380TripFuelPrediction.spec.ts. The A32NX predictions: not sim-tested.

What it is for

The A380X FMS predicted less fuel than the engines burn: 4 to 7 % less in cruise and about 3 % less on the TRIP (e.g. 19.8 t instead of 20.5 t on a 700 NM, FL340 RJTT-RKSI-like leg), so the DEST EFOB fell during the cruise. The cause was in the VNAV engine model: to find the N1 that gives a thrust, it inverted the engine thrust table (table 1506 of engines.cfg) column by column, which is not the inverse of the table the sim engine reads (bilinear in N1 and Mach). At M.85, exactly between the M.8 and M.9 columns, it took up to 2 % N1 too little. It now interpolates the table at the Mach first and inverts that single curve.

- The cruise fuel flow now matches the FADEC fuel flow at M.85 (A380TripFuelPrediction.spec.ts: the sim burn and the FCOM IN-FLT PERF examples).
- The same RJTT-RKSI-like leg (ZFW 342.6 t, 30 t at take-off, FL340, ISA, no wind) now gives a TRIP of 20.5 t.
- **A32NX:** `EngineModel.ts` is the shared FMGC code, so the A32NX VNAV predictions (climb, cruise, descent) use the corrected inverse too. They change only between the Mach columns of the A32NX thrust table.

Limitations and why

Limitation	Why	What to do
The TRIP is still below a SimBrief plan (23.5 t for the real RJTT-RKSI route).	The sim A380X itself burns less than the FCOM IN-FLT PERF examples (climb about 24 % less fuel, cruise about 5 % less); the FMS now matches the sim engines, not the FCOM tables.	-

How to use

1. Nothing to set: enter the flight plan, the CRZ FL and the FUEL&LOAD data as usual and read the TRIP and the EFOB predictions.

How to test

#	Do this	Expected result
1	A380X departed from RJTT with an RJTT-RKSI flight plan, MFD INIT CRZ FL 300, FUEL&LOAD done. In flight at FL100, clean, 250 kt, AP and A/THR ON, MSFS weather clear skies (ISA, no wind). FCU: altitude 30000, OP CLB, speed 300 kt.	The aircraft climbs and levels off at FL300.
2	FCU: push the SPD knob (managed M.85). After 3 min read the DEST EFOB on the MFD F-PLN page; read it again 20 min later.	The two DEST EFOB readings differ by at most 0.1 t (before the fix it fell by about 0.2 t).

GPS receivers, GPS deselection and ECAM STATUS

Real GPS receivers in MMR 1 and 2 for both aircraft, with the GPIRS position, GPS PRIMARY from the real accuracy, the GPS pages, the NAV GPS FAULT alerts, GPS deselection and the A32NX ECAM STATUS page; then the FCOM FM position (IRS plus a GPS or radio bias) with the POSITION MONITOR pages and the FCOM FMS messages

Aircraft	A32NX and A380X (shared systems code); the ECAM STATUS page and STS reminder are A32NX only
Where to find it	MCDU DATA > GPS MONITOR and SELECTED NAVAIDS (A32NX); MFD FMS > POSITION > GPS and NAVAIDS (A380X); E/WD and SD STATUS page; flyPad > Failures > ATA 34 Navigation; MCDU POSITION MONITOR, PROG and messages; MFD POSITION / MONITOR and FMS messages
Branch	`core/gps`, based on `core/a380x-mfd-fms` (stacked: needs core/surveillance and core/a380x-mfd-fms merged first). `core/a32nx-air-conditioning`, `core/a380x-elec` and `core/failures-a32nx` build on it.
Status	All six features tested in the sim by the user: OK: receivers and pages, A380X DIFF mode, GPS deselection on both aircraft, NAV GPS FAULT alerts, A32NX STATUS page and STS reminder, A380X INOP SYS (on a combined test build). Not sim-tested on this branch: the A32NX STS reminder, adapted here to the FlyByWire master CLR logic. A full MSFS restart is needed after installing (systems WASM). FM position and FMS messages: not sim-tested, unit tests only.
Pictures	One picture: the A380X POSITION/GPS page layout, taken in the browser harness before the receivers existed (the values on it are the old stand-in values). The other displays are text only.

1. GPS receivers in MMR 1 and 2
2. GPIRS position and GPS PRIMARY
3. GPS pages: MCDU GPS MONITOR and MFD POSITION/GPS
4. NAV GPS 1(2) FAULT alerts
5. GPS deselection and position uncertainty
6. A32NX ECAM STATUS page and STS reminder
7. FM position: IRS position plus a GPS or radio bias, POSITION MONITOR pages
8. FMS messages from the FCOM lists, and CHECK APPR SELECTION

1. GPS receivers in MMR 1 and 2

> **Aircraft:** A32NX and A380X | **Where:** No control: the receivers run when powered | **Status:** Tested in the sim by the user: OK. Degraded modes also covered by 21 Rust unit tests.

What it is for

There was no GPS receiver in either aircraft: the pages showed the sim GPS position for both receivers with random satellite counts and figures of merit, and GPS PRIMARY came from a random estimate. Now each Multi-Mode Receiver (MMR 1 and MMR 2) has a GPS receiver. A320 FCOM DSC-34: "Worldwide, 24 satellites broadcast accurate navigation data"; the two independent receivers process them and send their data to the ADIRUs.

- **Satellites:** the sim has none, so they are computed from the nominal 24-slot GPS constellation (six orbital planes, four satellites each) at the sim UTC date and time. A satellite counts when it is more than 5° above the horizon. Their geometry gives the dilution of precision, and from it the figure of merit (accuracy) and the integrity limit (HIL) of the position. Fewer than 5 satellites: no integrity limit.
- **Modes** (A320 FCOM DSC-22_20 GPS MONITOR, A380 FCOM DSC-34-10-40-10): **TEST** (self test at power up, 1.5 to 2.5 s), **INIT** (2.5 to 4.5 s), **ACQ** (satellite acquisition), **NAV** (4 or more satellites, data to the ADIRUs), **FAULT** (failed receiver).
- **Acquisition time:** 15 to 30 s after a short power cut (unpowered 15 min or less: warm start), 80 to 130 s from a cold start, 5 to 15 s to re-acquire after a loss of the signals.
- **Degraded modes** (the FCOMs only name them; the behaviour follows ARINC 743A, a design choice): **ALTAID** with 3 satellites (the ADR barometric altitude is the fourth measurement, no integrity limit); **AIDED** with fewer satellites,

the receiver coasts on the IR velocities for up to 2 min, its accuracy gets worse with time and it gives no integrity limit. After that: back to ACQ.

- **Antenna masking:** the antenna is on top of the fuselage. In a turn or a steep climb, satellites low on the far side of the aircraft are masked.
- **DIFF (A380X only):** the A380 receivers use satellite based augmentation. In NAV inside a WAAS (North America), EGNOS (Europe), MSAS (Japan) or GAGAN (India) service area, the mode shows DIFF and the position is several times more accurate. The service areas are coarse latitude/longitude boxes (design choice).
- **Power:** A32NX: the FCOM gives none. GPS 1 is on AC ESS and GPS 2 on AC 2, as the other side 1 / side 2 navigation receivers (design choice). A380X: as the FCOM supply table, GPS 1 (MMR 1) is on AC EMER and GPS 2 (MMR 2) on AC 3. The bus named AC ESS in the A380X code is the AC EMER busbar.
- **Failures:** flyPad > Failures > ATA 34 Navigation > **GPS 1** and **GPS 2** put the receiver in FAULT.

Limitations and why

Limitation	Why	What to do
The satellite positions are the nominal constellation, not the real satellites of the day.	No almanac file is used: the 24 nominal slots give a realistic count and geometry anywhere in the world.	None needed.
ALTAID and AIDED are rarely seen in flight.	With 24 satellites, 4 or more are almost always in view; masking in a steep turn seldom leaves fewer.	Covered by the Rust unit tests.
A full MSFS restart is needed after installing.	The receivers are in the systems WASM module.	Restart the sim.

How to use

1. Nothing to set: power the aircraft and the receivers start (TEST, INIT, ACQ, then NAV).
2. Watch them on the A32NX MCDU **DATA > GPS MONITOR** page or the A380X MFD **FMS > POSITION > GPS** page (see "GPS pages" below).

How to test

#	Do this	Expected result
1	Cold and dark (aircraft unpowered more than 15 min). Power the aircraft (batteries, external power or APU) and open the GPS page at once.	Both receivers show TEST briefly, then INIT, then ACQ with a rising number of satellites; NAV after about 80 to 130 s.
2	Remove the power for less than 15 min, then power again.	ACQ lasts only 15 to 30 s (warm start).
3	In NAV, read the satellites and the figure of merit.	Between 4 and 12 satellites (7 to 10 on average) and a merit of about 30 to 100 ft; the two receivers slightly different.
4	A380X in Europe or North America (for example EGLL or KJFK): open POSITION / GPS.	MODE shows **DIFF** on both receivers.
5	A380X outside the service areas (for example SBGR, FAOR or YSSY).	MODE shows **NAV**.
6	flyPad > Failures > ATA 34 Navigation > **GPS 1**.	GPS 1 shows FAULT, its data dashed; GPS 2 unchanged. Clear the failure: GPS 1 goes back through ACQ to NAV.

2. GPIRS position and GPS PRIMARY

> **Aircraft:** A32NX and A380X | **Where:** MCDU PROG page (A32NX); MFD POSITION / MONITOR (A380X); ND messages | **Status:** Tested in the sim by the user: OK.

What it is for

Each IR of the ADIRUs now mixes its inertial position with the position of one GPS receiver: the GPIRS position. The selection follows A320 FCOM DSC-22_20: IR 1 uses GPS 1, IR 2 uses GPS 2, IR 3 uses GPS 1 (GPS 2 with the ATT HDG selector on F/O 3). If one receiver is rejected, all the IRs use the other one. The same rule is used on the A380X (design choice: the A380 FCOM does not give it).

The FMS (the same code in the A32NX FMGC and the A380X FMC) uses the onside GPIRS, else GPIRS 3, else the offside one (A380 FCOM DSC-22-FMS-10-30-10). Its estimated position uncertainty (EPU) is the GPIRS figure of merit. Accuracy is **HIGH** when the EPU is within the required navigation performance (RNP), and **GPS PRIMARY** needs accuracy HIGH and the integrity limit within the RNP. Before, a random estimate gave GPS PRIMARY.

What you should see

Situation	What you should see
IRs aligned, receivers in NAV	GPS PRIMARY (A32NX PROG page, A380X POSITION / MONITOR), accuracy HIGH.
Receivers still in ACQ, or IRs not in NAV	No GPIRS position: no GPS PRIMARY; the FMS uses radio or IRS positions.
GPS 1 failed	IR 1 and IR 3 switch to GPS 2: GPS PRIMARY stays.
Both GPS failed	GPS PRIMARY is lost: the FMS shows GPS PRIMARY LOST.

How to use

1. Align the IRs as usual. GPS PRIMARY comes once the IRs are in NAV and the receivers are in NAV.
2. Check GPS PRIMARY on the A32NX MCDU PROG page or the A380X MFD POSITION / MONITOR page before an RNAV procedure.

How to test

#	Do this	Expected result
1	Cold and dark, power the aircraft and start the IR alignment. Open the A32NX PROG page (or the A380X POSITION / MONITOR page).	No GPS PRIMARY until the IRs are aligned and the receivers in NAV.
2	Wait for the end of the alignment.	GPS PRIMARY appears (green on the A32NX PROG page).
3	flyPad > Failures > ATA 34 Navigation > **GPS 1**.	GPS PRIMARY stays (the IRs use GPS 2).
4	Also fail **GPS 2**.	GPS PRIMARY disappears; GPS PRIMARY LOST is shown. Clear both failures: GPS PRIMARY comes back once a receiver is in NAV again.

3. GPS pages: MCDU GPS MONITOR and MFD POSITION/GPS

> **Aircraft:** A32NX (MCDU), A380X (MFD) | **Where:** A32NX MCDU DATA > GPS MONITOR; A380X MFD FMS > POSITION > GPS | **Status:** Tested in the sim by the user: OK.

What it is for

Both pages showed the sim GPS solution for both receivers with random satellite counts and figures of merit. They now show the data of each receiver.

A32NX MCDU GPS MONITOR

Per receiver (A320 FCOM DSC-22_20-50-10-28): GPS1(2) POSITION, TTRK (true track, in degrees: the old page showed the magnetic track in radians), GS, UTC, MERIT (figure of merit, FT), MODE/SAT (mode and number of satellites), GPS ALT. The data is dashed while the receiver is not in NAV (or ALTAID / AIDED), and for a receiver that no ADIRU has selected ("the data of the GPS that is not selected is dashed").

A380X MFD POSITION/GPS

Per receiver (A380 FCOM DSC-22-FMS-20-30): position, MODE (TEST, INIT, ACQ, NAV, ALTAID, AIDED, DIFF, FAULT), NBR OF SAT (while acquiring and navigating), ACCURACY (HFOM, in FT as on the FCOM figure), TRK, UTC, ALT and GND SPD. Dashes while the receiver does not compute a position.

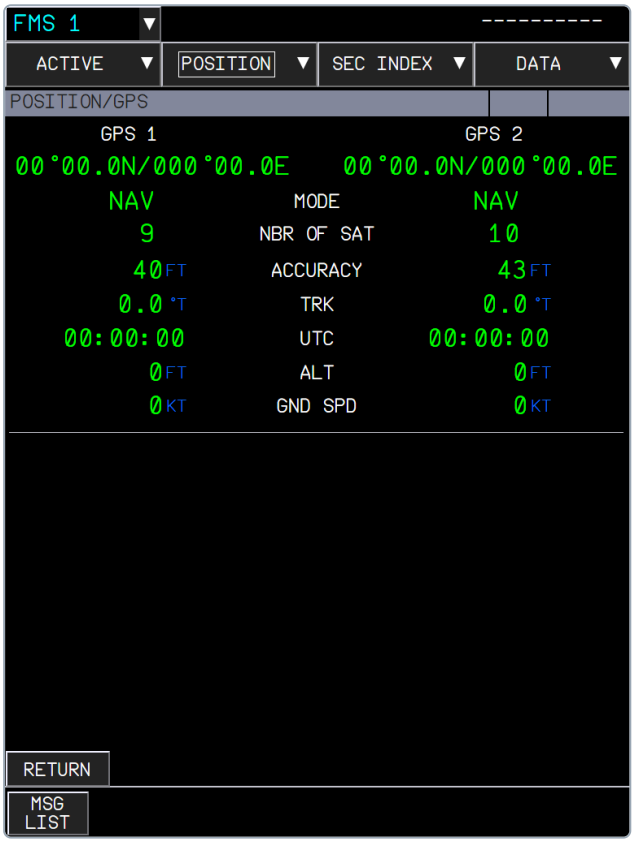


Figure 1. MFD POSITION/GPS: the page layout (picture from the browser harness, taken before the receivers existed: the values are the old stand-in values).

How to use

1. A32NX: MCDU **DATA** key > **GPS MONITOR**.
2. A380X: MFD **FMS** > **POSITION** menu > **GPS**.

How to test

#	Do this	Expected result
1	A32NX, powered, receivers in NAV: MCDU DATA > GPS MONITOR.	Two receivers with their own position, TTRK in degrees (same as the true track), GS, MERIT in FT and MODE/SAT such as NAV/9.

2	Fail GPS 1 (flyPad > Failures > ATA 34 Navigation > **GPS 1**).	GPS1 line: MODE/SAT shows FAULT, position, TTRK, GS, MERIT and GPS ALT dashed.
3	A380X, powered: MFD FMS > POSITION > GPS.	GPS 1 and GPS 2 columns with their own values; MODE NAV or DIFF.
4	Fail GPS 2.	GPS 2 column: MODE FAULT, data dashed, UTC --:--:--.

4. NAV GPS 1(2) FAULT alerts

> **Aircraft:** A32NX and A380X | **Where:** E/WD; SD STATUS page | **Status:** Tested in the sim by the user: OK.

What it is for

There was no GPS alert. A failed receiver (mode FAULT) now raises the FCOM alert:

- **A32NX: NAV GPS 1 FAULT** / **NAV GPS 2 FAULT** (FCOM PRO-ABN-NAV: "GPS 1(2) is failed"), amber caution, no procedure line. Flight phase inhibition as the radio altimeter faults (design choice). STATUS: INOP SYS GPS 1(2); with both: INOP SYS GPS 1+2 and the information FLS LIMITED TO F-APP + RAW (see "A32NX ECAM STATUS page").
- **A380X: NAV GPS 1 FAULT**, **NAV GPS 2 FAULT** or **NAV GPS 1+2 FAULT**, crew awareness (the A380 FCOM gives no procedure), inhibited as the ADR faults (design choice). STATUS (FCOM PRO-ABN-ECAM-10-34-10): INOP SYS GPS 1, GPS 2 or GPS 1+2; with both also BTV and ROW/ROP in APPR & LDG.

How to use

1. Read the alert on the E/WD and clear it with CLR.
2. Check the inoperative systems on the SD STATUS page (STS key).

How to test

#	Do this	Expected result
1	A32NX on the ground, engines off or at idle (not during the takeoff run). flyPad > Failures > ATA 34 Navigation > **GPS 1**.	E/WD: **NAV GPS 1 FAULT** in amber, MASTER CAUT.
2	Press CLR, then open the SD STATUS page.	INOP SYS: GPS 1.
3	Also fail **GPS 2**.	NAV GPS 2 FAULT; STATUS: INOP SYS GPS 1+2, and FLS LIMITED TO F-APP + RAW in green.
4	A380X on the ground: fail **GPS 1**.	E/WD: **NAV GPS 1 FAULT**. SD STATUS: INOP SYS GPS 1.
5	Also fail **GPS 2**.	The alert becomes **NAV GPS 1+2 FAULT**. STATUS: INOP SYS GPS 1+2; APPR & LDG: BTV, ROW/ROP.

5. GPS deselection and position uncertainty

> **Aircraft:** A32NX and A380X | **Where:** A32NX MCDU SELECTED NAVAIDS; A380X MFD POSITION > NAVAIDS | **Status:** Tested in the sim by the user: OK on both aircraft.

What it is for

The crew can tell the FMS not to use the GPS (for example when told GPS is unreliable in an area). The A32NX DESELECT key was inoperative and the A380X had no GPS option. Now:

- **A32NX** SELECTED NAVAIDS page, key [6L] (A320 FCOM DSC-22_20-50-10-28): GPS SELECTED / DESELECTED in blue small font, the action ({DESELECT or {SELECT) in white.

- **A380X** POSITION / NAVAIDS page: GPS SELECTED / DESELECTED radio buttons (A380 FCOM DSC-22-FMS-20-30).
- With the GPS deselected the GPIRS position is no longer used: the FMS goes to IRS/DME/DME, IRS/VOR/DME or IRS only, and GPS PRIMARY is lost.
- The deselection is reset at the end of the flight (DONE phase).
- **GPS IS DESELECTED** (A32NX, white MCDU message) / **GPS DESELECTED** (A380X, white MFD message) when the GPS is deselected and the aircraft is less than 80 NM from the top of descent, or in the approach phase. The message is removed when the GPS is selected again.
- **Position uncertainty without GPS** (A320 FCOM DSC-22_20-20-20 table): IRS/DME/DME 0.27 to 0.37 NM, IRS/VOR/DME 0.30 to 0.42 NM (growing with the distance to the navaids, up to 100 NM: design choice), IRS only growing from its starting value at 6 NM/h for 40 min, 0 for 50 min, 4 NM/h for 40 min, 0 for 45 min, then 2 NM/h; 5 NM after an IRS alignment.

Limitations and why

Limitation	Why	What to do
The 0.2 NM reset of the uncertainty at takeoff is not done.	Not part of this change.	None.

How to use

1. A32NX: MCDU **DATA** > **POSITION MONITOR** > **NAVAIDS>** [6R] > press [6L] **{DESELECT**. Press [6L] **{SELECT** to use the GPS again.
2. A380X: MFD **FMS** > **POSITION** > **NAVAIDS**, GPS: select **DESELECTED** (or **SELECTED**).

How to test

#	Do this	Expected result
1	A32NX, IRs aligned, GPS PRIMARY on the PROG page. MCDU DATA > POSITION MONITOR > NAVAIDS> [6R].	Line 6 left: GPS SELECTED in blue small font, {DESELECT in white.
2	Press [6L].	GPS DESELECTED, {SELECT. The PROG page loses GPS PRIMARY; the EPU becomes a radio or IRS only value (for example 0.27 to 0.42 NM near DMEs).
3	Type something in the scratchpad, then press [6L].	NOT ALLOWED; the selection does not change.
4	In flight with a flight plan, GPS deselected, fly to less than 80 NM from the T/D.	White MCDU message **GPS IS DESELECTED**.
5	Press [6L] {SELECT.	The message is removed; GPS PRIMARY comes back.
6	A380X: MFD FMS > POSITION > NAVAIDS, GPS: DESELECTED.	POSITION / MONITOR no longer shows GPS PRIMARY. Near the T/D (80 NM) or in approach: MFD message **GPS DESELECTED**.
7	Finish the flight (DONE phase).	The GPS is selected again.

6. A32NX ECAM STATUS page and STS reminder

> **Aircraft:** A32NX | **Where:** SD STATUS page (STS key on the ECAM control panel); bottom of the E/WD | **Status:** Tested in the sim by the user: OK. On this branch the STS reminder logic was adapted to the CLR logic of FlyByWire master (the fork version sits on a fork-only commit): that adaptation is not sim-tested.

What it is for

The A32NX STATUS page was a frame that always showed NORMAL. The FWC now fills it (A320 FCOM DSC-31-20 and DSC-31-25-20):

- Left column: the information lines in green (the limitations in blue are added by `core/a32nx-air-conditioning`). Right column: **INOP SYS** (white, underlined) and the inoperative systems in amber. An empty page shows NORMAL in green.
- The page appears by itself when it is not empty and the crew clears the last alert of the E/WD with CLR, or selects CONF 1 for the approach (flaps lever 0 to 1 in flight). The next CLR (or STS) push removes it.
- **STS reminder:** STS in white, framed, at the bottom of the E/WD, while the STATUS page is not empty and not shown and no alert is shown.
- On this branch only the GPS alerts fill it (INOP SYS GPS 1, GPS 2, GPS 1+2; FLS LIMITED TO F-APP + RAW). Other branches add their alerts.

How to use

1. After clearing the alerts with CLR, read the STATUS page that comes up, then press CLR to remove it.
2. When STS shows at the bottom of the E/WD, press **STS** to review the status.

How to test

#	Do this	Expected result
1	A32NX powered, no failure. Press STS.	STATUS page: NORMAL in green.
2	Fail GPS 1 (flyPad > Failures > ATA 34 Navigation > **GPS 1**).	NAV GPS 1 FAULT on the E/WD.
3	Press CLR.	The SD shows the STATUS page by itself: INOP SYS (underlined) and GPS 1 in amber.
4	Press CLR again.	The STATUS page goes; **STS**, white and framed, shows at the bottom of the E/WD.
5	Press STS.	The STATUS page shows; the STS reminder goes off while it is shown.
6	Clear the failure.	STATUS back to NORMAL; no STS reminder.
7	In flight with GPS 1 failed and its alert cleared, select flaps 1 for the approach.	The STATUS page comes up by itself.

7. FM position: IRS position plus a GPS or radio bias, POSITION MONITOR pages

> **Aircraft:** A32NX and A380X | **Where:** A32NX MCDU DATA > POSITION MONITOR, PROG, IRS MONITOR; A380X MFD FMS > POSITION > MONITOR; ND | **Status:** Not sim-tested (added; unit tests: FmPosition.spec 27, PositionUncertainty +3, PositionMonitorFields 7).

What it is for

The FMS position was the sim position. The FCOM FM position is now modelled (shared FMS code; A320 FCOM DSC-22_20-20 navigation, A380 FCOM FMS position computation):

- **FM position = MIX IRS position + a bias.** The bias follows the GPIRS position (on the ground and in flight) or the radio position (in flight only). In IRS only navigation the last bias is kept.
- Each FMS uses the GPIRS of its own side first, then ADIRU 3, then the other side.
- Without GPS PRIMARY, the position is updated at takeoff to the runway start plus the TO SHIFT.

- A manual position update (A32NX PROG **UPDATE AT**) moves the FM position; the EPU (estimated position uncertainty) follows the updates.
- The FMS writes `L:A32NX_FM_GPS_POS_DISAGREE`: the NAV FM/GPS POS DISAGREE alerts read it (their FWC and FWS parts are on `core/failures-a32nx` and `core/a380x-elec-fuel-alerts`).
- **A32NX MCDU:** POSITION MONITOR (and FROZEN) shows the real FMGC 1 and 2, GPIRS or radio, MIX IRS and IRS positions with the navigation mode and the deviations; PROG UPDATE AT works; the IRS page shows the GPIRS position and accuracy of its ADIRU.
- **A380X MFD POSITION / MONITOR:** the onside and offside FMS positions, the navigation mode, RADIO, MIXIRS, GPIRS, GPS 1 and GPS 2 from the receiver data.

Situation	What you should see
GPS PRIMARY (normal)	The FM position follows the GPIRS position; it carries the small GPS receiver error, so it differs a little from the sim position.
GPS deselected or failed, in flight, radio navaids in range	The bias follows the radio position (IRS/DME/DME or IRS/VOR/DME).
IRS only	The last bias is kept; the position drifts only as the IRS position does (see the limits).
Takeoff without GPS PRIMARY	The FM position jumps to the runway start plus the TO SHIFT at the takeoff thrust.

Limitations and why

Limitation	Why	What to do
No IRS drift and no radio position reasonableness check.	The sim ADIRS writes the sim position: there is no drift to correct.	-
No A380X POSITION UPDATE panel, no FM/IR POS DISAGREE.	Left for later.	-
No offsets, SEC / ALTN predictions, FLS beam or A32NX TRIP/EFOB table on these pages.	Left for later.	-

How to use

1. A32NX: MCDU DATA > POSITION MONITOR to compare the FMGC, GPIRS / radio, MIX IRS and IRS positions; FREEZE to hold them.
2. A32NX: for a manual update, PROG > UPDATE AT, enter the position (or a navaid ident) and confirm.
3. A380X: MFD FMS > POSITION > MONITOR.

How to test

#	Do this	Expected result
1	A32NX on the ground, IRS aligned, GPS PRIMARY. MCDU DATA > POSITION MONITOR.	FMGC1, FMGC2, GPIRS, MIX IRS and IRS 1-3 lines with real coordinates and the navigation mode; the deviations are small.
2	A380X: MFD FMS > POSITION > MONITOR.	FMS 1 and FMS 2 positions, the navigation mode, RADIO, MIXIRS, GPIRS, GPS 1 and GPS 2.
3	Both: deselect the GPS (A32NX SELECTED NAVAIDS, A380X POSITION / NAVAIDS) and take off.	At takeoff the FM position is updated to the runway start (plus the TO SHIFT); the ND symbol stays on the runway.

4	A32NX in flight: PROG > UPDATE AT a navaid ident, confirm.	The FM position moves to the update; the EPU changes.

What PR testers should check

- The FMS position now carries the GPS receiver error: watch LNAV and the ND for jumps, also when the GPS powers up.
- The takeoff update with the GPS deselected.
- The POSITION MONITOR pages of both aircraft.
- Not sim-tested yet.

8. FMS messages from the FCOM lists, and CHECK APPR SELECTION

> **Aircraft:** A32NX and A380X | **Where:** A32NX MCDU scratchpad messages; A380X MFD FMS message list | **Status:** Not sim-tested (added; unit tests: FmsMessageConditions.spec and CheckApproachSelectionLogic.spec, failing first).

What it is for

Messages of the A320 MCDU message list (FCOM DSC-22_20-90) and of the A380 MFD FMS message list that the FMS never raised:

Message	Aircraft	When (FCOM)
AREA RNP IS xx.xx	both	The RNP entered is larger than the default RNP of the flight area, and the database has no RNP for the active leg.
PROCEDURE RNP IS xx.xx (A380X: PROC RNP IS)	both	The RNP entered is larger than the database RNP of the active leg.
CHECK DATA BASE CYCLE (A380X: CHECK DATABASE CYCLE)	both	Today's date is outside the effectivity of the active navigation database (an unknown effectivity raises nothing).
CHECK IRS/AIRPORT POS (A380X: CHECK IRS / ARPT POSITION)	both	An IRS alignment position is 5 NM or more from the airport reference point.
IRS ONLY NAVIGATION	A32NX	In flight, en route, in IRS only navigation for more than 10 min, or at the start of the descent.
CHECK DEST DATA	A32NX	The destination data were received by uplink (wind not entered by the crew).
CRZ FL ABOVE MAX FL	A32NX	The crew enters a cruise level above the computed maximum level.
TIME TO EXIT	both	In the holding, the extra fuel is zero or less.
EXPECT TURN AREA EXCEEDANCE	A380X	The active or next leg is a procedure turn that exceeds the excursion distance of the database.
SPECIF NDB NOT AVAIL	A380X	The NDB of the NDB approach is deselected.
T.O TIME REACHED	A380X	The clock reaches the ETT; shown until the takeoff or a new ETT.
ADJUST DESIRED TRK OR HDG	A380X	A DIR TO with CRS IN or CRS OUT is inserted.
CHECK APPR SELECTION	A32NX	A non ILS approach in the F-PLN and an ILS tuned manually on the RAD NAV page, in cruise less than 100 NM from the T/D, in descent or in approach. Amber; it also sets the variable the PFD FMA line 3 shows.

The messages are type II: sent once when the condition starts and removed when it ends. A message with a value (AREA RNP IS 2.00) is replaced when the value changes.

Limitations and why

Limitation	Why	What to do
Other messages of the FCOM lists are still missing.	They need data or functions the FMS does not have yet.	-
CHECK APPR SELECTION on the PFD FMA needs `core/failures-a32nx`.	The PFD FMA part of the change is on that branch; here only the MCDU message and the variable.	Merge both.
The A380X MFD System Message Area (CHECK FMS MESSAGE) is on `core/a380x-atccom-datalink`.	It also shows the ATIS of the ATC COM datalink of that branch.	-

How to use

1. Read the message in the MCDU scratchpad (A32NX; CLR removes it) or in the MFD FMS message list (A380X).
2. Act as the FCOM says for each message (check the RNP, the database cycle, the IRS position, the approach selection...).

How to test

#	Do this	Expected result
1	A32NX with a current navigation database, cold and dark to cruise.	No CHECK DATA BASE CYCLE at any time.
2	A32NX in cruise less than 100 NM from the T/D, F-PLN with an RNAV (non ILS) approach. RAD NAV: type an ILS frequency in LS.	MCDU: **CHECK APPR SELECTION** in amber.
3	A32NX in flight, en route: PROG page, type 2.0 in the REQUIRED (RNP) field.	**AREA RNP IS 2.00** (or PROCEDURE RNP IS on a procedure leg with its own RNP).
4	A32NX: enter a CRZ FL above the MAX REC level on PROG.	**CRZ FL ABOVE MAX FL**.
5	A380X: F-PLN > DIR TO with CRS IN, INSERT.	FMS message **ADJUST DESIRED TRK OR HDG**.
6	A380X on the ground: F-PLN, vertical revision of the origin, enter an ETT a minute ahead and wait.	**T.O TIME REACHED** at that time, gone at the takeoff.

What PR testers should check

- CHECK DATA BASE CYCLE stays silent with a current database (the main risk of a wrong date check).
- Any of the new messages showing when the FCOM condition is not met.
- Not sim-tested yet.

A380X interactive ND

Revise the flight plan on the navigation display with the KCCU cursor, as in the A380 FCOM: DIR TO, INSERT NEXT WPT, DELETE and DATA from a click on the map, with PLACE/DIST entries

Aircraft	A380X
Where to find it	Captain and F/O ND (ARC, PLAN, ROSE-NAV) with the KCCU cursor; MFD FMS > ACTIVE > F-PLN > DIRECT TO
Branch	`core/a380x-interactive-nd`, based on `core/a380x-mfd-fms` (stacked: needs core/surveillance and core/a380x-mfd-fms merged first).
Status	Tested in the sim by the user: OK (before the commit). Unit tests: the DIR TO UTC estimate. PLACE/DIST entries on the ND (added): not sim-tested.
Pictures	No pictures: the ND pages follow the A380 FCOM figures (DSC-31-20-30-90).

1. Interactive ND

Aircraft: A380X | **Where:** Lateral ND, KCCU cursor; MFD DIRECT TO page (UTC) | **Status:** Tested in the sim by the user: OK.

What it is for

A380 FCOM DSC-31-20-30-90: with the KCCU cursor on the lateral ND in ARC, PLAN or ROSE-NAV mode, the flight crew revises the flight plan directly on the ND, without going to the MFD. The FBW ND was not interactive.

What you see

- **Interactive areas:** a blue box around every waypoint, NAVAID or airport under the cursor. A click gives the list of the elements there (P.POS for the aircraft mock-up), then the revision list of the chosen one.
- **Revision list:** the ident (green in the active flight plan, yellow in the temporary one), then: for a flight plan waypoint FROM P.POS DIR TO *, INSERT NEXT WPT, DELETE *; for a database element FROM P.POS DIR TO *; for an airport also DATA; for the aircraft mock-up DIR TO.
- **DIRECT TO page** at the bottom of the ND: the DIRECT TO target field (with a list of the flight plan waypoints, or a KCCU entry), UTC and DIST, the options DIRECT, DIRECT WITH ABEAM, CRS IN, CRS OUT, and the CRS field next to CRS IN / CRS OUT (default course from the flight plan, typed NNN, with M or T for magnetic or true). CANCEL closes the page when there is no temporary flight plan.
- **INSERT DIR TO * / ERASE DIR TO *** buttons either side of the aircraft while the cursor is in the ND; **INSERT TMPY * / ERASE TMPY *** for the other revisions.
- **INSERT NEXT WPT page:** FROM and the revised waypoint (green, or yellow with IN TMPY), its latitude / longitude, and the NEXT WPT field. A click on a blank area of the ND creates a latitude/longitude waypoint.
- **DUPLICATE page** (DIST(NM), LAT/LONG, RETURN, scroll arrows) when a typed ident is not unique, nearest first.
- **DATA** on an airport opens the DATA / AIRPORT page on the MFD of the same side.
- **DIR TO UTC:** the FMS has no predictions of the temporary flight plan, so the UTC at the target is estimated from the distance and the ground speed, on the ND and on the MFD DIRECT TO page. Below 30 kt ground speed it shows --:--.

Limitations and why

Limitation	Why	What to do

No revision on a DIR TO temporary flight plan; no DIRECT TO while the ND shows another temporary flight plan.	FCOM DSC-31-20-30-90 P 1 and P 9.	Insert or erase the temporary flight plan first.
The vertical display is not interactive.	Only the lateral ND is interactive in the FCOM.	None.
Pilot-defined entries (lat/long, typed places) stay as stored waypoints after ERASE.	They are stored waypoints, as on the FBW MFD.	Delete them on the MFD DATA pages if needed.
The UTC is an estimate (distance / ground speed).	The FMS computes no predictions for a temporary flight plan.	Read the UTC on the F-PLN page once the DIR TO is inserted.
Only the interactions listed above work.	The other ND interactions of the FCOM are not implemented yet (the A380X README says so).	Use the MFD pages for the other revisions.

How to use

1. Set the ND to ARC, PLAN or ROSE-NAV (map, not the OANS) and move the KCCU cursor onto the ND.
2. Click a waypoint, NAVAID or airport (or the aircraft symbol); pick it in the list if several are there.
3. Choose the revision: FROM P.POS DIR TO *, INSERT NEXT WPT, DELETE * or DATA.
4. Complete the page at the bottom of the ND (target, option, course, next waypoint) by clicks or KCCU entries.
5. Click **INSERT** (DIR TO * or TMPY *) to activate, or **ERASE** to discard.

How to test

#	Do this	Expected result
1	A380X in flight (or on the ground) with an active flight plan. Captain ND in ARC, range 20 or 40 NM. Move the KCCU cursor over a flight plan waypoint.	A blue box appears around the waypoint.
2	Click it.	Revision list: the ident in green, FROM P.POS DIR TO *, INSERT NEXT WPT, DELETE *.
3	Click FROM P.POS DIR TO *.	DIRECT TO page at the bottom of the ND: the waypoint as target, a yellow temporary flight plan, DIST in NM, UTC (estimated, in flight), DIRECT option selected; INSERT DIR TO * and ERASE DIR TO * either side of the aircraft.
4	Select CRS IN.	A CRS field appears next to it with the default course of the flight plan.
5	Click the CRS field, type a course with the KCCU (for example 090) and ENT.	The course shows; the temporary DIR TO follows it.
6	Click INSERT DIR TO *.	The DIR TO becomes active (green); the page closes.
7	Open the MFD FMS > ACTIVE > F-PLN > DIRECT TO page and select a waypoint.	The UTC shows an estimate (distance / ground speed) instead of dashes.
8	Click the aircraft symbol on the ND, then DIR TO.	DIRECT TO page with an empty target. Open the list (▼) and pick a waypoint, or click the target field and type an ident with the KCCU + ENT.
9	Type an ident that exists several times (a common NAVAID ident).	DUPLICATE page with DIST(NM) and LAT/LONG; pick one, or RETURN.

10	Click ERASE DIR TO *.	The temporary flight plan and the page go.
11	Click a flight plan waypoint > INSERT NEXT WPT, then click a blank area of the map.	INSERT NEXT WPT page with FROM and the waypoint; a latitude/longitude waypoint is created and inserted after it in a yellow temporary flight plan; INSERT TMPY * / ERASE TMPY * buttons.
12	Click INSERT TMPY *.	The new waypoint is in the active flight plan.
13	Click a flight plan waypoint > DELETE *.	Temporary flight plan without that waypoint; INSERT TMPY * keeps the change, ERASE TMPY * discards it.
14	Click an airport > DATA.	The Captain MFD opens the DATA / AIRPORT page of that airport.
15	Repeat one revision on the F/O ND with the F/O KCCU.	Same behaviour; DATA opens on the F/O MFD.

2. PLACE/DIST entries in the ND waypoint fields

> **Aircraft:** A380X | **Where:** ND INSERT NEXT WPT page, NEXT WPT field (KCCU entry) | **Status:** Not sim-tested (added; no test of its own: the PLACE/DIST code is tested on `core/a380x-mfd-fms`).

What it is for

The FMS now accepts PLACE/DIST waypoints (a PDnn waypoint at a distance along the flight plan from a flight plan waypoint, for example `<WPT>/10`): see the PLACE/DIST waypoints feature in the `core/a380x-mfd-fms` manual. The waypoint fields of the interactive ND go through the same shared entry code, which now needs the FMS function that creates the PLACE/DIST waypoint; the ND pages pass it on. Without this, a PLACE/DIST entry typed on the ND would not create the waypoint.

Limitations and why

Limitation	Why	What to do
Only the place of the revised waypoint is accepted (A380 FCOM, INSERT NEXT WPT).	Same rule as on the MFD: another place gives PLACE / WPT DISAGREE, a distance beyond the neighbouring waypoint ENTRY OUT OF RANGE.	-

How to use

1. Click a flight plan waypoint on the ND > INSERT NEXT WPT.
2. Click the NEXT WPT field and type `<that waypoint>/DIST` with the KCCU, then ENT.
3. Check the yellow temporary flight plan and click INSERT TMPY *.

How to test

#	Do this	Expected result
1	A380X in flight with an active flight plan. ND: click a waypoint ahead > INSERT NEXT WPT. Type `<that waypoint>/10` in NEXT WPT + ENT.	A PDnn waypoint 10 NM after it in a yellow temporary flight plan.
2	INSERT TMPY *.	PDnn is in the active flight plan (also on the MFD F-PLN page).

What PR testers should check

- PLACE/DIST entries from the ND fields give the same waypoint as from the MFD.

- Not sim-tested yet.

flyPad performance calculators

Takeoff (A380 new, A320 with MCDU uplink), landing and descent calculators on the flyPad, the company datalink reply time of the A380X FMS, and the later additions: A380 DERATED takeoff, BTV braking mode, 40 kt landing crosswind, descent target waypoint and T/D slider

Aircraft	A380X and A32NX (one shared flyPad screen per calculator; the company datalink reply time is A380X only)
Where to find it	flyPad > **Performance** (Takeoff, Landing, Top of Descent tabs); flyPad > Settings > Realism; A380X MFD PERF T.O / INIT / WIND; A32NX MCDU PERF TAKE OFF
Branch	`core/efb-performance`, based on `core/a380x-mfd-fms` (stacked: needs `core/surveillance` and `core/a380x-mfd-fms` merged first). `core/efb-ground` is stacked on it (the taxi page uses its runway exits).
Status	Landing calculator (both aircraft) with the BTV braking mode and the 40 kt crosswind, descent calculator (A32NX, A380X), descent target waypoint and the fix for an FMS without waypoints: tested in the sim by the user, OK. A380 takeoff calculator: used in the sim with the company T.O flow, TOW in tonnes and CG envelope edges OK; its DERATED takeoff (D04-D24): tested in the sim by the user, OK (flyPad to MFD and INSERT, E/WD level and N1, FMA MAN DTO, ENG THR LEVERS NOT SET at TOGA). Company datalink reply time: tested in the sim; the company wind part (reply time kept after the WIND page merge, one request per flight plan, NO COMPANY REPLY after 4 min) is covered by the WIND page sim test (PASS). A320 takeoff calculator: flyPad part used in the sim, MCDU uplink pages checked in the browser harness only. T/D slider: no separate sim record. All calculators have unit tests (back-filled and mutation-tested).
Pictures	flyPad, MFD and MCDU pages in the local browser harness, running the build installed in the sim (from the feature manuals of September 2026). The harness has no navigation database and no FMS: runways and inputs are typed, uplinks simulated. The DERATED takeoff pictures come from the same harness (current flyPad look). The other later additions (BTV mode, 40 kt, target waypoint) have no pictures.

1. A380 takeoff calculator and company T.O data
2. A380 takeoff calculator: DERATED takeoff
3. A320 takeoff calculator and MCDU uplink takeoff data
4. A380X company datalink reply time
5. Landing calculator
6. A380 landing calculator: BTV braking mode
7. A380 landing calculator: 40 kt crosswind on dry and wet runways
8. Descent calculator
9. Descent calculator: T/D distance slider
10. Descent calculator: target waypoint from the flight plan
11. Descent calculator with an FMS that sends no waypoints

1. A380 takeoff calculator and company T.O data

> **Aircraft:** A380X (the flyPad screen is shared with the A32NX, next feature) | **Where:** flyPad > Performance > Takeoff; MFD PERF T.O | **Status:** Used in the sim by the user with the company T.O data flow; TOW in tonnes and the CG envelope edges tested OK. Unit tests against the Airbus charts and the FCOM.

What it is for

Before each takeoff the crew needs the takeoff weight limit of the runway, the FLEX temperature (reduced thrust), the speeds V1, VR, V2 and the flaps. On the real A380 they come from the OIS takeoff application or from the airline (company takeoff data, sent to the FMS by datalink). The flyPad calculator does the computation; SEND TO FMS plays the airline: the MFD receives the data as company T.O data, and the crew checks and inserts it, as in FCOM DSC-22-FMS-20-30.

The flyPad screen

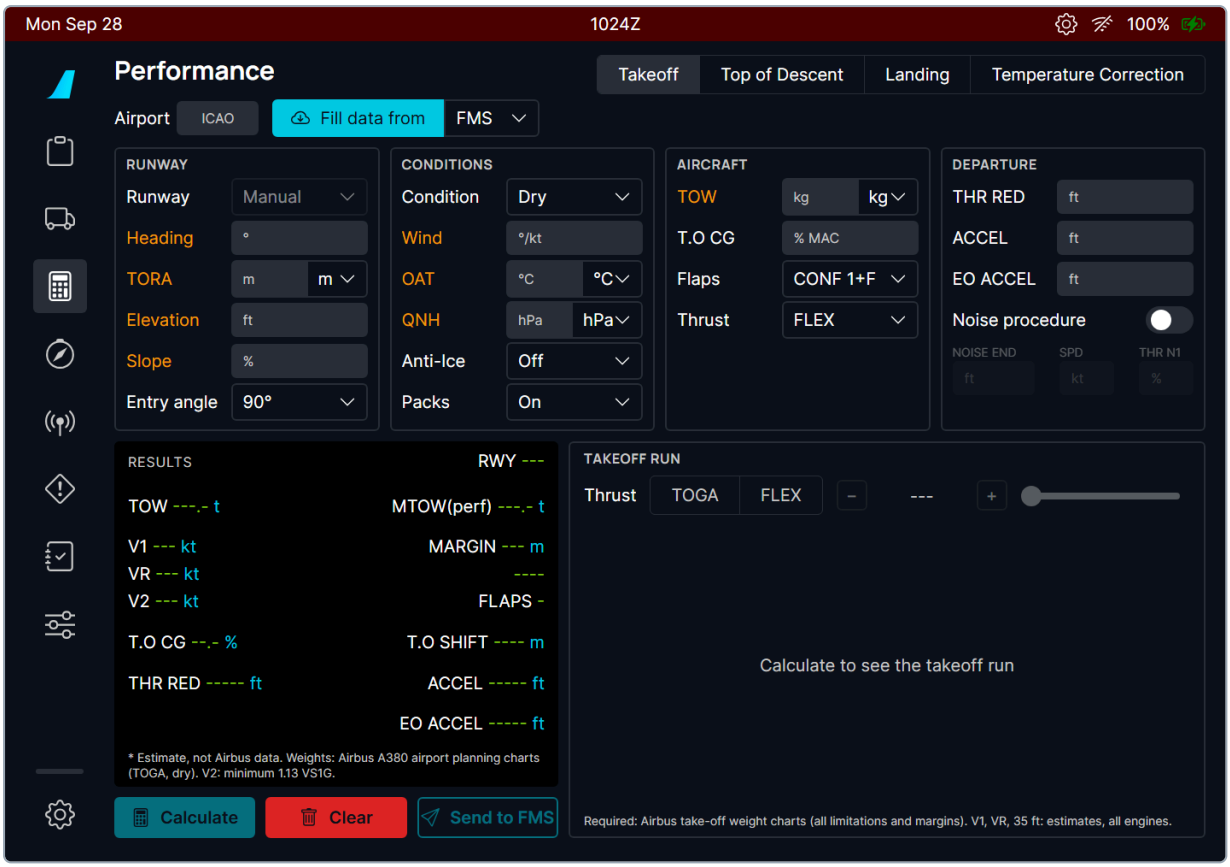


Figure 1. The takeoff screen. Amber labels: inputs still missing (Calculate stays dim).

Part	Contents
Airport, Fill data from	ICAO code; imports from the **FMS** (departure, runway, TOW, CG, THR RED / ACCEL), the **OFP** (SimBrief) or the **METAR** (wind, OAT, QNH).
RUNWAY	Runway from the list (or Manual), heading, TORA, elevation, slope, and the **entry angle** (90° or 180° turn onto the runway, or 0° when rolling straight on: it sets the line-up distance lost).
CONDITIONS	Runway condition, wind, OAT, QNH, anti-ice, packs.
AIRCRAFT	TOW in t (or klb), one decimal, T.O CG in % MAC, flaps (CONF 1+F, 2, 3 or OPT = the best one), thrust (TOGA, FLEX or DERATED: see "A380 takeoff calculator: DERATED takeoff" below).
DEPARTURE	THR RED, ACCEL, EO ACCEL altitudes (they default to the elevation + 1500 ft), and the noise abatement procedure (NOISE END altitude, speed, N1).
RESULTS	The results in the layout of the OIS takeoff application (FCOM PER-TOF-TOR-SRS).
TAKEOFF RUN	The runway with the V1, VR and 35 ft points, the required distance and the margin, for TOGA, any FLEX temperature of the allowed range, or a DERATED level the TOW permits.

A FLEX takeoff

1. Fill the airport and runway (or **Fill data from** FMS / OFP), the conditions (METAR), the TOW and the T.O CG.
2. Keep **Thrust FLEX** and press **Calculate**.
3. Read the FLEX temperature and the speeds in RESULTS; check the takeoff run and its margin.
4. **Send to FMS**, then insert the data on the MFD (see "On the MFD: company T.O data" below).

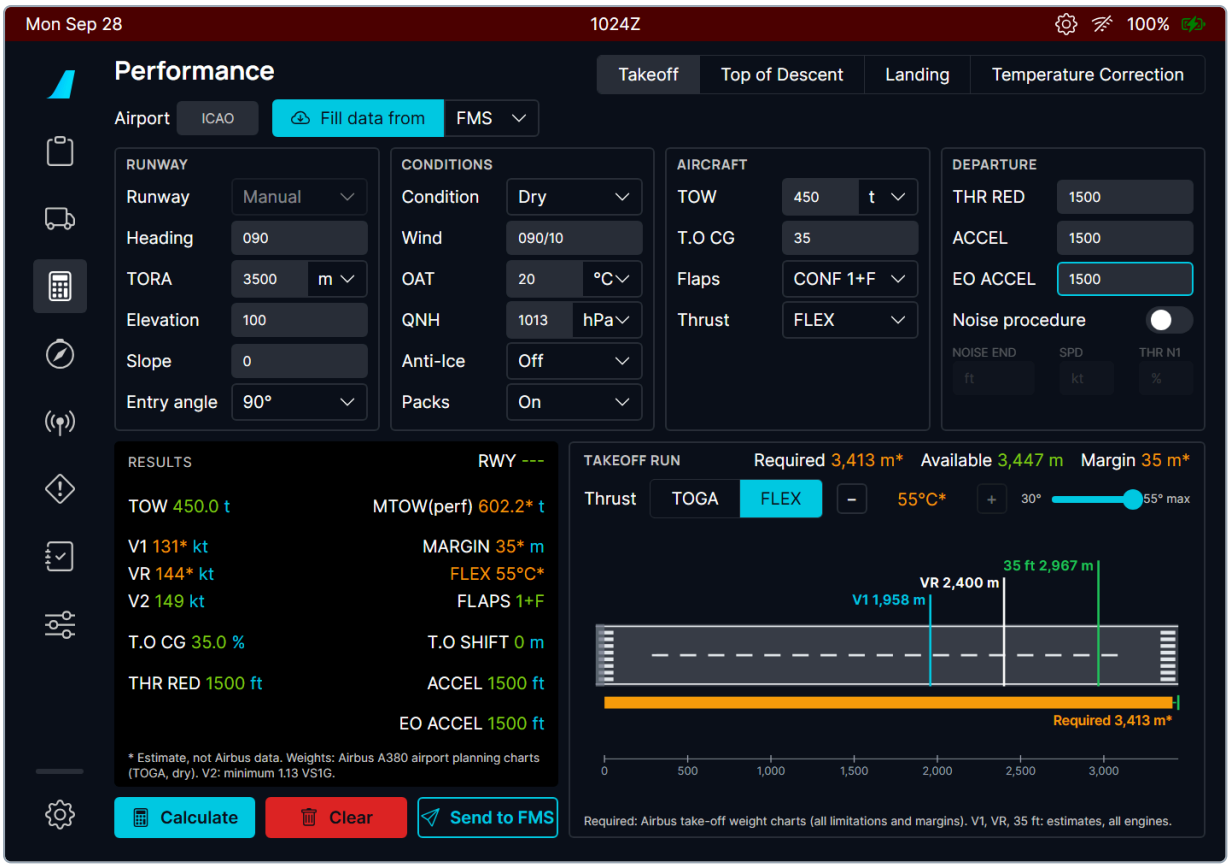


Figure 2. 450 t, 3500 m, 20 °C, CONF 1+F: FLEX 55 °C (the maximum, amber: estimate), V1 131 / VR 144 / V2 149 kt, MTOW(perf) 602 t. The takeoff run needs 3413 m of the 3447 m available after line-up: 35 m margin.

Reading the results

Line	Meaning
TOW / MTOW(perf)	Your takeoff weight, and the maximum the runway and conditions allow at TOGA.
V1, VR, V2	The takeoff speeds. V2 is at least 1.13 VS1G and 1.10 VMCA, VR at least 1.05 VMCA, V1 at least VMCG (FCOM PER-TOF-TOR-SRS).
MARGIN	Runway left beyond the required distance.
FLEX / TOGA	The FLEX temperature, or TOGA.
FLAPS, T.O CG, T.O SHIFT	The configuration, the CG entered, the takeoff shift (distance lost at line-up or an intersection).
THR RED, ACCEL, EO ACCEL	The altitudes of the DEPARTURE section.

> **NOTE**
> **Green** values come from Airbus data (the takeoff weight charts of the A380 Aircraft Characteristics); **amber with a star** are estimates of the FBW model (see Limitations).

TOGA or another FLEX temperature

The TAKEOFF RUN panel lets you look at the run for TOGA, or for any FLEX temperature between the OAT and the maximum (FCOM PER-TOF-THR-FLX). Use **-** / **+** or the slider; the speeds and the run follow.

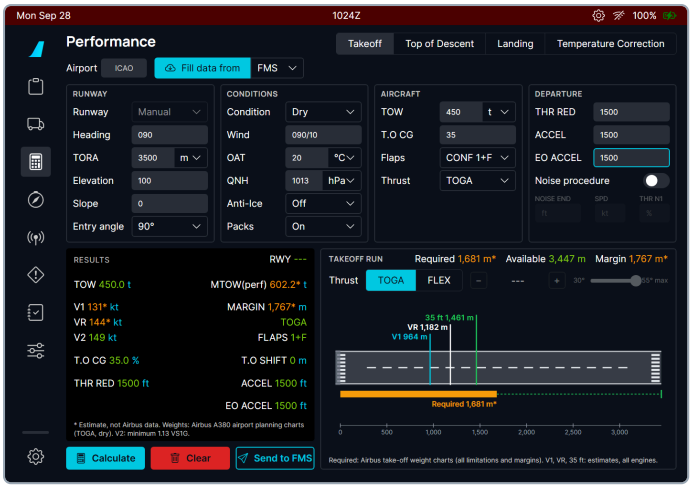


Figure 3. TOGA: the run needs 1681 m, margin 1767 m.

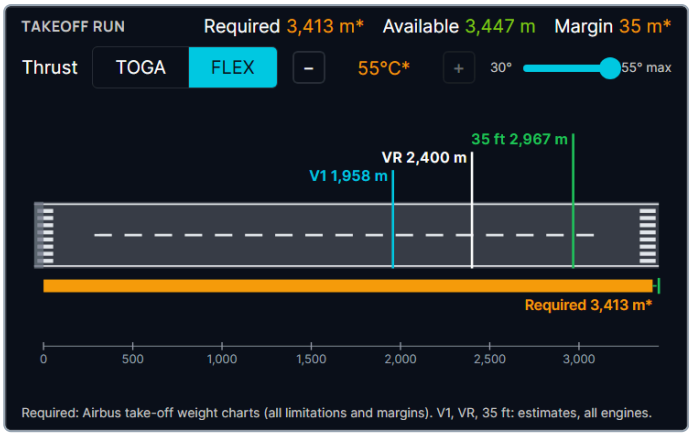


Figure 4. FLEX range 30 °C to 55 °C max: the selected temperature is the one sent to the FMS.

Send to FMS sends the thrust selected for the takeoff run only (TOGA, FLEX at its temperature or DERATED at its level), with its own speeds.

Options

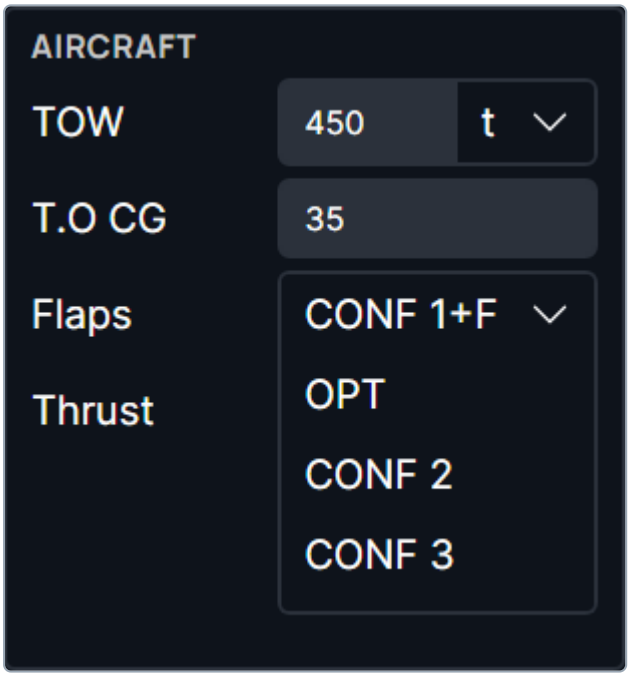


Figure 5. Flaps: CONF 1+F, 2, 3, or OPT (the calculator picks the best).

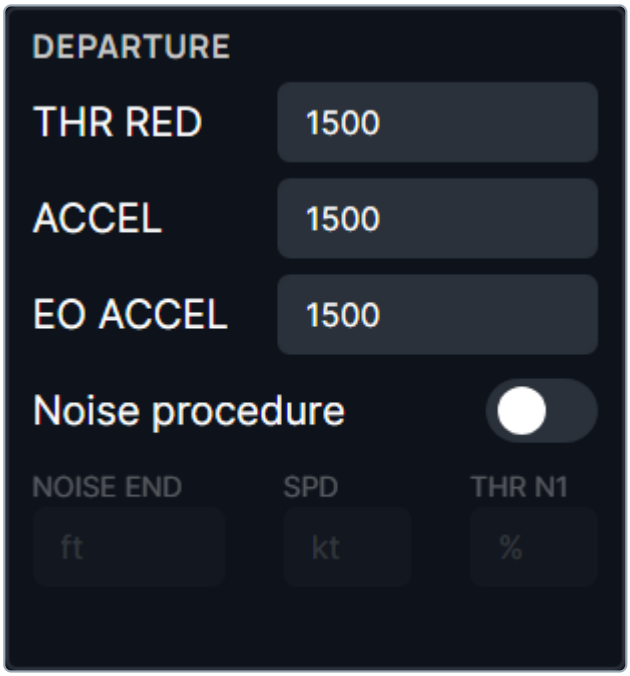


Figure 6. Noise procedure: NOISE END altitude, speed and N1 become available; they are sent with the data.

Errors

When the takeoff is not possible, a message says why and there is no result:

Figure 7. CG outside the takeoff envelope (here 30 % at 450 t: the forward limit moves aft with the weight).

Figure 8. 510 t on 2100 m at 40 °C: the maximum takeoff weight is 438.8 t.

Figure 9. A wet or contaminated runway: the A380 data covers dry runways only.

Message	Meaning
Beyond Takeoff Weight Limit - Max ...	TOW above the runway limit: reduce the weight or use another runway.
CG Outside Takeoff Envelope	The CG is outside the A380 takeoff CG envelope for this weight.
Above Structural MTOW	TOW above 510 t.
Only dry runways are covered...	The A380 data has no wet or contaminated takeoff.
Runway shorter than the A380 performance data (1700 m)	The charts start at 1700 m.
Above Maximum Tailwind / OAT / Pressure Altitude	Outside the A380 limits (FCOM LIM-12).
Flex Not Possible	No FLEX temperature is possible at this weight: take off at TOGA.

On the MFD: company T.O data

The MFD PERF T.O page gets two buttons: **CPNY T.O REQUEST** (ask the airline) and, once data has arrived, **RECEIVED CPNY T.O**.

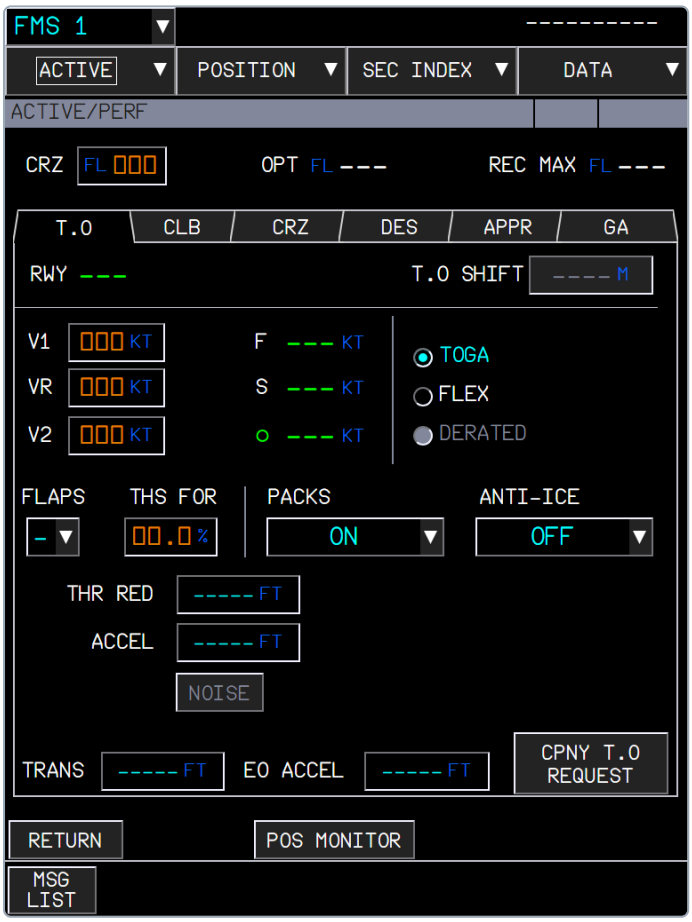


Figure 10. PERF T.O before any data: the CPNY T.O REQUEST button (bottom right).

Figure 11. COMPANY T.O DATA REQUEST: up to 2 runways, TOGA / FLEX / DERATED, free text; SEND T.O REQUEST.

- After SEND T.O REQUEST the page shows *REQUEST PENDING...*; without an answer after 4 minutes, *NO COMPANY REPLY*. The flyPad **Send to FMS** is that answer.
- When data arrives: the message *COMPANY T.O DATA RECEIVED WAITING FOR INSERTION* and the button **RECEIVED CPNY T.O** on PERF.

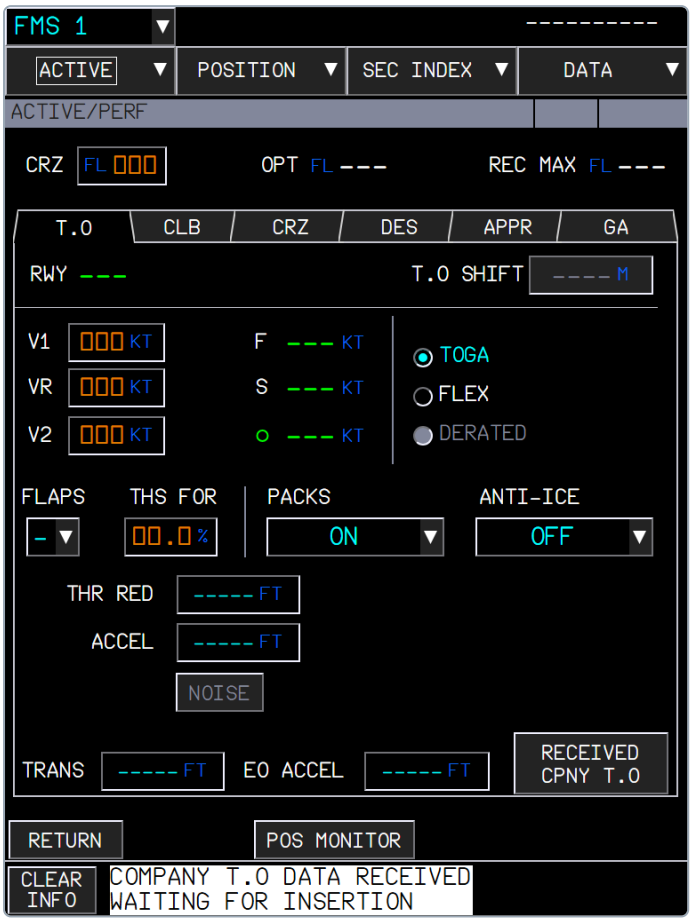


Figure 12. Data received: the message and the RECEIVED CPNY T.O button.

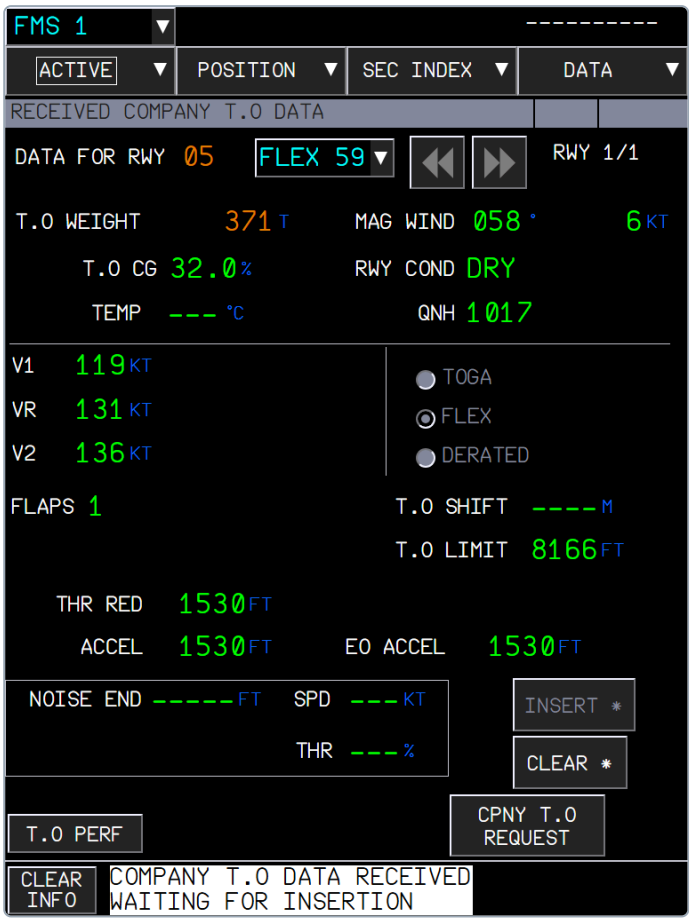


Figure 13. RECEIVED COMPANY T.O DATA: runway 05 in amber (not the flight plan runway), TOW in amber with UPLINK/ACTIVE TOW DISAGREE: INSERT is not available.

The RECEIVED COMPANY T.O DATA page

- Up to 4 runways (arrows, RWY 1/4) × 2 thrust ratings (the TOGA / FLEX list at the top).
- **Runway in amber**: not the departure runway of the active flight plan.
- **TOW in amber** with *UPLINK/ACTIVE TOW DISAGREE*: the uplink TOW is more than 2 t below or 7 t above the FMS TOW (FUEL&LOAD).
- TEMP shows dashes for a FLEX or DERATED takeoff (a derate reads `DRT D12` in the thrust list); T.O LIMIT is the TORA; the noise fields are filled when the procedure was sent.
- **INSERT** (only when runway and TOW agree) fills PERF T.O: speeds, flaps, thrust, shift, THR RED, ACCEL, EO ACCEL and noise. It does not fill THS FOR: the crew enters the CG there. **CLEAR** deletes the data.

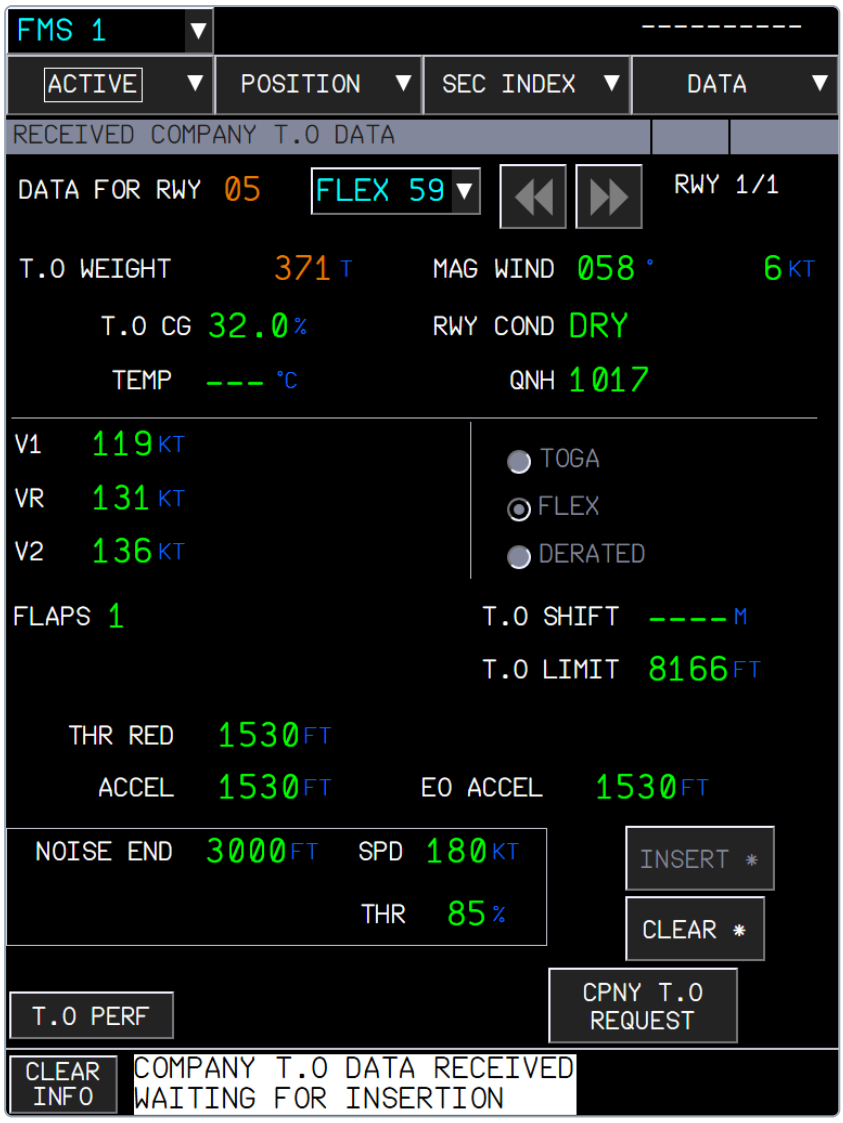


Figure 14. An uplink with the noise procedure: NOISE END 3000 ft, SPD 180 kt, THR 85 %.

> **TIP**
> The flyPad warns you right after Send to FMS if INSERT will not work: another departure runway in the FMS, no TOW in the FMS, or a TOW outside -2 t / +7 t of the FMS one.

What to expect

Situation	What you should see
Inputs missing	Amber labels; Calculate stays dim.
Calculate with valid inputs	RESULTS and the takeoff run; FLEX at its maximum by default.
TOW too high, CG out of envelope, wet runway...	A message with the reason; no result.
Send to FMS with a runway from the list	"Takeoff data sent to the FMS: RECEIVED CPNY T.O on the PERF page"; the MFD shows the data a moment later.
Send to FMS with a manual runway	Nothing is sent: the FMS needs the runway ident.
RECEIVED page, other runway than the flight plan	Runway in amber; INSERT not available.
INSERT	PERF T.O filled; THS FOR still to enter.

Limitations and why

Limitation	Why	What to do
Dry runways only (the DERATED takeoff too: see its own limitations below).	The only public A380 takeoff data is the takeoff weight charts of the Airbus A380 Aircraft Characteristics (dry, ISA and ISA+15, runway 1700-5100 m, altitude 0-8000 ft).	On a wet runway, use the dry result with extra margin, as a training estimate only.
Speeds, FLEX, wind, slope and temperature effects are estimates (amber *).	The charts give weights only. The rest is a model fitted on them and checked on the FCOM examples (FLEX 38 at 560 t / 614 t).	-
The run distances (V1, VR, 35 ft points) are all-engine estimates.	No A380 distance data is public.	-
Send to FMS needs a runway from the list.	The FMS checks the uplink against its departure runway.	Choose the runway in the list.
Not certified data.	For simulation and training.	-

How to use

1. On the MFD, enter the flight plan with its departure runway and fill FUEL&LOAD (ZFW, ZFWCG, block).
2. flyPad > **Performance** > **Takeoff**: **Fill data from** FMS, then METAR (or type the data).
3. Check TOW, T.O CG, flaps and thrust; press **Calculate**.
4. In TAKEOFF RUN, choose TOGA, the FLEX temperature or the DERATED level you will use; press **Send to FMS**.
5. On the MFD PERF T.O page: **RECEIVED CPNY T.O**, check the page, **INSERT**; enter the T.O CG in THS FOR.

How to test

#	Do this	Expected result
1	A380X on the ground, ADIRS aligned. MFD: flight plan with a departure runway, FUEL&LOAD filled. Open flyPad > Performance > Takeoff.	Empty screen: amber labels on the missing inputs, Calculate dim.
2	Choose FMS in **Fill data from** and press it, then METAR.	Airport, runway, TOW (in t, one decimal), CG, THR RED / ACCEL filled; wind, OAT and QNH from the METAR.
3	Press **Calculate**.	RESULTS: TOW, MTOW(perf), V1 / VR / V2, FLEX (or TOGA), flaps; chart values green, estimates amber with a star; TAKEOFF RUN with the V1 / VR / 35 ft marks and the margin.
4	On a long runway (e.g. 4000 m near sea level, 15 °C), type TOW 510.0 t with T.O CG 35.75 %, then 43 %, and Calculate each time.	Both accepted (envelope edges included).
5	T.O CG 35.7 % at 510 t; then 43.5 % at 400 t.	Message "CG Outside Takeoff Envelope", no result, each time.
6	Set the runway condition to Wet and Calculate.	Message that only dry runways are covered; no result.
7	Back to dry. In TAKEOFF RUN press TOGA, then FLEX and move the slider.	The run, the speeds and the margin follow; the FLEX range runs from the OAT up to the maximum.

8	Press **Send to FMS**.	Toast "Takeoff data sent to the FMS: RECEIVED CPNY T.O on the PERF page". After the transit time of the Company Datalink Reply Time setting (Real: 5 to 15 s), the MFD shows COMPANY T.O DATA RECEIVED WAITING FOR INSERTION and the RECEIVED CPNY T.O button on PERF T.O.
9	MFD PERF T.O > **RECEIVED CPNY T.O** > **INSERT**.	PERF T.O gets the speeds, flaps, thrust, T.O shift, THR RED / ACCEL / EO ACCEL (and the noise data when sent); THS FOR stays empty.
10	Change the FMS departure runway, then Send to FMS again.	flyPad warning about the runway; on the RECEIVED page the runway is amber and INSERT is not available.
11	PERF T.O > **CPNY T.O REQUEST** > **SEND T.O REQUEST**, and send nothing from the flyPad.	REQUEST PENDING..., then NO COMPANY REPLY after 4 minutes.

2. A380 takeoff calculator: DERATED takeoff

> **Aircraft:** A380X (the A32NX has no derate) | **Where:** flyPad > Performance > Takeoff > TAKEOFF RUN; MFD PERF T.O | **Status:** Tested in the sim by the user: OK (flyPad to MFD as DRT D08 and INSERT; E/WD D12 with N1 at the derated limit, 83.0 % against 88.3 % at TOGA; PFD FMA MAN DTO and the ENG THR LEVERS NOT SET caution at TOGA). Checked in the browser harness. Unit tests.

What it is for

A derate is a lower takeoff rating that is certified on its own (A380 FCOM PER-TOF-THR-DRT): the flight manual gives it its own takeoff limits and performance. The A380 has six levels, **D04, D08, D12, D16, D20, D24**: each one takes about 4 % more off the maximum takeoff thrust (D04 = about 4 % less, D24 = about 24 % less). The MFD PERF T.O page and the engines already had the derate; the flyPad calculator offered only TOGA and FLEX.

A derate is not FLEX. FLEX uses an assumed temperature and the crew can still push the levers to TOGA. A derated thrust depends on the real outside temperature, and the crew may not set TOGA during a derated takeoff below the speeds of the engine failure after V1 technique (the only exception is the windshear recovery). So the FCOM expects FLEX to give the better margin in most cases, and keeps the derate for **limiting runways**. The lower thrust also lowers the minimum control speeds VMCG and VMCA.

What you see

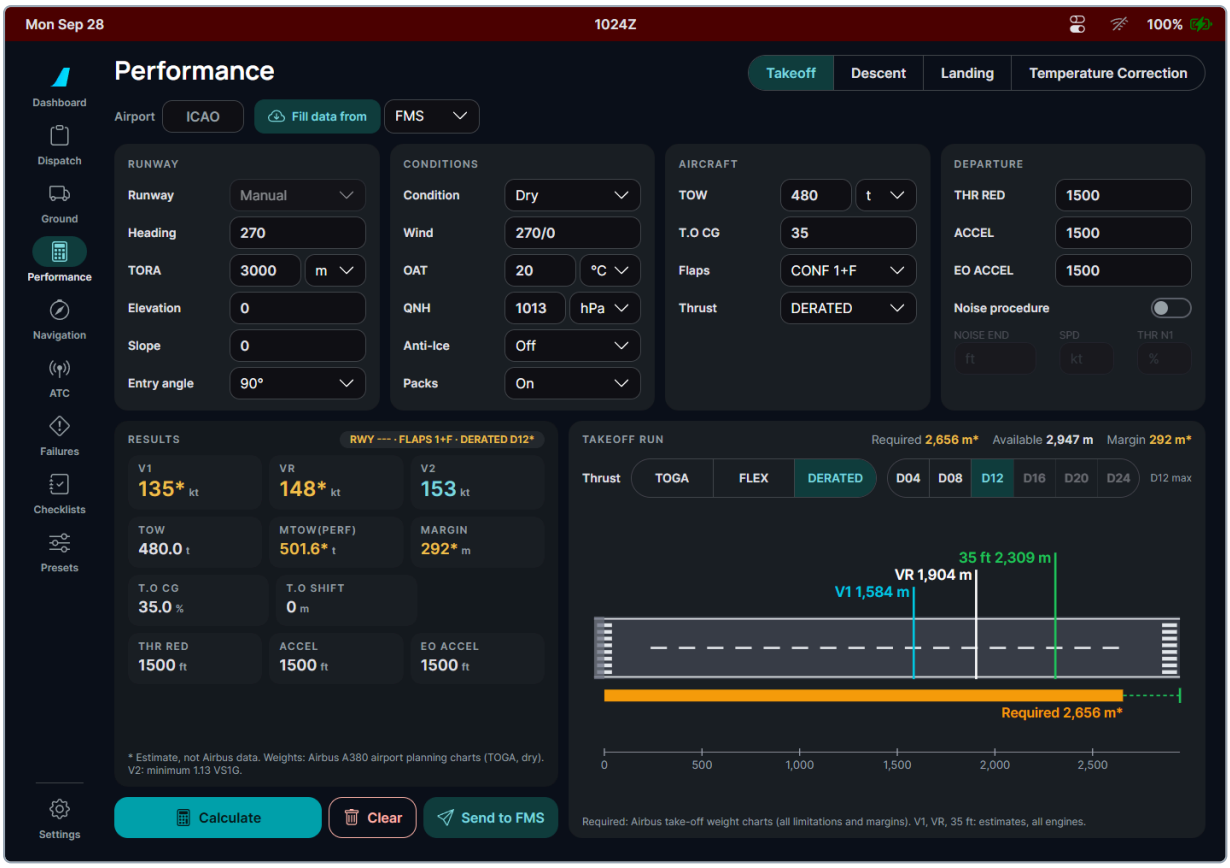


Figure 15. 480 t, 3000 m, sea level, 20 °C, CONF 1+F. TAKEOFF RUN > **DERATED**: D12 is chosen, the deepest level this TOW permits (D16, D20, D24 greyed). RESULTS reads DERATED D12*, MTOW(perf) 501.6* t is the D12 limit (the same take-off at FLEX: 44 °C, MTOW(perf) 570 t).

- **Three exclusive choices.** TOGA, FLEX and DERATED are three buttons of the TAKEOFF RUN panel: a derate and FLEX cannot be combined (FCOM: on the aircraft too, the MFD takes either a level or a FLEX temperature). The *Thrust* input of the AIRCRAFT section follows the same choice, both ways.
- **The level picker** D04 to D24 replaces the FLEX slider. The levels the TOW does not permit are greyed. DERATED starts at the **deepest permitted level**; the deepest one is written at the right ("D12 max").
- **RESULTS** shows `DERATED D12*` in its header; with a shallower level than the deepest one it adds the maximum, e.g. `DERATED D04* (max D12)`. The speeds, MTOW(perf), the margin and the run are those of the chosen level.
- If DERATED is chosen and no level is possible at this weight, the run goes back to TOGA with the message **Derate Not Possible**.

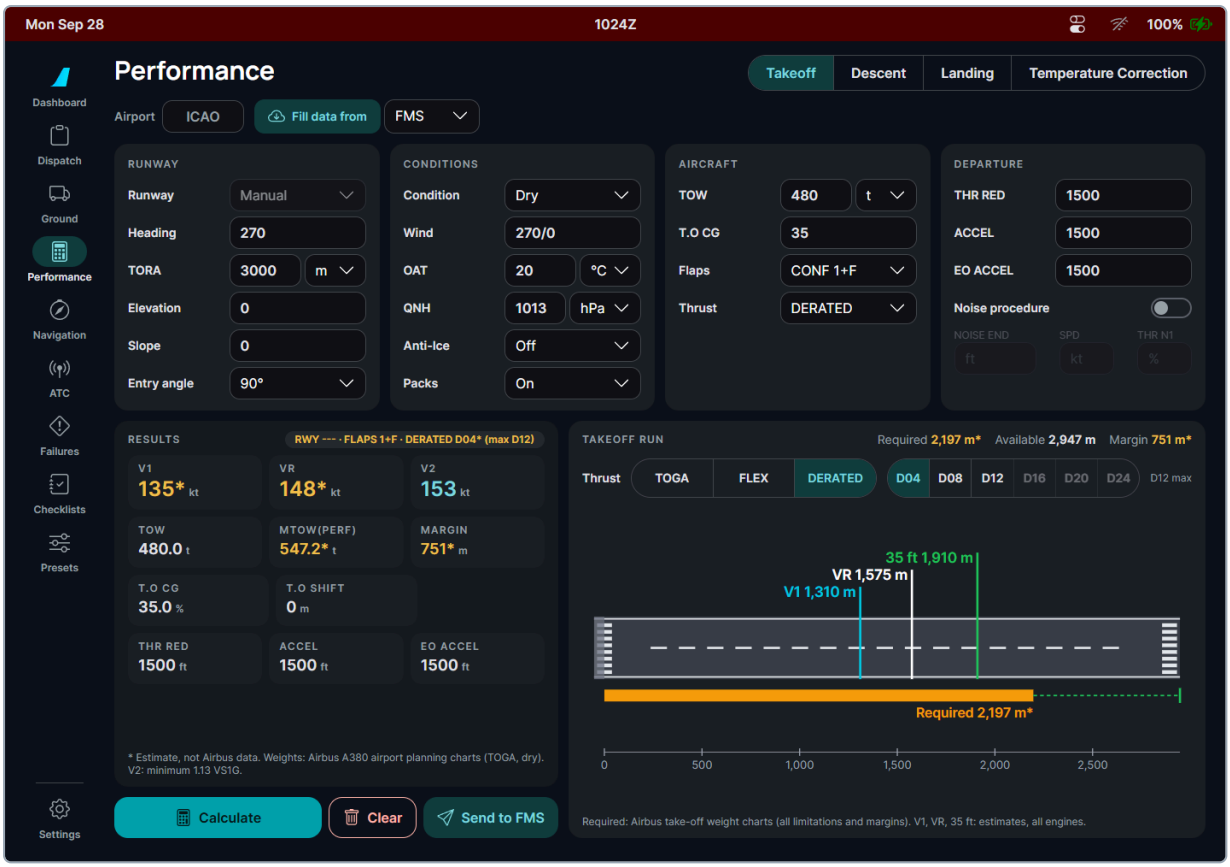


Figure 16. D04 picked by hand: header DERATED D04* (max D12), MTOW(perf) 547.2* t, the run needs 2197 m of the 2947 m available.

How it is calculated

All derate values are estimates of the FBW model (amber, with a star):

- **Weight limit** = the TOGA weight limit at the actual OAT x (1 - level %). In the pictures: TOGA limit 570 t, so D04 = 570 x 0.96 = 547.2 t and D12 = 570 x 0.88 = 501.6 t. In DERATED mode MTOW(perf) shows this derated limit.
- **Default level** = the deepest level whose limit is still at or above the TOW (D12 for 480 t above: D16 would allow only 478.8 t).
- **Speeds**: as for TOGA, with VMCG and VMCA lowered by the square root of the thrust ratio. The FCOM says that a lower thrust lowers them but gives no values: this factor is our estimate. When the speeds are set by the stall margin (V2 at least 1.13 VS1G), they do not change.
- **Takeoff run**: the V1, VR and 35 ft distances are computed for the thrust of the level.

Runway condition

FLEX is not permitted on a contaminated runway (FCOM PER-TOF-CTA); a derate is permitted on dry, wet and contaminated runways. So on a contaminated runway the flyPad offers **TOGA and DERATED** only. The A380 calculator still computes **dry runways only**: wet and contaminated runways still show the "runway condition not supported" message (see Limitations).

On the MFD: DERATED company T.O data

- **Send to FMS** with DERATED sends the thrust DERATED and its level with the speeds of that level.
- The RECEIVED COMPANY T.O DATA page shows the thrust as `DRT D12` (TEMP: dashes).
- **INSERT** sets PERF T.O to DERATED with the level. The engines then use the derated limit: the E/WD shows the level (e.g. D12) and the takeoff N1 is lower than the TOGA one.
- A **CPNY T.O REQUEST** for DERATED carries no level (the request page has none): the flyPad then picks the deepest permitted level.

- A320: no derate. The A320 calculator never offers it, and the MCDU refuses a DERATED uplink with *INVALID TAKEOFF UPLINK*.

On the PFD and the ECAM during the takeoff

- **FMA**: with a derated takeoff and the thrust levers at FLX/MCT, the first column of the PFD FMA shows `MAN` over `DTO` in a white box (not `MAN FLX +XX`, which is for a FLEX takeoff). This needs the fix e54ede63e (FMA shows MAN DTO for a derated takeoff).
- **No TOGA**: TOGA is not allowed for a derated takeoff (FCOM PRO-ABN-ECAM-10-70). With a derate selected, thrust levers at TOGA, even for a moment (e.g. pushed through TOGA on the way to FLX/MCT), trigger the master caution **ENG THR LEVERS NOT SET**, as on the real A380. The ECAM action is `THR LEVERS ... MCT/FLEX`. So set the levers straight to FLX/MCT.

Limitations and why

Limitation	Why	What to do
The derate is computed for dry runways only.	The Airbus charts behind the calculator are dry only, and the FCOM has no wet or contaminated A380 takeoff numbers (only the rules and the crosswind table).	Wet and contaminated runways show "runway condition not supported": no result.
Derate speeds, weight limits and distances are estimates (amber *).	The FCOM gives the levels and the rules, not the performance per level; the VMC decrease factor is our own.	Use them for simulation and training only.
No "optimum derate" for a VMCG-limited takeoff.	The FCOM caution not to use a derate deeper than the one giving the highest MTOW(perf) applies to VMCG-limited takeoffs. The chart model has no VMCG-limited weight, so it cannot find that optimum: the deepest level is simply the deepest one the TOW permits.	-

How to use

1. Calculate the takeoff as usual (see the A380 takeoff calculator above).
2. In TAKEOFF RUN press **DERATED** (or choose DERATED in the AIRCRAFT *Thrust* input): the deepest permitted level is selected.
3. Pick another level if you want a smaller reduction; check MTOW(perf), the speeds and the margin.
4. Press **Send to FMS**, then on the MFD PERF T.O: **RECEIVED CPNY T.O**, check `DRT Dnn`, **INSERT**.
5. For the takeoff, set the thrust levers **straight to FLX/MCT**: TOGA is not allowed with a derate, and even passing through TOGA triggers ENG THR LEVERS NOT SET. The PFD FMA shows MAN DTO.

How to test

#	Do this	Expected result
1	A380X on the ground. flyPad > Performance > Takeoff: heading 270, TORA 3000 m, elevation 0, slope 0, wind 270/0, OAT 20 °C, QNH 1013, dry, TOW 480 t, T.O CG 35 %, **Calculate**.	FLEX 44 °C, MTOW(perf) 570 t.
2	In TAKEOFF RUN press **DERATED**.	Level picker D04 to D24 with D12 selected and D16, D20, D24 greyed; "D12 max"; RESULTS header DERATED D12*; MTOW(perf) 501.6* t; AIRCRAFT Thrust reads DERATED.

3	Press **D04**.	Header DERATED D04* (max D12); MTOW(perf) 547.2* t; a shorter run.
4	Press FLEX, then DERATED again.	FLEX shows the slider (44 °C max); DERATED comes back at D12.
5	Set the runway condition to a contaminated one and open the AIRCRAFT Thrust list.	TOGA and DERATED only (no FLEX); Calculate still gives the "runway condition not supported" message.
6	Back to dry, DERATED D08, with a runway from the list and the flight plan on the MFD: **Send to FMS**.	After the reply time: MFD RECEIVED COMPANY T.O DATA with the thrust `DRT D08`, TEMP dashes.
7	**INSERT**, then look at PERF T.O and, with the engines running and the levers at FLX/MCT, the E/WD.	PERF T.O DERATED D08; the E/WD shows D08 and the takeoff N1 limit is lower than at TOGA.
8	Ready for takeoff with the derate inserted: set the thrust levers **straight to FLX/MCT** (not through TOGA) and look at the PFD FMA.	First FMA column: `MAN` / `DTO` in a white box (not MAN FLX +XX). No caution.
9	Still with the derate selected, push the thrust levers to **TOGA** (even briefly).	Master caution and the E/WD alert **ENG THR LEVERS NOT SET** with `THR LEVERS ... MCT/FLEX`.
10	A32NX: flyPad > Performance > Takeoff.	Only TOGA and FLEX; no DERATED.

3. A320 takeoff calculator and MCDU uplink takeoff data

> **Aircraft:** A32NX | **Where:** flyPad > Performance > Takeoff; MCDU PERF TAKE OFF > <TO DATA (6L) | **Status:** The flyPad calculation was used in the sim; the MCDU uplink pages are checked in the browser harness only; the full flow is not sim-tested. Unit tests of the takeoff distances.

What it is for

The A320 takeoff speeds, FLEX temperature and THS come from the airline or the crew's own tool. Here the flyPad computes them with the FlyByWire A320neo performance model (built from Airbus data) and sends them to the FMGC as an **uplink takeoff data** message, which the crew inserts from the MCDU PERF TAKE OFF page as on the real aircraft.

The flyPad screen (A320)

The screen is the one of the A380 (see "A380 takeoff calculator and company T.O data" above) with these A320 differences:

- No noise procedure fields; T.O CG is optional: below 27 % MAC the forward CG performance applies (A320 FCOM PER-LOD) and the THS is shown in RESULTS.
- All values come from the FBW A320 model: they are shown in green (no estimates).
- Wet and contaminated runways are computed (the A380 page accepts dry only).

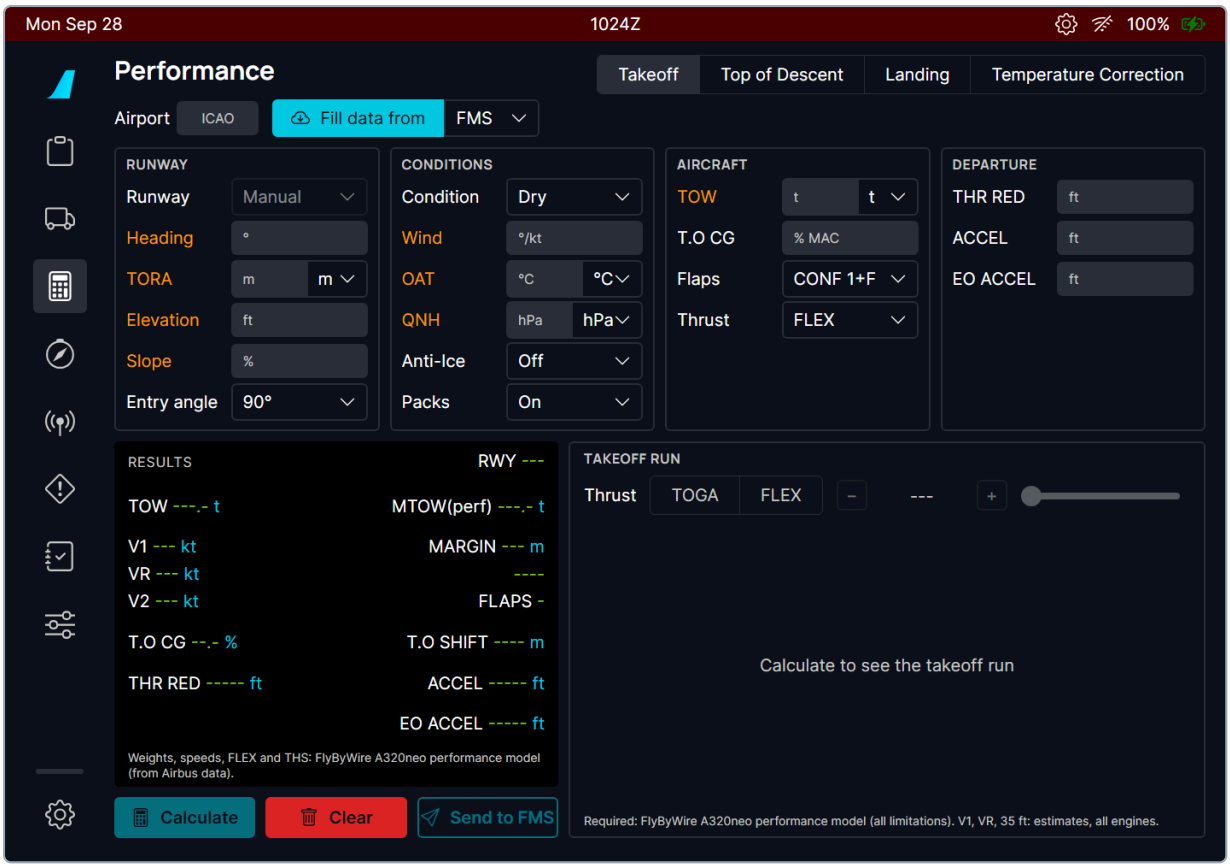


Figure 17. The A320 takeoff screen before any input.

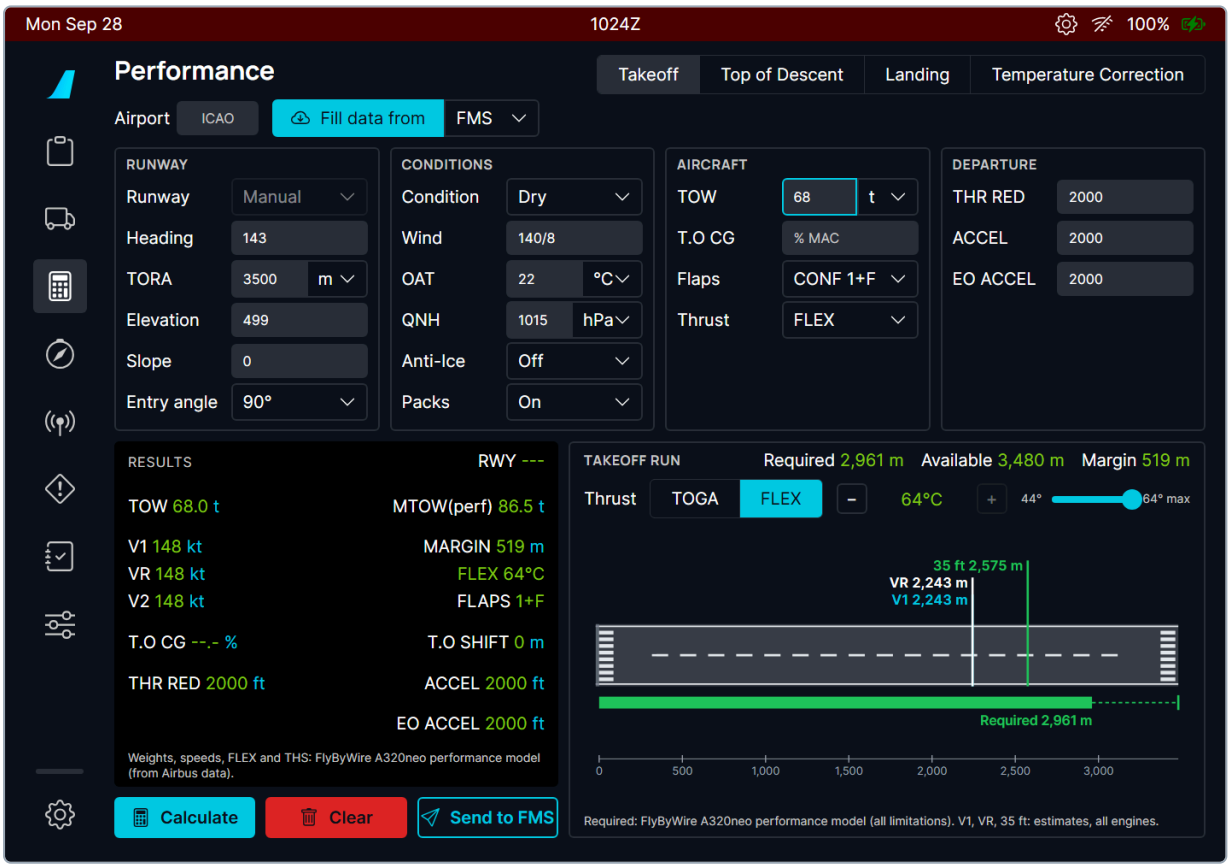


Figure 18. LFBO 14R-like runway (3500 m, 499 ft), 68 t, 22 °C: FLEX 64 °C (maximum), required 2961 m of 3480 m, margin 519 m.

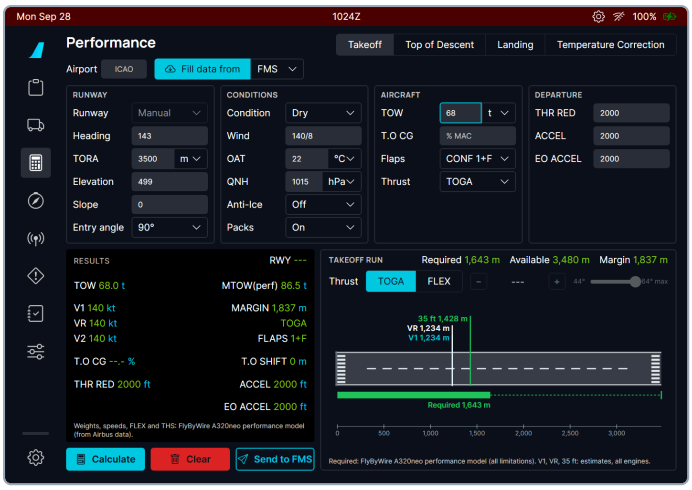


Figure 19. The TOGA run at the same weight: 1643 m.

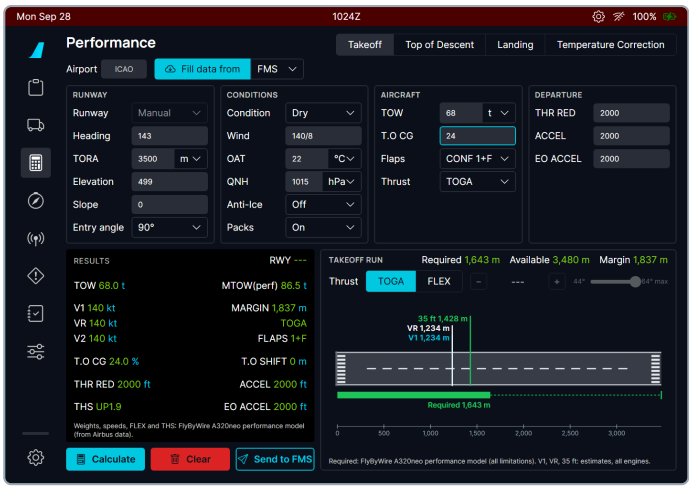


Figure 20. T.O CG 24 % (forward CG) at TOGA: the THS (UP1.9) appears in RESULTS.

> **NOTE**
>
> At light weights the FBW A320 model gives **V1 = VR = V2** (the pictures show 148/148/148 at FLEX and 140/140/140 at TOGA for 68 t). This comes from the FlyByWire model itself (the same with the FBW version before this change), not from this feature.

Sending to the MCDU

1. On the flyPad, choose the **runway from the list** (the uplink carries the runway ident), calculate, choose TOGA or the FLEX temperature for the takeoff run, and press **Send to FMS**.
2. The flyPad says "*Takeoff data sent to the FMS: MCDU PERF TAKE OFF, UPLINK TO DATA, RECEIVED TO DATA*", and warns right away if the FMGC departure runway or TOW will prevent the insertion.
3. On the MCDU the scratchpad shows **TAKEOFF DATA UPLINK**.
4. PERF TAKE OFF > **<TO DATA** (6L) > **RECEIVED <TO DATA** (6L): the UPLINK MAX TO DATA page (or FLX TO DATA with 4R). Check it, then **INSERT UPLINK**.

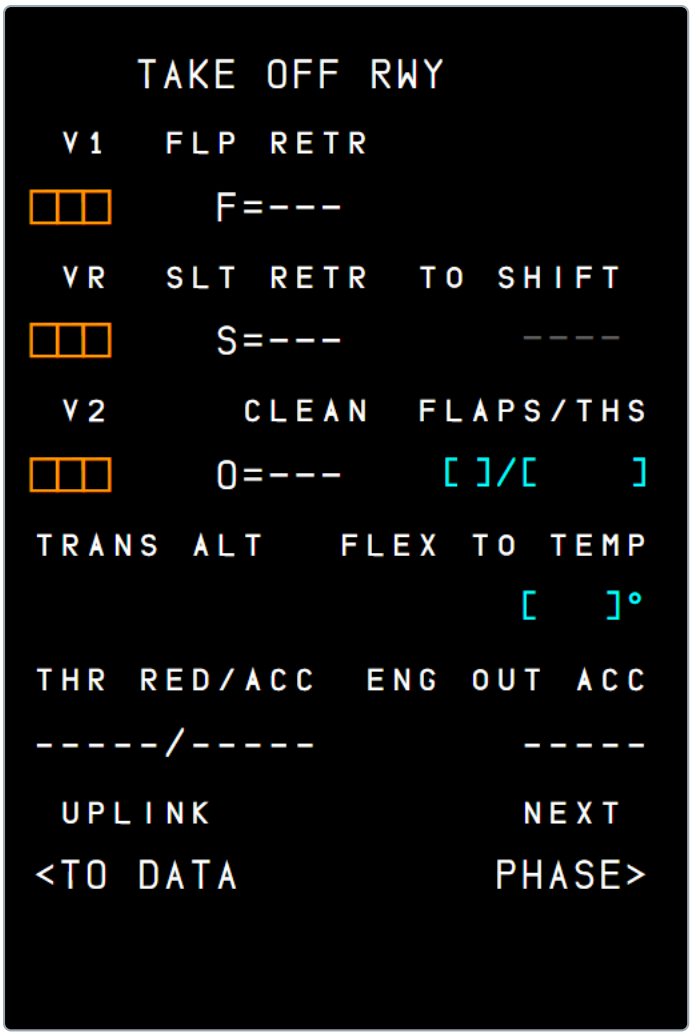


Figure 21. PERF TAKE OFF: the new UPLINK <TO DATA prompt (6L), in PREFLIGHT and DONE.

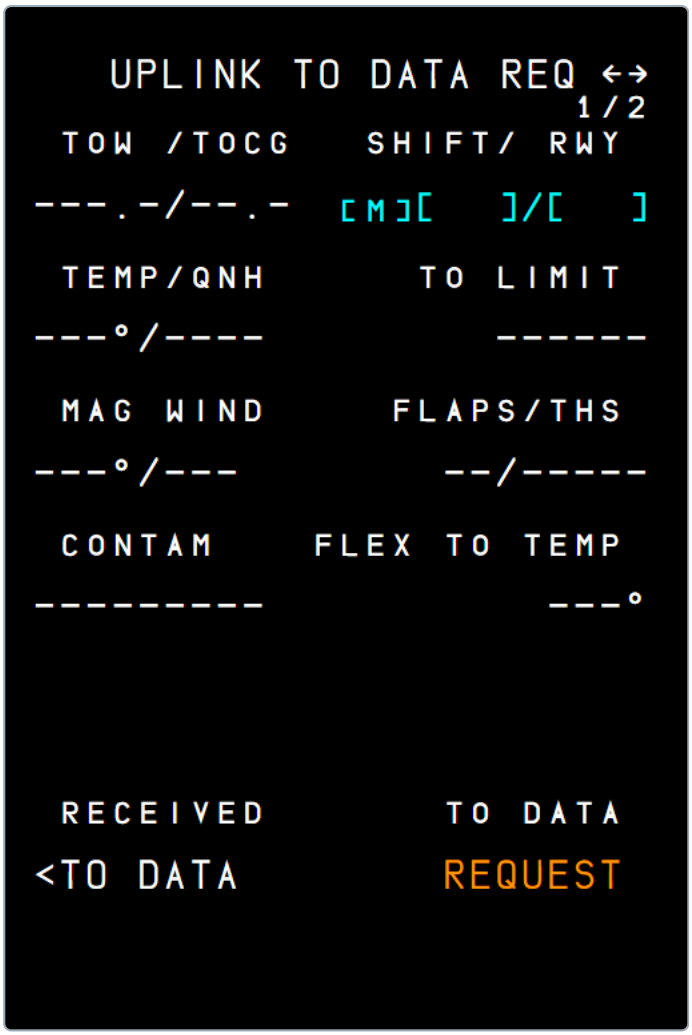


Figure 22. UPLINK TO DATA REQ: the request page (two runways, 1/2). RECEIVED <TO DATA (6L), TO DATA REQUEST (6R).

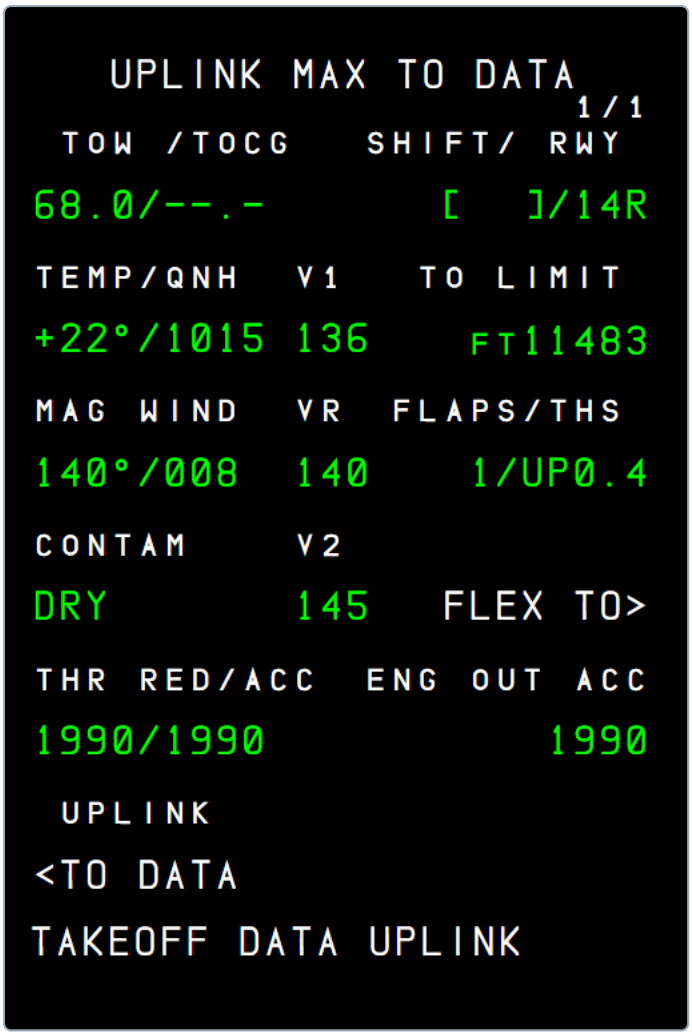


Figure 23. UPLINK MAX TO DATA after TAKEOFF DATA UPLINK: TOW, runway 14R, temperature and QNH, V1 / VR / V2, TO LIMIT, flaps / THS, THR RED / ACC, ENG OUT ACC. FLEX TO> (4R) shows the FLEX data.

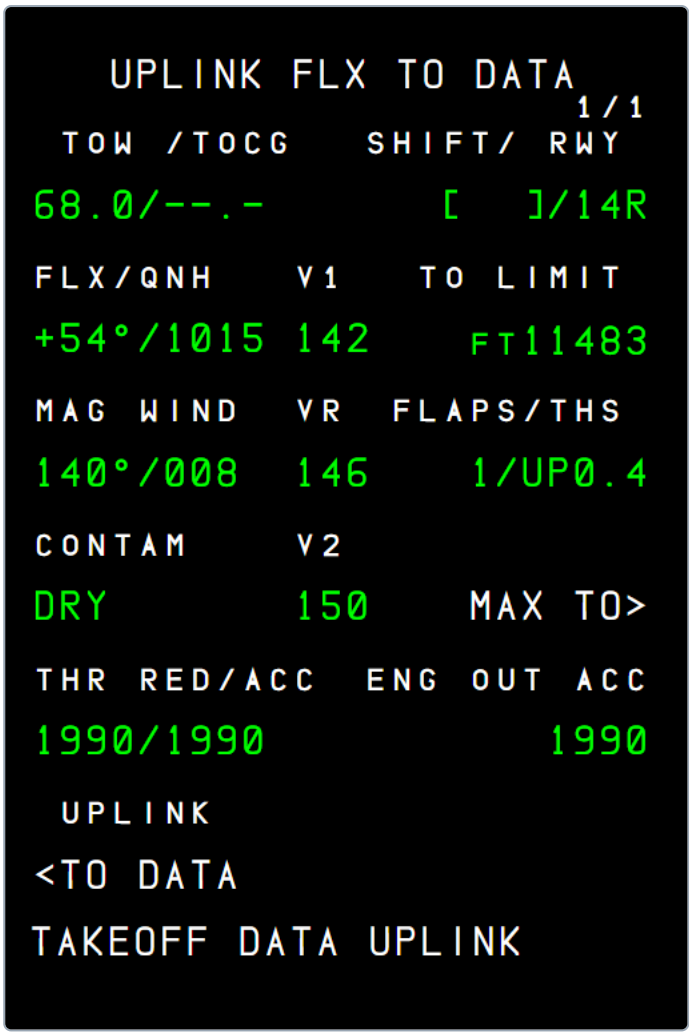


Figure 24. UPLINK FLX TO DATA: FLX +54 °C with its own speeds.

There is **no INSERT UPLINK prompt** in these pictures because the harness FMGC has no flight plan. The prompt appears only when the uplink is complete (V1 and VR present) and the checks below pass.

The insertion rules (A320 FCOM)

- The uplink runway must be the departure runway of the flight plan.
- The uplink TOW must be **not more than 1 t below or 3 t above** the FMGC TOW; otherwise there is no insertion.
- INSERT UPLINK fills V1, VR, V2, FLAPS/THS, FLEX TO TEMP (cleared for a MAX takeoff), THR RED/ACC, ENG OUT ACC and the TO SHIFT.
- An incomplete uplink (for example without V1 or VR) gives no INSERT UPLINK prompt (DSC-22_20-70 P 6).

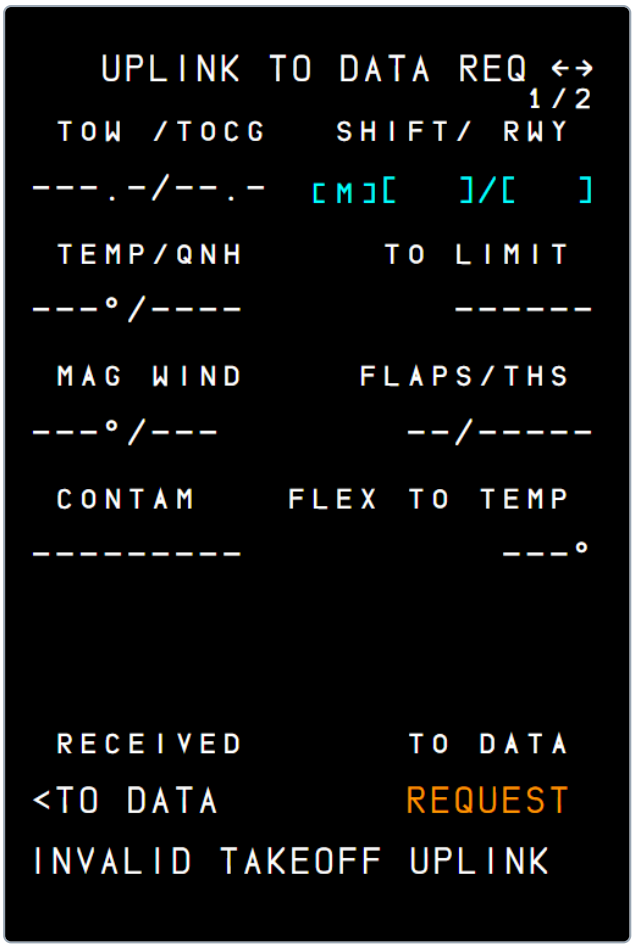


Figure 25. An uplink the FMGC cannot read: INVALID TAKEOFF UPLINK in the scratchpad.

MCDU message	Meaning
TAKEOFF DATA UPLINK	Uplink takeoff data received.
INVALID TAKEOFF UPLINK	The uplink is not valid (missing data).
REQUEST IS PENDING	A request was sent and has no answer yet.
NO ANSWER TO REQUEST	No answer 4 minutes after the request.

PERF TAKE OFF also gets the **TO SHIFT** field (1 m up to the runway length, CLR to delete): the distance lost when taking off from an intersection.

What to expect

Situation	What you should see
Send to FMS with a runway from the list	Success message; TAKEOFF DATA UPLINK on the MCDU.
Send to FMS with a manual runway	Nothing is sent: choose the runway in the list.
FMGC departure runway differs	flyPad warning; no INSERT UPLINK on the MCDU.
FMGC TOW differs by more than -1 t / +3 t	flyPad warning "INSERT UPLINK needs a TOW within -1 t / +3 t"; no INSERT UPLINK.
No TOW in the FMGC	flyPad warning "No TOW in the FMS (INIT B: ZFW and BLOCK)": fill INIT B first.
INSERT UPLINK	PERF TAKE OFF filled with the uplink.

Limitations and why

Limitation	Why	What to do
Only the thrust selected for the takeoff run is sent.	The airline sends what the crew asked for; the MCDU holds MAX and FLEX pages, the flyPad fills the one you use.	Select TOGA or FLEX before sending.
V1 = VR = V2 at light weights.	The FlyByWire A320 model (upstream).	Check against your own tools.
No real airline datalink.	The flyPad plays the airline.	-

How to use

1. On the MCDU, enter the flight plan with its departure runway and INIT B (ZFW, BLOCK).
2. flyPad > **Performance** > **Takeoff**: choose the **runway from the list**, fill the data, **Calculate**.
3. Choose TOGA or the FLEX temperature in TAKEOFF RUN, press **Send to FMS**.
4. MCDU PERF TAKE OFF > **<TO DATA** (6L) > **RECEIVED <TO DATA** (6L): check, then **INSERT UPLINK**.

How to test

#	Do this	Expected result
1	A32NX on the ground. MCDU: flight plan with a departure runway, INIT B ZFW and BLOCK. flyPad > Performance > Takeoff, runway from the list, conditions, TOW, Calculate.	RESULTS in green (FBW A320 model); THS shown when the T.O CG is below 27 %.
2	Press **Send to FMS**.	Toast "Takeoff data sent to the FMS: MCDU PERF TAKE OFF, UPLINK TO DATA, RECEIVED TO DATA"; MCDU scratchpad TAKEOFF DATA UPLINK.
3	MCDU PERF TAKE OFF > **<TO DATA** (6L) > **RECEIVED <TO DATA** (6L).	UPLINK MAX TO DATA (or FLX TO DATA with 4R): V1 / VR / V2 and their labels in column 11, not over the left labels.
4	Press **INSERT UPLINK**.	PERF TAKE OFF filled: V1, VR, V2, FLAPS/THS, FLEX TO TEMP (cleared for MAX), THR RED/ACC, ENG OUT ACC, TO SHIFT.
5	Change the INIT B block so that the TOW differs by more than +3 t, then Send to FMS again.	flyPad warning "INSERT UPLINK needs a TOW within -1 t / +3 t"; no INSERT UPLINK prompt.
6	Clear INIT B and send again.	flyPad warning "No TOW in the FMS (INIT B: ZFW and BLOCK)".
7	Choose a Manual runway on the flyPad and press Send to FMS.	Nothing is sent: the runway must come from the list.

4. A380X company datalink reply time

> **Aircraft:** A380X (the A32NX MCDU uplink stays instant) | **Where:** flyPad > Settings > Realism; MFD INIT, WIND, PERF T.O | **Status:** On the user's sim test list done ("datalink"); checked in the browser harness. Company winds with the FBW WIND page: covered by the sim test of the WIND page, PASS; unit tests CompanyReplyHold.spec and CompanyRequestTimeout.spec (plus the button rule specs).

What it is for

On the A380 the flight plan, the winds and the takeoff data come from the airline by datalink (FCOM DSC-22-FMS-10-40-90). A request takes time: the button shows **REQUEST PENDING...** until the answer, and **NO COMPANY REPLY**

appears when nothing comes within 4 minutes. FBW answered instantly; now the answer arrives after a reply time you choose.

The setting

Figure 26. flyPad > Settings > Realism: Company Datalink Reply Time, Real by default.

Setting	Answer to a request	Message sent without a request (e.g. SEND TO FMS)
Instant	immediately	immediately
Fast	5 to 15 s	1 to 3 s
Real (default)	60 to 120 s	5 to 15 s

Airbus gives no figure: these are typical airline values, below the 4 minute limit of the FCOM.

Requesting the flight plan

1. INIT > **CPNY F-PLN REQUEST** opens COMPANY F-PLN REQUEST (flight number, company route, free text).
2. Press **SEND F-PLN REQUEST**. The button becomes **REQUEST PENDING...**, here and on INIT; one request at a time.
3. After the reply time the SimBrief flight plan is delivered to the FMS, as before.
4. If SimBrief does not answer (no network, no flight plan), **NO COMPANY REPLY** appears after 4 minutes.

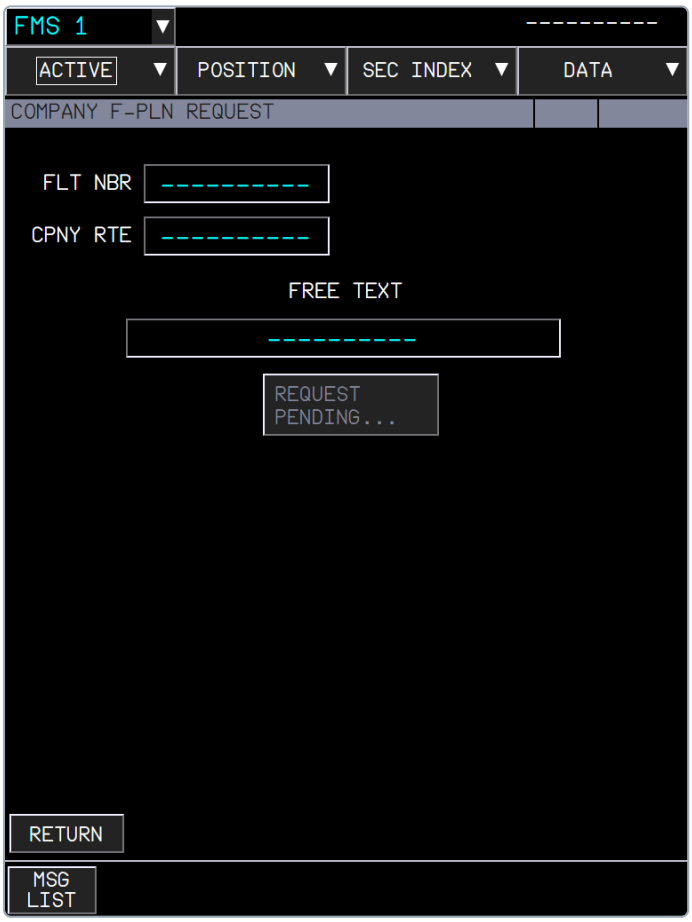


Figure 27. Request sent: REQUEST PENDING... (greyed until the answer).

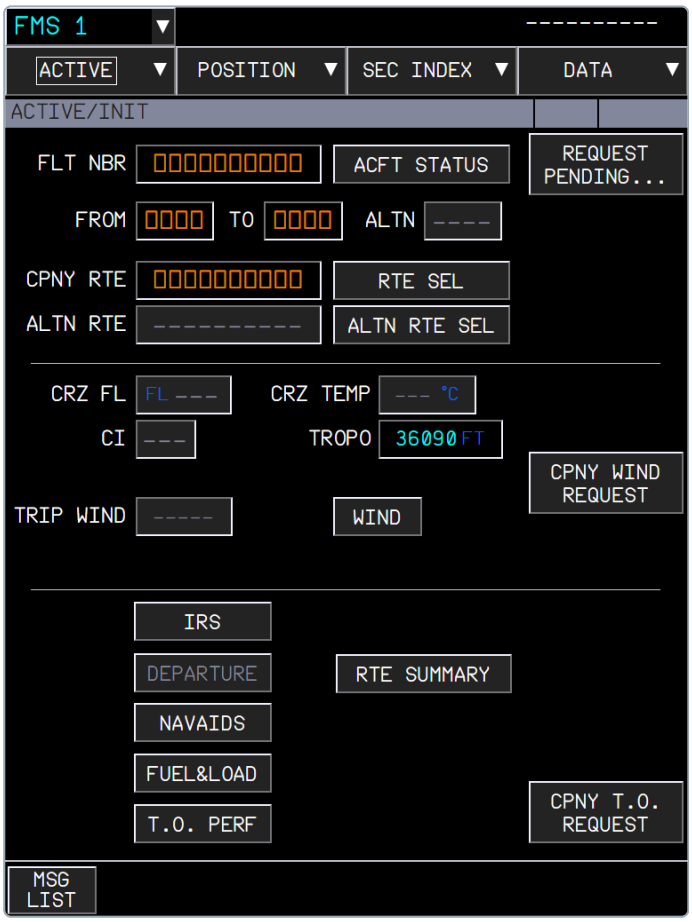


Figure 28. INIT shows the pending request too.

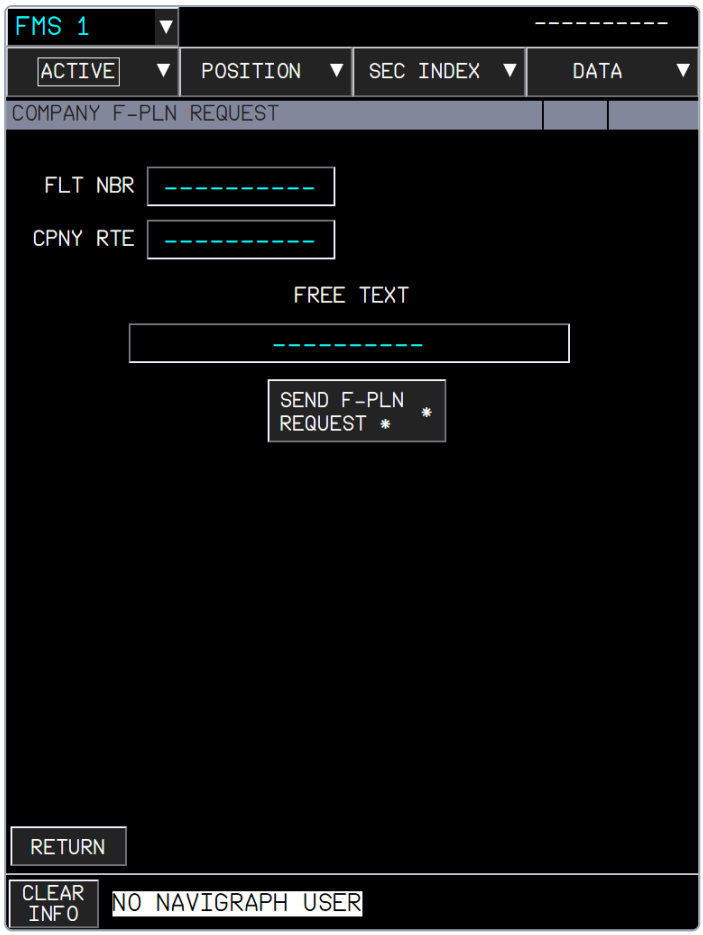


Figure 29. Without a Navigraph account or a SimBrief ID in the flyPad settings: NO NAVIGRAPH USER, no request.

Winds and takeoff data

- **CPNY WIND REQUEST** (INIT or WIND page, or SEND WIND REQUEST on the COMPANY WIND DATA REQUEST page): the winds are delivered after the reply time (see "Company winds" below).
- **Takeoff data** (PERF T.O > CPNY T.O REQUEST, see the A380 takeoff calculator above): the answer of the flyPad Send to FMS reaches the FMS not before the reply time of the request; without a request, after the transit time only.

What to expect

Situation	What you should see
Real, request sent	REQUEST PENDING... for one to two minutes, then the data.
Instant	The data at once, as before this change.
A second request while one is pending	Not possible: the button stays REQUEST PENDING.... (F-PLN request). For winds, the CPNY WIND REQUEST button of the plan that asked still opens the COMPANY WIND DATA REQUEST page, whose SEND WIND REQUEST button reads REQUEST PENDING.... and does nothing.
No answer	NO COMPANY REPLY after 4 minutes.
No Navigraph user / SimBrief ID	NO NAVIGRAPH USER; nothing is sent.

Limitations and why

Limitation	Why	What to do

The reply times are typical values.	Airbus publishes no figure for the airline answer.	Choose Fast or Instant if you prefer.
The A32NX MCDU uplink stays instant.	This change is for the A380 FMS; the A320 uplink has its own messages.	-

Company winds

The A380X WIND page and its company wind request come from FlyByWire master (merged into `core/a380x-mfd-fms`). FlyByWire's wind request goes through the ATSU and delivers the answer of SimBrief at once; two rules were put back on top of it, and a third closes a gap:

- **The reply time applies to the winds again.** The request time is recorded per flight plan and the answer is held until the reply time of the flyPad setting (Instant, Fast, Real) has passed; REQUEST PENDING... stays up meanwhile. Without this, only the company flight plan was delayed.
- **One wind request per flight plan, NO COMPANY REPLY after 4 minutes** (A380 FCOM DSC-22-FMS-10-40-90). The FMC remembers which flight plan asked: REQUEST PENDING... shows on the CPNY WIND REQUEST button and on the COMPANY WIND DATA REQUEST page of that plan only; on another plan the button is disabled (one company request at a time). If no answer comes within 4 minutes, NO COMPANY REPLY appears and a late answer is refused. Without this, a reload of the systems left REQUEST PENDING... up until a flight reload.
- **SEND F-PLN REQUEST is blocked while a wind request is pending** (the FMC dropped such a request silently).

How to use

1. flyPad > **Settings** > **Realism** > *Company Datalink Reply Time (F-PLN, Winds, T.O Data)*: Instant, Fast or Real (default).
2. Send the company requests as usual (CPNY F-PLN REQUEST, CPNY WIND REQUEST, CPNY T.O REQUEST) and wait for the answer while the button shows REQUEST PENDING....

How to test

#	Do this	Expected result
1	A380X on the ground, a SimBrief OFP available, Navigraph / SimBrief set in the flyPad. Set the reply time to **Real**.	Setting saved.
2	MFD INIT > **CPNY F-PLN REQUEST** > **SEND F-PLN REQUEST**.	REQUEST PENDING... (greyed) here and on INIT; the flight plan arrives 60 to 120 s later.
3	Press the request button again while it is pending.	Nothing happens: one request at a time.
4	Set **Instant** and request again (after clearing the plan).	The flight plan arrives at once.
5	INIT or WIND > **CPNY WIND REQUEST** with Real.	The winds arrive after the reply time.
6	Disconnect the network (or no OFP) and send a F-PLN request.	NO COMPANY REPLY after 4 minutes.
7	Reply time **Real**, a SimBrief flight loaded. WIND > **CPNY WIND REQUEST** (opens COMPANY WIND DATA REQUEST), SEND WIND REQUEST.	REQUEST PENDING... on the page and on the CPNY WIND button for the reply time (60 to 120 s), then the button reads RECEIVED CPNY WIND and COMPANY WIND DATA RECEIVED WAITING FOR INSERTION IN ACTIVE appears.
8	While the wind request is pending: MFD INIT > CPNY F-PLN REQUEST.	SEND F-PLN REQUEST is greyed out until the winds arrive.
9	While the wind request of the active plan is pending, open the WIND page of SEC 1.	The CPNY WIND REQUEST button there is disabled (one company request at a time).

10	Send a wind request and get no answer for 4 minutes (SimBrief unreachable).	NO COMPANY REPLY; an answer that arrives later is refused.
11	Remove the Navigraph user / SimBrief ID and press CPNY F-PLN REQUEST.	NO NAVIGRAPH USER; nothing is sent.

5. Landing calculator

> **Aircraft:** A380X and A32NX | **Where:** flyPad > Performance > Landing | **Status:** Tested in the sim by the user on both aircraft: OK, including the BTV exit advice on dry, wet and contaminated runways. Unit tests (mutation-tested).

What it is for

Before the descent, the crew checks that the aircraft can land on the planned runway: the landing distance for the runway condition and braking, the margin left, the approach speed VAPP and the go-around performance. On the real A380 this is the LDG PERF application of the OIS (FCOM PER-LND); on the A320 it is the QRH (in flight) and the FCOM tables (dispatch). This page does both, with the data of each aircraft.

- **IN-FLIGHT** (default): the actual landing distance for the braking mode you will use, as in the QRH / OIS, with the stop margin.
- **DISPATCH**: the regulatory (factored) required landing distance, to check a runway before departure.

The screen

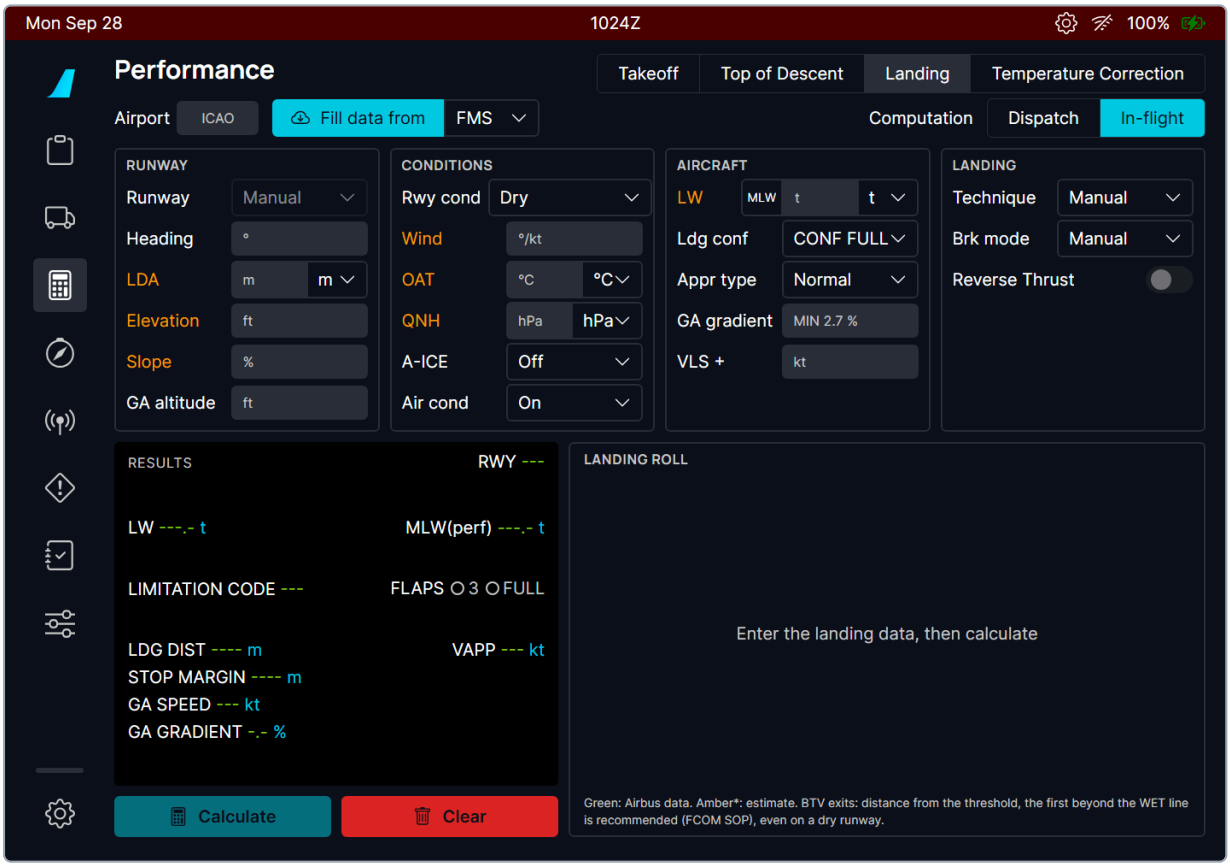


Figure 30. The A380 landing screen before any input. Amber labels are the inputs still missing for a calculation.

Part	Contents
Airport, Fill data from	The ICAO code; **Fill data from** imports the data from the **FMS** (A380 FMC / A320 MCDU), the **OFP** (SimBrief) or the **METAR**.
Computation	DISPATCH or IN-FLIGHT.

RUNWAY	Runway from the list (or Manual), heading, landing distance available (LDA), elevation, slope, and on the A380 the go-around altitude.
CONDITIONS	Runway condition, wind (direction/speed), OAT, QNH; on the A380 the anti-ice and air conditioning (they change the go-around gradient).
AIRCRAFT	Landing weight (MLW button = the maximum landing weight), landing configuration, and on the A380 the approach type, the required go-around gradient and the VLS increment.
LANDING	Manual landing or autoland, braking mode, reverse thrust, and on the A320 the overweight procedure.
RESULTS	The results in the OIS layout (below).
LANDING ROLL	A picture of the runway with the touchdown, the stop point of each braking mode, the stop margin, and on the A380 the BTV lines and the runway exits.

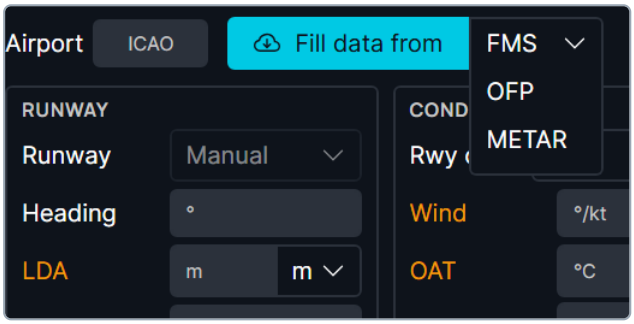


Figure 31. Fill data from: FMS (destination, runway, predicted landing weight and the PERF APPR QNH, temperature, wind and landing configuration), OFP or METAR.

A normal in-flight calculation (A380)

1. In flight, with the destination and the arrival runway in the flight plan and PERF APPR filled: choose **FMS** and press **Fill data from**. Otherwise type the airport, choose the runway and fill the conditions.
2. Check the landing weight (the FMS prediction) and the configuration.
3. Choose the braking mode you plan to use (manual, autobrake LO / 2 / 3 / HI, or BTV).
4. Press **Calculate**.

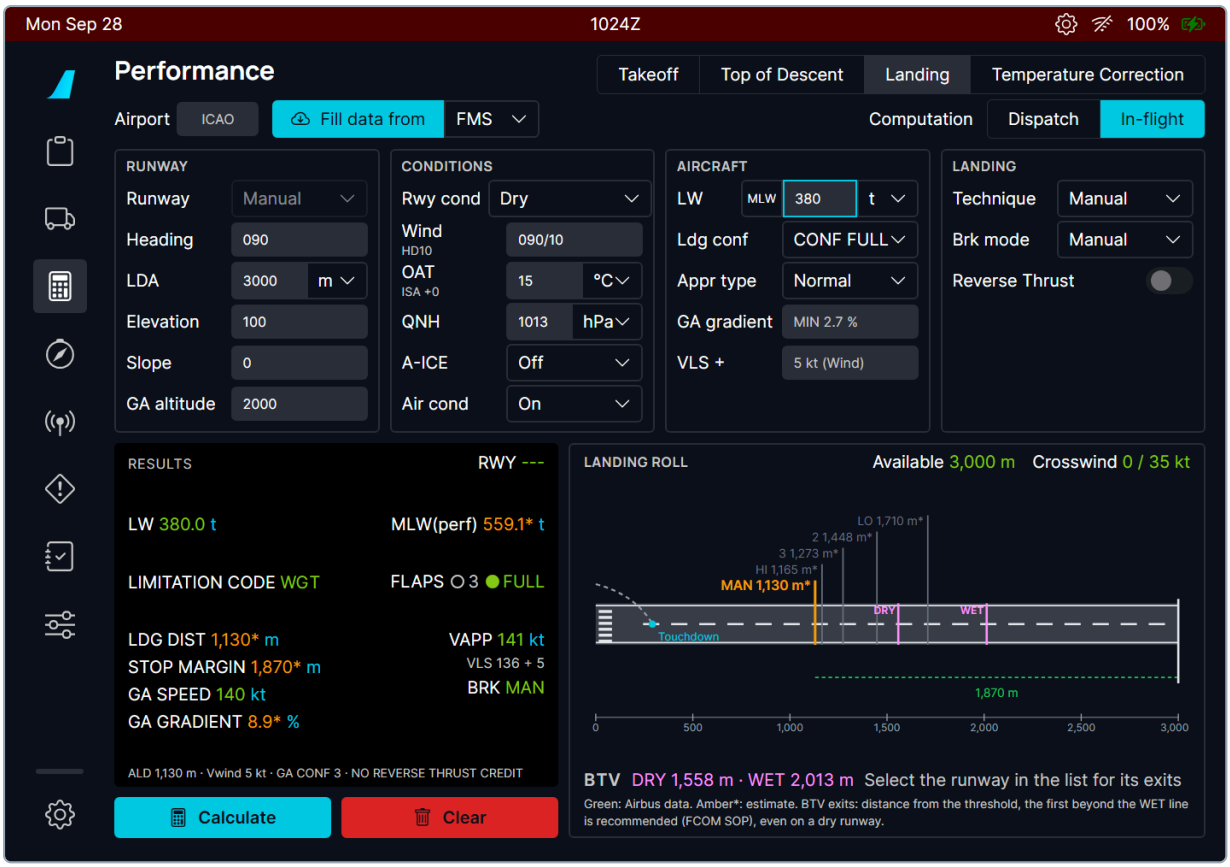


Figure 32. Dry runway, 3000 m, 380 t, CONF FULL, manual braking: landing distance 1130 m, stop margin 1870 m, VAPP 141 kt (VLS 136 + 5). The landing roll shows where each braking mode would stop.

Reading the results

Line	Meaning
LW	The landing weight used.
MLW(perf)	The **highest landing weight the runway and the go-around allow**: the lower of the weight whose landing distance just fits the LDA and the weight that still gives the required go-around gradient. It is a performance limit, not the structural maximum landing weight (395 t on the A380): it can be above it. Above the structural MLW the page shows OVERWEIGHT LANDING.
LIMITATION CODE	**WGT**: your weight is below MLW(perf), nothing limits you. **LDA** (red): the landing distance limits the weight. **ACG**: the go-around gradient limits it.
FLAPS	The landing configuration used (with AUTO CONF: the one the page chose).
LDG DIST / REG LDG DIST	In flight: the actual landing distance for the braking mode. Dispatch: the required (regulatory) distance.
STOP MARGIN	LDA minus the landing distance. Negative (red): the aircraft does not stop on the runway.
GA SPEED, GA GRADIENT	The go-around speed and the climb gradient of the go-around configuration.
VAPP	VLS of the landing configuration + the VLS increment (the wind increment, or what you enter).
Bottom line	The actual landing distance (ALD), the wind taken into account, the go-around configuration, and whether reverse thrust was credited.

Colours follow the flyPad takeoff calculator: **green** values come from Airbus data; values in **amber with a star (*)** are estimates of a model fitted on that data (see Limitations).

Braking modes

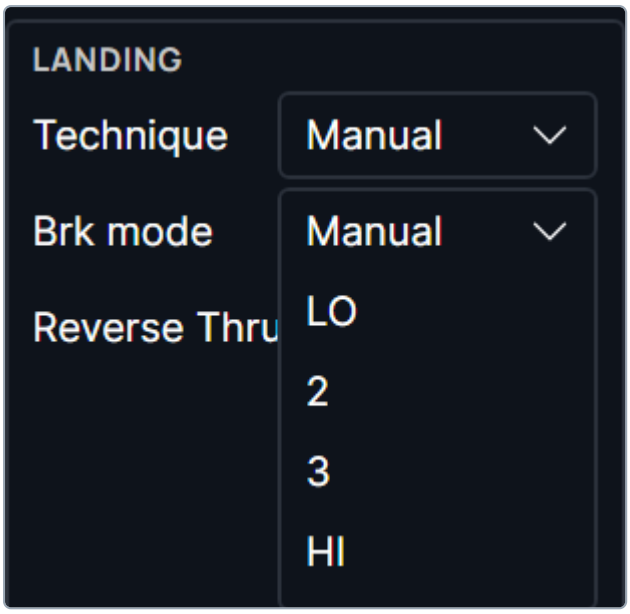


Figure 33. A380 braking modes: Manual, autobrake LO, 2, 3, HI (picture taken before the BTV mode was added: the list now ends with BTV, see "A380 landing calculator: BTV braking mode" below).

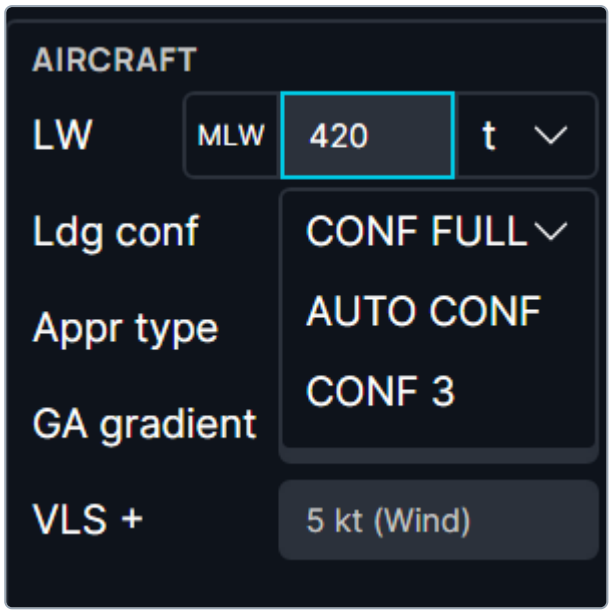


Figure 34. Landing configuration: CONF FULL, CONF 3 or AUTO CONF (the page chooses, as the OIS does).

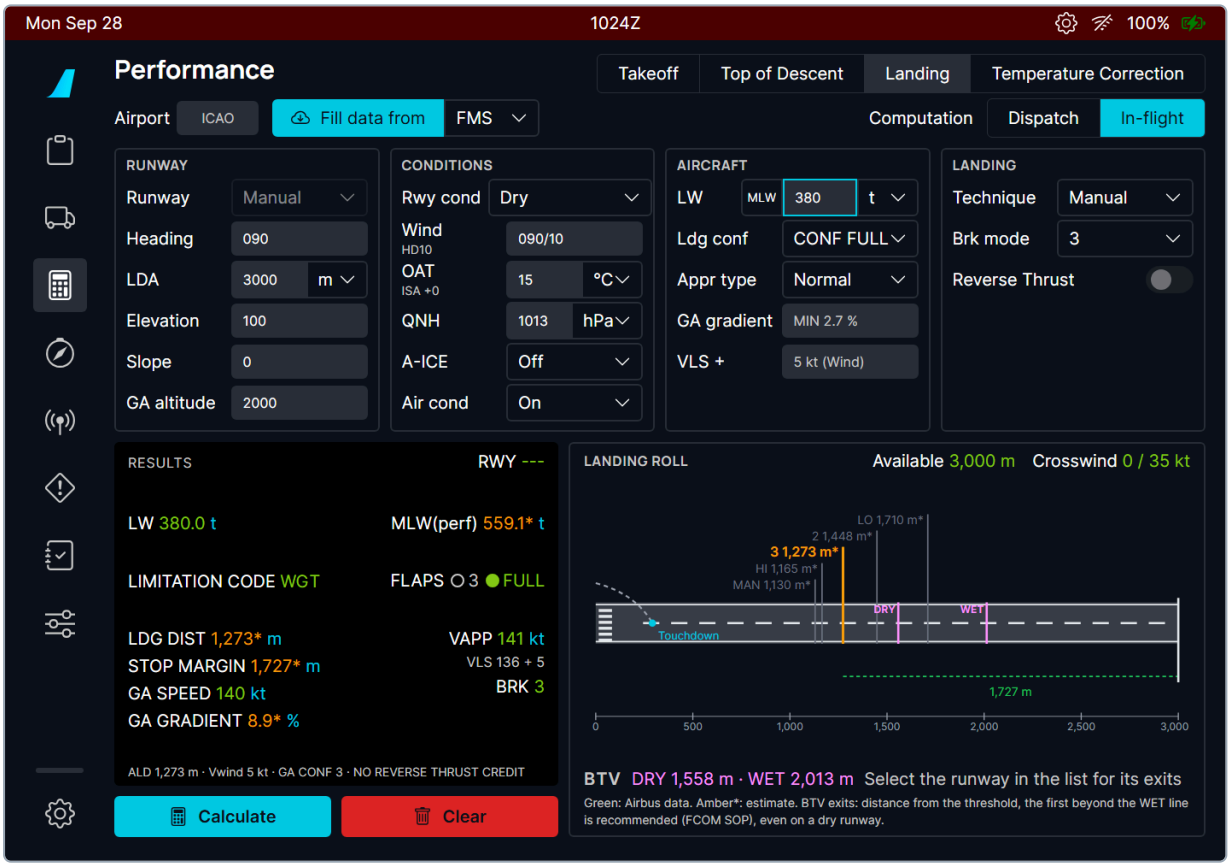


Figure 35. Same landing with autobrake 3: 1507 m. The stop point of the chosen mode is highlighted; the others stay visible for comparison.

AUTO CONF (A380, FCOM PER-LND): CONF FULL when its go-around in CONF 3 gives the required gradient, otherwise CONF 3 with a go-around in CONF 2; for an overweight landing, CONF 3 with a go-around in CONF 1+F.

Runway condition and reverse thrust

The runway condition follows the runway condition codes (RWYCC): dry, wet, water and slush 1/4" or 1/2", compacted snow, ice. The crosswind limit of the condition is shown at the top of the landing roll.

- **Reverse thrust** is credited only on a **contaminated** runway (FCOM: no credit on dry and wet runways); the result line says *REVERSE THRUST CREDIT* or *NO REVERSE THRUST CREDIT*.
- On the A380, **BTV is prohibited on a contaminated runway** (FCOM LIM-32-30): the BTV line says so and advises autobrake 3.
- With 1/2" of water or slush, a message reminds you that Airbus does not recommend these distances: a deeper contaminant gives a shorter distance in the data.

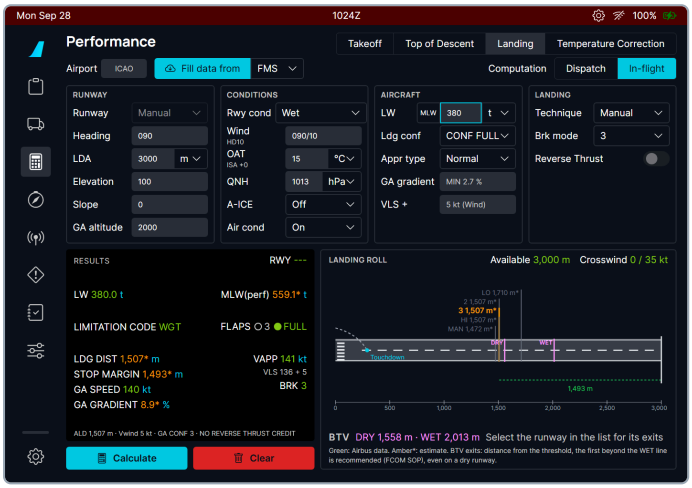


Figure 36. Wet runway, autobrake 3.

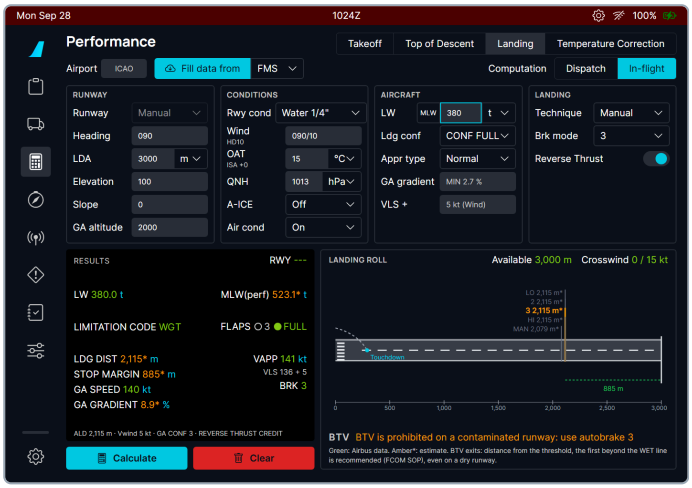


Figure 37. Water 1/4" with reverse thrust: credited, BTV prohibited, crosswind limit 15 kt.

BTV exits (A380)

On the A380, the page shows the **BTV DRY** and **WET** lines: where the aircraft stops with the BTV decelerations (FCOM DSC-32-10-30-20), from 400 m after the threshold plus 5 s of roll. When the runway comes from the list and a Navigraph account is linked, the runway exits of the airport map are drawn and rated:

Exit	Meaning
Recommended	The first exit beyond the WET line (the FCOM SOP choice, even on a dry runway).
Beyond WET	Achievable wet or dry.
Dry only	Between the DRY and WET lines: only on a dry runway.
Not achievable	Before the DRY line.

Without the runway list or the Navigraph data, the line says "*Select the runway in the list for its exits*". This is information for your briefing: the BTV selection itself is made on the OANS, as on the real aircraft.

Dispatch

DISPATCH gives the **required landing distance** used to check a runway before departure: the actual distance divided by 0.6 on a dry runway, ×1.15 on a wet one, and on a contaminated runway the higher of the wet value and ALD ×1.15 (A380 FCOM PER-LND). Braking is manual (the braking mode is greyed out). The orange bar is the required distance.

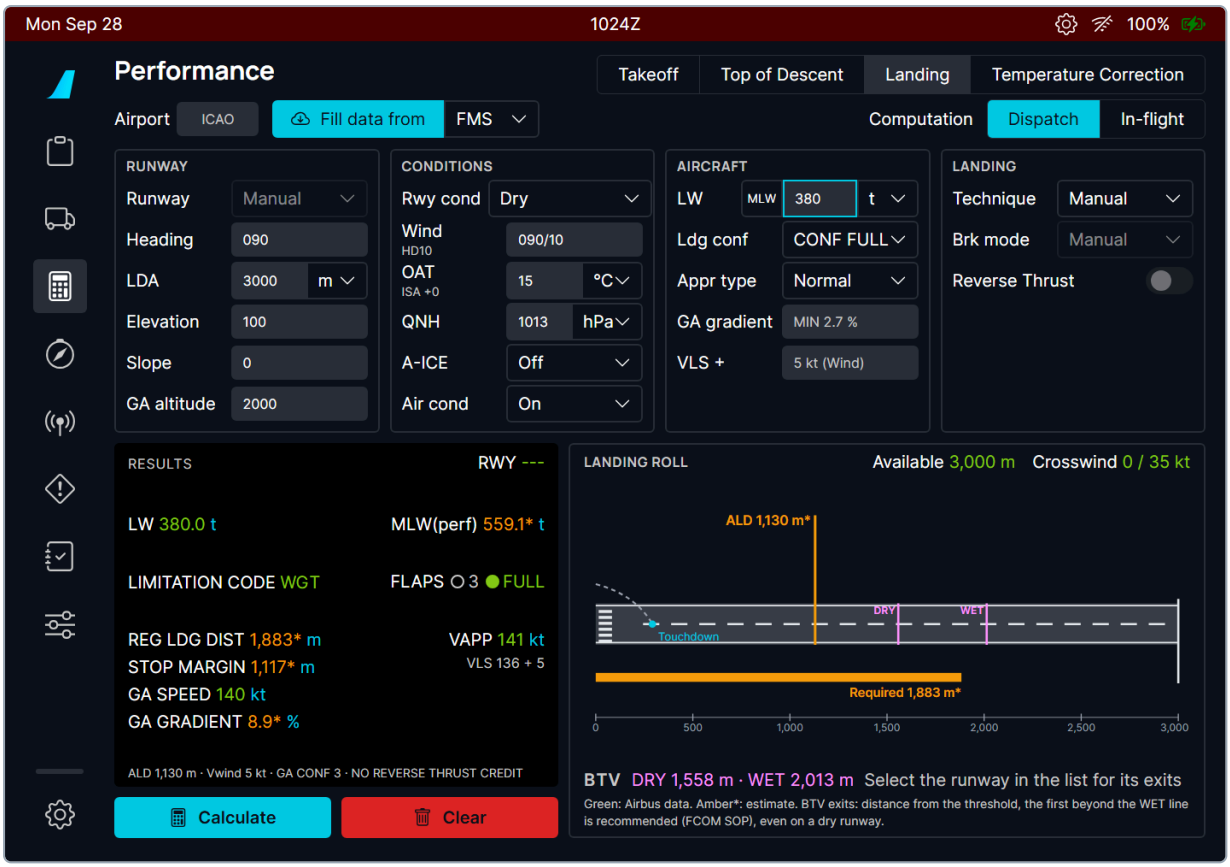


Figure 38. Dispatch on the same runway: required landing distance 1883 m (REG LDG DIST).

When the runway is too short or the aircraft too heavy

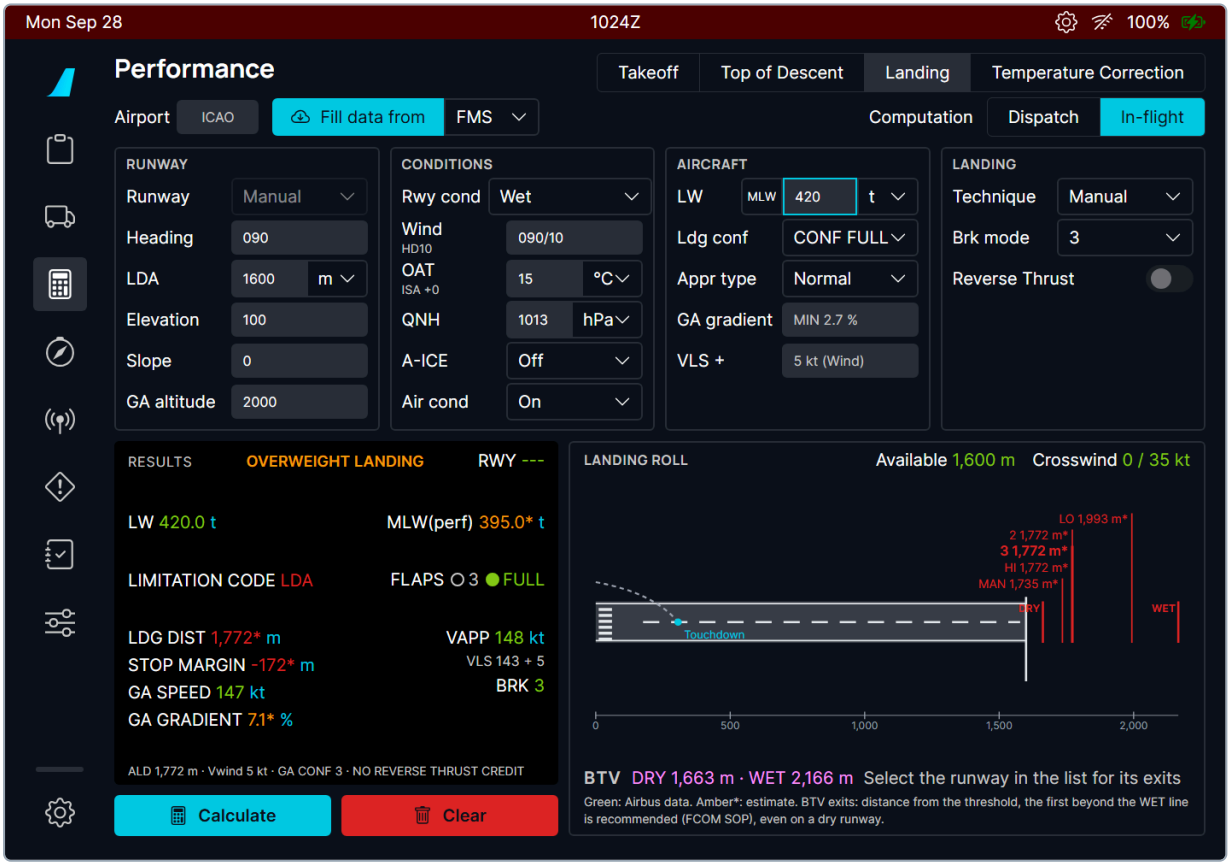


Figure 39. 1600 m wet runway at 420 t: OVERWEIGHT LANDING (above the 395 t MLW), limitation code LDA in red, negative stop margin (-172 m): the aircraft does not stop on this runway with autobrake 3.

The A320 version

The A320 uses the same screen with its own data: the **QRH in-flight landing distances** (shown ×1.15, the QRH margin) and the **FCOM dispatch tables** (PER-LDG-DIS-RLD) with the autoland correction and the wet minimum. Braking modes are Manual, LO and MED; the **Overweight Procedure** switch applies the QRH overweight landing corrections. There is no go-around section.

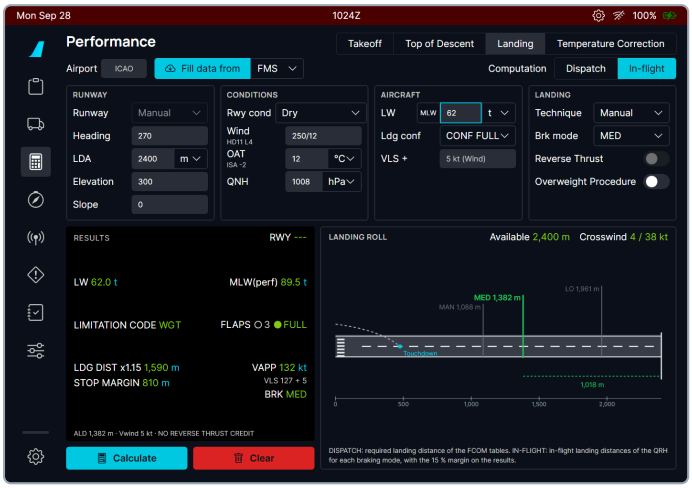


Figure 40. A320 in flight: 62 t, 2400 m dry, autobrake MED: 1382 m (1590 m with the 15 % margin).

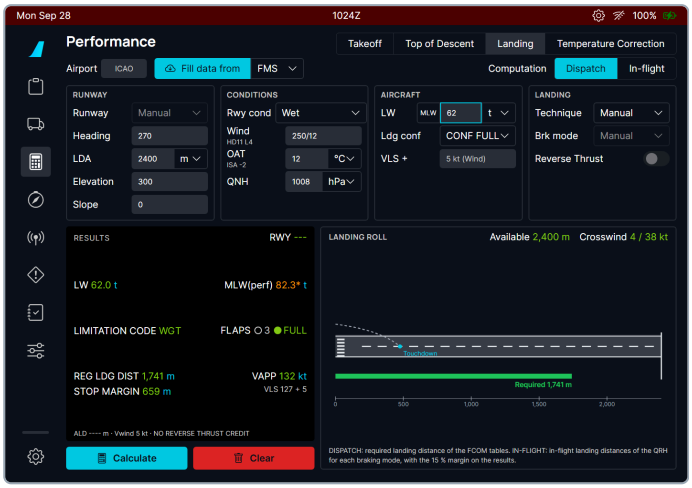


Figure 41. A320 dispatch on a wet runway: required 1741 m from the FCOM tables (green: Airbus data).

What to expect

Situation	What you should see
Inputs missing	Amber labels; the Calculate button stays dim until they are filled.
Fill data from FMS without a flight plan	Only what the FMS has is filled; complete the rest by hand.
Dry or wet runway, reverse thrust on	NO REVERSE THRUST CREDIT: the FCOM gives none there.
Contaminated runway	Longer distances, lower crosswind limit; A380: BTV prohibited.
Landing weight above the MLW	OVERWEIGHT LANDING in the RESULTS; A380 AUTO CONF lands in CONF 3.
LW above MLW(perf)	Limitation code LDA or ACG in red; the stop margin may be negative.
Change of an input after Calculate	Press Calculate again.

Limitations and why

Limitation	Why	What to do
Most A380 values are estimates (amber *).	The only public A380 landing data is the landing field length chart of the Airbus A380 Aircraft Characteristics (dry, manual, CONF FULL). The other cases use a model fitted on it (0.8 % error on the chart) and the FCOM rules.	Use the results for training and planning, not as certified data.
Autobrake and BTV distances are estimates.	They come from the autobrake decelerations of the FCOM, not from Airbus tables.	-
BTV exits need the runway from the list and a Navigraph account.	The exits come from the Navigraph airport map (the OANS data).	Choose the runway in the list; link Navigraph.
No BTV on the A320.	The A320 has no BTV.	-

The go-around gradient is calibrated on one FCOM example.	No A380 go-around tables are public.	-

How to use

1. In flight, with the destination, the arrival runway and PERF APPR (QNH, temperature, wind) in the FMS: flyPad > **Performance** > **Landing**, **Fill data from** FMS. Otherwise type the airport, choose the runway and enter the conditions.
2. Keep IN-FLIGHT (DISPATCH before departure), check the landing weight and the configuration, choose the braking mode and reverse thrust.
3. Press **Calculate**; read LDG DIST, STOP MARGIN, VAPP and the landing roll; on the A380 note the recommended BTV exit.

How to test

#	Do this	Expected result
1	A380X in flight (or on the ground with the destination entered), destination and arrival runway in the flight plan, PERF APPR filled. flyPad > Performance > Landing, Fill data from FMS.	Airport, runway, landing weight, QNH, temperature, wind and configuration filled.
2	Dry runway, manual braking, **Calculate**.	RESULTS: LW, MLW(perf), limitation code, FLAPS, LDG DIST, STOP MARGIN, GA SPEED / GRADIENT, VAPP; the landing roll with the stop point of each braking mode.
3	Choose autobrake 3 and Calculate.	A longer LDG DIST; the stop point of mode 3 highlighted.
4	With a Navigraph account linked and the runway from the list: look under the landing roll.	The BTV DRY and WET lines and the runway exits rated (recommended = first beyond WET; dry only; not achievable).
5	Set Water 1/4", reverse thrust on, Calculate.	REVERSE THRUST CREDIT; "BTV is prohibited on a contaminated runway: use autobrake 3"; lower crosswind limit at the top of the landing roll.
6	Dry runway with reverse thrust on.	NO REVERSE THRUST CREDIT.
7	Choose DISPATCH and Calculate.	REG LDG DIST (actual / 0.6 on dry); the braking mode is greyed.
8	IN-FLIGHT, landing weight 420 t on a short wet runway.	OVERWEIGHT LANDING, limitation code LDA in red, negative stop margin when the runway is too short.
9	A32NX: same page.	Braking modes Manual, LO, MED; Overweight Procedure switch; LDG DIST x1.15 in IN-FLIGHT; no go-around section.

6. A380 landing calculator: BTV braking mode

> **Aircraft:** A380X | **Where:** flyPad > Performance > Landing > LANDING > Brk mode | **Status:** Tested in the sim by the user: OK. Unit tests.

What it is for

The A380 can land with the Brake To Vacate (BTV) autobrake. The calculator gave the BTV DRY and WET lines and the exit advice, but BTV was not a braking mode. A380 FCOM DSC-32-20-220: the BTV landing distance is the BTV DRY line on a dry runway and the BTV WET line on a wet one (autoland plus autobrake distances until an exit is selected). **BTV** is now the last entry of the *Brk mode* list.

What you should see

Situation	What you should see
IN-FLIGHT, dry runway, Brk mode BTV	LDG DIST = the BTV DRY line of the landing roll; the BTV stop point is drawn with the other modes.
IN-FLIGHT, wet runway, BTV	LDG DIST = the BTV WET line.
Contaminated runway, BTV	Error "BTV: dry or wet runway only", no result (BTV is prohibited there, FCOM LIM-32-30). With another mode, the landing roll shows no BTV stop point.
DISPATCH	Braking is manual (the mode is greyed), as before.

Limitations and why

Limitation	Why	What to do
The BTV distance is an estimate (amber).	The BTV lines come from the BTV decelerations of the FCOM (DSC-32-10-30-20), not from Airbus tables.	Use it for planning only.
Selecting BTV here does not arm BTV in the cockpit.	On the aircraft the BTV runway and exit are selected on the OANS.	Select the exit on the OANS.

How to use

1. On the Landing page, in IN-FLIGHT, set *Brk mode* to **BTV** on a dry or wet runway.
2. Press **Calculate** and compare LDG DIST with the exits shown under the landing roll.

How to test

#	Do this	Expected result
1	A380X, flyPad > Performance > Landing, IN-FLIGHT, a runway from the list, dry, Brk mode **BTV**, Calculate.	LDG DIST equals the distance of the BTV DRY line; STOP MARGIN = LDA minus it.
2	Set the runway condition to Wet, Calculate.	LDG DIST equals the BTV WET line (longer).
3	Set Water 1/4" (or any contaminated condition) with BTV, Calculate.	Error "BTV: dry or wet runway only"; no result.
4	Keep the contaminated runway, choose autobrake 3, Calculate.	A result; the landing roll has no BTV stop point.

7. A380 landing calculator: 40 kt crosswind on dry and wet runways

> **Aircraft:** A380X | **Where:** flyPad > Performance > Landing | **Status:** Tested in the sim by the user on dry and wet runways: OK. Unit test.

What it is for

The landing calculator limited the crosswind to 35 kt on dry and wet runways (A380 FCOM PER-LND-CTA-CWD). The A380 FCOM revision for the EA engines (the GP7200 of the FBW A380X) increases the maximum certified crosswind at

landing from 35 kt to **40 kt**, gust included. The contaminated runway values are unchanged: 20 kt with slush or dry snow, 15 kt with standing water or compacted snow, 5 kt on ice. The 30 kt takeoff limit is unchanged too.

What you should see

Situation	What you should see
Dry or wet runway, crosswind up to 40 kt	The calculation runs without a question; the crosswind limit shown at the top of the landing roll is 40 kt.
Crosswind above 40 kt	A question "Crosswind above 40 kt": "The crosswind of N kt is above the maximum of 40 kt for this runway condition. Calculate anyway?"
Contaminated runway	The lower limit of the condition (20, 15 or 5 kt), as before.

How to use

1. Enter the wind as usual (or Fill data from FMS / METAR); the page computes the crosswind component.
2. If the question appears, answer it: the result is computed anyway when you accept.

How to test

#	Do this	Expected result
1	A380X, Landing page, a dry runway of heading 090, wind 180/38 kt, Calculate.	No crosswind question (before: question above 35 kt); limit 40 kt on the landing roll.
2	Wind 180/42 kt, Calculate.	Question "Crosswind above 40 kt ... Calculate anyway?".
3	Wet runway, wind 180/38 kt.	No question.
4	Water 1/4", wind 180/18 kt.	Question "Crosswind above 15 kt".

8. Descent calculator

> **Aircraft:** A380X and A32NX | **Where:** flyPad > Performance > Top of Descent | **Status:** Tested in the sim by the user: A32NX OK (after the fix of the last feature of this manual), A380X OK. Unit tests.

What it is for

The FMS computes the top of descent (T/D) for the managed descent. The crew still needs other answers: how long an emergency descent takes, where to start a descent at a given V/S, whether the aircraft is late on the profile after a shortcut, and a cross-check of the FMS T/D. The A380 OIS has an in-flight descent module for this (FCOM PER-IFT-DES); this page does the same on both aircraft, and replaces the old Top of Descent calculator.

It uses the **same flight and engine models as the FBW FMS predictions**, so its T/D agrees with the one drawn on the ND for the same inputs.

The screen

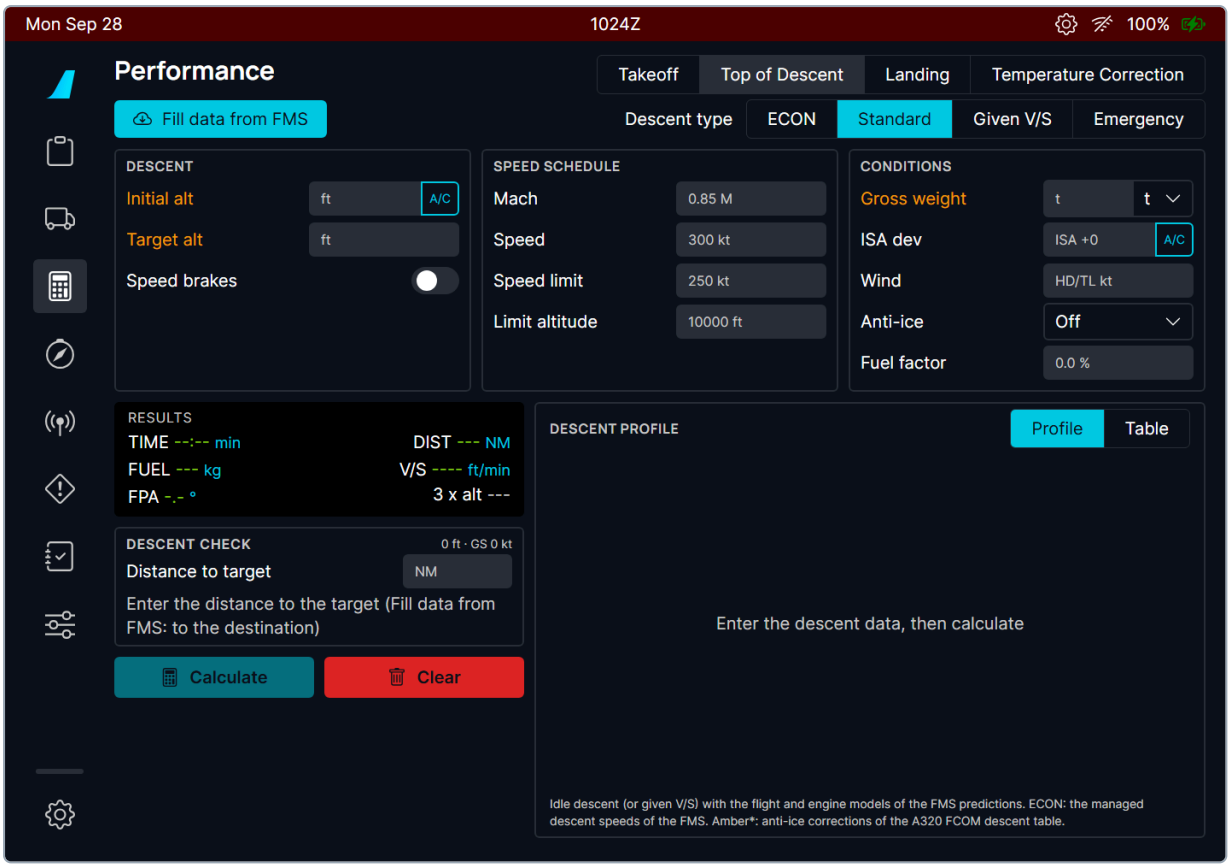


Figure 42. The descent calculator. Amber labels: inputs still needed.

Part	Contents
Fill data from FMS	Cruise level (or the current altitude when lower), destination elevation + 1500 ft, gross weight, the FMS speed limit, the ECON speeds and the distance to the destination.
Descent type	**ECON**: the managed descent speeds of the FMS. **Standard**: the FCOM speed schedule (A380 M.85 / 300 kt / 250 kt below FL100; A320 M.78 / 300 / 250). **Given V/S**: a constant vertical speed. **Emergency**: MMO / VMO with the speed brakes.
DESCENT	Initial altitude (**A/C**: the current one) and target altitude; speed brakes.
SPEED SCHEDULE	Mach, speed, speed limit and its altitude. Empty = the standard values shown in grey.
CONDITIONS	Gross weight, ISA deviation (**A/C**: from the current temperature), wind component (positive = headwind, negative = tailwind), anti-ice (off / engine / total), fuel factor.
RESULTS	Time, distance, fuel, mean V/S and flight path angle, and the "3 × altitude" rule of thumb for comparison; the crossover altitude and where the aircraft slows to the speed limit.
DESCENT CHECK	With the distance to the target: when to start down, or how late you are.
DESCENT PROFILE	Profile chart or table every 5000 ft.

Standard descent

1. Press **Fill data from FMS** in cruise with a flight plan, or type the initial altitude, the target altitude and the weight.
2. Choose the descent type and, if needed, the wind component and anti-ice.
3. Press **Calculate**.

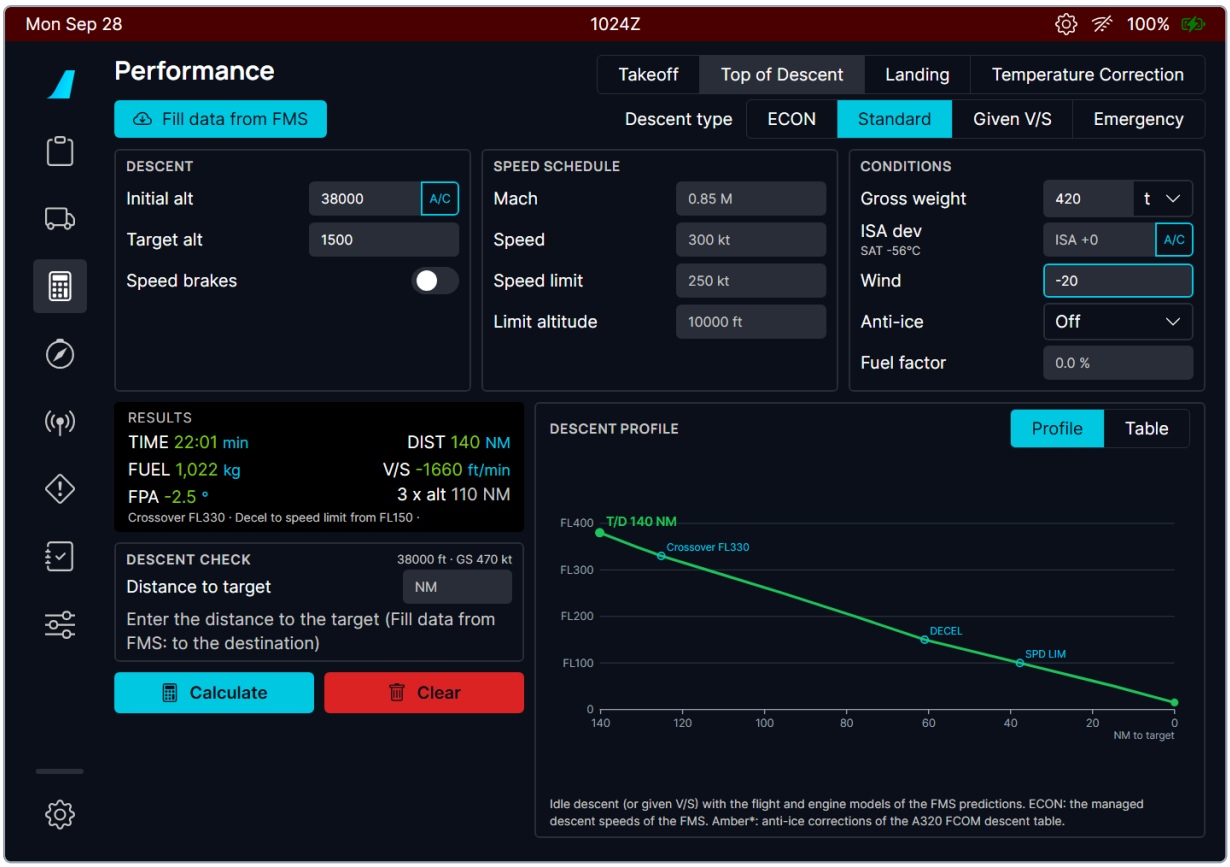


Figure 43. FL380 to 1500 ft, 420 t, 20 kt tailwind, standard speeds: 22 min, 140 NM, 1022 kg, -1660 ft/min on average (-2.5°). The chart shows the T/D, the crossover altitude (Mach to CAS), the deceleration (DECEL) and the speed limit (SPD LIM).

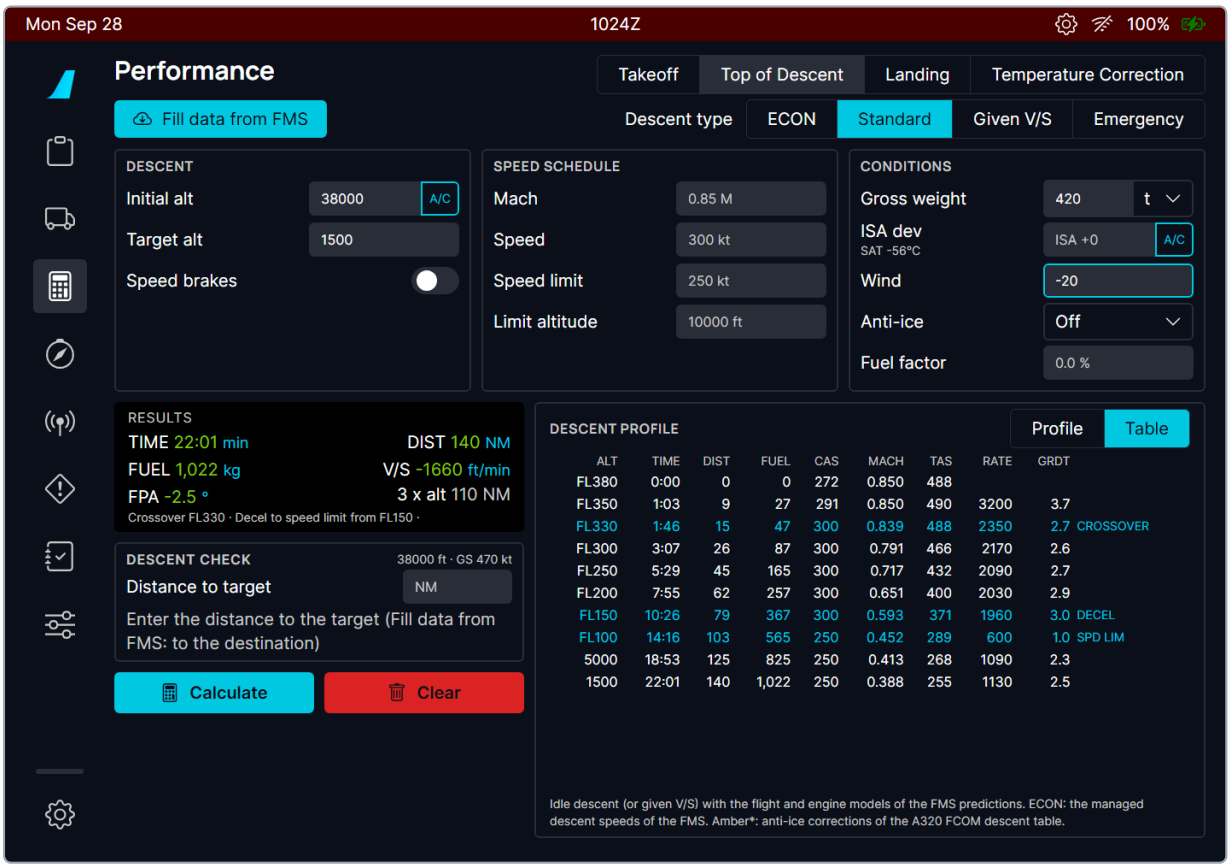


Figure 44. Table view: time, distance, fuel, CAS, Mach, TAS, RATE (rate of descent, ft/min) and GRDT (descent gradient, degrees) every 5000 ft, named and signed like the FCOM table results (PER-IFT-DES-DSR: both positive in descent); the crossover, DECEL and SPD LIM lines in blue. The RESULTS line gives V/S and FPA negative, as set on the FCU.

The deceleration to the speed limit is flown on the -1° idle path of the FMS descent strategy, which is why the profile flattens between DECEL and SPD LIM.

Descent check

Enter the **distance to the target** (Fill data from FMS enters the distance to the destination). The page reads the aircraft altitude and ground speed (shown top right of the box) and compares the aircraft with the profile. The aircraft is the triangle on the chart.

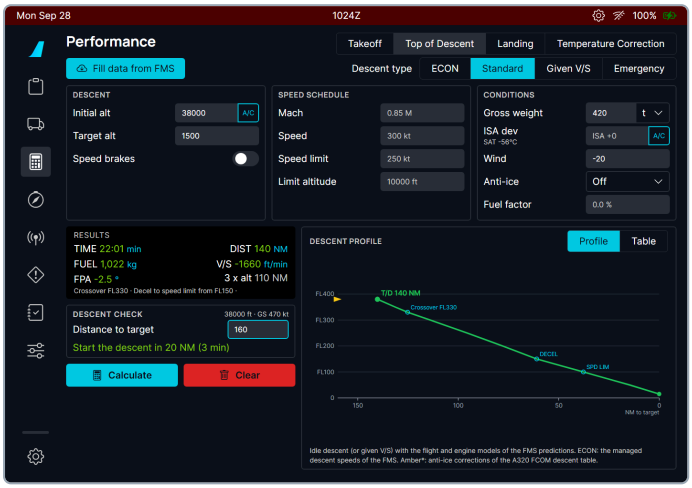


Figure 45. 160 NM to go, T/D at 140 NM: "Start the descent in 20 NM (3 min)".

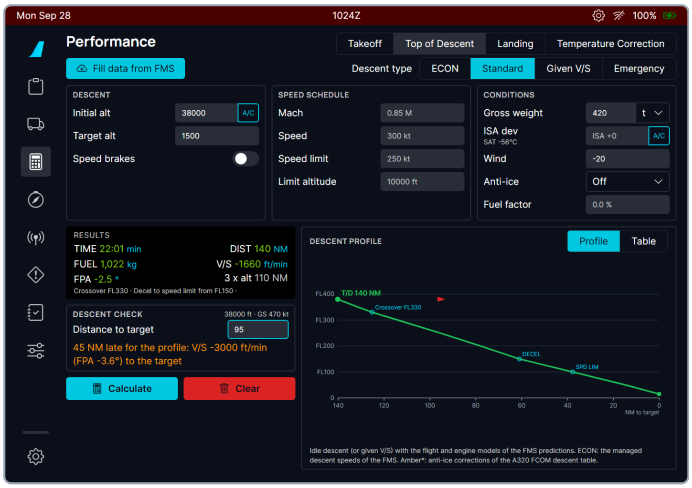


Figure 46. 95 NM to go at FL380: "45 NM late for the profile: V/S -3000 ft/min (FPA -3.6°) to the target".

> **TIP**
>
> After a shortcut, the late message gives the V/S (or FPA) that still makes the target, to set in V/S or FPA mode, or tells you to use the speed brakes.

Given V/S, emergency and ECON

Given V/S adds a V/S field. The page flies that V/S with the thrust it needs. Where the V/S is steeper than an idle descent can give, the page flies idle there and says so in amber (the aircraft cannot descend faster at idle without speed brakes).

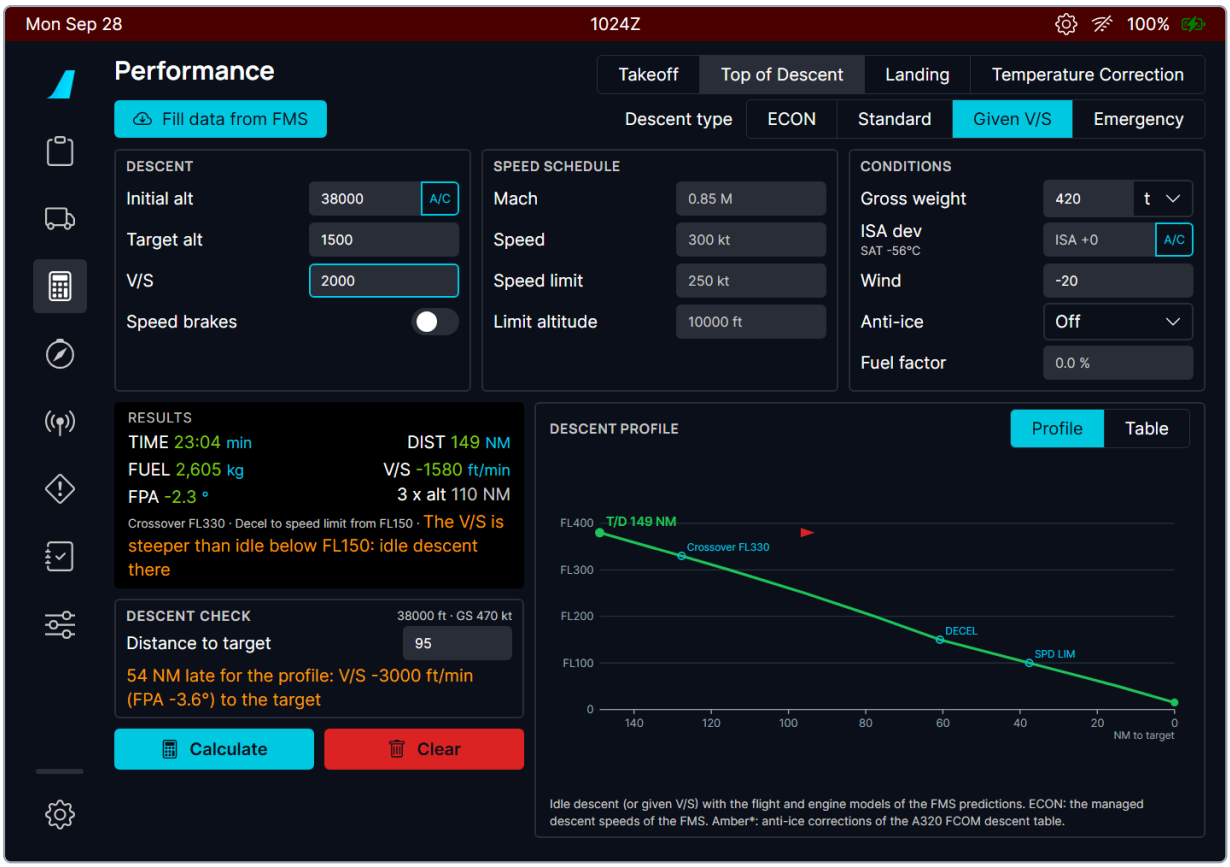


Figure 47. 2000 ft/min: shallower than idle at high level (149 NM, more fuel: 2605 kg), and steeper than idle below FL150, where the page flies idle.

Emergency uses MMO / VMO with the speed brakes extended (the schedule fields turn grey).

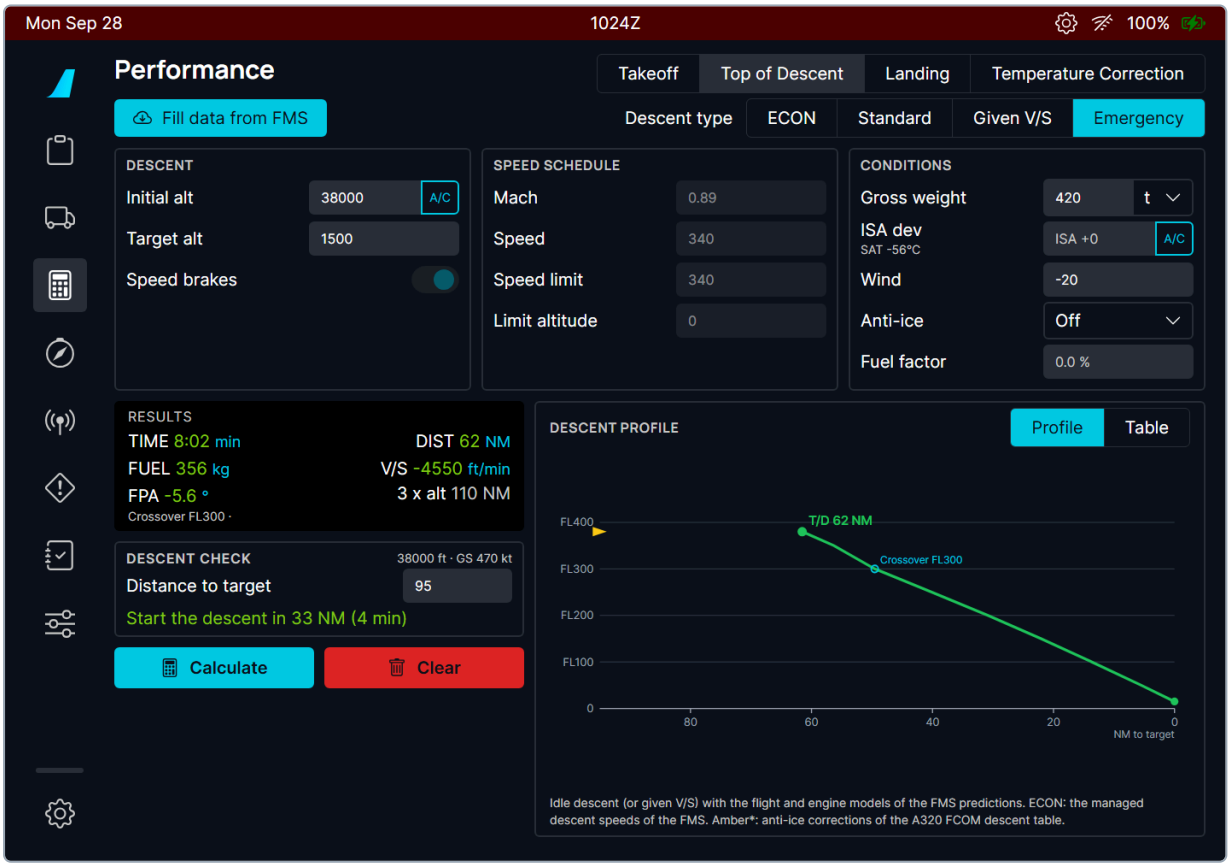


Figure 48. Emergency descent from FL380: 8 min, 62 NM, -4550 ft/min on average.

ECON needs the FMS: it asks the FMS for its managed descent speeds. Without a flight plan in the FMS, a message says so and the page stays on the previous type.

Figure 49. ECON without FMS data.

Anti-ice, fuel factor and estimates

Anti-ice (engine or total) raises the idle thrust and flattens the descent. Its effect uses the corrections of the A320 FCOM descent table (no A380 table is public): the values it changes turn **amber with a star**. The fuel factor (a percentage) scales the fuel burn, like the FMS PERF FACTOR.

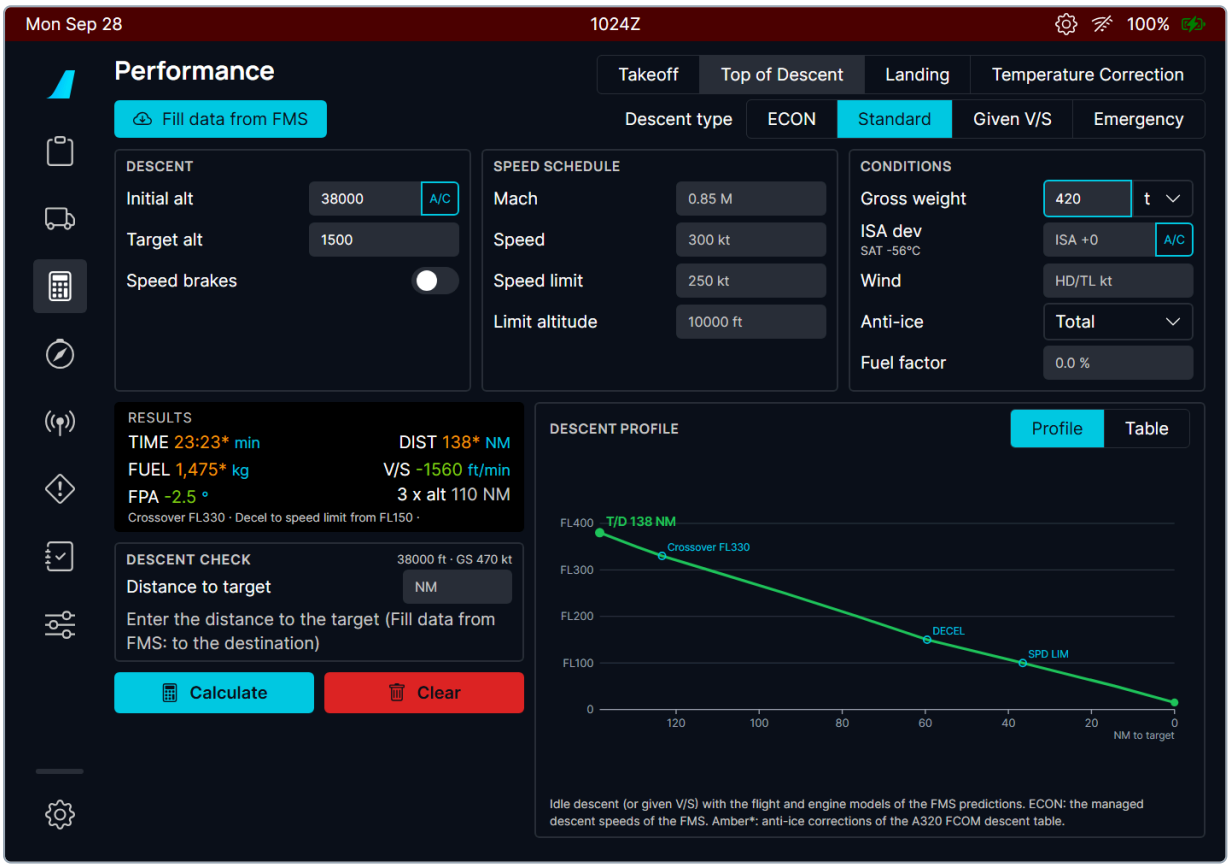


Figure 50. Total anti-ice: 138 NM, 23 min, 1475 kg, in amber (estimate).

What to expect

Situation	What you should see
Inputs missing	Amber labels; no results.
Any input changed	The results are cleared: press Calculate again.
Target above the initial altitude	Message "The target altitude must be below the initial altitude".
Initial altitude above the aircraft ceiling, or weight out of range	A message gives the limit (maximum altitude, or the weight range from the empty weight to the MTOW).
Given V/S steeper than idle	Amber line: idle descent below the altitude it names.
ECON without FMS flight plan	"No FMS data: enter a flight plan in the FMS".
Distance to target larger than the descent	Green "Start the descent in X NM (Y min)".

Distance shorter than the descent	Amber "N NM late for the profile" with the V/S and FPA needed.

Limitations and why

Limitation	Why	What to do
The descent is a still-air profile plus a mean wind.	The wind is one component for the whole descent, as in the FCOM and OIS; the FMS uses the wind of each altitude.	Use the mean wind of the descent.
About 8 % longer than the A320 FCOM table.	The FCOM table is for the CFM56 A320; the FBW A320neo engine model idles differently.	Compare with the FMS T/D, which uses the same model.
Anti-ice values are estimates.	Only the A320 FCOM has anti-ice descent corrections.	-
No descent path constraints or approach.	The page computes a clean descent to one target altitude, not the FMS path with its constraints.	Use it as a cross-check of the FMS.

How to use

1. In cruise with a flight plan: flyPad > **Performance** > **Top of Descent**, **Fill data from FMS** (or type the initial and target altitudes and the weight).
2. Choose the descent type (ECON, Standard, Given V/S, Emergency), the wind component and anti-ice.
3. Press **Calculate**; read TIME, DIST, FUEL, V/S and FPA, and the DESCENT CHECK line (start in X NM, or late with the V/S needed).

How to test

#	Do this	Expected result
1	A380X in cruise (e.g. FL380) with a flight plan to a destination. Top of Descent page, **Fill data from FMS**.	Initial altitude, target (destination elevation + 1500 ft), gross weight, speed limit, ECON speeds and the distance to the destination filled.
2	Descent type Standard, **Calculate**.	RESULTS TIME / DIST / FUEL / V/S / FPA and 3 x alt; the profile with T/D, crossover, DECEL and SPD LIM.
3	Switch to the Table view.	Rows every 5000 ft with RATE and GRDT positive; the RESULTS line keeps V/S and FPA negative.
4	Compare the T/D distance with the FMS T/D on the ND.	About the same (same models).
5	Enter a distance to target longer than DIST.	Green "Start the descent in X NM".
6	Enter a distance shorter than DIST.	Amber "N NM late for the profile: V/S ... (FPA ...) to the target".
7	Given V/S 2000 ft/min, Calculate.	Amber line "The V/S is steeper than idle below ...: idle descent there" where it applies.
8	Emergency, Calculate.	MMO / VMO with speed brakes; schedule fields grey.
9	Repeat on the A32NX.	Same page and behaviour with the A320 speeds (M.78 / 300 / 250).

9. Descent calculator: T/D distance slider

> **Aircraft:** A380X and A32NX | **Where:** Top of Descent page, under the profile | **Status:** No separate sim-test record of the slider (the descent page itself was tested OK, see above); checked in the browser harness.

What it is for

After a calculation, a quick answer to "what if I start down earlier (vectors) or later (a shortcut)?" without a new calculation: the V/S and path angle the moved descent needs, against the calculated ones.

What if the descent starts earlier or later?

After a calculation, the **T/D distance** slider under the profile moves the start of the descent: to the left farther from the target (earlier, e.g. vectors), to the right closer (later, e.g. a shortcut), up to 60 NM each way. The chart draws the moved top of descent dashed, straight to the target, and the line under the slider gives the V/S and the path angle it needs, to compare with the calculated ones.

- **Later**: steeper than the idle descent: the figures turn amber, "speed brakes or a higher speed needed" (red when the calculation already used the speed brakes: start earlier).
- **Earlier**: shallower than the idle descent: "thrust needed, more fuel".
- The figures keep the mean ground speed of the calculated descent: an estimate for the decision, not a new calculation.

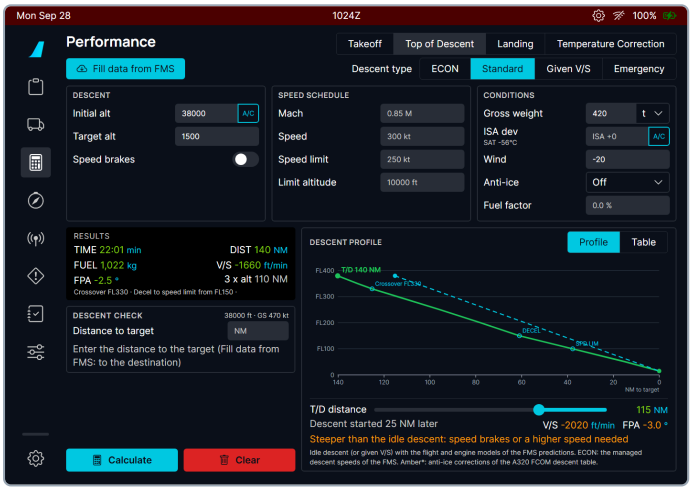


Figure 51. Slider 25 NM later: 115 NM, -2020 ft/min, -3.0°: steeper than idle.

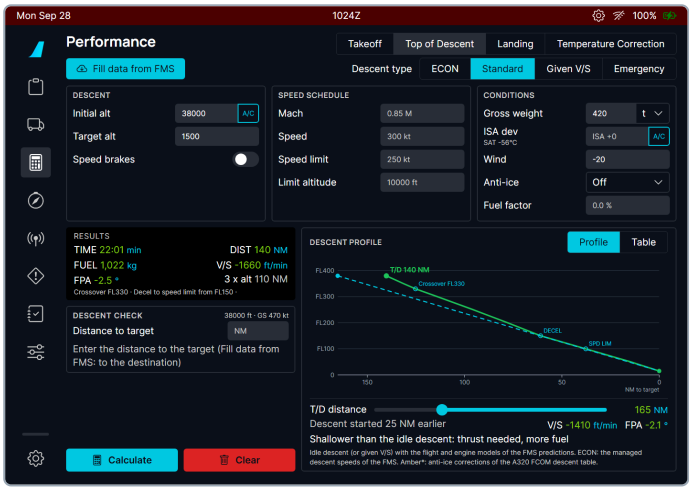


Figure 52. Slider 25 NM earlier: 165 NM, -1410 ft/min, -2.1°: shallower.

The **Calculate** and **Clear** buttons are at the bottom of the page, where they are on the takeoff and landing pages.

How to use

1. Calculate a descent.
2. Move the **T/D distance** slider under the profile: left = earlier (farther from the target), right = later.
3. Read the line under the slider: the V/S and FPA needed, and whether speed brakes or thrust are needed.

How to test

#	Do this	Expected result
1	Calculate a standard descent from FL380 (e.g. 420 t, 20 kt tailwind).	Slider centred: "Descent as calculated".
2	Move the slider 25 NM to the right.	"Descent started 25 NM later", steeper V/S and FPA in amber, "Steeper than the idle descent: speed brakes or a higher speed needed"; dashed line from the moved T/D to the target.
3	Move it 25 NM to the left.	"Descent started 25 NM earlier", shallower values, "Shallower than the idle descent: thrust needed, more fuel".

4	Emergency type (speed brakes in), slider to the right.	Red "Steeper than the descent with speed brakes: start the descent earlier".
5	Look at the buttons.	Calculate and Clear at the bottom of the page.

10. Descent calculator: target waypoint from the flight plan

> **Aircraft:** A380X and A32NX | **Where:** Top of Descent page > DESCENT > Target wpt | **Status:** Tested in the sim by the user: OK. Unit tests.

What it is for

The target of a descent is often a waypoint with an altitude constraint (the first constraint of the STAR, a crossing restriction from ATC), not the destination. The FMS (A380 FMC, A32NX MCDU) now sends the waypoints ahead of the aircraft, up to the missed approach, with their altitude constraints and their distance along the route. The **Target wpt** list of the flyPad picks one as the target.

What you see

- The list shows each waypoint with its constraint written as on the F-PLN: `9000` (at), `+9000` (at or above), `-9000` (at or below), `11000/9000` (window). A dash = no target waypoint (the destination).
- Picking a waypoint sets the **target altitude** from its constraint (the upper altitude of a window, to the nearest 100 ft; a waypoint without a constraint keeps the target altitude you entered) and the **distance to target** along the flight plan.
- The distance follows the aircraft: it is refreshed every 5 s. When the waypoint is sequenced, the target waypoint is cleared.
- Typing a distance yourself clears the target waypoint (the distance is no longer the waypoint's).

Limitations and why

Limitation	Why	What to do
The waypoints need the FMS.	They come from the active flight plan.	Without a flight plan the list holds only the dash.
Climb constraints of the SID are not used as targets.	An "at or above altitude 2" constraint is a climb constraint.	-

How to use

1. In cruise, with a STAR in the flight plan: Top of Descent page, **Fill data from FMS**.
2. Open **Target wpt** and pick the waypoint with the constraint you must meet.
3. Press **Calculate** and watch the DESCENT CHECK line as you fly.

How to test

#	Do this	Expected result
1	A380X or A32NX in cruise with a flight plan that has a STAR with altitude constraints. Top of Descent page.	The Target wpt list shows the waypoints ahead with their constraints (e.g. `ABCDE +11000`).
2	Pick a waypoint with an "at" constraint.	Target alt = the constraint; Distance to target = its distance along the flight plan (compare with the F-PLN).
3	Pick a waypoint with a window constraint.	Target alt = the upper altitude of the window.

4	Calculate and wait 10 s.	The distance to target decreases as the aircraft flies (refresh every 5 s).
5	Type a distance in the Distance to target field.	The Target wpt list goes back to the dash.

11. Descent calculator with an FMS that sends no waypoints

> **Aircraft:** A32NX (any aircraft whose FMS sends no waypoint list) | **Where:** Top of Descent page | **Status:** User-confirmed in the sim on the A32NX: no more crash. Unit tests.

What it is for

An FMS built before the waypoints were added to the descent data (for example an older A32NX MCDU) answers without a waypoint list. The page then crashed with the flyPad error screen ("fmsWaypoints.map is not a function"). The descent data now always carries a waypoint list, empty when the FMS sends none.

What you should see

Situation	What you should see
FMS without waypoints	The page opens and Fill data from FMS works; the Target wpt list holds only the dash.
Current FMS	The waypoints as in the previous feature.

How to use

1. Nothing to do: open the Top of Descent page as usual.
2. After an update, install the flyPad and the FMS (MCDU / MFD) together: they exchange the descent data.

How to test

#	Do this	Expected result
1	A32NX in flight with a flight plan. flyPad > Performance > Top of Descent.	The page opens (no error screen).
2	Press **Fill data from FMS**, then Calculate.	Data filled, results shown, no crash.
3	Delete the flight plan on the MCDU, open the page again and press Fill data from FMS.	No crash; a message says what the FMS lacks (e.g. "No FMS data: enter a flight plan in the FMS"); the Target wpt list holds only the dash.

Cockpit printers

The A380X MFD and ATC COM print on the paper of the pedestal printer, whose buttons work, and both aircraft get a printer paper door and no torn paper stuck in the slit

Aircraft	A380X (printing, buttons) and A32NX (paper door, no torn paper edge)
Where to find it	Pedestal printer; A380X MFD FMS > DATA > PRINTER, SEC INDEX, ATC COM ATIS and MSG RECORD; SD MAILBOX PRINT
Branch	`core/printer`, based on `core/a380x-atccom-datalink` (stacked: needs `core/surveillance`, `core/a380x-mfd-fms` and `core/a380x-atccom-datalink` merged first). The 3D parts are in four aircraft-large-files companion changes: pedestal paper 36b4bd9, A380X door 27d6123, A380X ON light 8c14d21, A32NX door 99237fb (fork TCdr/aircraft-large-files).
Status	FMS and ATIS printing: tested in the sim, FMS print again. Mailbox PRINT: OK (MSG RECORD print not sim-tested). Paper doors: A32NX OK; A380X looked at during the work and committed at the user's request. No torn paper edge: no sim test recorded. Printer buttons and ON light: OK.
Pictures	MFD pages in the local browser harness (build installed in the sim); the paper sheets rendered by the printer page of the build, with the printouts of the unit tests (test flight plan LFPG - EGLL). No pictures for the doors, the slit and the buttons.

1. A380X pedestal printer: FMS pages and the ATIS on paper
2. A380X ATC COM print: SD mailbox and MSG RECORD
3. Printer paper door (A380X and A32NX)
4. No torn paper edge left in the printer slit
5. A380X printer control panel: ON/OFF, SLEW, TEST, ABORT

1. A380X pedestal printer: FMS pages and the ATIS on paper

> **Aircraft:** A380X | **Where:** MFD FMS > DATA > PRINTER, SEC INDEX, ATC COM ATIS; pedestal printer | **Status:** Tested in the sim by the user with the paper added to the installed cockpit model (feed animation, tear-off, sheets on the pedestal and the CPT table, FMS pages and ATIS): OK; FMS print again OK. Unit tests: 26 FMS print + printer tests, ATIS print.

What it is for

The A380 has a printer on the pedestal. The FMS prints its data (initialization, takeoff, winds, flight reports) and the ATC COM prints the ATIS, so the crew has them on paper. The printer now works as on the FBW A320: the sheet comes out of the printer, you tear it off and read it.

Printing

From	What prints
FMS > DATA > PRINTER	F-PLN INIT, T.O DATA, WIND DATA; the active flight plan PRE-FLIGHT, IN-FLIGHT and POST-FLIGHT reports; the SEC 1 to 3 reports; the AUTO PRINT options (company data on reception, reports at engine start, takeoff and shutdown)
FMS > SEC INDEX	PRINT of a secondary flight plan
ATC COM > ATIS	PRINT of one ATIS, PRINT ALL (all the ATIS on one printout), and the auto print (printer icon) when an ATIS is received

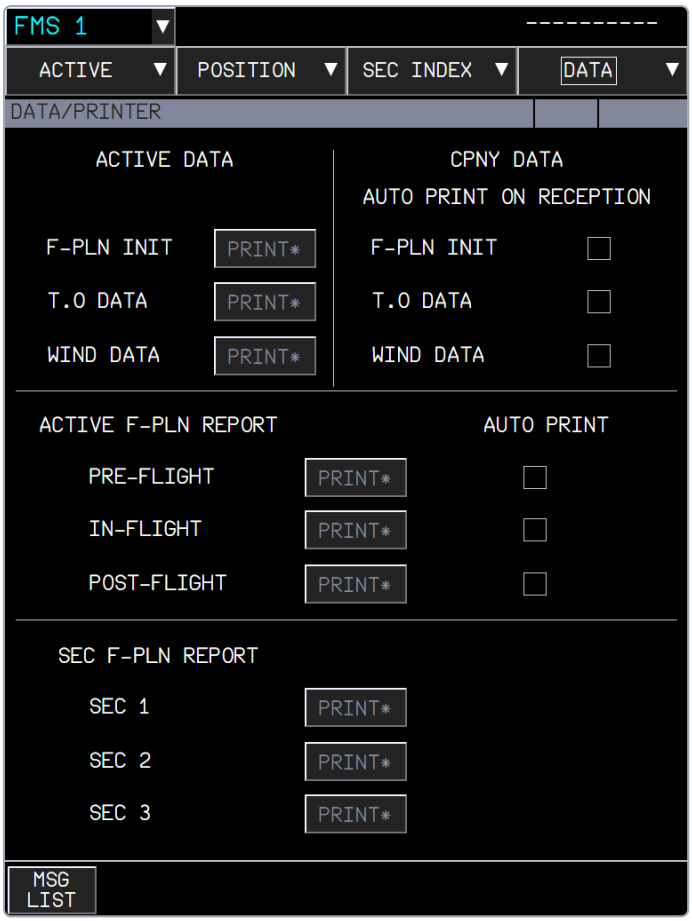


Figure 1. FMS DATA / PRINTER: the PRINT buttons and the AUTO PRINT options.

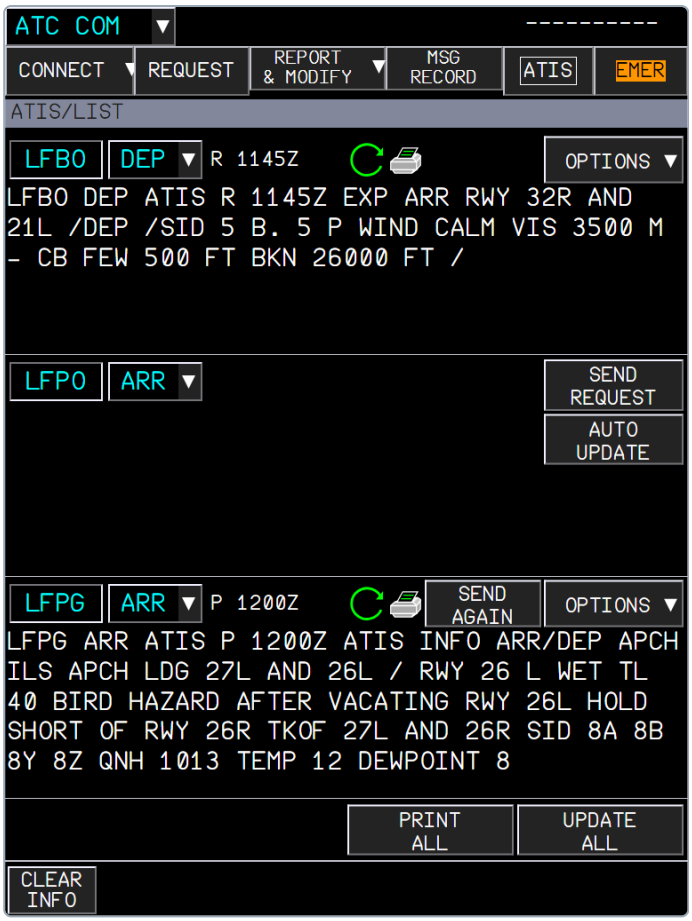
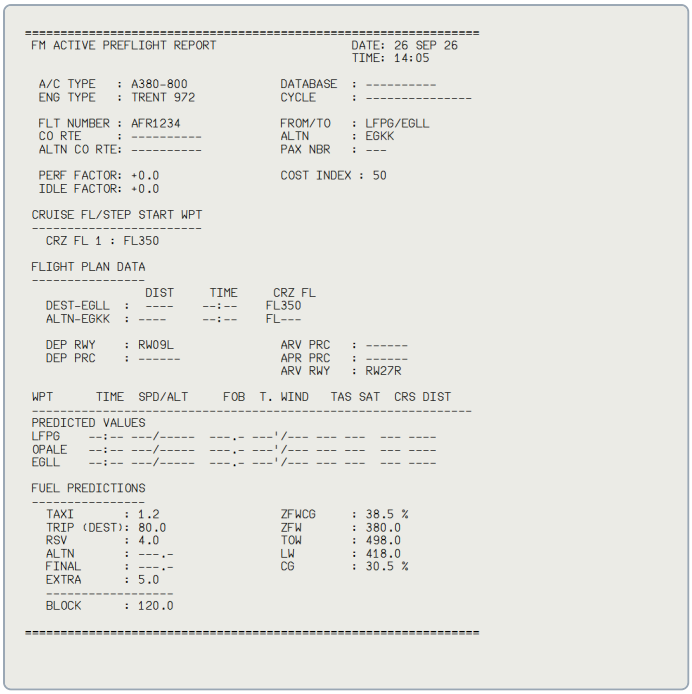


Figure 2. ATC COM ATIS: the printer icon (auto print) and PRINT ALL.

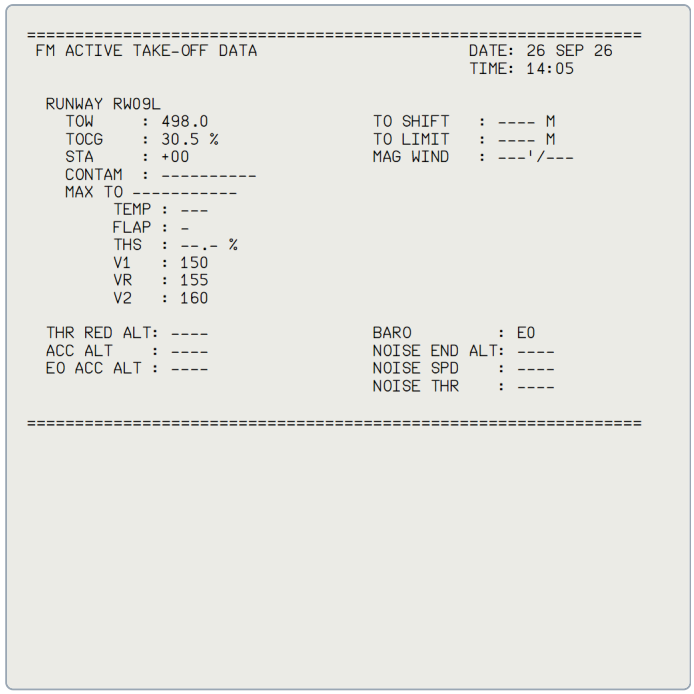
The paper

- Each printout starts with its title, the date and the time (UTC); an ATIS printout shows the time of the ATIS message.
- A sheet holds 45 lines: a longer page (e.g. a flight report) comes out on several sheets, one after the other.
- Click the sheet in the printer to tear it off: it goes on the pedestal (or the captain's pull-out table), where you can page through the torn sheets and discard them, as on the A320.



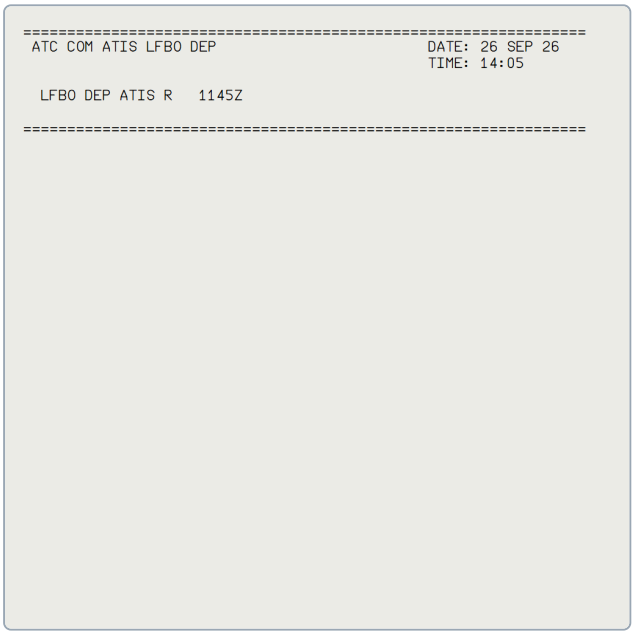
```
================================================================
 FM ACTIVE PREFLIGHT REPORT                  DATE: 26 SEP 26
                                             TIME: 14:05

  A/C TYPE   : A380-800            DATABASE  : ----------
  ENG TYPE   : TRENT 972           CYCLE     : ---------------

  FLT NUMBER : AFR1234             FROM/TO   : LFPG/EGLL
  CO RTE     : ----------          ALTN      : EGKK
  ALTN CO RTE: ----------          PAX NBR   : ---

  PERF FACTOR: +0.0                COST INDEX : 50
  IDLE FACTOR: +0.0

 CRUISE FL/STEP START WPT
 ------------------------
   CRZ FL 1 : FL350

 FLIGHT PLAN DATA
 ----------------
                 DIST     TIME     CRZ FL
   DEST-EGLL  :  ----     --:--    FL350
   ALTN-EGKK  :  ----     --:--    FL---

   DEP RWY    : RW09L               ARV PRC   : ------
   DEP PRC    : ------              APR PRC   : ------
                                    ARV RWY   : RW27R

 WPT      TIME  SPD/ALT     FOB  T. WIND   TAS SAT  CRS DIST
 ------------------------------------------------------------
 PREDICTED VALUES
 LFPG    --:-- ---/-----  ---.- ---'/--- --- ---  --- ----
 OPALE   --:-- ---/-----  ---.- ---'/--- --- ---  --- ----
 EGLL    --:-- ---/-----  ---.- ---'/--- --- ---  --- ----

 FUEL PREDICTIONS
 ----------------
   TAXI       : 1.2                 ZFWCG     : 38.5 %
   TRIP (DEST): 80.0                ZFW       : 380.0
   RSV        : 4.0                 TOW       : 498.0
   ALTN       : ---.-               LW        : 418.0
   FINAL      : ---.-               CG        : 30.5 %
   EXTRA      : 5.0
   ------------------
   BLOCK      : 120.0

================================================================
```

Figure 3. FM ACTIVE PREFLIGHT REPORT.



```
================================================================
 FM ACTIVE TAKE-OFF DATA                     DATE: 26 SEP 26
                                             TIME: 14:05

  RUNWAY RW09L
    TOW      : 498.0                 TO SHIFT   : ---- M
    TOCG     : 30.5 %                TO LIMIT   : ---- M
    STA      : +00                   MAG WIND   : ---'/---
    CONTAM   : ----------
    MAX TO ------------
          TEMP : ---
          FLAP : -
          THS  : --.- %
          V1   : 150
          VR   : 155
          V2   : 160

  THR RED ALT: ----                  BARO          : EO
  ACC ALT    : ----                  NOISE END ALT: ----
  EO ACC ALT : ----                  NOISE SPD     : ----
                                     NOISE THR     : ----

================================================================
```

Figure 4. FM ACTIVE TAKE-OFF DATA.



```
================================================================
 ATC COM ATIS LFBO DEP                       DATE: 26 SEP 26
                                             TIME: 14:05

   LFBO DEP ATIS R   1145Z

================================================================
```

Figure 5. ATC COM ATIS printout.

Limitations and why

Limitation	Why
The paper needs the aircraft-large-files companion	the FBW A380 cockpit model has no paper nodes: they are in the aircraft-large-files companion *pedestal printer paper* (36b4bd9 on the fork TCdr/aircraft-large-files). Without it the printouts still reach the flyPad Printouts page
The paper looks like the A320 one	the paper, its animation and its buttons come from the A320 printer

How to use

1. Printer powered (AC 1) and ON (green window on the printer control panel).
2. MFD FMS > DATA > **PRINTER**: press the PRINT of the page or report; or SEC INDEX > PRINT; or ATC COM > ATIS: PRINT (OPTIONS menu), PRINT ALL, or the printer icon for the auto print.

3. Watch the sheet come out of the printer, click it to tear it off.
4. Read the torn sheets on the pedestal (or the CPT pull-out table when it is out): page through them, discard them.

How to test

#	Do this	Expected result
1	A380X with the paper companion model installed, on the ground, AC power, flight plan entered (e.g. LFPG - EGLL). MFD FMS > DATA > PRINTER: press PRINT for F-PLN INIT.	A sheet comes out of the printer slit: title, date and UTC time, then the page in 64 columns; no characters cut at the right edge.
2	Print a long report (PRE-FLIGHT report).	Several sheets of 45 lines, one after the other (about 2.5 s apart).
3	Click the sheet in the printer.	It is torn off and lies on the pedestal beside the printer; the page buttons page through the torn sheets; discard removes them.
4	Pull out the CPT table, print again and tear the sheet off.	The sheet lies on the table and slides aft with the meal table.
5	ATC COM > ATIS with an ATIS received: OPTIONS > PRINT, then PRINT ALL with two areas filled.	PRINT: the ATIS of that area, with the time written in the ATIS. PRINT ALL: all the ATIS on ONE printout, one after the other.
6	Tick the printer icon (AUTO PRINT) of an area and wait for a new ATIS version.	The new version prints when it is received.
7	MFD FMS > SEC INDEX: PRINT of a secondary flight plan.	The SEC report prints.

2. A380X ATC COM print: SD mailbox and MSG RECORD

> **Aircraft:** A380X | **Where:** SD MAILBOX PRINT; MFD ATC COM > MSG RECORD | **Status:** Mailbox PRINT tested in the sim by the user (a route clearance printed): OK. MSG RECORD PRINT ALL / PRINT not sim-tested (unit tests).

What it is for

The FCOM lets the crew print an ATC message from the mailbox (DSC-46-10-20-60) and the message record from the MFD (DSC-46-10-20-30 P 29, DSC-46-10-40 P 30). These buttons said PRINTER NOT AVAIL; they now print on the pedestal printer, through the same queue as the FMS prints.

What you see

From	What prints	Printout title
SD MAILBOX: PRINT	The message displayed in the mailbox; PRINTING in the information area while it prints	ATC COM MAILBOX
MFD ATC COM > MSG RECORD (list): PRINT ALL	All the recorded messages: for each one its time, direction, station and status, then its text, in 64 columns	ATC COM MSG RECORD
MFD ATC COM > MSG RECORD (zoom): PRINT	The message shown in the zoom	ATC COM MSG RECORD

A stray closing brace at the end of the printed CPDLC text is gone. With the printer off or unpowered, the buttons show PRINTER NOT AVAIL (see the printer buttons below).

How to use

1. SD mailbox: with the message displayed, press **PRINT**.

2. MFD ATC COM > **MSG RECORD**: PRINT ALL for the whole record, or ZOOM a message and PRINT it.
3. Tear the sheet off the pedestal printer as for any printout.

How to test

#	Do this	Expected result
1	A380X, datalink connected, printer ON. Receive an uplink (a live controller, or the self-sent route clearance of `core/a380x-atccom-datalink`). Press **PRINT** in the SD mailbox.	PRINTING in the mailbox information area; a sheet titled ATC COM MAILBOX with the message comes out of the printer, without a } at the end.
2	Close a few dialogues, open MFD ATC COM > MSG RECORD, press **PRINT ALL**.	One printout ATC COM MSG RECORD: each message with its time, direction, station and status, then its text.
3	ZOOM one message and press **PRINT**.	A printout of that message only.
4	Set the printer OFF (ON/OFF button) and press PRINT again.	PRINTER NOT AVAIL; nothing prints.

3. Printer paper door (A380X and A32NX)

> **Aircraft:** A32NX and A380X | **Where:** Pedestal printer: paper door and its latch / handle | **Status:** A32NX: tested in the sim by the user, OK. A380X: looked at in the sim during the work and committed at the user's request after the model crash fix; no separate test record of the final version.

What it is for

Neither cockpit model had an animated printer door. A click on the paper door (or its latch on the A380X, its handle on the A32NX) now opens or closes it: the door swings up about its forward edge and shows the paper roll and the roller unit under the slit. The A320 FCOM has a PRINTER DOOR LATCH (DSC-45-30); the A380 FCOM does not describe the door: its look and travel are design choices. Opening the door has no effect on printing.

What you see

Situation	What you should see
Click on the closed door (or latch / handle)	The door swings up (about 80 degrees on the A380X): paper roll, roller unit and the paper going up into the slit.
Click again	The door closes.
Printing with the door open (A380X)	The sheet still comes out of the slit; its lower part shows in the open compartment.

Limitations and why

Limitation	Why	What to do
Needs the aircraft-large-files companion models.	The door parts, the compartment and the mechanism are in the cockpit models: A380X 27d6123 (on the pedestal paper 36b4bd9), A32NX 99237fb.	Install them with this branch; a full MSFS restart is needed after a model change.

How to use

1. Click the printer paper door (or its latch / handle) to open it; click again to close it.

How to test

#	Do this	Expected result
1	A380X with the companion models installed (MSFS restarted). Click the printer paper door on the pedestal.	The door swings up about its forward edge; paper roll and roller unit visible; variable `L:A380X_PRINTER_DOOR` = 1.
2	Click the latch.	The door closes (variable back to 0).
3	Print an FMS page with the door open.	Printing works as with the door closed.
4	A32NX with its companion model installed. Click the printer door, then its handle.	The door opens, then closes (`L:A32NX_PRINTER_DOOR`).

4. No torn paper edge left in the printer slit

> **Aircraft:** A32NX and A380X | **Where:** Pedestal printer slit | **Status:** No sim test recorded (behaviour change only).

What it is for

The torn paper edge of the FBW printer (the PRINT_TORN part) stood in the slit whenever the printer was not printing: it looked like a sheet stuck in the printer. It is no longer shown on either aircraft: paper only comes out of the slit while printing.

What you should see

Situation	What you should see
Printer idle	Empty slit, no paper strip.
Printing	The sheet comes out of the slit as before; once torn off, the slit is empty again.

How to use

1. Nothing to set: print and tear off sheets as usual.

How to test

#	Do this	Expected result
1	A32NX cold and dark, look at the pedestal printer.	No paper strip in the slit.
2	Print a page (e.g. MCDU or flyPad print), tear the sheet off.	The sheet comes out while printing; after the tear-off the slit is empty.
3	Same on the A380X (print an FMS page from MFD DATA > PRINTER).	Empty slit before and after the print.

5. A380X printer control panel: ON/OFF, SLEW, TEST, ABORT

> **Aircraft:** A380X | **Where:** Pedestal printer control panel | **Status:** Tested in the sim by the user: OK, the green ON window after the emissive fix of the companion model. Unit tests: PedestalPrinter.spec.ts, FmsPrinter.spec.ts.

What it is for

The A380 cockpit model has the printer control panel (ON/OFF with a green window, SLEW, TEST, ABORT), but nothing was clickable. The printer is supplied by AC 1 (FCOM DSC-46-20-70). The A380 FCOM does not describe the buttons: SLEW, TEST and ABORT follow the usual printer functions (design choices).

What you see

Button	What it does
ON/OFF	Switches the printer on or off. The green window shows while the printer is on and AC 1 is powered. Off or unpowered: an FMS print shows **PRINTER NOT AVAIL** (type II, FCOM DSC-22-FMS-20-110) and nothing prints; the ATC COM prints show their PRINTER NOT AVAIL.
SLEW	Feeds out one blank sheet.
TEST	Prints the test page: PRINTER TEST, date and time, lines of the character set, END OF TEST.
ABORT	Drops the sheets waiting to be printed.

Limitations and why

Limitation	Why	What to do
The green window needs the aircraft-large-files companion.	Its emissive material is in the companion change 8c14d21 (on the door 27d6123).	Install it; restart MSFS.
No buttons on the A32NX printer.	The A32NX model has no printer button parts (the FCOM SLEW switch, DSC-45-30): it needs model work.	-

How to use

1. Keep the printer ON (green window) to print.
2. SLEW to feed a blank sheet, TEST to check the printer, ABORT to cancel a long printout.

How to test

#	Do this	Expected result
1	A380X with the companion models installed, cold and dark. Look at the printer control panel.	Green window off.
2	Switch the batteries and AC power on (AC 1 powered).	Green window on.
3	Press **ON/OFF**.	Green window off (`L:A380X_PRINTER_OFF` = 1).
4	MFD FMS > DATA > PRINTER: PRINT F-PLN INIT.	MFD message **PRINTER NOT AVAIL**; no sheet.
5	Press ON/OFF again, then **TEST**.	Green window on; the test page prints: PRINTER TEST with the date and time, the character lines, END OF TEST.
6	Press **SLEW**.	One blank sheet comes out.
7	Print a long report (PRE-FLIGHT) and press **ABORT** after the first sheet.	The remaining sheets are not printed.

flyPad ground: taxi route, ATC frequencies and GSX

A taxi route page on the Navigraph airport map (with green flags on the A380X OANS), the airport ATC frequencies on it, the Services page linked to GSX, and the GSX external power sync fixed when it is enabled; the taxi route now reads real ATC clearances, and the external power follows the sim's ground power

Aircraft	A32NX and A380X (the OANS flags are A380X only)
Where to find it	flyPad > **Ground** > **Taxi** and **Services**; flyPad > Settings > 3rd Party Options; A380X ND (OANS); overhead ELEC panel EXT PWR
Branch	`core/efb-ground`, based on `core/efb-performance` (stacked: needs `core/surveillance`, `core/a380x-mfd-fms` and `core/efb-performance` merged first; the taxi page uses the runway exits of the landing calculator). `core/efb-redesign` is stacked on it.
Status	Taxi route tested in the sim by the user at RJTT and CYUL, and on both aircraft with the A380X OANS flags: OK. ATC frequencies: OK. GSX power sync: OK. GSX services link: first version tested (four bugs found); the fixes of those bugs on this branch are not sim-tested here (the reworked version on core/efb-redesign was sim-tested). Added (not sim-tested, unit tests only): taxi route along real ATC clearances, GSX power sync setting re-read, external power following the sim ground power.
Pictures	Taxi pages in the local browser harness, running the build installed in the sim, on a small test airport (XTST) made for the pictures; in the sim, the page uses the Navigraph map of the real airport. No pictures for the frequencies and GSX (they need the sim data and GSX).

1. Taxi route page
2. A380X: the accepted taxi route on the OANS
3. Airport ATC frequencies on the taxi page
4. Services page linked to GSX
5. GSX external power sync when it is enabled
6. Taxi route along real ATC clearances
7. GSX power sync: the setting is re-read, so enabling it takes effect
8. External power follows the sim's ground power (GSX GPU, sim GPU, jetway)

1. Taxi route page

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Taxi | **Status:** Tested in the sim by the user at RJTT and Montreal CYUL (the fixes in "Locate the aircraft..."), and on both aircraft: OK. Unit tests of the route finder.

What it is for

Before taxiing, the crew gets a taxi clearance from ATC such as "*taxi to holding point runway 09 via M, K, A*". On an unknown airport, following it on the chart is where mistakes happen: a wrong turn, a runway crossed without clearance. This page draws the route of the clearance on the airport map, lists the runways it crosses and stops it at the runway holding point.

The real A380 has no taxi routing: its OANS (the airport map on the ND) only lets the crew place green flags and white crosses (FCOM DSC-34-10-70-20), and MSFS has no datalink taxi clearance (D-TAXI). So the route is built on the flyPad, and the cockpit displays stay as the FCOM describes them. On the A380, the accepted route is marked on the OANS with those green flags (see "A380X: the accepted taxi route on the OANS" below).

- **Departure** (default): from a gate, a ramp or cargo stand, or the aircraft position, to a runway entry.
- **Arrival**: from the runway exit you plan to take (or the aircraft position) to your gate.
- The page **suggests** the shortest route; you type the taxiways of the ATC clearance and **accept** it.

Before you start

- **Navigraph account** linked in the flyPad settings (third party options). The airport map comes from the Navigraph airport database, the same data as the A380 OANS. Without it the page says "*No airport database*".
- **Airport:** type the ICAO code, or press **ORIG** / **DEST** to take the airports of the SimBrief flight plan loaded in the flyPad. The line below shows how many gates, ramp and cargo stands the map has.
- An airport that is not in the Navigraph airport database shows "*Not in the Navigraph airport database*".

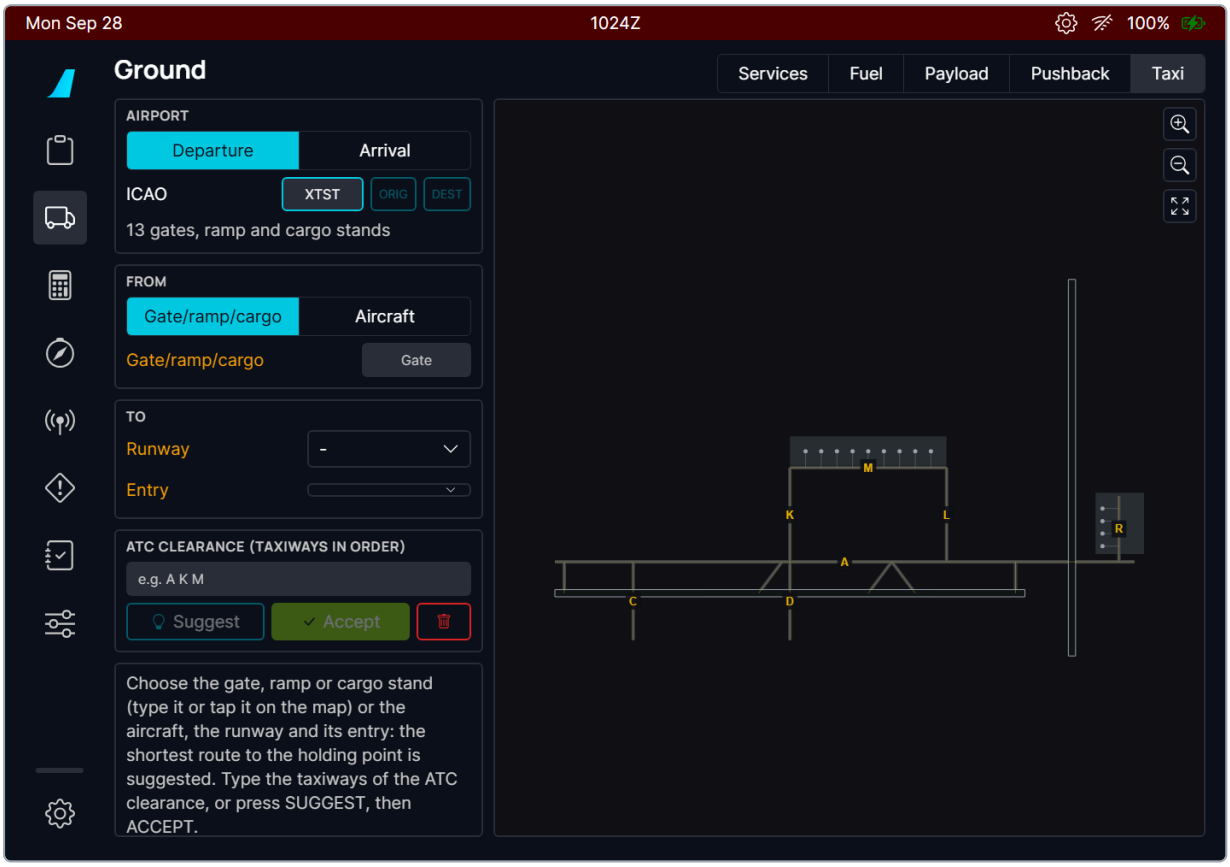


Figure 1. The Taxi page after entering the airport: the airport map on the right, the route inputs on the left. Amber labels show what is still missing.

Part of the page	What it does
AIRPORT	Departure or Arrival, the ICAO code, ORIG / DEST buttons, the number of stands on the map.
FROM	Where the taxi starts: a gate / ramp / cargo stand (departure), a runway exit (arrival), or the aircraft.
TO	Where it ends: the runway and its entry (departure), or the gate (arrival).
ATC CLEARANCE	The taxiways of the clearance, in order. SUGGEST fills it with the shortest route, ACCEPT confirms it, the red bin clears the page.
Route box	The route as a clearance reads, its length, the taxiways and the runway crossings in red.
Map	The airport with the route, the runway entries (cyan), the holding point (amber bar), the aircraft (yellow). Drag to move, buttons to zoom and to fit the route.

Departure: from the gate to the runway

1. Keep **Departure** selected and enter the airport.
2. In FROM, keep **Gate/ramp/cargo** and type your gate. While you type, the gates that match appear as buttons under the field: tap one. You can also **tap the gate on the map** (zoom in to see the names).

3. In TO, choose the **runway**. The **entry** is set to the full length entry; choose another one for an intersection departure.
4. The shortest route appears on the map as a **dashed magenta line** ending at the **HOLD** bar of the runway holding point.

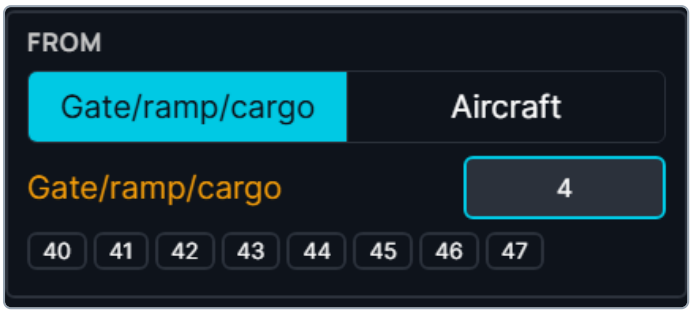


Figure 2. Typing "4" lists the gates 40 to 47: tap one.

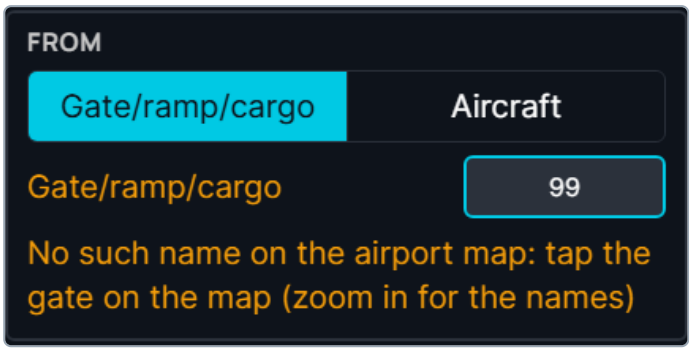


Figure 3. A name that is not on the map: the amber line says so.

Gate names are found even when they are written differently on the map: **"16" finds "016"**, spaces and dashes do not matter. If nothing matches, the airport map uses another name (for example a letter prefix): tap the gate on the map instead.

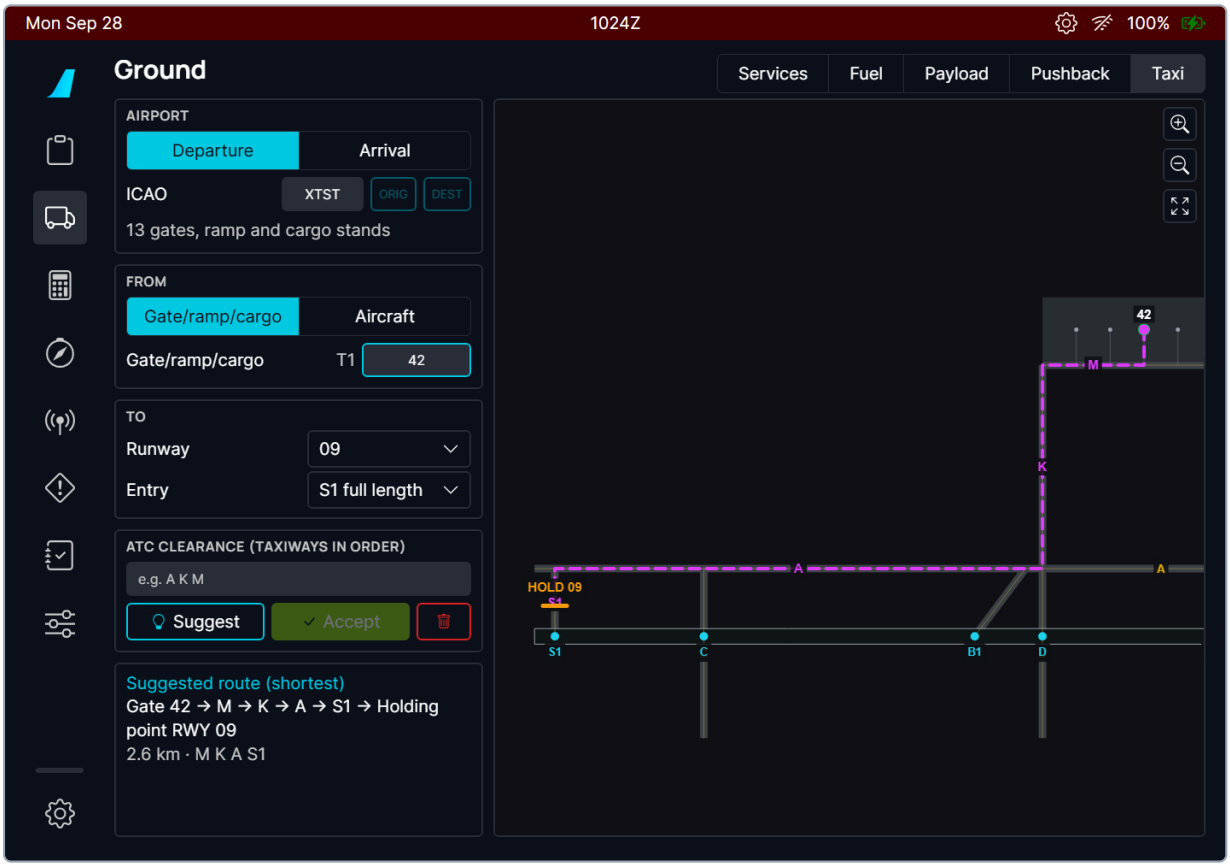


Figure 4. Gate 42 to runway 09, full length: the suggested route (dashed) along M, K and A to entry S1, stopped at the holding point (HOLD 09). The route box reads it as a clearance: Gate 42 → M → K → A → S1 → Holding point RWY 09, 2.6 km.

Runway entries

The **Entry** list shows every taxiway that enters the runway, in order from the threshold. The first one is the **full length** entry; the others show the **runway length remaining** from them, to check an intersection departure against your takeoff performance. On the map, the entries of the selected runway are the cyan dots with their names.

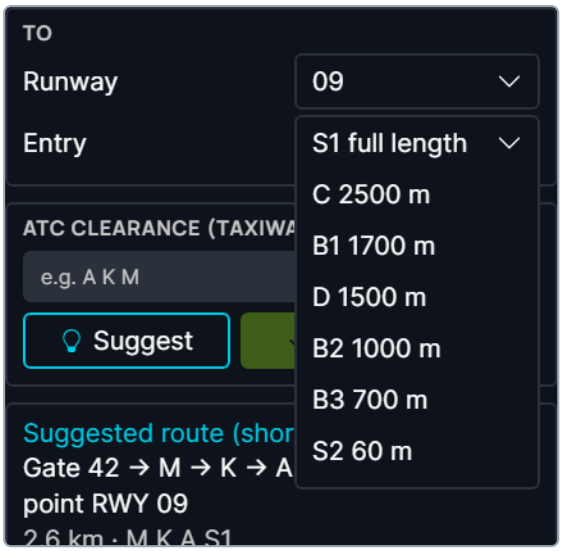


Figure 5. The entries of runway 09: S1 full length, then C (2500 m left), B1 (1700 m), D (1500 m) and so on.

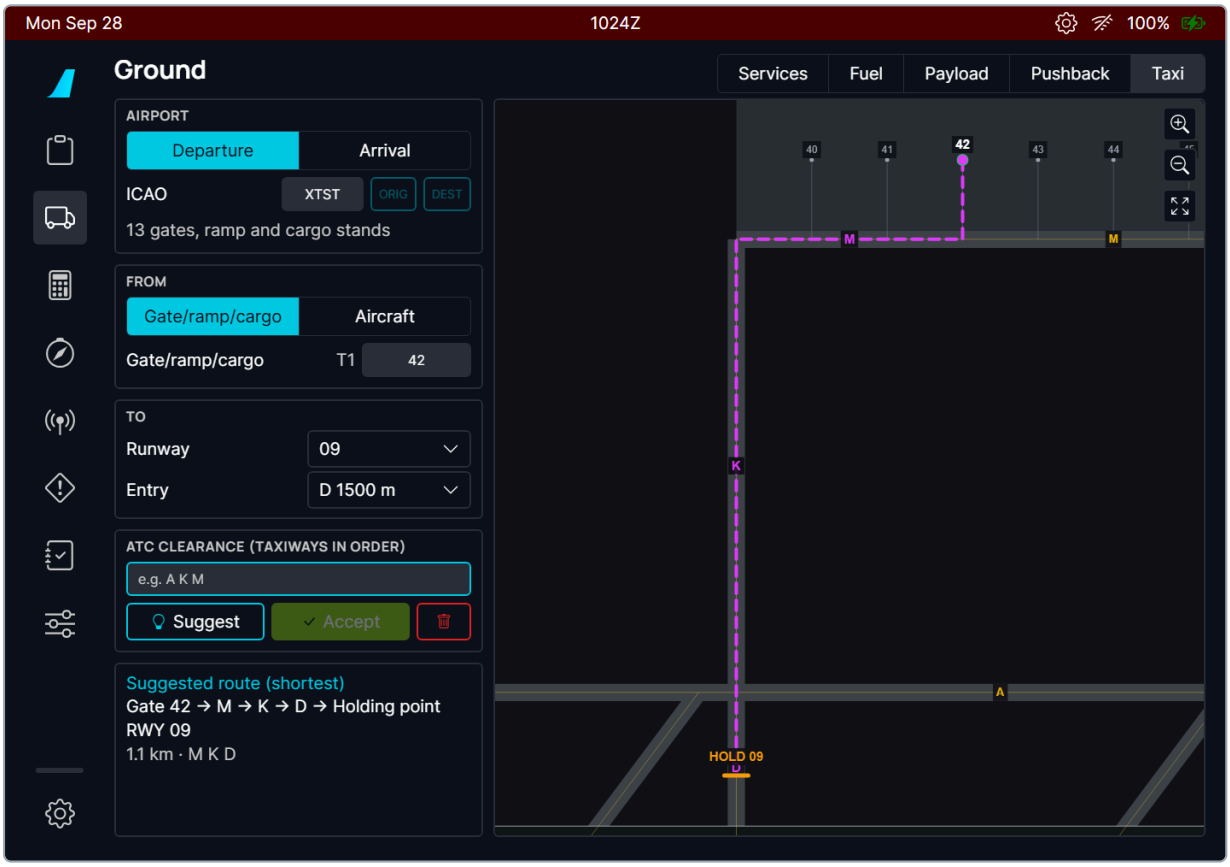


Figure 6. Intersection departure from D: the route stops at the holding point of D, with 1500 m of runway ahead.

The ATC clearance

Type the taxiways of your clearance in the order ATC gives them, separated by spaces (commas and dashes work too). The route then follows them strictly, in that order:

- Short unnamed connecting lines between two cleared taxiways are allowed.
- The runway entry can be left out: "*via M, K, A*" to a holding point still ends on the entry you chose.
- While the typed route is not accepted, the box reads "*Route along the clearance (not accepted)*" and the line stays dashed.

SUGGEST copies the taxiways of the suggested route into the clearance field: a quick start when ATC clears you along the obvious way. **ACCEPT** confirms the route: the line turns **solid**. Any change (gate, runway, entry, clearance) takes the acceptance away.

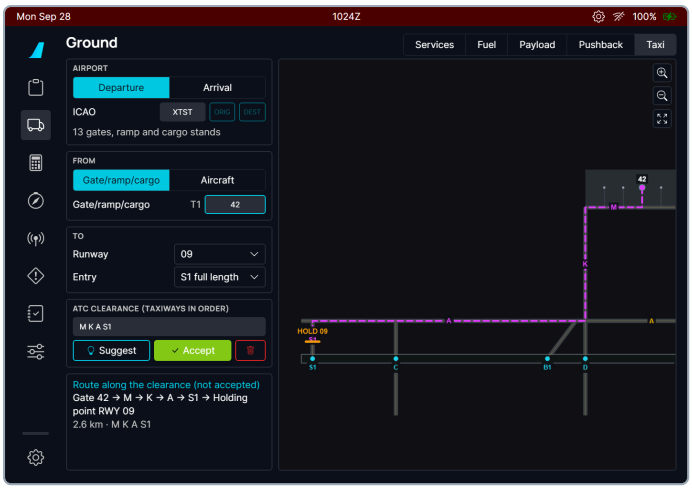


Figure 7. After SUGGEST: the clearance field holds M K A S1, the route is a preview.

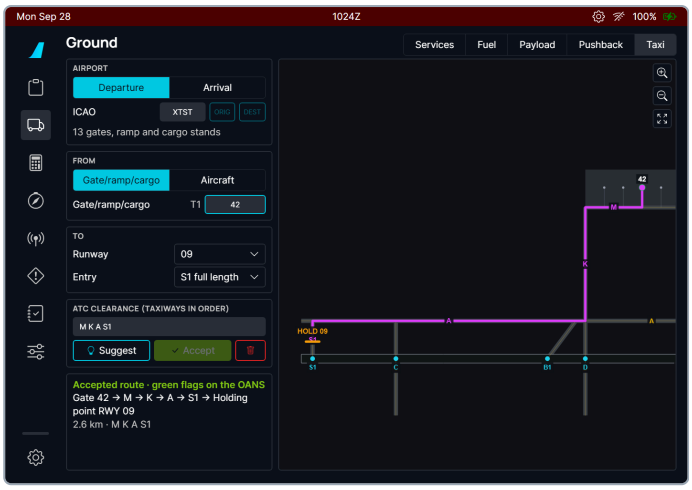


Figure 8. After ACCEPT: solid line, "Accepted route". On the A380 the route is also flagged on the OANS.

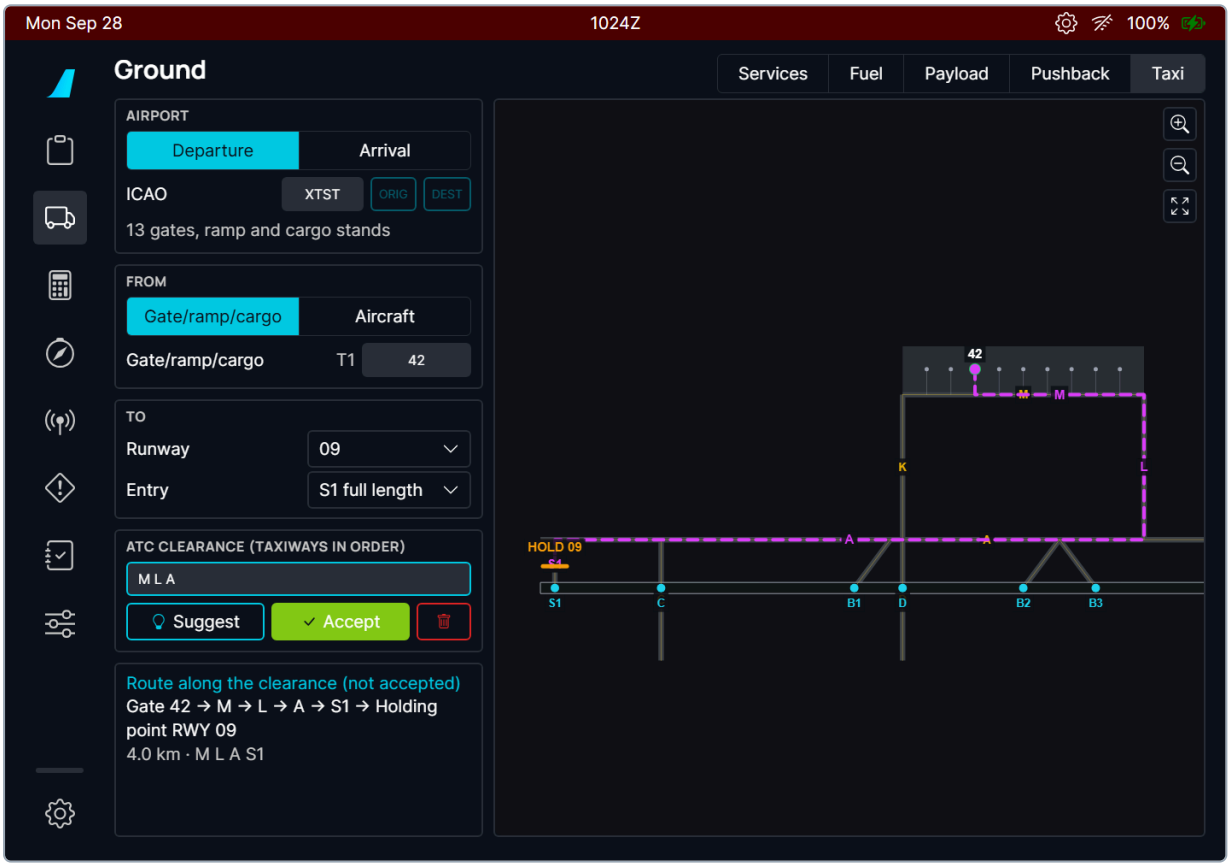


Figure 9. A different clearance, "M L A": the route goes the long way round via L (4.0 km instead of 2.6 km), as ATC asked.

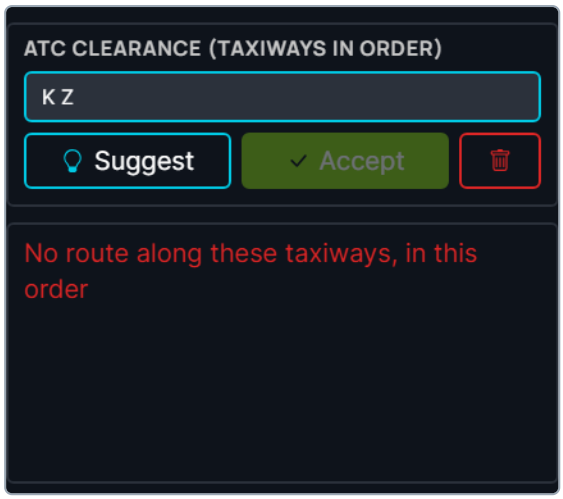


Figure 10. A clearance with a taxiway that does not exist here, or taxiways that do not connect in that order: "No route along these taxiways, in this order". Check the spelling and the order.

Runway crossings

Every runway the route crosses is circled in **red** on the map with its name, and listed in red under the route ("*Crossing runway 01/19*"). A runway crossing always needs its own ATC clearance: stop at the holding point before it and ask if you do not have it.

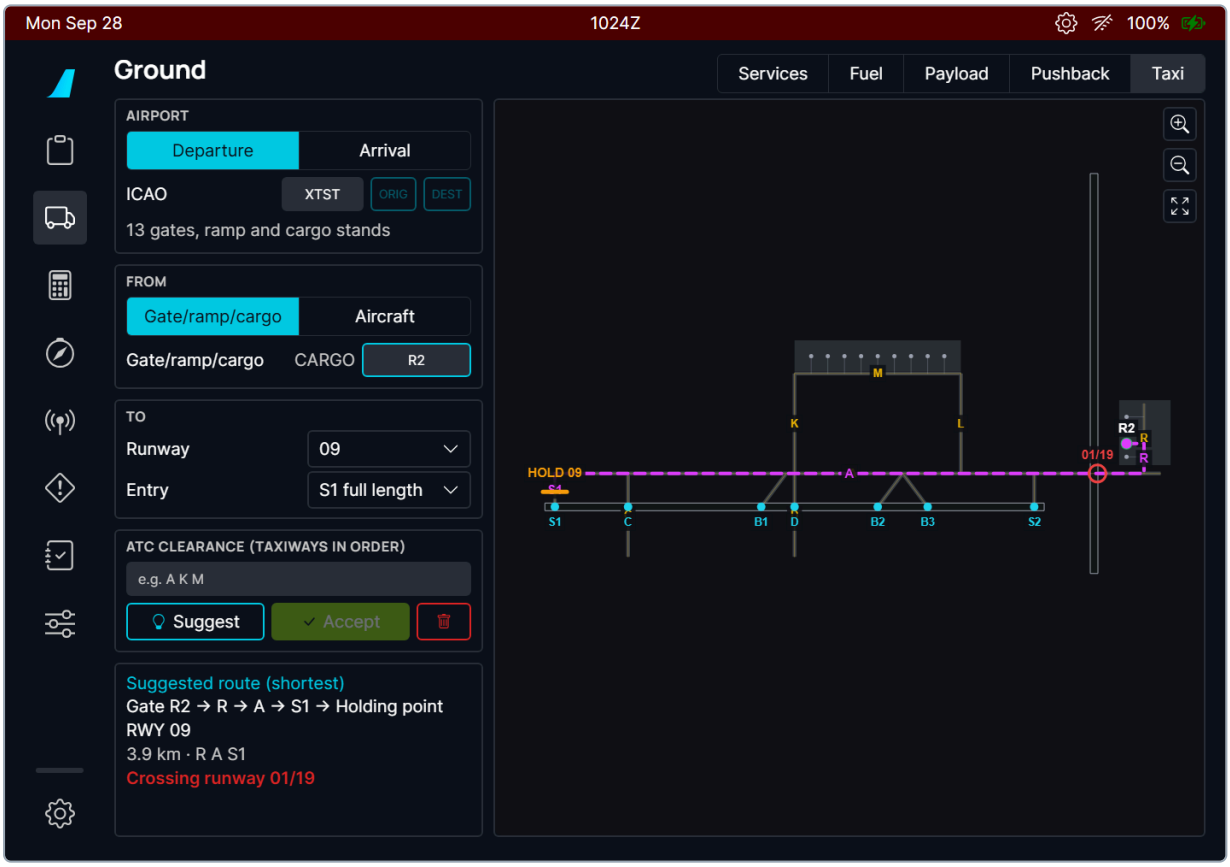


Figure 11. From cargo stand R2 to runway 09: taxiway A crosses runway 01/19 (red circle, red line in the route box).

Arrival: from the runway exit to the gate

1. Select **Arrival** and enter the airport (DEST fills it from the flight plan).
2. In FROM, keep **Runway exit**, choose the landing runway and the exit you plan to take. The exits are those of the landing calculator, with their distance from the threshold.
3. In TO, type or tap the gate.
4. Check the route, type the clearance if ATC gives another one, and ACCEPT.

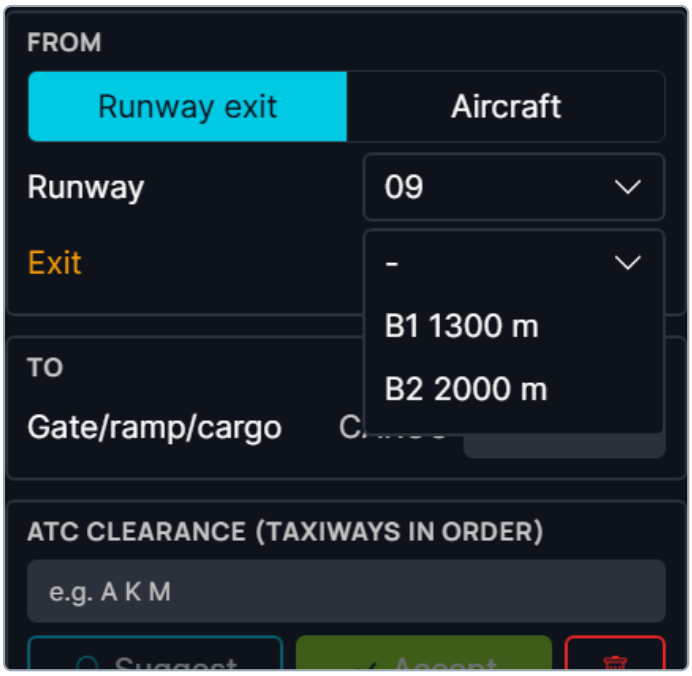


Figure 12. The exits of runway 09 with their distance from the threshold.

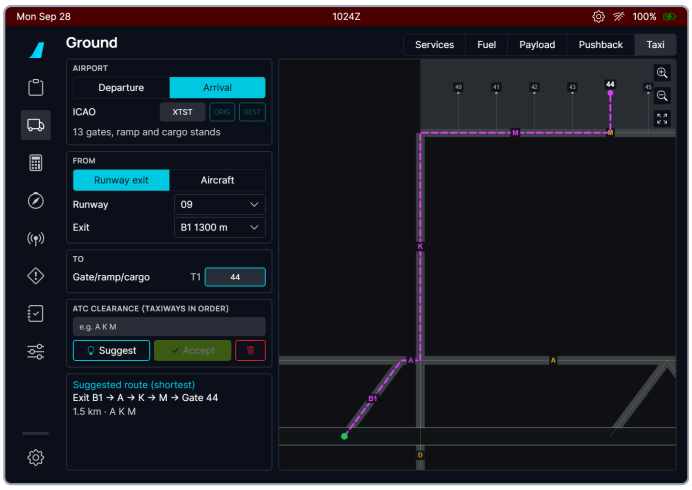


Figure 13. Exit B1 to gate 44: Exit B1 → A → K → M → Gate 44. The green dot is the start.

> **TIP**
> Prepare the arrival route during the descent briefing with the exit from the landing calculator: after landing, the page is ready. If you vacate by another exit, choose it (or use the aircraft position) and the route updates.

From the aircraft position

Choose **Aircraft** in FROM to start the route where the aircraft is, for example after a pushback, or when ATC changes the route while you taxi. The page takes the position when you select it; press **Update** to take it again. The route starts on the taxi line nearest to the aircraft and **prefers its heading**: it does not start by turning back if there is a way ahead.

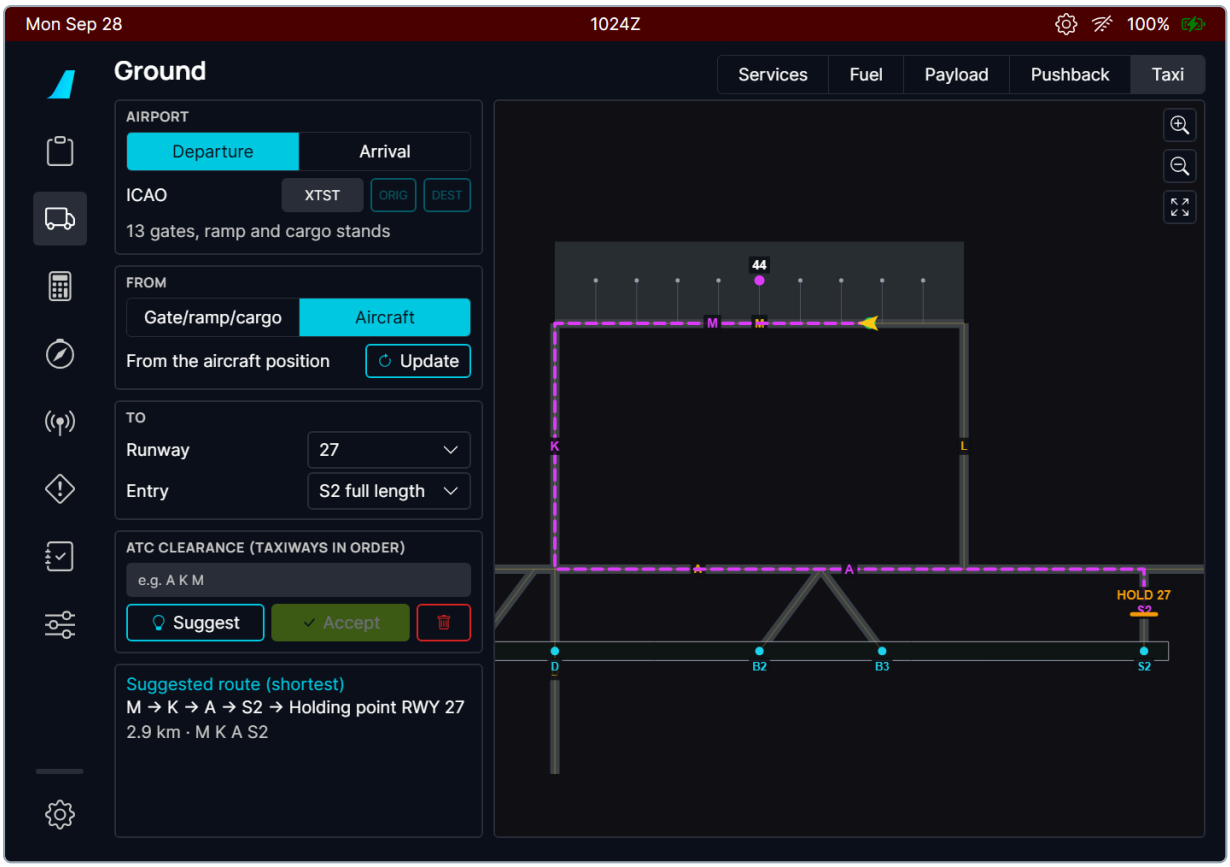


Figure 14. The aircraft (yellow) on M facing west, for runway 27: the route goes ahead via K and A (2.9 km) rather than turning back on M.

"*The aircraft is not at this airport*" means the aircraft is more than 8 km from the airport of the page.

Locate the aircraft, taxiway names of several words, proposed entry

- **Locate button** (pin, under the zoom buttons): centres the map on the aircraft and zooms in until the stand names show. It is greyed when the aircraft is not at this airport.
- Taxiways named with **several words** (e.g. SOUTH RAMP, NORTH RAMP at Montreal CYUL) are read as one taxiway in the clearance: type them as the map shows them, or press SUGGEST.
- The **runway entry** proposed first is the first one from the threshold you can reach **without crossing a runway**. At CYUL, runway 24L full length (R) is across 06R/24L from the terminal: the entry proposed is the first one on the terminal side; choose R in the Entry list if ATC clears you across.

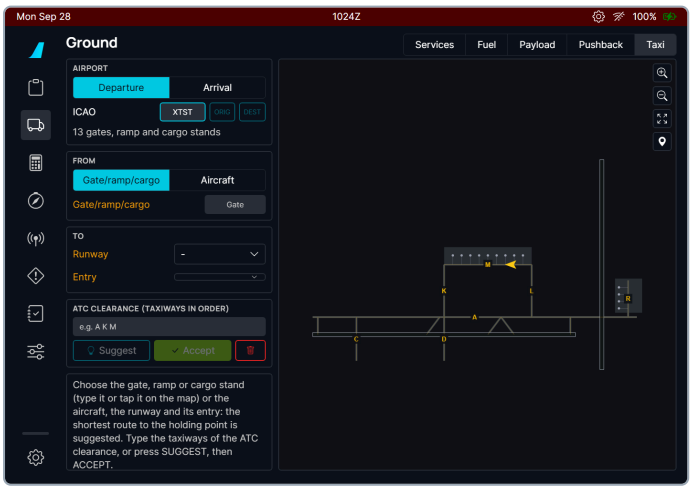


Figure 15. The whole airport, the aircraft on M.

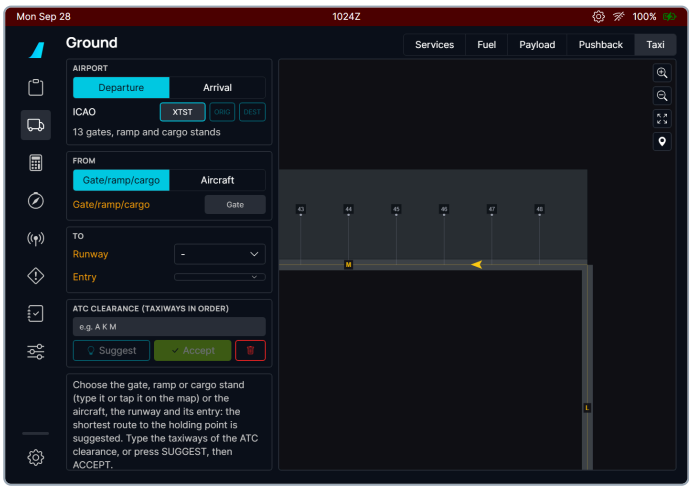


Figure 16. After the locate button: centred on the aircraft, stand names shown.

The map

- **Drag** the map to move it; **+ / -** zoom; the **frame** button fits the route (or the whole airport).
- The map is north up. Aprons are lighter grey, taxiways dark grey, runways black with a white edge. The thin yellow lines are the taxi guidance lines the route follows.
- Taxiway names are shown in yellow; the taxiways of the route in magenta. Gate names appear when you zoom in.
- **Tap a gate** on the map to choose it (when the page asks for a gate).

Figure 17. Zoomed on the apron: the gate names, the route to gate 46 along M.

What to expect

Situation	What you should see
Airport entered, nothing else	The airport map, amber labels on the inputs still missing, the help text in the route box.
Start and end chosen	The suggested route, dashed, with its clearance text, length and crossings.
SUGGEST	The clearance field holds the taxiways of the suggested route; the route is a preview (dashed).
Clearance typed	The route follows the typed taxiways in order (dashed until accepted), or the red message "No route along these taxiways, in this order".
ACCEPT	Solid line, "Accepted route" in green (A380: green flags on the OANS).
Any input changed after ACCEPT	Back to a preview: accept again.
Departure to a runway with holding position lines	The route stops at the HOLD bar before the runway.
Two entries behind the same HOLD bar (e.g. RJTT 34R C2 and C3)	The same route to the holding point: the entries only differ after it.
Gate whose lead-in is another stand's line (e.g. RJTT gate 5 behind 5R)	The route leaves along that line: "Gate 5 → Gate 5R → W ...".
Runway without holding position lines on the map	The route ends on the runway entry itself and the box says "Runway 09" instead of "Holding point RWY 09".
The route crosses a runway	Red circle with the runway name on the map, red "Crossing runway" line.
Aircraft start, aircraft far from the airport	"The aircraft is not at this airport"; no route.
Leaving the page and coming back	Everything is kept, except an Aircraft start: press Aircraft or Update again.

Limitations and why

Limitation	Why	What to do
Needs a Navigraph account.	The airport map (taxi lines, stands, holding points) exists only in the Navigraph airport database; MSFS does not give it to the aircraft.	Link your account in the flyPad settings.
Closed taxiways, one-way taxiways and NOTAMs are unknown.	The airport database has the lines painted on the ground, not their status of the day.	Type the taxiways of the ATC clearance: the route follows them.
The suggestion is the shortest route, not the usual one.	Airports have preferred flows (ATC habits, one-way loops) that are not in the data.	Always check against the clearance and the airport chart.
Some real airports have gaps in their taxi lines.	A lead-in line may stop a few metres short of the taxi line. Gaps up to 15 m between parts that are otherwise not connected are joined; larger ones are not.	If "No taxi line of the airport map connects the start to the end" appears, choose another start (the nearest taxiway, or the aircraft) or end.
Gate names follow the airport database.	The database may use names that differ from the gate signs.	Tap the gate on the map.
No route on the A380 OANS, only flags.	The A380 FCOM has no taxi route display; the flags are the FCOM way to mark points on the airport map.	-
The aircraft position is taken once.	The route is a plan to check, not a moving map that follows you.	Press Update after a change.

Troubleshooting

What you see	Why	What to do
SUGGEST and ACCEPT are greyed	There is no route yet: the gate is not found (amber label, no terminal next to the name), or the runway / entry / exit is missing.	Pick the gate from the suggestion buttons or tap it on the map; check the amber labels.
"No such name on the airport map"	No stand of the map has this name.	Tap the gate on the map.
"No taxi line of the airport map connects the start to the end"	The start and the end are on parts of the map that are not connected.	Start from the nearest taxiway (Aircraft), or choose another gate or entry.
"No route along these taxiways, in this order"	A taxiway of the clearance is misspelt, missing, or they do not connect in that order.	Check the clearance; press SUGGEST to see the names the map uses.
"Not in the Navigraph airport database"	Navigraph has no airport map for this airport.	-

How to use

1. Link your Navigraph account (flyPad > Settings > 3rd Party Options).
2. flyPad > **Ground** > **Taxi**: Departure or Arrival, the airport (or ORIG / DEST).
3. FROM: your gate (departure), the runway exit (arrival) or the aircraft; TO: the runway and its entry, or the gate.
4. Type the taxiways of the ATC clearance in order (or SUGGEST), check the route and the runway crossings, press **ACCEPT**.

How to test

#	Do this	Expected result
1	Navigraph account linked. A32NX or A380X parked at a gate of a large airport (e.g. RJTT). flyPad > Ground > Taxi, Departure, airport RJTT.	The airport map; the line under the airport gives the number of stands; amber labels on the missing inputs.
2	In FROM type your gate number.	Matching gates as buttons under the field ("16" also finds "016"); tap one, or tap the gate on the map.
3	In TO choose a runway; keep the proposed entry.	Dashed magenta route to the HOLD bar; the proposed entry is the first one reached without crossing a runway.
4	Open the Entry list.	All entries from the threshold: full length first, then the remaining length of each.
5	Press SUGGEST, then ACCEPT.	The clearance field holds the taxiways; after ACCEPT the line is solid and the box says "Accepted route".
6	Type another clearance with existing taxiways (e.g. a longer way round), without ACCEPT.	The route follows them in order, dashed, "Route along the clearance (not accepted)".
7	Type a taxiway that does not exist.	"No route along these taxiways, in this order".
8	Choose a runway the route must cross to reach.	Red circle with the runway name on the map and a red "Crossing runway" line.
9	FROM = Aircraft.	The route starts at the aircraft and goes ahead along its heading (no U-turn through the aircraft).
10	Press the locate button (pin) under the zoom buttons.	The map centres on the aircraft, stand names shown.
11	Arrival: DEST, runway, exit, gate.	Route from the exit to the gate; exits with their distance from the threshold.
12	Leave the page and come back.	Everything kept, except an Aircraft start.

2. A380X: the accepted taxi route on the OANS

> **Aircraft:** A380X | **Where:** ND (OANS) after ACCEPT on the flyPad Taxi page | **Status:** Tested in the sim by the user: OK.

What it is for

The A380 FCOM has no taxi route display: the OANS lets the crew mark points of the airport with green flags (DSC-34-10-70-20). When the crew accepts a route on the flyPad, the A380X marks it with those flags, so the route can be followed on the ND.

On the A380: the route on the OANS

When you **accept** the route on the A380 flyPad, green flags appear on the OANS (the airport map on the ND) at the start of each taxiway of the route and at the gate or holding point. They are the crew flags of the FCOM ("*to locate or mark a given point on the airport*"), so you can remove them like any flag: click one and DELETE, or ERASE ALL SYMBOLS in the ND menu. Flags you placed yourself are never touched.

- The flags appear when the OANS displays that airport (they come back if the OANS reloads it).
- A new accepted route replaces the flags of the previous one; a route that is no longer accepted removes them.
- The A320 has no OANS: nothing else changes on it.

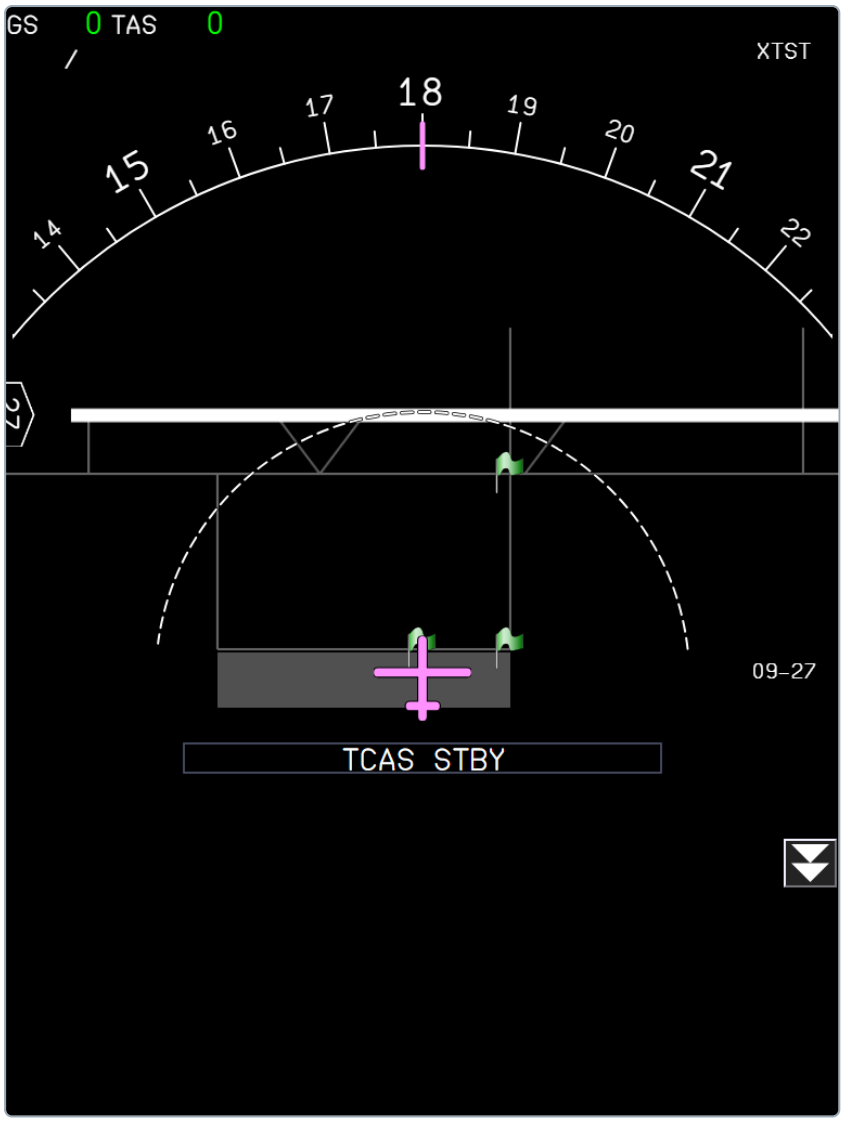


Figure 18. The A380 ND (OANS, ARC, zoom range) at gate 42 facing south, after ACCEPT of the departure route above (gate 42 to runway 09): green flags where M, K and A start (S1 and the holding point are further west, off the picture). Harness picture: the test airport has no buildings.

> **NOTE**
> Tested in the sim by the user: OK: the flags are set on ACCEPT and removed when the route changes.

Limitations and why

Limitation	Why	What to do
Flags only, no route line.	The A380 FCOM has no route display on the OANS; flags are its way to mark points.	-
An Aircraft start is lost when leaving the Taxi page, so its flags go away when you come back.	The aircraft start is local to the page.	Select Aircraft again and ACCEPT.

How to use

1. Accept a route on the flyPad Taxi page (A380X).
2. On the ND, select the OANS (ARC or ROSE, a zoom range) at that airport: the flags show the start of each taxiway and the end of the route.

How to test

#	Do this	Expected result
1	A380X at a gate, Navigraph linked, OANS showing the airport on the ND. flyPad Taxi page: route to a runway, **ACCEPT**.	Green flags on the OANS at the start of each taxiway of the route and at the holding point; the flyPad status says "green flags on the OANS".
2	Place your own flag on the OANS, then change the clearance on the flyPad (route no longer accepted).	The route flags go away; your own flag stays.
3	ACCEPT a new route.	Its flags replace those of the previous route.
4	Click a route flag on the OANS and DELETE, or ERASE ALL SYMBOLS in the ND menu.	The flags are removed like crew flags.

3. Airport ATC frequencies on the taxi page

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Taxi, bottom left of the map | **Status:** Tested in the sim by the user: OK.

What it is for

Before taxi the crew needs the ground, tower and delivery frequencies of the airport. The taxi page shows them in a box laid out like the frequency box of an airport chart, for the origin, the destination or the airport of the page. The Navigraph airport map has no frequencies: they come from the sim's own airport data.

What you see

- A green **Frequencies** button at the bottom left of the map; it opens the **FREQUENCIES** box (X closes it).
- Airport buttons at the top: **ORIG** and **DEST** of the SimBrief flight plan loaded in the flyPad, and **Airport** when the page airport is another one.
- One row per chart label, in the chart order: ATIS, AWOS, ASOS, CTL, APP, DEP, TWR, GND, DLV, CTAF, UNICOM, MULTICOM, FSS; the frequencies in MHz, two columns, without duplicates, in the order of the sim data. Other types (ILS...) are left out.
- Messages: "Enter an airport", "Loading the frequencies...", "Not in the sim's airport data", "No frequency for this airport in the sim's airport data".

Limitations and why

Limitation	Why	What to do
The list is the sim's, not the chart's.	The chart's selection of frequencies is in no data the aircraft can read; MSFS lists all its frequencies.	Check the chart for the one to use.
No station names, no tuning.	The box is a reference, like the chart box.	Tune the RMP as usual.

How to use

1. On the Taxi page, press **Frequencies** (bottom left of the map).
2. Choose ORIG, DEST or the page airport; read the frequency you need.

How to test

#	Do this	Expected result
1	SimBrief OFP loaded in the flyPad (e.g. RJTT to another airport). Ground > Taxi, airport RJTT. Press **Frequencies**.	The FREQUENCIES box with ORIG RJTT selected and DEST; rows ATIS, APP, DEP, TWR, GND, DLV... with MHz values (three decimals).

2	Press DEST.	The frequencies of the destination.
3	Change the page airport to a third airport.	An "Airport" button appears and is selected; its frequencies.
4	Compare with the airport chart or the sim world map.	The same frequencies (the sim may list more).
5	Press X.	Back to the Frequencies button.

4. Services page linked to GSX

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, GSX panel | **Status:** First version tested in the sim by the user with GSX 4 (A380X, RJTT): four bugs found. Their fixes on this branch (panel position, counters, greyed switch, status lines) are not sim-tested on this classic page. The Material 3 Services page of `core/efb-redesign`, which carries them with further GSX fixes, was tested in the sim (A32NX: 170 passengers given to GSX, count 0 to 170, Done).

What it is for

With GSX installed, the ground services are GSX's, but the flyPad Services buttons called the sim's own vehicles. A **GSX services** switch now links the page to GSX through the GSX Remote API (version 2, local server on port 8744): the service buttons request the GSX services and show what GSX is doing. With the switch off the page works as before.

What you see

- The **GSX panel** under the right-hand group of buttons (GPU, cargo door): the switch and a status line: "GSX not found: start GSX, with its Remote control server on (GSX settings)" (switch greyed), "GSX is starting...", "GSX found" (switch off) or "Service buttons linked to GSX, GSX menus and messages shown here".
- Linked: **Jet Bridge**, **Stairs**, **Fuel Truck**, **GPU** and **Catering Truck** request the GSX services. The **Baggage Truck** button is hidden: GSX loads the bags during boarding.
- The panel adds the services without an aircraft button: Deboarding, Boarding, Pushback (Departure), De-ice, Water, Lavatory, Cleaning, named as GSX names them.
- Under each linked button a status line: *On its way*, *Waiting*, GSX's own phase (or *In progress*), *Leaving*, *Done*, with its count; a progress bar for the fuel truck from the "x / y kg" refuelling status. Button colours follow the service: amber when called or waiting, green while performed, amber and locked while leaving, dim when GSX cannot serve it.
- Linking switches on the flyPad **GSX Fueling** and **GSX Payload Synchronization** (Settings > 3rd Party Options), so GSX boards the passengers and loads the fuel set on the Payload and Fuel pages.
- While linked, GSX is under remote control: no GSX pop-up menu or notification in the sim. GSX's questions (menus that stay open, e.g. the choice of operator) and its messages appear on the panel; tap an answer, or "Close the GSX menu".
- A request GSX refuses (e.g. boarding before the doors are open) gives the message "GSX did not accept <service>" with GSX's reason.

Limitations and why

Limitation	Why	What to do
Needs GSX with its Remote control server on.	The link uses the GSX Remote API (GSX 4).	Turn on the Remote control server in the GSX settings.

The four bugs of the first sim test (panel over the fuselage drawing, Board counter stuck at 0, passenger count not moving, switch not greyed without GSX) are fixed here but not checked in the sim on this page.	The fixes came after the test. `core/efb-redesign` reworks the counter (the flyPad counts the GSX passengers itself and gives GSX its passenger number) and the greyed switch again; that version is the one tested in the sim.	Merge `core/efb-redesign`; report any of them if seen again.
Turning the link off leaves the GSX fuel and payload syncs on.	Only linking changes them.	Turn them off in Settings > 3rd Party Options if wanted.
The page keeps the classic FBW look on this branch.	The Material 3 redesign of the Services page is in `core/efb-redesign`.	-

How to use

1. Start GSX with its Remote control server on.
2. flyPad > **Ground** > **Services**: when the panel says "GSX found", turn on **GSX services**.
3. Request the services with the buttons and the GSX panel; answer GSX's questions on the panel.
4. Turn the switch off to go back to the FBW services.

How to test

#	Do this	Expected result
1	A380X (or A32NX) at a gate, GSX not running. flyPad > Ground > Services.	GSX panel under the GPU / cargo door group; switch greyed; amber "GSX not found: start GSX, with its Remote control server on (GSX settings)".
2	Start GSX with the Remote control server on.	"GSX is starting...", then "GSX found"; the switch becomes active.
3	Turn on **GSX services**.	Green "Service buttons linked to GSX..."; the Baggage Truck button disappears; the panel lists Deboarding, Boarding, Pushback, De-ice, Water, Lavatory, Cleaning; Settings > 3rd Party Options shows the GSX fuel and payload syncs on.
4	Press **Jet Bridge** (or Stairs).	Amber, status "On its way", then green while performed, "Done" at the end. No GSX menu pops up in the sim.
5	Press **Fuel Truck**.	Status and progress bar following "x / y kg"; the fuel set on the Fuel page.
6	Press **Boarding** on the panel (doors open).	The count of boarded passengers rises to the number set on the Payload page.
7	When GSX asks a question (e.g. the operator choice), answer it on the panel.	GSX continues with that answer.
8	Turn the switch off.	The Baggage Truck button is back; the buttons call the FBW services again.

5. GSX external power sync when it is enabled

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Settings > 3rd Party Options > GSX External Power Synchronization | **Status:** Tested in the sim by the user (A380X, GSX 4, RJTT): EXT AVAIL on the ELEC panel with the GSX GPU, OK.

What it is for

The FBW option *GSX External Power Synchronization* makes the external power available (`L:A32NX_EXT_PWR_AVAIL`) when the GSX GPU is connected or the jetway is requested or connected, until pushback. It only looked at GSX when a GSX state changed: enabling the option with the GPU or the jetway already connected did nothing until the next GSX change. It now checks GSX at once when it is enabled.

What you should see

Situation	What you should see
Option enabled with the GSX GPU or jetway already connected	External power available at once: EXT AVAIL on the overhead ELEC panel.
Option enabled before GSX connects	As before: available when the GPU connects.

Limitations and why

Limitation	Why	What to do
With GSX 4, EXT AVAIL may already show before the option is enabled.	GSX 4 also sets the sim's own external power variable.	Watch `L:A32NX_EXT_PWR_AVAIL:1` with a SimVar tool to see the sync itself.

How to use

1. flyPad > **Settings** > **3rd Party Options**: turn on **GSX External Power Synchronization**, before or after GSX connects its GPU or jetway.
2. Press the EXT PWR pushbutton when AVAIL shows.

How to test

#	Do this	Expected result
1	A380X (or A32NX) cold and dark at a GSX gate, the option OFF. Call the GSX GPU (or the jetway) and wait until it is connected.	`L:A32NX_EXT_PWR_AVAIL:1` stays 0 (unless GSX 4 already made EXT AVAIL show, see Limitations).
2	Turn on **GSX External Power Synchronization**.	At once: `L:A32NX_EXT_PWR_AVAIL:1` = 1, EXT AVAIL on the overhead ELEC panel (before: nothing until the next GSX state change).
3	Press EXT PWR.	The aircraft is powered by the ground power.
4	Request the GSX pushback.	The external power becomes unavailable, as before.

6. Taxi route along real ATC clearances

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Taxi, clearance field | **Status:** Not sim-tested (added after the user's RJTT report of the same day; unit tests in taxiRoute.spec.ts "Taxi route along an ATC clearance", on a synthetic RJTT-like map).

What it is for

At RJTT the clearance `W A E C1 C1` typed from the aircraft after pushback gave "No route along these taxiways", and a full BeyondATC instruction ("via L4 cross runway 34L, taxi to holding point C1, runway 34R, via L4, A4B, W4, W, A, E, C1") could not be used. Real clearances leave out the apron taxiways, repeat the holding point, name runway exits and come as sentences. The route finder now accepts them:

- **Typed list or pasted instruction:** the clearance field reads a list of taxiways or a pasted ATC instruction. Phraseology words (VIA, TAXI, TO, HOLDING POINT...), runway designators, the stand after STAND or GATE and frequencies are left out. The runways after CROSS (RUNWAY) are the cleared crossings. The clearance limit (after TO or HOLDING POINT) is the last taxiway. HOLD SHORT (OF) a taxiway is left out. "A-4B" is read as the airport name A4B. A name said twice in a row is one taxiway.
- **Apron taxiways before the clearance:** before the first cleared taxiway, taxiways that are not in the clearance may lead to it, within 2000 m of the start. They cost more (x1.5), so the route joins the clearance as soon as it can. A cleared taxiway is never used out of order.
- **Uncleared runway crossings:** in that first part, a runway crossing that the clearance does not mention costs a large penalty: the route goes round when it can; the crossing still shows on the map. "cross runway 34L" in the clearance removes the penalty.
- **Runway exit lines** named in the clearance are cleared taxiways; a clearance that starts with the landing exit ("B1 A K M") works. Exit lines are listed in the route taxiways (SUGGEST), except the start exit.
- **Unknown names:** the page lists the cleared names that are not a taxiway of the airport map, under the clearance field in amber ("Not a taxiway of the airport map: X9"). The field placeholder now says "e.g. A K M, or paste the ATC instruction".

Situation	What you should see
From the aircraft after pushback, clearance `W A E C1 C1`	A route: the apron taxiways from the aircraft to W are added, then W A E C1.
A full ATC instruction pasted in the field	The taxiways in order (for the RJTT example L4 A4B W4 W A E C1); "Crossing runway 34L" shown if the route crosses it.
A name in the clearance that the airport map does not have	"Not a taxiway of the airport map: <name>" under the field.
A clearance taxiway used out of order would be shorter	Not used: the clearance order is kept.

Limitations and why

Limitation	Why	What to do
The RJTT taxiway names A4B, W4 and L4 are not checked against the real Navigraph map.	No real airport map data offline: the tests use a synthetic map.	Report any name listed as "Not a taxiway of the airport map".
The apron taxiways before the clearance are searched within 2000 m of the start only.	Keeps the route from wandering on other taxiways.	Add the first taxiway to the clearance if it is further.

How to use

1. flyPad > Ground > Taxi: set FROM (gate, exit or Aircraft) and TO as before.
2. Type the cleared taxiways in order, or paste the whole ATC instruction (for example from BeyondATC) in the clearance field.
3. Check the route, any "Crossing runway" line and any "Not a taxiway of the airport map" line, then ACCEPT.

How to test

#	Do this	Expected result
1	RJTT, Departure, FROM Aircraft after pushback near gate 67, TO runway 34R entry C1. Type `W A E C1 C1`.	A route (not "No route along these taxiways"), with the first taxiways added from the aircraft.

2	Paste the full instruction "via L4 cross runway 34L, taxi to holding point C1, runway 34R, via L4, A4B, W4, W, A, E, C1".	Route L4 A4B W4 W A E C1; "Crossing runway 34L" shown if L4 crosses it.
3	Type a clearance with a name that does not exist (for example X9).	"Not a taxiway of the airport map: X9" in amber under the field.
4	Arrival: clearance starting with the landing exit (for example `B1 A K M` at an airport that has them).	A route from the exit along the clearance.

What PR testers should check

- At RJTT (Departure, From Aircraft after pushback near gate 67, runway 34R entry C1): "W A E C1 C1" gives a route with the first taxiways added from the aircraft.
- The pasted BeyondATC instruction gives L4 A4B W4 W A E C1, with "Crossing runway 34L" if L4 crosses it.
- Note any name listed as "Not a taxiway of the airport map".
- Not sim-tested yet.

7. GSX power sync: the setting is re-read, so enabling it takes effect

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Settings > 3rd Party Options > GSX External Power Synchronization | **Status:** Not sim-tested (added; unit tests: GsxSync.spec.ts, 3, all failing without the change).

What it is for

Follow-up of the feature above. The flyPad switch stores the setting and signals the change through a listener that the extras host (the code that runs the GSX sync) never registers, so the change did not always reach it: turning the sync on with the GSX GPU or jetway already connected could still leave EXT PWR unavailable. The extras host now re-reads the stored setting (at most once a second) and checks the GSX power source again when the setting goes from off to on.

Limitations and why

Limitation	Why	What to do
Up to one second between the switch and EXT AVAIL.	The setting is polled once a second.	-

How to use

1. As above: turn on GSX External Power Synchronization, before or after GSX connects its GPU or jetway.

How to test

#	Do this	Expected result
1	Cold and dark at a GSX gate, sync OFF, GSX GPU connected.	No EXT AVAIL from the sync.
2	Turn the flyPad GSX power sync on.	EXT PWR AVAIL within a second.

What PR testers should check

- Turn the flyPad GSX power sync on with the GSX GPU connected: EXT PWR AVAIL at once.
- Not sim-tested yet.

8. External power follows the sim's ground power (GSX GPU, sim GPU, jetway)

> **Aircraft:** A32NX and A380X | **Where:** Overhead ELEC panel EXT PWR; flyPad > Ground > Services GPU button | **Status:** Not sim-tested (added; unit tests: GPUManagement.spec.ts, 10, failing first).

What it is for

The aircraft read the sim's external power only on the first frame and when the flyPad GPU button was used. A GPU or a jetway connected after the flight loaded (GSX 4, the sim GPU) did nothing until the GPU door moved. Now the external power follows the sim's ground power, whoever connects it.

Situation	What you should see
Sim ground power appears (GSX GPU, sim GPU or jetway power), aircraft at or below 0.3 kt	EXT PWR AVAIL on the overhead (if the external power was not already available).
That sim ground power goes	EXT PWR no longer available, but only if it was connected from the sim power.
Crew connection from the flyPad GPU button, or EXT PWR set by the GSX sync	Kept: the sim power going does not remove it.

Limitations and why

Limitation	Why	What to do
Not above 0.3 kt.	A moving aircraft cannot be on ground power.	-

How to use

1. Connect the ground power any way (GSX GPU or jetway, the sim GPU, or the flyPad GPU button) and press EXT PWR when AVAIL shows.

How to test

#	Do this	Expected result
1	Load the flight cold and dark at a gate, then call the GSX GPU (or connect the jetway).	EXT AVAIL on the overhead as soon as the GPU is connected (no wait for the GPU door).
2	Disconnect the GSX GPU.	EXT AVAIL goes.
3	Connect the GPU with the flyPad Services GPU button, then make the sim power go.	EXT AVAIL stays (crew connection kept).

What PR testers should check

- GSX GPU and jetway connect / disconnect after the flight is loaded.
- The flyPad GPU button.
- Not sim-tested yet.

flyPad Material 3 redesign and independent flyPads

A new Material 3 look for every flyPad page, a flyPad per pilot on both aircraft, a custom flyPad theme, the A32NX buffet onset (coffin corner) page, GSX services link fixes, flyPad fixes, a landing calculator that keeps calculating above the wind limits, the RAT stow and cargo door rows of the Services page, pushback and A380X door button fixes, the A32NX MCDU engine-out REC MAX, the A380 taxiway width check and the A380X smart FIVE HUNDRED pin option

Aircraft	A32NX and A380X (the Buffet tab: A32NX only)
Where to find it	The captain and first officer flyPads (all pages)
Branch	`core/efb-redesign`, based on `core/efb-ground` (stacked: needs `core/surveillance`, `core/a380x-mfd-fms`, `core/efb-performance` and `core/efb-ground` merged first). The independent flyPads need two companion commits in the `aircraft-large-files` repository: 24f7b02 (A380X) and d3060ea (A32NX). The RAT stow row needs `core/rat-ground-stow` (the systems side of the stow); the A380X cargo door row reads the inhibition written by `core/a380x-cargo-doors` (merge both). The smart FIVE HUNDRED option needs the callout of `core/a380x-elec-fuel-alerts` (merge both).
Status	Sim-tested by the user: Services and Pushback look, quick controls fix, A32NX independent flyPads, frame and Dashboard / Dispatch / Performance / other pages on both aircraft. Fuel / Payload / Taxi look approved by the user. Not sim-tested (browser harness only): the flyPad UI fixes, the Payload LOAD row change, the checklist and lighting preset fixes, the landing wind check. RAT stow row: sim-tested on both aircraft. Tested in the sim by the user (a combined test build, not this core branch itself): Buffet tab on the A32NX and its Live / What if pill row fix (A32NX), custom theme (A32NX, user word), the first colour wheel ("the wheel works"), GSX boarding passenger count (A32NX), GSX turnaround Done memory, Baggage row Done, Turnaround scrollbar and tug slider fixes (A380X). A380X cargo door row: sim-tested with the systems part (PASS, recorder-checked). Pushback map speed-up and pushback going on off the page: confirmed in the sim by the user. A380X Main 5 R / Main 4 R door buttons: sim tests passed. A32NX MCDU engine-out REC MAX: tested in the sim by the user: OK. The vivid colour wheel that replaced the first one: OK in the sim (A380X). Not sim-tested: the Light theme label fix, the greyed GSX switch, the GSX deboarding passenger number, and the A380 taxiway width check and smart FIVE HUNDRED option (added).
Pictures	No pictures: the flyPad pictures of the older manuals show the look before this redesign.

1. Material 3 look of the Ground pages
2. Material 3 look of the frame and of the other pages
3. flyPad UI fixes after the redesign
4. Payload page: LOAD title row and balance chart
5. Independent captain and first officer flyPads
6. Quick controls clicks no longer open the troubleshooting page
7. Checklists: always loaded, green check when marked complete
8. Automatic lighting preset loading lists all eight presets
9. Error details on the flyPad error screen
10. Landing calculator above the wind limits, with a wind check
11. RAT stow (maintenance) row on the Services page
12. A380X cargo door row shows the real doors
13. Custom flyPad theme, and a readable Light theme
14. Performance > Buffet: A32NX buffet onset envelope (coffin corner)
15. A32NX MCDU PROG: engine-out REC MAX
16. GSX services link: passenger counts, request chips, greyed switch
17. GSX turnaround: completed services remembered, Baggage steps
18. Taxi page: A380 taxiway width check
19. A380X smart FIVE HUNDRED pin option back in the flyPad

1. Material 3 look of the Ground pages

Aircraft: A32NX and A380X | **Where:** flyPad > Ground | **Status:** Services and Pushback tested in the sim by the user: OK. Fuel, Payload and Taxi: approved by the user, checked in the browser harness. Pushback map drag fix: not sim-tested on its own; the later map speed-up (only the visible airport, at most 10 Hz) and the pushback going on off the page: confirmed in the sim by the user ("pushback fixed"). Tug speed slider fix: confirmed in the sim by the user (A380X). A380X Main 5 R / Main 4 R door buttons:

> sim tests N1 and P1 passed. Unit tests: pushbackAirportDraw.spec, pushbackControllerInput.spec, A380ServiceDoors.spec, groundServicePage.spec.

What it is for

The flyPad pages get a Material 3 dark look on the flyPad cyan: tonal cards, dense list rows with switches and chips, a segmented control for the tabs. A shared Material kit (page, cards, section headers, list rows, status and action chips, switch, segmented control, buttons, progress bar, banner) on new theme tokens for the blue, dark and light themes is used by every redesigned page. The Ground pages came first. The functions of the pages do not change.

What you see

Page	New look
Services	Three columns: the doors and the ground equipment as scrolling lists (switch or Request / Release chip, GSX status and fuel progress on the rows); the aircraft from above, nose up and zoomed on the fuselage, its doors coloured with their state, the holds and service tags (A380X two decks, A32NX one); the GSX card (switch, turnaround timeline of the departure or arrival with its Baggage step, other GSX services as chips, GSX questions and messages; see "GSX services link" and "GSX turnaround" below). Status chips: GPU, tug, doors open, chocks and cones (A32NX).
Pushback	The map fills the left card. A control rail: system switch, Call / Release tug button, parking brake row, Backward / Paused / Forward, a steering dial over the direction slider, the speed slider, Left / Straight / Right and the controller switch. When the aircraft is at the SimBrief origin or destination, the Navigraph airport map (aprons, taxiways, runways, guidance lines, stands) is drawn under the pushback map, heading up.
Fuel	The aircraft with its tanks on a wide card and a refuel rail (target, slider, start and stop, duration, plan summary). A380X: the whole aircraft outline with the tank quantities in tables (each feed tank now shows its own quantity; before, the four feed tanks of the drawing read feed tank 3). A32NX: the over-wing outline with its tanks filled, the quantities as tiles.
Payload	The cabin and seat map on a full-width card with the holds under it (load over capacity, click to plan), then the load table, the boarding controls and the balance chart.
Taxi	The route form as cards, the runway crossings as banners, the map buttons and the frequency box in the kit.

Pushback map fix: the airport under the pushback map redrew only after the map centre had moved about 100 m, so the aircraft slid over a frozen airport while the map was dragged. It now follows every move.

Pushback map: only the visible airport, at most 10 times a second

The airport layer drew the whole Navigraph airport (thousands of points) on every position update, 20 times a second: the A380X ran at 21 FPS on the Pushback page against 42 FPS on another page (capture). It now draws only the shapes and lines that reach the map window, and redraws only when the picture moves by about a pixel, at most 10 times a second.

The pushback goes on when another flyPad page is opened

Leaving the Pushback page stopped the tug (with a short "Pausing Pushback" message), because the rudder pedal and sidestick steering lived in the page. The steering now runs in the captain flyPad itself, whatever page is shown (when the controller switch of the Pushback rail is on). The page resets the tug commands only when it opens with no tug attached (captain flyPad), so a pushback in progress goes on when the page is left and opened again.

A380X Services door buttons: Main 5 R and a new Main 4 R

The aft right door button said "Main 4 R" but opened MAIN 5R, the door the SD DOOR page and the FWS name. It is now labelled **Main 5 R** and its marker on the planform sits on door 5 R. A new **Main 4 R** button opens and closes MAIN 4R. The Doors card lists Main 1 L, Main 2 L, Upper 1 L, Main 4 R, Main 5 R and the cargo row: six rows without a

scrollbar. The catering truck still docks at MAIN 5R: a button only moves its door; the door a ground vehicle (or GSX) docks at comes from the aircraft configuration.

Tug speed slider fix: the handle of the vertical tug speed slider of the Pushback rail sat 5 px right of its rail (the slider library offset suits its own 14 px handle, the flyPad handle is 24 px). It is now centred on the rail.

Limitations and why

Limitation	Why	What to do
The airport under the pushback map needs the Navigraph airport map.	It is drawn from the Navigraph airport database, only at the SimBrief origin or destination.	Link a Navigraph account and import a SimBrief flight plan.
Spacing is made with margins, not the CSS gap.	The sim browser (Coherent GT) ignores the CSS gap property.	-

How to use

1. Open **flyPad > Ground** and pick the tab (Services, Fuel, Payload, Pushback, Taxi) at the top right.
2. Use the switches, chips and buttons as before: open or close a door, request a service, refuel, board, push back.

How to test

#	Do this	Expected result
1	A380X at a gate, aircraft powered, SimBrief flight plan imported. Open flyPad > Ground > Services.	Three columns: DOORS and ground equipment lists, the aircraft from above (two decks) with its doors, the GSX card. Nothing cut at the edges.
2	Open a door with its switch in the list.	The row status changes and the door shape on the aircraft drawing changes colour.
3	Repeat on the A32NX.	Same layout with the single-deck A320 drawing and the chocks and cones chips.
4	Open Ground > Pushback (Navigraph linked, aircraft at the SimBrief origin).	The map on the left with the airport under it; the control rail on the right.
5	Drag the pushback map.	The airport moves with the aircraft symbol at once (no sliding over a frozen airport).
6	A380X: compare the frame rate (MSFS FPS display) on the Pushback page and on the Dashboard.	About the same frame rate on both pages (before the fix, about half on the Pushback page).
7	Call the tug and push back with the controller switch on (sidestick back, rudder pedals to steer). During the push open the Dashboard and steer with the rudder pedals.	The pushback goes on and follows the rudder pedals; no "Pausing Pushback" message. Open Pushback again: the push goes on.
8	A380X at a stand, engines off: Ground > Services, Doors card. Switch Main 5 R on, then ECAM control panel DOOR key.	The SD DOOR page shows MAIN 5R open (the same door); the 5 R marker on the planform lights up. Switch it off.
9	Switch Main 4 R on, wait until open, then off.	The 4th right main deck door opens and closes; Main 5 R stays closed; the Doors card shows six rows, no scrollbar.
10	Look at the vertical tug speed slider on the Pushback rail and move it.	The handle is centred on its rail over the whole travel.

11	Open Ground > Fuel on the A380X, then on the A32NX.	A380X: whole aircraft with tank tables, each feed tank with its own quantity. A32NX: wing tanks filled, quantities as tiles. The refuel rail on the right.
12	Open Ground > Payload, then Ground > Taxi.	Payload: cabin card across the page, holds under it, load table, boarding and balance cards. Taxi: route cards and map in the new look.

2. Material 3 look of the frame and of the other pages

> **Aircraft:** A32NX and A380X | **Where:** flyPad frame; Dashboard, Dispatch, Performance, Navigation, ATC, Failures, Checklists, Presets, Settings | **Status:** Tested in the sim by the user on both aircraft: OK, incl. the quick controls on both tablets.

What it is for

The rest of the flyPad in the same Material look, so that every page matches. The shared switch, segmented selects, text fields, drop-downs and tooltips take the new look, so every page follows.

What you see

Part	New look
Frame	A navigation rail with the icon names (the open section in a tonal pill), the status bar, the quick controls as a card (brightness sliders with their Auto switch, state tiles, sim rate tile at the bottom right), the section tabs as a segmented control.
Dashboard	The flight card (airports, progress, figure tiles, route, SimBrief import) and the Important Information column (weather readings or raw report, pinned charts, maintenance and checklists as rows).
Dispatch	Overview: the nose drawing on a wide card and the aircraft figures in three lists; OFP and Printouts in kit cards and buttons.
Performance	A shared calculator kit (form cards, labelled rows, result tiles, the fill-from button): takeoff, landing and descent results as tiles; temperature correction with one row per published altitude and its corrected value beside it. A32NX: a new Buffet tab (see "Performance > Buffet" below).
Navigation	Chart list, viewer controls and pinned charts in the kit.
ATC	Type filter chips, controller cards with their buttons always shown, the COM 1 card.
Failures	Chapter cards with failure and active counts, failure tiles.
Checklists	The list with done / next marks and counts, the open checklist with its progress.
Presets	State tiles with the loading card; lighting presets beside the automatic loading.
Settings	Two panes: the categories beside the open page. Settings > flyPad has the theme choice, with a custom theme (see "Custom flyPad theme" below).

How to use

1. Pick a section on the navigation rail on the left, then a tab in the segmented control at the top right.
2. Open the quick controls from the status bar (brightness, Auto, state tiles, sim rate).

How to test

#	Do this	Expected result
1	Either aircraft, flyPad on. Look at the left rail.	Icons with their names; the open section in a cyan pill.

2	Open the quick controls from the status bar.	A card with the brightness slider and its Auto switch, state tiles and the sim rate tile at the bottom right.
3	Open Dashboard, then Dispatch > Overview, OFP, Printouts.	Cards in the new look; the A380X / A320 nose drawing on the Overview.
4	Open Performance > Takeoff, Descent, Landing, Temperature Correction.	Form cards and result tiles; the temperature correction lists one row per altitude with its corrected value.
5	Open Navigation, ATC, Failures, Checklists, Presets.	Every page in the new look, no old grey boxes.
6	Open Settings and pick a category.	Two panes: categories on the left, the open page on the right.
7	Repeat on the first officer flyPad (see "Independent captain and first officer flyPads").	Same look; the quick controls work there too.

3. flyPad UI fixes after the redesign

> **Aircraft:** A32NX and A380X | **Where:** Settings, dialogs, toasts, Ground pages, texts | **Status:** Checked in the browser harness on both aircraft. Not sim-tested. Unit tests: detent gauge, Settings categories, fuel percentage, failure counts, descent notes.

What it is for

A check of every flyPad page on both aircraft and both tablets after the redesign found 23 problems: three layout bugs and parts still in the old look or with poor texts. They are fixed:

Area	Fix
Throttle calibration	Opened inside the two-pane Settings it was taller and wider than the pane: the button bar was cut and, with 4 axes on the A380X, Axis 4 and Save and Apply were off screen. It is now one card that fits, the axes in one row, the button bar always visible, a new detent gauge. The in-detent highlight also works at exactly 0 (its colour is now the light primary, not green).
Settings	About: the content covered the title, now in the page flow with build info rows. The open Settings category gets its icon colour and bold label (the active class was lost). Navigraph sign-in, SimBrief trash, Aircraft Options Select, Automatic Call Outs Reset and the Sim Options Calibrate button and SimBridge status in the kit look; language and keyboard selects sized like the others.
Dialogs and shared parts	Confirmation and alert dialogs, scroll buttons, progress bar, card, divider, the critical error page and the toasts (each toast type its own colour, errors red) no longer use the old colours.
Ground pages	Services legend at its small size; Taxi frequencies button and panel, empty-state text, one-line database chip, Entry chevron; Payload balance chart in the new colours and font; Fuel: no "NaN %" or full bars while the tank capacities are still 0.
Texts	Presets help in cards, with the A380X wording (MFD with KCCU, AFS CP); Performance tab "Descent"; "1 active failure", "1 failure"; descent notes without a trailing separator; the A380X descent legend names the A320 FCOM anti-ice source; chart list and pinned charts empty texts; ATC network name in capitals and an empty controller list message.

Limitations and why

Limitation	Why	What to do
A tooltip ("Click Text to Change the Preset's Name") can stay on screen after leaving the Presets page under the mouse.	An older flyPad bug, not part of these fixes.	Move the mouse over the flyPad again.

How to use

1. Nothing to set. Calibrate the throttle from **flyPad > Settings > Sim Options > Calibrate** as before.

How to test

#	Do this	Expected result
1	A380X: flyPad > Settings > Sim Options > Calibrate.	The calibration fits the pane: four axes in one row, the button bar with Save and Apply visible.
2	Same on the A32NX.	Two axes, the button bar visible.
3	Move a throttle into a detent, incl. the 0 (idle) detent.	The detent is highlighted, also at exactly 0.
4	Settings > About.	The title is not covered; build info as rows.
5	Click several Settings categories.	The open category has a coloured icon and a bold label.
6	Trigger a confirmation dialog (e.g. Ground > Payload deboarding) and a toast.	Both in the Material colours; an error toast is red.
7	Ground > Fuel right after loading the aircraft.	No "NaN %" and no full bars before the tank capacities are known.
8	Failures page with one failure active.	The text reads "1 active failure" (singular).
9	Performance tabs.	The descent calculator tab is named "Descent".

4. Payload page: LOAD title row and balance chart

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Payload | **Status:** Checked in the browser harness on both aircraft; the user reviewed the first version in the sim (screenshot) and chose the grid end. Final version not sim-tested. Unit tests: chart fit.

What it is for

On the redesigned Payload page the LOAD title sat on its own line above the load table and the balance chart left an empty band in its card. Now:

- The LOAD title sits on the PLANNED / CURRENT header row of the load table.
- The SimBrief fill action is a full-width button under the table, as on the Fuel page. It shows with the same rule as before (an OFP loaded that differs from the planned load, no boarding or GSX payload sync running).
- The balance chart fills the height of its card (never smaller than the former 280 px).
- The weight grid ends at the lowest labelled weight (A32NX 40 t, A380X 270 t) instead of leaving an empty band under the envelope; the weight labels are centred on their grid lines, the unit sits under the lowest label, and the MTOW / MLDW / MZFW names are re-placed for the new scale.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed after an update.	The chart scale lives in the aircraft config files (flypad-payload.json5).	Restart the sim after installing.

How to use

1. Open **flyPad > Ground > Payload**; plan the load in the table or with the SimBrief button under it.
2. Read the centre of gravity on the balance chart.

How to test

#	Do this	Expected result
1	A32NX at a gate, flyPad > Ground > Payload.	LOAD is on the PLANNED / CURRENT row of the load table.
2	Import a SimBrief flight plan whose load differs from the planned one (no boarding running).	A full-width SimBrief button appears under the table.
3	Look at the balance chart.	It fills its card; the lowest grid line is 40 t, labels centred on their lines, the unit under the lowest label.
4	Repeat on the A380X.	Same layout; the lowest grid line is 270 t.

5. Independent captain and first officer flyPads

> **Aircraft:** A32NX and A380X | **Where:** Captain and first officer tablets | **Status:** A32NX tested in the sim by the user: OK. A380X: no separate record; both tablets were used in the sim test of the redesigned pages (quick controls on both tablets OK). Unit tests: tablet index.

What it is for

Both tablets showed the same flyPad: their screens share one material in the cockpit model and one gauge drew on it, so one pilot changing a page changed it for the other. Now the first officer tablet has its own flyPad gauge (`efb.html?Index=2`), with a new material on its screen in the cockpit model.

What is per tablet and what is shared

Per tablet	Shared
The page shown and the navigation	The flyPad settings
The loaded OFP (SimBrief import)	The aircraft state (doors, services, fuel, payload)
The checklists and their progress	
The brightness and its Auto setting (first officer: `L:A32NX_EFB_2_BRIGHTNESS`, settings `EFB_2_*`)	
A32NX: the power button (first officer: `H:A32NX_EFB_2_POWER`; before, both buttons sent the same event, so one press toggled both tablets)	

The actions that change the shared aircraft state by themselves run on the captain flyPad only: the tug release after a pushback, the automatic loading of lighting presets, GSX remote control, the rudder pedal and sidestick steering of the pushback, and the reset of the tug commands when the Pushback page opens with no tug attached. A flyPad clears only the OANS taxi flags it has set itself.

Limitations and why

Limitation	Why	What to do
Without the aircraft-large-files companion, the first officer tablet keeps showing the captain flyPad.	The new screen material is in the high-detail cockpit model, which lives in aircraft-large-files (A380X 24f7b02, A32NX d3060ea); the large-files pointer of this branch is not bumped yet.	Merge the two large-files commits, then bump the `large-files` pointer.
A full MSFS restart is needed after an update.	panel.cfg, the cockpit behaviour and the model change.	Restart the sim after installing.

Lighting presets load automatically from the captain flyPad only.	Two flyPads would load them twice.	Keep the captain flyPad on.

How to use

1. Use each tablet on its own: open any page, import the OFP, run the checklists.
2. Set each tablet brightness from its own quick controls.
3. A32NX: each tablet has its own power button.

How to test

#	Do this	Expected result
1	A380X with this branch and the large-files companion, MSFS restarted. Turn both tablets on. Open Dashboard on the captain tablet and Ground > Fuel on the first officer tablet.	Each tablet shows its own page.
2	Import the SimBrief OFP on the captain tablet only.	The captain Dispatch > OFP shows it; the first officer OFP stays empty until imported there.
3	Change the brightness in the quick controls of the first officer tablet.	Only that tablet changes.
4	Open a checklist and check items on one tablet.	The other tablet checklist is unchanged.
5	Repeat on the A32NX; then press the power button of the right tablet.	Only the right tablet turns off.
6	Push back with the flyPad, then open and leave the Pushback page on the first officer tablet during the push.	The pushback does not stop (leaving the page stops it on neither tablet).

6. Quick controls clicks no longer open the troubleshooting page

> **Aircraft:** A32NX and A380X | **Where:** Status bar > quick controls | **Status:** Tested in the sim by the user: OK.

What it is for

The quick controls pane is drawn inside the status bar. When the status bar shows an issue (red), a click on it opens the troubleshooting page, so every click in the quick controls, and the click that closes them, also opened that page. The pane and its backdrop now keep their clicks, and the backdrop closes the pane on the click itself (closing on mouse down let the click land on the status bar under it).

How to use

1. Open the quick controls from the status bar, use them, close them with a click outside.
2. To open the troubleshooting page, click the red status bar itself.

How to test

#	Do this	Expected result
1	A flyPad with a red status bar (an issue listed in Settings > About > Troubleshooting). Open the quick controls.	The quick controls open.
2	Click a switch or slider in the quick controls.	The control works; the page under it stays (no troubleshooting page).
3	Click outside the quick controls.	They close; the troubleshooting page does not open.

4	Click the red status bar.	The troubleshooting page opens, as before.

7. Checklists: always loaded, green check when marked complete

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Checklists | **Status:** Checked in the browser harness. Not sim-tested. Unit tests: checklist marks.

What it is for

- **Checklists sometimes missing.** The checklist file was read without waiting for its text, so an empty list was returned; the flyPad stored (and froze) it and the checklists parsed later could not be added. The file is now read and parsed before the list is returned; a second call waits for the same read, and a failed read is tried again by the next call.
- **Green check.** The list showed the done mark only for the checklists of the current or previous flight phases, so with the aircraft unpowered (flight phase 0) a completed checklist kept an empty circle. A checklist marked with **Mark checklist as complete** now shows a filled green check, whatever the flight phase. One with all items done but not marked keeps the empty circle.

How to use

1. Open **flyPad > Checklists**, pick a checklist, check its items.
2. Press **Mark checklist as complete** at the end.

How to test

#	Do this	Expected result
1	Load the aircraft several times (new flights) and open flyPad > Checklists each time.	The checklists are always listed.
2	Aircraft cold and dark (unpowered). Open a checklist, check all its items, but do not mark it.	The list shows an empty circle for it.
3	Press Mark checklist as complete.	The list shows a filled green check for it, although the aircraft is unpowered.

8. Automatic lighting preset loading lists all eight presets

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Presets > Interior Lighting | **Status:** Checked in the browser harness. Not sim-tested. Unit tests: preset list.

What it is for

The Day, Dawn/Dusk and Night drop-downs of the automatic preset loading offered only the presets that had a name: with unnamed presets they offered only "None". They now list the eight presets as "<number> - <name>" ("No Name" when unnamed).

How to use

1. Open **flyPad > Presets > Interior Lighting** and save the cockpit lighting into a preset.
2. Turn on the automatic loading and pick a preset for Day, Dawn/Dusk and Night.

How to test

#	Do this	Expected result

1	flyPad > Presets > Interior Lighting with no preset named. Turn on the automatic loading and open the Day drop-down.	None, then 1 - No Name ... 8 - No Name.
2	Name preset 2 "Taxi" and open the drop-down again.	It lists 2 - Taxi among the eight.
3	Pick 2 for Day, reload the flight in daytime with the aircraft powered.	Preset 2 is loaded by the captain flyPad.

9. Error details on the flyPad error screen

> **Aircraft:** A32NX and A380X | **Where:** flyPad error screen | **Status:** Used in the sim by the user: it showed the error of the Top of Descent crash, which was then fixed.

What it is for

The flyPad error screen only said that something went wrong. It now shows the error (name and message), the first lines of its stack and the React components it was thrown in, so a crash can be reported and found.

How to use

1. If the flyPad shows the error screen, take a screenshot of the error box and report it.
2. Press **Reset Display** on the error screen to recover.

How to test

#	Do this	Expected result
1	There is no normal way to crash the flyPad on purpose. When the error screen appears (or in the browser harness with a forced error), look at it.	Under "A critical error has been encountered.": a box with the error name and message, up to 6 lines of its stack and up to 6 lines of the component stack.

10. Landing calculator above the wind limits, with a wind check

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Performance > Landing | **Status:** Checked in the browser harness on both aircraft. Not sim-tested. Unit tests: wind ranking, calculators beyond the tailwind limit.

What it is for

A crosswind above the maximum asked "Calculate anyway?" and a tailwind above the limit gave an error and no result. Now the calculation always runs: the pilot sees the figures with a warning and can compare the other runways and the alternate. This is on this branch because it changes the redesigned performance widgets.

What you should see

Situation	What you should see
Crosswind above the maximum	Results shown, with a red banner "Crosswind *n* kt, maximum *m* kt": above the maximum for this runway condition, figures for information.
Tailwind above the limit	Red banner "Tailwind *n* kt, limit *m* kt". The distances, stop margin and MLW(PERF) continue the tailwind correction of the data and are marked extrapolated (~ and a dagger, amber), with a note under the results.
Wind check tab (beside Landing roll, red dot when the wind is above a limit)	The destination runways ranked by wind: within limits first, most headwind on top; LDA, HEAD/TAIL, CROSS and LIMITS columns; each with a **Use** button that loads it.

Alternate card in the wind check	The OFP alternate (ICAO editable), its METAR from the weather source of the settings and its best runway, with **Calculate for <ICAO> <runway>**.
Wind entered without a direction	The ranking asks for the wind with its direction (e.g. 330/45).

The limits used

	A32NX	A380X
Maximum crosswind, gust included	Dry, good or wet 38 kt; 29, 25, 20 or 15 kt on contaminated runways (A320 FCOM PER-LDG-DIS-MAT)	Dry / wet 40 kt; 20 kt slush or dry snow; 15 kt standing water or compacted snow; 5 kt icy
Tailwind limit	15 kt	10 kt

Gusts are included in the components as in FCOM LIM; a variable wind is taken from the worst direction.

Limitations and why

Limitation	Why	What to do
The wind check is not an Airbus function.	It is a design choice of the flyPad: advice only, the flight crew decides.	-
Figures beyond the tailwind limit are extrapolated.	The Airbus data stop at the limit; the tailwind correction is continued.	Use them as an indication only.
The alternate check needs an internet connection and the OFP.	The METAR comes from the weather source set in the flyPad settings; the alternate from the SimBrief OFP.	Enter the alternate ICAO by hand if the OFP has none.

How to use

1. Fill the landing calculator as usual and press Calculate.
2. If a red wind banner shows, open the **Wind check** tab beside the landing roll picture.
3. Load a better runway with **Use**, or the alternate with **Calculate for**, and calculate again.

How to test

#	Do this	Expected result
1	A380X, SimBrief OFP with an alternate loaded. flyPad > Performance > Landing. Enter a runway, a dry runway condition and a wind giving about 45 kt of crosswind. Calculate.	Results shown; red banner "Crosswind 45 kt, maximum 40 kt"; red dot on the Wind check tab. No Yes/No prompt.
2	Enter a wind giving a 15 kt tailwind. Calculate.	Results shown with "Tailwind 15 kt, limit 10 kt"; distances marked ~ and a dagger in amber, and the extrapolation note.
3	Open the Wind check tab.	The destination runways ranked, those within limits first; LIMITS column says "Within limits" or which limit is exceeded.
4	Press Use on the best runway.	The runway is loaded into the calculator; calculate again: no banner if it is within limits.
5	Look at the alternate card.	The OFP alternate, "Fetching the METAR..." then its METAR and best runway; "Calculate for <ICAO> <runway>" loads it.

6	Repeat on the A32NX with a 16 kt tailwind.	Banner "Tailwind 16 kt, limit 15 kt", extrapolated figures.

11. RAT stow (maintenance) row on the Services page

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, equipment list | **Status:** Tested in the sim by the user: OK on both aircraft. Unit test: ratStow.spec (row look and status text).

What it is for

The flyPad side of the RAT ground stow: a maintenance row that asks the systems to stow the RAT after the landing (A320 FCOM DSC-29-10-20: the RAT "can be stowed only when the aircraft is on the ground"). The systems side (`L:A32NX_RAT_STOW_REQUEST`, the shared Rust RAT model) is on `core/rat-ground-stow`; see that manual for the rules, the limitations and the full test.

> **CAUTION**
> Merge both branches: without `core/rat-ground-stow` the row shows while the RAT is out but pressing Stow has no effect.

What you should see

Situation	What you should see
RAT stowed	No RAT row.
RAT out, aircraft stopped on the ground	"RAT stow (maintenance)", status "RAT extended", button **Stow**.
RAT out, in flight or moving	Row greyed, status "RAT extended · stow when stopped on the ground".
Stow pressed	Status "Stowing" for about 3 s; the row disappears when the systems have stowed the RAT, or stays if they refused (an extension still commanded).

How to use

1. Stop on the ground with the RAT out and normal electrical power back.
2. flyPad > Ground > Services: press **Stow** on the row "RAT stow (maintenance)".

How to test

#	Do this	Expected result
1	A380X on the ground, engines running: EMER ELEC PWR panel RAT MAN ON (then release it). flyPad Services.	Row "RAT stow (maintenance)", active.
2	Press Stow.	"Stowing", the RAT rig retracts, the row disappears.
3	A32NX on the ground: HYD panel RAT MAN ON (or EMER ELEC PWR MAN ON), then Stow on the flyPad.	The HYD page RAT goes back to stowed, HYD RAT FAULT clears, the row disappears.

12. A380X cargo door row shows the real doors

> **Aircraft:** A380X | **Where:** flyPad > Ground > Services, Doors list (row "Cargo fwd + aft") and the aircraft planform | **Status:** Sim-tested by the user with the systems part of `core/a380x-cargo-doors` on a combined test build: PASS, recorder-checked. Unit test: cargoDoor.spec (11 cases).

What it is for

The cargo door row of the Services page read the open request of the ground crew, not the doors. With an engine running the request could be on while the doors did not move, and the row said "open". The row now shows the real

forward and aft doors (position and lock, from the systems) and says when the request waits for hydraulic power.

> **CAUTION**
> Merge both branches: the row reads `L:A32NX_HYD_CARGO_DOOR_OPERATION_INHIBITED`, which only `core/a380x-cargo-doors` writes (without it the row reads 0 = not inhibited and shows the real doors only). That manual describes the systems rules (A380 FCOM DSC-29-10, DSC-52-30-40-10) and the full test.

What you should see

Situation	What you should see
Both doors closed and locked, hydraulic power available	Row "Cargo fwd + aft", status "Closed", switch off.
Engine running, doors closed	Status "Closed · no hydraulic power, engine running"; the switch is greyed out.
Engine running, an open request is on and the doors are closed	Status "Closed · opens once the engines are stopped". The switch can still cancel the request.
Doors moving	Status "Moving" with a progress bar; the FWD and AFT holds on the planform follow the position of each door.
Doors open (from 90 %)	Status "Open".
A door stopped open or unlocked by an engine start	Amber row, status "Not locked · no hydraulic power, engine running"; the switch can still ask the door to close (it follows once the engines are stopped).

The row stays one row for both doors (the request moves both); it is closed only when both doors are locked and open only when both are open. The planform now shows the FWD and the AFT cargo door as two holds, each with its own position.

How to use

1. Stop the engines.
2. flyPad > Ground > Services > Doors: switch "Cargo fwd + aft" on to open, off to close; read the status line.

How to test

Use the same sim test as `core/a380x-cargo-doors`, with both branches installed.

#	Do this	Expected result
1	A380X on the ground, APU on, engine 1 running at idle. flyPad > Ground > Services > Doors.	"Cargo fwd + aft": "Closed · no hydraulic power, engine running", switch greyed out.
2	ENG MASTER 1 OFF, wait for the stop. Switch "Cargo fwd + aft" on; wait 40 s.	"Moving" with a progress bar, then "Open"; the FWD and AFT holds light up on the planform.
3	Doors open: start engine 1 and switch the row off.	The doors hold; amber "Not locked · no hydraulic power, engine running"; the switch stays usable.
4	ENG MASTER 1 OFF and wait until the engine has stopped.	The doors close; the row reads "Closed".

13. Custom flyPad theme, and a readable Light theme

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Settings > flyPad > Theme | **Status:** Custom theme: tested in the sim by the user: OK by the user's word (A32NX, no recording). The first colour wheel (12 hues on 4 rings, alert hues hatched): confirmed in the sim by the user ("the wheel works"). The vivid colour wheel that replaced it: OK in the sim (A380X, "the wheel works"). The Light theme label fix: not sim-tested (harness only). Unit tests: themePalette.spec, colourWheel.spec (contrast, alert hue note, stored red / yellow themes kept, presets unchanged).

What it is for

The flyPad had three fixed themes: Blue, Dark and Light. A fourth choice, **Custom**, lets the pilot pick the colours: a base (the surfaces and text of the Gray, Black or Light theme) plus a primary and a secondary colour. The Blue, Dark and Light themes keep their exact colours.

- **Primary colour**: buttons, switches, progress bars, chart lines.
- **Secondary colour**: selected tabs, segmented buttons, chips and the navigation rail. "Same" uses the primary colour.
- The flyPad adjusts only the lightness of the picked colours until every text and control pair reaches the WCAG 2.1 AA contrast (4.5:1 for text, 3:1 for controls). A contrast list beside a preview card shows each pair with "AA" (or "FAIL").
- Amber and red are the alert colours: the preset discs offer none, and a colour close to their hues (picked on the wheel or typed) is accepted with a note that buttons and switches in it may look like warnings.
- The pencil button at the end of the colour discs opens the **colour wheel**: the 12 pure colours of the painter's wheel (yellow at the top, then clockwise yellow-green, green, sky blue, blue, violet, magenta, crimson, red, red-orange, orange, amber) on 5 rings, from the pure colour at the rim to pale tints next to the centre. Every wedge can be picked, red and yellow included; a theme stored in red or yellow is kept. Beside the wheel: the picked colour, a hex field (type a code, then Enter), a preview, the contrast checks of that colour, and, when the flyPad must adjust the colour for the chosen base, the shade it will really use. Nothing changes before **Apply**; **Cancel** keeps the theme as it was. The secondary wheel also offers "Same"; only the hue and colourfulness of a secondary colour count (the base sets its lightness).
- The theme applies to both tablets and to the Payload balance chart. An existing user keeps their theme after the update (the old setting is carried over once).

Light theme fix: several titles and labels were hard-coded white and disappeared on the Light theme (and on a Light base): the Settings page title, the ATC frequency cards and callsign, the checklist and failure (ATA) titles, the preset names, the aircraft symbol of the pushback map, the battery level and the slider labels. They now use the theme text colour; a selected ATC frequency card uses its own text colour.

Limitations and why

Limitation	Why	What to do
Orange, amber or red theme colours look like alerts.	These hues mean caution and warning on the flyPad; the warn and error colours are never themed. The flyPad allows them with a note.	Prefer another hue.
A picked colour can look lighter or darker than its disc or hex code.	Its lightness is moved until the contrast reaches AA on the chosen base.	-

How to use

1. Open **flyPad > Settings > flyPad** and scroll to **Theme**.
2. Press Blue, Dark or Light for a standard theme, or **Custom**.
3. Custom: pick the **Base** (Gray, Black, Light), then a **Primary color** and a **Secondary color** disc, or press the pencil button to pick it on the colour wheel (or type a hex code, then Enter), then **Apply**.
4. Check the contrast list beside the preview card; press **Reset to Blue** to go back to the usual cyan flyPad.

How to test

#	Do this	Expected result

1	Either aircraft, cold and dark at the gate, flyPad on. Note the theme used before the update. Open flyPad > Settings > flyPad, scroll to the bottom.	Theme shows the same theme as before (Blue, Dark or Light), the same colours; no Base / Primary color / Secondary color rows.
2	Press Custom, Base Black, the violet Primary color disc (4th) and the teal Secondary color disc (6th after "Same").	Theme Custom, Base Black; filled buttons and switches violet; the selected tabs (left rail, segmented buttons) teal; the contrast list shows "AA" on every line.
3	Open flyPad > Ground > Payload.	The MTOW line of the balance chart and the "Boarded" seats are violet.
4	Reload the flight, then open Settings > flyPad.	The same custom theme (Black, violet, teal) is back.
5	Primary color: press the pencil button.	The colour wheel: 12 pure colours at the rim (yellow at the top) and 5 rings down to pale tints; no hatched wedge.
6	Press the red wedge of the rim, then Apply.	The note "Close to the amber and red of the alerts: buttons and switches in this color may look like warnings"; after Apply the primary is red. Reload the flight: still red.
7	Pencil button, type 2E86AB and press Enter, then Apply.	No note; the primary becomes a steel blue.
8	Press Reset to Blue.	Theme Blue, the usual cyan flyPad.
9	Choose the Light theme. Open Settings, ATC, Checklists, Failures (an ATA chapter), Presets and Ground > Pushback; look at the battery level in the status bar and at a slider label.	Every title, callsign, preset name, the pushback aircraft symbol, the battery level and the slider labels are dark and readable on the light background (none white).

14. Performance > Buffet: A32NX buffet onset envelope (coffin corner)

> **Aircraft:** A32NX | **Where:** flyPad > Performance > Buffet | **Status:** Tested in the sim by the user: Live page in cruise and in a turn OK, REC MAX 398 as on the MCDU; What if checked from a user screenshot; Live / What if pill row fix OK. Unit tests: 41 (buffet, buffetLive, buffetUnits, RecMaxFlightLevel).

What it is for

A new Performance tab, on the A32NX only: the buffet onset envelope (the coffin corner) from the A320 FCOM LIM-13 BUFFET ONSET chart, for the aircraft now (**Live**) or for any weight, CG, level, Mach and bank (**What if**). It shows how much margin is left before the low speed and high speed buffet, how far the aircraft can bank, the 1.3 g ceiling and the FMS REC MAX.

- The digitised FCOM chart (Mach, pressure altitude, CG, weight) is reduced to one lift table scaled by the ISA pressure ratio, with the chart CG correction (within 1.5 t of the 18 curves; within about 1 % of the QRH M .78 1.3 / 1.4 g lines).
- REC MAX: the FMS value of the MCDU PROG page (A320 FCOM DSC-22_20-40-30: lowest of the 0.3 g buffet margin, MAX CRZ level flight, 300 ft/min at MAX CLB, green dot to VMO/MMO, the certified FL 398). The MCDU now publishes it in `L:A32NX_FM_REC_MAX_FL` (0 without a gross weight) for the flyPad. With one engine out it is the engine-out REC MAX (see the next feature).
- The A380X has no Buffet tab: the A380 FCOM has no buffet onset chart.

What you see

Part	Content
Live / What if	A segmented pill at the top. Live reads the aircraft every second; What if takes typed values (the **A/C** button copies the live values).

Inputs	Gross weight (tag FMS GW, or W&B without an FMS weight; t or klb as set in the flyPad), CG (% MAC), Altitude and Mach (ADR 1, or SIM when ADR 1 is not valid), Bank (IR 1 or SIM).
Chart	Pressure altitude against Mach: the grey 1.0 g line (coffin corner), the cyan 1.3 g line (0.3 g margin), the 1.4 g line (QRH turbulence margin, off by default: chip 1.4 g above the chart), the red VMO / MMO line, the dashed REC MAX line (Live), the aircraft or the entered point; in a turn of 3 degrees or more a dashed amber "NN° bank" line.
Results	The big tile "Buffet margin" (wings level) or "Margin in the turn" with the Level margin (bank 3 degrees or more), then tiles such as Max bank, 1.3 g ceiling, 1.3 g Mach range, REC MAX (Live), Max operating (Mach, MMO or VMO) and 1.0 g ceiling. Margins amber under 0.3 g, red under 0.2 g.
Banners	Red: buffet onset in the turn ("Buffet onset in a 30° turn: max bank 27°..."), level margin below the FMS 0.2 g limit. Amber: level margin below the 0.3 g of REC MAX. Neutral: no gross weight, Mach outside the chart (M .50 to .82).

Pill row fix: the row of the Live / What if pill was 2 px shorter than the pill, so the top of the pill was cut off. The row now takes the height of the pill.

Limitations and why

Limitation	Why	What to do
A32NX only.	No A380 buffet onset chart in the FCOM (its BUFFET CEILING figure is qualitative).	-
No REC MAX tile and line in What if, and none in Live without a gross weight in the FMS.	REC MAX is the FMS value for the aircraft now.	Complete MCDU INIT A and INIT B (ZFW / ZFWCG, block fuel).
Live values from ADR 1 and IR 1.	Design choice: the real flyPad has no ADIRS link; ADR 1 / IR 1 are the captain side sources.	The tag beside each value says where it comes from.
The envelope is advice only.	It is a flyPad design choice built on the FCOM chart, not an Airbus flyPad function.	The FMS REC MAX and the FCOM stay the reference.

How to use

1. Open **flyPad > Performance**, then the **Buffet** tab (5th tab, top right).
2. In cruise, keep **Live**: read the buffet margin, the max bank and REC MAX.
3. To plan a level or a turn, press **What if**, enter Gross weight, CG, Altitude, Mach and Bank, and read the margin and the banner.
4. Turn on the **1.4 g** chip to see the turbulence margin line.

How to test

#	Do this	Expected result
1	A32NX in cruise at FL350 to FL390, about M .78, wings level, autopilot ON, MCDU INIT A and INIT B done. On the MCDU press PROG and note the REC MAX flight level.	A REC MAX FL on the PROG page.
2	flyPad > Performance > Buffet.	Buffet is the 5th tab; Live selected (the pill not cut at the top); Gross weight tagged FMS GW, Altitude and Mach ADR 1, Bank 0 with IR 1; chart with the grey 1.0 g, cyan 1.3 g and red VMO/MMO lines and the aircraft at your FL and Mach; the REC MAX tile and dashed line show the MCDU value; no red banner.

3	Turn the FCU HDG knob 90 degrees left (about 25 degrees of bank) and watch the page.	While the bank is 3 degrees or more: Bank follows the roll, a dashed amber "NN° bank" line, the big tile reads "Margin in the turn", Max bank and Level margin shown. Wings level again: "Buffet margin", no amber line.
4	Press What if. Enter Gross weight 76 t (167.6 klb), CG 24, Altitude 39000, Mach 0.76, Bank 30.	Red banner "Buffet onset in a 30° turn: max bank 27° at FL390, M .760. Level margin 0.13 g is below the FMS 0.2 g limit."; Margin in the turn -0.02 g in red; 1.3 g Mach range "none"; 1.3 g ceiling FL361; no REC MAX tile or line.
5	Press the 1.4 g chip above the chart.	A light cyan 1.4 g line and its legend line appear.
6	Open flyPad > Performance on the A380X.	No Buffet tab.

15. A32NX MCDU PROG: engine-out REC MAX

> **Aircraft:** A32NX | **Where:** MCDU PROG page, REC MAX; flyPad > Performance > Buffet | **Status:** Tested in the sim by the user: OK (user values). Unit test: RecMaxFlightLevel.spec.

What it is for

With one engine out the PROG page REC MAX still showed the all-engines value (about FL380). The FCOM says it shows the engine-out value then.

> **NOTE**
> A320 FCOM, MCDU PROG page : "If one engine is out, this field displays the recommended maximum engine-out altitude, that is computed based on the long-range cruise speed and assuming that anti-ice is off."

When the FMS is in its engine-out condition, REC MAX is now the one engine inoperative long-range cruise ceiling: a straight line through the two FCOM examples, FL182 at 70 t and FL203 at 62 t (PER-OEI-GEN-10 and -15), limited to FL398. With both engines REC MAX is unchanged. The flyPad Buffet page reads the same value. This branch holds only the REC MAX part of the A32NX (the engine, ignition, gravity feed and IR parts are on `core/failures-a32nx`).

Limitations and why

Limitation	Why	What to do
No ISA deviation or anti-ice effect on the engine-out value.	The ceiling table is in the QRH, not in the FCOM text; the FCOM gives ISA examples only, and the PROG value assumes anti-ice off.	Use the QRH for an exact ceiling.

How to use

1. After an engine failure in cruise, open the MCDU PROG page and read REC MAX.

How to test

#	Do this	Expected result
1	A32NX in cruise, both engines running, MCDU INIT A and INIT B done. MCDU PROG page: note REC MAX.	The all-engines value (about FL380).
2	Pedestal: ENG MASTER 2 OFF. MCDU PROG page.	REC MAX about FL180 to FL205 (depends on the weight: about FL182 at 70 t, FL203 at 62 t).
3	flyPad > Performance > Buffet, Live.	The REC MAX tile shows the same engine-out value.

4	ENG MASTER 2 ON (relight), wait for idle. MCDU PROG page.	REC MAX back to the all-engines value.

16. GSX services link: passenger counts, request chips, greyed switch

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, GSX services card; Ground > Payload | **Status:** GSX boarding passenger number and counter: tested in the sim by the user: OK (A32NX: GSX given 170 passengers at switch-on, count 0 to 170, Done). Greyed switch, request chips and the deboarding number: not sim-tested. Unit tests: GsxRemote, GsxServicesPanel, ServicesLayout, gsxPassengers, gsxPassengerSync, Material.

What it is for

Fixes of the GSX link of the Services page (the GSX services card, from `core/efb-ground`), found in the sim (A380X at RJTT, GSX Remote API). The link needs the third-party add-on GSX Pro (FSDreamTeam) with its Remote control server on.

- **Passenger number.** GSX boarded its own number of passengers and the Board counter stayed at 0. The flyPad now gives GSX its passenger number (`L:FSDT_GSX_NUMPASSENGERS`, which GSX uses in place of its own figure when set before the boarding or deboarding, GSX manual DEVELOPERS): the passengers planned on the Payload page, or the passengers on board once a deboarding is requested or after landing. The Services page and the Payload pages of both aircraft use the same rule; before, a GSX deboarding started from the Payload page was given 0 passengers.
- **Passenger counter.** The flyPad counts the GSX boarding and deboarding passengers itself: the Board line shows boarded over planned (e.g. 45/120) and its bar fills; before, it showed GSX's own figure twice.
- **Request chips.** A turnaround step offers **Request** only while GSX lets the service be triggered, and **Stop** while it runs. A completed service no longer offers Request (also the Ground equipment rows, e.g. the Catering Truck).
- **Greyed switch.** Without GSX the GSX services switch is faded and does not move. The disabled look of the Material controls is now set by the flyPad itself (the sim browser does not reliably support the CSS :disabled state).

Limitations and why

Limitation	Why	What to do
Needs GSX Pro.	A third-party add-on (FSDreamTeam); the flyPad only drives it.	Install GSX Pro and turn on its Remote control server (GSX settings).
GSX resets its passenger number to 0 when it restarts.	GSX behaviour (GSX manual).	The flyPad gives the number again while the Services page is linked, and just before a Board / Deboard request.

How to use

1. Start GSX Pro with its Remote control server on.
2. Plan the passengers on **flyPad > Ground > Payload** (do not press Board).
3. On **Ground > Services**, turn the **GSX services** switch on when the card says "GSX found".
4. Request the jet bridge or stairs, then **Board > Request** in the Turnaround card; after landing, **Deboard > Request**.

How to test

#	Do this	Expected result

1	Needs GSX Pro (third-party) with its Remote control server on. A380X (or A32NX) at a gate with a jet bridge, GSX not started yet (start the flight before Couatl, or Couatl tray icon > Exit). Open flyPad > Ground > Services.	In the GSX services card: the switch faded and it does not move when tapped; red text "GSX not found: start GSX, with its Remote control server on (GSX settings)".
2	Start GSX (Couatl) and wait.	The card says "GSX found"; the switch is at full brightness and can be turned on.
3	Ground > Payload: set 120 passengers (do not press Board). Back on Services, turn the GSX services switch on.	Within 2 s GSX is given 120 passengers (`L:FSDT_GSX_NUMPASSENGERS` = 120).
4	Ground equipment > Jet Bridge > Request; when connected, Turnaround > Board > Request. Watch the Board line for 2 minutes.	GSX boards 120 passengers; the Board line reads boarded/120, starting near 0 and rising, its bar filling; at the end "Done 120/120"; Ground > Payload shows the same passengers.
5	Look at the Board line, and at Catering (Turnaround and Ground equipment) if catering is done.	No Request chip on a Done service.
6	After the next landing at a gate with GSX: Turnaround > Deboard > Request (Payload page open or not).	GSX is given the passengers on board (not 0) and deboards them; the Deboard line counts them.

17. GSX turnaround: completed services remembered, Baggage steps

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, GSX services card > Turnaround | **Status:** Tested in the sim by the user (A380X): Done kept after GSX reset the boarding to available (recording), Baggage row Done (recording: boarding cargo 50 to 100 %), Turnaround horizontal scrollbar gone. Unit tests: gsxTurnaround, GsxServicesPanel.

What it is for

Needs the third-party add-on GSX Pro. GSX gives one completion signal, the service state 6 "completed", and does not keep it: in the sim (A32NX) GSX set the boarding state back to 1 (available) 14 s after the end of the boarding, so the turnaround offered the boarding again as if it had not been done.

- **Completed services remembered.** The flyPad itself remembers the completed boarding, deboarding, catering and refuelling of the current turnaround (also across flyPad page changes). A service requested or running again is forgotten; the boarding is forgotten when a deboarding starts, and the reverse; everything is forgotten at the end of the turnaround (an engine running or the pushback requested). The jet bridge, stairs and GPU follow the sim.
- **Baggage steps.** A Baggage row after Board in the departure turnaround (catering, boarding, baggage, pushback) and after Deboard in the arrival turnaround (deboarding, baggage, catering). GSX has no baggage service to request: it loads the baggage during the boarding and unloads it during the deboarding, so the row has no Request button ("Loaded with the boarding" / "Unloaded with the deboarding" while idle) and shows the GSX baggage progress (`L:FSDT_GSX_BOARDING_CARGO_PERCENT` / `L:FSDT_GSX_DEBOARDING_CARGO_PERCENT`). It is Done when 100 % is reached after the loading was seen running in this turnaround (a 100 % left from an earlier loading does not count), remembered and forgotten with its boarding or deboarding.
- **List fixes.** The Turnaround list scrolls vertically only (no horizontal scrollbar), and the progress bar of a step sits in its own full-width box under the texts (it was pushed out of the card).

Limitations and why

Limitation	Why	What to do
Needs GSX Pro.	A third-party add-on; the states come from its variables and Remote API.	Install GSX Pro with its Remote control server on.

The Done marks last for the current turnaround only.	GSX does not keep its completion; the flyPad clears them when an engine runs or the pushback is requested.	-
No Request on the Baggage rows.	GSX has no separate baggage service.	Request Board or Deboard.

How to use

1. On **Ground > Services**, link GSX with the GSX services switch.
2. Run the turnaround steps in order with their Request chips; a Done step keeps its check mark.
3. Follow the baggage on the Baggage row during the boarding or deboarding.

How to test

#	Do this	Expected result
1	Needs GSX Pro (third-party) with its Remote control server on. A380X or A32NX at a gate, GSX linked on Ground > Services, passengers planned. Look at the Turnaround card.	The departure steps in this order: catering, boarding, Baggage, pushback (service names as GSX gives them); the Baggage row reads "Loaded with the boarding", with no Request button. No horizontal scrollbar in the list.
2	Request Board and watch the Board and Baggage rows.	The progress bars stay inside the card, under the texts; the Baggage progress rises with the GSX baggage loading.
3	Wait for the end of the boarding and the baggage loading.	Board and Baggage both Done with a check mark.
4	Wait about 30 s (GSX resets the boarding to available), then open Ground > Payload and come back to Services.	Board and Baggage are still Done; Board offers no Request.
5	Start an engine (or request the pushback).	The turnaround is cleared: the Done marks go.
6	After landing at a gate with GSX, look at the Turnaround card, then Request Deboard.	The arrival steps: deboarding, Baggage ("Unloaded with the deboarding"), catering; at the end the deboarding and Baggage are Done.

18. Taxi page: A380 taxiway width check

> **Aircraft:** A380X | **Where:** flyPad > Ground > Taxi (A380X) | **Status:** Not sim-tested (added; unit tests: taxiWidth.spec.ts; checked in the browser harness on the synthetic test airport XTST with an 18 m taxiway N: the suggestion avoids N, a clearance via N is drawn and flagged).

What it is for

The taxi route page did not know whether a taxiway suits the A380 (wingspan 79.75 m, aerodrome reference code letter F). ICAO Annex 14 Volume I, 3.9.3 (taxiway width of the straight portion) gives 25 m for code letter F. On the A380 flyPad, each taxi line of the Navigraph airport map is now rated:

- **Wingspan limit of the line** in the airport map, when the map has one, against 79.75 m (not seen in the Navigraph data so far; read only when present).
- **Else the pavement width under the line:** measured across the line (60 m each side) through the taxiway, apron and runway areas of the map (adjacent areas joined, holes left out), the narrowest of samples every 30 m away from the junction fillets, against 25 m with 1 m tolerance.
- **Else "not verified"** (no data), which is not treated as unsafe.

Situation	What you should see

SUGGEST, or the proposed runway entry	The route keeps off the lines too narrow when it can.
A typed clearance that needs a line too narrow	The route still follows the clearance and is drawn, but flagged: the route text marks the line, and a chip and a banner list the narrow taxiways (ATC may know better: the crew decides).
The map	Lines too narrow in orange, lines not verified dashed grey, with a legend.
A32NX	No width check (the A320 is code C).

Limitations and why

Limitation	Why	What to do
Pavement width is not wingtip clearance.	Buildings and parked aircraft are not in the check.	Follow the airport charts.
Lines on aprons measure wide.	The apron area around them counts as pavement.	-
An airport may clear the A380 on narrower taxiways with restrictions.	The flag means "check the airport A380 notes", not "forbidden".	Check the airport A380 notes.
Airport loading may take longer.	The runway intersection layer is now loaded for the pavement, and every line is measured.	-

How to use

1. A380X flyPad > Ground > Taxi: plan the route as usual.
2. Look at the map legend: orange lines are too narrow, dashed grey ones not verified.
3. If the route chip or banner lists narrow taxiways, check the airport A380 notes or ask ATC for another route.

How to test

#	Do this	Expected result
1	A380X at RJTT (Navigraph linked). flyPad > Ground > Taxi, Departure, airport RJTT.	The map shows a legend; some lines in orange (too narrow) or dashed grey (not verified).
2	SUGGEST a route to a runway.	The route avoids the orange lines when another way exists.
3	Type a clearance through an orange taxiway.	The route follows it and is drawn; the route text marks the line; a chip and a banner list it.
4	Same page on the A32NX.	No width colours, no banner.

What PR testers should check

- At RJTT (A380): which taxiways turn orange or dashed, and whether known A380 routes show as too narrow.
- The airport load time.
- Not sim-tested yet.

19. A380X smart FIVE HUNDRED pin option back in the flyPad

> **Aircraft:** A380X | **Where:** flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs | **Status:** Not sim-tested (added; the callout logic it sets has a unit test on `core/a380x-elec-fuel-alerts`).

What it is for

The A380 FCOM automatic callouts (DSC-34-10-50) have a "smart FIVE HUNDRED" pin programming: the FIVE HUNDRED callout is given only when the aircraft is more than half a dot off the glide slope, or without a valid glide slope. The flyPad option was commented out. The Automatic Call Outs page shows it again, as a sub-option **Inhibited When Glideslope Captured** under **Five Hundred** (greyed while Five Hundred is off).

Limitations and why

Limitation	Why	What to do
The option does nothing without `core/a380x-elec-fuel-alerts`.	The FWS callout that reads the pin is on that branch.	Merge both.

How to use

1. flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs: Five Hundred on, then **Inhibited When Glideslope Captured** on.

How to test

#	Do this	Expected result
1	A380X flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs.	Under Five Hundred the sub-option Inhibited When Glideslope Captured; greyed while Five Hundred is off.
2	Turn both on; fly an ILS approach on the glide slope (with `core/a380x-elec-fuel-alerts`).	No FIVE HUNDRED callout at 500 ft.
3	Repeat more than half a dot off the glide slope (or a non-ILS approach).	FIVE HUNDRED is called.

What PR testers should check

- The option is saved and greyed correctly; the callout follows it with the FWS branch merged.
- Not sim-tested yet.

A32NX air conditioning per the FCOM

The A32NX air conditioning controllers, alerts, STATUS lines and SD COND / BLEED pages follow the A320 FCOM (DSC-21-10, PRO-ABN-AIR, PRO-ABN-COND)

Aircraft	A32NX
Where to find it	Overhead AIR COND panel (PACK, HOT AIR, temperature selectors), ENG FIRE pushbuttons; E/WD; SD COND, BLEED and STATUS pages
Branch	`core/a32nx-air-conditioning`, based on `core/gps` (stacked: needs core/surveillance, core/a380x-mfd-fms and core/gps merged first; the STATUS lines use the A32NX STATUS page of core/gps).
Status	All four features tested in the sim by the user: OK. Two triggers cannot be produced in the sim (fourth duct excursion above 80 °C, pack valve position disagree): covered by Rust and vitest unit tests. A full MSFS restart is needed after installing (systems WASM).
Pictures	No pictures: the COND and BLEED pages are the stock FBW pages with the changes described.

1. Air conditioning controller per the FCOM
2. COND page trim air valve XX and air conditioning STATUS lines
3. Packs regulate cockpit and cabin separately with the hot air valve closed
4. BLEED page: each pack's own outlet temperature

1. Air conditioning controller per the FCOM

> **Aircraft:** A32NX | **Where:** Overhead AIR COND panel, ENG FIRE pb; SD BLEED / COND | **Status:** Tested in the sim by the user: OK: trim air valves, galley fan, FIRE pb per side, 30 s reopening. Fourth excursion and valve disagree: unit tests only.

What it is for

An audit of the air conditioning system controllers (ACSC) against A320 FCOM DSC-21-10 and PRO-ABN-AIR / PRO-ABN-COND found six deviations. All six are fixed:

- **ENG FIRE pushbutton:** each one closes the pack of its own side (DSC-21-10-50). Before, the pack flow controller read the ENG 1 FIRE pb for both packs: ENG 2 FIRE closed pack 1.
- **Pack valves after an engine start:** on the ground they reopen 30 s after the engine start (DSC-21-10-50 PACK pb: "reopening of the valves is delayed for 30 s to avoid a supplementary pack closure cycle during second engine start"); in flight at once.
- **Hot air valve and trim air valve failures:** the hot air pressure regulating valve closes when the cockpit trim air valve fails or both cabin ones fail, and stays open with one cabin trim air valve failed (DSC-21-10-20). Before, it closed on any trim air valve failure.
- **LAV + GALLEY FAN fault:** the cabin temperature sensors are no longer ventilated, so the FWD / AFT CABIN selectors set the cabin duct temperature directly (it was fixed at 15 °C). With ACSC 2 also lost, the cabin gets the cockpit duct air (PRO-ABN-COND).
- **DUCT OVHT** also triggers on the fourth excursion of a duct temperature above 80 °C in one flight, not only above 88 °C (PRO-ABN-COND). The count restarts at each takeoff run.
- **PACK 1(2) FAULT** also triggers when the pack flow control valve position disagrees with its command, not only on the loss of both lanes of the ACSC (PRO-ABN-AIR).

What you should see

Situation	What you should see
ENG 2 FIRE pb pushed	SD BLEED: pack 2 valve closes, pack 1 stays open (before: pack 1 closed).
Engine start on the ground, PACK pbs ON	The pack valves stay closed for 30 s after the start, then open.

Forward (or aft) cabin trim air valve failed	The hot air valve stays open; only that zone loses its trim.
Cockpit trim air valve failed, or both cabin ones	The hot air valve closes (SD COND: hot air valve symbol closed). The HOT AIR pb shows no FAULT for this automatic closure.
Lavatory and galley extraction fan failed	COND LAV+GALLEY FAN FAULT; turning the FWD / AFT CABIN selectors moves the cabin duct temperature directly.

Limitations and why

Limitation	Why	What to do
The fourth-excursion DUCT OVHT and the valve-disagree PACK FAULT cannot be produced in the sim.	There is no pack valve failure, and the valve closes by itself on low pressure; the duct temperature cannot be driven above 80 °C four times on purpose.	Covered by Rust tests and the AcscPackFaults vitest spec.
Not simulated: PACK REGUL FAULT, bypass valve and ram air inlet, pack overheat.	The packs are a placeholder in the FBW model.	None.
A full MSFS restart is needed after installing.	The controllers are in the systems WASM module.	Restart the sim.

How to use

1. Nothing new to set: use the AIR COND panel as usual (PACK 1 / 2, HOT AIR, COCKPIT / FWD CABIN / AFT CABIN).
2. In an engine fire the ENG FIRE pb closes the pack of that side only.

How to test

#	Do this	Expected result
1	A32NX on the ground, both engines running, PACK 1 and 2 ON. Open the SD **BLEED** page. Push the **ENG 2 FIRE** pb.	Pack 2 valve closes; pack 1 stays open. (Without `core/fire` the stock FBW also shows an ENG 2 FIRE warning when the pb is pushed: not part of this test.) Reset the pb.
2	Engines off, APU bleed ON, PACKs ON. Start engine 1 and watch the pack valves on the BLEED page.	The pack valves close during the start and reopen about 30 s after the engine is started.
3	Open the SD **COND** page. flyPad > Failures > ATA 21 Air Conditioning > **Forward Zone Trim Air Valve**.	The hot air valve stays open; the FWD valve shows amber XX (see the next feature).
4	Add **Aft Zone Trim Air Valve**.	The hot air valve closes (both cabin valves failed).
5	Clear them; fail **Cockpit Trim Air Valve** alone.	The hot air valve closes.
6	Clear the failures. flyPad > Failures > ATA 21 Air Conditioning > **Extraction Fan of lavatory and galley**. Turn the FWD CABIN selector from cold to hot.	E/WD: COND LAV+GALLEY FAN FAULT. The FWD cabin duct temperature on the COND page follows the selector directly.
7	Also fail **ACSC 2 Lane 1** and **ACSC 2 Lane 2**.	The cabin ducts take the cockpit duct temperature.

2. COND page trim air valve XX and air conditioning STATUS lines

> **Aircraft:** A32NX | **Where:** SD COND page; E/WD; SD STATUS page | **Status:** Tested in the sim by the user: OK.

What it is for

FCOM DSC-21-10-50 (COND page, zone trim air valve): "The arrow is green. It is replaced by amber crosses ("XX") if the valve fails". The FBW page had a TODO there. It now shows amber XX for a valve that either ACSC reports failed.

The air conditioning alerts now fill the ECAM STATUS page (PRO-ABN-AIR, PRO-ABN-COND). The STATUS page gains a limitations part (blue, before the information):

Alert	STATUS left column	INOP SYS
AIR PACK 1(2) FAULT	CKPT (CAB) AT FIXED TEMP when ACSC 1 (2) is lost	PACK 1(2), COND CTL 1(2) when the ACSC is lost
AIR PACK 1+2 FAULT	MAX FL.....100/MEA-MORA (blue, first), CKPT AT FIXED TEMP, CAB AT FIXED TEMP	PACK 1+2, COND CTL 1, COND CTL 2
AIR PACK 1(2) OFF	-	PACK 1(2)
AIR COND CTL 1(2)-A(B) FAULT	-	COND CTL 1(2)-A(B)
COND CKPT / FWD CAB / AFT CAB DUCT OVHT	CAB TEMP BY PACK ONLY	HOT AIR
COND HOT AIR FAULT	CAB TEMP BY PACK ONLY (hot air closed, packs on)	HOT AIR; PACK 1+2 with both packs off
COND L+R CAB FAN FAULT	-	L+R CAB FAN
COND LAV+GALLEY FAN FAULT	CAB TEMP CKPT CTL ONLY (ACSC 2 working) or CAB AT FIXED TEMP (ACSC 2 lost)	GALLEY FAN (+ PACK 2, COND CTL 2 with ACSC 2 lost)

- COND HOT AIR FAULT gets the full FCOM procedure: -HOT AIR...OFF, then ".IF HOT AIR STILL OPEN AND DUCT OVHT PERSISTS:" -PACK 1 OFF, -PACK 2 OFF, -DES TO FL 100/MEA-MORA, ".WHEN DIFF PR <1 PSI AND FL BELOW 100:" -RAM AIR ON, MAX FL 100/MEA-MORA (the descent lines in flight only).
- The TRIM AIR SYS FAULT line "CPKT TRIM VALVE" is now spelled **CKPT TRIM VALVE**.
- The FCOM FWD CRG HEAT line is left out: the FBW A320 has no cargo heating.

How to use

1. After an air conditioning alert, clear it with CLR and read the STATUS page (it comes up by itself, or press STS).
2. On the SD COND page, an amber XX in place of a zone arrow means that trim air valve has failed.

How to test

#	Do this	Expected result
1	A32NX powered, packs on. SD COND page. flyPad > Failures > ATA 21 Air Conditioning > **Cockpit Trim Air Valve**.	COND page: amber XX in place of the cockpit valve arrow. E/WD: COND TRIM AIR SYS FAULT, -CKPT TRIM VALVE (not CPKT).
2	Clear it. Fail **Cabin Fan 1** and **Cabin Fan 2**.	COND L+R CAB FAN FAULT, -PACK FLOW...HI. After CLR, STATUS: INOP SYS L+R CAB FAN.
3	Clear them. Fail **ACSC 1 Lane 1**.	AIR COND CTL 1-A FAULT; STATUS: INOP SYS COND CTL 1-A.
4	Also fail **ACSC 1 Lane 2**.	AIR PACK 1 FAULT; STATUS: CKPT AT FIXED TEMP (green), INOP SYS PACK 1 and COND CTL 1.

5	Also fail **ACSC 2 Lane 1** and **ACSC 2 Lane 2**.	AIR PACK 1+2 FAULT; STATUS: MAX FL.....100/MEA-MORA in blue on top, then CKPT AT FIXED TEMP and CAB AT FIXED TEMP; INOP SYS PACK 1+2, COND CTL 1, COND CTL 2.
6	Clear all. Fail **Cockpit Duct Overheat**.	COND CKPT DUCT OVHT; STATUS: CAB TEMP BY PACK ONLY, INOP SYS HOT AIR.
7	In flight, with **Trim Air Pressure Regulating Valve** failed (the valve stays where it is) and a duct overheat failure, watch COND HOT AIR FAULT if it is raised with the valve still open.	The lines .IF HOT AIR STILL OPEN AND DUCT OVHT PERSISTS:, -PACK 1 / -PACK 2 OFF, -DES TO FL 100/MEA-MORA, .WHEN DIFF PR <1 PSI AND FL BELOW 100:, -RAM AIR ON, MAX FL 100/MEA-MORA.

3. Packs regulate cockpit and cabin separately with the hot air valve closed

> **Aircraft:** A32NX | **Where:** Overhead HOT AIR pb and temperature selectors; SD BLEED | **Status:** Tested in the sim by the user: OK: HOT AIR OFF pack 1 16 °C / pack 2 29 °C; ON both back to the same value within 60 s.

What it is for

With the hot air pressure regulating valve closed (failure, HOT AIR pb OFF or the automatic closure), the trim air valves cannot mix the zones. FCOM DSC-21-10-40 (hot air valve failed closed): pack 1 regulates to the cockpit demand and pack 2 to the mean of the FWD and AFT cabin demands. With ACSC 2 lost, pack 2 follows the cockpit demand (PRO-ABN-COND). With the valve open, both packs still supply the lowest zone demand (DSC-21-10-20). Before, both packs always followed the lowest demand.

Limitations and why

Limitation	Why	What to do
The zone temperatures do not separate as much as the pack outlets.	The FBW air mixer gives every zone the same air, so pack 1 cannot cool the cockpit alone.	Read the effect on the BLEED page pack outlet temperatures.

How to use

1. With the HOT AIR pb OFF (or the valve failed closed), set the COCKPIT selector for the cockpit and the FWD / AFT CABIN selectors for the cabin: pack 1 serves the cockpit, pack 2 the cabin.

How to test

#	Do this	Expected result
1	A32NX, packs running, SD **BLEED** page. HOT AIR pb ON. Set COCKPIT cold, FWD and AFT CABIN hot.	Both pack outlet temperatures are the same (the lowest demand).
2	Set the HOT AIR pb OFF and wait about a minute.	Pack 1 outlet colder (cockpit demand), pack 2 warmer (cabin demand): for example 16 °C and 29 °C.
3	Set the HOT AIR pb ON again.	Both pack outlets go back to the same value within about 60 s.

4. BLEED page: each pack's own outlet temperature

> **Aircraft:** A32NX | **Where:** SD BLEED page | **Status:** Tested in the sim by the user: OK: HOT AIR ON 10 / 10 °C, OFF 15 / 30 °C.

What it is for

The pack outlet temperatures of the SD BLEED page were a placeholder showing the FWD cabin zone temperature on both packs. They now show the outlet temperature the air conditioning system computes for each pack, in 5 °C steps, amber above 90 °C (FCOM DSC-21-10-50, ECAM BLEED page, item 1).

Limitations and why

Limitation	Why	What to do
The pack bypass valve and the compressor outlet temperature are still placeholders.	The packs are not modelled internally yet.	None.

How to use

1. Open the SD **BLEED** page: the temperature above each pack is that pack's outlet temperature.

How to test

#	Do this	Expected result
1	A32NX, packs running, SD BLEED page, HOT AIR pb ON.	Both pack outlet values equal (for example 10), in steps of 5.
2	HOT AIR pb OFF, COCKPIT cold and cabin selectors hot.	The two values differ (for example 15 and 30): they are each pack's own outlet, no longer the FWD cabin temperature.

A380X cockpit buttons, lights and probes

Dead A380X cockpit pushbuttons now work, the 52 RESET panel buttons can be clicked, the FMC-C reset button is checked, the AUTOLAND lights flash, the side slip and angle of attack vanes on the nose move, the ACD keys beside the OITs work and the F/O laptop keyboard has its keys

Aircraft	A380X
Where to find it	Thrust levers, overhead EVAC and RESET panels, right pedestal, glareshield AUTOLAND lights, nose probes (outside view)
Branch	`core/a380x-cockpit`, based on `master` (FlyByWire master 76db0aef1 merged in); independent of the other core branches. The MFP vanes and the ACD / F/O keyboard also need their companion aircraft-large-files changes (fork TCdr/aircraft-large-files, branches feat/a380x-mfp-vane-animations, b8fb739, and feat/a380x-acd-keys-fo-keyboard, 7f56bcb).
Status	All six features tested in the sim by the user: OK. Pushbuttons, RESET panel and side slip vanes, FMC-C reset check; AUTOLAND light and MFP vanes; ACD keys and F/O keyboard.
Pictures	No pictures: the buttons and the vanes are the existing cockpit and fuselage models.

1. A/THR disconnect, EVAC HORN, DFDR EVENT and ACMS TRIGGER pushbuttons work
2. Clickable RESET panel buttons and side slip probe vanes
3. FMC-C reset button checked in the all-FMCs-reset test
4. AUTOLAND light press-to-test and flashing
5. MFP vanes follow the angle of attack
6. ACD keys beside the OITs work; F/O laptop keyboard keys

1. A/THR disconnect, EVAC HORN, DFDR EVENT and ACMS TRIGGER pushbuttons work

> **Aircraft:** A380X | **Where:** Thrust levers 1 and 4, overhead EVAC panel, right pedestal | **Status:** Tested in the sim by the user: OK.

What it is for

The cockpit behaviour XML named model nodes or animations that are not in the A380X cockpit model, so these pushbuttons could not be clicked or did not move. Checked against the cockpit model:

Pushbutton	Before	Now
A/THR instinctive disconnect pbs (thrust levers 1 and 4)	Not clickable (PUSH_THROTTLE_1/2)	Clickable (ATHR_DISCONNECT_ENG1/2): they disconnect the A/THR through the existing FBW interface
EVAC HORN SHUT OFF	Not clickable (PUSH_OVHD_EVAC_HORN)	Clickable (PUSH_OVHD_EVAC_HORN_OFF)
DFDR EVENT, ACMS TRIGGER (right pedestal)	Did not move	Move when pressed (ANIM_PUSH_DFDR_EVENT / ANIM_PUSH_ACMS_TRIGGER)

Limitations and why

Limitation	Why	What to do
DFDR EVENT and ACMS TRIGGER only move.	There is no flight data recorder or ACMS function behind them; their tooltips still say (Inop.).	None.

How to use

1. Press an A/THR instinctive disconnect pushbutton on thrust lever 1 or 4 to disconnect the A/THR.
2. Press EVAC HORN SHUT OFF on the overhead EVAC panel, DFDR EVENT or ACMS TRIGGER on the right pedestal as on the aircraft.

How to test

#	Do this	Expected result
1	A380X in flight, A/THR active. Click the instinctive disconnect pushbutton on thrust lever 1.	The pushbutton moves and the A/THR disconnects (A/THR leaves the FMA).
2	Engage the A/THR again and click the pushbutton on thrust lever 4.	Same result.
3	Click EVAC HORN SHUT OFF on the overhead EVAC panel.	The pushbutton can be clicked and moves.
4	Click DFDR EVENT and ACMS TRIGGER on the right pedestal.	Each one moves in and out when pressed.

2. Clickable RESET panel buttons and side slip probe vanes

> **Aircraft:** A380X | **Where:** Overhead RESET panels; nose side slip probes | **Status:** Tested in the sim by the user: OK; the user accepted that 42 buttons only drive their variable, and confirmed the vane direction.

What it is for

All 52 circuit breakers of the overhead RESET panels can now be pulled and pushed, with their FCOM label as tooltip. Each one writes `L:A32NX_RESET_PANEL_<name>` (true = pulled, computer reset).

- Read by the avionics: ARPT NAV, FMC A, FMC B, FMC C, FWS 1, FWS 2, AESU 1, AESU 2, NSS AVNCS, NSS FLT OPS.
- The other 42 (AICU, ATC, AVS, BSCS, CIDS, CPCS, DSMS, DTLNK ROUTER, EIPMs, ESS TR, TR 1, TR 2A, FLAPS, FQMS, GCU, LGCIS, PACK CTL, PAX BBAND, SCS, SDF, SLATS, TCS, VCS) are clickable but no system reads them yet.

The three side slip probe vanes on the nose now follow the side slip angle (sim `INCIDENCE BETA`, up to 60 degrees each side): they line up with the airflow.

Limitations and why

Limitation	Why	What to do
42 RESET buttons do nothing in the systems.	The systems they reset are not wired to the RESET panel yet.	Use the 10 wired ones to test a reset.

How to use

1. Hover a RESET panel button to read its label; click to pull it (reset), click again to push it back.
2. Look at the nose from the outside view to see the side slip vanes.

How to test

#	Do this	Expected result
1	Hover the buttons of the overhead RESET panels.	Each one shows its FCOM label as tooltip.
2	Click a button.	It pulls out; click again: it pushes in.
3	Click one of the 42 not wired buttons, for example a PACK CTL button.	It moves; no system reacts.
4	In flight, outside view on the nose. Fly a steady sideslip with the rudder (for example right rudder).	The three side slip vanes turn to line up with the airflow; centred in coordinated flight.

3. FMC-C reset button checked in the all-FMCs-reset test

> **Aircraft:** A380X | **Where:** Overhead RESET panels: FMC A, FMC B, FMC C | **Status:** Tested in the sim by the user: OK: FMC-A and FMC-B pulled, the FMS kept the plan with FMC-C in. Unit tests: FmcResetPanel.spec.ts.

What it is for

The FMS stops and resets when all the FMC reset buttons are pulled. The check read FMC-B twice and never FMC-C, so pulling FMC-A and FMC-B alone reset the FMS and lost the flight plan while FMC-C was still in. Per A380 FCOM DSC-22-FMS-30 FMC-C is the standby FMC, and FMS 1 and FMS 2 are only lost when FMC-A, FMC-B and FMC-C all fail (PRO-ABN-ECAM-10-22-FMS AUTO FLT FMS 1+2 FAULT). The check now covers A, B and C.

What you should see

Situation	What you should see
FMC-A and FMC-B pulled, FMC-C in	The FMS keeps running; the flight plan stays.
FMC-A, FMC-B and FMC-C pulled	The FMS stops and resets (the flight plan is lost), as before.

How to use

1. Pull FMC A, FMC B and / or FMC C on the overhead RESET panels to reset those FMCs; push them back in to restore.

How to test

#	Do this	Expected result
1	A380X with a flight plan in the MFD F-PLN page.	The plan is shown.
2	Pull the **FMC A** and **FMC B** reset buttons.	The flight plan stays on the MFD (FMC-C still in).
3	Pull **FMC C** as well.	The FMS resets: the flight plan is lost.

4. AUTOLAND light press-to-test and flashing

> **Aircraft:** A380X | **Where:** Glareshield, captain and F/O AUTOLAND lights | **Status:** Tested in the sim by the user: OK.

What it is for

A380 FCOM DSC-22-FG-80-100: the AUTOLAND light flashes in autoland below 200 ft RA when the autoland conditions are lost, and "as long as the AUTOLAND light is pressed, the AUTOLAND light flashes" (light test). The cockpit lamp code cannot read the sim time, so the old flashing expression never lit the light. The flashing (1 Hz: half on, half off) is now computed in the systems host, for the light test and for the autoland warning.

What you should see

Situation	What you should see
AUTOLAND light held pressed	It flashes red at 1 Hz until released.
Autoland below 200 ft RA, autoland conditions lost	Both AUTOLAND lights flash (the warning itself is the existing FBW logic).

How to use

1. Press and hold the AUTOLAND light on the glareshield to test it (cockpit preparation).

How to test

#	Do this	Expected result
1	A380X powered, on the ground. Press and hold the captain AUTOLAND light.	It flashes red, about once per second.
2	Release it.	It goes off.
3	Repeat with the F/O AUTOLAND light.	Same.

5. MFP vanes follow the angle of attack

> **Aircraft:** A380X | **Where:** Nose multifunction probes (outside view) | **Status:** Tested in the sim by the user: OK.

What it is for

The angle of attack vanes of the three multifunction probes on the nose (one on the left, two on the right, A380 FCOM DSC-34-10-30-10) now follow the angle of attack (sim `INCIDENCE ALPHA`, up to 60 degrees each way, smoothed so that the jumps at very low speed on the ground do not show).

Limitations and why

Limitation	Why	What to do
The vanes do not move without the companion model change.	The MFP_ANGLE1..3 animations are in the A380_FUSELAGE model of the aircraft-large-files repository (fork branch feat/a380x-mfp-vane-animations, b8fb739).	Install that model change with this branch.

How to use

1. Look at the nose probes from the outside view in flight.

How to test

#	Do this	Expected result
1	A380X in flight, outside view on the nose probes. Pitch up slowly to raise the angle of attack.	The three MFP vanes turn: their trailing tip rises with the angle of attack.
2	Lower the nose to a small angle of attack.	The vanes come back near their neutral position.

6. ACD keys beside the OITs work; F/O laptop keyboard keys

> **Aircraft:** A380X | **Where:** Captain and F/O side consoles: the 8-key Additional Control Device below each OIT; F/O laptop keyboard | **Status:** Tested in the sim by the user: OK: F/O laptop keyboard, all 8 captain ACD keys hover and click with the tiller centred (recorded), OIS XFR and OIS MENU. Unit tests: LaptopKeyboard.spec.ts.

What it is for

A380 FCOM DSC-46-20-20-60: each pilot has an Additional Control Device (ACD), a pad with OIS XFR, OIS MENU, ESC, ENTER and four arrow keys, for the OIT when the keyboard is stowed. Its keys were in the model but could not be clicked. They now send the same event as the laptop key of the same name, so the OIT takes ESC, ENTER and the arrows like keyboard keys, and OIS XFR / OIS MENU like the laptop OIS keys. Their legends were painted into the side console texture at about 3.5 texels per mm (a smear): each key now has a sharp, lit legend. The F/O laptop keyboard had no clickable keys and no legends: it now has the same keys, legends and press animation as the captain keyboard.

What you should see

Situation	What you should see
Mouse over an ACD key	The key highlights and shows its name (ESC, ENTER, Up, Down, Left, Right, OIS MENU, OIS XFR).
ACD key clicked	The key moves in and out; the OIT gets the key.
F/O laptop keyboard pulled out	The same legends as the captain keyboard; the keys move when clicked.

Limitations and why

Limitation	Why	What to do
OIS XFR and OIS MENU do nothing on this branch.	The OIT handling of those two keys (X-OIS, FLT OPS menu) is on core/fcom-gap-fixes.	Merge core/fcom-gap-fixes too.
The legends, click areas, F/O keyboard keys and tiller fix need the companion model change.	They are in the A380_COCKPIT_LOD00 model and two new textures of the aircraft-large-files repository (fork branch feat/a380x-acd-keys-fo-keyboard, 7f56bcb).	Install that model change with this branch.
Some keys can be hard to reach with the mouse from some head positions.	The flyPad (when mounted) and the sidestick stand between the seat and the pad. The tiller handle was split into small pieces so that it no longer hides the OIS and ENTER keys.	Move the view slightly.

How to use

1. Click the ACD keys below your OIT as on the aircraft: ENTER validates, ESC cancels, the arrows move, OIS MENU shows the FLT OPS menu, OIS XFR shows the other OIT (FLT OPS side, laptop on, logged in).
2. Pull out the F/O meal table, then the F/O keyboard, to use the F/O laptop keyboard.

How to test

#	Do this	Expected result
1	A380X powered. Pull out the F/O meal table, then the F/O keyboard; compare with the captain keyboard.	Same legends on both; F/O keys move when clicked.
2	Tiller centred. Move the mouse over each key of the captain ACD.	Each key highlights and shows its name.
3	Click ESC, ENTER and each arrow.	Each key moves.
4	With the companion OIT change (core/fcom-gap-fixes): F/O OIT on FLT OPS STS, captain OIT on the FLT OPS menu, both logged in. Click OIS XFR on the captain ACD.	The captain OIT shows the F/O STATUS PAGE and "X-OIS Activated"; OIS XFR again goes back.

A32NX exterior: spoilers by panel group and visible RAT

The A32NX exterior spoilers move by panel group from their own deflection instead of the wing average, and the ram air turbine (RAT) is now visible on the exterior model

Aircraft	A32NX
Where to find it	Outside view of the wings and of the left belly fairing
Branch	`core/a32nx-exterior`, based on `master` (FlyByWire master 76db0aef1 merged in); independent of the other core branches. The exterior model and the new RAT model (fbw-a32nx/src/model/rat, generated by generate-rat.js) are in the main repository (no aircraft-large-files change). The RAT goes back in only after a ground stow, which is on `core/rat-ground-stow` (flyPad row on `core/efb-redesign`).
Status	Spoilers: tested in the sim by the user: OK. RAT model: tested in the sim by the user: OK (deploy and stow; with the fixed model the aircraft loads and the RAT deploys and turns).
Pictures	No pictures: the movements do not show well on a still picture.

1. Exterior spoilers move by panel group

> **Aircraft:** A32NX | **Where:** Outside view of the wings | **Status:** Tested in the sim by the user: OK.

What it is for

Each wing has five spoiler panels: 2 to 5 help the ailerons in roll, 2 to 4 are the speedbrakes, and all five are ground spoilers (A320 FCOM DSC-27). The exterior model moved all the panels of a wing together from the wing average, so spoiler 1 rose in turns and with the speedbrakes. The panels now follow their own deflection from the hydraulic model, by group, and the angle is shown in true degrees.

- Three groups per wing: panel 1, panels 2 + 3, panels 4 + 5. Panels 2 and 3 share their skin joints in the model, and so do panels 4 and 5, so a group shows the mean of its two panels.
- True degrees: the model animation goes to 45 degrees; the ground spoilers (50 degrees) show as the model maximum.

What you should see

Situation	What you should see
Roll in flight	The panels 2 to 5 of the down-going wing rise (as two groups); spoiler 1 stays down.
Speedbrakes extended	Panels 2 to 4 rise; spoiler 1 stays down; the 4 + 5 group shows the mean of panels 4 and 5 (half way when only panel 4 is out).
Ground spoilers at landing or rejected takeoff	All three groups rise to the model maximum.

Limitations and why

Limitation	Why	What to do
Panels 2 / 3 and 4 / 5 move together.	Their vertices are blended between two skin joints in the model; five separate panels need an edit of the model skin weights.	None for now.

A full MSFS restart is needed after an update.	The exterior model and its behaviour XML change.	Restart the sim after installing.

How to use

1. Nothing to set: fly as usual and watch the wings from the outside view or the cabin.

How to test

#	Do this	Expected result
1	A32NX in flight, outside view behind a wing. Roll left with the sidestick.	On the left wing, the spoiler groups 2 + 3 and 4 + 5 rise; spoiler 1 (inboard) stays down. The right wing spoilers stay down.
2	Level the wings and extend the speedbrakes fully.	Panels 2 to 4 rise on both wings; spoiler 1 stays down; the outboard 4 + 5 group is lower than the 2 + 3 group.
3	Retract the speedbrakes. Land with the ground spoilers armed.	At touchdown all spoilers of both wings, spoiler 1 included, rise fully.

2. Visible ram air turbine (RAT)

> **Aircraft:** A32NX | **Where:** Outside view, under the left belly fairing | **Status:** Tested in the sim by the user: OK (deploy and stow, with the fixed model the aircraft loads and the RAT deploys and turns, RAT flight R1). Unit tests: RatModel.spec.ts.

What it is for

The A32NX RAT was only a system: the HYD page showed it out, but nothing showed on the aircraft. A RAT model is now part of the exterior (the two most detailed levels of detail, LOD 0 and 1), driven by the hydraulic system values.

- Stowed: the fuselage skin closes the bay; the doors, the bay and the turbine are hidden.
- Deploying: the doors under the left belly fairing open over the first 20 % of the travel, then the strut swings down.
- The propeller (blades 0.749 m across, the real 29.5 in RAT) turns in the airflow at the system RAT speed. Until the RAT is 80 % out the blades stay lined up with the strut, so they pass through the slot.
- Above 500 rpm a blur disc replaces the blades (a turning blade would only strobe on screen).
- The blade pitch follows the RAT governor (fine at low speed, coarser when governed).
- After a ground stow the strut swings back in and the doors close.

What you should see

Situation	What you should see
RAT stowed	No RAT and no doors: a closed belly fairing.
RAT extending (both AC buses lost, RAT MAN ON or EMER ELEC PWR MAN ON)	The doors open, the strut swings down, the propeller starts to turn once the RAT is almost fully out.
RAT out in flight	A blur disc at the propeller.
RAT stowed on the ground (flyPad Ground > Services > RAT stow)	The strut swings in and the bay closes.

Limitations and why

Limitation	Why	What to do

The RAT goes back in only after a ground stow.	The stow request is on `core/rat-ground-stow` (systems) and `core/efb-redesign` (flyPad row); without them the RAT stays out for the rest of the session, as on master.	Merge both branches.
Only LOD 0 and 1 have the RAT.	It is small and under the aircraft: the farther levels of detail leave it out.	-
Position and size were measured on another A320 model.	No Airbus drawing was available.	-
A full MSFS restart is needed after an update.	The exterior model, models.json and the behaviour XML change.	Restart the sim after installing.
The first version made the whole A32NX invisible.	MSFS refuses an animation buffer view that mixes time and rotation data. The generator now gives each animation its own buffer view, and RatModel.spec.ts checks it.	-

How to use

1. Nothing to set: when the RAT extends, look at the left belly fairing from the outside view.
2. After the landing, stow it from flyPad > Ground > Services (needs `core/rat-ground-stow` and `core/efb-redesign`).

How to test

#	Do this	Expected result
1	Load the A32NX after a full MSFS restart. Outside view.	The aircraft is visible; no RAT and no doors under the belly.
2	In flight, about 5,000 ft, 200 kt. Overhead HYD panel: open the guard and press RAT MAN ON. Outside view of the left belly fairing.	The doors open, the strut swings down, then the propeller turns and shows as a blur disc. The ECAM HYD page shows the RAT out.
3	Land and stop. flyPad > Ground > Services: press Stow on "RAT stow (maintenance)" (needs `core/rat-ground-stow` and `core/efb-redesign`).	The strut swings back in and the bay closes.

A380X electrical alerts and AC ESS loads

The A380X ECAM raises the FCOM ELEC alerts from the electrical network, the AC ESS loads are on the AC ESS busbar, the two transponders and the second NSS ANSU are wired, and new battery, EXT PWR, RAT generator, TR monitoring and drive disconnect failures come with their ELEC alerts

Aircraft	A380X
Where to find it	Overhead ELEC panel; flyPad > Failures > ATA 24; E/WD and SD (ELEC AC / DC, STATUS); pedestal SURV panel and MFD SURV > STATUS & SWITCHING; OIT
Branch	`core/a380x-elec`, based on `core/gps` (stacked: needs core/surveillance, core/a380x-mfd-fms and core/gps merged first).
Status	Tested in the sim by the user: ELEC AC BUS 2 and AC BUS 4 FAULT OK; ELEC AC ESS BUS FAULT with its STATUS, GEN OFF, the AC ESS and AC EMER failures, XPDR 1 / XPDR 2 with the SURV alerts and ANSU 1 / 2 OK. Battery and EXT PWR failures (pick): OK on a combined test build; the RAT generator, TR monitoring and drive disconnect failures are not sim-tested (unit and Rust tests). The other ELEC alerts are not sim-tested (unit tests: 30). A full MSFS restart is needed after installing (systems WASM, cockpit XML).
Pictures	No pictures: the alerts use the stock E/WD and SD STATUS pages.

1. ELEC alerts from the electrical network
2. AC ESS and AC EMER loads
3. XPDR 1 / XPDR 2, per-system TCAS faults and SURV XPDR alerts
4. NSS AVNCS ANSU 2
5. Battery, EXT PWR, RAT generator, TR monitoring and drive disconnect failures

1. ELEC alerts from the electrical network

> **Aircraft:** A380X | **Where:** E/WD; SD ELEC AC / DC and STATUS | **Status:** Tested in the sim by the user: AC BUS 2 / AC BUS 4 FAULT OK, AC ESS BUS FAULT with STATUS and GEN OFF OK. Other alerts not sim-tested (unit tests: 30).

What it is for

The A380X ECAM had the ELEC alert texts but nothing raised them. The flight warning system now computes them from the electrical network (busbars, engine and APU generators, TRs, pushbuttons), with the procedures, the flight phase inhibitions and the master lights of A380 FCOM PRO-ABN-ECAM-10-24:

- **ELEC AC BUS 1(2)(3)(4) FAULT**, **AC ESS BUS FAULT**, **AC ESS BUS ALTN** (AC ESS supplied through AC 4: AC ESS FEED in ALTN), **DC BUS 1(2) FAULT**, **DC ESS BUS FAULT**, **EMER CONFIG** (all four AC busbars lost, master warning).
- **ELEC GEN 1(2)(3)(4) FAULT** (stays, with its GEN OFF line ticked, after the generator is switched off, until it works again), **GEN 1(2)(3)(4) OFF** (GEN pb-sw OFF with its engine running, after 5 s), **DRIVE 1(2)(3)(4) DISCONNECTED**, **APU GEN A(B) FAULT**, **TR 1(2)(ESS) FAULT** (no master light, no chime), **APU TR FAULT**, **BUS TIE OFF**.
- Procedure lines are sensed where the FCOM says so (switches set, OFF THEN ON resets, AIR DATA SWTG, WXR & TAWS / XPDR & TCAS SYS 2...). Lines that the FBW texts had from another FCOM standard and that the FCOM procedure does not contain are hidden.
- ELEC AC BUS 2 FAULT shows **LDG DIST: AFFECTED** (FCOM: LDG DIST IMPACT ON WET/CONTAM RWY ONLY).
- **ELEC AC ESS BUS FAULT** has its FCOM STATUS: LIMITATIONS (APPR & LDG) SLATS SLOW; INOP SYS all phases ADR 1, GPS 1, WXR 1, XPDR 1, TCAS 1, HF 1, FEED TK 2 MAIN PMP, STBY PITOT HEATG, L WINDSHIELD HEATG, DFDR; INOP SYS APPR & LDG SLAT SYS 1, CAT 2, GLS AUTOLAND, LS 1, TAWS 1; REDUND LOSS (STATUS MORE) RA SYS C, FEED TK 3 STBY PMP, TRIM TK L PMP, ENG 1+2+3+4 IGN A, PACK 1 CTL 1, PACK 2 CTL 1. GPWS 1, TERR SYS 1 and ADS-B RPTG 1 are not shown as separate alerts while it is active.

> **NOTE**
> The XPDR 1 INOP SYS text is defined once for the ELEC and the SURV XPDR alerts, and the L WINDSHIELD HEATG INOP SYS text of this STATUS is added here (the same text and code as on `core/ecam-fws`).

Limitations and why

Limitation	Why	What to do
A bus fault alert needs at least one AC busbar powered.	Design choice: with all four AC busbars off the aircraft is unpowered on the ground or in ELEC EMER CONFIG, whose procedure covers the lost busbars.	None.
"Failed" means "no normal output although it should give one".	The sim gives no health word for the generators and TRs, only their output.	None.
Not done here: combined busbar faults, DRIVE OIL, units that are not simulated, T.O CONFIG.	First part of the ELEC alerts. The battery, EXT PWR, RAT, TR monitoring and drive disconnect alerts and the STATUS of the GEN, DRIVE, APU GEN and TR alerts came later (next feature); the combined busbar alerts and the T.O CONFIG check of the GEN / DRIVE alerts are on `core/a380x-elec-fuel-alerts`.	Use that branch for them.
VENT AVNCS EXTRACT lines are never ticked.	There is no avionics ventilation pushbutton in the sim.	None.
REDUND LOSS needs the ECP MORE key fix.	The MORE key and light fix is on `core/ecam-fws`.	Merge that branch to page to STATUS MORE.

How to use

1. When an ELEC alert shows, follow the E/WD procedure (the sensed lines tick by themselves as you act).
2. Check the ELEC AC / DC SD page (called with the alert) and the STATUS page afterwards.

How to test

#	Do this	Expected result
1	A380X on the ground, four engines running, all GEN pb-sw ON, BUS TIE AUTO. flyPad > Failures > ATA 24 Electrical Power > **AC 2**.	E/WD: **ELEC AC BUS 2 FAULT** with its procedure, including LDG DIST: AFFECTED; SD ELEC AC page.
2	Clear it. Fail **AC 4**.	**ELEC AC BUS 4 FAULT**; on the ground the line FOR ENG STOP : KEEP ENG 1 LAST.
3	Clear it. Fail **AC ESS** (this label is the A380 AC ESS busbar).	**ELEC AC ESS BUS FAULT**, first line AC ESS FEED ..... ALTN; FAULT light in the AC ESS FEED pb-sw. No separate GPWS 1, TERR SYS 1 or ADS-B RPTG 1 alert.
4	Press CLR until the STATUS page shows (or press STS).	SLATS SLOW; INOP SYS ADR 1, GPS 1, WXR 1, XPDR 1, TCAS 1, HF 1, FEED TK 2 MAIN PMP, STBY PITOT HEATG, L WINDSHIELD HEATG, DFDR; APPR & LDG SLAT SYS 1, CAT 2, GLS AUTOLAND, LS 1, TAWS 1. With `core/ecam-fws`: MORE shows the REDUND LOSS list.
5	Clear it. Set the **GEN 2** pb-sw OFF (engine 2 running).	After 5 s: **ELEC GEN 2 OFF**. GEN 2 ON: the alert goes.

6	Fail **Generator 3**.	**ELEC GEN 3 FAULT** with GEN 3 ... OFF. Set GEN 3 OFF: the line ticks and the alert stays. Clear the failure and set GEN 3 ON: the alert goes.
7	APU running, APU GEN A pb-sw ON. Fail **APU Generator 1**.	**ELEC APU GEN A FAULT** with APU GEN A ... OFF.
8	Fail **TR 1**, then **TR APU**.	**ELEC TR 1 FAULT** (no master caution, no chime), then **ELEC APU TR FAULT**.
9	Fail **DC 1**, then (cleared) **DC ESS**.	**ELEC DC BUS 1 FAULT**, then **ELEC DC ESS BUS FAULT**, with their procedures.
10	No failure. Set the BUS TIE pb-sw OFF.	**ELEC BUS TIE OFF**.
11	Set the AC ESS FEED pb-sw to ALTN.	**ELEC AC ESS BUS ALTN**.
12	In flight, fail **AC 1**, **AC 2**, **AC 3** and **AC 4**.	**ELEC EMER CONFIG** in red (master warning) with its procedure; the descent / APU lines depend on whether it triggered above or below FL 230.

2. AC ESS and AC EMER loads

> **Aircraft:** A380X | **Where:** Overhead ELEC panel (AC ESS FEED); flyPad failures AC ESS / AC EMER | **Status:** Tested in the sim by the user: OK with the AC ESS and AC EMER failures.

What it is for

The A380X electrical model uses the A320 bus names: the real A380 **AC ESS** busbar (400XP) is the A320 "AC ESS SHED" bus in the code, and the **AC EMER** busbar (491XP) is the A320 "AC ESS" bus. About twenty consumers read the wrong one, so they were on AC EMER where the FCOM puts them on AC ESS. They now use the AC ESS busbar (TypeScript, cockpit XML and Rust):

- SURV SYS 1 (XPDR 1, TCAS 1, TAWS 1, WXR 1), RA SYS C, the slat motor, the 427XP EHAs, the feed tank 2 main, feed tank 3 standby and trim tank left pumps, channel 1 of both pack controllers, the F/O OIT and the FLT OPS ANSU (A380 FCOM DSC-24, DSC-34-20-100, DSC-46-20-70, DSC-21).
- The **AC ESS FEED** FAULT light shows a loss of the AC ESS busbar (FCOM DSC-24-20: "The AC ESS busbar is not supplied"). An AC EMER failure leaves AC ESS supplied: no FAULT light.
- The manual crossbleed control is powered by DC ESS (FCOM DSC-36-30) and the outflow valve manual control by its controller supplies DC 1 / 2 or DC ESS (FCOM DSC-21-30-10). Before, both were on A320 buses that the A380 does not have, so they had no power.
- With only the batteries (static inverter on AC EMER), the AC ESS loads (427XP EHAs, TAWS 1, XPDR 1...) are unpowered and the AC EMER loads (VOR 1, LS 1, GPS 1...) remain.

Limitations and why

Limitation	Why	What to do
The flyPad labels are AC ESS (the real AC ESS busbar) and AC EMER; the code names stay the A320 ones.	Renaming the buses in the code would touch every system.	Use the flyPad labels.

How to use

1. Nothing new to set. In an AC ESS loss, follow ELEC AC ESS BUS FAULT (AC ESS FEED ALTN).

How to test

#	Do this	Expected result
1	A380X powered, engines running. flyPad > Failures > ATA 24 Electrical Power > **AC ESS**.	AC ESS FEED pb-sw FAULT light on; ELEC AC ESS BUS FAULT; the F/O OIT goes dark; MFD SURV STATUS & SWITCHING shows XPDR 1 and TCAS 1 failed.
2	Clear it. Fail **AC EMER**.	No AC ESS FEED FAULT light, no ELEC AC ESS BUS FAULT; XPDR 1, TCAS 1 and the F/O OIT keep working.

3. XPDR 1 / XPDR 2, per-system TCAS faults and SURV XPDR alerts

> **Aircraft:** A380X | **Where:** Pedestal SURV panel; MFD SURV > STATUS & SWITCHING; E/WD; SD STATUS | **Status:** Tested in the sim by the user: OK with XPDR 1, XPDR 2 and both failed.

What it is for

The A380 has two surveillance systems, each with its transponder and TCAS: SURV SYS 1 on AC ESS and SURV SYS 2 on AC 4 (FCOM DSC-34-20-100). MSFS has one transponder: it now acts as the transponder of the selected system (XPDR & TCAS SYS 1 / SYS 2 on the SURV panel or the MFD SURV page). The XPDR 2 failure did nothing before.

- XPDR 1 failed or unpowered: TCAS 1 is inoperative too; same for 2. The sim TCAS computer fault counts for the selected system only.
- **SURV TCAS 1(2) FAULT** is per system: no false TCAS 2 FAULT when AC ESS is lost (TCAS 2 is on AC 4). They need the busbar of that system powered (ELEC AC ESS BUS FAULT lists XPDR 1 / TCAS 1 itself).
- New **SURV XPDR 1 FAULT**, **XPDR 2 FAULT** and **XPDR 1+2 FAULT** (FCOM PRO-ABN-ECAM-10-34-20): caution, line XPDR & TCAS ..... SYS 2 (1) when the failed system is selected and the other one works; STATUS INOP SYS XPDR 1(2) and TCAS 1(2), plus AP/FD TCAS MODE when the failed system is selected; XPDR 1+2 FAULT: XPDR 1+2, TCAS 1+2. The TCAS FAULT alert of the same system is not shown with it.
- MFD SURV STATUS & SWITCHING: XPDR 1(2) and TCAS 1(2) shown failed when that XPDR is failed or unpowered.

How to use

1. Select the XPDR & TCAS system on the pedestal SURV panel or MFD SURV > STATUS & SWITCHING.
2. On SURV XPDR 1(2) FAULT, select the other system as the E/WD line asks.

How to test

#	Do this	Expected result
1	A380X on the ground, powered, XPDR & TCAS SYS 1 selected. flyPad > Failures > ATA 34 Navigation > **XPDR 1**.	E/WD: **SURV XPDR 1 FAULT** with XPDR & TCAS ..... SYS 2. MFD SURV STATUS & SWITCHING: XPDR 1 and TCAS 1 failed. STATUS: INOP SYS XPDR 1, TCAS 1, AP/FD TCAS MODE. No separate TCAS 1 FAULT.
2	Select SYS 2.	The line ticks; AP/FD TCAS MODE leaves the STATUS; the transponder works again (as XPDR 2).
3	Also fail **XPDR 2**.	**SURV XPDR 1+2 FAULT**; STATUS: XPDR 1+2, TCAS 1+2.
4	Clear both; with SYS 1 selected fail only **XPDR 2**.	**SURV XPDR 2 FAULT** without the switching line; STATUS XPDR 2, TCAS 2, no AP/FD TCAS MODE.
5	Clear it. flyPad > Failures > ATA 24 Electrical Power > **AC ESS**.	No SURV TCAS 2 FAULT (before: a false one); XPDR 1 / TCAS 1 are listed by ELEC AC ESS BUS FAULT.

4. NSS AVNCS ANSU 2

> **Aircraft:** A380X | **Where:** OIT (NSS AVNCS applications) | **Status:** Tested in the sim by the user: OK.

What it is for

A380 FCOM DSC-46-20-30: the avionics side of the network server system (NSS AVNCS) has two servers; ANSU 2 provides the functions when ANSU 1 fails. Only ANSU 1 existed (the ANSU 2 failure did nothing). ANSU 2 is now built and takes over; with both failed the NSS AVNCS applications (AOC, eLogbook, maintenance) are lost. Supplies (FCOM DSC-46-20-70): ANSU 1 AC 2 or BAT 2, ANSU 2 AC 1 or AC EMER or BAT 1, FLT OPS ANSU AC ESS or BAT 1.

How to use

1. Nothing to set: the OIT avionics applications use whichever NSS AVNCS ANSU works.

How to test

#	Do this	Expected result
1	A380X powered, OIT on an NSS AVNCS page. flyPad > Failures > ATA 46 Information Systems > **NSS AVNCS ANSU 1**.	The OIT avionics applications stay available (ANSU 2).
2	Also fail **NSS AVNCS ANSU 2**.	The NSS AVNCS applications are lost on the OIT.
3	Clear both failures.	The applications become available again.

5. Battery, EXT PWR, RAT generator, TR monitoring and drive disconnect failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 24 Electrical Power; overhead ELEC panel (BAT, EXT PWR, DRIVE pbs); E/WD, STATUS | **Status:** Tested in the sim by the user on a combined test build: BAT 1 and EXT PWR 1 OK. The RAT generator, TR monitoring and drive disconnect failures and the new STATUS lines: not sim-tested. Unit tests FwsElecSystemAlerts.spec.ts, a380ElecFailures.spec.ts; Rust tests. The core branch itself was not sim-tested.

What it is for

Sixteen new flyPad failures, simulated in the Rust electrical system, with the ELEC alerts of A380 FCOM PRO-ABN-ECAM-10-24:

flyPad failure (ATA 24)	What it does	ECAM
BAT 1, BAT 2, BAT ESS, APU BAT	The battery disconnects from its DC busbar; FAULT light in its BAT pb-sw (in either position, while failed).	ELEC BAT 1 (2)(ESS) FAULT or APU BAT FAULT, line BAT ... OFF; master caution and single chime; STATUS INOP SYS BAT 1 (2) (ESS), BAT 1+2 with both, APU BAT (INFO APU BAT START NOT AVAIL with the APU TR failed).
EXT PWR 1, 2, 3, 4	The GGPCU of that external power keeps its contactors open: no external power on that side.	ELEC EXT PWR 1 (2)(3)(4) FAULT while the ground power unit is connected, line EXT PWR n P/B RECYCLE (done after OFF then ON); no master light, no chime; INOP SYS EXT PWR n.
RAT Generator	The RAT generator gives no output: the RAT does not supply the AC ESS busbar. Detected RAT stowed or deployed.	ELEC RAT FAULT (crew awareness, no master light, no chime); INOP SYS RAT.
TR 1 Monitoring, TR 2 Monitoring, ESS TR Monitoring	The monitoring function fails; the TR keeps working.	ELEC TR 1 (2)(ESS) MONITORING FAULT (crew awareness, no master light, no chime, no STATUS).

DRIVE 1 (2)(3)(4) Disconnect Function	The DRIVE pb no longer disconnects that generator drive.	ELEC DRIVE 1 (2)(3)(4) DISC FAULT (master caution, single chime, crew awareness, no STATUS).

Also in this feature

- **ELEC BAT 1 (2)(ESS)(APU) OFF**: a battery pb-sw abnormally OFF for 5 s (line BAT ... ON). Not for a failed battery (its procedure asks OFF), not for the APU BAT while ELEC APU TR FAULT asks it OFF, and not with all four batteries OFF together (aircraft secured, battery check).
- The GEN, DRIVE, APU GEN and TR alerts now list their STATUS: INOP SYS GEN n, APU GEN A (B), TR 1 (2)(ESS), APU TR (INFO APU START NOT AVAIL with the APU BAT failed); for a lost engine generator (GEN n FAULT, GEN n OFF, DRIVE n DISCONNECTED) with the APU off: APPR & LDG INOP SYS CAT 3 DUAL and INFO CAT 3 SINGLE ONLY.
- Flight phase inhibitions of the FCOM figures: BAT FAULT / OFF phases 3 to 7, 9 and 10; EXT PWR FAULT 3 to 10; RAT FAULT 3 to 11; TR MONITORING FAULT and DRIVE DISC FAULT 2 to 11.

Limitations and why

Limitation	Why	What to do
ELEC RAT FAULT covers only the RAT generator.	There is no RAT uplock or heater in the simulation.	-
ELEC EXT PWR FAULT shows only with the ground power unit connected.	Design choice: without one there is nothing to supply.	Connect ground power first.
The TR monitoring alerts read the flyPad failures directly.	The failure changes nothing in the network (design choice).	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 24 Electrical Power and activate one of the new failures.
2. Follow the E/WD (BAT ... OFF, EXT PWR ... RECYCLE) and read the STATUS page.
3. Deactivate the failure and set the pb-sw back.

How to test

Setup: cold and dark at the gate, all four BAT pbs AUTO, EXT 1 ON, APU off.

#	Do this	Expected result
1	flyPad > Failures > ATA 24 Electrical Power > **BAT 1**.	ELEC BAT 1 FAULT (amber, single chime), FAULT light in the BAT 1 pb-sw, line BAT 1 ... OFF.
2	Overhead ELEC panel: BAT 1 pb-sw OFF.	The line ticks; no ELEC BAT 1 OFF caution.
3	Deactivate BAT 1; BAT 1 pb-sw AUTO. Start the APU, APU GEN A and B ON. flyPad > Failures > ATA 24 Electrical Power > **EXT PWR 1**.	ELEC EXT PWR 1 FAULT (no chime); AC 1 is taken over by the APU.
4	EXT 1 pb OFF, then ON. Deactivate EXT PWR 1.	The EXT PWR 1 P/B RECYCLE line ticks after OFF / ON; the alert goes.
5	No failure, engines off, batteries AUTO: set the BAT 2 pb-sw OFF, wait 5 s.	ELEC BAT 2 OFF with BAT 2 ... ON. BAT 2 AUTO: the alert goes.

6	Still on the ground with the engines off (flight phase 1: these alerts are inhibited from phase 2). flyPad > Failures > ATA 24 Electrical Power > **DRIVE 3 Disconnect Function**.	ELEC DRIVE 3 DISC FAULT (master caution, single chime). Clear it.
7	flyPad > Failures > ATA 24 Electrical Power > **TR 1 Monitoring**, then **RAT Generator**.	ELEC TR 1 MONITORING FAULT and ELEC RAT FAULT, no master caution, no chime; STATUS INOP SYS RAT. Clear both.
8	Engines running, APU off. Fail **Generator 3**.	ELEC GEN 3 FAULT; STATUS: INOP SYS GEN 3, APPR & LDG CAT 3 DUAL, INFO CAT 3 SINGLE ONLY.

A380X flyPad failures

The A380X flyPad failure list says what each failure really fails, lists the landing gear sensor and actuator failures, and adds brake, trim air, feed tank pump, crossfeed valve and transfer pump failures, with the trim air duct overheat and fuel pump alerts of the FCOM, and the A380X engine failures: flameout and seizure with the in-flight relight envelope and the FCOM auto relight, ignition and start, oil, stall, EGT and vibration, FADEC, thrust lever, reverser and fuel filter clog failures, with their ECAM alerts and SD ENGINE page indications, the THR LEVERS NOT SET and OIL TEMP LO alerts, the thrust loss and FLEX TEMP not received FADEC failures, the E/WD N1 red limit with its red cross, the crossbleed engine start at X BLEED AUTO, the engine alert levels and procedures of the FCOM, braking and steering control failures, erroneous ADR airspeed and AOA failures with the PRIM air data monitoring, and the RCL / RCL LAST, T.O CONFIG NORMAL, automatic SD page and BTV INOP rules of the A380 FCOM

Aircraft	A380X
Where to find it	flyPad > Failures (ATA 21, 24, 26, 28, 32, 34, 72, 73, 74, 77, 78, 79, 80); overhead FUEL, AIR and ENG MAN START panels; ENG MASTER levers and ENG START selector; thrust levers; brake pedals and tiller; ECAM control panel (RCL, RCL LAST, T.O CONFIG); E/WD, STATUS, PFD and the SD FUEL, COND, BLEED and ENGINE pages
Branch	`core/failures-a380x`, based on `master` (FlyByWire master 76db0aef1 merged in), with `core/fire` and `core/ecam-fws` merged in (for the engine failures: they cut the fuel on the LP valve starvation of core/fire and complete the ENG FAIL ECAM procedures of core/ecam-fws). Merge core/fire and core/ecam-fws first, or with it. It also carries, as identical files: the shared engine failure modules of the A32NX engine failures (systems engine failure, start, oil, malfunction and engine control modules, the FADEC oil failures and the in-flight start throttle hold; the fbw-common part of `core/failures-a32nx`), the EWD EGT limits part of a `core/fcom-audit-fixes` fix, and the A380X crossbleed valve supply fix of `core/a380x-elec`. The ELEC EMER CONFIG alert itself is on `core/a380x-elec`: merged with it, that alert must read the FWS signal of this branch (inhibited by ALL ENG FLAME OUT). The fire and ECAM features are described in the core/fire and core/ecam-fws manuals. The RCL / RCL LAST part needs the EMER CANC pb and ECP MORE key of `core/ecam-fws` (merged here). Since it also carries, as identical files, the shared engine start and engine failure modules of the A32NX and the shared overspeed fields of the A32NX LEAP-1A red line fix: the A380X sets the new values to its old behaviour, so they change nothing on the A380X. Two adaptations: the ENG TWO ENGS OUT deferred procedures read the FUEL JETTISON ARM / ACTIVE pbs directly (the jettison FWS is `core/fuel`), and T.O CONFIG NORMAL sees no ELEC GEN / DRIVE alert here (they are `core/a380x-elec`'s; both branches together are `core/a380x-elec-fuel-alerts`).
Status	Gear failures tested in the sim by the user: OK. Brake / trim air failures and feed pump / crossfeed valve failures tested in the sim by the user: OK; the trim air duct overheat / jammed valve alerts and the transfer pump failures with the fuel follow-ups too. Clearer names: not sim-tested (labels only, checked by the a380.spec.ts unit tests). Engine failures: all tested, the core branch itself was not sim-tested. Flameout / seizure / relight: OK (two and four engines out). Auto relight, igniter indication, ground start with a failed igniter, stall / EGT / vibration, oil failures, FADEC network link and reverser unlocked: tested; the three faults found then (restart after a shutdown, FADEC network link thrust, oil overheat) are fixed on this branch. HP fuel valve, restart after shutdown and reverse thrust: OK. Core spike guard (E/WD N3 at the end of a relight), fuel filter clog, ENG MASTER FAULT light, SD start valve and the starter failure fix: OK (recorded). The other start, FADEC, thrust lever and reverser failures: not sim-tested (Rust, native and unit tests). Engine alerts (feature branch, recorded): THR LEVERS NOT SET, THRUST LOSS and T.O THRUST DISAGREE OK; OIL TEMP LO, the E/WD N1 and SD N3 red crosses, and the stall during a relight fix: not sim-tested; the N1 / FF / EGT part of the core spike guard: OK. picks, tested on a combined test build: crossbleed start X1 / X2 and engine alert

	levels A1 / A2 OK, ADR air data A1 OK (flight), relight thrust pulse S1 and EGT OVER LIMIT with FADEC FAULT A3 OK; brake / steering B1 and RCL, T.O CONFIG NORMAL, SD pages N3 / N4 / N6 recorded as passed in the commit messages; BTV INOP not sim-tested. The core branch itself was not sim-tested.
Pictures	No pictures: the failures page, the ECAM alerts and the SD ENGINE page are the stock FBW displays.

1. Landing gear sensor and actuator failures listed, ROLLOUT removed
2. Clearer failure names
3. Brake and trim air failures
4. Trim air duct overheat and jammed valve alerts
5. Feed tank pump and crossfeed valve failures
6. Transfer pump failures and fuel follow-ups
7. A380X engine failures (flameout, seizure, relight)
8. Automatic relight with both igniters, and the engine feed tank fuel draw
9. SD ENGINE page igniter indication
10. Engine ignition and start failures, FCOM ground start
11. HP fuel valve: no engine burning without a FADEC start
12. Engine oil failures (leak, filter clog, overheat)
13. Engine fuel filter clog, ENG MASTER FAULT light, SD start valve and N3 red cross
14. Engine stall, EGT overtemperature and vibration failures
15. FADEC, thrust lever and reverser failures
16. ENG THR LEVERS NOT SET and ENG OIL TEMP LO
17. FADEC thrust failures: ENG THRUST LOSS and ENG T.O THRUST DISAGREE
18. E/WD N1 red above 111 % with a red cross
19. Crossbleed engine start with X BLEED at AUTO
20. Engine alerts as in the FCOM
21. Braking and steering control failures
22. Erroneous ADR airspeed and AOA failures, PRIM air data monitoring
23. FCOM : RCL / RCL LAST, T.O CONFIG NORMAL, automatic SD pages, BTV INOP

1. Landing gear sensor and actuator failures listed, ROLLOUT removed

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 32 Landing Gear | **Status:** Tested in the sim by the user: OK. Unit tests: a380.spec.ts.

What it is for

The landing gear proximity sensor and jammed actuator failures (ids 32004 to 32025) were already simulated by the common FBW gear system and mapped in the A380X systems, but the flyPad listed only one of them, and with the wrong name: "uplock nose gear #1" failed the **right main gear** uplock sensor #1. All 18 failures are now listed, with the ids and names of the Rust failure map (the same as on the A32NX).

The **ROLLOUT** failure (ATA 22) only set an FCDC input that nothing reads (there is no steering control system yet). It is no longer listed; its id stays in the code so the C++ failure list keeps in step.

What changed in the list

Id	Before	Now
32004	not listed	Proximity sensor damage uplock nose gear #1
32005	not listed	Proximity sensor damage downlock nose gear #2
32006	Proximity sensor damage uplock nose gear #1 (but failed the right gear)	Proximity sensor damage uplock right gear #1
32007	not listed	Proximity sensor damage downlock right gear #2
32008	not listed	Proximity sensor damage uplock left gear #2
32009	not listed	Proximity sensor damage downlock left gear #1
32010 to 32015	not listed	Proximity sensor damage closed / opened nose, right, left gear door #1 / #2 (six entries)
32020	not listed	Nose gear jammed actuator

32021, 32022	not listed	Left / Right main gears (body + wing) jammed actuator
32023	not listed	Nose gear door jammed actuator
32024, 32025	not listed	Left / Right main gear doors jammed actuator
22001	ROLLOUT (ATA 22)	not listed

> **NOTE**
> A door *uplock* sensor ("closed ... door") detects the door closed, a door *downlock* sensor ("opened ... door") the door open. The 32021 / 32022 and 32024 / 32025 names come from the clearer names feature below: one failure jams the whole left or right main gear group.

What you should see

Situation	What you should see
A jammed gear actuator failure set, gear lever moved	That gear (or gear door) group stays where it was: the common FBW gear system freezes its actuator.
A proximity sensor failure set	That sensor gives its LGCIS a wrong signal. The gear logic and its indications are the existing FBW gear model: this branch only lists the failures.
ATA 22 list	No ROLLOUT entry.

How to use

1. Open flyPad > Failures > ATA 32 Landing Gear.
2. Pick the sensor or actuator failure by its name (the table above gives what each one fails) and activate it.
3. Deactivate it to clear the failure.

How to test

#	Do this	Expected result
1	Open flyPad > Failures > ATA 32 Landing Gear.	The 12 "Proximity sensor damage ..." and 6 "... jammed actuator" entries are listed, among them "Proximity sensor damage uplock right gear #1" and "Proximity sensor damage uplock nose gear #1".
2	Open flyPad > Failures > ATA 22 Auto Flight.	No ROLLOUT entry (FMC-A, FMC-B, FMC-C, FCU 1, FCU 2 are still there).
3	In flight, gear down, activate **Nose gear jammed actuator**, then set the gear lever UP.	The nose gear does not retract; the main gears do.
4	Deactivate the failure.	The nose gear retracts.

2. Clearer failure names

> **Aircraft:** A380X | **Where:** flyPad > Failures | **Status:** Not sim-tested (labels only). Unit tests: a380.spec.ts.

What it is for

The failure names were audited ATA by ATA against what the systems code fails (the Rust failure map, the C++ failure lists and the TypeScript consumers). Names that were misleading or misspelt now say what is failed. Only the labels change: the ids and the failures themselves are the same.

Old name and new name

ATA	Before	Now
21	Foward Cargo Isolation Valve / Foward Cargo Extraction Fan	Forward Cargo Isolation Valve / Forward Cargo Extraction Fan
24	TR ESS	ESS TR
24	TR APU	APU TR
24	AC 247XP	AC EHA (247XP)
24	DC 247PP	DC EHA (247PP)
24	DC 309PP	DC APU (309PP)
26	Engine 1 Loop A ... Engine 4 Loop B	Engine 1 fire detection loop A ... Engine 4 fire detection loop B
26	APU Loop A / APU Loop B	APU fire detection loop A / APU fire detection loop B
26	Main Landing Gear Bay Loop A / B	MLG bay fire detection loop A / B
32	LGCIU 1 Power supply, LGCIU 2 Power supply	LGCIS 1 power supply, LGCIS 2 power supply
32	LGCIU 1 Internal error, LGCIU 2 Internal error	LGCIS 1 internal error, LGCIS 2 internal error
32	Main left / right gear jammed actuator	Left / Right main gears (body + wing) jammed actuator
32	Main left / right gear door jammed actuator	Left / Right main gear doors jammed actuator
34	RA SYS A Interrupted (B, C)	RA SYS A antenna interrupted (B, C)
34	RA SYS A Direct Coupling (B, C)	RA SYS A antenna direct coupling (B, C)
34	TERR 1 / TERR 2	TERR SYS 1 / TERR SYS 2

Limitations and why

Limitation	Why	What to do
"AC ESS" (24105) and "AC EMER" (24104) keep their names.	The A380X electrical model calls the real AC ESS busbar "AC ESS SHED" and the AC EMER busbar "AC ESS"; the labels already follow the real busbars.	Use "AC ESS" for the AC ESS busbar, "AC EMER" for the AC EMER busbar.
"XPDR 2" (34004) and "NSS AVNCS ANSU 2" (46002) keep their names but fail nothing on this branch.	Nothing reads them on master. They are wired on `core/a380x-elec`.	Test them with that branch.

How to use

1. Open flyPad > Failures and pick the failure by its new name (table above).

How to test

#	Do this	Expected result
1	Open flyPad > Failures > ATA 24 Electrical Power.	The entries read "ESS TR", "APU TR", "AC EHA (247XP)", "DC EHA (247PP)", "DC APU (309PP)".
2	Open ATA 26 Fire Protection.	The loop failures read "Engine 1 fire detection loop A" ... "MLG bay fire detection loop B".

3	Open ATA 32 Landing Gear and ATA 34 Navigation.	"LGCIS 1 power supply" ... "LGCIS 2 internal error"; "RA SYS A antenna interrupted", "RA SYS A antenna direct coupling", "TERR SYS 1", "TERR SYS 2".
4	Open ATA 21 Air Conditioning.	"Forward Cargo Isolation Valve" and "Forward Cargo Extraction Fan" are spelt right.
5	Activate "ESS TR" with the aircraft powered.	The ESS TR fails, as "TR ESS" did before (same id 24002).

3. Brake and trim air failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 32 and ATA 21 | **Status:** Tested in the sim by the user: OK; the brake leak works but shows nothing on the SD HYD page.

What it is for

The A380X hydraulic and air conditioning models already had brake leak and trim air faults, but the flyPad had no failure for them. Eleven failures are now listed (same ids as on the A32NX for the brakes), and a spec checks that every id of the systems failure map is listed with its label.

flyPad name (ATA)	What it fails
Brakes NORM circuit leak (green hydraulic) (32)	The NORM brakes are fed by the GREEN system in this model: GREEN fluid leaks out (0.1 gal/s) while the circuit is pressurised.
Brakes ALTN circuit leak (yellow hydraulic) (32)	The ALTN brakes (with the brake accumulator) are fed by the YELLOW system: YELLOW fluid leaks out while pressurised.
Brakes ALTN accumulator gas leak (32)	The ALTN brake accumulator loses its gas charge: the stored brake pressure drops when the YELLOW system is not pressurised.
Cockpit / Main deck zone 1 / Upper deck zone 1 / Forward cargo trim air valve jammed (21)	The trim air valve of that zone stays where it was when the failure was set.
Cockpit / Main deck zone 1 / Upper deck zone 1 / Forward cargo duct overheat (hot trim air) (21)	200 deg C air is blown into the zone duct while its hot air valve is open. The zone gets too hot and the controller closes the trim air valves; HOT AIR OFF stops the overheat (A380 FCOM COND DUCT OVHT procedure).

Limitations and why

Limitation	Why	What to do
No ECAM alert and no SD indication for the brake failures.	The SD HYD page has no brake leak indication. The trim air failures now have their alerts and SD marks: see "Trim air duct overheat and jammed valve alerts" below.	Watch the hydraulic quantity / pressure and the brake pressure.
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Brakes: flyPad > Failures > ATA 32 Landing Gear, the three "Brakes ..." entries.
2. Trim air: flyPad > Failures > ATA 21 Air Conditioning, the "... trim air valve jammed" and "... duct overheat (hot trim air)" entries.

How to test

#	Do this	Expected result
1	A380X on the ground, four engines running (GREEN and YELLOW pressurised). Activate **Brakes NORM circuit leak (green hydraulic)**.	The GREEN system loses fluid; YELLOW is not affected. Nothing is shown for the leak itself on the SD HYD page.
2	Deactivate it, activate **Brakes ALTN circuit leak (yellow hydraulic)**.	The YELLOW system loses fluid; GREEN is not affected.
3	Set the parking brake, then stop the engines (YELLOW no longer pressurised). Activate **Brakes ALTN accumulator gas leak**.	The brake pressure held by the accumulator drops.
4	In flight, HOT AIR pushbuttons ON, activate **Cockpit duct overheat (hot trim air)**. Open the SD COND page.	The cockpit zone gets hotter; then the trim air valves close. Setting HOT AIR OFF stops the overheat.
5	Activate **Cockpit trim air valve jammed**, then change the cockpit temperature selection.	The cockpit trim air valve no longer moves: the cockpit temperature stops following the selection; the other zones still regulate.

4. Trim air duct overheat and jammed valve alerts

> **Aircraft:** A380X | **Where:** overhead AIR panel (HOT AIR 1 / 2, PACK 1 / 2); E/WD, STATUS; SD COND page | **Status:** Tested in the sim by the user: OK. Rust tests (air_conditioning), unit tests: CondTrimAirFlags.spec.ts, CondTrimAirMarks.spec.ts.

What it is for

The trim air failures above now give the alerts of the A380 FCOM (PRO-ABN-ECAM-10-21-10). The temperature controller monitors each trim air duct and valve: a duct above 70 deg C for 1.2 s is an overheat, latched until its HOT AIR pushbutton is set OFF; a valve that does not follow its demand is jammed. The FWS reads both from the controller and raises:

- **COND DUCT OVHT** (master caution, single chime, FAULT light in the HOT AIR pushbutton, SD COND page) with the CKPT DUCT OVHT, CABIN DUCT OVHT and FWD CARGO DUCT OVHT cases, the deferred procedure **WHEN DUCT OVHT OUT**, and its STATUS lines (INOP SYS, CABIN TEMP REGUL DEGRADED, FWD CRG TEMP REGUL DEGRADED).
- **COND FWD CARGO TEMP REGUL FAULT** for a jammed forward cargo trim air valve: CARGO TRIM AIR VLV FAULT, CARGO TEMP ... MONITOR, STATUS INFO FWD CRG TEMP REGUL DEGRADED (SD COND page only: no chime, no master light).
- SD COND page: the amber OVHT mark of an overheated cockpit or cabin zone duct and the trim air valve mark of a jammed valve (before: never shown).

What you should see

Situation	What you should see
Cockpit duct overheat	COND DUCT OVHT, CKPT DUCT OVHT: HOT AIR 2 ... OFF; if the hot air valve stays open (HOT AIR JAMMED OPEN): PACK 2 ... OFF. HOT AIR 2 FAULT light, OVHT on the SD COND page.
Main deck zone duct overheat	CABIN DUCT OVHT: HOT AIR 1 ... OFF (HOT AIR 2 for an upper deck zone); if jammed open, PACK 1 (2) ... OFF; CARGO TEMP ... MONITOR.
Forward cargo duct overheat	FWD CARGO DUCT OVHT: HOT AIR 1 ... OFF; if jammed open, PACK 1 ... OFF.

Overheat gone (duct cooled, HOT AIR OFF)	Deferred procedure WHEN DUCT OVHT OUT: HOT AIR 1 (2) ... ON, and PACK 1 (2) ... ON if it was switched off.
Forward cargo trim air valve jammed	COND FWD CARGO TEMP REGUL FAULT; STATUS INFO FWD CRG TEMP REGUL DEGRADED; the valve mark on the SD COND page.

Limitations and why

Limitation	Why	What to do
Which HOT AIR valve supplies which zone is a design choice.	The FCOM does not say it: the main deck zones and the forward cargo are on HOT AIR 1, the cockpit and the upper deck on HOT AIR 2.	-
A jammed forward cargo valve counts as a degraded regulation (no INOP SYS).	Design choice: the aircraft has two forward cargo trim air valves and the FCOM says a jammed valve loses the "optimized" regulation.	-
The alerts are inhibited during the takeoff and the landing (FWS phases 3 to 7, 9, 10).	The FCOM gives the inhibition as a figure: that of COND HOT AIR FAULT / FWD CARGO ISOL FAULT is used.	Test on the ground before takeoff or in cruise.
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 21 Air Conditioning and activate a "... duct overheat (hot trim air)" or a "... trim air valve jammed" failure.
2. Follow the ECAM with the HOT AIR and PACK pushbuttons of the overhead AIR panel; check the SD COND page.
3. Deactivate the failure; when the duct has cooled, follow WHEN DUCT OVHT OUT.

How to test

#	Do this	Expected result
1	A380X in cruise (or on the ground before takeoff), packs and HOT AIR 1 / 2 ON. Activate **Cockpit duct overheat (hot trim air)**.	After the duct passes 70 deg C: master caution, E/WD COND DUCT OVHT, CKPT DUCT OVHT with HOT AIR 2 ... OFF; HOT AIR 2 FAULT light; SD COND page with the amber OVHT mark on the cockpit.
2	Set HOT AIR 2 OFF. Open the STATUS page.	The line is done; STATUS INOP SYS and INFO lines of COND DUCT OVHT.
3	Deactivate the failure and wait for the duct to cool.	The alert goes; WHEN DUCT OVHT OUT asks HOT AIR 2 ... ON.
4	Activate **Main deck zone 1 duct overheat (hot trim air)**.	COND DUCT OVHT, CABIN DUCT OVHT with HOT AIR 1 ... OFF and CARGO TEMP ... MONITOR.
5	Deactivate it. Activate **Forward cargo trim air valve jammed**, then change the cargo temperature selection.	COND FWD CARGO TEMP REGUL FAULT (no chime): CARGO TRIM AIR VLV FAULT, CARGO TEMP ... MONITOR; STATUS INFO FWD CRG TEMP REGUL DEGRADED.
6	Activate **Cockpit trim air valve jammed** and change the cockpit temperature selection.	The trim air valve mark appears on the cockpit zone of the SD COND page.

5. Feed tank pump and crossfeed valve failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL | **Status:** Tested in the sim by the user: OK, after the fix that holds the MSFS pump / valve switch.

What it is for

Twelve new failures: the main and standby pump of each feed tank and the four crossfeed valves. The FUEL panel pushbuttons now set a selection, and the systems host moves the MSFS pump or valve switch from it: a failed pump is held off, a jammed valve is held where it was. The FAULT lights, the SD FUEL page and the FCOM alerts compare the selection with the real state (A380 FCOM PRO-ABN-ECAM-10-28, DSC-28-20).

flyPad name (ATA 28)	What it fails
Feed tank 1 main pump ... Feed tank 4 main pump	That main pump stops (held off).
Feed tank 1 standby pump ... Feed tank 4 standby pump	That standby pump stops (held off).
Crossfeed valve 1 jammed in place ... Crossfeed valve 4 jammed in place	That crossfeed valve stays where it was.

What you should see

Situation	What you should see
Feed pump failed, its pushbutton ON	FAULT light in the pushbutton; the pump amber on the SD FUEL page.
One main pump failed	**FUEL FEED TK 1 MAIN PMP FAULT** (tank 1 to 4), SD FUEL page; procedure FEED TK 1 MAIN PMP ... OFF; STATUS INOP SYS FEED TK 1 MAIN PMP.
One standby pump failed	**FUEL FEED TK 1 STBY PMP FAULT**, crew awareness; STATUS MORE: REDUND LOSS FEED TK 1 STBY PMP.
Main and standby pumps of one tank failed	**FUEL FEED TK 1 MAIN + STBY PMPs FAULT** (replaces the two single alerts). After takeoff: CROSSFEED 1 ON, CROSSFEED 2 ON (CROSSFEED 3 and 4 for tanks 3 and 4), FEED TK 1 MAIN PMP OFF, FEED TK 1 STBY PMP OFF, AVOID NEG G LOAD; before takeoff only the two pump lines. STATUS: INOP SYS FEED TK 1 PMPs, FEED TK 1 BY GRVTY ONLY.
Crossfeed valve jammed and its pushbutton moved	The valve disagrees with its pushbutton: FAULT light, valve amber on the SD FUEL page, **FUEL CROSSFEED VLV 1 FAULT** (crew awareness), STATUS INOP SYS CROSSFEED VLV 1; if the valve is stuck closed and engine 1 is not running, FEED TK 1 NOT USABLE.

Limitations and why

Limitation	Why	What to do
The alerts are inhibited during the takeoff and the landing.	The FCOM gives the inhibition as a figure: the one of JETTISON VLV NOT CLOSED is used (FWS flight phases 4 to 7, 9 and 10).	Test on the ground before takeoff or in cruise.
A pump turned OFF by the crew also gives the FAULT alert.	FCOM: a pump is faulty when it runs at low pressure, is abnormally not running, or is turned off by the crew (except in flight phases 1 and 12).	Expected behaviour.
The gravity feed ceiling (MAX FL 280, 50/MEA) is not simulated.	No gravity feed model.	None.
A full MSFS restart is needed after an update.	The cockpit behaviour XML changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate a pump or crossfeed valve failure.
2. Follow the ECAM procedure with the overhead FUEL panel pushbuttons; check the SD FUEL page.
3. Deactivate the failure to restore the pump or valve.

How to test

#	Do this	Expected result
1	A380X on the ground, engines running, all feed pumps ON, crossfeed valves in their normal position. Open the SD FUEL page.	No FAULT light, no fuel alert.
2	Activate **Feed tank 1 main pump**.	FAULT light in the FEED TK 1 MAIN pushbutton, pump amber on the SD, E/WD: FUEL FEED TK 1 MAIN PMP FAULT with FEED TK 1 MAIN PMP ... OFF.
3	Set the FEED TK 1 MAIN pushbutton OFF. Open the STATUS page.	The procedure line is done; STATUS INOP SYS: FEED TK 1 MAIN PMP.
4	Set it ON again and also activate **Feed tank 1 standby pump**.	The single alerts are replaced by FUEL FEED TK 1 MAIN + STBY PMPs FAULT (on the ground: the two pump OFF lines); STATUS: FEED TK 1 PMPs, FEED TK 1 BY GRVTY ONLY.
5	Deactivate both. Activate **Feed tank 2 standby pump** only.	FUEL FEED TK 2 STBY PMP FAULT; STATUS MORE: REDUND LOSS FEED TK 2 STBY PMP.
6	Deactivate it. Activate **Crossfeed valve 1 jammed in place**, then move the CROSSFEED 1 pushbutton.	The valve does not move: FAULT light, valve amber on the SD, FUEL CROSSFEED VLV 1 FAULT, STATUS INOP SYS CROSSFEED VLV 1.
7	Deactivate the failure.	The valve follows its pushbutton again; the alert clears.

6. Transfer pump failures and fuel follow-ups

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; E/WD, STATUS; SD FUEL page | **Status:** Tested in the sim by the user: OK. Unit tests: FuelPumpValveFlags.spec.ts, FuelPumpsAndValves.spec.ts, failable-fuel-switches.spec.ts.

What it is for

Eight new failures of the transfer pumps that the FBW fuel transfers use: **L outer tank pump**, **R outer tank pump**, **L mid tank fwd pump**, **R mid tank fwd pump**, **L inner tank fwd pump**, **R inner tank fwd pump**, **Trim tank L pump**, **Trim tank R pump**. A failed pump is held off whatever the fuel system commands. The pushbutton FAULT light, the amber pump on the SD FUEL page and the FUEL ... PMP FAULT alerts of the A380 FCOM (PRO-ABN-ECAM-10-28) come as soon as a pump fails, runs in an empty tank with its pushbutton ON, or is switched OFF by the crew (except in flight phases 1 and 12). A repaired pump follows the automatic transfers again.

- **FUEL FEED TK 1(2)(3)(4) LEVEL LO** now triggers for a single low feed tank: the four tanks shared one 30 s confirmation, so only all four low together could trigger it.
- The **CROSSFEED** pushbutton OPEN light now follows its valve.
- The sensed CROSSFEED ... ON and ALL FEED TKs PMPs ON items of the ECAM procedures tick on the pushbuttons, even with a jammed valve or a failed pump.
- Corrupted degree signs in the ECAM texts are fixed.

What you should see

Situation	What you should see
L outer tank pump failed	FAULT light; **FUEL L OUTR TK PMP FAULT**: L OUTR TK PMP ... OFF; STATUS: OUTR TKs XFR BY GRVTY ONLY, INOP SYS L OUTR TK PMP.
L inner (mid) tank fwd pump failed	**FUEL L INR (MID) TK FWD PMP FAULT**: L INR (MID) TK FWD PMP ... OFF; STATUS INOP SYS L INR (MID) TK FWD PMP.
Its aft pump switched OFF too	**FUEL L INR (MID) TK FWD+AFT PMPs FAULT** replaces the single alerts: both pumps ... OFF; L INR (MID) TK NOT USABLE; INOP SYS L INR (MID) TK PMPs.
Trim tank L and R pumps failed	**FUEL TRIM TK L+R PMPs FAULT**: both pumps ... OFF, PITCH ATT < 3° FOR GRVTY FWD XFR, TRIM TK FEED ... OPEN, TRIM TK : 2300 KG MAX NOT USABLE; STATUS: TRIM TK XFR BY GRVTY ONLY, TRIM TK GRVTY XFR : AROUND 3000 KG / HR, INOP SYS TRIM TK PMPs.
One feed tank low	FUEL FEED TK n LEVEL LO for that tank after 30 s.
CROSSFEED pushbutton ON	The OPEN light comes on when the valve is open.

Limitations and why

Limitation	Why	What to do
The aft gallery pumps (inner and mid tank aft) cannot fail.	They never run in the simulation; switching them OFF still gives their alert.	-
No gravity transfer.	Not simulated: the STATUS gravity lines are information only.	None.
The alerts are inhibited during the takeoff and the landing.	Same inhibition as the feed pump alerts above.	Test on the ground before takeoff or in cruise.
A full MSFS restart is needed after an update.	The cockpit behaviour XML changes (OPEN light).	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate one of the eight transfer pump failures.
2. Follow the ECAM with the overhead FUEL panel; check the SD FUEL page.
3. Deactivate the failure to restore the pump; it follows the automatic transfers again.

How to test

#	Do this	Expected result
1	A380X on the ground before takeoff (or in cruise) with fuel in the outer tanks, all fuel pumps ON. Activate **L outer tank pump**.	FAULT light in the L OUTR TK pushbutton, pump amber on the SD FUEL page, master caution, E/WD FUEL L OUTR TK PMP FAULT with L OUTR TK PMP ... OFF.
2	Set the L OUTR TK pushbutton OFF. Open STATUS.	The line is done; STATUS: OUTR TKs XFR BY GRVTY ONLY, INOP SYS L OUTR TK PMP.
3	Deactivate it and set the pushbutton ON. Activate **L inner tank fwd pump**, then set the L INR TK AFT pushbutton OFF.	FUEL L INR TK FWD+AFT PMPs FAULT replaces the single alert; L INR TK NOT USABLE.
4	Restore everything. Activate **Trim tank L pump** and **Trim tank R pump**.	FUEL TRIM TK L+R PMPs FAULT with its procedure; the degree sign of PITCH ATT < 3° is shown correctly.
5	Restore everything. Set a CROSSFEED pushbutton ON.	Its OPEN light comes on with the valve.

7. A380X engine failures (flameout, seizure, relight)

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 72 Engine; pedestal ENG MASTER 1-4 and ENG START selector; overhead AIR panel X BLEED selector; E/WD, STATUS, SD ENGINE page | **Status:** Sim-tested OK, recorded: on the ground and in flight (windmilling, quick and crossbleed relights without APU), relight with APU bleed, quick relight of engine 4 with the levers in CL (stops at idle, then accelerates), starter duct vents after the start, two engines out (windmilling relight at 255 kt) and all engines out (ALL ENG FLAME OUT, RAT, no ELEC EMER CONFIG, relight). The core spike guard (the MSFS core jump at the end of a relight, below): sim-tested OK (E/WD N3 held at 61.6 % while MSFS jumped to 96.8 %); its N1 / FF / EGT hold: OK (FL150 relight: E/WD N1 held at 26.4 % and FF at 615 while MSFS N1 went to 62.5 %). The softened thrust pulse (MSFS core limiter, pick): tested in the sim by the user on a combined test build, OK (engine 4 relight at FL150), native test msfs_core_limiter_test. The stall during a relight fix: not sim-tested (installed, no visible effect expected; native). Rust tests (a380_systems engine_failure.rs, engine_out_aircraft_tests.rs, pneumatic.rs, hydraulic), native tests (relight_start_test, engine_start_throttle_hold_test), unit tests EngineFailAlerts.spec.ts, FwsAbnormalSensedItems.spec.ts, a380.spec.ts. The core branch itself was not sim-tested.

What it is for

The A380X port of the A32NX engine failures. Eight new failures in the flyPad chapter ATA 72 Engine. A **flameout** cuts the engine fuel once: the crew relights the engine with ENG MASTER OFF then ON inside the in-flight relight envelope of the A380 FCOM. A **seizure** also stops the core: no relight while the failure is active. The FWS gives the FCOM ENG FAIL, ENG SHUT DOWN, TWO ENGS OUT and ALL ENG FLAME OUT alerts for the four engines.

Id	flyPad failure (ATA 72 Engine)	What the code does
72000-72003	Engine 1 (2)(3)(4) flameout (crew relight possible)	Cuts the engine fuel (latched, MSFS fuel valve 60-63). The engine runs down and windmills. ENG MASTER OFF then ON inside the envelope relights it; it then runs normally while the failure stays listed. Deactivate and activate it again for a new flameout.
72010-72013	Engine 1 (2)(3)(4) seizure (no relight)	Cuts the fuel and stops the core (N3). No relight while the failure is active; once it is cleared the engine relights by the normal procedure.

What changed after the first sim tests

- **No N3 overshoot at the end of an in-flight start.** The three in-flight relights of took N3 from 61 % to 97.5 % for 5 to 7 s with the levers in CL: while the engine was out the FADEC thrust loop wound the MSFS throttle up to 100 %. Now the MSFS throttle stays at idle while the engine is out or starting and until it runs at idle, then the engine accelerates to the lever or A/THR demand (A380 FCOM DSC-70-30 "FADEC stabilizes the engine at idle speed"; ENG RELIGHT IN FLIGHT: THR LEVER IDLE, then WHEN IDLE REACHED).
- **A compressor stall no longer ends that idle hold early.** The hold ends when the commanded N1 reaches idle; with the "Engine n compressor stall" failure active at the end of a relight, the stall N1 loss (up to 15 %) was counted in the commanded N1 and the hold ended about 0.8 s early. The stall N1 loss is now taken out of that check (native : FL200, 300 kt relight, levers in CL, with a stall).
- **TWO ENGS OUT STATUS** lists the FCOM lines PART SPLRs, PART L/G RETRACTION, G (Y) HYD SYS, PACK 1 (2) and one GEN line for the two generators lost (GEN 1+2, 3+4 on the same side; GEN 1+3, 1+4, 2+3, 2+4 on opposite sides).
- **ALL ENG FLAME OUT**: its RAT MAN ON line is sensed (it turns green when the RAT is deployed), it has the missing L/G GRVTY (EXTN 2 MIN) DOWN line before a forced landing, and it inhibits ELEC EMER CONFIG as in the FCOM. The ELEC EMER CONFIG alert itself is on `core/a380x-elec`: this branch only computes the inhibited signal, so the inhibition shows once both branches are merged.
- **ENG RELIGHT IN FLIGHT** shows the single engine limits MAX GUARANTEED ALTITUDE : 30000 FT and MIN SPEED FOR WINDML RELIGHT : 260 KT (FCOM).
- **No starter air without a source.** A starter assisted relight was granted with no APU or crossbleed air: the air trapped in a stopped engine bleed duct and the slowly venting starter duct (about 45 s) kept the starter

pressurised. The duct of an engine with no air source now empties within about 3 s, and the starter duct is empty (below 5 PSIG) about 3 s after the start valve closes (design values: the FCOM gives none).

- **X BLEED selector OPEN opens the crossbleed valves.** Their manual control used the A320 DC ESS SHED busbar, which the A380 does not have; it now uses DC ESS (A380 FCOM DSC-36-30). The crossbleed starter air of a starter assisted relight depends on it.
- **The FADECs read the ENG START selector.** They read a non-existent variable (always 0) and never saw the selector at IGN START; a static check now guards the variable names.
- **Each FADEC reads the LGCIU of its side**: engines 1 and 2 LGCIU 1, engines 3 and 4 LGCIU 2 (design choice, the FCOM does not say). Before, the on-ground input of engines 3 and 4 was read past the end of the two-LGCIU list.
- **ENG MASTER OFF closes the HP fuel valve** and the engine stops burning at once: see "HP fuel valve" below.
- **The thrust pulse at the end of a relight is softened**. At the end of an in-flight start MSFS jumps its core to its top speed for a few seconds (recorded: 57.7 to 96.2 % in 0.5 s, thrust from -2 100 to +11 800 lb on the relit engine, a yaw kick). The core spike guard only hid it from the indications. A new FADEC limiter now holds the MSFS core near idle until it has settled, so the real thrust pulse is gone.

FCOM references

A380 FCOM	What it says	Used for
PRO-ABN-ECAM-10-70 ENG RELIGHT IN FLIGHT, chart	Single engine: "MAX GUARANTEED ALTITUDE : 30000 FT", "MIN SPEED FOR WINDML RELIGHT : 260 KT"; multiple engines: 28 000 ft and 250 kt; ENG START SEL IGN START; "Engine must relight within 30 s after fuel flow increases".	The relight envelope and the 30 s attempt
DSC-70-30 IN FLIGHT,	Engine start valve opens "if N2 is below 11 %, or if the aircraft airspeed (CAS) is below 260 kt"; it closes above 58.4 %.	The start valve in flight and for a restart
DSC-70-80-30-20,	Quick relight: MASTER ON-OFF-ON within 30 s, "provided that N2 is greater than 45 %"; starter assisted below 11 % N2, windmilling at or above 260 kt; during a windmilling start the FADEC disconnects both hydraulic pumps.	Quick relight, start type, engine pumps
ENG 1(2)(3)(4) FAIL,	Triggered when the engine is not running; "If more than two engines failed: ENG (AFFECTED) RELIGHT PROC ... APPLY" .	ENG FAIL and RELIGHT PROC APPLY
ENG 1(2)(3)(4) SHUT DOWN,	Shown when the ENG MASTER is OFF or the ENG FIRE pb pushed; STATUS INOP SYS BTV (engine 2 or 3), CAT 3 DUAL and CAT 3 SINGLE ONLY if the APU is off.	ENG SHUT DOWN STATUS
ENG TWO ENGS OUT ON SAME / OPPOSITE SIDE,	Two engines failed on the same side / on opposite sides, LAND ANSA, STATUS (CAT 2 / CAT 3 INOP, CAT 2 ONLY).	The two cautions
ENG ALL ENG FLAME OUT,	The title of the all-engines procedure.	Title of the old ALL ENGINES FAILURE

The relight envelope, read off the chart (IAS, flight level):

Zone	Where	Needs
3, windmilling start	260-340 kt up to FL300 (multiple engines out: from 250 kt, up to FL280)	ENG START IGN START, ENG MASTER OFF then ON; the start valve stays closed
1 and 2, starter assisted relight	150-260 kt up to FL200; 185-260 kt from FL200 to FL250	ENG START IGN START, ENG MASTER OFF then ON, and starter air: APU BLEED, or X BLEED OPEN with the other engines
2 (and 3), quick relight	220-260 kt up to FL200 (and zone 3), N3 above 45 %	ENG MASTER OFF then ON within 30 s, at any ENG START position

What you should see

Situation	What you should see
Flameout in flight	N3, EGT and FF run down, FF 0, N1 / N3 settle at the windmill values. Amber **ENG n FAIL** with its FCOM procedure lines (THR LEVER n IDLE reads its own lever; ENG n RELIGHT PROC CONSIDER).
ENG MASTER OFF in flight	ENG n FAIL goes, **ENG n SHUT DOWN** comes with its STATUS: INOP SYS BTV for engine 2 or 3, CAT 3 DUAL and CAT 3 SINGLE ONLY with the APU off; the engine 4 bleed INOP comes from engine 4 (it came from another engine before).
Windmilling relight (zone 3, no APU)	At the MASTER ON the fuel comes back; engine at idle in about 35 s; the start valve stays closed; both engine pumps of that engine are off-loaded during the start.
Crossbleed relight (zone 1, X BLEED OPEN)	The start valve opens and the starter is pressurised in about 0.5 s; engine at idle in about 35 s; the start valve closes at N3 about 58 %.
Quick relight (ENG START at NORM)	MASTER OFF then ON at once: the engine relights from about 75 % N3, at idle in about 25 s.
Any in-flight relight with the levers in CL (A/THR ON)	N3 reaches idle (about 62 %) and holds there until the engine runs, then rises in about 5 s to the N3 of the other engines; it does not go more than 1-2 % above them.
ENG RELIGHT IN FLIGHT procedure	Single engine: MAX GUARANTEED ALTITUDE : 30000 FT, MIN SPEED FOR WINDML RELIGHT : 260 KT; multiple engines: 28000 FT and 250 KT.
Two engines failed	ENG TWO ENGS OUT ON SAME SIDE or ON OPPOSITE SIDE caution with LAND ANSA; the multiple-engine envelope applies. STATUS: PART SPLRs, PART L/G RETRACTION, G (Y) HYD SYS, PACK 1 (2), the GEN line of the two generators lost, CAT 2 / CAT 3 INOP, CAT 2 ONLY.
Three engines failed	ENG FAIL shows RELIGHT PROC APPLY instead of CONSIDER.
All four engines failed	Red ENG ALL ENG FLAME OUT (the FCOM title; it was ALL ENGINES FAILURE). RAT MAN ON turns green once the RAT is out; L/G GRVTY (EXTN 2 MIN) DOWN before a forced landing. With `core/a380x-elec` merged: no ELEC EMER CONFIG while it is shown.
Starter assisted relight below 260 kt without APU BLEED and with X BLEED at AUTO	No starter air: the engine does not relight (the starter duct is empty). Set X BLEED OPEN.
On the ground	A failed engine starts again as a normal start, with APU air (ENG START IGN START, MASTER OFF then ON). Two engines shut down for taxi give no TWO ENGS OUT caution.

Design choices

- The envelope governs **every** in-flight relight, also after a crew shutdown without any failure: MSFS would otherwise relight anywhere.
- The quick relight also works in zone 3, where the windmill alone lights the engine.
- A windmilling engine keeps N3 = 11 % x IAS / 260 kt and N1 = 1.5 x N3 (the FCOM gives no values).
- FBW models the Trent 972 (three spools): the FCOM N2 (GP7200 core) is read on the Trent N3. ENG FAIL keeps the N3 < 50 % threshold of FBW (the FCOM says N2 below 61 %).
- TWO ENGS OUT is shown only with exactly two engines out (with three, ENG FAIL RELIGHT PROC APPLY; with four, ALL ENG FLAME OUT).

- During a windmilling start the two engine pumps are off-loaded (depressurised), not disconnected: the ENG PUMPS DISC clutch cannot be reconnected in flight.
- Once a relight is granted in flight the FADEC holds the MSFS core at 25 % N2: MSFS only burns above 20 % N2, and its starter only turns with the APU bleed, so without it a windmilling or crossbleed relight would never burn.
- The throttle hold of an in-flight start (idle until the engine runs) works in flight only; a fallback ends it for a start that settles just below the FBW idle N3.
- With the ENG MASTER OFF in flight the MSFS core runs down to the windmill speed: also so, not a fault.

Limitations and why

Limitation	Why	What to do
ENG TWO ENGS OUT ON OPPOSITE SIDE has no PACK line in its procedure.	Not added yet (FCOM); its STATUS has the PACK 1 (2) INOP SYS line.	-
The MSFS core jump at the end of a relight is held, not removed.	At the end of a start the MSFS engine model jumps its core from its low idle (60 % corrected) to the top of its fuel flow table (about 97 %) for 2 to 6 s. The FADEC filters this jump for the indications (core spike guard): the E/WD N1/N3, fuel flow, EGT and the FADEC logic stay at idle (sim-tested: MSFS 96.8 %, E/WD 61.6 %); the guard also holds the N1 (so the fuel flow and EGT) while the MSFS fan follows the core jump a few frames late. Since the MSFS core limiter also holds the MSFS core near idle, so the thrust pulse of about 10 s recorded (yaw rate at most 0.2 deg/s) is gone (sim : passed). A ground start shows no jump (recorded).	None.
Multiple engines out: the windmilling relight needs at least 250 kt.	FCOM envelope (sim-confirmed).	Descend to keep the speed.
A full MSFS restart is needed after an update.	The systems, FADEC and fbw WASM modules change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 72 Engine and activate a flameout or a seizure.
2. Follow the ECAM. To relight: get inside the envelope (table above), ENG START selector (pedestal, engine panel) IGN START, ENG MASTER OFF for 30 s, then ON. Below 260 kt without the APU, set the X BLEED selector (overhead AIR panel) to OPEN for starter air.
3. For a quick relight, set the ENG MASTER OFF then ON at once (within 30 s, N3 above 45 %), at any ENG START position.
4. When the engine is at idle: ENG START NORM, X BLEED AUTO; clear the failure in the flyPad (a seizure must be cleared before a relight).

How to test

Needs one MSFS restart after installing the branch.

Windmilling relight: engine 1 relights by the airflow alone, no APU

Setup: in flight at FL200, 300 kt, AP and A/THR ON, APU OFF, X BLEED selector AUTO (overhead AIR panel), ENG START selector NORM (pedestal, engine panel with the four ENG MASTER levers).

#	Do this	Expected result
1	flyPad > Failures > ATA 72 > "Engine 1 flameout (crew relight possible)": activate it.	Amber ENG 1 FAIL; engine 1 FF 0, N3 and EGT fall.
2	ENG START selector: IGN START. ENG MASTER 1: OFF, wait 30 s.	ENG 1 SHUT DOWN replaces ENG 1 FAIL.
3	ENG MASTER 1: ON.	FF appears within a few seconds, engine 1 at idle within about 60 s, the alerts clear; SD ENGINE page: the engine 1 start valve stays closed. Lever in CL: engine 1 N3 holds at idle until the engine runs, then rises to the N3 of the other engines without passing them by more than 1-2 %.
4	ENG START selector NORM; flyPad: deactivate the failure.	-

Quick relight: engine 4 relights at NORM

Setup: FL150, 240 kt, ENG START selector NORM.

#	Do this	Expected result
1	Activate "Engine 4 flameout (crew relight possible)"; at once (within 5 s) ENG MASTER 4 OFF, then ON.	Engine 4 FF and EGT rise within a few seconds, back at idle, ENG 4 FAIL goes; the selector stays at NORM. No yaw kick when the relight ends (about 40 s later); about 10 s after that engine 4 joins the others with no overshoot.
2	flyPad: deactivate the failure.	-

Starter assisted relight with crossbleed air: engine 3

Setup: FL150, 210 kt, APU OFF.

#	Do this	Expected result
1	X BLEED selector OPEN. Activate "Engine 3 flameout (crew relight possible)".	ENG 3 FAIL.
2	ENG START selector IGN START; ENG MASTER 3 OFF, wait 30 s; ENG MASTER 3 ON.	SD ENGINE page: engine 3 start valve open, FF within a few seconds, engine 3 at idle within about 60 s, start valve closed again, ENG 3 FAIL goes.
3	ENG START NORM, X BLEED AUTO, deactivate the failure.	-

On the ground: relight with APU air, and two engines out for taxi

Setup: on the ground, four engines at idle, parking brake ON, APU AVAIL, APU BLEED pb ON (overhead AIR panel), ENG START NORM.

#	Do this	Expected result
1	Activate "Engine 1 flameout (crew relight possible)"; when N1 is below 5 %: ENG MASTER 1 OFF, ENG START IGN START, wait 30 s, ENG MASTER 1 ON.	ENG 1 FAIL, then a normal start: start valve open, engine 1 at idle within about 60 s, ENG 1 FAIL goes.
2	ENG START NORM, deactivate the failure. ENG MASTER 1 OFF and ENG MASTER 4 OFF, wait 20 s.	No TWO ENGS OUT ON OPPOSITE SIDE caution, no ENG 1 / ENG 4 FAIL.

Two engines out at 255 kt: engine 1 relights by windmilling

Setup: FL100, 255 kt (FCU speed), flaps 0, AP and A/THR ON, APU OFF, APU BLEED OFF, X BLEED AUTO. Keep at least 250 kt: descend if needed.

#	Do this	Expected result
1	Activate "Engine 1 flameout (crew relight possible)", then "Engine 3 flameout (crew relight possible)". Wait 30 s.	ENG TWO ENGS OUT ON OPPOSITE SIDE with LAND ANSA; STATUS INOP SYS GEN 1+3.
2	ENG START selector IGN START. ENG MASTER 1 OFF, wait 30 s, ENG MASTER 1 ON.	Engine 1 lights up (fuel flow, EGT rise) and reaches idle; engine 3 stays out. SD ENGINE page: "A B" on engine 1 during the relight.
3	Deactivate both failures; relight engine 3 the same way; ENG START NORM.	-

All engines out at FL200

Setup: FL200, 300 kt, APU OFF, ENG START selector NORM, four engines running.

#	Do this	Expected result
1	Activate the four "Engine n flameout (crew relight possible)" failures one after the other. Wait 20 s.	Red ENG ALL ENG FLAME OUT; the RAT deploys; AC ESS powered by the emergency generator. With `core/a380x-elec` merged: no ELEC EMER CONFIG.
2	Press the RAT MAN ON pb (overhead EMER ELEC PWR panel, guarded).	The RAT MAN ON line turns green.
3	ENG START selector IGN START. ENG MASTER 4 OFF, wait 30 s, ENG MASTER 4 ON.	Engine 4 relights and ALL ENG FLAME OUT goes; the ENG FAIL alerts of the other engines follow.
4	Deactivate the failures and relight the other engines.	-

> **NOTE**
> Recording used for the tests: per engine FUEL_CUT, RELIGHT_ATTEMPT, RELIGHT_IGNITION, ENGINE_STATE, ENGINE_N3, PNEU starter valve / pressurized, MSFS TURB ENG N2, GENERAL ENG COMBUSTION and fuel flow, with the airspeed, altitude, ENG START selector, X BLEED and APU BLEED.

8. Automatic relight with both igniters, and the engine feed tank fuel draw

> **Aircraft:** A380X | **Where:** Automatic (FADEC); pedestal ENG MASTER levers; E/WD | **Status:** Auto relight sim-tested OK (recorded: ignition 0.5 s after the combustion loss with the ENG START selector at NORM, relit when the fuel came back, ignition off 60 s later). Feed tank draw: passive check in a test flight climb (four engines kept burning), no dedicated reproduction. Native tests auto_relight_test (recorded flameout replay), feed_tank_draw_test. The core branch itself was not sim-tested.

What it is for

In a flight of MSFS dropped the combustion of all four engines by itself at 7 900 ft. With the ENG START selector at NORM nothing relit them: the engines windmilled for 150 s until the crew selected IGN START. Two changes:

- **AUTO RELIGHT** (A380 FCOM DSC-70-80-30-20 QUICK AND AUTO RELIGHT: "If the FADEC detects an engine flame out, on ground or in flight, continuous ignition with both igniters is automatically selected. The ignition is maintained for 60 s after engine relight."). When a running engine (ENG MASTER ON, FADEC supplied, fuel not cut) loses its combustion for 0.5 s, its FADEC turns on both igniters whatever the ENG START selector position, until 60 s of continuous combustion after the relight.
- **Feed tank draw.** Each engine is fed by MSFS through a 1 gallon buffer tank that the FADEC drains by the fuel burned. The FADEC took the whole burn of an update at once and compared kilograms with pounds, so one long sim frame could empty the four buffer tanks together and leave no fuel at any engine (the suspected cause of the four-engine flameout). Now, while the feed tank holds fuel, the FADEC leaves a small quantity in the buffer and carries the rest of the burn over to the next updates, so the fuel burned is still taken in full. A dry feed tank still empties the buffer and the engine flames out (fuel exhaustion).

What you should see

Situation	What you should see
MSFS loses the combustion of a running engine, ENG START at NORM	Within about 1 s the FADEC turns on the ignition; the engine burns again as soon as it has fuel, back at the thrust of the others within about 30 s. SD ENGINE page: "A B" on that engine. 60 s after the relight the ignition goes off.
ENG MASTER OFF, ENG FIRE pb released, or a flyPad flameout / seizure failure	No auto relight: a manual abort, an unpowered FADEC or a failure the crew must relight (section above).
Long sim frame (stutter) in a climb	The four engines keep burning; the fuel used rises steadily.

Limitations and why

Limitation	Why	What to do
There is no flyPad failure for an MSFS flameout.	The flyPad flameout cuts the fuel on purpose and asks for a crew relight; the auto relight is for a flameout the FADEC did not cause.	The test closed the MSFS fuel valve of engine 4 for 5 s with a SimConnect tool. Since the HP fuel valve hold (below) the systems reopen an MSFS valve that disagrees with them, so that way of testing probably no longer cuts the fuel.
A full MSFS restart is needed after an update.	The FADEC WASM module changes.	Restart the sim after installing.

How to use

1. Nothing to set: keep the ENG MASTER levers ON and the ENG START selector at NORM in flight.
2. If an engine flames out by itself, watch it relight on the E/WD; follow ENG FAIL if it does not.

How to test

The flameout needs MSFS to lose the combustion by itself, or a SimConnect tool that interrupts the engine fuel (see the limitation above).

#	Do this	Expected result
1	A380X at FL150, 240 kt, flaps 0, four engines running, levers in CL, A/THR ON, ENG START selector NORM, ENG 4 ANTI ICE and WING ANTI ICE OFF (so the MSFS automatic ignition is off).	-
2	Interrupt the fuel of engine 4 for 5 s (SimConnect tool). Touch nothing for 90 s.	Engine 4 N1 and N3 start to fall; within about 1 s of the fuel coming back it burns again with the selector still at NORM, and it is back at the thrust of the others within about 30 s. A brief ENG 4 FAIL is possible.
3	Climb from FL50 to FL100 at 250 kt, levers in CL, without touching the engine controls.	No ENG FAIL, no N1 / N3 drop; the FUEL USED rises steadily on the four engines.

9. SD ENGINE page igniter indication

> **Aircraft:** A380X | **Where:** SD ENGINE page, start parameters of each engine | **Status:** Sim-tested OK (ground start with igniter A failed: "A", nothing during the crank, then "A B"; "A B" on all engines at IGN START in flight; "A B" during a relight; a second ground

> start of the same engine shows "B"). Native test igniter_selection_test, unit test StartParameters.spec.ts. The core branch itself was not sim-tested.

What it is for

The SD ENGINE page drew an "A" from the N2 during a start, whatever the FADEC did. It now shows the igniters the FADEC really energizes (A380 FCOM DSC-70-90: "The igniter A(B) is energized." / "Both igniters A and B are energized."), in this priority order:

1. None with the ENG MASTER OFF or the FADEC unpowered (ENG FIRE pb released).
2. "A B" when the FADEC selects both igniters itself: auto relight, an in-flight relight that lights up.
3. "A B" for the quick relight (ENG MASTER OFF then ON within 30 s above 45 % N3), for 60 s.
4. Start: in flight "A B"; on the ground the igniters of the start sequence (next section): one igniter from 20 % to 58 % N3, A or B alternated at each start, "A B" for a new automatic attempt, none during the automatic dry crank.
5. Continuous ignition (ENG START at IGN START, engine running): "A B" at once in flight; on the ground only after the selector was set to NORM then back to IGN START after the start, and not at low power (N1 below 53 % for more than 30 s), as in FCOM DSC-70-80-20.

Limitations and why

Limitation	Why	What to do
A failed igniter is still shown when it is energized ("A" during the first attempt with igniter A failed).	The FCOM shows the igniter energized and has no line for a failed one; kept by decision.	-
The continuous ignition inhibition at high combustion chamber pressure is not modelled.	MSFS has no combustion chamber pressure.	-
A full MSFS restart is needed after an update.	The FADEC WASM module changes.	Restart the sim after installing.

How to use

1. Open the SD ENGINE page (ECP ENG key); the start parameters show the igniter letters above the start valve of each engine during a start, a relight or the continuous ignition.

How to test

#	Do this	Expected result
1	A380X on the ground, APU AVAIL, APU BLEED ON, ENG START selector IGN START, SD ENGINE page. ENG MASTER 2 ON.	A green "A" (or "B") above the engine 2 start valve from about 20 % N3 to about 58 % N3, then no letter; no letter below 20 % N3.
2	ENG MASTER 2 OFF; wait until N3 is below 10 %; ENG MASTER 2 ON.	The other letter, in the same way (A, then B).
3	In flight at FL100, all engines running, ENG START selector NORM. Set IGN START, wait 5 s.	On all four engines the start parameters replace the nacelle temperatures and show "A B".
4	Set NORM, wait 5 s.	The letters go out and the nacelle temperatures come back.

10. Engine ignition and start failures, FCOM ground start

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 74 Ignition and ATA 80 Starting; ENG START selector, ENG MASTER levers, ENG MAN START pbs; E/WD, STATUS, SD ENGINE page | **Status:** Sim-tested OK (test flight, recorded): dry crank with the ENG MAN START pb, then "Engine 1 igniter A": no light up, START FAULT NO LIGHT UP, automatic crank, light up at the second attempt. The other start failures (start valve, hot, hung, stall, starter) are not sim-tested. Rust tests (engine_start.rs, a380_systems engine_failure.rs), native test start_sequence_test, unit tests EngineStartAlerts.spec.ts, a380.spec.ts. The core branch itself was not sim-tested.

What it is for

The FBW FADEC ran the engine model but not what a real FADEC does during a start. The ground start now follows the A380 FCOM (DSC-70-80-20 IGNITION, DSC-70-80-30 STARTING, DSC-70-80-40 CRANKING): ENG START selector to IGN START, then ENG MASTER ON; one igniter (alternated between starts) and fuel at 20 % N3; start valve closed and ignition off at about 58 % N3. A start that does not light up within 20 s, stalls, hangs or goes above 745 °C EGT is aborted, dry cranked for 30 s and tried again with both igniters, up to three attempts. A start valve stuck closed, no starter air or a failed starter leaves the core at rest. The ENG MAN START pbs work: dry crank (ENG START at CRANK, MAN START ON, ENG MASTER OFF), wet crank and manual start.

flyPad failure (ATA)	What it does
Engine 1 (2)(3)(4) igniter A / igniter B (74)	That igniter does not light the engine. The first automatic attempt with only that igniter does not light up: START FAULT NO LIGHT UP, then a new attempt with both. ENG n IGN A (B) FAULT; both failed: ENG n IGN A+B FAULT.
Engine 1 (2)(3)(4) start valve stuck closed (80)	No starter air: the core stays at rest. ENG n START VLV FAULT (NOT OPEN) with VLV STUCK CLOSED.
Engine 1 (2)(3)(4) start valve stuck open (80)	The valve does not close after the start: ENG n START VLV FAULT (NOT CLOSED).
Engine 1 (2)(3)(4) hot start (start EGT over limit) (80)	The start EGT overshoots: START FAULT EGT OVERLIMIT, the start is aborted.
Engine 1 (2)(3)(4) hung start (N3 stops below idle) (80)	The core stops accelerating below idle: START FAULT HUNG START.
Engine 1 (2)(3)(4) stall during the start (80)	START FAULT ENG STALL.
Engine 1 (2)(3)(4) starter shaft shear (80)	The starter does not turn the core although it gets air: START FAULT LOW N2, no further attempt.

ECAM alerts

Alert	Lines (A380 FCOM PRO-ABN-ECAM-10-70)
ENG n START FAULT	The sub-case (NO STARTER AIR PRESSURE, LOW N2, HUNG START, EGT OVERLIMIT, NO LIGHT UP, ENG STALL, STARTER TIME EXCEEDED, THR LEVERS NOT AT IDLE), then: no starter air: XBLEED OPEN, APU BLEED ON (in flight MIN SPEED FOR WINDML RELIGHT : 260 KT); automatic start on the ground: NEW START IN PROGRESS, then ENG n MASTER OFF once aborted; manual start: ENG n MASTER OFF, ENG START SEL CRANK, AFTER 30S: ENG START SEL NORM, ENG n MAN START OFF; levers not at idle: ALL THR LEVERS IDLE.
ENG n START VLV FAULT (NOT CLOSED)	ENG n BLEED OFF (in flight), APU BLEED OFF (engine 1), X BLEED CLOSE, ENG n MAN START OFF (manual start), ENG n MASTER OFF (on the ground), WING A-ICE OFF and AVOID ICING CONDs (engines 1 and 4).

ENG n START VLV FAULT (NOT OPEN)	VLV STUCK CLOSED; on the ground ENG n MAN START OFF, ENG n MASTER OFF; in flight MIN SPEED FOR WINDML RELIGHT : 260 KT, WINDMILL START INITIATED, ENG PARAMETERS MONITOR WITH CARE.
ENG n IGN A (B) FAULT	STATUS MORE: REDUND LOSS ENG n IGN A (B).
ENG n IGN A+B FAULT	STATUS INOP SYS ENG n IGNITION.

> **NOTE**
> The start alerts are inhibited in the flight phases 3 to 7, 9 and 10 (IGN A (B) FAULT also in phase 8), per the FCOM figures. START FAULT and IGN A (B) FAULT come without chime and MASTER CAUT; IGN A+B FAULT keeps them (see "Engine alerts as in the FCOM" below for the in-flight START FAULT cases).

Limitations and why

Limitation	Why	What to do
Hung start, stall and starter detection times are design values.	The FCOM gives the symptoms, not the times; the FBW engine model has no compressor pressure.	-
START FAULT LOW N1 is not modelled.	No N1 rotor failure.	-
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules and the pedestal behaviour XML change.	Restart the sim after installing.

How to use

1. Normal start: APU BLEED ON, ENG START selector IGN START, ENG MASTER ON; the FADEC does the rest (ENG START NORM when the engines are running).
2. Failures: flyPad > Failures > ATA 74 Ignition or ATA 80 Starting; activate one, start the engine and follow the ECAM.
3. Dry crank: ENG START selector CRANK, ENG MAN START n pb ON (ENG MASTER OFF), then OFF.

How to test

Setup: A380X cold and dark, batteries ON, APU AVAIL, APU BLEED ON, all ENG MASTER levers OFF, thrust levers at IDLE, SD ENGINE page.

#	Do this	Expected result
1	ENG START selector CRANK. ENG MAN START 2 pb (overhead ENG MAN START panel) ON, wait 20 s, then OFF.	Engine 2 N3 rises to about 20-25 % and stays there; no fuel flow, no EGT rise; it does not start. After MAN START OFF its N3 runs down. No ENG 2 START FAULT.
2	flyPad > Failures > ATA 74 Ignition > "Engine 1 igniter A": activate. ENG START selector IGN START. ENG MASTER 1 ON, wait 80 s.	About 22 s after MASTER ON: amber ENG 1 START FAULT with NO LIGHT UP and NEW START IN PROGRESS, and amber ENG 1 IGN A FAULT; N3 at 20-30 %, no EGT, no fuel flow. SD: "A" during the first attempt, nothing during the crank, "A B" for the second attempt. About 55 s after MASTER ON: EGT rises, engine 1 starts, START FAULT goes.
3	Deactivate the failure. Activate "Engine 3 start valve stuck closed"; ENG MASTER 3 ON.	Engine 3 stays at rest; ENG 3 START VLV FAULT (NOT OPEN) with VLV STUCK CLOSED. ENG MASTER 3 OFF, deactivate.

4	Activate "Engine 3 hot start (start EGT over limit)"; ENG MASTER 3 ON.	The EGT overshoots: ENG 3 START FAULT EGT OVERLIMIT, the start is aborted and cranked, then a new attempt. ENG MASTER 3 OFF, deactivate.

11. HP fuel valve: no engine burning without a FADEC start

> **Aircraft:** A380X | **Where:** Pedestal ENG MASTER levers and ENG START selector | **Status:** Sim-tested: the fault found (test flight, a restart 34 s after a shutdown skipped the start sequence); restart after shutdown OK (full start sequence, igniter B), ENG MASTER ON at NORM stays dark, a start at IGN START opens the HP valve at the fuel-on point; the HP valves stayed closed from load through power-up (more than 200 s). Rust tests (engine_failure.rs HpFuelValve, sim_fuel_valve.rs), native start_sequence tests. The core branch itself was not sim-tested.

What it is for

A380 FCOM DSC-70-30 ENGINE SHUTDOWN: with the ENG MASTER OFF "The FADEC closes the LP and HP fuel valves. The engine decelerates and stops." The ENG MASTER only closed an MSFS valve upstream of the engine buffer tank, so MSFS kept burning a shut down engine for 20 to 60 s, and a restart in that time skipped the start sequence. And with the ENG START selector at NORM, ENG MASTER ON on the ground let the MSFS starter turn the core and MSFS lit it at 22 % N2 without any FADEC start (the FCOM: IGN START first, fuel at 20 %; at NORM the MASTER ON starts nothing).

- The HP fuel valve of each engine (the MSFS fuel valves 60-63 after the buffer tanks) is now closed with the ENG MASTER OFF or the fuel cut, and on the ground until the FADEC start sequence reaches its fuel-on point (IGN START then MASTER ON) or the quick relight opens it (MASTER OFF then ON within 30 s above 45 % N3). It is open for the aircraft presets and a flight loaded with the master ON. In flight the relight logic keeps the fuel cut until the engine may light up. Once open it stays open until the MASTER OFF or a fuel cut, so a running engine keeps its fuel at NORM and its auto relight has fuel.
- MSFS could reopen a closed valve by itself a few seconds later (flight load, power-up of the valve circuit). The systems now watch the MSFS valve switch and command it again whenever it disagrees, and once on the first update of a flight.
- An engine is shown running (ON) only when MSFS really burns in it: a core turned past 20 % by the MSFS starter at NORM no longer shows ON unlit (and no longer triggers the auto relight).

What you should see

Situation	What you should see
ENG MASTER OFF	The engine stops burning at once (FF 0, EGT falls) and runs down.
ENG MASTER ON again soon after a shutdown, ENG START at IGN START	A full start sequence: start valve, one igniter, fuel at 20 % N3.
ENG MASTER ON on the ground with the ENG START selector at NORM	Nothing lights up: no fuel flow, no EGT.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Start an engine the FCOM way: ENG START selector IGN START first, then ENG MASTER ON.
2. Shut it down with ENG MASTER OFF.

How to test

#	Do this	Expected result
1	A380X cold and dark on the ground, APU AVAIL, APU BLEED ON, ENG START selector NORM. ENG MASTER 1 ON, wait 60 s.	Engine 1 stays dark (no fuel flow, no EGT), also after the aircraft power-up.
2	ENG MASTER 1 OFF. ENG START selector IGN START, ENG MASTER 1 ON.	A normal start: fuel at about 20 % N3, engine 1 at idle.
3	ENG MASTER 1 OFF; after about 30 s (engine still turning) ENG MASTER 1 ON.	Fuel flow goes to 0 at once at the MASTER OFF; the restart goes through the start sequence again (start valve opens, one igniter on the SD).

12. Engine oil failures (leak, filter clog, overheat)

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 79 Oil; E/WD; SD ENGINE page | **Status:** Sim-tested (test flight): filter clog and leak to OIL PRESS LO and the shutdown OK; the overheat stopped at 128 °C (fixed: the FADEC keeps its own temperature). : the overheat reached the 196 °C limit only after 4 min (then retuned), the cool-down after clearing the failure OK. The retuned overheat curve (OIL TEMP HI about 2 min at cruise) and the oil pressure from the whole oil system are not sim-tested on the A380X (the latter checked on the A32NX). Native tests engine_oil_failures_test, oil_system_test, unit tests OilIndications.spec.ts, EngineOilAlerts.spec.ts. The core branch itself was not sim-tested.

What it is for

Twelve new failures, three per engine, computed by the FADEC on the oil quantity, pressure and temperature, with the SD ENGINE page indications and the ECAM alerts of the A380 FCOM (DSC-70-90, PRO-ABN-ECAM-10-70).

flyPad failure (ATA 79 Oil)	What it does
Engine 1 (2)(3)(4) oil leak	The oil system empties at 2 qt/min (the tank reading falls about 1.6 qt/min); the pump pressure stays full down to 1.3 qt in the system and is gone at 0.3 qt, so OIL PRESS LO follows the low oil quantity indication.
Engine 1 (2)(3)(4) oil filter clog	CLOGGED on the SD ENGINE page and ENG n OIL FILTER CLOGGED; no other effect.
Engine 1 (2)(3)(4) oil overheat	The oil temperature rises towards a target that grows with the core speed (170 °C at idle, 230 °C at 100 % N3, 60 s time constant): OIL TEMP HI about 2 min after the failure at cruise; reducing thrust brings it below 196 °C, at IDLE within about 1 min. Once cleared, the oil cools down gradually.

What you should see

Situation	What you should see
SD ENGINE page, oil quantity	Pulses below the 1.2 qt advisory (first white dash).
SD ENGINE page, oil temperature	Pulses above 163 °C.
SD ENGINE page, oil pressure	FCOM scale 0-100 / 100-440 PSI, red at or below 25 PSI; CLOGGED below it for a clogged filter.
Oil pressure below 25 PSI	Red **ENG n OIL PRESS LO** (MASTER WARN, continuous chime): THR LEVER n IDLE, ENG n MASTER OFF; then ENG n SHUT DOWN.
Oil temperature above 196 °C	Amber **ENG n OIL TEMP HI**: THR LEVER n REDUCE BELOW OIL TEMP LIM, IF HI TEMP PERSISTS: ENG n MASTER OFF.
Clogged filter	Amber **ENG n OIL FILTER CLOGGED** on the ground (phases 1, 2, 11 and 12), without chime or MASTER CAUT; in flight only the SD CLOGGED.

Limitations and why

Limitation	Why	What to do
The leak and overheat rates are design values.	The FCOM gives no rates.	-
The alerts are monitored only with the engine running and its ENG MASTER ON.	Design choice: a stopped engine has no oil pressure, and the procedures end with the MASTER OFF.	-
Flight phase inhibitions.	OIL PRESS LO: phases 1, 5, 6, 12; OIL TEMP HI: 1, 4-7, 9, 10, 12; OIL FILTER CLOGGED: 3 to 10 (FCOM figures).	Test in cruise.
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 79 Oil and activate a leak, a filter clog or an overheat.
2. Watch the SD ENGINE page; follow the ECAM (reduce thrust, or IDLE and ENG MASTER OFF).
3. Deactivate the failure.

How to test

Setup: level at FL200, 300 kt, A/THR ON, four engines running, SD ENGINE page.

#	Do this	Expected result
1	Activate "Engine 2 oil filter clog", "Engine 4 oil overheat" and "Engine 1 oil leak".	SD: amber CLOGGED under the engine 2 oil pressure (no E/WD alert in flight). Engine 4 oil temperature rises and pulses above 163; then amber ENG 4 OIL TEMP HI with THR LEVER 4 ... REDUCE BELOW OIL TEMP LIM.
2	When ENG 4 OIL TEMP HI shows: A/THR off, THR LEVER 4 IDLE; fly the speed with the other levers.	Engine 4 oil temperature falls below 196 and the alert goes within about 1-2 minutes.
3	Wait for engine 1 (several minutes).	Engine 1 oil quantity falls and pulses below 1.2 qt; then its oil pressure falls and turns red at 25 PSI: red ENG 1 OIL PRESS LO with MASTER WARN, THR LEVER 1 ... IDLE, ENG 1 MASTER ... OFF.
4	THR LEVER 1 IDLE, ENG MASTER 1 OFF. Deactivate the three failures.	OIL PRESS LO goes, ENG 1 SHUT DOWN comes. Engine 4 oil cools down gradually.

13. Engine fuel filter clog, ENG MASTER FAULT light, SD start valve and N3 red cross

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 73 and ATA 80; pedestal ENG MASTER levers; SD ENGINE page; E/WD | **Status:** Sim-tested by the user, data-checked OK: fuel filter clog on engine 3 at idle; start valve stuck closed (START VALVE FAULT 5 s after ENG MASTER ON); starter shaft shear (start aborted and ENG 1 FAULT light 9.8 s after ENG MASTER ON, N3 at most 0.2 %). The N3 red cross is unit-tested only (not reachable on the ground).

What it is for

- **Fuel filter clog:** new failures Engine 1-4 fuel filter clog (ATA 73). Amber **CLOGGED** below the fuel flow on the SD ENGINE page while the engine runs, and the **ENG n FUEL FILTER CLOGGED** caution (FCOM: pressure drop above 35 PSI, crew awareness, inhibited in flight phases 3 to 10). No effect on the engine: the filter is bypassed.

- **ENG MASTER FAULT light:** it comes on when the FADEC aborts an automatic start (FCOM DSC-70-90), and goes off with the ENG MASTER OFF. It followed the MSFS engine failure before, which a start abort does not set.
- **SD start valve:** amber when the start valve is abnormally open or closed (START VLV FAULT); with a valve fault the start parameters show even at NORM, so a valve stuck open after the start is seen.
- **N3 red cross:** above the 118.7 % red limit a red "+" shows next to N3 until the next start of that engine on the ground (FCOM: N2 of the GP7200 core, read on the Trent N3).
- **Starter failure fix:** with a sheared starter the MSFS starter could still turn the core to about 6 %, past the 5 % starter fault detection, so the start never aborted. The FADEC now keeps the core at rest and aborts the start without a new attempt (FCOM DSC-70-80-30-20).

Limitations and why

Limitation	Why	What to do
The FAULT light has no HP fuel valve and no overthrust cause.	FBW has no HP fuel valve failure (the valve always follows its command) and no overthrust protection.	-
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules and the cockpit behaviour change.	Restart the sim after installing.

How to use

1. Call the SD ENGINE page (ECP ENG key).
2. Open flyPad > Failures > ATA 73 (fuel filter clog) or ATA 80 (starter shaft shear, start valve stuck) and activate the failure, then start or run the engine as below.
3. Deactivate the failures after the test.

How to test

Setup: at the gate, APU running, APU BLEED ON, all ENG MASTER levers OFF, ENG START selector IGN START.

#	Do this	Expected result
1	flyPad ATA 80 **Engine 1 starter shaft shear** ON; ENG MASTER 1 ON, wait 30 s; then OFF.	N3 stays near 0; about 10 s after the MASTER ON the start aborts: E/WD ENG 1 START FAULT and the amber FAULT light in the ENG 1 MASTER lever (the other three stay off); the light goes off with the MASTER OFF.
2	flyPad ATA 80 **Engine 2 start valve stuck closed** ON; ENG MASTER 2 ON, wait 15 s; then OFF.	The engine 2 start valve circle and bar are amber on the SD; E/WD ENG 2 START VLV FAULT.
3	ENG MASTER 3 ON, wait until idle; flyPad ATA 73 **Engine 3 fuel filter clog** ON.	Amber CLOGGED below the engine 3 fuel flow; E/WD amber ENG 3 FUEL FILTER CLOGGED, without chime or MASTER CAUT (FCOM level); engine 3 keeps running.

14. Engine stall, EGT overtemperature and vibration failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 72 Engine and ATA 77 Engine Indicating; E/WD; SD ENGINE page | **Status:** Sim-tested OK (test flight: engine 3 stall, engine 2 EGT overtemperature and high vibration). The retuned vibration (square of the rotor speed) is not sim-tested on the A380X. EWD EGT amber limit from the thrust limit: tested. Rust tests (engine_malfunction.rs), native test egt_failure_offset_test, unit tests EngineParameterAlerts, EgtLimits, EngineVibration specs. The core branch itself was not sim-tested.

What it is for

Twelve new failures where the engine keeps running. What each one does is a design choice: the FCOM gives the symptoms, not values.

flyPad failure (ATA)	What it does
Engine 1 (2)(3)(4) compressor stall (above 60 % N1) (72)	Above 60 % N1 command the engine stalls: it loses thrust (the thrust loop aims lower), the EGT rises quickly, N1 / N3 fluctuate. It recovers with the lever at idle and stalls again above the threshold.
Engine 1 (2)(3)(4) EGT overtemperature (72)	The EGT is higher, more so at high thrust: ENG EGT OVER LIMIT at climb thrust.
Engine 1 (2)(3)(4) high vibration (77)	N1, N2 and N3 vibrations growing with the square of the rotor speed: above the SD advisory (5 units) at climb and cruise thrust, below it at reduced thrust and at idle. At a low cruise power the advisory can show on N3 only (fan 60 %: N1 4.7; core 81 %: N3 5.3), as the FCOM advisory is per rotor; reducing thrust clears it. No ECAM alert (ABN PROC).

What you should see

Situation	What you should see
Stall	Amber **ENG n STALL** (SD ENGINE page): THR LEVER n IDLE, ENG n PARAMETERS CHECK, IF ABNORMAL: ENG n MASTER OFF, ENG n RELIGHT PROC CONSIDER. Inhibited from V1 to 400 ft unless two or more engines stall.
EGT above its limit	Amber **ENG n EGT OVER LIMIT**: THR LEVER n REDUCE BELOW EGT LIMIT, IF OVER LIMIT PERSISTS: ENG n MASTER OFF; the amber attention box on the engine.
N1 or N2 above the red limit	Red **ENG n N1/N2 OVER LIMIT**: THR LEVER n REDUCE BELOW N1 LIMIT, IF OVER LIMIT PERSISTS: ENG n MASTER OFF.
E/WD EGT gauge	A red mark at the highest EGT reached above the red line. The amber limit comes from the active thrust limit (none at FLEX / TOGA or in reverse, the CLB / MCT limit otherwise), not from the lever angle.
SD ENGINE page	Separate N1, N2 and N3 vibrations that pulse above 5 units; N3 red above its limit.

Limitations and why

Limitation	Why	What to do
The A380X has no overspeed failure (the A32NX has one).	Not part of this stage. The shared overspeed ceiling fields added for the A32NX LEAP-1A red lines are set to none on the A380X: no change here.	-
A full MSFS restart is needed after an update.	The systems, FADEC and fbw WASM modules change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 72 Engine (stall, EGT) or ATA 77 Engine Indicating (vibration) and activate the failure at climb or cruise thrust.
2. Follow the ECAM: reduce the thrust of that engine; deactivate the failure.

How to test

Setup: four engines running, A/THR ON, CLB, climbing at 280 kt.

#	Do this	Expected result

1	Climbing through FL150: activate "Engine 3 compressor stall (above 60 % N1)", wait 20 s.	Amber ENG 3 STALL with its lines; SD ENGINE page shown; engine 3 THR / N1 drops and fluctuates, EGT 3 rises.
2	Thrust lever 3 IDLE, wait 20 s; deactivate the stall; lever 3 back to CL.	The stall alert goes.
3	Activate "Engine 2 EGT overtemperature" and "Engine 2 high vibration", wait 30 s.	EGT 2 above 850 °C, amber, with amber ENG 2 EGT OVER LIMIT and the attention box on engine 2; SD VIB N1, N2 and N3 of engine 2 above 5, pulsing.
4	Deactivate both.	The EGT and vibrations come back to normal; the alert goes.

15. FADEC, thrust lever and reverser failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 73 Engine - Fuel And Control and ATA 78 Exhaust; thrust levers; E/WD, STATUS, SD ENGINE page | **Status:** Sim-tested (test flight): "Engine 1 FADEC network link" alerts and displays OK but the engine stayed at its thrust (fixed: the engine now follows its lever, not re-tested in the sim); "Reverser 2 unlocked (engine at idle)" OK. Reverse thrust OK (engines 2 and 3). The other FADEC, thrust lever and reverser failures and ENG REVERSER SELECTED are not sim-tested. Rust tests (engine_control_failure.rs, reverser), native tests (fadec_failure_inputs_test, fadec_fault_manual_thrust_test), unit tests FadecReverserAlerts.spec.ts, EngineParametersAvailability.spec.ts. The core branch itself was not sim-tested.

What it is for

Twenty-six new failures with the alerts, procedures and STATUS of the A380 FCOM (PRO-ABN-ECAM-10-70). Reversers exist on engines 2 and 3 only.

flyPad failure (ATA)	What it does	ECAM
Engine 1 (2)(3)(4) FADEC network link (73)	The FADEC still controls the engine, but the A/THR and the SD ENGINE page lose it (XX on the SD; the E/WD keeps N1, THR, EGT). The engine follows its thrust lever.	ENG n FADEC FAULT: THR LEVER n MAN ADJUST, ENG n PARAMETERS MONITOR, IF ABNORMAL: THR LEVER n IDLE, ENG n MASTER OFF; STATUS INOP SYS ENG n A/THR (A/THR engaged).
Engine 1 (2)(3)(4) FADEC system fault (73)	Alert only (crew awareness).	ENG n FADEC SYS FAULT
Engine 1 (2)(3)(4) FADEC overheat (73)	Alert only (crew awareness).	ENG n FADEC TEMP HI
Thrust lever 1 (2)(3)(4) resolvers (both) (73)	The FADEC has lost the lever: the engine is at idle on the ground and limited to CLB in flight (the A/THR still manages it between idle and CLB).	ENG n THR LEVER FAULT: on the ground ENG n IDLE ONLY, THR LEVER n IDLE; in flight ENG n CLB ONLY, THR LEVER n CLB; STATUS INOP SYS ENG n THR LEVER, INFO ENG n IDLE (CLB) ONLY.
Reverser 2 (3) fault (does not deploy) (78)	The reverser does not deploy; the engine stays at reverse idle.	ENG 2 (3) REVERSER FAULT: T.O PERF CHECK (on the ground), LDG DIST IMPACT ON WET/CONTAM RWY ONLY.
Reverser 2 (3) unlocked (engine at idle) (78)	The cowls stay stowed (no reverse thrust, no buffet); the FADEC sets the engine at idle.	ENG 2 (3) REVERSER UNLOCKED: ENG n IDLE ONLY, THR LEVER n IDLE, ENG n MASTER OFF on the ground; in flight IF BUFFET: ENG n MASTER OFF; STATUS INFO ENG n IDLE ONLY.

Reverser 2 (3) energized (tertiary lock unlocked) (78)	The tertiary lock stays released; the primary locks hold the cowls stowed.	ENG 2 (3) REVERSER ENERGIZED: THR LEVER n IDLE.
Reverser 2 (3) failed locked (does not deploy) (78)	The locks are not released: no deployment.	ENG 2 (3) REV LOCKED: T.O PERF CHECK, LDG DIST IMPACT ON WET/CONTAM RWY ONLY.
Reverser 2 (3) control fault (ETRAC) (78)	The reverser is inoperative.	ENG 2 (3) REVERSER CTL FAULT: T.O PERF CHECK, LDG DIST IMPACT ON WET/CONTAM RWY ONLY.

Also in this feature

- **Reverse thrust stays at reverse idle until the reverser is fully deployed** (FCOM DSC-70-70): a reverser that does not deploy no longer gives forward thrust with the lever in reverse.
- **Reverse thrust reaches MSFS.** The in-flight start throttle hold (engine failures section) clamped the MSFS throttle at 0 and so blocked all reverse thrust; the reverse throttle now goes through.
- **An engine with a FADEC FAULT follows its thrust lever.** In the test the engine stayed at its N1 with the levers in CL: the FADEC THRUST LOCK latched when the lost link took the A/THR away. The A380 ENG FADEC FAULT does not disconnect the A/THR (only that engine is lost, INOP SYS ENG n A/THR), and the procedure says THR LEVER ... MAN ADJUST, so the lock no longer latches for it.
- **ENG REVERSER SELECTED** comes up when the lever of engine 2 or 3 is in reverse in flight: THR LEVER 2 (3)(2+3) FWD THR.

Limitations and why

Limitation	Why	What to do
ENG 2 (3) REVERSER INHIBITED and REVERSER MINOR FAULT are not wired.	No flyPad failure for them (maintenance action / no effect).	-
"Before takeoff" lines (T.O PERF CHECK) show on the ground.	Design choice: before takeoff = on the ground.	-
A full MSFS restart is needed after an update.	The systems and fbw WASM modules change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 73 Engine - Fuel And Control (FADEC, thrust lever) or ATA 78 Exhaust (reversers 2 and 3) and activate a failure.
2. Follow the ECAM; for a FADEC FAULT adjust that thrust lever by hand.
3. Deactivate the failure.

How to test

Setup: FL100, 250 kt, flaps 0, AP and A/THR ON, four levers in CL.

#	Do this	Expected result
1	Activate "Engine 1 FADEC network link".	Amber ENG 1 FADEC FAULT with THR LEVER 1 .... MAN ADJUST; SD ENGINE page XX for engine 1 while the E/WD keeps its N1 / THR / EGT; STATUS INOP SYS ENG 1 A/THR. Engine 1 goes to the CLB thrust of its lever; the A/THR still manages engines 2 to 4.

2	Move thrust lever 1 out of CL and back.	Engine 1 follows its lever (it does not stay frozen).
3	Activate "Reverser 2 unlocked (engine at idle)".	Amber ENG 2 REVERSER UNLOCKED with ENG 2 IDLE ONLY, THR LEVER 2 .... IDLE, .IF BUFFET, ENG 2 MASTER .... OFF; engine 2 N1 goes to idle; STATUS INFO ENG 2 IDLE ONLY.
4	Deactivate both. Activate "Thrust lever 4 resolvers (both)".	ENG 4 THR LEVER FAULT with ENG 4 CLB ONLY, THR LEVER 4 CLB; engine 4 limited to CLB. Deactivate it.
5	After landing: all thrust levers to full reverse until 70 kt, then IDLE.	REV on engines 2 and 3 only; their N1 rises above idle once the reversers are deployed.

16. ENG THR LEVERS NOT SET and ENG OIL TEMP LO

> **Aircraft:** A380X | **Where:** E/WD; thrust levers; T.O CONFIG pb; MFD FMS PERF T.O (FLEX) | **Status:** THR LEVERS NOT SET: tested in the sim by the user: OK, data-confirmed (on the feature branch). OIL TEMP LO: not sim-tested. Unit tests EngineTakeoffAlerts.spec.ts.

What it is for

Two take-off alerts of the A380 FCOM (PRO-ABN-ECAM-10-70) that the A380X did not trigger. Both are amber cautions (MASTER CAUT, single chime), shown in flight phases 2 and 3 only.

- **ENG THR LEVERS NOT SET** : a thrust lever of a running engine is at or between CL and FLX/MCT, or at FLX/MCT without a FLEX TEMP, for 3 s. Line: ALL THR LEVERS ... TOGA (no FLEX TEMP: TOGA take-off) or THR LEVERS ... MCT/FLEX (FLEX TEMP entered, levers below FLX/MCT).
- **ENG 1(2)(3)(4) OIL TEMP LO** : on the ground the engine oil is below 50 °C 30 s after the engine is running, or after the T.O CONFIG pb, or at take-off power. Lines: THR LEVER n ... IDLE, DELAY T.O FOR WARM UP.

Limitations and why

Limitation	Why	What to do
No derated take-off case.	FBW has no derated take-off.	-
A T.O CONFIG press keeps the oil monitor armed until the engine stops.	Design choice: the FCOM says "after the flight crew presses".	-

How to use

1. Nothing to set. Set all thrust levers to FLX/MCT (FLEX TEMP entered) or TOGA for take-off.
2. After an engine start in the cold, wait until the oil is warm before take-off.

How to test

#	Do this	Expected result
1	A380X on the ground, PARKING BRK ON, engine 4 running at idle, all levers at IDLE, no FLEX TEMP. Thrust lever 4 only to the CL detent.	After 3 s: amber ENG THR LEVERS NOT SET with ALL THR LEVERS ... TOGA, MASTER CAUT.
2	Lever 4 back to IDLE.	The caution goes.
3	Cold and dark at a cold airport (oil at the outside temperature, below 50 °C): start engine 1 and wait 30 s.	Amber ENG 1 OIL TEMP LO with THR LEVER 1 ... IDLE and DELAY T.O FOR WARM UP while the oil is below 50 °C (SD ENGINE page).

17. FADEC thrust failures: ENG THRUST LOSS and ENG T.O THRUST DISAGREE

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 73; E/WD; thrust levers | **Status:** Tested in the sim by the user on the feature branch: T.O THRUST DISAGREE OK (FADEC 4 take-off mode recorded); THRUST LOSS OK with levers 3 and 4 at FLX/MCT (max THR 85 recorded, the caution seen). Unit tests EngineThrustAlerts.spec.ts, a380.spec.ts; native test fadec_thrust_failures_test.cpp.

What it is for

Two FCOM cautions that need a FADEC fault the sim cannot produce, with the failures that make them possible (they act on the FADEC models of the fbw WASM):

- New flyPad failures, ATA 73: **Engine 1 (2, 3, 4) max thrust miscalculated (thrust loss)** (73050-73053): that FADEC computes a maximum THR 15 % lower, so the engine gives less thrust above idle (N1 command = IDLE + 0.85 x (limit - IDLE)).
- **ENG 1(2)(3)(4) THRUST LOSS** (amber, phase 3 only): the maximum THR of an engine is 10 % lower than that of at least two other engines. Lines: THR LEVER n ... IDLE, ENG n MASTER ... OFF.
- New flyPad failures, ATA 73: **FADEC 1 (2, 3, 4) FLEX TEMP not received (T.O mode TOGA)** (73060-73063): that FADEC takes off in TOGA mode (MCT thrust in the FLX/MCT detent).
- **ENG T.O THRUST DISAGREE** (amber, phases 2 and 11): the FADECs do not all have the same take-off mode. No line (crew awareness).

Limitations and why

Limitation	Why	What to do
THRUST LOSS shows in phase 3 only.	FCOM inhibition; the A380X FWS needs two levers at take-off power for phase 3.	Set two levers (for example 3 and 4) to FLX/MCT.
Use FADEC 2, 3 or 4 for the FLEX TEMP failure.	The E/WD thrust rating is the one of FADEC 1.	-
A full MSFS restart is needed after an update.	The fbw WASM module changes.	Restart the sim after installing.

How to use

1. flyPad > Failures > ATA 73 Engine - Fuel And Control: activate one of the failures before take-off.
2. Follow the caution; clear the failure to end it.

How to test

#	Do this	Expected result
1	A380X on the ground, PARKING BRK ON, engines running at idle, all levers at IDLE, FLEX TEMP 60 on the MFD FMS ACTIVE/PERF T.O page. flyPad > Failures > ATA 73: activate **FADEC 4 FLEX TEMP not received (T.O mode TOGA)**.	Amber ENG T.O THRUST DISAGREE (no line), single chime, MASTER CAUT.
2	Clear the failure.	The caution goes.
3	Activate **Engine 4 max thrust miscalculated (thrust loss)**; thrust levers 3 and 4 to FLX/MCT, hold 5 s.	Phase 3: amber ENG 4 THRUST LOSS with THR LEVER 4 ... IDLE and ENG 4 MASTER ... OFF (a red CONFIG PARK BRK ON also shows: normal with take-off power and the parking brake).
4	Levers back to IDLE, clear the failure.	The cautions go.

18. E/WD N1 red above 111 % with a red cross

> **Aircraft:** A380X | **Where:** E/WD N1 indication | **Status:** Not sim-tested. Unit .spec.ts.

What it is for

FCOM DSC-70-90 N1 INDICATIONS: the N1 value turns red above the 111 % red limit, and a red cross stays next to it after an exceedance until the next engine start on the ground. The A380X E/WD N1 stayed green.

Limitations and why

Limitation	Why	What to do
Hard to see in the sim.	The A380X has no N1 overspeed failure; the N1 rarely passes 111 %.	-

How to use

1. Nothing to set. After an N1 exceedance the red cross stays until the next engine start on the ground.

How to test

#	Do this	Expected result
1	A380X, engines started on the ground, normal take-off thrust.	N1 values green, no red cross.
2	Run `npx vitest run fbw-a380x/src/systems/instruments/src/EWD/elements/N1.spec.ts`.	All tests pass (red above 111 %, the cross latched until the next ground start).

19. Crossbleed engine start with X BLEED at AUTO

> **Aircraft:** A380X | **Where:** Overhead AIR panel (X BLEED selector, APU BLEED and ENG BLEED pbs); ENG MASTER levers; SD BLEED page | **Status:** Tested in the sim by the user on a combined test build: OK, checked in the recorder log (FCOM card tests X1 and X2). Rust tests (pneumatic.rs, engine_out_aircraft_tests.rs). The core branch itself was not sim-tested.

What it is for

A380 FCOM DSC-36-10-50 : the crossbleed valves "are normally closed. They open automatically, and at the same time: On ground, at engine start; When the APU bleed valve is selected open; In case any engine bleed system fails." The first case was missing: with the X BLEED selector at AUTO the valves stayed closed during an engine start, so the crossbleed engine start of PRO-SUP-70-10 (ALL ENG BLEEDS ON, X BLEED AUTO, SECOND ENG START) had no starter air without the APU. Now the three crossbleed valves open together on the ground while the FADEC commands the start valve of any engine open, and the running engines supply the starter air.

Limitations and why

Limitation	Why	What to do
"At engine start" is from the ENG MASTER ON (or MAN START ON, crank) to the start valve closing at 58.4 % N3 or the end of the start.	The FCOM gives no opening or closing point (design choice).	-
On the ground only.	FCOM rule. In flight a starter assisted relight without the APU still needs X BLEED OPEN (engine failures section).	Set X BLEED OPEN in flight.

An engine bleed failure is modelled as an ENG BLEED pb OFF.	There is no engine bleed failure with the pb ON yet.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Start the first engine with APU air as usual.
2. Set the APU BLEED pb OFF, keep all ENG BLEED pbs ON and the X BLEED selector at AUTO.
3. ENG START selector IGN START, then ENG MASTER ON for the next engine: the crossbleed valves open by themselves and close again after the start.

How to test

Setup: on the ground, PARKING BRK ON, APU running with the APU BLEED pb ON, engine 1 started and at idle, ENG MASTER 2, 3 and 4 OFF, ENG START selector IGN START, all four ENG BLEED pbs ON, X BLEED selector AUTO.

#	Do this	Expected result
1	ECAM control panel: BLEED key. APU BLEED pb OFF; PACK 1 and PACK 2 pbs OFF; wait 10 s.	SD BLEED page: the three crossbleed valves closed.
2	ENG MASTER 2 ON; wait until engine 2 shows AVAIL (1 to 2 min).	Within 2 s the three crossbleed valves open; engine 2 N3 rises; no ENG 2 START FAULT. After the start the crossbleed valves are closed again.
3	ENG MASTER 3 and ENG MASTER 4 ON together; wait until both are AVAIL. PACK 1 and 2 ON.	The crossbleed valves open within 2 s, engines 3 and 4 start without START FAULT, the valves close again.

20. Engine alerts as in the FCOM

> **Aircraft:** A380X | **Where:** E/WD, STATUS, MASTER CAUT lights; SD ENGINE page | **Status:** Tested in the sim by the user on a combined test build: OK (FCOM card tests A1 IGN / OIL FILTER levels and A2 START FAULT; A3 EGT OVER LIMIT with FADEC FAULT). The in-flight START FAULT cases and the deferred procedures: not sim-tested (unit tests EngineFailAlerts, EngineStartAlerts, EngineParameterAlerts, EngineOilAlerts, EngineFuelFilterAlerts specs; Rust tests). The core branch itself was not sim-tested.

What it is for

The engine alerts of this branch are aligned with A380 FCOM PRO-ABN-ECAM-10-70 and its figures (levels, inhibitions, SD pages, lines):

- **ENG n FAIL** is no longer inhibited from V1 to 1500 ft (the FCOM figure has no inhibition). Its line AUTO RELIGHT IN PROGRESS shows only while the FADEC auto relight runs.
- The **IGNITION** memo also shows for the continuous ignition the FADEC selects by itself (auto relight, quick relight).
- **ENG n START FAULT**, **IGN A (B) FAULT**, **OIL FILTER CLOGGED** and **FUEL FILTER CLOGGED** come without chime and without MASTER CAUT (crew awareness). IGN A+B FAULT keeps the single chime and MASTER CAUT.
- **ENG n EGT OVER LIMIT** and **N1/N2 OVER LIMIT** call no SD page. EGT OVER LIMIT does not come up with an ENG n FADEC FAULT (FCOM ENG FADEC FAULT: the EGT OVER LIMIT alert will not trigger in that case).
- In flight, **START FAULT NO STARTER AIR PRESSURE** comes up with its windmill start lines (WINDMILL START INITIATED, ENG PARAMETERS MONITOR, IF ABNORMAL: ENG n MAN START OFF, ENG n MASTER OFF), and **NO LIGHT UP** comes after the 30 s of the relight (ENG RELIGHT IN FLIGHT), not after the 20 s of the ground start. The shared engine start schedule got two fields for this; the A32NX keeps its values.

- New deferred procedures: **ENG TWO ENGS OUT** (ALL PHASES: IF GW DECREASE RQRD, JETTISON ARM ON, JETTISON ACTIVE ON), **TWO ENGS OUT ON SAME SIDE** (FOR APPROACH) and **ON OPPOSITE SIDE** (APPR & LDG), shown under their heading on the STATUS page. **ENG n START VLV FAULT (NOT CLOSED)** gets its LIMITATIONS, INOP SYS and, for engine 1 or 4, the FOR APPROACH deferred procedure IF SEVERE ICE ACCRETION: FOR LDG : FLAP LVR 3.

> **NOTE**
> On this branch the JETTISON ARM / ACTIVE lines of the TWO ENGS OUT deferred procedures read the FUEL JETTISON pbs directly; with `core/fuel` merged they read its jettison FWS states.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed after an update.	The systems WASM module changes (engine start schedule).	Restart the sim after installing.

How to use

1. Nothing to set: follow the E/WD as usual. An alert without chime or MASTER CAUT is a crew awareness alert.
2. With two engines out, check the deferred procedures on the STATUS page before the approach.

How to test

#	Do this	Expected result
1	On the ground, PARKING BRK ON, four engines at idle, no alert. flyPad > Failures > ATA 74 > "Engine 1 igniter A": activate, wait 10 s.	Amber ENG 1 IGN A FAULT, no single chime, both MASTER CAUT lights off.
2	ATA 79 > "Engine 1 oil filter clog": activate, wait 10 s.	Amber ENG 1 OIL FILTER CLOGGED, no chime, no MASTER CAUT.
3	ATA 74 > "Engine 1 igniter B": activate.	Amber ENG 1 IGN A+B FAULT with a single chime and both MASTER CAUT lights. Press a MASTER CAUT pb and deactivate the three failures.
4	APU BLEED ON, engine 1 running, ENG MASTER 2 OFF, ENG START IGN START. ATA 80 > "Engine 2 hot start (start EGT over limit)": activate; ENG MASTER 2 ON; wait about 60 s.	Amber ENG 2 START FAULT with EGT OVERLIMIT, no chime, MASTER CAUT off. ENG MASTER 2 OFF, deactivate the failure.
5	In flight above FL100, A/THR ON, levers in CL. ATA 73 > "Engine 1 FADEC network link": activate; press a MASTER CAUT pb. ATA 72 > "Engine 1 EGT overtemperature": activate; wait 60 s.	ENG 1 FADEC FAULT with chime and MASTER CAUT; then the engine 1 EGT rises and turns amber, but no ENG 1 EGT OVER LIMIT (no new chime). Deactivate both.

21. Braking and steering control failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 32 Landing Gear; brake pedals, tiller, PARK BRK; E/WD, STATUS; triple brake pressure indicator | **Status:** Recorded as passed in the commit message (combined test build); the test notes have no separate entry for it. Unit tests FwsBrakeAlerts.spec.ts, a380.spec.ts; Rust tests. The core branch itself was not sim-tested.

What it is for

Eight new failures of the braking control system (BCS), the emergency brake control unit (EBCU) and the steering control system (SCS), with the BRAKES and STEER alerts of A380 FCOM PRO-ABN-ECAM-10-32. The Rust hydraulics choose the braking mode the faults leave (DSC-32-10-30) and the nose wheel steering supply (DSC-32-10-40).

flyPad failure (ATA 32)	What it does	ECAM
Brakes normal braking (BCS NORM BRK)	The pedals brake in alternate braking (ALTN circuit, A-SKID available).	BRAKES NORM BRK FAULT
Brakes alternate braking (BCS ALTN BRK)	Alone: no effect while normal braking works. With the normal braking failed: emergency braking through the EBCU (no A-SKID, pressure limited).	BRAKES ALTN BRK FAULT; with the EBCU failed too: ALTN + EMER BRK FAULT
Brakes emergency braking (EBCU)	The EBCU is lost; with normal and alternate braking failed too, no pedal braking: the parking brake is the only braking.	BRAKES EMER BRK FAULT; all three: BRAKES PEDAL BRAKING FAULT (replaces the single alerts)
Steering normal nose wheel steering (NORM N/W STEER)	The nose wheel steers on the alternate system (steering LEHGS), at most 10 deg/s instead of 15 deg/s.	STEER NORM N/W STEER FAULT
Steering alternate nose wheel steering (steering LEHGS)	No alternate NWS: with the normal NWS also lost, the nose wheel casters.	STEER ALTN N/W STEER FAULT
Steering body wheel steering (B/W STEER)	The body wheel steering stays centred and locked.	STEER B/W STEER FAULT (crew awareness)
Steering control system 1 (STEER CTL 1) / 2 (STEER CTL 2)	One SCS is enough (the backup takes over); with both lost neither NWS nor BWS works.	STEER CTL 1 (2) FAULT; both: STEER N/W + B/W STEER FAULT

Also in this feature

- **BRAKES ACCU PRESS LO** (accumulator below 3 000 psi for 10 s, design value) and **PARK BRK PRESS LO** (park brake on and pressure below 1 700 psi for 5 s, design value).
- The second B/W STEER FAULT text is renamed **B/W STEER NOT LOCKED**, as in the FCOM.
- With only engines 3 and 4 running (no GREEN), the nose wheel now steers on the alternate system instead of not at all, as the FCOM asks.

Limitations and why

Limitation	Why	What to do
A brake fault acts on all wheel groups at once.	The model has one NORM circuit (GREEN) and one ALTN circuit (YELLOW, one accumulator) for all wheel groups; the real A380 splits them per gear.	-
The nose wheel steering request keeps the A-SKID condition of this base.	The change that drops it is on `core/fcom-audit-fixes`.	Merge that branch for it.
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 32 Landing Gear and activate a brake or steering failure on the ground.
2. Follow the E/WD; brake with the pedals (watch the triple brake pressure indicator) and steer with the tiller.
3. Deactivate the failure.

How to test

Setup: on the ground, four engines running, park brake OFF, below 10 kt.

#	Do this	Expected result
1	Activate "Steering normal nose wheel steering (NORM N/W STEER)"; turn the tiller fully left.	Amber STEER NORM N/W STEER FAULT with a chime; the nose wheel turns visibly slower.
2	Deactivate it; stop. Activate "Brakes normal braking (BCS NORM BRK)"; press both brake pedals fully.	BRAKES NORM BRK FAULT (no chime); triple indicator L/R about 2 500 psi.
3	Also activate "Brakes alternate braking (BCS ALTN BRK)"; press both pedals.	BRAKES ALTN BRK FAULT too; L/R about 1 160 psi (emergency braking).
4	Deactivate both; PARK BRK ON.	The alerts go; normal braking.

22. Erroneous ADR airspeed and AOA failures, PRIM air data monitoring

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 34 Navigation; PFD; E/WD, STATUS | **Status:** Tested in the sim by the user on a combined test build: OK (flight: one, then two ADRs wrong; the AP could not be re-engaged afterwards, as the FCOM says: the FD needs two ADRs). The AOA failures: not sim-tested. Unit tests FwsAirDataAlerts.spec.ts, FwsLimitations.spec.ts, native air_data_rejection_test, Rust tests. The core branch itself was not sim-tested.

What it is for

Six new failures: **ADR 1 (2)(3) airspeed erroneous (pitot)** and **ADR 1 (2)(3) angle of attack erroneous**. The ADR then sends a wrong value with a valid status (airspeed 40 kt low, AOA 10 deg high; design values above the FCOM monitoring thresholds), which the PRIM model could never reject, as it only votes on the status.

A new systems module compares the three ADRs with the standby air data as the PRIMs do (A380 FCOM DSC-34-10-30-10 / -40) and rejects a diverging source until the end of the flight, or all of them when it cannot tell which one is wrong. A rejected ADR goes to the PRIM and SEC models as failed: alternate law with two ADRs out, direct law with all of them.

Alert (PRO-ABN-ECAM-10-34-10)	When	Lines
NAV AIR DATA DISAGREE	One or more sources rejected (not all).	ADR n (or STBY INSTRUMENTS) REJECTED BY PRIMs, RISK OF UNDUE OVERSPEED WARNING (two sources), AIR DATA XCHECK, AIR DATA SWTG AS RQRD.
NAV ALL AIR DATA DISAGREE	All air data rejected.	ALL AIR DATA REJECTED BY PRIMs, the thrust and pitch target lines of the FCOM tables for the altitude, FLAPS & L/G MAINTAIN CURRENT CONF, ... FLY THE GREEN, FLY PITCH AND THRUST.
NAV AOA DISAGREE	Two or more AOAs rejected (one alone gives no NAV alert).	RISK OF UNDUE STALL WARNING.

Also: F/CTL DIRECT LAW MAX SPEED 310/.86 and MANEUVER WITH CARE are now APPR & LDG limitations, as in the FCOM (PRO-ABN-ECAM-10-27-10); a limitation shown in ALL PHASES by another alert is not repeated in APPR & LDG.

Limitations and why

Limitation	Why	What to do
A rejected ADR stays rejected until the end of the flight, also after the failure is cleared.	As the PRIMs do (FCOM).	A new flight.

The cabin altitude table, the MFP heating lines and the level-off thrust table above FL 250 of ALL AIR DATA DISAGREE are not shown.	Not modelled.	-
A full MSFS restart is needed after an update.	The systems and fbw WASM modules change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 34 Navigation and activate an ADR airspeed or AOA failure in flight.
2. Follow the E/WD (cross-check the air data, switch the AIR DATA source as required).
3. Deactivate the failure; the PRIMs keep the ADR rejected for the rest of the flight.

How to test

Setup: cruise above FL250, AP ON.

#	Do this	Expected result
1	Activate "ADR 1 airspeed erroneous (pitot)"; wait 5 s.	The CAPT PFD speed reads 40 kt low; amber NAV AIR DATA DISAGREE with ADR 1 REJECTED BY PRIMs; normal law stays.
2	Activate "ADR 2 airspeed erroneous (pitot)"; wait 5 s.	F/CTL ALTN LAW (PROT LOST); the AP disconnects.
3	Deactivate both.	The ADRs stay rejected until the end of the flight (normal).
4	New flight, cruise. Activate "ADR 1 angle of attack erroneous" and "ADR 2 angle of attack erroneous".	Amber NAV AOA DISAGREE with RISK OF UNDUE STALL WARNING.

23. FCOM : RCL / RCL LAST, T.O CONFIG NORMAL, automatic SD pages, BTV INOP

> **Aircraft:** A380X | **Where:** ECAM control panel (RCL, RCL LAST, T.O CONFIG); E/WD; SD; auto brake | **Status:** Recorded as passed in the commit message on a combined test build (tests N3 T.O CONFIG NORMAL, N4 RCL / RCL LAST, N6 automatic SD pages, sessions; the session notes give N6 CRUISE at the first take-off). BTV INOP: not sim-tested. Unit tests FwsRecall.spec.ts, ToConfigNormal.spec.ts, FwsSystemDisplayLogic.spec.ts; Rust autobrake tests. The core branch itself was not sim-tested.

What it is for

- **RCL LAST** brings back the alert of the last CLR action (both pbs wrote the same variable before). **RCL** brings back every cleared alert at once, or NORMAL for 3 s when there is none; only a press of more than 3 s shows the alerts inhibited by the flight phase, so a short press no longer removes the T.O INHIBIT memo for the flight (A380 FCOM DSC-31-40-20).
- **T.O CONFIG NORMAL** follows the A380 checks of DSC-31-40-10 (the FCOM list of T.O CONFIG alerts: surfaces and trims, doors, BRAKES HOT, the generators and drives of the four engines, engine oil temperature, PITCH TRIM/FMS/CG and T.O FLAPS/FMS DISAGREE, T.O speeds) instead of the A320 GEN 1+2 / two doors condition.
- **Automatic SD pages** (DSC-31-40-30): CRUISE from 1500 ft after take-off (it came 60 s after the flaps were up), WHEEL at the landing gear extension at any altitude (only below 16000 ft before).
- **BTV INOP**: while an ECAM alert lists BTV in STATUS INOP SYS (ENG 2 (3) SHUT DOWN, TWO ENGS OUT, ENG 2 (3) REVERSER faults), BTV does not arm, and an armed BTV reverts to BRK HI (DSC-32-20-220, LIM-32).

Limitations and why

Limitation	Why	What to do
T.O CONFIG NORMAL does not check the ELEC GEN / DRIVE alerts on this branch.	Those alerts are on `core/a380x-elec`.	Use `core/a380x-elec-fuel-alerts` (both branches merged).
CONFIG L (R) SIDESTICK FAULT (BY TAKE-OVER) is not checked.	The sim has no sidestick take-over fault.	-
A full MSFS restart is needed after an update.	The systems WASM module (autobrakes) and the ECAM control panel XML change.	Restart the sim after installing.

How to use

1. RCL LAST: bring back the last cleared alert. RCL: bring back all cleared alerts; hold it more than 3 s to see the alerts inhibited by the flight phase.
2. Before take-off press T.O CONFIG: T.O CONFIG NORMAL shows when no alert of the FCOM list is active.
3. The SD changes page by itself; with an alert that lists BTV INOP, select another auto brake mode.

How to test

Adapted from tests N3, N4 and N6 : the card used the generator failures, whose ELEC alerts are not on this branch; the feed pump failures of this branch give two cautions instead.

#	Do this	Expected result
1	On the ground, engines running and warm, T.O speeds entered, pitch trim in range, doors closed, flaps 0. ECAM control panel: T.O CONFIG key.	A CONFIG alert for the flaps; no T.O CONFIG NORMAL.
2	Set the take-off flaps of the FMS; T.O CONFIG key again.	T.O CONFIG NORMAL.
3	flyPad > Failures > ATA 28: "Feed tank 1 main pump", then "Feed tank 2 main pump". CLR until both cautions are cleared. Press RCL LAST.	Only the last cleared caution comes back.
4	CLR. Press RCL (short).	Both cautions come back at once.
5	CLR both, deactivate both failures, wait 5 s, press RCL.	NORMAL for 3 s.
6	Take off with the SD on its automatic page.	The T.O INHIBIT memo still shows at take-off after the RCL presses; the SD shows CRUISE as soon as passing 1500 ft.
7	On the approach, extend the landing gear.	The SD shows WHEEL at once, also above 16000 ft.
8	Optional, on landing with BTV armed: set ENG MASTER 2 OFF in flight before.	STATUS INOP SYS BTV; BTV does not arm (an armed BTV reverts to BRK HI).

A32NX flyPad failures

Clearer A32NX failure names and new failures with their FCOM alerts: transponders, RMPs and VHF radios, FWC 1 and 2, engine and APU fires with a real fire detection model and fire detection unit failures, wing tank pumps, the X FEED valve, the centre tank transfer valves, engine flameouts and seizures with the in-flight relight envelope, and the engine oil, ignition and start, stall, EGT, overspeed, vibration, FADEC, thrust lever and reverser failures with the FCOM start sequence, the fan blocked failure with ENG LOW N1, and the THR LEVER ABV IDLE and SAT ABOVE FLEX TEMP alerts; then engine and APU fire visual effects, the electrical (battery, BCL, IDG) and pneumatic (bleed valves, leaks, BMC) failures, the FCOM ELEC, AIR bleed, HYD, APU shutdown and F/CTL alerts fixes (FADEC relight, automatic ignition, gravity feed, IR alignment), the DC ESS supply in DC BUS 1+2 fault and the LEAP-1A N1 / N2 red lines; and the crew recovery procedures after failures (FAC reset, AC BUS 1 transfer, SEC test, IDG, manual cabin pressure, ignition indications, engine fire run-on, A/THR without V2); then the A32NX parts (FAC actuator, ADR / IR, brake control unit, cargo and avionics smoke, APU ECB, SFCC channel, HP fuel valve, FADEC fault, LP valve, FQI, engine anti-ice valve, probe and window heat and TCAS computer failures with their alerts; alerts on existing data, PFD FMA messages, SD advisory mode, CVR power, CPC landing elevation, BLEED page valves, gravity feed ceiling, GPWS self-test, reactive windshear, FM/GPS and LS tuning disagree), the A380X smoke detection function, the sidestick priority / intermediate RA / WINDSHEAR voices, the exceedance marks fix and the 5 s DC BUS 1 transfer

Aircraft	A32NX (the fire visual effects, the smoke detection function and the anti-ice systems part also A380X)
Where to find it	flyPad > Failures (ATA 21, 22, 23, 24, 26, 27, 28, 29, 30, 31, 32, 34, 36, 49, 72, 73, 74, 77, 78, 79, 80); flyPad > Settings > Sim Options; E/WD, STATUS and SD pages; PFD FMA; pedestal ATC and RMP panels; overhead FIRE, CARGO SMOKE, FUEL, ELEC, AIR COND, HYD, ADIRS, ANTI ICE, RCDR and ENG MAN START panels; glareshield GPWS pb; A/SKID & N/W STRG and autobrake; ENG MASTER and ENG MODE controls; thrust levers; exterior model
Branch	`core/failures-a32nx`, based on `core/gps` (stacked: needs core/surveillance, core/a380x-mfd-fms and core/gps merged first), with `core/fire` merged in (the fire failures use its fire warnings, and the engine failures use its ENG FIRE pb LP valve starvation). Since it also merges `core/fcom-audit-fixes` and `core/a32nx-air-conditioning`: the new FWC alerts build on their code (F/CTL law lines, APU memos, GPWS lamp test, air conditioning STATUS lines). Merge them into this branch when they change. Only the features below belong to this branch; the GPS, fire warning, FCOM audit and air conditioning features are described in the core/gps, core/fire, core/fcom-audit-fixes and core/a32nx-air-conditioning manuals. Some engine commits also carry A380X files (they are on `core/failures-a380x`); this manual describes only what the A32NX gets, except the engine and APU fire visual effects, whose A380X part is on this branch too (the fire model of both aircraft lives here). It now also carries the A380X smoke detection function (SDF) and the A380X anti-ice systems and cockpit part, whose A380X alerts are on `core/a380x-elec-fuel-alerts`: merge both.
Status	Transponder, RMP/VHF, FWC, fire and fuel failures tested in the sim by the user: OK; the fire detection unit failures, the FIRE TEST light logic, the APU fire horn and the centre tank transfer valve failures too (the horn itself is silent in MSFS 2024); the engine flameout and seizure failures: OK, tests; their LEAP-1A relight envelope: sim-tested OK (recorded); the in-flight start without N2 overshoot: OK (recorded). Engine failures A2-A5 (test flight and retest, recorded): oil failures, ground starts with a failed igniter and the dry crank, stall, EGT overtemperature and vibration, thrust lever resolvers and FADEC FAULT, normal reverse thrust: OK; the other engine failures, the later tuning and the T.O CONFIG NORMAL / WING TK LO LVL fixes: not sim-tested. Clearer names: not sim-tested (labels only). Engine alerts (feature branch): SAT ABOVE FLEX TEMP OK (recorded); THR LEVER ABV IDLE and the fan blocked / LOW N1 OK (recorded). The fuel filter clog and the stall during a relight fix: not sim-tested. Later commits, tested by the user on a combined test build (recorded):

	fire visual effects OK; cabin pressure fixes OK; FADEC relight and the 10 min IR alignment OK, but the automatic continuous ignition FAILED (no IGNITION memo); ELEC alerts OK; electrical failures OK; pneumatic failures OK; AIR bleed (BLEED 1 OFF part) OK; HYD and APU alerts OK; F/CTL alerts OK; DC ESS supply OK; LEAP-1A red lines OK. Nothing is sim-tested on this branch itself. Recovery procedures (checked against the code and the FCOM): FAC reset OK, manual cabin V/S OK, the others per the two last sections. The later FCOM alert additions: not sim-tested; unit/Rust tests only.
Pictures	No pictures: the failures page, the ECAM alerts and the panels are the stock FBW displays.

1. Clearer failure names
2. Transponder failures
3. RMP and VHF failures
4. FWC 1 and FWC 2 failures
5. Fire detection model and fire failures
6. Fire detection unit failures, FIRE TEST light logic, APU fire horn
7. Wing tank pump and X FEED valve failures
8. Centre tank transfer valve failures and fuel follow-ups
9. Engine flameout and seizure failures, in-flight relight
10. Engine oil failures: leak, filter clog, overheat
11. Engine fuel filter clog
12. Engine ignition and start failures, FCOM start sequence
13. Engine stall, EGT overtemperature, overspeed and vibration failures
14. FADEC, thrust lever and reverser failures
15. T.O CONFIG NORMAL and FUEL WING TK LO LVL fixes
16. ENG THR LEVER ABV IDLE and ENG SAT ABOVE FLEX TEMP
17. Engine fan blocked failure and ENG LOW N1
18. Engine and APU fire visual effects
19. FCOM fixes: DIRECT LAW trim, GPWS mode 2, CAB PRESS and oil temperature indications
20. FCOM fixes: FADEC relight, automatic ignition, start faults, gravity feed, IR alignment
21. ELEC bus, ESS bus and TR alerts; DC ESS supply in DC BUS 1+2 fault
22. Electrical failures: battery thermal runaway, BCL, IDG oil
23. Pneumatic failures with BMC leak and bleed monitoring
24. AIR bleed cautions
25. HYD and APU shutdown alerts
26. F/CTL surface cautions and STATUS
27. FAC actuator failures, YAW DAMPER / RUD TRIM / RUD TRV LIM cautions
28. ADR and IR failures, ADIRU power supplies, NAV ADR / IR / RA alerts
29. alerts on existing data (CAB PR, THRUST LOCKED, fuel, WAI, doors, L/G, GPWS)
30. PFD FMA messages, AVAIL, SD FUEL imbalance, SD BLEED crossbleed valve
31. SD advisory mode
32. CVR automatic power and CVR TEST
33. CPC landing elevation without FMGS data, EXCESS CAB ALT threshold
34. brake and steering control units (BSCU, ABCU), BRAKES alerts
35. APU automatic shutdowns, cargo and avionics smoke, cargo extinguishing
36. A380X smoke detection function (systems part)
37. BLEED page HP valve and bleed valve disagree
38. HP fuel valve, FADEC fault failures, pre-start motoring
39. gravity feed ceiling, LP valve and FQI failures
40. engine anti-ice valves, probe and window heat, anti-ice key events
41. SFCC flap and slat channel failures
42. GPWS self-test, TCAS computer failure, reactive windshear
43. NAV FM/GPS POS DISAGREE and NAV LS TUNING DISAGREE
44. Voices: sidestick priority, intermediate RA callouts, WINDSHEAR
45. Engine exceedance marks only on displayed values
46. DC BUS 2 takes over DC BUS 1 after 5 s
47. Recovery after electrical failures: FAC reset, AC BUS 1 transfer, SEC test, IDG
48. Crew procedures: manual cabin pressure, ignition indications, engine fire, A/THR without V2

1. Clearer failure names

> **Aircraft:** A32NX | **Where:** flyPad > Failures | **Status:** Not sim-tested (labels only).

What it is for

The failure names were audited ATA by ATA against what the systems code fails. Three names were misleading: the ATA 21 valve is the hot air pressure regulating valve, the FMGC failures only fail the flight guidance part of the FMGC, and the RA failures are antenna failures. Only the labels change.

ATA	Before	Now
21	Trim Air Pressure Regulating Valve	Hot Air Pressure Regulating Valve
22	FMGC 1 / FMGC 2	FMGC 1 (flight guidance) / FMGC 2 (flight guidance)

34	RA 1 Interrupted / RA 2 Interrupted	RA 1 antenna interrupted / RA 2 antenna interrupted
34	RA 1 Direct Coupling / RA 2 Direct Coupling	RA 1 antenna direct coupling / RA 2 antenna direct coupling

How to use

1. Open flyPad > Failures and pick the failure by its new name (table above).

How to test

#	Do this	Expected result
1	Open flyPad > Failures > ATA 21 Air Conditioning.	"Hot Air Pressure Regulating Valve" is listed.
2	Open ATA 22 Auto Flight.	"FMGC 1 (flight guidance)" and "FMGC 2 (flight guidance)".
3	Open ATA 34 Navigation.	"RA 1 antenna interrupted", "RA 2 antenna interrupted", "RA 1 antenna direct coupling", "RA 2 antenna direct coupling".

2. Transponder failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 34 Navigation; pedestal ATC/TCAS panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK.

What it is for

Two new failures, **ATC/XPDR 1** and **ATC/XPDR 2**. MSFS has a single transponder: it stands for the transponder selected on the ATC panel (SYS 1 / SYS 2). When the selected one fails, the sim transponder stops replying and the ATC FAIL light comes on; selecting the other system brings the replies back. With both lost, the TCAS goes to standby. The ECAM alerts and STATUS follow the A320 FCOM PRO-ABN-NAV procedures.

Fix on the way: the ATC/XPDR STBY indication read a second sim transponder that does not exist; the standby state of each transponder now comes from the system selection and the mode selector.

What you should see

Situation	What you should see
ATC/XPDR 1 failed, SYS 1 selected	ATC FAIL light; no transponder reply. E/WD amber **NAV ATC/XPDR 1 FAULT** with -ATC/XPDR ..... SYS 2 (only while XPDR 1 is selected and XPDR 2 works). STATUS INOP SYS: ATC/XPDR 1, ADS-B RPTG 1.
SYS 2 selected	ATC FAIL off, replies again; the procedure line is done.
ATC/XPDR 2 failed	The same with NAV ATC/XPDR 2 FAULT, -ATC/XPDR ..... SYS 1, INOP SYS ATC/XPDR 2, ADS-B RPTG 2.
Both failed	**NAV ATC/XPDR 1+2 FAULT** (crew awareness); STATUS INOP SYS: TCAS, ATC/XPDR 1, ATC/XPDR 2, ADS-B RPTG 1, ADS-B RPTG 2; the TCAS goes to standby.
A transponder loses its bus (AC ESS SHED for XPDR 1, AC 2 for XPDR 2)	The transponder is lost, but no ATC/XPDR FAULT: a bus loss is an ELEC alert.

Limitations and why

Limitation	Why	What to do

The ATC/XPDR alerts are inhibited during the takeoff and below 800 ft in approach to the landing roll.	The FCOM gives the inhibition as a figure: the one of the RA faults is used (flight phases 3, 4, 5, 7, 8).	Test at the gate or in cruise.

How to use

1. Open flyPad > Failures > ATA 34 Navigation and activate **ATC/XPDR 1** or **ATC/XPDR 2**.
2. Follow the ECAM: select the other system on the ATC/TCAS panel.

How to test

#	Do this	Expected result
1	A32NX at the gate, powered, ATC/TCAS panel: SYS 1, mode AUTO (or ON).	No ATC FAIL light.
2	Activate **ATC/XPDR 1**.	ATC FAIL light on; master caution and single chime; E/WD NAV ATC/XPDR 1 FAULT with -ATC/XPDR ..... SYS 2. ATC (or a traffic tool) no longer sees your transponder.
3	Open the STATUS page.	INOP SYS: ATC/XPDR 1, ADS-B RPTG 1.
4	Select SYS 2 on the ATC panel.	ATC FAIL off, the transponder replies again, the SYS 2 line is done.
5	Activate **ATC/XPDR 2** as well.	NAV ATC/XPDR 1+2 FAULT; STATUS INOP SYS: TCAS, ATC/XPDR 1, ATC/XPDR 2, ADS-B RPTG 1, ADS-B RPTG 2; the TCAS is in standby.
6	Deactivate both failures.	The alerts clear; the TCAS works again.

3. RMP and VHF failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 23 Communication; pedestal RMP 1 and RMP 2 | **Status:** Tested in the sim by the user: OK.

What it is for

Five new ATA 23 failures: **RMP 1**, **RMP 2**, **VHF 1**, **VHF 2** and **VHF 3** (same ids as the A380X ATA 23 failures). A failed RMP goes blank and stops tuning; the other RMP still tunes every VHF (A320 FCOM DSC-23-10-20: "If one RMP fails, the remaining one controls all the VHF transceivers"). A failed VHF switches its MSFS COM radio off: it neither receives nor transmits.

What you should see

Situation	What you should see
RMP 1 (or 2) failed	Both frequency displays blank, the green key lights and SEL off, the keys and knobs do nothing (FCOM DSC-23-60 failure cases).
The other RMP	Still tunes VHF 1, 2 and 3.
VHF 1 (2, 3) failed	COM 1 (2, 3) silent: no reception, no transmission. The RMPs still show and change its frequency.

Limitations and why

Limitation	Why	What to do
No ECAM alert for these failures.	Not part of this change.	Check the RMP and the radio.
A NAV key left on on a failed RMP still holds the FMGC navaid tuning.	As with an RMP switched off: the navaid tuner (shared with the A380X) reads the NAV key only.	Avoid the NAV key before failing an RMP.
No RMP 3 and no audio control panel / AMU failure.	RMP 3 is not in the A32NX cockpit, and the MSFS radios have no audio channel to fail.	None.

How to use

1. Open flyPad > Failures > ATA 23 Communication and activate RMP 1, RMP 2, VHF 1, VHF 2 or VHF 3.
2. With an RMP failed, tune the radios from the other RMP.

How to test

#	Do this	Expected result
1	A32NX powered, an ATIS frequency on VHF 1, audible.	The ATIS is heard.
2	Activate **RMP 1**.	RMP 1 goes blank, its key lights go off, its knobs do nothing. The ATIS is still heard (VHF 1 itself works).
3	On RMP 2, press the VHF 1 key and tune another frequency, transfer it.	VHF 1 changes frequency: RMP 2 tunes VHF 1.
4	Deactivate RMP 1. Activate **VHF 1**.	COM 1 goes silent; transmitting on VHF 1 is not possible. RMP 1 still shows the frequency.
5	Deactivate VHF 1.	The ATIS is heard again.

4. FWC 1 and FWC 2 failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 31 Indicating / Recording System; E/WD, STATUS | **Status:** Tested in the sim by the user: OK.

What it is for

Two new failures, **FWC 1** and **FWC 2** (A320 FCOM DSC-31-05-30, PRO-ABN-FWS). With one FWC failed, the other one keeps the warning system working and shows the fault. With both failed, the ECAM alerts, memos, STATUS, MASTER WARN / CAUT lights, aurals (including the STALL aural) and the altitude callouts are lost, and the E/WD shows the FWC 1+2 FAULT display. Each FWC also follows its own electrical supply.

What you should see

Situation	What you should see
FWC 1 failed	E/WD **FWS FWC 1 FAULT** (shown by FWC 2), level 1: no MASTER CAUT, no chime. STATUS: CAT 3 SINGLE ONLY; INOP SYS: CAT 3 DUAL, FWC 1.
FWC 2 failed	The same with FWS FWC 2 FAULT and FWC 2.
Both failed	E/WD: **FWS FWC 1+2 FAULT**, -MONITOR SYS, -MONITOR OVERHEAD PANEL; right side NOT AVAIL: ECAM WARN, ALTI ALERT, STATUS, A/CALL OUT, MEMO. No other alert, memo, STATUS, STS reminder, master light or aural.
One FWC restored	Every active alert is shown again, as a new one.

Limitations and why

Limitation	Why	What to do
FWS FWC 1(2) FAULT is level 1 and inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the level and the inhibition in figures only: those of the other computer faults are used.	Test at the gate or in cruise.

How to use

1. Open flyPad > Failures > ATA 31 Indicating / Recording System and activate FWC 1 and / or FWC 2.
2. With both failed, monitor the systems and the overhead panel yourself.

How to test

#	Do this	Expected result
1	A32NX at the gate, powered, no alert. Activate **FWC 1**.	E/WD: FWS FWC 1 FAULT, no master caution, no chime.
2	Open the STATUS page.	CAT 3 SINGLE ONLY; INOP SYS: CAT 3 DUAL, FWC 1.
3	Activate **FWC 2** as well.	E/WD: FWS FWC 1+2 FAULT, -MONITOR SYS, -MONITOR OVERHEAD PANEL, and NOT AVAIL: ECAM WARN, ALTI ALERT, STATUS, A/CALL OUT, MEMO. The memos are gone.
4	Press the STS key on the ECAM control panel.	No STATUS page (STATUS is in the NOT AVAIL list).
5	Deactivate FWC 2.	FWS FWC 1 FAULT again; the active alerts come back as new ones.

5. Fire detection model and fire failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 26 Fire Protection; overhead FIRE panel; ENG MASTER panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK. An agent putting out an MSFS failures menu fire: not sim-tested (Rust fire protection tests).

What it is for

Nine new failures: **Engine 1 fire**, **Engine 2 fire**, **APU fire**, and **Engine 1 fire detection loop A**, **Engine 1 fire detection loop B**, **Engine 2 fire detection loop A**, **Engine 2 fire detection loop B**, **APU fire detection loop A**, **APU fire detection loop B**.

Each zone now has the two detection loops and the fire detection unit of the aircraft (A320 FCOM DSC-26-20-10). A fire warning comes when both loops detect the fire, when one loop detects it and the other one is failed, or when both loops break within 5 s (flame effect). The logic moved from the A380X into the common code; the A380X behaviour is unchanged. The fires of the MSFS failures menu still give the warnings, through the loops; an AGENT discharge (or the automatic APU discharge) can also put such a fire out, which ends the ENG / APU FIRE warning, and the fire stays out while the MSFS failure stays on (both aircraft). The FIRE TEST is unchanged.

- The ENG MASTER panel FIRE light comes on with the engine fire warning (before: FIRE TEST only).
- APU fire on the ground: the APU shuts down and the APU AGENT discharges automatically 3 s after the warning (FCOM DSC-26-20-20).
- Each AGENT discharge has one chance in two to put a flyPad fire (or an MSFS failures menu fire) out.
- Loop supplies (FCOM PRO-ABN-ELEC INOP SYS): DC BUS 2 for ENG 1 LOOP B and ENG 2 LOOP A, DC ESS BUS for ENG 1 LOOP A and ENG 2 LOOP B, DC BAT BUS for both APU loops. An unpowered loop is a failed loop.

What you should see

Situation	What you should see
Engine 1 fire	ENG 1 FIRE warning (from `core/fire`), red ENG 1 FIRE pushbutton light and red FIRE light on the ENG MASTER panel.
APU fire on the ground	APU FIRE warning; the APU shuts down; 3 s later the APU AGENT discharges (DISCH light).
AGENT discharged	The fire goes out, or not (one chance in two per discharge).
One loop failed	Amber **ENG 1 FIRE LOOP A FAULT** (LOOP B, ENG 2, APU FIRE LOOP A / B); STATUS INOP SYS ENG 1 LOOP A (ENG 1 LOOP B, ENG 2 LOOP A / B, APU LOOP A / B). A fire is still detected by the other loop.
Both loops of a zone failed	**ENG 1 FIRE DET FAULT** (ENG 2, APU FIRE DET FAULT) instead of the two loop cautions; STATUS INOP SYS FIRE DET 1 (FIRE DET 2, APU FIRE DET).

Limitations and why

Limitation	Why	What to do
FIRE DET FAULT also comes from a failed fire detection unit.	See the next feature, "Fire detection unit failures".	-
The LOOP and DET FAULT cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the inhibition as a figure: that of the other crew awareness cautions is used.	Test at the gate or in cruise.
A full MSFS restart is needed after an update.	The systems WASM module and the cockpit behaviour XML change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 26 Fire Protection and activate a fire or a loop failure.
2. Fight a fire with the ECAM procedure (FIRE pushbutton, AGENT 1, AGENT 2). Deactivating the failure also ends the fire.

How to test

#	Do this	Expected result
1	A32NX on the ground, engines running. Activate **Engine 1 fire**.	ENG 1 FIRE warning, master warning, continuous chime, red light in the ENG 1 FIRE pushbutton and on the ENG MASTER 1 panel.
2	Follow the ECAM: ENG MASTER 1 OFF, push the ENG 1 FIRE pushbutton, press AGENT 1.	AGENT 1 DISCH; the fire goes out or not (one chance in two). If not, AGENT 2 gives another chance.
3	Deactivate the failure, reset the FIRE pushbutton. With the APU running, activate **APU fire**.	APU FIRE warning; the APU shuts down; about 3 s later the APU AGENT discharges by itself.
4	Deactivate it. Activate **Engine 2 fire detection loop A**.	Amber ENG 2 FIRE LOOP A FAULT; STATUS INOP SYS ENG 2 LOOP A.
5	Activate **Engine 2 fire** with loop A still failed.	ENG 2 FIRE warning: loop B alone detects the fire.
6	Deactivate the fire, activate **Engine 2 fire detection loop B** as well.	The loop cautions are replaced by ENG 2 FIRE DET FAULT; STATUS INOP SYS FIRE DET 2.

7	Optional: start an engine fire from the MSFS failures menu.	The ENG FIRE warning still comes (through the loops).
8	Optional (not sim-tested): with that MSFS menu fire still on, push the ENG FIRE pb and press AGENT 1 (AGENT 2 if the FIRE pb stays red).	One agent in two puts the fire out: the ENG FIRE warning and the FIRE pb light go, and stay off while the MSFS failure stays on.

6. Fire detection unit failures, FIRE TEST light logic, APU fire horn

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 26 Fire Protection; overhead FIRE panel; ENG MASTER panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK; the horn sound itself is silent in MSFS 2024. Rust tests (fire_protection), unit tests: FireDetectionFaults.spec.ts, EngineMasterFireLight.spec.ts.

What it is for

Three follow-ups of the fire detection model above:

- **Fire detection unit failures:** three new failures, **Engine 1 fire detection unit (FDU)**, **Engine 2 fire detection unit (FDU)** and **APU fire detection unit (FDU)**. A failed FDU gives the FIRE DET FAULT caution of its zone even with healthy loops, and detects no fire. Clearing the failure with a fire present gives the fire warning at once.
- **ENG MASTER panel FIRE light during the FIRE TEST:** the A320 FCOM (DSC-26-20-20, ENG 1(2) FIRE TEST pb) says that the FIRE lights do not come on with an engine 1 loop A fault or an engine 2 loop B fault. The light now stays dark in those cases (a loop fault is its failure or the loss of its DC ESS supply).
- **APU fire external horn:** an APU fire on the ground sounds the horn in the nose gear bay (FCOM DSC-26-20-20 APU FIRE light, DSC-49-10-20), on the same sound as the MECH call; not during the FIRE TEST.

What you should see

Situation	What you should see
Engine 1 FDU failed	Amber **ENG 1 FIRE DET FAULT** (ENG 2 FIRE DET FAULT, APU FIRE DET FAULT for the other units), no LOOP FAULT caution (a failed FDU cannot tell which loop is failed); STATUS INOP SYS FIRE DET 1 (FIRE DET 2, APU FIRE DET).
A fire in a zone with its FDU failed	No fire warning.
FDU failure cleared while the fire burns	The fire warning comes at once.
ENG 1 FIRE TEST with Engine 1 fire detection loop A failed	The ENG MASTER 1 FIRE light stays dark; the rest of the test as before.
ENG 2 FIRE TEST with Engine 2 fire detection loop B failed	The ENG MASTER 2 FIRE light stays dark.
APU fire on the ground	The external horn variable is set (L:A32NX_FIRE_APU_EXTERNAL_HORN); the horn sound plays at the nose gear.

Limitations and why

Limitation	Why	What to do
The APU fire horn is not heard in MSFS 2024.	The Asobo horn sound it uses is silent in MSFS 2024 (the MECH call too).	Check the L:var with a SimConnect / variable tool.
No APU SHUT OFF pb and no APU FIRE light on the external power panel.	The exterior model has no such panel.	None.
A full MSFS restart is needed after an update.	The systems WASM module, the cockpit behaviour XML and the sound.xml change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 26 Fire Protection and activate an "... fire detection unit (FDU)" failure.
2. Follow the ECAM (crew awareness); deactivate the failure to restore the unit.

How to test

#	Do this	Expected result
1	A32NX at the gate, powered, no failure. Activate **Engine 1 fire detection unit (FDU)**.	Amber ENG 1 FIRE DET FAULT, no ENG 1 FIRE LOOP A / B FAULT; STATUS INOP SYS FIRE DET 1.
2	Also activate **Engine 1 fire**.	No ENG 1 FIRE warning.
3	Deactivate **Engine 1 fire detection unit (FDU)**.	The ENG 1 FIRE warning comes at once. Deactivate the fire.
4	Repeat with **Engine 2 fire detection unit (FDU)** and **APU fire detection unit (FDU)**.	ENG 2 FIRE DET FAULT / STATUS FIRE DET 2; APU FIRE DET FAULT / STATUS APU FIRE DET.
5	Activate **Engine 1 fire detection loop A**, then press and hold the ENG 1 FIRE TEST pb.	The ENG 1 FIRE pushbutton lights as in the normal test, but the FIRE light on the ENG MASTER 1 panel stays dark.
6	Deactivate it, activate **Engine 2 fire detection loop B**, hold the ENG 2 FIRE TEST pb.	The ENG MASTER 2 FIRE light stays dark. With loop A of engine 2 failed instead, it comes on.
7	On the ground with the APU running, activate **APU fire**.	APU FIRE warning; L:A32NX_FIRE_APU_EXTERNAL_HORN is 1 (the horn itself is silent in MSFS 2024).

7. Wing tank pump and X FEED valve failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL | **Status:** Tested in the sim by the user: OK, after the fix that holds the MSFS pump / valve switch.

What it is for

Five new failures: **L TK pump 1**, **L TK pump 2**, **R TK pump 1**, **R TK pump 2** and **X FEED valve jammed in place**. The FUEL panel pushbuttons now set a selection, and the systems host moves the MSFS pump or valve switch from it: a failed pump is held off, the jammed valve is held where it was. The FAULT lights, the SD FUEL page and the cautions compare the selection with the real state (A320 FCOM PRO-ABN-FUEL).

- SD FUEL page: amber LO only for a pump that is ON with a low pressure; the X FEED valve is amber when it disagrees with its pushbutton (FCOM colours).
- The FUEL X FEED memo now needs the X FEED pushbutton ON and the valve not fully closed.

What you should see

Situation	What you should see
L TK pump 1 failed, pushbutton ON	FAULT light; amber LO on the SD FUEL page (page called). E/WD **FUEL L TK PUMP 1 LO PR** with -L TK PUMP 1 ..... OFF (shown while the pushbutton is ON). STATUS INOP SYS L TK PUMP 1.

Both pumps of one wing failed	**FUEL L TK PUMP 1+2 LO PR** (R for the right wing): .IF NO FUEL LEAK: -FUEL X FEED ON (above FL 150 only), -ENG MODE SEL IGN, -L TK PUMP 1 OFF, -L TK PUMP 2 OFF, .WHEN L TK FUEL RQRD: L TK FEED GRVTY ONLY, -FUEL X FEED OFF (below FL 150, X FEED ON), PROC:GRVTY FUEL FEEDING, AVOID NEGATIVE G FACTOR. Each action line goes once done. STATUS INOP SYS L TK PUMPS.
X FEED valve jammed, pushbutton moved	The valve stays: amber on the SD FUEL page, **FUEL X FEED VALVE FAULT** (crew awareness), STATUS INOP SYS FUEL X FEED. With the valve stuck closed, no FUEL X FEED memo.

Limitations and why

Limitation	Why	What to do
The cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the inhibition as a figure: that of the other single system faults is used.	Test at the gate or in cruise.
The centre tank has no pump failure.	The A320neo centre tank has no pumps: its fuel is moved by the jet pumps through the CTR TK L / R XFR valves, whose jams are a feature of their own (below).	None.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate a pump or the X FEED valve failure.
2. Follow the ECAM with the overhead FUEL panel; check the SD FUEL page.

How to test

#	Do this	Expected result
1	A32NX at the gate, APU or engines running, all fuel pumps ON. Activate **L TK pump 1**.	After a few seconds: FAULT light in the L TK PUMP 1 pushbutton, SD FUEL page with amber LO, master caution, E/WD FUEL L TK PUMP 1 LO PR with -L TK PUMP 1 ..... OFF.
2	Set the L TK PUMP 1 pushbutton OFF. Open STATUS.	The FAULT light and the OFF line go; no LO on the SD; STATUS INOP SYS L TK PUMP 1.
3	Set it ON again and also activate **L TK pump 2**.	FUEL L TK PUMP 1+2 LO PR with its procedure (on the ground, below FL 150: .IF NO FUEL LEAK:, -ENG MODE SEL IGN, -L TK PUMP 1 OFF, -L TK PUMP 2 OFF, ...); STATUS INOP SYS L TK PUMPS.
4	Deactivate both. With the X FEED valve closed, activate **X FEED valve jammed in place**, then set the X FEED pushbutton ON.	The valve stays closed: amber X FEED valve on the SD, FUEL X FEED VALVE FAULT, STATUS INOP SYS FUEL X FEED, no FUEL X FEED memo.
5	Deactivate the failure.	The valve opens; the FUEL X FEED memo comes; the caution clears.

8. Centre tank transfer valve failures and fuel follow-ups

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL; flyPad checklists | **Status:** Tested in the sim by the user: OK. Unit tests: FuelPumpValveFaults.spec.ts, FuelPumpsAndValves.spec.ts, FuelPumpValveIndications.spec.ts, CheckItemStates.spec.ts.

What it is for

Two new failures, **CTR TK L XFR valve jammed in place** and **CTR TK R XFR valve jammed in place**: the centre tank transfer valve of that side stays open or closed, whatever its CTR TK XFR pushbutton commands. The FWC raises the A320 FCOM PRO-ABN-FUEL cautions: FUEL CTR L(R) XFR FAULT (valve not fully closed, or not fully open) and, with both valves stuck the same way, FUEL CTR L + R XFR FAULT, which replaces the two single ones. The SD FUEL page shows the valve amber and boxes the centre tank quantity.

- The centre tank fuel only transfers while an inner tank pump of the same wing runs: the jet pumps are driven by the wing tank pumps (FCOM DSC-28-10). Switching the wing pumps off stops the centre tank transfer.
- flyPad checklist: the securing / parking checklist item **FUEL PUMPS ... OFF** now reads the four wing tank pump pushbuttons, not whether the pumps run (the CTR TK XFR pushbuttons are items of their own in the FCOM).

What you should see

Situation	What you should see
L valve stuck open (CTR TK L XFR pb OFF)	E/WD **FUEL CTR L XFR FAULT**: -CTR TK L XFR ... OFF, -FUEL X FEED ... ON, -L TK PUMP 1 ... OFF, -L TK PUMP 2 ... OFF, .WHEN CTR TK EMPTY: -L TK PUMP 1 ... ON, -L TK PUMP 2 ... ON, -FUEL X FEED ... OFF (each action line goes once done). STATUS INOP SYS CTR TK L XFR.
L valve stuck closed (CTR TK L XFR pb ON)	**FUEL CTR L XFR FAULT**: -FUEL MODE SEL ... MAN, .IF UNSUCCESSFUL: -FUEL X FEED ... ON, -L TK PUMP 1 / 2 ... OFF, .WHEN CTR TK EMPTY: pumps ON, X FEED OFF. STATUS INOP SYS CTR TK L XFR.
Both valves stuck open	**FUEL CTR L+R XFR FAULT**: -CTR TK L XFR ... OFF, -CTR TK R XFR ... OFF; STATUS INOP SYS CTR TK XFR.
Both valves stuck closed	**FUEL CTR L+R XFR FAULT**: -FUEL MODE SEL ... MAN, .IF UNSUCCESSFUL: CTR AVAIL BY GRAVITY, 2T (4400LBS) UNUSABLE; STATUS: CTR TK USABLE BY GRAVITY, 2T (4400LBS) UNUSABLE, INOP SYS CTR TK XFR.
SD FUEL page	The jammed valve amber; the centre tank quantity boxed.

Limitations and why

Limitation	Why	What to do
The cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the inhibition as a figure: that of the other fuel cautions is used.	Test at the gate or in cruise.
No gravity transfer from the centre tank.	Not simulated: with both valves stuck closed, the centre tank fuel stays there.	Plan with the centre tank fuel as unusable.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate **CTR TK L XFR valve jammed in place** or **CTR TK R XFR valve jammed in place**.
2. Move the CTR TK XFR pushbutton (or let the automatic control move the valve), then follow the ECAM with the overhead FUEL panel.

How to test

#	Do this	Expected result

1	A32NX at the gate with fuel in the centre tank, engines or APU running, all FUEL pushbuttons ON. Activate **CTR TK L XFR valve jammed in place** while the valve is open, then set the CTR TK L XFR pushbutton OFF.	The valve stays open: master caution, E/WD FUEL CTR L XFR FAULT with -CTR TK L XFR ... OFF done, -FUEL X FEED ... ON, -L TK PUMP 1 ... OFF, -L TK PUMP 2 ... OFF; SD FUEL with the valve amber.
2	Open the STATUS page.	INOP SYS: CTR TK L XFR.
3	Also activate **CTR TK R XFR valve jammed in place** and set the CTR TK R XFR pushbutton OFF.	The two single cautions are replaced by FUEL CTR L+R XFR FAULT; STATUS INOP SYS CTR TK XFR.
4	Deactivate both failures, CTR TK XFR pushbuttons ON. Set L TK PUMP 1 and 2 OFF.	The left centre tank transfer stops (no jet pump drive) while the right one goes on.
5	Set every wing tank pump pushbutton OFF and open the flyPad checklist with the FUEL PUMPS ... OFF item.	The item is checked from the four wing tank pump pushbuttons.

9. Engine flameout and seizure failures, in-flight relight

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 72 Engine; ENG MASTER 1 / 2, ENG MODE selector; E/WD, STATUS, SD ENG | **Status:** Tested in the sim by the user: OK; the LEAP-1A relight envelope (a27ba6c35): sim-tested OK (recorded). Rust tests (engine_failure.rs of systems and a320_systems, incl. the LEAP-1A envelope corners and two test-bed relights; full-aircraft tests incl. the ENG FIRE pb LP valve starvation), unit tests: EngineFailAlerts.spec.ts, A32NX_FADEC.spec.ts, Eng.spec.ts, a320.spec.ts. The end of an in-flight start without N2 overshoot: sim-tested OK (relight, recorded), native test engine_start_throttle_hold_test. The engine fuel valve hold: not sim-tested on the A32NX (Rust tests sim_fuel_valve.rs). The stall during a relight fix: not sim-tested (installed, no visible effect expected; native test). The FADEC automatic relight after a flameout: OK on a combined test build (recorded). Not sim-tested on this branch.

What it is for

Four new failures in a new flyPad chapter, ATA 72 Engine. A **flameout** cuts the engine fuel once: the crew can relight the engine with the ENG MASTER OFF then ON, inside the in-flight relight envelope. A **seizure** also stops the core: no relight while the failure is active. The FWC gives the A320 FCOM ENG 1(2) FAIL and ENG 1(2) SHUT DOWN alerts with their STATUS, and the flyPad failures page now describes the ATA 28 Fuel and ATA 72 Engine chapters like the others.

Id	flyPad failure (ATA 72 Engine)	What the code does
72000	Engine 1 flameout (crew relight possible)	Cuts the engine 1 fuel (latched). The engine runs down and windmills. In flight the FADEC makes one automatic relight attempt for 30 s (see "FCOM fixes: FADEC relight, automatic ignition..."): it relights the engine inside the relight envelope once the continuous ignition is on, without an ENG MASTER cycle. Otherwise ENG MASTER 1 OFF then ON inside the relight envelope relights it; it then runs normally while the failure stays listed. Deactivate and activate it again for a new flameout.
72001	Engine 2 flameout (crew relight possible)	The same for engine 2.
72010	Engine 1 seizure (no relight)	Cuts the fuel; the core (N2) stops in about 3 s while N1 windmills. No relight while the failure is active; once it is cleared the engine relights by the normal procedure.
72011	Engine 2 seizure (no relight)	The same for engine 2.

The **relight envelope** comes from the LEAP-1A chart ENG RELIGHT IN FLIGHT of the Air Arabia A320 QRH (18-Aug-21, ABN-19.06A, A6-ATA to ATF, CFM LEAP-1A engines, the engines FBW models). It replaces the PW1100G chart of the IndiGo A320neo FCOM, used while no LEAP-1A chart was available. Vc is the indicated airspeed, Zp the pressure altitude.

Zone	Where	Needs
3, stabilized windmill relight (N2 > 8 %)	270-350 kt up to 27 000 ft; the ceiling falls from 27 000 ft at 330 kt to 24 000 ft at 350 kt	ENG MODE IGN, ENG MASTER OFF then ON
1, starter assisted relight	135-270 kt up to 15 000 ft; 200-270 kt from 15 000 to 27 000 ft	ENG MODE IGN, ENG MASTER OFF then ON, and starter air (APU BLEED, or X BLEED OPEN with the other engine)
2, windmill quick relight (t < 20 s)	230-270 kt up to 20 000 ft	ENG MASTER OFF then ON within 20 s, with N2 above 15 % (design choice, not on the LEAP-1A chart), at any ENG MODE position

- Above these limits (e.g. in cruise at FL 350) no relight lights up.
- A windmilling engine (no combustion, in flight) keeps N2 = 8 % x IAS / 270 kt and N1 = 1.5 x N2 (at 300 kt: N2 8.9 %, N1 13.3 %); on the ground it runs down to 0. Its fuel flow shows 0 and its oil pressure falls with N2.
- During an in-flight relight both igniters are supplied (A320 FCOM DSC-70-80-30), whatever the ENG MODE selector: the SD ENG page shows A B.
- The SD ENG page reads the N2 the FADEC shows, so a seized core reads 0 there as on the E/WD.
- ENG ALL ENGINES FAILURE gets its FCOM flight phase inhibition (phases 1-4 and 8-10). Its OPT RELIGHT SPD line reads 270/.77, the LEAP-1A value of the same QRH (ABN-19.01A ALL ENG FAIL; the CFM56 aircraft use 300/.77); the removal of the unused 280/300/260 lines is on core/fcom-audit-fixes.
- **End of an in-flight start (fix):** an engine started or relit in flight used to overshoot to 110-114 % N2 for 3-7 s at the end of the start, because the thrust loop wound the MSFS throttle up while the engine was out. In flight, the MSFS throttle is now held at idle while the engine is out or starting, until the engine runs at idle; then the engine accelerates to the lever or A/THR demand (A320 FCOM DSC-70-80-40, QRH ENG RELIGHT IN FLIGHT "When idle reached"). Ground starts are unchanged, and reverse thrust passes through this hold (a first version of it blocked the reverse thrust).
- **Stall during a relight (fix):** the hold ends when the commanded N1 reaches idle. With the "Engine 1(2) compressor stall" failure active at the end of a relight, the stall N1 loss (up to 15 %) was counted in the commanded N1 and the hold ended about 0.8 s early; the stall N1 loss is now taken out of that check (native : a relight with a stall).
- **ENG MODE selector (fix):** the FADEC read a selector variable that does not exist (always NORM), so it never saw IGN/START; it now reads the real selector.
- **Engine fuel valve hold (fix):** MSFS can reopen the engine fuel valve (valves 13 and 14 of the A32NX engine fuel cut) on its own after the systems closed it. The systems now command the valve again whenever MSFS disagrees, and once when a flight is loaded.

What you should see

Situation	What you should see
Flameout in flight	N2, EGT and FF run down, FF 0, N1 / N2 settle at the windmill values. Amber **ENG 1 FAIL**, single chime, MASTER CAUT, SD ENG page: -ENG MODE SEL ..... IGN, -THR LEVER 1 ..... IDLE, .IF NO RELIGHT AFT 30S:; 30 s after the alert -ENG MASTER 1 ..... OFF, .IF DAMAGE: -ENG 1 FIRE P/B ..... PUSH, -AGEN AFTER 10S ..... DISCH, .IF NO DAMAGE: -ENG 1 RELIGHT ..... CONSIDER (A320 FCOM PRO-ABN-ENG ENG 1(2) FAIL). LAND ASAP amber. The ENG 1 FAULT light stays off (FCOM: a flameout does not light it).

ENG MASTER 1 OFF in flight (after a failure or not)	ENG 1 FAIL goes, **ENG 1 SHUT DOWN** comes: .IF NO FUEL LEAK:, -IMBALANCE ..... MONITOR, -TCAS MODE SEL ..... TA (with the wing anti-ice on: the pack of the affected side OFF and X BLEED OPEN; with the ENG FIRE pb pushed: X BLEED SHUT, WING ANTI ICE OFF, AVOID ICING CONDITIONS). STATUS: IF PERF PERMITS: X BLEED ..... OPEN, IF NO ENG 1 DAMAGE: CONSIDER ENG 1 RELIGHT, CAT 3 SINGLE ONLY, ONE PACK ONLY IF WAI ON; INOP SYS: CAT 3 DUAL, ENG 1 BLEED, PACK 1, MAIN GALLEY, GEN 1, G ENG 1 PUMP (WING A. ICE with the ENG FIRE pb pushed).
Relight inside the envelope	Light-up: FF, N2 and EGT rise, A B igniters on the SD ENG page, AVAIL; the engine alerts and LAND ASAP clear.
Relight outside the envelope	No light-up: N2 stays at the windmill value, FF 0.
Seizure	N2 to 0 in about 3 s, N1 windmills, oil pressure 0 (red), ENG 2 FAIL as above; GEN 2 and the yellow engine pump lost.
Both engines failed	Red ENG ALL ENGINES FAILURE; the single ENG 1 / ENG 2 FAIL alerts do not show while it is active. Relighting one engine brings the single alert of the other one.
On the ground	ENG 1 FAIL with the ground lines (-THR LEVER 1 ..... IDLE, -ENG MASTER 1 ..... OFF, .IF DAMAGE: ..., -AGENT 1 ..... DISCH, .IF NO DAMAGE: ...), no LAND ASAP; N1 / N2 run down to 0. ENG MASTER OFF gives no ENG SHUT DOWN in flight phases 1-2 and 9-10 (FCOM: phases 3-8 only).

Limitations and design choices

Limitation	Why	What to do
The relight envelope applies to every in-flight relight, also after a crew shutdown without any failure.	Design choice: MSFS would otherwise relight anywhere; the QRH chart is the limit of any relight.	Relight inside the envelope (descend and adjust the speed).
The envelope corners are read off the QRH chart grid.	The chart gives no table; the calibrated airspeed is taken as the indicated airspeed.	-
The quick relight also needs N2 above 15 %.	Design choice kept from the PW1100G chart (the LEAP-1A chart only says t < 20 s): the sign of an engine that was running before the MASTER cycle.	An engine that has windmilled for a while needs ENG MODE IGN (zone 1 or 3).
The windmill N1 / N2 values are design values.	The chart only marks zone 3 "N2 > 8 %"; N2 is taken proportional to the airspeed, 8 % at 270 kt, and N1 = 1.5 x N2.	-
The seizure stops the core in about 3 s.	Design choice: the FCOM gives no rate.	-
ENG 1(2) FAIL does not come back after the ENG MASTER OFF, even if the relight does not light up.	Design choice: the MASTER OFF disarms the alert; ENG SHUT DOWN is shown meanwhile.	After a failed attempt (no light-up within 30 s), MASTER OFF and try again.
While ENG ALL ENGINES FAILURE is active, the single engine alerts are not shown.	Design choice: its procedure replaces them.	-

In flight, the HP fuel valve opening at N2 > 15 % is not modelled.	The fuel comes back when the engine lights up inside the envelope (on the ground the start sequence of "Engine ignition and start failures" opens the fuel at 22 % N2).	-
The idle hold after an in-flight start ends after a fixed time if MSFS settles below the FBW idle.	Design choice (no FCOM value): MSFS can end its start slightly below the FBW idle core speed.	-
A full MSFS restart is needed after an update.	The systems WASM module, the FADEC WASM module and the fbw WASM module (throttle hold) change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 72 Engine and activate a flameout or a seizure.
2. Follow the ECAM (ENG FAIL, then ENG SHUT DOWN). To relight, get inside the envelope (table above), ENG MODE IGN, ENG MASTER OFF for 30 s, then ON. For a quick relight in zone 2, set the MASTER OFF then ON at once (within 20 s, N2 above 15 %, 230-270 kt up to 20 000 ft).
3. Clear the failure in the flyPad after the test; a seizure must be cleared before a relight.

How to test

Setups: S1 cruise 60 t, FL 350, M.78; S2 60 t, 10 000 ft, 300 kt; S3 60 t, 15 000 ft, 210 kt, APU running with APU BLEED ON; S4 on the ground, both engines at idle.

#	Do this	Expected result
1	Activate **Engine 1 flameout (crew relight possible)**.	ENG 1 FAIL with its lines, LAND ASAP, windmill N1 / N2, FF 0, oil pressure falling with N2; the 30 s lines come after 30 s. ENG MODE IGN and THR LEVER 1 IDLE clear their lines.
2	After 30 s, ENG MASTER 1 OFF; then MASTER ON at FL 350.	ENG 1 SHUT DOWN with its STATUS and INOP SYS; at FL 350 no light-up (outside the envelope), A B igniters on the SD ENG page.
3	Descend to 20 000 ft or below at 280-300 kt; ENG MODE IGN, MASTER 1 OFF 30 s, ON.	Light-up at once; the engine back at idle in about 30-60 s; all engine 1 alerts and LAND ASAP clear.
4	Flameout; MASTER 1 OFF; APU BLEED OFF, X BLEED not open; 30 s; MASTER 1 ON.	No light-up (210 kt is below zone 3, no starter air). MASTER OFF, APU BLEED ON, 30 s, MASTER ON: the start valve opens and the engine lights up.
5	At 245 kt below 13 000 ft, ENG MODE NORM: flameout, then at once MASTER 1 OFF then ON within 20 s.	Quick relight at NORM; the ENG MODE knob stays at NORM and still works afterwards. Waiting until N2 is below 15 % before the MASTER OFF / ON: no light-up. Since the code makes the FADEC relight a flameout in this quick relight zone by itself, at once and at NORM, before the MASTER cycle (not sim-tested): the crew quick relight is then best seen with the next step (MASTER OFF / ON without a failure).
6	Activate **Engine 2 seizure (no relight)**.	N2 to 0 in about 3 s, N1 windmills, oil pressure 0, ENG 2 FAIL; MASTER 2 OFF 30 s / ON at 300 kt: no light-up. Clear the seizure, MASTER 2 OFF 30 s / ON: light-up.

7	Optional: activate both flameouts.	ENG ALL ENGINES FAILURE without the single ENG FAIL alerts; relighting engine 1 brings ENG 2 FAIL.
8	No failure, MASTER 1 OFF at FL 350, then ON with ENG MODE IGN.	ENG 1 SHUT DOWN; no light-up at FL 350; at FL 200 / 290 kt, MASTER OFF 30 s then ON: light-up.
9	Activate **Engine 1 flameout (crew relight possible)** on the ground.	ENG 1 FAIL with the ground lines, no LAND ASAP, N1 / N2 to 0; MASTER 1 OFF gives no ENG 1 SHUT DOWN; a normal start works.
10	Normal flight (starts, single-engine taxi, takeoff to descent, shutdown); on the ground at idle, push the ENG 1 FIRE pb with MASTER 1 ON.	No ENG FAIL / SHUT DOWN alert in the normal flight. With the FIRE pb: ENG 1 SHUT DOWN at once, no ENG 1 FAIL; the engine stops after about 100 s on its residual fuel (LP valve starvation of core/fire).

LEAP-1A envelope re-test (recorded over SimConnect, OK): two points the old PW1100G envelope refused. L1 setup: FL 150, 250 kt, AP and A/THR ON, APU OFF, ENG MODE NORM, both engines running. L2 setup: climbed to FL 265 at 290 kt (279 kt at the relight), AP and A/THR ON, APU OFF, X BLEED AUTO, ENG MODE NORM.

#	Do this	Expected result
1	L1 (quick relight at FL 150): ENG MASTER 1 OFF, count 5 s, ENG MASTER 1 ON.	Engine 1 fuel flow comes back within a few seconds, N2 and EGT rise, engine 1 back at idle; no ENG 1 FAIL stays; the ENG MODE selector stays at NORM.
2	L2 (windmill relight at FL 265): activate **Engine 1 flameout (crew relight possible)**; THR LEVER 1 IDLE; ENG MODE IGN/START; ENG MASTER 1 OFF, count 10 s; ENG MASTER 1 ON. At idle: ENG MODE NORM, THR LEVER 1 CL, deactivate the failure.	Engine 1 fuel flow appears within a few seconds, N2 and EGT rise, engine 1 at idle within about 60 s, ENG 1 FAIL goes away (the aircraft may sink a few hundred feet meanwhile).

End of the in-flight start (recorded, OK). Setup: FL 150, 250 kt, AP and A/THR ON, APU OFF, ENG MODE NORM, both engines running, both thrust levers in CL.

#	Do this	Expected result
1	ENG MASTER 1 OFF, count 5 s, ENG MASTER 1 ON. Leave both levers in CL and watch engine 1 N2 for 40 s.	Engine 1 relights within about 10 s; its N2 rises to idle (about 70 %) and holds there until the engine runs, then rises in about 5 s to the N2 of engine 2 (about 93 %). Engine 1 N2 never goes more than 1-2 % above engine 2 N2 (before the fix: 110-114 %).

10. Engine oil failures: leak, filter clog, overheat

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 79 Oil; SD ENGINE and CRUISE pages; E/WD | **Status:** Sim-tested by the user, data-checked OK (re: no false oil alert in a climb to FL 300; the three oil failures together in cruise: overheat 81 to 158 °C with ENG 2 OIL HI TEMP, leak to the red oil pressure below 13 PSI with ENG 1 OIL LO PR, then the shutdown). The later tuning (gradual cool-down after the overheat, overheat times): not sim-tested on the A32NX. Rust tests oil_failure.rs, native tests engine_oil_failures_test and oil_system_test, unit tests EngineOilAlerts.spec.ts, EngOilIndications.spec.ts.

What it is for

Six new failures: **Engine 1 oil leak**, **Engine 2 oil leak**, **Engine 1 oil filter clog**, **Engine 2 oil filter clog**, **Engine 1 oil overheat** and **Engine 2 oil overheat**, with the oil indications and the ECAM alerts of the A320 FCOM (DSC-70-90-40, PRO-ABN-ENG).

The FADEC computes the oil quantity, pressure and temperature again: its oil model had been dead since a FADEC migration and is restored (from the A380X FADEC, with the A32NX engine imbalance effect on the oil pressure). The

failures act on that model:

- **Leak:** the oil system empties at 2 qt per minute of simulation time (the SD tank reading falls about 1.6 qt/min). The pump keeps its normal pressure down to 2.5 qt in the whole system and gives none at 0.6 qt, so the pressure falls once the tank reads nearly empty.
- **Filter clog:** CLOG on the SD ENGINE page and the ENG OIL FILTER CLOG caution, no other effect.
- **Overheat:** the oil goes to 135 °C at idle and 175 °C at 100 % N2 (about 165 °C at cruise), with a 60 s time constant: ENG OIL HI TEMP after about 2 minutes in cruise; THR LEVER IDLE brings it below 155 °C within about 30 s. Once the failure is cleared the oil cools down gradually to its normal temperature.
- **SD ENGINE page:** the FCOM 0-22 QT scale; the quantity pulses below 3.25 QT (until 4.75 QT); the oil pressure red below 13 PSI, pulsing between 13 and 16 PSI with N2 above 75 % (until it is back above 20 PSI); amber CLOG under the oil pressure. The CRUISE page shows the same FADEC oil quantity.
- **Fixes found in the sim tests:** the SD oil quantity read nearly empty in every climb (the oil gulping formula got the thrust in newtons instead of pounds), and the pressure followed the tank reading, which gave a false ENG OIL LO PR in a normal climb: the pump pressure now follows the oil of the whole system. The overheat stopped far below its target because MSFS pulls the oil temperature back to its own model: the FADEC now keeps its own overheat temperature.
- The start N2 and fuel flow of the FADEC model also blend smoothly at low N2 now (they overshot below about 10 % N2).

What you should see

Situation	What you should see
Oil leak, after several minutes	Oil quantity falling, then pulsing below 3.25 QT; the oil pressure falls and turns red below 13 PSI; red **ENG 1 OIL LO PR**, MASTER WARN and continuous chime: -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF.
Oil filter clog	Amber **ENG 2 OIL FILTER CLOG** (title only, single chime) and amber CLOG on the SD ENGINE page.
Oil overheat	The oil temperature rises from about 85 °C, pulses above 140 °C and is amber above 155 °C; amber **ENG 2 OIL HI TEMP**: -THR LEVER 2 ... IDLE, -ENG MASTER 2 ... OFF. It also comes after more than 15 minutes above 140 °C, and the SD value then turns amber too (fix).
Lever at idle	The -THR LEVER line goes.
ENG MASTER OFF	The oil alert ends and ENG SHUT DOWN takes over (FCOM associated procedure).

Limitations and why

Limitation	Why	What to do
Inhibitions: OIL LO PR in flight phases 1 and 10, OIL HI TEMP in 4, 5, 7, 8, OIL FILTER CLOG in 3, 4, 5, 7, 8.	The FCOM figures.	Test in cruise.
The oil alerts and CLOG only show with the engine running and its ENG MASTER ON.	Design choice: a stopped engine has no oil pressure, and the procedures end with the ENG MASTER OFF.	-
The leak rate, the pressure loss and the overheat temperatures are design values.	The FCOM gives no rates; the overheat cause is not modelled either.	-

The leak rate is per simulation minute.	At 4x sim rate the oil goes four times faster on the real clock.	-
The upper oil pressure scale stays the FBW one (130 PSI, advisory from 129 PSI).	The FCOM 0-100 PSI scale is the CFM56 one; the FBW LEAP-1A oil pressure reaches about 90 PSI at take-off thrust.	-
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules change.	Restart the sim after installing.

How to use

1. Call the SD ENGINE page (ECAM control panel ENG key).
2. Open flyPad > Failures > ATA 79 Oil and activate a leak, a filter clog or an overheat.
3. Follow the ECAM (THR LEVER IDLE, then ENG MASTER OFF if needed). Deactivate the failure after the test.

How to test

Setup: normal departure, takeoff and climb at CL with A/THR ON, then cruise at FL 300-390.

#	Do this	Expected result
1	Climb normally and watch the oil indications.	No OIL LO PR, no pulsing oil quantity; oil quantity about 10-15 qt in the climb.
2	In cruise, SD ENGINE page. Activate **Engine 2 oil filter clog**, **Engine 2 oil overheat** and **Engine 1 oil leak**.	Amber ENG 2 OIL FILTER CLOG and CLOG on the SD; engine 2 oil temperature rises past 155 °C within about 3 minutes, then amber ENG 2 OIL HI TEMP with -THR LEVER 2 ... IDLE, -ENG MASTER 2 ... OFF.
3	Deactivate **Engine 2 oil filter clog** and **Engine 2 oil overheat**.	Both engine 2 alerts and CLOG go; the temperature falls gradually.
4	Wait for the leak (about 6 to 10 minutes after it began).	Engine 1 oil quantity falls about 2 QT/min and pulses below 3.25 QT; the pressure falls and turns red below 13 PSI; red ENG 1 OIL LO PR with MASTER WARN.
5	THR LEVER 1 IDLE, ENG MASTER 1 OFF. Deactivate **Engine 1 oil leak**.	The lever line goes; after the MASTER OFF, ENG 1 SHUT DOWN replaces the oil alert.

11. Engine fuel filter clog

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 73 Engine - Fuel And Control; SD ENGINE page; E/WD | **Status:** Not sim-tested (offline: Rust fuel_filter_failure.rs tests, EngFuelFilterIndication.spec, EngineFuelFilterAlerts.spec, failure list specs). The A380X version of the same code path was sim-tested.

What it is for

Two new failures, **Engine 1 fuel filter clog** and **Engine 2 fuel filter clog**: "the pressure loss across the fuel filter is excessive" (FCOM DSC-70-90-40). Amber **CLOG** below FUEL USED on the SD ENGINE page while the engine runs, and the **ENG 1(2) FUEL FILTER CLOG** caution (crew awareness, "short term engine operation is not affected"). No effect on the engine: the filter is bypassed.

Limitations and why

Limitation	Why	What to do
Inhibited in flight phases 3, 4, 5, 7 and 8.	The FCOM procedure page.	Test on the ground or in cruise.

No fuel system contamination failure.	Not modelled; the clog only gives the indication.	-

How to use

1. Call the SD ENGINE page (ECAM control panel ENG key).
2. Open flyPad > Failures > ATA 73 and activate **Engine 1 fuel filter clog** with engine 1 running.
3. Deactivate it after the test.

How to test

On the ground with engine 1 running at idle (flight phase 1 or 2), SD ENGINE page.

#	Do this	Expected result
1	Activate **Engine 1 fuel filter clog**.	Amber CLOG below FUEL USED of engine 1; amber ENG 1 FUEL FILTER CLOG with MASTER CAUT. Engine 1 keeps running normally.
2	Deactivate it.	CLOG and the caution go.

12. Engine ignition and start failures, FCOM start sequence

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 74 Ignition, ATA 80 Starting; ENG MODE, ENG MASTER, ENG MAN START; E/WD, STATUS, SD ENGINE | **Status:** Sim-tested by the user: OK (test flight S7, recorded: a ground start with Engine 1 igniter A failed, the automatic dry crank and the new start with both igniters; the dry crank with ENG MAN START). The other start failures: not sim-tested (Rust tests). Rust tests engine_start.rs, native test start_sequence_test, unit tests EngineStartAlerts.spec.ts, StartIndications.spec.ts.

What it is for

The FBW FADEC ran the engine model but not what a real FADEC does during a start. The systems now run the start sequence of the A320 FCOM (DSC-70-80-30, DSC-70-80-40):

- **Igniters:** one igniter for a first automatic start on the ground, alternated between starts; both for a new attempt, a manual start and an in-flight start. Continuous ignition with the ENG MODE selector at IGN/START (one igniter, both if it has failed), with the green IGNITION memo. Since the FADEC also selects the continuous ignition by itself while the engine anti-ice is on (see "FCOM fixes: FADEC relight, automatic ignition..."). The SD igniter indication now comes from this sequence.
- **Automatic start:** ignition from 16 % N2, fuel (HP valve) at 22 % N2, ignition off and start valve closed at 50 % N2. On the ground, a start that does not light up within 18 s, stalls, hangs or goes above 725 °C EGT is aborted and tried again with both igniters, up to three attempts; then the ENG FAULT light (ENG MASTER panel) until ENG MASTER OFF. A start that does not light up or hangs is first dry cranked for 30 s; after a stall or an EGT above the limit the FADEC runs a new start first and dry cranks only when no new start is left.
- **ENG MAN START pushbuttons:** manual start, dry crank (ENG MODE CRANK, MAN START ON, MASTER OFF) and wet crank. The MSFS starter turns the core while the sequence keeps the fuel cut, with no EGT rise. A manual start lights up from 20 % N2.
- **Starter shaft shear in flight**: a starter that fails during an in-flight relight gives ENG 1(2) START FAULT STARTER SHAFT SHEAR with WINDMILL START ONLY, for the rest of the flight.
- The start code shared with the A380X got two settings (A380X engine alerts); the A32NX keeps its values (no light-up after 18 s also in flight, LO START AIR PRESS for a ground start only): nothing changes for the A32NX.
- The SD ENGINE page shows the start indications at CRANK too.
- Tuning: an engine that is OFF only shows running once MSFS really burns fuel in it.

flyPad failure (both engines)	ATA	What the code does
Engine 1(2) igniter A / igniter B	74 Ignition	The igniter does not light the engine: ENG IGN A (B) FAULT; a start that uses it does not light up.
Engine 1(2) start valve stuck closed / stuck open	80 Starting	The start valve stays as it is whatever the FADEC commands: ENG START VALVE FAULT.
Engine 1(2) hot start (start EGT over limit)	80 Starting	The start EGT overshoots: START FAULT EGT OVERLIMIT, abort and new attempts.
Engine 1(2) hung start (N2 stops below idle)	80 Starting	N2 hangs below idle: START FAULT HUNG START.
Engine 1(2) stall during the start	80 Starting	N2 hangs and the EGT overshoots: START FAULT STALL.
Engine 1(2) starter shaft shear	80 Starting	The starter gets air but does not turn the engine: no start, no starter assisted relight; START FAULT STARTER SHAFT SHEAR.

What you should see

Situation	What you should see
Start fault on the ground (automatic start)	Amber **ENG 1 START FAULT** with the cause (IGNITION FAULT, STALL, EGT OVERLIMIT, HUNG START, STARTER SHAFT SHEAR, LO START AIR PRESS or THR LEVER NOT AT IDLE), then AUTO CRANK IN PROGRESS during the dry crank, NEW START IN PROGRESS during the next attempt (after a STALL or EGT OVERLIMIT: NEW START IN PROGRESS first, AUTO CRANK IN PROGRESS only after the last new start), and -ENG MASTER 1 ... OFF once the start is aborted.
Start fault, manual start or in flight	Manual start on the ground: -ENG MASTER 1 ... OFF, -MAN START 1 ... OFF. In flight: -ENG MASTER 1 ... OFF. Starter shaft shear in flight: WINDMILL START ONLY, and STATUS ENG 1 WINDMILL START ONLY. Low start air: -BLEED AIR SUPPLY ... CHECK; lever not at idle: -THR LEVER 1 ... IDLE.
Start valve stuck open after the start	**ENG 1 START VALVE FAULT**, START VALVE NOT CLOSED: -APU BLEED ... OFF (engine 1), -X BLEED ... SHUT; in flight -ENG 1 BLEED ... OFF, -MAN START 1 ... OFF, -WING ANTI ICE ... OFF, AVOID ICING CONDITIONS (with its STATUS); on the ground -ENG MASTER 1 ... OFF.
Start valve stuck closed at a start	ENG 1 START VALVE FAULT, START VALVE NOT OPEN: -X BLEED ... OPEN (other engine running), -APU BLEED ... ON (APU available, below FL 200), .IF UNSUCCESSFUL: MAN START / ENG MASTER OFF.
Igniter failed	**ENG 1 IGN A FAULT** (IGN B FAULT) crew awareness, STATUS INOP SYS ENG 1 IGN A (B); both failed: ENG 1 IGN A+B FAULT, AVOID ADVERSE CONDITIONS, INOP SYS ENG 1 IGN.
ENG MODE IGN/START with the engines running	Green IGNITION memo.

Limitations and why

Limitation	Why	What to do
The numbers are those of the CFM56-5B FCOM.	FBW models the LEAP-1A, for which no FCOM is available.	-
The 30 s automatic dry crank is a design value.	The FCOM gives no length (30 s is its minimum ventilation).	-

No starter time limit.	The A320 FCOM gives none.	-
The SD still shows the selected igniter when that igniter has failed.	Design choice: the FCOM shows the igniter energized.	-
Inhibitions: the three cautions in flight phases 3, 4, 5, 7, 8; IGN A or B FAULT also in 6; START FAULT also in 6 when due to the lever not at idle.	The FCOM figures.	Test on the ground or in cruise.
The "EEC control of start valve failed" lines are not modelled.	They are for IAE and PW engines.	-
An aborted automatic start takes about 2.5 minutes in the sim.	Up to three attempts with 30 s dry cranks.	Use the igniter test below; the other abort paths are covered by Rust tests.
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules and the cockpit behaviour XML (ENG FAULT light) change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 74 Ignition or ATA 80 Starting and activate a failure before the start.
2. Start the engine as usual (ENG MODE IGN/START, ENG MASTER ON) and follow the ECAM; the FADEC aborts and retries a failed automatic start on the ground by itself.
3. For a dry crank: ENG MODE CRANK, ENG MAN START ON with the ENG MASTER OFF.

How to test

Setup: A32NX cold and dark at the gate, freshly loaded, external power or batteries ON, APU AVAIL, APU BLEED ON, parking brake ON, thrust levers IDLE.

#	Do this	Expected result
1	flyPad > Failures > ATA 74 Ignition: activate **Engine 1 igniter A**. ENG MODE IGN/START, ENG MASTER 1 ON. Wait 75 s.	About 20 s after MASTER ON: amber ENG 1 START FAULT, IGNITION FAULT, AUTO CRANK IN PROGRESS and amber ENG 1 IGN A FAULT; SD ENG: N2 about 20-30 %, no EGT rise, start valve open, igniter A shown then gone.
2	Keep waiting.	About 50 s after MASTER ON: NEW START IN PROGRESS, SD igniters A B, EGT rises, engine 1 starts to idle, START FAULT goes; the ENG 1 FAULT light stays off. Deactivate the failure.
3	ENG MODE CRANK; ENG MAN START 2 pb ON (overhead, guarded) for 20 s, then OFF; ENG MODE NORM.	SD ENG: engine 2 start valve open, N2 about 20-25 % and steady, no fuel flow, no EGT rise, no igniter; engine 2 does not start. Then the valve closes and N2 runs down. No ENG 2 START FAULT.
4	Optional (not sim-tested): activate **Engine 2 starter shaft shear**, ENG MODE IGN/START, ENG MASTER 2 ON.	N2 does not rise; ENG 2 START FAULT with STARTER SHAFT SHEAR, -ENG MASTER 2 ... OFF, -MAN START 2 ... OFF. ENG MASTER 2 OFF and deactivate the failure.
5	Optional (not sim-tested): engine 1 off, activate **Engine 1 start valve stuck open**, then start engine 1 (ENG MODE IGN/START, ENG MASTER 1 ON).	Above 50 % N2 the valve stays open: ENG 1 START VALVE FAULT, START VALVE NOT CLOSED with its ground lines (-APU BLEED ... OFF, -X BLEED ... SHUT, -ENG MASTER 1 ... OFF).

13. Engine stall, EGT overtemperature, overspeed and vibration failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 72 Engine, ATA 77 Engine Indicating; E/WD, SD ENGINE | **Status:** Sim-tested by the user: OK (test flight, recorded: stall in the climb, EGT overtemperature with high vibration). VIB N2 then stayed at 4.6 at idle, above the 4.3 advisory: the vibration tuning that brings it below at idle is not sim-tested. The LEAP-1A N1 / N2 red lines and the retuned overspeed indication: tested in the sim by the user on a combined test build: OK (TOGA take-off without an N2 caution, N2 105.3 %; the overspeed failure gives ENG N1 OVER LIMIT with THR LVR BELOW LIMIT; recorded). Rust tests engine_malfunction.rs, unit tests EngineParameterAlerts.spec.ts, EngineLimits.spec.ts, EngVibration.spec.ts, Eng.spec.ts.

What it is for

Eight new failures; the engine keeps running:

flyPad failure (both engines)	ATA	What the code does
Engine 1(2) compressor stall (above 60 % N1)	72 Engine	Above 60 % N1 command the engine stalls: it loses N1 (about 15 %), the EGT rises (about 100 °C or more), N1 and N2 fluctuate, N1 vibrates. It recovers with the lever at IDLE and stalls again above the threshold (QRH: reduce thrust below it).
Engine 1(2) EGT overtemperature	72 Engine	The EGT is higher in proportion to N1 (about 100 °C at climb thrust): ENG EGT OVER LIMIT at climb thrust.
Engine 1(2) N1/N2 overspeed indication	72 Engine	The N1 indication reads 20 % high, up to 101.9 %, and the N2 indication 6 % high, up to 116.9 % (retuned for the LEAP-1A red lines): ENG N1 OVER LIMIT at take-off and climb thrust (above the 101 % red line, below the 102.8 % second band). Reducing the thrust below about 84 % N1 ends the alert. The thrust is unchanged; at idle the N1 reads about 24 % instead of 20 %.
Engine 1(2) high vibration	77 Engine Indicating	N1 and N2 vibrations growing with the square of the rotor speed: above the SD advisory (6 / 4.3) at climb and cruise thrust, below it at reduced thrust and clearly below at idle. No ECAM alert (QRH procedure).

- **E/WD limits** (now in one shared place): EGT amber above 940 °C at TOGA/FLX, 905 °C at CLB/MCT, 725 °C without a rated limit or in the start; red above 975 °C, with a red mark at the highest EGT reached (it stays until the next engine start on the ground); these EGT values are the CFM56-5B TCDS values the user chose. N1 amber above its limit, red above the LEAP-1A red line 101 % (red arc and red mark); N2 red above the LEAP-1A 116.5 %, with a red cross. The N1 / N2 red lines come from the CFM LEAP-1A General Familiarization, page 34 (before, the CFM56 104 % / 105 % of the FCOM, which gave ENG N2 OVER LIMIT at every TOGA take-off, as the LEAP-1A runs above 105 % N2 there).
- **SD ENGINE page:** separate N1 and N2 vibrations, pulsing above the advisory.

What you should see

Situation	What you should see
Stall in flight	Amber **ENG 1 STALL**: -THR LEVER 1 ... IDLE, -ENG 1 PARAMETERS ... CHECK, -ENG 1 STALL PROC ... APPLY; SD ENGINE page. On the ground: -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF.
A parameter above its limit	Amber **ENG 1 EGT OVER LIMIT** (N1, N2 OVER LIMIT; the title names the parameter): -THR LVR 1 .. BELOW LIMIT; once the EGT went above 975 °C (or N1 above 102.8 %, N2 above 117.3 %): -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF. N1 is over its limit above 101 %, N2 above 116.5 %.
TOGA take-off, no failure	No ENG N2 OVER LIMIT (the LEAP-1A N2 stays below 116.5 %).
High vibration at climb thrust	VIB N1 6 or more and VIB N2 4.3 or more, pulsing green.

Limitations and why

Limitation	Why	What to do
What each failure does (N1 loss, EGT and indication offsets, vibration levels) is a design choice.	The FCOM gives the symptoms, not values.	-
The EGT limits are the CFM56-5B TCDS values.	The LEAP-1A document gives 1060 °C "To Be Confirmed", so it is not used (user decision); the FCOM text gives 950 / 915 °C.	-
The second bands of OVER LIMIT (N1 102.8 %, N2 117.3 %) are design values.	The LEAP-1A document gives no second band: each keeps the FCOM gap above the red line (N1 1.8 %, N2 0.8 %).	-
Inhibitions: STALL in flight phases 3, 4, 5, 7, 8; OVER LIMIT in 4 and 8.	The FCOM figures.	Test the stall in the climb above 1 500 ft or in cruise.
A full MSFS restart is needed after an update.	The systems, FADEC and fbw WASM modules change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 72 Engine or ATA 77 Engine Indicating and activate a failure at climb or cruise thrust.
2. Follow the ECAM (THR LEVER IDLE or below the limit); deactivate the failure after the test.

How to test

Setup: climbing through FL 150 at about 280 kt, A/THR ON at CLB, AP ON, both engines running.

#	Do this	Expected result
1	Activate **Engine 1 compressor stall (above 60 % N1)**. Wait 20 s.	Amber ENG 1 STALL with its three lines; SD ENGINE page; N1 1 fluctuates and settles about 15 % below N1 2; EGT 1 rises by about 100 °C or more.
2	Thrust lever 1 to IDLE, wait 20 s.	Within about 10 s the alert goes; N1 1, EGT 1 and the vibration back to normal.
3	Thrust lever 1 back to CL, wait 10 s; deactivate the failure.	The stall comes back, then ends with the failure.
4	At CL, activate **Engine 1 EGT overtemperature** and **Engine 1 high vibration**. Wait 30 s.	EGT 1 about 100 °C above EGT 2, amber above 905 °C with ENG 1 EGT OVER LIMIT and -THR LVR 1 .. BELOW LIMIT (above 975 °C: red, -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF, and a red mark stays); SD VIB N1 6 or more and VIB N2 4.3 or more, pulsing.
5	Thrust lever 1 to IDLE, wait 30 s; deactivate both failures, lever back to CL.	The alert goes; EGT 1 green; both vibrations steady green below the advisory (VIB N2 stayed above it at idle before the tuning). A red EGT mark stays.

LEAP-1A red lines and overspeed (recorded OK on a combined test build). Setup: A32NX on the runway, V2 entered in the MCDU (A/THR armed at TOGA).

#	Do this	Expected result
1	TOGA take-off, A/THR armed; climb to the acceleration altitude.	N2 about 105 %, not red; no ENG N2 OVER LIMIT (before it came at every TOGA take-off).

2	In the climb at CL, activate **Engine 1 N1/N2 overspeed indication**.	N1 1 reads about 101.9 %, red; amber ENG 1 N1 OVER LIMIT with -THR LVR 1 .. BELOW LIMIT (no ENG MASTER OFF line).
3	Thrust lever 1 back below about 84 % N1, then deactivate the failure.	The alert goes; N1 1 back to its normal value (a red mark stays).

14. FADEC, thrust lever and reverser failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 73 Engine - Fuel And Control, ATA 78 Exhaust; thrust levers; E/WD, STATUS, SD | **Status:** Sim-tested by the user: OK (test flight, recorded: Thrust lever 1 resolvers (both), then Engine 2 FADEC channel A and B). The reverser fault test of that flight found no reverse thrust at all (the in-flight start hold clamped it): fixed, normal reverse on the ground data-checked OK; the reverser failures were not re-tested after the fix. FADEC overheat, resolver disagree, reverser unlocked and pressurized: not sim-tested. Rust tests engine_control_failure.rs and hydraulic, native tests fadec_failure_inputs_test and reverse_throttle_test, unit tests FadecReverserAlerts.spec.ts, A32NX_FADEC.spec.ts.

What it is for

flyPad failure (both engines / levers / reversers)	ATA	What the code does
Engine 1(2) FADEC channel A / channel B	73	One channel lost: the other takes control (ENG FADEC A (B) FAULT). Both lost: ENG FADEC FAULT, the engine data are lost (XX on the E/WD and SD), the A/THR disconnects, no reverser; the engine keeps following its lever.
Engine 1(2) FADEC overheat	73	ENG FADEC HI TEMP (ECU above 105 °C).
Thrust lever 1(2) resolvers (both)	73	The FADEC selects idle, CLB or frozen takeoff thrust by the FCOM rules: in flight it keeps CLB (or the takeoff thrust if the lever failed in TO/FLX) until the slats are out, then idle; idle on the ground; no reverser.
Thrust lever 1(2) resolver disagree	73	ENG THR LEVER DISAGREE: idle on the ground, CLB as the maximum in cruise, idle in approach.
Reverser 1(2) fault (does not deploy)	78 Exhaust	The reverser stays stowed; its engine stays at reverse idle: ENG REVERSER FAULT.
Reverser 1(2) unlocked (engine at idle)	78 Exhaust	The doors leave their stowed and locked position (no reverse thrust); the FADEC sets the engine at idle: ENG REVERSE UNLOCKED.
Reverser 1(2) pressurized (shutoff valve open)	78 Exhaust	ENG REV PRESSURIZED.

- Reverse thrust stays at reverse idle until the reverser is fully deployed (FCOM DSC-70-70): a reverser that does not deploy no longer gives forward thrust with the lever in reverse. The hydraulic reversers report their status to the FADECs and stop deploying without a FADEC command.
- Fix: the in-flight start throttle hold (feature "Engine flameout ...") clamped the MSFS throttle at 0, which blocked all reverse thrust on the ground; the reverse throttle now reaches MSFS.

What you should see

Situation	What you should see
One FADEC channel failed	Amber **ENG 1 FADEC A FAULT** (B), crew awareness.

Both channels failed	**ENG 2 FADEC FAULT**: in flight -THR LEVER 2 ... IDLE, -ENG 2 PARAMETERS ... CHECK, .IF ABN ENG OPERATION: -ENG MASTER 2 ... OFF (on the ground THR LVR 2 NOT ABV IDLE); engine 2 N1, EGT, N2, FF XX; A/THR OFF; STATUS THR LVR 2 NOT ABV IDLE.
FADEC overheat	**ENG 1 FADEC HI TEMP**, FADEC OVHT; in flight -ENG 1 PARAMETERS ... CHECK, .IF ABN ENG OPERATION: -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF; on the ground also -ENG MODE SEL ... NORM, -FADEC GND PWR ... CHK OFF.
Thrust lever resolvers lost, in flight at CL	**ENG 1 THR LEVER FAULT** with -A/THR ... KEEP ON (A/THR engaged) or ENG 1 HI PWR IN MAN THR; LAND ASAP amber; STATUS WHEN SLATS OUT: ENG 1 AT IDLE, INOP SYS REVERSER 1, ENG 1 THR. With the slats out: ENG 1 AT IDLE.
Thrust lever resolvers lost, on the ground	ENG 1 THR LEVER FAULT with ENG 1 IDLE POWER ONLY (-THR LEVER 1 ... IDLE with the lever above idle), as in the FCOM (fix, not sim-tested).
Resolver disagree	**ENG 1 THR LEVER DISAGREE**: on the ground ENG 1 IDLE POWER ONLY; in cruise AVAIL MAX POWER : CLB with -A/THR ... KEEP ON (or ON); in approach ENG 1 AT IDLE. STATUS ENG 1 AVAIL MAX PWR:CLB, GND ENG 1 MAX PWR: IDLE, INOP SYS ENG 1 THR.
Reverser fault	Amber **ENG 1 REVERSER FAULT** (title only); STATUS INOP SYS REVERSER 1; with the levers in reverse, REV only on the other engine.
Reverser unlocked in flight	**ENG 1 REVERSE UNLOCKED**: ENG 1 AT IDLE, -THR LEVER 1 ... IDLE, -MAX SPEED ... 300/.78, .IF BUFFET: -MAX SPEED ... 240 KT, -ENG MASTER 1 ... OFF; STATUS MAX SPEED 300/.78.
Reverser pressurized	**ENG 1 REV PRESSURIZED**: in flight -THR LEVER 1 ... IDLE; on the ground THR LVR 1 NOT ABV IDLE.

Limitations and why

Limitation	Why	What to do
Inhibitions (FCOM figures): FADEC A(B) FAULT and FADEC HI TEMP phases 3, 4, 5, 7, 8; FADEC FAULT 4, 5, 7, 8; REV PRESSURIZED 4, 5, 8; REVERSE UNLOCKED 4, 5, 8; REVERSER FAULT 3, 4, 5; THR LEVER FAULT / DISAGREE 4, 5. Some are not inhibited when the FADEC sets the engine at idle.	The FCOM gives them as figures.	Test on the ground or in cruise.
The flyPad reverser fault gives the title only.	It is not a position fault: the FCOM lines are for a position fault with the reverser pressurized.	-
A reverser that really deploys in flight is not modelled.	The unlocked doors stay in the stowed position.	-
FADEC OVHT is always shown with ENG FADEC HI TEMP.	Design choice: the flyPad overheat is an ECU above 105 °C.	-
A full MSFS restart is needed after an update.	The systems and fbw WASM modules change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 73 Engine - Fuel And Control or ATA 78 Exhaust and activate a failure.
2. Follow the ECAM and STATUS; deactivate the failure after the test.

How to test

Check the flyPad throttle calibration (reverse and idle detents) first.

#	Do this	Expected result
1	On the runway, both engines at idle, parking brake ON. Activate **Reverser 1 fault (does not deploy)**.	Amber ENG 1 REVERSER FAULT; STATUS INOP SYS REVERSER 1.
2	Both thrust levers to full reverse for 5 s, then IDLE. Deactivate the failure.	REV green on engine 2 only; engine 2 N1 rises, engine 1 N1 stays near idle; at IDLE the reverser stows, no other caution.
3	After a landing (no failure): both levers to full reverse until 70 kt, then IDLE.	REV green on both engines, N1 well above idle, the aircraft decelerates; no forward thrust jump at IDLE (OK).
4	FL 100, 220 kt, flaps 0, AP and A/THR ON, levers in CL. Activate **Thrust lever 1 resolvers (both)**, then thrust lever 1 to IDLE.	ENG 1 THR LEVER FAULT with -A/THR ... KEEP ON and LAND ASAP; STATUS WHEN SLATS OUT: ENG 1 AT IDLE, INOP SYS REVERSER 1, ENG 1 THR; engine 1 N1 does not drop to idle.
5	FLAPS 1.	Engine 1 N1 goes to idle and stays there.
6	Activate **Engine 2 FADEC channel A**, then **Engine 2 FADEC channel B**.	ENG 2 FADEC FAULT; engine 2 N1, EGT, N2, FF XX on the E/WD and the SD ENGINE page; the A/THR disconnects; thrust lever 2 still changes engine 2 thrust.
7	Deactivate all failures, FLAPS 0, lever 1 to CL, A/THR ON.	All alerts clear.

15. T.O CONFIG NORMAL and FUEL WING TK LO LVL fixes

> **Aircraft:** A32NX | **Where:** ECAM control panel T.O CONFIG key; E/WD, STATUS | **Status:** Not sim-tested. Unit tests ToConfigNormal.spec.ts, WingTankLowLevel.spec.ts.

What it is for

Two deviations from the A320 FCOM found by the FCOM gap audit (the other fixes of that set are on `core/fcom-audit-fixes`). They use the wing tank pump alerts of this branch.

- **T.O CONFIG NORMAL** is no longer given with a FUEL L(R) TK PUMP 1+2 LO PR alert (FCOM DSC-31-15: the T.O CONFIG test triggers it, and NORMAL needs no such alert).
- **FUEL L(R) WING TK LO LVL** (inner tank below 750 kg for 30 s) asks for -FUEL MODE SEL ... MAN only if the centre tank is not empty, then .IF NO FUEL LEAK AND FUEL IMBALANCE: -FUEL X FEED ... ON, -L TK PUMP 1 / 2 ... OFF. It now gives its STATUS: INOP SYS L (R) TK PUMPS and, with fuel in the centre tank, CTR TK FEED: MAN ONLY (FCOM PRO-ABN-FUEL).

Limitations and why

Limitation	Why	What to do
FUEL WING TK LO LVL is inhibited in flight phases 3, 4, 5, 7, 8 and 9.	FBW inhibition, unchanged.	Test at the gate or in cruise.

A low level in one wing only needs a fuel tool.	The flyPad fuel page fills both wings alike.	Set the tank quantity with a SimConnect variable tool (FUEL TANK LEFT MAIN QUANTITY).

How to use

1. Nothing to set: the alerts follow the FCOM by themselves.
2. Press the T.O CONFIG key before takeoff as usual; follow FUEL WING TK LO LVL and its STATUS if it comes.

How to test

#	Do this	Expected result
1	A32NX at the gate, engines running, all FUEL pumps ON. flyPad > Failures > ATA 28 Fuel: activate **L TK pump 1** and **L TK pump 2**.	FUEL L TK PUMP 1+2 LO PR.
2	Press the T.O CONFIG key (ECAM control panel).	No T.O CONFIG NORMAL in the T.O memo. Deactivate the failures: T.O CONFIG gives T.O CONFIG NORMAL again (with a takeoff configuration set).
3	Centre tank empty; set the left inner tank to about 600 kg with a SimConnect tool (right inner tank above 750 kg). Wait 30 s.	FUEL L WING TK LO LVL without the -FUEL MODE SEL ... MAN line; STATUS INOP SYS L TK PUMPS.
4	Put fuel in the centre tank (FUEL MODE SEL at AUTO).	The -FUEL MODE SEL ... MAN line shows; STATUS also CTR TK FEED: MAN ONLY.

16. ENG THR LEVER ABV IDLE and ENG SAT ABOVE FLEX TEMP

> **Aircraft:** A32NX | **Where:** E/WD; thrust levers; MCDU PERF TAKE OFF page (FLEX TO TEMP) | **Status:** SAT ABOVE FLEX TEMP: tested in the sim by the user: OK, data-confirmed (on the feature branch). THR LEVER ABV IDLE: OK after a real landing (phase 9, lever 1 at CL with lever 2 in reverse: red warning and its line within 0.4 s; it stays with lever 2 back at idle and goes with lever 1 out of CL; recorded). Unit tests EngineThrustSettingAlerts.spec.ts.

What it is for

Two thrust setting alerts of the A320 FCOM (PRO-ABN-ENG) that the A32NX did not have. They need no failure: they come from the thrust levers and the FLEX TEMP.

- **ENG 1(2) THR LEVER ABV IDLE** (red, MASTER WARN, repetitive "RETARD-RETARD"; FCOM): during the landing roll (FWC phases 8 and 9), one thrust lever is above idle while the other one is in reverse, or stays above idle after the other one came out of reverse. Line: -THR LEVER 1(2) ... IDLE. The voice is heard above 40 kt and stops when all levers are at IDLE or REVERSE.
- **ENG SAT ABOVE FLEX TEMP** (amber, MASTER CAUT; FCOM): before take-off (phase 2 only) the outside air temperature is above the FLEX TEMP entered in the MCDU. Line: T.O DATA ... CHECK. No FLEX TEMP entered gives no alert.

Limitations and why

Limitation	Why	What to do
THR LEVER ABV IDLE only after a real landing.	The FWC goes to phase 9 only after a touchdown or a take-off aborted above 80 kt; a take-off power cut at a standstill goes back to phase 2.	Test it after a landing.
"At reverser deselection" has no time limit in the FCOM.	Design choice: the memory of a reverse selection is kept until both levers are at idle or the landing roll ends.	-

How to use

1. Nothing to set. Before take-off, enter a FLEX TEMP above the outside temperature (or none for TOGA).
2. After landing, bring both thrust levers to REVERSE or IDLE together; follow the RETARD call if it comes.

How to test

#	Do this	Expected result
1	A32NX at the gate, engines running, FWC phase 2. MCDU PERF TAKE OFF: enter a FLEX TO TEMP 10 °C below the outside temperature (SAT).	Within a second: amber ENG SAT ABOVE FLEX TEMP with T.O DATA ... CHECK, single chime, MASTER CAUT.
2	Enter a FLEX TO TEMP of 60.	The caution goes; the T.O memo is back.
3	After a landing, stopped on the runway or a taxiway (phase 9), both levers at IDLE, PARKING BRK ON: lever 2 to REVERSE IDLE, then lever 1 to the CL detent.	Red ENG 1 THR LEVER ABV IDLE with -THR LEVER 1 ... IDLE, MASTER WARN.
4	Lever 2 forward to IDLE (lever 1 stays at CL), then lever 1 to IDLE.	The warning stays while lever 1 is at CL, and goes once it is at IDLE.

17. Engine fan blocked failure and ENG LOW N1

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 72; ENG MASTER and ENG MODE; E/WD | **Status:** OK (recorded: N1 0 % while N2 rose; ENG 1 LOW N1 with .IF CONFIRMED and ENG MASTER 1 OFF 2.5 s after N2 35 %; core hung at 41.5 %, start aborted; alert gone at ENG MASTER OFF; normal restart without it). Unit tests EngineLowN1Alerts.spec.ts, a320.spec.ts; native test fan_blocked_start_test.cpp; Rust failure type.

What it is for

The A320 FCOM alert **ENG 1(2) LOW N1** (on ground, PRO-ABN-ENG) is for a start in which the fan does not turn (a blocked fan, for example ice). The sim cannot produce that by itself, so the branch adds a failure for it.

- New flyPad failures, ATA 72: **Engine 1 fan blocked (no N1 rotation at start)** and **Engine 2 fan blocked (no N1 rotation at start)** (ids 72020, 72021). During a start on the ground the FADEC holds the N1 at 0 and the core hangs below idle (about 41 % N2).
- ENG 1(2) LOW N1 (amber, MASTER CAUT, single chime): N2 at or above 35 % with N1 below 2 % for 2 s, on the ground; lines .IF CONFIRMED:, -THR LEVER 1(2) ... IDLE (only with the lever above idle), -ENG MASTER 1(2) ... OFF. It stays until the ENG MASTER is set OFF; inhibited in flight phases 4 to 9.

Limitations and why

Limitation	Why	What to do
The 35 % N2, 2 % N1 and 2 s values are not in the FCOM.	Design choices for "no N1 rotation".	-
ENG START FAULT may follow.	The FADEC aborts an automatic start whose core hangs, about 20 s later.	Both alerts are expected; LOW N1 stays.
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules change.	Restart the sim after installing.

How to use

1. flyPad > Failures > ATA 72 Engine: activate "Engine 1 (2) fan blocked (no N1 rotation at start)" before the engine start.
2. Start the engine as usual; follow ENG LOW N1 (ENG MASTER OFF). Clear the failure before the next start.

How to test

#	Do this	Expected result
1	A32NX on the ground, PARKING BRK ON, APU BLEED ON, engine 1 OFF, both levers at IDLE. flyPad > Failures > ATA 72: activate **Engine 1 fan blocked (no N1 rotation at start)**.	No alert yet.
2	ENG MODE selector IGN/START, ENG MASTER 1 ON. Watch engine 1 for 60 s.	N2 rises, N1 stays at 0. Once N2 is above 35 % for 2 s: amber ENG 1 LOW N1 with .IF CONFIRMED: and -ENG MASTER 1 ... OFF, single chime, MASTER CAUT. N2 hangs near 41 %; ENG 1 START FAULT may follow.
3	ENG MASTER 1 OFF.	ENG 1 LOW N1 goes (and START FAULT with it).
4	Clear the failure, wait 30 s, ENG MASTER 1 ON.	A normal start: N1 rises with N2, no LOW N1.

18. Engine and APU fire visual effects

> **Aircraft:** A32NX, A380X | **Where:** Exterior model (engine and APU exhausts); flyPad > Settings > Sim Options; flyPad > Failures > ATA 26, ATA 79 | **Status:** A32NX: tested in the sim by the user on a combined test build: OK (tests F1 engine fire, F2 fire put out by the agent, F3 APU fire smoke). A380X: tests F4, O1, K1 and L1 passed (per the commit). The engine damage smoke (MSFS failures menu) and the flyPad switch on the A32NX: not sim-tested. Unit tests EngineFireEffects.spec.ts (both aircraft), sync.spec.ts; Rust fire protection tests. Not sim-tested on this branch.

What it is for

A fire you can see from outside: flames at the exhaust of an engine on fire, smoke at the APU exhaust for an APU fire, until the fire is out. Also smoke for an engine failed from the MSFS failures menu and for an engine oil leak. A flyPad switch turns all these effects off.

- **A32NX:** flames and smoke at the exhaust of the engine on fire (flyPad **Engine 1 fire** / **Engine 2 fire**, or a fire of the MSFS failures menu); smoke at the APU exhaust for an **APU fire**. The APU exhaust point of the model moved to the tail cone tip, where the real APU exhaust is.
- The effects follow the fire itself, not the detection: nothing during a FIRE TEST, and the effect stays after the engine shutdown until the fire is out (deactivated failure, or put out by an AGENT discharge).
- Smoke at the exhaust of an engine failed from the MSFS failures menu (engine damage), and of an engine with the flyPad **Engine 1(2) oil leak** (ATA 79) while the oil flows out.
- **A380X:** the same for engines 1 to 4 and the APU; the stock MSFS engine fire, engine damage and oil leak effects are turned off so they do not show twice; the engine 2 effects (heat haze, contrails, fire) no longer vanish at the medium model detail (LOD01).
- **Fire model (both aircraft):** an AGENT discharge, or the automatic APU discharge, can also put out a fire set from the MSFS failures menu, which then ends the ENG / APU FIRE warning (see "Fire detection model and fire failures").
- **flyPad > Settings > Sim Options > "Situation Visual Effects (Engine and APU Fire, Engine Damage, Oil Leak)"**: on by default; OFF hides all these effects (L:A32NX_CONFIG_SITUATION_FX, both flyPads).

Limitations and why

Limitation	Why	What to do
Each AGENT discharge puts the fire out one time in two.	The FBW fire model (user decision).	Use AGENT 2 if the FIRE pb stays red.

After the ENG FIRE pb the engine runs on for up to 2 min 30 s.	The FIRE pb closes the LP fuel valve; the engine stops once the fuel downstream is burnt (FCOM).	-
On this branch the A380X oil leak smoke has nothing to start it.	The A380X flyPad engine oil failures (ATA 79) are on `core/failures-a380x`; this branch has only the model part.	Use a build with core/failures-a380x merged.
The A380X engine 2 effects at LOD01 / LOD02 need the exhaust nodes of the wing LOD models.	The model files are in the large-files repository, not in the code repository.	Install the matching A380X wing models.
A full MSFS restart is needed after an update.	The exterior models, aircraft.cfg, the model behaviour XML and the systems WASM modules change.	Restart the sim after installing.

How to use

1. Keep flyPad > Settings > Sim Options > Situation Visual Effects ON (default).
2. Set a fire (flyPad > Failures > ATA 26, or the MSFS failures menu) and look at the aircraft from outside; fight the fire with the ECAM procedure.
3. Set the switch OFF to hide the effects (the fire itself and its warnings are not changed).

How to test

Setup: A32NX on a stand, both engines at idle, PARKING BRK ON, APU running; outside (drone) view of the left engine from behind.

#	Do this	Expected result
1	F1: flyPad > Failures > ATA 26: activate **Engine 1 fire**; wait 20 s. Then deactivate it; wait 10 s.	Flames and smoke at the left engine exhaust; gone within a few seconds of the deactivation, with the ENG 1 FIRE warning.
2	F2: activate **Engine 1 fire** again. ENG 1 FIRE pb: lift the guard and push it; AGENT 1; wait 30 s (AGENT 2 if the FIRE pb is still red). Then deactivate the failure, FIRE pb back in.	The flames stop within a few seconds of the agent that puts the fire out, together with the ENG 1 FIRE pb red light. Engine 1 runs on for up to 2 min 30 s (normal).
3	F3: view of the tail from the side. Activate **APU fire**; look 20 s; deactivate it.	Smoke from the tip of the tail cone; gone after the deactivation. On the ground the APU bottle discharges by itself and puts the fire out one time in two: smoke stopping early is correct.
4	Optional (not sim-tested): MSFS menu (Esc > Failures > Engines > Engine 1): fail engine 1. Then flyPad > Settings > Sim Options: Situation Visual Effects OFF, then ON.	Smoke at the engine 1 exhaust; gone with the switch OFF, back with it ON.
5	A380X: engines off, APU running. flyPad > Failures > ATA 26: **Fire - Engine 1**; wait 15 s; deactivate; activate again, then ENG 1 FIRE pb and AGENT 1 (AGENT 2 if still red).	Flames and smoke at the left outer engine; gone with the deactivation; they stop with the FIRE pb red light when the agent puts the fire out.

19. FCOM fixes: DIRECT LAW trim, GPWS mode 2, CAB PRESS and oil temperature indications

> **Aircraft:** A32NX | **Where:** E/WD, STATUS; SD CAB PRESS and ENGINE pages; GPWS; overhead RAT & EMER GEN MAN ON pb | **Status:** Tested in the sim by the user on a combined test build: OK, recorded: CAB PRESS pulses (redone with short taps), ENG SHUT DOWN with the reverser unlocked, oil temperature amber after 15 min (caution 900.6 s after 140 °C). DIRECT LAW trim line,

> GPWS mode 2 hold, THR LEVER FAULT ground lines and RAT tooltip: not sim-tested (unit tests PressIndications.spec.ts, EngOilIndications.spec.ts, EngineFailAlerts.spec.ts, FadecReverserAlerts.spec.ts, FlightControlLawAlerts.spec.ts; Rust egpws tests).

What it is for

Deviations from the A320 FCOM found by the FCOM gap audit:

- **F/CTL DIRECT LAW** no longer asks for MAN PITCH TRIM USE when the green and yellow hydraulic systems are both at low pressure: the trim cannot move (FCOM "EXCEPT IF HYD Y + G SYS LO PR").
- **GPWS mode 2:** after a PULL UP, leaving the warning envelope keeps the TERRAIN aural and the red PULL UP lights until the aircraft has climbed 300 ft (45 s at most).
- **SD CAB PRESS page:** the CAB ALT value pulses from 8800 ft until back at 8600 ft, the V/S value pulses above 1750 ft/min until back at 1650 ft/min, delta P pulses above 1.5 PSI in approach; the needles no longer pulse.
- **SD ENGINE page:** the oil temperature turns amber after 15 min between 140 and 155 °C (the timer restarted at every temperature change and never ran out).
- **ENG 1(2) SHUT DOWN** with the reverser unlocked: .IF BUFFET: -MAX SPEED ... 240 KT on the E/WD; STATUS MAX SPEED 300/.78 and the approach procedure (IF BUFFET:, FOR LDG USE FLAP 3, GPWS LDG FLAP 3 ON, APPR SPD VREF+10 KT, LDG DIST PROC APPLY). **ENG THR LEVER FAULT** on the ground: ENG 1(2) IDLE POWER ONLY.
- The RAT & EMER GEN MAN ON pushbutton tooltip no longer says "(Inop.)".

Limitations and why

Limitation	Why	What to do
The cabin altitude must stay below 9550 ft in the CAB PRESS test.	Above it the CAB PR EXCESS CAB ALT warning comes.	Use short taps of the MAN V/S CTL switch.
A full MSFS restart is needed after an update.	The systems WASM module (GPWS) and the cockpit XML (tooltip) change.	Restart the sim after installing.

How to use

1. Nothing to set: the alerts and the SD pages follow the FCOM by themselves.
2. Call the SD CAB PRESS (PRESS key) or ENGINE (ENG key) page to watch the pulsing values.

How to test

#	Do this	Expected result
1	Cruise at FL 250 or above, AP ON, pressurization AUTO. SD CAB PRESS page. MODE SEL pb (MAN light on); MAN V/S CTL UP in short taps until CAB V/S about +2000 ft/min; wait until CAB ALT 8900 ft; DN until about -500 ft/min; wait until CAB ALT 8500 ft; MODE SEL pb back.	The V/S digits pulse green above 1750 ft/min and stop at 1650 ft/min or less; the CAB ALT digits pulse from 8800 ft and stop at 8600 ft; the needles stay steady green.
2	Cruise above FL 200, A/THR ON. flyPad > Failures > ATA 78: **Reverser 1 unlocked (engine at idle)**; thrust lever 1 IDLE; ENG MASTER 1 OFF; STS key.	ENG 1 SHUT DOWN with .IF BUFFET: -MAX SPEED ... 240 KT after -TCAS MODE SEL ... TA (ENG 1 REVERSE UNLOCKED also shown). STATUS: MAX SPEED 300/.78, APPR PROC:, IF BUFFET:, FOR LDG USE FLAP 3, GPWS LDG FLAP 3 ON, APPR SPD VREF+10 KT, LDG DIST PROC APPLY. Deactivate the failure, relight.
3	On the ground, engines at idle, SD ENGINE page. flyPad > Failures > ATA 79: **Engine 1 oil overheat**; thrust lever 1 forward until the oil reads 145-150 °C; wait 16 min.	The oil temperature pulses green, then about 15 min after 140 °C turns steady amber together with ENG 1 OIL HI TEMP. Lever IDLE, deactivate the failure.

4	Optional (not sim-tested): hover over the RAT & EMER GEN MAN ON pb (overhead EMER ELEC PWR panel).	The tooltip has no "(Inop.)".

20. FCOM fixes: FADEC relight, automatic ignition, start faults, gravity feed, IR alignment

> **Aircraft:** A32NX | **Where:** ENG MODE selector, ENG MASTER, overhead ANTI ICE ENG pb, ENG MAN START; overhead FUEL and ADIRS panels; E/WD, SD ENGINE and FUEL | **Status:** Tested in the sim by the user on a combined test build, recorded: FADEC relight after a flameout OK (relight at the IGN selection, idle in 35 s); IR alignment 10 min OK (601 s). Automatic continuous ignition with the engine anti-ice: FAILED : L:A32NX_ENGINE_1_CONTINUOUS_IGNITION went to 1 but the E/WD IGNITION memo did not show; cause open. Start fault order, manual start at 20 % N2, starter shaft shear in flight and gravity feed: not sim-tested (Rust tests, native cross_feed_burn_test.cpp).

What it is for

Engine, fuel and ADIRS deviations from the A320 FCOM:

- **FADEC relight:** in flight, a flameout of a running engine starts a FADEC relight attempt at once (FCOM PRO-ABN-ENG ENG 1(2) FAIL: "Selection of continuous ignition confirms the immediate relight attempt made by the FADEC"). The attempt lasts 30 s from the flameout and lights up inside the relight envelope (see "Engine flameout and seizure failures") once the continuous ignition is on, with ENG MODE IGN or selected automatically: no ENG MASTER cycle. In the quick relight zone (230-270 kt up to 20 000 ft, N2 above 15 %) it lights up at once at any ENG MODE position (design choice: the time since the flameout stands for the MASTER OFF time). No light-up within 30 s: the engine stays out until a crew relight (MASTER OFF then ON). Not on the ground, not for a seizure.
- **Automatic continuous ignition:** the FADEC selects the continuous ignition by itself while the engine anti-ice is on (DSC-70-80-30); the IGNITION memo and the SD igniters should show it.
- **Start faults:** on the ground, after a stall or an EGT over limit the FADEC runs a new start first and dry cranks only when no new start is left; a manual start lights up from 20 % N2; a starter that fails during an in-flight relight gives ENG START FAULT STARTER SHAFT SHEAR with WINDMILL START ONLY for the rest of the flight (details in "Engine ignition and start failures").
- **Gravity feed:** with the X FEED open and all four wing tank pumps off, each engine burns the fuel of its own wing tank (before, the burned fuel was taken from no tank).
- **IR alignment:** a complete IR alignment takes 10 min at any latitude, as in the A320 FCOM (before, 5 to 17 min with the latitude, like the A380).
- The MCDU PROG REC MAX with one engine out is not on this branch (it is on `core/efb-redesign`).

Limitations and why

Limitation	Why	What to do
The IGNITION memo did not show with the automatic continuous ignition (sim).	Cause open: the ignition variable was set, and the FWC review found nothing in this code that removes the memo.	Select ENG MODE IGN for the memo; the relight itself works with the automatic ignition (Rust tests).
The automatic ignition only follows the engine anti-ice.	Design choice: the MSFS ENG ANTI ICE stands for the valve; the other conditions of the FCOM automatic selection are a figure, not in the text.	-
In the quick relight zone a flameout relights at once by itself.	Design choice of the FADEC attempt (see above).	Test the crew relight outside that zone.
A full MSFS restart is needed after an update.	The systems and FADEC WASM modules change.	Restart the sim after installing.

How to use

1. After an ENG FAIL in flight, follow the ECAM: ENG MODE SEL IGN within 30 s lets the FADEC relight the engine inside the envelope; after 30 s use the crew relight (ENG MASTER OFF then ON).
2. For the IR alignment, flyPad > Settings > Realism > ADIRS align time: Real; IR selectors to NAV.
3. For gravity feeding, FUEL X FEED pb ON with the wing tank pumps off.

How to test

#	Do this	Expected result
1	G9: A32NX cold and dark, ADIRS align time Real. IR 1, 2 and 3 selectors to NAV.	IRS IN ALIGN memo counting down; the alignment ends after about 10 min at any latitude (OK: 601 s).
2	G1: about FL 100, 290 kt, A/THR ON, ENG MODE NORM, engine anti-ice OFF. flyPad > Failures > ATA 72: **Engine 1 flameout (crew relight possible)**; wait 10 s; ENG MODE IGN/START.	Amber ENG 1 FAIL; within about 2 s of IGN/START engine 1 N2 rises back to idle without touching ENG MASTER 1, and ENG 1 FAIL clears (OK). Deactivate the failure, ENG MODE NORM.
3	G3 (FAILED): cruise, ENG MODE NORM, SD ENGINE page. Overhead ANTI ICE ENG 1 pb ON; wait 10 s; then OFF.	Expected: IGN A (or B) for engine 1 on the SD and the green IGNITION memo, gone at the pb OFF. In the test the IGNITION memo did not show.
4	Optional G2 (not sim-tested): as G1, but wait 35 s before IGN/START.	No relight; ENG MASTER 1 OFF then ON still relights.
5	Optional G4 (not sim-tested): ground, APU BLEED ON, ENG MODE IGN/START. ATA 80: **Engine 1 stall during the start**; ENG MASTER 1 ON.	ENG 1 START FAULT STALL with NEW START IN PROGRESS first, then AUTO CRANK IN PROGRESS, then -ENG MASTER 1 ... OFF.
6	Optional G5 (not sim-tested): ground, APU BLEED ON, ENG MODE IGN/START. ENG MAN START 1 ON; when N2 stops rising, ENG MASTER 1 ON.	Light-up (FF and EGT rise) at 20-21 % N2, no IGNITION FAULT.
7	Optional G6 (not sim-tested): FL 100, 200 kt, APU BLEED ON. **Engine 1 starter shaft shear**, then **Engine 1 flameout (crew relight possible)**; ENG MODE IGN/START; ENG MASTER 1 OFF, 30 s, ON; wait 12 s.	ENG 1 START FAULT STARTER SHAFT SHEAR; STATUS ENG 1 WINDMILL START ONLY.
8	Optional G8 (not sim-tested): note both wing tank quantities (SD FUEL). ATA 28: **L TK pump 1**, **L TK pump 2**, **R TK pump 1**, **R TK pump 2**; FUEL X FEED pb ON; wait 5 min.	Both wing tanks go down (before: neither moved).

21. ELEC bus, ESS bus and TR alerts; DC ESS supply in DC BUS 1+2 fault

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 24 Electrical Power; overhead ELEC panel; E/WD, STATUS, SD ELEC | **Status:** ELEC alerts: tested in the sim by the user on a combined test build: OK (no nuisance alert in a start-up; AC BUS 1 FAULT, DC EMER CONFIG and DC BUS 1+2 FAULT; recorded). DC ESS supply: OK (recorded). Unit tests ElecBusAlerts.spec.ts; Rust tests electrical/mod.rs.

What it is for

New ELEC alerts of the A320 FCOM (PRO-ABN-ELEC) with their procedure and STATUS lines, triggered by the Rust electrical network: **AC BUS 1(2) FAULT**, **AC ESS BUS FAULT**, **AC ESS BUS SHED**, **AC ESS BUS ALTN**, **DC BUS 1(2) FAULT**, **DC BUS 1+2 FAULT**, **DC ESS BUS FAULT**, **DC ESS BUS SHED**, **DC BAT BUS FAULT**, **DC EMER CONFIG**, **ESS BUSES ON BAT**, **TR 1(2) FAULT**, **GEN 1(2) OFF**, **BAT 1(2) OFF** and **EMER GEN 1 LINE OFF**.

- DC EMER CONFIG covers DC BUS 1, DC BUS 2, DC BUS 1+2 and DC ESS BUS FAULT / SHED; DC BUS 1+2 FAULT covers DC BUS 1 and DC BUS 2 FAULT; an ESS BUS FAULT covers its SHED alert; with the ESS buses on the batteries

no SHED alert (the shedding is normal). In ELEC EMER CONFIG the bus losses that make that configuration are not shown again.

- **DC ESS supply (fix):** with DC BUS 1 and 2 lost (DC BUS 1+2 FAULT) the ESS TR now supplies the DC ESS BUS (FCOM DSC-24-10-30-30): FAC 1, FMGC 1 / AP 1, FCDC 1, FCU channel 1 and the other DC ESS users stay available; only FAC 2 / AP 2 / FMGC 2 are lost. Before, the DC ESS BUS was lost too.
- **No power break (fix):** the DC BAT BUS and the DC ESS BUS behind it lost power for one update each time a DC bus was lost or came back: the FCU restarted and dropped the AP and A/THR. They now stay powered.

Limitations and why

Limitation	Why	What to do
The flight phase inhibitions and alert levels are general Airbus values.	The FCOM gives them as figures that the text copy drops (design choice).	-
No FUEL CONSUMPT INCRSD / FMS PRED UNRELIABLE lines in these procedures.	The condition ("affects the aerodynamic characteristics") is not computed.	-
After DC BUS 1+2 is restored in flight, FAC 2 stays in fault.	As on the aircraft: the FCOM asks for FAC OFF then ON.	FAC 2 pb OFF then ON.
In DC EMER CONFIG the cabin climbs.	The packs and the cabin pressure controllers lose power (seen in the sim, as the FCOM gives).	Descend as the procedure asks.
The STATUS of DC BUS 1+2 FAULT does not page over two pages here.	The STATUS page paging needs the fork-only CLR / RCL rework, which this branch does not have.	-
The DC ESS test ran with a SEC power-up self-test fix that is not on this branch.	That fix needs code this branch does not have.	A SEC FAULT after the DC buses come back is not checked here.
A full MSFS restart is needed after an update.	The systems WASM module changes (DC ESS fix).	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 24 Electrical Power and activate a bus failure ("AC 1", "DC 1", "DC 2", "DC ESS"...).
2. Follow the ECAM and STATUS; deactivate the failure after the test (FAC OFF then ON if it stays in fault).

How to test

#	Do this	Expected result
1	E1: cold and dark at the gate. BAT 1 and 2 AUTO, EXT PWR ON; APU MASTER SW ON, START, APU BLEED ON at AVAIL; EXT PWR OFF; start engine 2, then engine 1; APU BLEED OFF, APU MASTER SW OFF.	Never an AC / DC BUS, ESS, SHED, TR, ESS BUSES ON BAT, APU BLEED FAULT, X BLEED FAULT, ENG BLEED NOT CLSD or BLEED n OFF alert.
2	E2: cruise, AP ON. ATA 24: **AC 1**; wait 10 s; VENT BLOWER pb OVRD; STS key. Then deactivate **AC 1**, BLOWER AUTO.	After about 1 s amber ELEC AC BUS 1 FAULT, chime, MASTER CAUT, SD ELEC, -BLOWER ... OVRD; then AC ESS BUS ALTN (no chime); never AC / DC ESS BUS FAULT. STATUS: LDG DIST PROC APPLY, SLATS SLOW, CAT 2 ONLY, INOP SYS. The AP may disconnect during the AC ESS transfer.

3	Cruise. ATA 24: **DC 1**, then **DC 2**, then **DC ESS**; wait 10 s; then deactivate the three.	Amber DC EMER CONFIG with -EMER ELEC PWR MAN ON and the DC BUS 1+2 lines, DC BAT BUS FAULT, amber LAND ASAP; no separate DC BUS 1 / DC BUS 2 / DC BUS 1+2 / DC ESS BUS FAULT.
4	Cruise, AP and A/THR ON. ATA 24: **DC 1** and **DC 2**; wait 30 s; then deactivate both.	ELEC DC BUS 1+2 FAULT; the DC ESS BUS stays powered (SD ELEC), FAC 1 healthy, normal law, AP 1 and A/THR stay engaged; only FAC 2 FAULT. After the restore: no AP OFF; FAC 2 FAULT until FAC 2 pb OFF then ON (OK).

22. Electrical failures: battery thermal runaway, BCL, IDG oil

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 24 Electrical Power; overhead ELEC panel (BAT, IDG, GALY & CAB pb); E/WD, STATUS, SD ELEC | **Status:** Tested in the sim by the user on a combined test build: OK (P1 battery thermal runaway, recorded; P2 IDG oil overheat, judged by the user, no IDG variable recorded). BCL, IDG oil leak, GEN OVERLOAD and STATIC INV FAULT: not sim-tested. Unit tests ElecSystemAlerts.spec.ts, IdgIndications.spec.ts; Rust tests.

What it is for

flyPad failure (ATA 24)	What the code does
Battery 1(2) (thermal runaway)	A cell short: the charging current keeps rising (2 A per second); the BCL sees it above 12 A for 2 s, opens the battery contactor and latches the fault: ELEC BAT 1(2) FAULT, BAT pb FAULT light. BAT pb OFF resets it.
BCL 1(2)	The battery charge limiter fails and keeps the battery contactor open: ELEC BCL 1(2) FAULT, BAT pb FAULT light.
IDG 1(2) oil leak	The drive loses 1 % of its oil each second while it turns: IDG LO PR about 50 s later (below half of the oil), then the oil heats up.
IDG 1(2) oil overheat	The oil cooling is lost: the IDG temperature rises 0.5 °C/s; ELEC IDG 1(2) OIL OVHT above 180 °C (about 3.5 min), and the IDG disconnects by itself at 200 °C (about 40 s later).

- Alerts (FCOM PRO-ABN-ELEC): **ELEC BAT 1(2) FAULT** and **BCL 1(2) FAULT** (crew awareness; STATUS APU BAT START NOT AVAIL, INOP SYS BAT 1(2) / BCL 1(2)); **ELEC IDG 1(2) OIL LO PR** / **OIL OVHT** (-IDG 1(2) ... OFF; STATUS CAT 3 SINGLE ONLY, INOP SYS MAIN GALLEY if the APU GEN is not online, GEN 1(2), CAT 3 DUAL); **ELEC GEN 1(2) / APU GEN OVERLOAD** (a generator above 100 % of its rated load: -GALY/CAB ... OFF, INOP SYS GALY/CAB); **ELEC STATIC INV FAULT** (the existing **Static Inverter** failure).
- Cockpit: the BAT, IDG and GALY & CAB pb FAULT lights.
- SD ELEC page: IDG LO PR, the IDG temperature pulsing green from 147 °C and amber above 180 °C. The normal IDG oil now reads about 50-95 °C (it read 160-210 °C at high power).

Limitations and why

Limitation	Why	What to do
Inhibitions: BAT, BCL, GEN OVERLOAD and STATIC INV FAULT in flight phases 3, 4, 5, 7, 8; IDG OIL LO PR / OVHT in 1, 4, 5, 7, 8, 10.	The FCOM figures (2019 FCOM PDF).	Test with the engines running at the gate or in cruise.
The runaway, leak and heating rates are design values.	The FCOM gives none.	-

A disconnected IDG cannot be reconnected in flight.	As on the aircraft (fusible device).	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 24 Electrical Power and activate a battery, BCL or IDG failure with the engines running.
2. Follow the ECAM; after a battery fault set the BAT pb OFF then AUTO once the failure is cleared.

How to test

Setup: on the ground, both engines running, BAT 1 and 2 AUTO.

#	Do this	Expected result
1	P1: ATA 24: **Battery 1 (thermal runaway)**; wait 15 s; STS key.	Amber ELEC BAT 1 FAULT, chime, BAT 1 pb FAULT light, BAT 1 contactor open on the SD ELEC page. STATUS: APU BAT START NOT AVAIL, INOP SYS BAT 1.
2	Deactivate it; BAT 1 pb OFF, then AUTO.	The light and the alert go.
3	P2: ATA 24: **IDG 2 oil overheat**; watch the SD ELEC page for 5 min.	The IDG 2 temperature rises, pulses green from 147 °C, turns amber above 180 °C with ELEC IDG 2 OIL OVHT and the IDG 2 FAULT light, then the IDG disconnects by itself at 200 °C (DISC on the SD).
4	Optional (not sim-tested): ATA 24: **IDG 1 oil leak**; wait 1 min.	ELEC IDG 1 OIL LO PR, IDG 1 FAULT light, LO PR on the SD ELEC page.

23. Pneumatic failures with BMC leak and bleed monitoring

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 36 Pneumatic; overhead AIR COND panel; E/WD, SD BLEED | **Status:** Tested in the sim by the user on a combined test build: OK (left wing bleed leak; engine bleed overheat in the climb; recorded). The other ATA 36 failures: not sim-tested. Unit tests BleedLeakMonitoringAlerts.spec.ts, a320.spec.ts; Rust tests pneumatic.rs.

What it is for

flyPad failures (ATA 36)	What the code does
Engine 1(2) bleed valve stuck closed / stuck open (no regulation)	The valve stays closed / fully open whatever the BMC commands; stuck open: ENG BLEED FAULT by overpressure at climb power, ENG BLEED NOT CLSD at engine start, with APU BLEED ON or the engine stopped.
Engine 1(2) HP valve stuck closed	No HP bleed at low thrust: ENG HP VALVE FAULT, low bleed pressure at idle.
Engine 1(2) precooler fan air valve stuck closed (overheat) / stuck open (low temperature)	Closed: ENG BLEED HI TEMP, then FAULT above climb power. Open: ENG BLEED LO TEMP in flight with the wing anti-ice at low thrust.
X BLEED valve stuck closed / stuck open; APU bleed valve stuck closed / stuck open	The valve stays where it is: X BLEED FAULT or APU BLEED FAULT (see "AIR bleed cautions").
Engine 1(2) pylon bleed leak, Left / Right wing bleed leak, APU bleed duct leak	Hot air leaks while the duct is pressurised; the leak detection loops close the valves that feed it (the pylon loop at most thrust settings, not always at ground idle; the wing loops at any thrust; the APU loop with APU BLEED ON).

Engine 1(2) pylon, Left / Right wing loop A / B, APU leak detection loop	LEAK DET FAULT (a wing: both loops of that wing failed).
BMC 1 (2) (bleed monitoring computer)	BLEED MONIT SYS 1(2) FAULT; the other BMC keeps the overpressure, overheat and wing leak monitoring; the engine bleed valve of the failed BMC no longer closes by itself and its FAULT light is lost. Both: BLEED MONITORING FAULT.

- The BMCs now detect leaks, HP valve faults and an overpressure above 57 PSI, light the ENG BLEED and APU BLEED FAULT lights and close the affected valves (for a wing leak, the engine bleed and X BLEED valves).
- New AIR cautions (amber, SD BLEED page): **L(R) WING LEAK** and **ENG 1(2) BLEED LEAK** (-ENG 1(2) BLEED ... OFF, -APU BLEED ... OFF on the left side, -X BLEED ... SHUT, -WING ANTI ICE ... OFF, AVOID ICING CONDITIONS), **APU BLEED LEAK** (-APU BLEED ... OFF), **ENG 1(2) / L(R) WNG / APU LEAK DET FAULT**, **ENG 1(2) HP VALVE FAULT**, **BLEED MONIT SYS 1(2) FAULT** and **BLEED MONITORING FAULT**.

Limitations and why

Limitation	Why	What to do
No APU BLEED LEAK "APU LEAK FED BY ENG" variant, no LEFT / RIGHT LEAK variants of ENG 1+2 BLEED FAULT.	Not modelled: the Rust APU duct leak only blows APU air.	-
The flight phase inhibitions are the FCOM PDF figures.	Read on the FCOM PDF pages.	Test on the ground or in cruise.
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 36 Pneumatic and activate a valve, leak, loop or BMC failure.
2. Follow the ECAM (bleed pb OFF, X BLEED SHUT...); deactivate the failure and set the bleed pb back ON.

How to test

#	Do this	Expected result
1	K1: on the ground, both engines at idle, APU off. ATA 36: **Left wing bleed leak**; wait 10 s.	Amber AIR L WING LEAK, ENG 1 BLEED FAULT light, engine 1 bleed valve and X BLEED closed (SD BLEED).
2	ENG 1 BLEED pb OFF; deactivate the leak; wait 15 s; ENG 1 BLEED pb ON.	The alert and the FAULT light go out.
3	K2: climbing above 1 500 ft at CLB thrust. ATA 36: **Engine 2 precooler fan air valve stuck closed (overheat)**; wait 20 s.	ENG 2 BLEED HI TEMP for a few seconds, then AIR ENG 2 BLEED FAULT, FAULT light on, engine 2 bleed valve closed.
4	ENG 2 BLEED pb OFF; deactivate the failure; wait 30 s; ENG 2 BLEED pb ON.	The FAULT clears and the valve reopens.

24. AIR bleed cautions

> **Aircraft:** A32NX | **Where:** Overhead AIR COND panel (ENG BLEED, APU BLEED pb, X BLEED selector); E/WD, STATUS, SD BLEED | **Status:** Tested in the sim by the user on a combined test build: OK (tests E1: no nuisance bleed alert at start-up, and B1: AIR BLEED 1 OFF; recorded). The other bleed cautions: tested through K1 / K2 of the pneumatic failures, otherwise not sim-tested. Unit tests BleedAlerts.spec.ts, ApuMemos.spec.ts.

What it is for

New AIR cautions of the A320 FCOM (PRO-ABN-AIR) with their procedures and STATUS, from the flags of the Rust pneumatic system: **ENG 1(2) BLEED FAULT** and **ENG 1+2 BLEED FAULT**, **ENG 1(2) BLEED LO TEMP** and **ENG 1+2 BLEED LO TEMP**, **ENG 1(2) BLEED HI TEMP**, **ENG 1(2) BLEED NOT CLSD**, **BLEED 1(2) OFF**, **APU BLEED FAULT** and **X BLEED FAULT**, with the X BLEED and APU BLEED STATUS lines.

- ENG BLEED FAULT: engine running with a bleed pressure above 57 PSI, or a bleed temperature above 257 °C for 55 s, 270 °C for 15 s or 290 °C for 5 s (the BMC overheat monitor). ENG BLEED HI TEMP: precooler outlet above 240 °C. ENG BLEED LO TEMP: bleed air below 150 °C in flight with the wing anti-ice ON.
- BLEED 1(2) OFF: the ENG BLEED pb OFF with the engine running, shown in flight phases 2 and 6 only.
- Fix: the APU BLEED memo and the bleed alerts read the APU BLEED pb variable the cockpit writes (L:A32NX_OVHD_PNEU_APU_BLEED_PB_IS_ON); the old name was never written, so the APU BLEED memo never showed.

Limitations and why

Limitation	Why	What to do
The FCOM EMER CANC and RCL behaviour of the same change is not on this branch.	It needs the fork-only E/WD CLR / RCL rework, which no core branch has: EMER CANC and RCL keep the FBW behaviour here.	-
The valve confirmation times (APU BLEED FAULT 10 s, X BLEED FAULT 5 s, BLEED NOT CLSD 10 s) are design values.	The FCOM gives none.	-
The flight phase inhibitions are the FCOM PDF figures.	Read on the FCOM PDF pages.	Test on the ground with the engines running, or in cruise.

How to use

1. Nothing to set: the cautions follow the bleed system. Use the ATA 36 failures (previous feature) to make the bleed faults.
2. Follow the ECAM and STATUS.

How to test

#	Do this	Expected result
1	On the ground, both engines running, no alert (FWC phase 2). ENG 1 BLEED pb OFF; wait 3 s.	Amber AIR BLEED 1 OFF, single chime, MASTER CAUT, SD BLEED page.
2	ENG 1 BLEED pb back ON.	The caution goes.
3	Optional (not sim-tested on its own): APU AVAIL, APU BLEED pb ON.	Green APU BLEED memo on the E/WD in place of APU AVAIL (it never showed before the fix).

25. HYD and APU shutdown alerts

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 29 Hydraulic Power; overhead HYD and APU panels, APU FIRE pb; E/WD, STATUS, SD HYD | **Status:** Tested in the sim by the user on a combined test build: OK (engine start without nuisance HYD alerts and APU EMER SHUT DOWN;: blue then yellow reservoir leak; recorded). Unit tests HydraulicAlerts.spec.ts, ApuShutdownAlerts.spec.ts, StatusMessages.spec.ts.

What it is for

- New HYD cautions and warnings of the A320 FCOM (PRO-ABN-HYD) with their procedure and STATUS lines: **B (Y) ELEC PUMP LO PR / OVHT**, **G ENG 1 (Y ENG 2) PUMP LO PR**, **B / G / Y RSVR LO AIR PR**, **RSVR LO LVL** and **SYS LO PR**, **PTU FAULT**, and the red **B+Y**, **G+B**, **G+Y SYS LO PR** warnings (with red LAND ASAP) that replace the single system alerts.
- The RSVR OVHT procedures show the "x SYS LO PR" lines; RSVR OVHT, RAT FAULT and PTU FAULT now have their STATUS (INOP SYS, APPR PROC). No pump caution on the ground with its engine stopped.
- New **APU AUTO SHUT DOWN** (an automatic shutdown other than a fire) and **APU EMER SHUT DOWN** (APU FIRE pb pushed, or an APU fire on the ground) cautions: -MASTER SW ... OFF, STATUS INOP SYS APU; they end when the APU has stopped after the MASTER SW OFF.
- A STATUS line given by two alerts is shown once.

Limitations and why

Limitation	Why	What to do
The flight phase inhibitions and alert levels are general Airbus values.	The FCOM gives them as figures that the text copy drops (design choice), except the notes the text keeps.	-
No SECONDARY FAILURES lines (*F/CTL, *WHEEL) and no option lines in STATUS.	Not modelled (design choice).	-
No APU EMER SHUT DOWN while the APU FIRE warning shows.	Design choice: the fire procedure has the MASTER SW OFF line too.	-
A reservoir leak does not refill.	The leaked fluid stays lost.	End or reload the flight after the test.
Watch items: an ENG FAIL in flight with the ENG MASTER still ON also brings the G ENG 1 (Y ENG 2) PUMP LO PR; AC BUS 1 FAULT also brings HYD B SYS LO PR.	The FCOM inhibits the pump caution only on the ground.	-

How to use

1. Open flyPad > Failures > ATA 29 Hydraulic Power and activate a reservoir or pump failure; or push the APU FIRE pb with the APU running.
2. Follow the ECAM and STATUS.

How to test

#	Do this	Expected result
1	Y1: cold and dark at the gate, APU running, APU BLEED ON. Start engine 2, then engine 1; ENG MODE NORM.	No HYD caution at any moment (no ENG PUMP, ELEC PUMP, SYS LO PR or PTU FAULT).
2	Push the APU FIRE pb (no fire); STS key; then APU MASTER SW OFF; APU FIRE pb back in.	Amber APU EMER SHUT DOWN with -MASTER SW ... OFF, no APU FIRE warning; STATUS INOP SYS APU. The line goes at MASTER SW OFF and the caution ends when the APU has stopped.

3	Y2 (last in the flight): cruise, AP ON. ATA 29: **Blue reservoir leak**; wait for the caution; BLUE ELEC PUMP pb OFF; STS key.	HYD B RSVR LO LVL, single chime, -BLUE ELEC PUMP ... OFF; no separate HYD B SYS LO PR. After the pb OFF: B SYS LO PR, FUEL CONSUMPT INCRSD, FMS PRED UNRELIABLE. STATUS: LDG DIST PROC APPLY, SLATS SLOW, CAT 3 SINGLE ONLY; INOP SYS BLUE HYD, B ELEC PUMP.
4	ATA 29: **Yellow reservoir leak**; wait for the warning.	Red HYD B+Y SYS LO PR with the continuous chime and red LAND ASAP, MAX SPEED 320/.77, MANEUVER WITH CARE; the B and Y RSVR LO LVL cautions go.

26. F/CTL surface cautions and STATUS

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 27 Flight Controls; E/WD, STATUS, SD F/CTL | **Status:** Tested in the sim by the user on a combined test build: OK (tests C1 left elevator and aileron lost, C2 ELAC 1+2 FAULT; recorded). Unit tests FlightControlSurfaceAlerts.spec.ts, FlightControlStatus.spec.ts.

What it is for

- New F/CTL cautions from the flight control computers: **L(R) ELEV FAULT**, **L(R) AIL FAULT**, **SPLR FAULT**, **ELEV SERVO FAULT**, **AIL SERVO FAULT** and **ELAC 1(2) PITCH FAULT**, with the FCOM phase inhibitions. A loss caused by a hydraulic system or a failed computer stays with the HYD or ELAC / SEC alert.
- Both ELACs failed: one **ELAC 1+2 FAULT** procedure instead of two ELAC FAULT alerts. ALTN LAW shows SPD BRK DO NOT USE for an L or R elevator fault, as in the FCOM. FCDC 1+2 FAULT is no longer inhibited in phase 3.
- STATUS: ALTN LAW, DIRECT LAW, ELAC, SEC, FCDC, elevator, aileron and spoiler faults show their FCOM limitations, approach procedure, information and INOP SYS lines.

Limitations and why

Limitation	Why	What to do
SPLR FAULT, AIL / ELEV SERVO FAULT and ELAC PITCH FAULT cannot be made in the sim.	No flyPad failure for the servos.	-

How to use

1. Open flyPad > Failures > ATA 27 Flight Controls and activate ELAC, SEC or FCDC failures.
2. Follow the ECAM and STATUS; an ELAC may need its pb OFF then ON after the failure is cleared.

How to test

Setup: on the ground, both engines running.

#	Do this	Expected result
1	C1: overhead HYD panel PTU pb OFF, then ENG 1 PUMP pb OFF.	Amber HYD G ENG 1 PUMP LO PR with the G SYS LO PR line (no separate HYD G SYS LO PR).
2	ATA 27: **ELAC 1**, then **SEC 1**; STS key.	ELAC 1 FAULT, SEC 1 FAULT, then amber F/CTL L ELEV FAULT and F/CTL L AIL FAULT; no R ELEV, no SERVO FAULT. INOP SYS lists L ELEV and L AIL.
3	Deactivate ELAC 1 and SEC 1; ENG 1 PUMP and PTU pb back ON.	Everything goes (ELAC 1 may need its pb OFF then ON).

4	C2: ATA 27: **ELAC 1**, then **ELAC 2**; STS key.	One amber F/CTL ELAC 1+2 FAULT (-ELAC 1 OFF THEN ON, -ELAC 2 OFF THEN ON, .IF BOTH ELAC FAULT : ...). STATUS: MAX SPEED 320 KT, ALTN LAW : PROT LOST, WHEN L/G DN : DIRECT LAW; INOP SYS F/CTL PROT, L+R AIL, ELAC 1+2.
5	Deactivate ELAC 2, then ELAC 1.	Back to F/CTL ELAC 1 FAULT only, then no alert.

27. FAC actuator failures, YAW DAMPER / RUD TRIM / RUD TRV LIM cautions

Aircraft: A32NX | **Where:** flyPad > Failures > ATA 22 Auto Flight; overhead FLT CTL panel (FAC pbs); E/WD, STATUS | **Status:** Not sim-tested (unit tests: FacActuatorAlerts.spec.ts, 17 tests; native fac_actuator_failure_test.cpp).

> **NOTE**
> This feature and the next ones up to "DC BUS 2 takes over DC BUS 1 after 5 s" came to this branch later: the A32NX parts of the FCOM alert additions, the new voices, the exceedance marks fix and the DC BUS 1 transfer. **None of them is sim-tested**: they are checked by unit tests and Rust tests only. Three commits of that set change nothing for the user: a CHANGELOG move (the HYD and APU SHUT DOWN lines into the current list), a spec fix (the anti-ice cautions spec sets the normal fuel, cargo door and landing elevation state) and a Rust lint fix-up (an unused host stub of the systems WASM removed).

What it is for

The FCOM gap audit found the FAC actuator failures missing: the FAC inputs for its actuators were always "healthy". Most other P1 items (HYD single and dual system cautions, PTU FAULT, ELAC and elevator faults, F/CTL STATUS, APU AUTO / EMER SHUT DOWN) were already on this branch: see "HYD and APU shutdown alerts" and "F/CTL surface cautions and STATUS".

- Six new failures, ATA 22: the yaw damper actuators 1 and 2, the rudder trim actuator motors 1 and 2 and the rudder travel limiter actuator motors 1 and 2. FAC 1 drives the green yaw damper actuator and motor 1 of the trim and limiter actuators, FAC 2 the yellow actuator and motor 2. A failed actuator is reported to its FAC and the other FAC takes the function over (FAC 1 has the priority).
- New AUTO FLT cautions with their procedures and STATUS (A320 FCOM PRO-ABN-AUTO_FLT, the flight phase inhibitions are read on the FCOM PDF figures).

What the crew sees

Situation	What you should see
One actuator failed	AUTO FLT YAW DAMPER 1(2), AUTO FLT RUD TRIM 1(2) FAULT or AUTO FLT RUD TRV LIM 1(2). STATUS INOP SYS YAW DAMPER 1(2), RUD TRIM 1(2) or RUD TRV LIM 1(2). The other FAC keeps the function.
Both actuators of a function failed	AUTO FLT YAW DAMPER SYS, RUD TRIM SYS or RUD TRV LIM SYS with -FAC 1 ... OFF THEN ON and -FAC 2 ... OFF THEN ON. STATUS INOP SYS YAW DAMPER, RUD TRIM or RUD TRV LIM. Both yaw dampers lost: the ELACs go to alternate law (F/CTL ALTN LAW).
RUD TRV LIM SYS	RUD WITH CARE ABV 160 KT. When the limiter stays locked at high speed with the slats extended (design choice: limit below 24 degrees): MAX X WIND FOR LDG 15 KT, -AUTO BRK ... DO NOT USE, .AT LDG ROLL: -DIFF BRAKING ... AS RQRD, also on the STATUS.
Function lost by the FACs or the hydraulics (FAC 1+2 FAULT, FAC pbs OFF, HYD G+Y SYS LO PR)	No SYS caution: those alerts already list the lost function (design choice).

Limitations and why

Limitation	Why	What to do
No F/CTL GND SPLR or SPD BRK FAULT cautions.	They need a SEC model change: the fault bit and the lever status are constants in the generated SEC code.	-
A failed FAC reports no actuator fault.	The FAC 1(2) FAULT alert covers it.	-
A full MSFS restart is needed after an update.	The fly-by-wire WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 22 Auto Flight and activate an actuator failure.
2. Follow the ECAM (FAC OFF THEN ON for a SYS caution) and read the STATUS page.
3. Deactivate the failure after the test.

How to test

#	Do this	Expected result
1	Cruise, AP ON. flyPad > Failures > ATA 22: **Yaw damper actuator 1 (FAC 1, green)**; STS key.	Amber AUTO FLT YAW DAMPER 1; STATUS INOP SYS YAW DAMPER 1; the yaw damper stays active through FAC 2.
2	Add **Yaw damper actuator 2 (FAC 2, yellow)**.	AUTO FLT YAW DAMPER SYS with -FAC 1 ... OFF THEN ON, -FAC 2 ... OFF THEN ON (no separate YAW DAMPER 1 / 2); F/CTL ALTN LAW; STATUS INOP SYS YAW DAMPER.
3	Deactivate both failures; FAC 1 and FAC 2 pb OFF then ON.	The cautions go.
4	Repeat with **Rudder trim actuator motor 1 (FAC 1)** and **Rudder trim actuator motor 2 (FAC 2)**.	AUTO FLT RUD TRIM 1 FAULT, then AUTO FLT RUD TRIM SYS; STATUS INOP SYS RUD TRIM.
5	Repeat with **Rudder travel limiter actuator motor 1 (FAC 1)** and **Rudder travel limiter actuator motor 2 (FAC 2)**.	AUTO FLT RUD TRV LIM 1, then AUTO FLT RUD TRV LIM SYS with RUD WITH CARE ABV 160 KT; STATUS INOP SYS RUD TRV LIM.

What PR testers should check

- One actuator failed: the 1(2) caution and the other FAC takes over.
- Both actuators failed: the SYS caution; for the yaw dampers, alternate law too.

28. ADR and IR failures, ADIRU power supplies, NAV ADR / IR / RA alerts

> **Aircraft:** A32NX (A380X ADIRU supplies too) | **Where:** flyPad > Failures > ATA 34 Navigation; overhead ADIRS panel; E/WD, STATUS | **Status:** Not sim-tested (unit tests: adirs.rs 17 Rust tests, AdirsAlerts.spec.ts 22, the failure list specs).

What it is for

- **ADR and IR fault failures** (ATA 34): the ADR (IR) outputs go to failure warning and its FAULT light comes on (A320 FCOM DSC-34-NAV-10-20). **Erroneous failures:** an ADR with an erroneous pitot reads a low speed (40 kt below), an ADR with an erroneous AOA probe a high angle of attack (10 degrees above), both with a valid status; an IR can have an erroneous heading. These values are design values (the FCOM gives none).

- **Per-ADIRU power supplies:** ADR 1 on AC ESS, ADR 2 on AC BUS 2, ADR 3 on AC BUS 1; the IRs fall back to the batteries (IR 2 and 3 for 5 min), with the ON BAT light and, on the ground, the external horn (FCOM). Power cuts up to 200 ms and the first 5 s after loading are ignored, so a flight loaded on the runway or in the air keeps its alignment.
- **New NAV alerts** (A320 FCOM PRO-ABN-NAV): ADR 1(2)(3) FAULT, ADR 1+2 (1+3)(2+3) FAULT, ADR 1+2+3 FAULT, ADR DISAGREE, IR 1(2)(3) FAULT, IR 1+2 (1+3)(2+3) FAULT, IR DISAGREE, RA 1 AND 2 FAULT, RA DEGRADED, and the RA 1(2) FAULT STATUS (CAT 2 ONLY; INOP SYS RA 1(2), CAT 3).
- The shared ADIRS model now takes the supplies of each aircraft, so the A380X ADIRU supplies (from the A380 FCOM ELEC INOP SYS lists) are on this branch too. The erroneous ADR model comes from a prerequisite commit of the shared ADIRS code (the A380X air data alerts that use it are on `core/failures-a380x`).

What the crew sees

Failure	E/WD	STATUS
ADR 1 fault	NAV ADR 1 FAULT: -AIR DATA SWTG ... CAPT 3, -ADR 1 P/B ... OFF	-
ADR 2 fault	NAV ADR 2 FAULT: -AIR DATA SWTG ... F/O 3, -ADR 2 P/B ... OFF, -BARO REF ... CHECK	-
Two ADRs	NAV ADR 1+2 (1+3)(2+3) FAULT with the switching and ATC/XPDR lines and the ADR P/B OFF lines	INOP SYS ADR 1+2 (1+3)(2+3)
Three ADRs	NAV ADR 1+2+3 FAULT: -AP+FD OFF, -A/THR OFF, -PROBE/WINDOW HEAT ON, -ADR 1+2+3 P/B OFF, -SPD FLY THE GREEN, -STBY INST MAY BE UNREL, -ALL ADR OFF PROC APPLY	INOP SYS ADR 1+2+3
Two remaining ADRs disagree	NAV ADR DISAGREE: .IF SPD DISAGREE: -ADR CHECK PROC APPLY; .IF NO SPD DISAGREE: AOA DISCREPANCY	Approach procedure (FLAP 3), IF NO SPD DISAGREE: RISK OF UNDUE STALL WARN
IR 1 (2) fault	NAV IR 1 (2) FAULT: -ATT HDG SWTG ... CAPT 3 (F/O 3), -ATC/XPDR ... SYS 2 (SYS 1)	INOP SYS IR 1 (2)
RA 1 and RA 2	NAV RA 1 AND 2 FAULT	INOP SYS RA 1+2

Limitations and why

Limitation	Why	What to do
No ELAC alternate law on ADR DISAGREE.	Not done yet.	-
No flashing IR FAULT light (ATT mode), no ADIRU and AVNCS external power panel light, no ATT HDG CAPT 3 supply switching.	Not done yet.	-
The ADR DISAGREE (16 kt, 3.6 degrees AOA in flight) and IR DISAGREE (5 degrees) thresholds and the RA DEGRADED threshold (15 ft or 10 %) are design choices.	The FCOM gives no value.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 34 Navigation and activate an ADR, IR or erroneous failure.
2. Follow the ECAM: AIR DATA / ATT HDG switching and the ADR or IR pb OFF on the overhead ADIRS panel.
3. Deactivate the failure and set the switching back to NORM.

How to test

#	Do this	Expected result
1	Cold and dark, BAT 1 and 2 ON, no external power; ADIRS IR 1, 2, 3 selectors to NAV.	The ON BAT light comes on and the external horn sounds on the ground (IRs on the batteries).
2	Cruise, AP ON. flyPad > Failures > ATA 34: **ADR 1 (air data lost, FAULT light)**.	ADR 1 FAULT light on the ADIRS panel; NAV ADR 1 FAULT with -AIR DATA SWTG ... CAPT 3, -ADR 1 P/B ... OFF.
3	Add **ADR 2 (air data lost, FAULT light)**; STS key.	NAV ADR 1+2 FAULT; STATUS INOP SYS ADR 1+2. Deactivate both.
4	ADR 3 pb OFF; then **ADR 1 airspeed erroneous (pitot)**.	Expected from the code: NAV ADR DISAGREE (ADR 1 reads 40 kt low against ADR 2). Deactivate, ADR 3 pb ON.
5	**IR 1 (inertial data lost, FAULT light)**.	IR 1 FAULT light; NAV IR 1 FAULT with -ATT HDG SWTG ... CAPT 3, -ATC/XPDR ... SYS 2; STATUS INOP SYS IR 1.
6	**RA 1** and **RA 2**.	NAV RA 1 AND 2 FAULT; STATUS INOP SYS RA 1+2.

What PR testers should check

- Cold and dark with the ADIRS in NAV: ON BAT light and the horn.
- Flights loaded on the runway and in cruise keep the IR alignment.
- ADR double and triple faults against the ELAC law.

29. alerts on existing data (CAB PR, THRUST LOCKED, fuel, WAI, doors, L/G, GPWS)

> **Aircraft:** A32NX | **Where:** E/WD, STATUS; overhead AIR COND, FUEL, ANTI ICE and GPWS panels; FCU A/THR pb; doors | **Status:** Not sim-tested (unit tests: 11 spec files, AirConditioningAlerts, AllEnginesFailure, AutoFlightStatus, DoorAlerts, FuelTransferAlerts, GpwsAlerts, LandingGearAlerts, ThrustLockedAlert, WingAntiIceAlerts, WingTankLowLevel and PaxOxygenMasks specs; 50 tests failing before the wiring).

What it is for

FCOM alerts whose data already existed in the sim but that the FWC did not show (A320 FCOM line numbers):

- **RAM AIR ON** green memo with the RAM AIR pb ON (DSC-21-10-50).
- **CAB PR LDG ELEV FAULT**: LDG ELEV selector at AUTO and no FMGS landing elevation, for example no destination in the flight plan; -LDG ELEV ... ADJUST. New STATUS lines of CAB PR EXCESS CAB ALT, SAFETY VALVE OPEN, SYS 1(2) FAULT and SYS 1+2 FAULT, with the MAN CAB PR CTL table.
- **ENG THRUST LOCKED**: the levers are not moved within 5 s after an involuntary A/THR disconnection or a disconnection by the FCU pb; the title flashes, the single chime and MASTER CAUT come again every 5 s; -THR LEVERS ... MOVE. Shown in flight phases 5, 6 and 7.
- **AUTO FLT:** STATUS of an involuntary A/THR OFF; FAC 1+2 FAULT now says RUD WITH CARE ABV 160 KT (was 140 KT).

- **FUEL:** the L+R WING TK LO LVL lines with LAND ASAP; **FUEL AUTO TRANSFER FAULT** and **FUEL CTR TK XFR OFF** (both CTR TK XFR pbs OFF in automatic mode).
- **WING A.ICE SYS FAULT** (WING ANTI ICE pb ON and one valve stays closed; with an engine bleed lost the -X BLEED ... OPEN variant) and **WING A.ICE L(R) HI PR** with THRUST LIM PENALTY.
- The passenger oxygen masks drop when the cabin altitude passes 14 000 ft.
- **DOOR** cautions for the four cabin doors, the two emergency exits and the two cargo doors; in flight .IF ABN CAB V/S: MAX FL ... 100/MEA-MORA, on the ground title only. Shown in phases 2, 3, 6 and 9.
- **L/G DOORS NOT CLOSED** (with MAX SPEED 250/.60 if the recycle fails) and **L/G SYS DISAGREE**, with the ATA 32 STATUS lines.
- **NAV GPWS FAULT** (-GPWS ... OFF, line kept after the pb is OFF) and **NAV GPWS TERR DET FAULT** (-GPWS TERR ... OFF).
- The STATUS of **ENG ALL ENGINES FAILURE** .

Limitations and why

Limitation	Why	What to do
No FMGC CAT 2/3 inop and AP OFF STATUS lines.	They need an FMGC model.	-
No WING A.ICE L(R) VALVE OPEN / OPEN ON GND.	No wing anti-ice valve failure in the sim.	-
No DOOR AVIONICS cautions.	The avionics doors are not in the sim.	-
The STATUS of ANTI ICE ENG 1 VALVE CLSD lists INOP SYS "WAI REGUL".	Known bug, the same: the WAI REGUL line and the ENG 1 A. ICE line have the same code.	See "Engine anti-ice valves, probe and window heat, anti-ice key events".

How to use

1. Nothing to set: the alerts come from the normal cockpit state.
2. Follow the ECAM and the STATUS page.

How to test

#	Do this	Expected result
1	At the gate, powered. Overhead RAM AIR pb ON.	Green RAM AIR ON memo on the E/WD. RAM AIR back OFF.
2	Engines running, no destination in the MCDU flight plan, pressurization LDG ELEV AUTO.	Amber CAB PR LDG ELEV FAULT with -LDG ELEV ... ADJUST. Enter a destination: the caution goes.
3	Engines running. Open the L FWD cabin door (flyPad or the MSFS door control).	Amber DOOR L FWD CABIN, title only (on the ground). Close the door.
4	Engines running, FUEL MODE SEL AUTO. CTR TK L XFR and CTR TK R XFR pbs OFF.	Amber FUEL CTR TK XFR OFF with -CTR TK L XFR ... ON, -CTR TK R XFR ... ON. Both pbs back ON.
5	Cruise, A/THR active, thrust levers in CL. Push the FCU A/THR pb; do not move the levers.	After 5 s amber ENG THRUST LOCKED, title flashing, with -THR LEVERS ... MOVE; the single chime and MASTER CAUT come again every 5 s. Move the levers: the caution goes.
6	Engines running, ATA 34: **EGPWC**.	Amber NAV GPWS FAULT with -GPWS ... OFF (and NAV GPWS TERR DET FAULT with -GPWS TERR ... OFF). Deactivate the failure.

What PR testers should check

- CAB PR LDG ELEV FAULT without a destination.
- FUEL AUTO TRANSFER FAULT with odd fuel loads at spawn.
- The DOOR cautions.
- ENG THRUST LOCKED flashing.
- The position of the RAM AIR ON memo.

30. PFD FMA messages, AVAIL, SD FUEL imbalance, SD BLEED crossbleed valve

> **Aircraft:** A32NX | **Where:** PFD FMA; E/WD N1 gauges; SD FUEL and BLEED pages | **Status:** Not sim-tested (unit tests: AvailIndication.spec.ts, FmaLine3Messages.spec.ts, CrossBleedValveIndication.spec.ts, FuelImbalanceIndication.spec.ts).

What it is for

- **PFD FMA line 3 messages** (A320 FCOM DSC-22_30-100): **DISCONNECT AP FOR LDG** (non-precision approach, AP still on at the minimum -50 ft, or 400 ft RA without a minimum), **FOR GA: SET TOGA** (approach vertical mode, below 1000 ft RA, all levers at or above MCT but below TOGA), **SET GREEN DOT SPEED** (engine out, selected speed 10 kt or more away from green dot, not in ALT* / ALT) and **CHECK APPR SELECTION** (the FMS message that sets it is on `core/gps`, merged here).
- **AVAIL** on the E/WD N1 gauge after a start: steady for 10 s on the ground, pulsing for 60 s after a relight in flight; it goes when the lever leaves the idle detent.
- **SD FUEL:** the wing quantities pulse when the imbalance is above 1500 kg.
- **SD BLEED:** the crossbleed valve is amber in transit or when it disagrees with its selector.

Limitations and why

Limitation	Why	What to do
No MORE DRAG FMA message.	It needs VNAV outputs.	-
No WHEEL UP LOCK indication.	The gear model has no uplock state.	-
No bleed lines in ENG SHUT DOWN.	Not done yet.	-
DISCONNECT AP FOR LDG takes FINAL, FPA or V/S in the approach phase as a non-precision approach.	Design choice: the PFD does not receive the FMS approach type.	-

How to use

1. Nothing to set: read the PFD FMA, the E/WD and the SD pages as usual.

How to test

#	Do this	Expected result
1	On the ground, start engine 1.	At the end of the start AVAIL shows steady under N1 for 10 s, then goes.
2	Engines running or in cruise, with more than 1500 kg difference between the left and right wing tanks; SD FUEL page.	The quantities of the heavier wing pulse.

3	Engines running, SD BLEED page. flyPad > Failures > ATA 36: **X BLEED valve stuck closed**; X BLEED selector OPEN.	The crossbleed valve symbol is amber (disagree). Selector AUTO, deactivate the failure.
4	ILS approach, AP and A/THR ON, below 1000 ft RA. Thrust levers to MCT (not TOGA).	Amber FOR GA: SET TOGA on FMA line 3. Levers back to CL or TOGA.
5	Non-precision approach (FINAL APP), AP ON, BARO minimum set; continue below the minimum -50 ft.	Amber DISCONNECT AP FOR LDG on FMA line 3. Disconnect the AP.

What PR testers should check

- The width of the fuel imbalance underline.
- The AVAIL and fuel pulse animations.
- The colour of the MCDU CHECK APPR SELECTION message.

31. SD advisory mode

> **Aircraft:** A32NX | **Where:** SD (lower ECAM display); ECAM control panel | **Status:** Not sim-tested (unit tests: SdAdvisories.spec.ts, SdAdvisoryMode.spec.ts, FakeDmc.spec.ts).

What it is for

The ECAM advisory mode of the A320 (FCOM DSC-31-20, PRO-ABN-ABN-ADV): when a parameter drifts out of its normal range, its value pulses green and its system page is called on the SD by itself. A failure page has priority, and the crew can still select another page.

- Sources: APU EGT (above EGT MAX -33 C, not during the start), APU flap not closed 3 min after MASTER OFF, cabin V/S, cabin altitude and delta P (CAB PRESS page), IDG oil temperature (ELEC page), engine oil and vibrations (ENGINE page), fuel imbalance above 1500 kg in phases 2 and 6 (FUEL page).
- The values are the same that pulse on the SD pages, so the page called shows the pulsing value.

Limitations and why

Limitation	Why	What to do
With one ECAM display the page is called anyway; the ECP pb light does not pulse.	The single display flag is written by a display reconfiguration that is not on this branch; nothing writes it on this branch, so the advisory always behaves as with two displays.	-
No APU oil quantity, nacelle temperature, fuel temperature or CKPT OXY advisory.	No data for them in the sim, or the page does not pulse that value.	-

How to use

1. Nothing to set. When a page comes by itself, read the pulsing value; select another page with the ECP if needed.

How to test

#	Do this	Expected result
1	Cruise, AP ON, SD on the CRUISE page. Pressurization MODE SEL MAN; MAN V/S CTL UP in short taps until the cabin V/S is about +2000 ft/min (keep the cabin below 9550 ft).	The SD changes to the CAB PRESS page by itself; the V/S value pulses green.
2	MAN V/S CTL DN until the V/S is back to normal; MODE SEL back to AUTO.	The pulsing stops and the SD goes back to the flight phase page.

What PR testers should check

- The advisory page in normal operations: no page called without a reason.

32. CVR automatic power and CVR TEST

> **Aircraft:** A32NX | **Where:** Overhead RCDR panel (GND CTL and CVR TEST pbs) | **Status:** Not sim-tested (unit tests: CockpitVoiceRecorder.spec.ts, 11 tests).

What it is for

The cockpit voice recorder runs by itself as in the A320 FCOM (DSC-23-10-40): on the ground during the first 5 min after the electrical network is powered, on the ground with an engine running, in flight, and 5 min more after the last engine shutdown; on the ground the GND CTL pb runs it too. The CVR TEST pb, pressed and held, gives the test tone when the CVR runs and the parking brake is on.

Limitations and why

Limitation	Why	What to do
The CVR always stops 5 min after the last engine shutdown.	The CVR jack is not simulated.	GND CTL pb ON.
A full MSFS restart is needed after an update.	The sound file list (cvr_test) changes.	Restart the sim after installing.

How to use

1. Parking brake ON.
2. Press and hold the CVR TEST pb: the test tone is heard while the CVR runs.
3. After the first 5 min on the ground without an engine running, push RCDR GND CTL ON first.

How to test

#	Do this	Expected result
1	Cold and dark: BAT 1 and 2 ON, EXT PWR ON, parking brake ON. Within 5 min, press and hold CVR TEST.	The test tone is heard.
2	Wait more than 5 min (no engine, GND CTL OFF); press and hold CVR TEST.	No tone (the CVR stopped).
3	RCDR GND CTL pb ON; press and hold CVR TEST.	The test tone is heard again.

What PR testers should check

- CVR TEST without GND CTL in the first 5 min after power-up.

33. CPC landing elevation without FMGS data, EXCESS CAB ALT threshold

> **Aircraft:** A32NX | **Where:** SD CAB PRESS page; pressurization LDG ELEV selector; E/WD | **Status:** Not sim-tested (Rust tests of cabin_pressure_controller.rs).

What it is for

- With the LDG ELEV selector at AUTO and no FMGS landing elevation (no destination), the cabin pressure controller has no landing elevation: the SD CAB PRESS page shows XX for it (A320 FCOM DSC-21-20-30, PRO-ABN-CAB_PR). It goes with CAB PR LDG ELEV FAULT (see "Alerts on existing data").

- **CAB PR EXCESS CAB ALT** in cruise triggers above a cabin altitude of 9550 ft only, also with a high destination airport; in climb and descent above the higher of 9550 ft and 1000 ft above the airfield.

Limitations and why

Limitation	Why	What to do
Without FMGS data the controller keeps a landing elevation of 0 ft until the crew sets LDG ELEV.	The FCOM gives no value for this case.	Turn the LDG ELEV selector (procedure LDG ELEV ... ADJUST).
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Enter a destination in the flight plan, or set the LDG ELEV selector by hand.

How to test

#	Do this	Expected result
1	Engines running, no destination in the flight plan, LDG ELEV AUTO, SD CAB PRESS page.	LDG ELEV shows XX; CAB PR LDG ELEV FAULT on the E/WD.
2	Turn the LDG ELEV selector out of AUTO.	The caution goes. Back to AUTO and enter a destination: the FMGS landing elevation replaces XX.

What PR testers should check

- No CAB PR EXCESS CAB ALT in cruise below 9550 ft cabin altitude with a high destination airport.

34. brake and steering control units (BSCU, ABCU), BRAKES alerts

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 32 Landing Gear; A/SKID & N/W STRG switch; autobrake pbs; E/WD, STATUS | **Status:** Not sim-tested (Rust: 15 test bed tests; unit tests: BrakeSystemAlerts.spec.ts 18, the failure list specs).

What it is for

- A new model of the BSCU with two channels and of the ABCU (A320 FCOM): channel 1 on AC BUS 1 and DC BUS 1, channel 2 on AC BUS 2 and DC BUS 2, the ABCU on DC ESS. The active channel changes at each gear DOWN selection and when it fails. Both channels lost: no antiskid, alternate braking limited to 1000 PSI, no nose wheel steering. ABCU lost too: parking brake only.
- The autobrake arms only with green pressure, the antiskid powered, no braking failure and one ADIRU. MAX activates only above 40 kt wheel speed and keeps braking below it.
- New cautions with their STATUS: **BRAKES A/SKID N/WS FAULT** (MAX BRK PR ... 1000PSI), **BRAKES ALTN BRK FAULT**, **BRAKES AUTO BRK FAULT**, **BRAKES NORM + ALTN FAULT** (PARK BRK ONLY), **BRAKES NORM BRK FAULT**, **BRAKES SYS 1(2) FAULT** and **WHEEL N/W STRG FAULT**.
- Six new failures, ATA 32. The brake and steering failure types come from a prerequisite commit of shared code (the A380X brake and steering failures that use them are on `core/failures-a380x`).

Limitations and why

Limitation	Why	What to do

The flight phase inhibitions are design choices.	The FCOM figures are missing from the FCOM text.	-
A unit lost only because its bus is unpowered gives no BRAKES caution.	Design choice: the ELEC bus alert already shows it with the same INOP SYS.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 32 Landing Gear and activate a BSCU, ABCU, braking, autobrake or steering failure.
2. Follow the ECAM and the STATUS; brake with care.

How to test

#	Do this	Expected result
1	On the ground, engines running, parking brake OFF. flyPad > Failures > ATA 32: **BSCU channel 1**.	BRAKES SYS 1 FAULT; braking and steering still work on channel 2.
2	Add **BSCU channel 2**; taxi slowly.	BRAKES A/SKID N/WS FAULT with MAX BRK PR ... 1000PSI; no nose wheel steering; the brakes give about 1000 PSI at most.
3	Add **ABCU (alternate braking)**; stop with the parking brake.	BRAKES NORM + ALTN FAULT with PARK BRK ONLY; STATUS PARK BRK ONLY. Deactivate the three failures.
4	**Nose wheel steering**.	WHEEL N/W STRG FAULT; no steering with the tiller.
5	In flight, autobrake LO armed. **Autobrake**.	BRAKES AUTO BRK FAULT.
6	Take-off run, autobrake MAX armed; reject above 40 kt, then repeat below 40 kt.	Above 40 kt the autobrake brakes to a stop; below 40 kt it does not activate.

What PR testers should check

- MAX rejected take-off above and below 40 kt.
- Taxi with BSCU channels 1 and 2 failed: no steering, about 1000 PSI.
- ABCU failure: parking brake only.
- AUTO BRK FAULT after arming LO in flight and failing the autobrake.

35. APU automatic shutdowns, cargo and avionics smoke, cargo extinguishing

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 26 Fire Protection and ATA 49 Airborne Auxiliary Power; overhead APU, CARGO SMOKE, VENTILATION and EMER ELEC panels; E/WD, STATUS | **Status:** Not sim-tested (Rust tests of the ECB, smoke detection, cargo extinguishing and the A320 fire tests; unit tests: SmokeAlerts.spec.ts, the failure list specs).

What it is for

- **APU automatic shutdowns** by the ECB (A320 FCOM DSC-49-20 FAULT light): ECB failure, overspeed, low oil pressure, high oil temperature, and the loss of its DC supply above N 70 % (confirmed 1 s). The MASTER SW FAULT light comes on and APU AUTO SHUT DOWN shows (see "HYD and APU shutdown alerts"). Four new failures, ATA 49.
- **Cargo smoke detection** (FWD and AFT, two loops each on the SDCU channels 1 and 2): **SMOKE FWD (AFT) CARGO SMOKE** with -CAB FANS ... OFF, .IF FWD (AFT) CARG CLOSED: -AGENT ... DISCH, and the ground lines; the

red SMOKE light of the CARGO SMOKE panel. One loop failed is tolerated; both loops failed: **SMOKE FWD (AFT) CRG DET FAULT**, STATUS INOP SYS FWD (AFT) CRG DET.

- **Cargo extinguishing** : one bottle with two discharge heads; the cargo DISCH pbs now work and their DISCH light comes on when the bottle is empty. A bottle at low pressure gives **SMOKE FWD CRG BTL 1 FAULT** and **SMOKE AFT CRG BTL 1 FAULT**.
- **Avionics smoke** (ventilation extract duct detector): **AVIONICS SMOKE** after 5 s with -SMOKE/FUMES PROC ... APPLY; the SMOKE light of the GEN 1 LINE pb and the FAULT lights of the BLOWER and EXTRACT pbs come on.
- Nine new failures, ATA 26. The CARGO SMOKE TEST pb still shows the FWD CARGO SMOKE warning, as before.

Limitations and why

Limitation	Why	What to do
Other APU shutdown causes (inlet flap, slow start, EGT overtemperature, sensors...) do not exist.	No model behind them.	-
The -FWD (AFT) ISOL VALVE ... OFF line never shows.	The A320 here has no cargo ventilation.	-
The cargo bottle squibs cannot fail; no bottle diverter or control alerts.	Not modelled.	-
A full MSFS restart is needed after an update.	The systems WASM module and the cockpit XML (lights) change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 26 Fire Protection (smoke, detection loops, bottle) or ATA 49 Airborne Auxiliary Power (APU).
2. Follow the ECAM: for cargo smoke, CAB FANS OFF and the cargo DISCH pb of the compartment (AGENT DISCH).

How to test

#	Do this	Expected result
1	On the ground, APU running (AVAIL). flyPad > Failures > ATA 49: **APU overspeed (automatic shutdown)**.	The APU shuts down; APU MASTER SW FAULT light; APU AUTO SHUT DOWN with -MASTER SW ... OFF. MASTER SW OFF, deactivate the failure.
2	Repeat with **APU low oil pressure (automatic shutdown)**, **APU high oil temperature (automatic shutdown)** and **APU electronic control box (ECB)**.	The same automatic shutdown with the FAULT light.
3	Cruise. ATA 26: **Aft cargo smoke**.	Red SMOKE AFT CARGO SMOKE with -CAB FANS ... OFF, .IF AFT CARG CLOSED: -AGENT ... DISCH; red SMOKE light on the CARGO SMOKE panel.
4	Push the AFT DISCH pb.	After the discharge the DISCH lights come on and the AGENT line goes. Deactivate the failure.
5	ATA 26: **Forward cargo smoke detection loop 1**, then **Forward cargo smoke detection loop 2**.	Nothing with one loop; with both, amber SMOKE FWD CRG DET FAULT; STATUS INOP SYS FWD CRG DET.
6	ATA 26: **Cargo fire bottle low pressure**.	SMOKE FWD CRG BTL 1 FAULT and SMOKE AFT CRG BTL 1 FAULT.

7	ATA 26: **Avionics smoke (ventilation extract duct)**; wait 5 s.	AVIONICS SMOKE with -SMOKE/FUMES PROC ... APPLY; GEN 1 LINE SMOKE light, BLOWER and EXTRACT FAULT lights. Deactivate the failure.

What PR testers should check

- On batteries only, the APU shuts down when its DC supply goes (but not with the APU GEN or EXT PWR on).
- No APU shutdown on electrical power transfers.
- The DISCH, AGENT and BTL light emissives.
- The smoke failures from the flyPad.

36. A380X smoke detection function (systems part)

> **Aircraft:** A380X | **Where:** A380X overhead CARGO SMOKE panel; flyPad > Failures > ATA 26 (with core/a380x-elec-fuel-alerts) | **Status:** Not sim-tested (6 Rust tests in fire_and_smoke_protection.rs).

What it is for

The A380X Smoke Detection Function of the CIDS for the cargo compartments (FWD, AFT, bulk), the avionics bays and the IFE bay, each with two smoke detectors (A380 FCOM DSC-26-50, DSC-26-60). On the CARGO SMOKE panel the AGENT TO FWD / AFT pbs show SMOKE and DISCH, and the BTL 1 and BTL 2 lights come on when a bottle has discharged: bottle 1 first (about one minute), then bottle 2 slowly. It lives on this branch because it extends the shared fire model of both aircraft, which is only here.

> **CAUTION**
> The A380X SMOKE, SMOKE DET FAULT and CARGO BTL FAULT alerts, the A380X flyPad failure list and the simvar docs are on `core/a380x-elec-fuel-alerts`. **Merge both branches** for the A380X smoke alerts; this branch alone only has the systems side and the panel lights.

Limitations and why

Limitation	Why	What to do
The crew rests, lavatories, CWS / RCC, showers and NSS have no smoke detection; the SDF cannot fail as a computer.	Not modelled.	-
Detector 1 is on the DC ESS bus and detector 2 on the DC 2 bus.	Design choice: the FCOM gives no supply.	-
A full MSFS restart is needed after an update.	The A380X systems WASM module and the cockpit XML change.	Restart the sim after installing.

How to use

1. With `core/a380x-elec-fuel-alerts` merged: open the A380X flyPad > Failures > ATA 26 and activate a smoke or detector failure.
2. Follow the ECAM; push AGENT TO FWD or AFT on the CARGO SMOKE panel for a cargo smoke.

How to test

#	Do this	Expected result
1	A380X in cruise, both branches merged. flyPad > Failures > ATA 26: **Aft cargo smoke**.	The SMOKE light of the AGENT TO AFT pb comes on (and the SMOKE alert of core/a380x-elec-fuel-alerts).

2	Push AGENT TO AFT.	BTL 1 light after about one minute, then BTL 2 later. Deactivate the failure.
3	**Forward cargo smoke detector 1**.	The SMOKE DET FAULT alert of the forward compartment (alerts on core/a380x-elec-fuel-alerts).

What PR testers should check

- The A380X smoke failures from the flyPad and the CARGO SMOKE panel lights.

37. BLEED page HP valve and bleed valve disagree

> **Aircraft:** A32NX | **Where:** SD BLEED page; flyPad > Failures > ATA 36 Pneumatic | **Status:** Not sim-tested (Rust ata_36_failures tests, 2; unit tests: EngineBleedValveIndication.spec.ts).

What it is for

On the SD BLEED page (A320 FCOM): the HP valve is amber when it is closed with an HP valve fault, and the engine bleed valve is amber when it has been out of its commanded position for 5 s. This extends "Pneumatic failures with BMC leak and bleed monitoring". The A380X bleed monitoring and leak alerts are on `core/a380x-elec-fuel-alerts`.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 36 Pneumatic, activate a valve failure and watch the SD BLEED page.

How to test

#	Do this	Expected result
1	Engines running at idle, SD BLEED page. flyPad > Failures > ATA 36: **Engine 1 HP valve stuck closed**.	ENG 1 HP VALVE FAULT; the engine 1 HP valve symbol is amber, closed.
2	Deactivate it. **Engine 1 bleed valve stuck closed** with ENG 1 BLEED pb ON; wait 5 s.	The engine 1 bleed valve symbol turns amber (closed while commanded open). Deactivate the failure.

What PR testers should check

- No false bleed valve disagree on a fast thrust increase at take-off.
- Leak failures from the flyPad (a pylon leak needs high power to be detected).

38. HP fuel valve, FADEC fault failures, pre-start motoring

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 73 Engine - Fuel And Control and ATA 78 Exhaust; ENG panel (FAULT lights); E/WD | **Status:** Not sim-tested (Rust tests of the systems and a320 crates; unit tests: EngineMonitoringAlerts.spec.ts, the failure list specs).

What it is for

- **HP fuel valve stuck** (A320 FCOM PRO-ABN-ENG): stuck closed on a stopped engine there is no fuel and no light-up: **ENG 1(2) HP FUEL VALVE** with HP FUEL VALVE NOT OPEN, -MAN START ... OFF, -ENG MASTER ... OFF; stuck

open on a running engine: HP FUEL POS SWT FAULT at the shutdown. The valve position must disagree with its command for 2 s.

- **FADEC fault failures** with crew awareness alerts: **ENG 1(2) FADEC ALTERNATOR**, **OVSPD PROT FAULT**, **PROBES FAULT**, **SENSOR FAULT** (on the ground -THR LEVER ... IDLE, -ENG MASTER ... OFF; in flight AVOID RAPID THR CHANGES, also on the STATUS), **ONE TLA FAULT** and **REV SWITCH FAULT** (the reverser still works). 14 new failures, ATA 73 and 78.
- The **ENG FAULT light** also comes on for a thrust control malfunction (both FADEC channels or both lever resolvers lost) and an abnormal HP valve position.
- **Pre-start motoring:** an automatic ground start of a hot engine dry cranks it for up to 60 s first, by its thermal state (the schedule is a design choice, the FCOM gives none). See also "Engine ignition and start failures, FCOM start sequence".

Limitations and why

Limitation	Why	What to do
Most FADEC faults have no effect on the engine model.	They are crew awareness alerts; the other ENG alerts need engine models.	-
No approach idle.	No FCOM value.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 73 Engine - Fuel And Control (or ATA 78 Exhaust for the reverser switch) and activate a failure.
2. Follow the ECAM. Most of these alerts are inhibited from take-off power to 1500 ft and in approach, so test them on the ground or in cruise.

How to test

#	Do this	Expected result
1	On the ground, APU BLEED ON, engine 1 shut down. flyPad > Failures > ATA 73: **Engine 1 HP fuel valve stuck**; ENG MODE IGN/START, ENG MASTER 1 ON.	No light-up (no fuel flow); amber ENG 1 HP FUEL VALVE with HP FUEL VALVE NOT OPEN and -ENG MASTER 1 ... OFF; ENG 1 FAULT light. MASTER OFF, deactivate.
2	Engines running on the ground. **Engine 1 FADEC alternator**.	Amber ENG 1 FADEC ALTERNATOR, title only. Deactivate.
3	**Engine 1 core sensors (PS3)**.	ENG 1 SENSOR FAULT with -THR LEVER 1 ... IDLE, -ENG MASTER 1 ... OFF (on the ground). Deactivate.
4	**Thrust lever 1 one resolver**, then ATA 78: **Reverser 1 permission switch**.	ENG 1 ONE TLA FAULT; then ENG 1 REV SWITCH FAULT. Deactivate both.
5	Shut engine 1 down after a long run, then start it again at once with the automatic start.	Dry cranking (N2 steady, no fuel) for up to 60 s before the ignition and fuel.

What PR testers should check

- Hot restart motoring.

39. gravity feed ceiling, LP valve and FQI failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; E/WD | **Status:** Not sim-tested (Rust: gravity_feed 16 and engine_lp_valve 5 tests; unit tests: FuelLpValveFqiAlerts.spec.ts, FuelPumpsAndValves.spec.ts).

What it is for

- **Gravity feed ceiling** (A320 FCOM GRAVITY FUEL FEEDING): FL 360 after more than 30 min above FL 300, FL 260 otherwise, and the higher of FL 150 and take-off elevation + 7000 ft if FL 300 was never passed. An engine fed by gravity (both pumps of its wing tank off or failed, and no pump pushing through the X FEED) above the ceiling is starved after 3 min (design choice) and flames out; it stays out until ENG MASTER OFF below the ceiling, then MASTER ON relights it inside the relight envelope. This extends the gravity feed of "FCOM fixes: FADEC relight, automatic ignition, start faults, gravity feed, IR alignment".
- **LP fuel valve jammed** failures: jammed open the engine still shuts down with its ENG MASTER, and **FUEL ENG 1(2) LP VALVE OPEN** shows; jammed closed the engine is starved.
- **FQI channel** failures: **FUEL FQI CH 1(2) FAULT** .

Limitations and why

Limitation	Why	What to do
The 3 min before the starvation and the pressure altitude as flight level are design choices.	The FCOM gives neither a time nor a partial thrust loss.	Descend below the ceiling within 3 min.
Restoring the pumps or descending alone does not relight a starved engine.	As the FCOM relight procedure: ENG MASTER OFF then ON.	-
No fuel alerts that need temperature, IDG, overflow or inerting models.	Not modelled.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. With all wing tank pumps of one side lost, follow the ECAM and descend below the gravity feed ceiling.
2. Open flyPad > Failures > ATA 28 Fuel for the LP valve and FQI failures.

How to test

#	Do this	Expected result
1	On the ground, engines running. flyPad > Failures > ATA 28: **ENG 1 LP fuel valve jammed in place**; ENG MASTER 1 OFF.	Engine 1 shuts down; amber FUEL ENG 1 LP VALVE OPEN. Deactivate.
2	**FQI channel 1**.	Amber FUEL FQI CH 1 FAULT. Deactivate.
3	Cruise at FL 330 less than 30 min after passing FL 300, X FEED closed. **L TK pump 1** and **L TK pump 2**; wait 3 min.	Engine 1 is fed by gravity above the FL 260 ceiling and flames out after 3 min (ENG 1 FAIL).
4	Descend below FL 260; ENG MASTER 1 OFF then ON inside the relight envelope; deactivate the pump failures.	Engine 1 relights.

What PR testers should check

- ELEC EMER CONFIG in cruise: pumps off, flameout after 3 min above the ceiling unless descending.

- No false FUEL alert.

40. engine anti-ice valves, probe and window heat, anti-ice key events

> **Aircraft:** A32NX (A380X anti-ice systems and cockpit part too) | **Where:** flyPad > Failures > ATA 30 Ice And Rain Protection; overhead ANTI ICE panel; E/WD, STATUS | **Status:** Not sim-tested (Rust tests engine_anti_ice, probe_window_heat, A380 ice_rain_protection and oxygen; unit tests: EngineAntiIceAlerts, ProbeWindowHeatAlerts, AntiIceCautions, FadecWingAntiIce specs, the failure list specs).

What it is for

- **Engine anti-ice valve** (shared model, A320 FCOM DSC-30-30): it opens with the pb ON (or without power) once the engine supplies HP air above 10 psig, in 2 s, with the FAULT light in transit; the MSFS engine anti-ice follows the valve. New cautions **ANTI ICE ENG 1(2) VALVE CLSD** (AVOID ICING CONDITIONS) and **ANTI ICE ENG 1(2) VALVE OPEN** (THRUST LIM PENALTY) with STATUS. VALVE CLSD needs the engine running.
- **Probe and window heat computers** and all 14 probe and window heat cautions with their STATUS (DSC-30-40-10, DSC-30-50-10): ANTI ICE ALL PITOT, CAPT + F/O PITOT, CAPT + STBY PITOT, F/O + STBY PITOT, CAPT (F/O) (STBY) PROBES, the single PITOT, L STAT, R STAT and AOA cautions, CAPT (F/O) TAT, L + R WINDSHIELD, L (R) WINDSHIELD and L (R) WINDOW.
- **MSFS anti-ice key events** (ANTI_ICE_*_ENGn, ANTI_ICE_ON / OFF / SET / TOGGLE, TOGGLE_STRUCTURAL_DEICE) now move the cockpit pbs on both aircraft, so key bindings and hardware panels keep working; the sim variables follow the valves.
- About 25 new A32NX failures, ATA 30. Because the key event mapping serves both aircraft, the A380X anti-ice systems and cockpit side (ENG and WING ANTI ICE pb behaviour, oxygen, FADEC wing anti-ice data) is on this branch too; the A380X ECAM alerts, flyPad failures and SD oxygen page are on `core/a380x-elec-fuel-alerts`.

Limitations and why

Limitation	Why	What to do
ANTI ICE ENG 1 VALVE CLSD lists INOP SYS "WAI REGUL" instead of "ENG 1 A. ICE".	Known bug, the same: two STATUS entries have the same code (300300002) and the later one, WAI REGUL, wins. This branch keeps that order so its specs agree; one of them needs a new code.	-
No ADR probe heat bits; no wing anti-ice indication on the SD BLEED page.	Not done yet.	-
The key event mapping tests cannot run on the host.	They need the WASM target; they compile only.	-
A full MSFS restart is needed after an update.	The systems WASM modules and the cockpit XML change.	Restart the sim after installing.

How to use

1. Use the overhead ANTI ICE pbs, or the MSFS anti-ice key bindings: both move the pbs.
2. Open flyPad > Failures > ATA 30 Ice And Rain Protection to fail a valve, a heat computer or a heater.

How to test

#	Do this	Expected result

1	On the ground, engines running. flyPad > Failures > ATA 30: **Engine 1 anti-ice valve stuck closed**; ANTI ICE ENG 1 pb ON; STS key.	FAULT light on the pb; amber ANTI ICE ENG 1 VALVE CLSD with AVOID ICING CONDITIONS. STATUS INOP SYS shows "WAI REGUL" (known bug, see above). Pb OFF, deactivate.
2	**Engine 2 anti-ice valve stuck open**; ANTI ICE ENG 2 pb stays OFF.	ANTI ICE ENG 2 VALVE OPEN with THRUST LIM PENALTY; STATUS THRUST LIM PENALTY. Deactivate.
3	**CAPT pitot heating**.	ANTI ICE CAPT PITOT with -AIR DATA SWTG ... CAPT 3; STATUS INOP SYS CAPT PITOT. Add **F/O pitot heating**: one ANTI ICE CAPT + F/O PITOT with its procedure.
4	**CAPT probe heat computer**.	ANTI ICE CAPT PROBES with -AIR DATA SWTG ... CAPT 3.
5	**Left windshield heating**, then **Right windshield heating**.	ANTI ICE L WINDSHIELD, then one ANTI ICE L + R WINDSHIELD; STATUS INOP SYS WSHLD HEAT. Deactivate all.
6	Bind a key to the MSFS engine 1 anti-ice toggle and press it.	The ANTI ICE ENG 1 pb moves; no loop and no repeated toggling.

What PR testers should check

- Key binding and hardware panel toggles of ENG and WING anti-ice (no loop, no toggle storm at load).
- A32NX ENG A.ICE FAULT light before about 25 % N2.
- A380X wing anti-ice pb test and take-off inhibition (with core/a380x-elec-fuel-alerts).

41. SFCC flap and slat channel failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 27 Flight Controls; E/WD, STATUS, SD F/CTL | **Status:** Not sim-tested (Rust SFCC channel tests; unit tests: SlatFlapChannelAlerts.spec.ts, the failure list specs).

What it is for

Four new failures, ATA 27: the flap and slat channels of SFCC 1 and 2. A failed channel stops its motor, so the surfaces move at half speed (A320 FCOM), and sets its fault bit. New crew awareness cautions **F/CTL FLAP SYS 1(2) FAULT** and **F/CTL SLAT SYS 1(2) FAULT**. STATUS: FLAP SYS 1 FAULT gives APPR PROC: GPWS FLAP MODE ... OFF; FLAP SYS 1(2) FAULT ENG 1(2) APPR IDLE ONLY and FLAPS SLOW; SLAT SYS FAULT SLATS SLOW. The A380X part is on `core/a380x-elec-fuel-alerts`.

Limitations and why

Limitation	Why	What to do
No F/CTL FLAPS (SLATS) FAULT / LOCKED, tip brake or flap attach sensor alerts.	The FCOM text gives no trigger for the first; the model has no tip brakes nor attach sensors.	-
An unpowered SFCC gives no channel caution.	The ELEC bus alerts cover it.	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 27 Flight Controls and activate an SFCC channel failure.
2. Expect slow flaps or slats; read the STATUS before the approach.

How to test

#	Do this	Expected result
1	Cruise or on the ground with engines running. flyPad > Failures > ATA 27: **SFCC 1 flap channel**; STS key.	Amber F/CTL FLAP SYS 1 FAULT; STATUS APPR PROC: GPWS FLAP MODE ... OFF, ENG 1 APPR IDLE ONLY, FLAPS SLOW.
2	Select flaps 1 then 2 (below their speeds).	The flaps move at about half the normal speed.
3	**SFCC 2 slat channel**.	F/CTL SLAT SYS 2 FAULT; STATUS SLATS SLOW. Deactivate both.

What PR testers should check

- Half-speed flaps and slats with a channel failure.

42. GPWS self-test, TCAS computer failure, reactive windshear

> **Aircraft:** A32NX | **Where:** Glareshield GPWS pb; PFD; ND; flyPad > Failures > ATA 34 Navigation | **Status:** Not sim-tested (Rust egpws tests, 3 new; unit tests: TcasFault.spec.ts 5, ReactiveWindshearDetection.spec.ts 13).

What it is for

- **GPWS self-test** (A320 FCOM, 48990): on the ground, pressing the GPWS pb runs the test: GLIDESLOPE with the amber light, PULL UP with the red light, then the terrain test pattern with TERRAIN TERRAIN PULL UP. Not with GPWS SYS OFF; it stops at lift-off. The sequence is a design choice (Honeywell EGPWS short test).
- **TCAS computer** failure (ATA 34): out of standby, amber TCAS on the PFD and the ND, NAV TCAS FAULT, no traffic and no RA.
- **Reactive windshear** (FAC 1 and 2): red WINDSHEAR on both PFDs for at least 15 s. The shared detector and PFD warning serve both aircraft; the A380X PRIM part is on `core/a380x-elec-fuel-alerts`. The WINDSHEAR voice is in "Voices: sidestick priority, intermediate RA callouts, WINDSHEAR".

Limitations and why

Limitation	Why	What to do
The windshear threshold (F-factor 0.13, filtered) is a design choice.	The FCOM gives no formula.	-
No REAC W/S DET FAULT alert, no SRS escape guidance, no predictive windshear, no FLS modes.	They need models, terrain or wind data.	-
A full MSFS restart is needed after an update.	The systems WASM module (EGPWS) changes.	Restart the sim after installing.

How to use

1. GPWS test: on the ground, GPWS SYS ON, press the GPWS pb on the glareshield.
2. TCAS: flyPad > Failures > ATA 34 Navigation, **TCAS computer**.
3. Windshear: nothing to set; follow the FCOM windshear procedure if WINDSHEAR shows.

How to test

#	Do this	Expected result
1	On the ground, powered, GPWS SYS ON. Press the GPWS pb on the glareshield.	GLIDESLOPE with the amber light, PULL UP with the red light, then TERRAIN TERRAIN PULL UP (the terrain test pattern step).
2	GPWS SYS pb OFF; press the GPWS pb again.	No test.
3	TCAS mode AUTO (not STBY). flyPad > Failures > ATA 34: **TCAS computer**.	Amber TCAS on the PFD and the ND; NAV TCAS FAULT; no traffic. Deactivate.

What PR testers should check

- No false WINDSHEAR in gusts and turns.
- The GPWS test lights and aurals.

43. NAV FM/GPS POS DISAGREE and NAV LS TUNING DISAGREE

> **Aircraft:** A32NX | **Where:** E/WD, STATUS | **Status:** Not sim-tested (unit tests: NavPositionAlerts.spec.ts).

What it is for

- **NAV FM/GPS POS DISAGREE** (A320 FCOM): the FM and GPS positions disagree; amber with MASTER CAUT and single chime, -A/C POS ... CHECK. The FMS computes the condition: the FM position model is on `core/gps`, merged here. Inhibited for the first 15 s of phase 5.
- **NAV LS TUNING DISAGREE** : MMR 1 and MMR 2 tuned differently for 3 s (design choice); crew awareness, STATUS INOP SYS CAT 2; shown in phases 6 and 7 only.

Limitations and why

Limitation	Why	What to do
LS TUNING DISAGREE has no AP / FD effect (APPR arming).	That belongs to the flight guidance.	-

How to use

1. Nothing to set. On FM/GPS POS DISAGREE check the aircraft position (MCDU POSITION MONITOR, ND).

How to test

#	Do this	Expected result
1	Fly a normal flight with an ILS approach tuned by the FMS.	No NAV LS TUNING DISAGREE and no NAV FM/GPS POS DISAGREE.
2	There is no flyPad failure that makes the FM and GPS positions disagree on purpose.	The caution comes only from the FM position model of core/gps (L:A32NX_FM_GPS_POS_DISAGREE): check that it never shows in normal flights.

What PR testers should check

- No nuisance LS TUNING DISAGREE while the FMS tunes the two receivers.

44. Voices: sidestick priority, intermediate RA callouts, WINDSHEAR

> **Aircraft:** A32NX | **Where:** Sidestick take over pbs; FWC voices | **Status:** Not sim-tested (unit tests: SidestickPriorityAurals, IntermediateHeightCallout, FwsVoiceCallouts specs).

What it is for

- **DUAL INPUT** every 5 s while both sidesticks are deflected, and **PRIORITY LEFT / PRIORITY RIGHT** once the other sidestick has been deactivated by the take over pb for 1 s (A320 FCOM). The 1 s is a design choice so that a short press to disconnect the AP does not call it.
- **Intermediate radio height callout:** below 410 ft, when no height callout came for 11 s (above 50 ft) or 4 s (below 50 ft), the present height is said, rounded down to a callout word, then every 4 s.
- **WINDSHEAR** x3 for the reactive windshear warning.
- The new clips are synthetic voices made with a public-domain voice (tools/fws-voice).

Limitations and why

Limitation	Why	What to do
DUAL INPUT cannot sound.	The F/O sidestick sends no input to the flight controls in the sim; the logic is ready.	-
The voices wait while a GPWS or TCAS aural plays.	As the radio height callouts do (design choice).	-
A full MSFS restart is needed after an update.	New sound files and sound.xml.	Restart the sim after installing.

How to use

1. Bind a key or button to the captain sidestick take over (AP disconnect) pb; press and hold it.
2. Listen to the callouts in a slow final approach.

How to test

#	Do this	Expected result
1	In flight, AP OFF. Press and hold the captain take over pb for 3 s.	PRIORITY LEFT once, after about 1 s.
2	Short press of the take over pb.	No PRIORITY voice.
3	Final approach: level off or descend very slowly between 300 and 200 ft RA for more than 15 s.	About 11 s after the last callout: TWO HUNDRED (or THREE HUNDRED), then every 4 s.

What PR testers should check

- The WAV volume against the other voices, and the retrigger.
- The intermediate callouts are not too chatty.

45. Engine exceedance marks only on displayed values

> **Aircraft:** A32NX | **Where:** E/WD N1, N2 and EGT gauges | **Status:** Not sim-tested (unit tests: EngineLimits.spec.ts).

What it is for

The red N1, N2 and EGT exceedance marks and the N2 red cross latched on any value, also while the gauge showed XX (FADEC not powered) and on invalid numbers. FCOM: a value that is not available shows XX instead (A320 FCOM). They now only latch on valid, displayed values and keep the highest value. The red crosses seen in flight came from the old CFM56 N2 limit of 105 %, already replaced by the LEAP-1A 116.5 % (see "Engine stall, EGT overtemperature, overspeed and vibration failures"). The A380X part is on `core/a380x-elec-fuel-alerts`.

Limitations and why

Limitation	Why	What to do
The marks follow the FADEC power flag (L:A32NX_FADEC_POWERED_ENGn) for the validity.	A value without FADEC power is shown as XX, not as a value (FCOM).	-

How to use

1. Nothing to set.

How to test

#	Do this	Expected result
1	Cold and dark, then power up and start both engines normally.	No red mark and no red cross on the N1, N2 and EGT gauges.
2	Before the start, on batteries only (FADEC not powered: XX on the gauges); then power up and start.	No red mark appears while XX shows or when the values come.

What PR testers should check

- No red exceedance mark or cross after a normal flight.

46. DC BUS 2 takes over DC BUS 1 after 5 s

> **Aircraft:** A32NX | **Where:** SD ELEC page; flyPad > Failures > ATA 24 Electrical Power | **Status:** Not sim-tested (Rust: 3 new test bed tests; the TR 1 fault distribution test now waits 5 s).

What it is for

A320 FCOM DSC-24-10-30-30, FAILURE OF AC BUS 1 : "DC BUS 2 supplies DC BUS 1 and DC BAT BUS automatically after 5 s." The model switched at once. Now, after a TR 1 or AC BUS 1 loss, DC BUS 1 and the DC BAT BUS stay unpowered for 5 s, then DC BUS 2 supplies them. The DC ESS BUS stays on the ESS TR as before. A failed DC BUS 1 busbar still transfers at once, and the TR 2 case stays immediate (no FCOM delay). This changes the AC BUS 1 FAULT case of "ELEC bus, ESS bus and TR alerts" and of the recovery section below.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Nothing to set; after an AC BUS 1 or TR 1 loss, wait 5 s for DC BUS 1 to come back.

How to test

#	Do this	Expected result
1	Cruise, SD ELEC page. flyPad > Failures > ATA 24: **TR 1**.	DC BUS 1 and DC BAT BUS amber for about 5 s, then green again (supplied through DC BUS 2). Deactivate.
2	**AC 1**.	ELEC AC BUS 1 FAULT; DC BUS 1 and DC BAT BUS dark for 5 s, then back. Deactivate.

What PR testers should check

- After a TR 1 or AC BUS 1 failure, DC BUS 1 and DC BAT BUS are dark for 5 s, then recover.
- No odd ECAM flicker in that window.

47. Recovery after electrical failures: FAC reset, AC BUS 1 transfer, SEC test, IDG

> **Aircraft:** A32NX | **Where:** Overhead FLT CTL panel (FAC, SEC pbs), ELEC panel (IDG pbs); flyPad > Failures > ATA 24 | **Status:** FAC reset tested in the sim by the user: OK (DC BUS 1+2 FAULT before the DC ESS fix, FAC 1 and FAC 2 OFF THEN ON gave normal law back; on the DC ESS fix: only FAC 2 FAULT, cleared by FAC 2 OFF THEN ON). AC BUS 1 transfer: OK (before the DC ESS fix). SEC power-up test and IDG: not sim-tested on this branch.

What it is for

Removing a flyPad electrical failure gives the power back, but in flight some computers stay failed until the crew resets them, as on the aircraft. This section says what stays failed after the failures of this branch and how to recover it. Every statement was checked against the code of this branch and the A320 FCOM.

FAC FAULT stays after the power comes back

Each FAC watches its power supply (`Fac.cpp`). A cut of 10 to 200 ms stops it and it restarts by itself; after a cut of more than **200 ms** its memory is lost and it latches FAULT. On the ground (nose gear compressed) it comes back by itself; **in flight only its pushbutton brings it back: FAC OFF then ON**. This is the FCOM procedure: AUTO FLT FAC 1(2) FAULT "FAC (AFFECTED) ... OFF THEN ON" and AUTO FLT FAC 1+2 FAULT "FAC 1 ... OFF THEN ON, FAC 2 ... OFF THEN ON" . FAC 1 is supplied by the DC ESS (shed) bus, FAC 2 by DC BUS 2.

Situation	What you should see
DC BUS 1+2 FAULT (flyPad *DC 1* + *DC 2*), then the failures removed	The ESS TR keeps the DC ESS BUS powered (this branch, FCOM DSC-24-10-30-30), so FAC 1, AP 1 and normal law stay. FAC 2 lost its power: **AUTO FLT FAC 2 FAULT** stays after DC BUS 2 is back (the FCOM INOP SYS of DC BUS 1+2 FAULT lists FAC 2, AP 2 and FCU 2). Recover with **FAC 2 OFF then ON**.
DC EMER CONFIG (flyPad *DC 1* + *DC 2* + *DC ESS*), then the failures removed	Both FACs lost their power: **AUTO FLT FAC 1+2 FAULT** stays (alternate law, no AP, no A/THR, rudder with care). Recover with **FAC 1 OFF then ON, then FAC 2 OFF then ON**; normal law, AP and A/THR come back.
Any other FAC FAULT after a power cut in flight	Same action: the affected FAC OFF then ON. If it does not come back, leave it OFF (FCOM).

AC BUS 1 FAULT: a few seconds on the AC ESS BUS transfer

AC BUS 1 normally supplies the AC ESS BUS. With AC BUS 1 lost, the AC ESS FEED automatic switching moves the AC ESS BUS to AC BUS 2 (FCOM); on this branch that takes **3 s** (`AC_ESS_FEED_TO_AC_BUS_2_DELAY_IN_SECONDS`). FCOM: "the AC ESS BUS and the DC ESS BUS will automatically recover, due to the fact that the AC BUS 2 will automatically supply the AC ESS BUS" . During the transfer the captain PFD, the captain ND and the E/WD, which read the AC ESS BUS, go blank for **up to 3 s** and then come back by themselves.

> **NOTE**
> Observation, not rechecked: in an earlier sim test the AP disconnected (AP OFF warning) during this transfer. That test was flown before this branch's fix of the DC ESS power break on DC bus changes, which also dropped the AP and A/THR. If the AP drops, re-engage it once the displays are back.

SEC power-up test after a power cut

After a power cut of more than 20 ms the SEC runs its 30 s power-up test when its power comes back, also in flight (`Sec.cpp`, the FlyByWire logic): its FAULT light comes on during the first 2.5 s and again from 27 to 29.5 s, and the E/WD can show F/CTL SEC FAULT for a moment. No action: the SEC is back after the test. (A fork change that skips this test in flight is only, not on this branch.) The ELACs and FCDCs restart by themselves when their power comes back.

IDG disconnection is final

The IDG pushbutton disconnects the drive from the engine for good: on the aircraft "only maintenance personnel can reconnect it" (FCOM). On this branch the drive is never reconnected for the rest of the flight, in the air or on the ground (`engine_generator.rs`); an IDG oil overheat above 200 °C also disconnects it by itself (thermal disconnect). That engine generator stays lost: the other generator, or the APU GEN, supplies its side.

> **NOTE**
> Disconnect the IDG only with the engine running or windmilling, and do not hold the IDG pushbutton for more than 3 s (FCOM).

How to use

1. After an electrical failure is over, read the E/WD: an **AUTO FLT FAC 1(2) FAULT** or **FAC 1+2 FAULT** that stays means the FAC latched. Overhead FLT CTL panel: that **FAC pushbutton OFF, then ON** (FAC 1 first when both).
2. Check the FAULT light goes off and the E/WD alert goes; re-engage the AP and the A/THR (FCU).
3. With AC BUS 1 lost: wait for the captain displays to come back (3 s), then re-engage the AP if it dropped.
4. A short F/CTL SEC FAULT about 28 s after the power came back is the SEC power-up test: no action.
5. Before pushing an IDG pushbutton, remember the generator is lost for the rest of the flight.

How to test

#	Do this	Expected result
1	A32NX in cruise, AP 1 and A/THR on. flyPad > Failures > ATA 24: activate *DC 1*, then *DC 2*.	ELEC DC BUS 1+2 FAULT. The DC ESS BUS stays powered: normal law, AP 1 and A/THR stay engaged. AUTO FLT FAC 2 FAULT shows (FAC 2 on DC BUS 2).
2	flyPad: deactivate *DC 1* and *DC 2*. Wait 30 s.	DC BUS 1 and 2 are back; AUTO FLT FAC 2 FAULT stays, FAC 2 FAULT light on.
3	Overhead FLT CTL panel: **FAC 2** pushbutton OFF, then ON.	The FAC 2 FAULT light goes off, the alert clears; AP 2 is available again.
4	flyPad: activate *DC 1*, *DC 2* and *DC ESS* (DC EMER CONFIG), wait 10 s, then deactivate the three.	AUTO FLT FAC 1+2 FAULT stays after the failures are removed; alternate law, no AP, no A/THR.
5	FAC 1 OFF then ON, then FAC 2 OFF then ON.	Both FAULT lights off, the alert clears, normal law; AP and A/THR can be engaged again.
6	flyPad: activate *AC 1*.	ELEC AC BUS 1 FAULT; the captain PFD, ND and the E/WD blank for up to 3 s and come back (AC ESS BUS on AC BUS 2). If the AP dropped, re-engage it.

7	Within about 30 s after a DC bus came back, watch the SEC pushbuttons.	A SEC FAULT light may come on briefly (power-up test, around 27 s) and go off by itself.

48. Crew procedures: manual cabin pressure, ignition indications, engine fire, A/THR without V2

> **Aircraft:** A32NX | **Where:** Overhead CABIN PRESS panel, ENG FIRE pbs; pedestal ENG MODE selector; FCU | **Status:** Manual cabin V/S tested in the sim by the user: OK (- with short taps: cabin V/S at most 2,070 ft/min). Engine fire run-on: OK. IGNITION memo: FAILED (no memo with continuous ignition on), open. A/THR with V2: OK; the no-V2 procedure is not sim-tested.

What it is for

Procedures and indications that surprised during the sim tests: how to fly the cabin pressure by hand, why the SD shows no IGN with the continuous ignition on, why an engine keeps running after its FIRE pushbutton, and how to get the A/THR without a V2. Checked against the code of this branch and the A320 FCOM.

Manual cabin pressure control (MAN V/S CTL)

When both pressure controllers are lost (CAB PR SYS 1+2 FAULT: MODE SEL MAN, MAN V/S CTL AS RQRD, FCOM; in a DC EMER CONFIG test the cabin climbed at about 5,500 ft/min with CAB PR EXCESS CAB ALT), the crew drives the outflow valve with the MAN V/S CTL switch: **UP opens** the valve (cabin climbs), **DN closes** it (cabin descends) (FCOM). The manual motor, on the DC BAT BUS, moves the valve over its full travel in **55 s** (`pressure_valve.rs`). In cruise the valve is only a few percent open, so the cabin V/S reacts strongly: about **1,800 ft/min per second** of input in the model (an estimate).

- Use **short taps of 0.2 to 0.5 s**, then wait and watch the cabin V/S on the SD CAB PRESS page: "only short inputs should be applied" (FCOM), and the valve change can take 10 s to show.
- Never hold the switch to drive the valve to a stop in cruise: the cabin climbs or dives by thousands of ft/min.
- Targets from the STATUS page: climb 500 ft/min, descent 300 ft/min; cabin altitude by flight level (FL390 8,000 ft ... FL250 3,000 ft); in final approach MAN V/S CTL FULL UP, delta P zero before opening the doors.
- To recover: **MODE SEL back to AUTO** once a controller works again (e.g. after the electrical failure is removed): the automatic system takes the valve back. MAN for at least 10 s then AUTO also selects the other system (FCOM).

IGNITION memo and the SD IGN indication

The green **IGNITION** memo shows "when continuous ignition is activated on any engine" (FCOM): ENG MODE selector at IGN/START with the engine running, the FADEC automatic ignition, or the engine anti-ice on this branch. The SD ENGINE page shows the igniters (IGN A / B), the start valve and the PSI only with the ENG MODE selector at **IGN/START or CRANK** (`StartIndications.ts`; the ENGINE page comes at the start sequence or at CRANK, FCOM). So with the selector at NORM and the continuous ignition on (engine anti-ice), the SD shows no IGN letters: the E/WD memo is the indication.

> **NOTE**
> Known issue: in a sim test the IGNITION memo did not show although the continuous ignition variable was on. The cause is not found yet.

An engine keeps running after its ENG FIRE pushbutton

The ENG FIRE pushbutton closes the LP fuel valve (FCOM); the engine then runs on the fuel left between the valve and the nozzles and stops by itself, "after a time delay up to 2 min 30 s" (FCOM, on this branch through the LP valve model of `core/fire`). How long it runs on depends on the engine state (thrust, airport altitude): in the model about 1 min 40 s at a sea-level idle, never more than 2 min 30 s. The FCOM ENG FIRE procedure sets the **ENG MASTER OFF first**, which closes the LP and HP valves and stops the combustion without run-on, then pushes the FIRE pushbutton.

A/THR without a V2

The A/THR arms at FLX/TOGA only when SRS engages, and SRS needs a V2: "If V2 is not inserted, the speed reference system (SRS) will not engage for takeoff" (FCOM). Without V2 the A/THR stays off at take-off, V/S engages 5 s after lift-off and the current speed becomes the FCU target. This is the FlyByWire FMGC code (`FmgcComputer.cpp`, unchanged from FlyByWire master), the same on every branch.

1. Normal: enter V2 on the MCDU PERF TAKE OFF page (with the other speeds): the A/THR arms at FLX/TOGA.
2. Without V2, at ACC ALT (FCOM TAKEOFF WITH NO V2 ENTRY): push the **A/THR** pushbutton on the FCU, set the thrust levers to the **CL** detent, push the SPD knob for a managed speed (or select the climb speed and pull it).
3. If the A/THR still does not engage (seen after a quick runway start without V2): push or pull the FCU SPD knob first so that a speed target exists, then push the A/THR pushbutton.

How to use

1. Manual pressure: MODE SEL MAN, short MAN V/S CTL taps (0.2-0.5 s) towards the STATUS target V/S, watch the SD CAB PRESS page; MODE SEL AUTO when a controller is back.
2. Continuous ignition: look for the green IGNITION memo on the E/WD; the SD IGN letters only show at IGN/START or CRANK.
3. Engine fire: follow the ECAM order (THR LEVER IDLE, ENG MASTER OFF, ENG FIRE pb PUSH, AGENT 1 after 10 s).
4. Take-off: always enter V2; if it was not entered, follow the no-V2 steps above at ACC ALT.

How to test

#	Do this	Expected result
1	A32NX in cruise at FL250 or above, cabin pressure AUTO. Overhead CABIN PRESS panel: MODE SEL **MAN**. Open the SD CAB PRESS page.	MAN shows in the MODE SEL pushbutton; the outflow valve stays where it was.
2	MAN V/S CTL: one short tap (0.2-0.5 s) UP, then wait 10 s.	The cabin V/S increases (climb) by a few hundred ft/min (about 1,800 ft/min per second of input); a longer input gives a much larger change.
3	Short taps DN until the cabin V/S is near 0, then MODE SEL **AUTO**.	The automatic system takes the valve back; the cabin V/S returns to its normal schedule.
4	Engines running, ENG MODE selector at NORM. Overhead ANTI ICE panel: ENG 1 pb ON; SD ENGINE page.	E/WD green IGNITION memo (known issue: it did not show in a sim test). The SD shows no IGN letters (selector at NORM).
5	ENG MODE selector to IGN/START.	The SD ENGINE page shows the IGN letters, the start valves and the PSI.
6	On the ground, engines at idle, no fire: ENG MASTER 1 left ON, push the ENG 1 FIRE pushbutton.	The SD FUEL LP valve 1 closes (amber crossline); engine 1 keeps running, then stops by itself within 2 min 30 s.
7	Take-off with V2 entered, thrust levers FLX or TOGA.	FMA: A/THR armed (blue), SRS engaged.
8	Take-off without V2 (PERF page V2 empty), thrust levers FLX or TOGA. At ACC ALT: A/THR pushbutton, thrust levers CL, SPD knob pushed.	No A/THR at take-off; V/S engages 5 s after lift-off; after the pushbutton the A/THR is active in the CL detent.

Display unit failures and reconfiguration

flyPad failures for the A32NX ND and ECAM display units and the eight A380X CDS display units: the failed display goes blank, with the A380X CDS ... DU FAULT alert; both aircraft move a lost display to another display unit (A32NX: PFD/ND XFR pbs, ECAM/ND XFR selector, held system page keys; A380X: PFD/ND and DU RECONF pushbuttons, the EWD and the SD also on the ND display unit); the A380X hosted EWD and PFD now hidden when not shown, and the CAPT PFD on the CAPT FMS side

Aircraft	A32NX and A380X
Where to find it	flyPad > Failures > ATA 31 Indicating / Recording System; the display units; A32NX PFD/ND XFR pbs, brightness knobs, SWITCHING panel ECAM/ND XFR, ECAM control panel; A380X E/WD and SD STATUS; A380X PFD/ND and DU RECONF pushbuttons
Branch	`core/display-unit-failures`, based on `core/nd-weather-terrain` (stacked: needs core/nd-weather-terrain merged first, the radar / terrain gauge goes blank with a failed ND). The A32NX reconfiguration also needs the cockpit model companion in aircraft-large-files (fork branch `feat/a32nx-du-reconfiguration`, b790e8d).
Status	All four features tested in the sim by the user: the failures and the A380X reconfiguration OK, the EWD and SD on the ND display unit included (load, CAPT, F/O, both sides); the A32NX reconfiguration OK on all steps. The A380X mouse fix for DUs holding hidden displays: OANS clicks on the F/O ND OK on a combined test build, not sim-tested on this core branch. Hosted display gate and CAPT PFD FMS side (added): not sim-tested, unit tests only.
Pictures	No pictures: a failed display unit is blank, the alerts are the stock FBW E/WD and SD displays.

1. A32NX ND and ECAM display unit failures
2. A32NX display unit reconfiguration
3. A380X CDS display unit failures
4. A380X display unit reconfiguration
5. A380X hosted EWD and PFD: hidden when not shown
6. A380X CAPT PFD reads the CAPT side FMS data

1. A32NX ND and ECAM display unit failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 31; CPT / F/O ND, upper and lower ECAM | **Status:** Tested in the sim by the user: OK.

What it is for

The flyPad could fail the two PFDs only. Four more display units (DUs) can now be failed: the CAPT and F/O NDs and the upper and lower ECAM DUs. A failed DU goes blank, the way a failed PFD already did. The A320 FCOM (DSC-31-05-60, FAILURE OF A DU) lists a blank screen with an amber F, a distorted display or a blank screen with INVALID DISPLAY UNIT; the blank (OFF) screen is used here.

- The weather radar and terrain picture of the ND (the native gauge over it) goes blank with a failed ND.
- It is also no longer drawn on an ND that is switched off (brightness knob OFF) or still in its self-test.
- The lower ECAM failure blanks the whole SD (both of its display layers). Before, the SD display unit ignored a failure change.
- When the failure is cleared, the DU runs its self-test, then shows its picture again.

The flyPad failures (ATA 31 Indicating / Recording System)

flyPad failure name	Display unit that goes blank
Captain PFD display	CAPT PFD (existing failure)
F/O PFD display	F/O PFD (existing failure)
Captain ND display	CAPT ND, with its weather / terrain picture

F/O ND display	F/O ND, with its weather / terrain picture
Upper ECAM display	Upper ECAM DU (E/WD)
Lower ECAM display	Lower ECAM DU (SD)

Limitations and why

Limitation	Why	What to do
No ECAM alert for a failed DU.	The branch only blanks the DU.	-
A lost DU's picture moves only as the FCOM says.	See "A32NX display unit reconfiguration" below: the PFD goes to the ND DU and the E/WD to the lower ECAM DU by themselves; the ND and the SD need the PFD/ND XFR pb, the ECAM/ND XFR selector or a held system page key (FCOM DSC-31-05-60).	Use those controls.
Always a blank screen.	The amber F, the distorted picture and INVALID DISPLAY UNIT of the FCOM are not drawn.	-

How to use

1. Open **flyPad > Failures**, chapter **ATA 31 Indicating / Recording System**.
2. Activate one of the display failures (table above): that DU goes blank.
3. Deactivate it to get the DU back (after its self-test).

How to test

#	Do this	Expected result
1	A32NX, any phase, all displays on. CPT ND in ARC with the weather radar on (weather around) or TERR ON ND. flyPad > Failures > ATA 31: activate **Captain ND display**.	The CPT ND goes blank, with no weather or terrain left on it. The F/O ND is not affected.
2	Deactivate **Captain ND display**.	The CPT ND shows its self-test, then the ND picture with the weather / terrain.
3	Activate **F/O ND display**, then deactivate it.	Same on the F/O ND.
4	Activate **Upper ECAM display**.	The upper ECAM DU goes blank; the E/WD moves to the lower ECAM DU (automatic reconfiguration, next section).
5	Deactivate it; activate **Lower ECAM display**.	The E/WD comes back on the upper DU after its self-test; the lower ECAM DU (SD) goes blank, whatever SD page was shown.
6	Deactivate it.	The SD comes back after its self-test.
7	No failure. With the weather radar picture on the CPT ND, turn the CPT ND brightness knob fully OFF, then back on.	The ND and its weather picture go dark together (no weather left on the dark ND) and come back together.

2. A32NX display unit reconfiguration

> **Aircraft:** A32NX | **Where:** PFD/ND XFR pbs; PFD and ND brightness knobs; SWITCHING panel ECAM/ND XFR; ECAM control panel page keys | **Status:** Tested in the sim by the user on all steps: OK. Unit tests: DisplayReconfiguration.spec.ts (shared), DmcDisplayReconfiguration.spec.ts (systems host), displayUnit.spec.tsx (common DU and legacy SD).

What it is for

The A320 moves the picture of a lost display unit (DU) to another one (A320 FCOM DSC-31-05-60 RECONFIGURING DUS). Before, a failed or switched-off A32NX DU simply lost its picture. Now the PFD, the ND, the E/WD and the SD can be shown on another DU, automatically or with the cockpit controls, as the FCOM describes.

> **NOTE**
> A320 FCOM DSC-31-05-60 : if the upper ECAM display fails or is switched off, "the engine/warning page automatically replaces the system/status page on the lower ECAM DU"; "If a PFDU fails, the system automatically transfers the PFD image to the NDU."

> **NOTE**
> A320 FCOM DSC-31-50 : turning the PFD brightness knob fully counterclockwise switches off the PFD, its image goes to the NDU, and the pilot can recover the ND with the PFD/ND XFR pushbutton. DSC-31-30 : the ECAM/ND XFR selector moves the system/status display to the CAPT or F/O ND with "ECAM ON ND" on the lower ECAM display; with both ECAM DUs failed it moves the E/WD, without that message.

What you should see

Situation	What you should see
PFD DU lost (flyPad failure, PFD brightness knob OFF, or no power)	The PFD DU is blank and the PFD is shown on the ND DU of the same side (the ND is lost until the PFD/ND XFR pb is pressed).
PFD/ND XFR pb pressed	The PFD and the ND of that side swap DUs; press again to swap back. After an automatic transfer, the pb brings the ND back on the ND DU (FCOM DSC-31-50).
ND DU lost	Nothing automatic; the PFD/ND XFR pb puts the ND on the PFD DU (FCOM).
Upper ECAM DU lost	The E/WD replaces the SD on the lower ECAM DU.
Lower ECAM DU lost	Nothing automatic; the SD is lost until ECAM/ND XFR or a held page key is used.
ECAM/ND XFR on CAPT (F/O)	The SD is shown on the CAPT (F/O) ND DU; the lower ECAM DU shows **ECAM ON ND**. The SWITCHG PNL memo comes on only if a PFD/ND XFR pb is pressed as well (FCOM DSC-31-30 MEMO DISPLAY, the EIS DMC selector alone still shows it).
Both ECAM DUs lost, ECAM/ND XFR on CAPT (F/O)	The E/WD is shown on that ND DU, with no ECAM ON ND message.
A system page key held (0.5 s) with one ECAM DU lost	The SD (the page of that key) is shown on the remaining ECAM DU instead of the E/WD, for at most 3 minutes; releasing the key brings the E/WD back. With both ECAM DUs lost, it shows the SD on the ND chosen with ECAM/ND XFR.
Lost DU recovered	Each picture goes back to its own DU; the recovered DU runs its self-test first.

How it is done

No extra copies of the instruments. MSFS draws a gauge on the model material of the same name, so every DU has one glass per picture it can show (17 overlay glasses carrying the material of that picture, an ECAM ON ND glass and a blank glass). The DMC logic of the systems host writes the picture of each DU, and the cockpit behaviour shows the matching glass. The brightness stays the DU's own knob. A new small gauge (`DuMessage`) draws the ECAM ON ND message.

Design choices

- The DU that receives a picture does not run a self-test: the picture is already live (the FCOM gives none).

- A system page key must be held 0.5 s before the SD appears: a click that only selects or deselects a page does not flash the SD, and the SD has its new page drawn before it is shown.
- ECAM ON ND is drawn in the style of the SELF TEST IN PROGRESS message: the FCOM gives no colour.
- A lost DU shows dark glass (a blank screen, which the FCOM allows for a failed DU).
- An unpowered DU counts as unavailable, like a failed one or one switched off with its knob, so its picture moves.
- A change of the PFD DU (lost or back) restarts from the automatic configuration: the PFD always goes to an available DU first.

Limitations and why

Limitation	Why	What to do
No DMC failures, no INVALID DATA, the EIS DMC selector only feeds the memo.	The three DMCs are not modelled (FCOM DSC-31-05-50).	-
No STS re-press within 2 s for the next STATUS page and no advisory key-light pulse in the single-display mode.	Not part of this change; there is no ECAM advisory logic yet.	-
An ECAM picture on an ND DU is replaced by the PFD when that side's PFD DU is lost or its PFD/ND XFR pb is pressed.	Design choice: the PFD keeps the priority on the ND DU; the picture moved to the PFD DU is always the ND.	Use the other ND with ECAM/ND XFR.
Needs the cockpit model companion and an MSFS restart.	The glasses are in the A32NX interior model (aircraft-large-files fork branch `feat/a32nx-du-reconfiguration`, b790e8d, stacked on `feat/a32nx-printer-door` 99237fb and the F/O flyPad screen d3060ea); panel.cfg (new `DuMessage` gauge, `[VCockpit21]` on this branch) and the behaviour XML change.	Install the model companion and restart the sim. With `core/efb-redesign` also merged, renumber one of the two `[VCockpit21]` sections.

How to use

1. Nothing to do for the automatic transfers: a lost PFD DU or upper ECAM DU moves its picture by itself.
2. Press the **PFD/ND XFR** pushbutton of your side to swap your PFD and ND (or to get the ND back after an automatic PFD transfer).
3. To see the SD with an ECAM DU lost: turn the **ECAM/ND XFR** selector on the SWITCHING panel to CAPT or F/O, or hold the system page key on the ECAM control panel.
4. Turn the ECAM/ND XFR selector back to NORM to get the ND back.

How to test

#	Do this	Expected result
1	A32NX powered, all displays on. flyPad > Failures > ATA 31: activate **Captain PFD display**.	The CAPT PFD DU goes blank; the PFD appears on the CAPT ND DU at once (no self-test there).
2	Press the CAPT **PFD/ND XFR** pushbutton.	The ND comes back on the CAPT ND DU; the PFD DU stays blank.
3	Deactivate the failure.	The CAPT PFD DU runs its self-test and shows the PFD; the ND DU shows the ND.
4	Press the CAPT PFD/ND XFR pb, then again.	The PFD and ND swap DUs, then swap back.

5	Turn the F/O PFD brightness knob fully OFF, then back on.	The PFD appears on the F/O ND DU, then goes back to the PFD DU.
6	Activate **Upper ECAM display**.	The upper ECAM DU goes blank; the E/WD replaces the SD on the lower ECAM DU.
7	Hold a system page key (e.g. **HYD**) on the ECAM control panel, then release it.	After about half a second the HYD page is shown on the lower ECAM DU; on release the E/WD comes back. A short click does not flash the SD.
8	Turn the **ECAM/ND XFR** selector to CAPT.	The SD is shown on the CAPT ND DU; the lower ECAM DU keeps the E/WD (the upper DU is still lost).
9	Deactivate the failure, keep ECAM/ND XFR on CAPT.	The E/WD is back on the upper DU; the lower ECAM DU shows **ECAM ON ND**; no SWITCHG PNL memo.
10	Press the CAPT PFD/ND XFR pb.	The CAPT PFD and ND swap DUs (the PFD replaces the SD on the CAPT ND DU, design choice) and the SWITCHG PNL memo appears in green. Press it again: the SD is back on the CAPT ND DU and the memo goes.
11	Activate **Upper ECAM display** and **Lower ECAM display**, ECAM/ND XFR on F/O.	The E/WD is shown on the F/O ND DU, without ECAM ON ND. Holding a page key shows the SD there instead.
12	Deactivate all failures, ECAM/ND XFR to NORM.	Every DU shows its own picture again.

3. A380X CDS display unit failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 31; the 8 CDS DUs; E/WD; SD STATUS | **Status:** Tested in the sim by the user: OK.

What it is for

The eight display units of the A380 CDS (A380 FCOM DSC-31-15-10: CAPT PFD, CAPT ND, CAPT MFD, E/WD, SD, F/O MFD, F/O ND, F/O PFD) can be failed from the flyPad. The failed DU goes blank (on an ND, the weather radar and terrain picture too, the VD included) and the FWS raises the **CDS ... DU FAULT** alert with its FCOM procedure and its INOP SYS line (FCOM PRO-ABN-ECAM-10-31).

The flyPad failures (ATA 31 Indicating / Recording System)

flyPad failure name	ECAM alert (amber)	STATUS INOP SYS
CAPT PFD display unit	CDS CAPT PFD DU FAULT	CAPT PFD DU
CAPT ND display unit	CDS CAPT ND DU FAULT	CAPT ND DU
CAPT MFD display unit	CDS CAPT MFD DU FAULT	CAPT MFD DU
EWD display unit	CDS EWD DU FAULT	EWD DU
SD display unit	CDS SD DU FAULT	(none: the FCOM lists no SD line)
F/O MFD display unit	CDS F/O MFD DU FAULT	F/O MFD DU
F/O ND display unit	CDS F/O ND DU FAULT	F/O ND DU
F/O PFD display unit	CDS F/O PFD DU FAULT	F/O PFD DU

What you should see

Situation	What you should see

Failure activated	The DU goes blank. Single chime, MASTER CAUT, and the amber alert on the E/WD.
ECAM procedure (each DU)	*xx* DU ..... OFF 5 S THEN ON; IF NOT SUCCESSFUL: *xx* DU ..... OFF; DU RECONF P/B AVAIL (green). The SD DU FAULT adds: IF SD DU OFF : F/O DU RECONF P/B FOR ATC MAILBOX.
SD STATUS page	INOP SYS (all phases): the DU line of the table above.
Takeoff and landing	The alert is inhibited in the FWS flight phases 3 to 7, 9 and 10; the DU is blank anyway.
DU without power	No DU FAULT alert (design choice): the lost busbar has its own ELEC alert.
Failure cleared	The DU runs its self-test and shows its picture again; the alert and the INOP SYS line go.

Limitations and why

Limitation	Why	What to do
The reconfiguration has limits.	See "A380X display unit reconfiguration" below: the PFD and the EWD move to another DU by themselves, the MFD never does, and the SD only on an ND DU with the DU RECONF pb, without the ATC mailbox.	With the SD DU failed, press the F/O DU RECONF pb (it acts on the F/O ND DU) until the SD pages (STATUS included) show there.
The OFF 5 S THEN ON item does not bring the DU back.	The flyPad failure stays until it is cleared, which is the procedure's NOT SUCCESSFUL branch.	Clear the failure in the flyPad.
The second FCOM trigger (a DU that declared two other DUs faulty in turn) is not modelled.	It needs the DU monitoring loops, which are not simulated.	-

How to use

1. Open **flyPad > Failures**, chapter **ATA 31 Indicating / Recording System**.
2. Activate one of the eight **... display unit** failures.
3. Follow the ECAM procedure, then check the SD STATUS page.
4. Deactivate the failure to get the DU back.

How to test

#	Do this	Expected result
1	A380X on the ground (before the takeoff power) or in cruise, all DUs on. CPT ND in ARC with the WX overlay and weather around. flyPad > Failures > ATA 31: activate **CAPT ND display unit**.	The CAPT ND and its VD go blank, with no weather or terrain left. Single chime, MASTER CAUT. E/WD: **CDS CAPT ND DU FAULT** with CAPT ND DU ..... OFF 5 S THEN ON, IF NOT SUCCESSFUL, CAPT ND DU ..... OFF, DU RECONF P/B AVAIL in green.
2	Open the SD **STATUS** page.	INOP SYS: **CAPT ND DU**.
3	Deactivate the failure.	The CAPT ND runs its self-test, then shows the ND, the VD and the weather again; the alert and the INOP SYS line are gone.
4	Repeat with **CAPT PFD display unit**, **CAPT MFD display unit**, **F/O PFD display unit**, **F/O ND display unit** and **F/O MFD display unit**.	Each time the named DU goes blank and the E/WD shows the matching CDS ... DU FAULT, with its INOP SYS line on the STATUS.

5	Activate **SD display unit**.	The SD goes blank; E/WD: **CDS SD DU FAULT** with the extra green line IF SD DU OFF : F/O DU RECONF P/B FOR ATC MAILBOX (the FCOM gives this alert no INOP SYS line).
6	Deactivate it; activate **EWD display unit**.	The EWD DU goes blank and the E/WD moves to the SD DU (automatic reconfiguration), where CDS EWD DU FAULT can be read; single chime and MASTER CAUT; INOP SYS **EWD DU** on the STATUS.
7	Activate any DU failure during the takeoff roll (above takeoff power).	The DU goes blank but no alert until the end of the inhibition.

4. A380X display unit reconfiguration

> **Aircraft:** A380X | **Where:** CAPT and F/O PFD/ND and DU RECONF pushbuttons; the 8 CDS DUs | **Status:** Tested in the sim by the user on all steps: OK. Mouse fix: OANS clicks on the F/O ND OK and the ATC mailbox on the F/O ND (needs `core/a380x-mailbox-du-reconf`) OK, on a combined test build; not sim-tested on this core branch. Unit tests: CdsReconfiguration.spec.ts (shared and systems host), HostedDisplay.spec.ts, hostedInstrument.spec.ts; native ndwxr test cds_display_test.

What it is for

The A380 CDS moves a lost display to another display unit (A380 FCOM DSC-31-15-20 CDS abnormal operations, DSC-31-20-50 PFD/ND pb). Before, a failed or switched-off DU simply lost its picture. Now:

- **Automatic:** when a PFD DU fails, loses its power or is switched off, the PFD is shown on the ND DU of the same side (the ND is lost). When the EWD DU is lost, the E/WD is shown on the SD DU.
- **PFD/ND pushbutton:** exchanges the on-side PFD and ND. With the exchange selected, the PFD is on the ND DU and comes back to the PFD DU if the ND DU fails (the PFD keeps the priority).
- **DU RECONF pushbutton:** active only with a failed on-side DU; it acts on the DU the FCOM gives (CAPT: the MFD DU, then the ND DU if the MFD DU is failed, then the PFD DU if the ND and MFD DUs are failed; F/O: the ND DU, the MFD DU if the PFD or ND DU is failed, then the PFD DU) and steps through the displays of the FCOM order that this DU can draw, then back to its normal display.
- **EWD and SD on the ND DU:** with the CAPT (F/O) PFD and MFD DUs failed, the pb acts on the ND DU, which shows the PFD automatically; it now cycles **PFD > EWD > SD > PFD** (the MFD of the FCOM sequence cannot be drawn there). With only the MFD DU (CAPT) or the SD DU (F/O) failed, the ND DU cycles ND > EWD > SD > ND the same way. The EWD (SD) is never shown on both sides at once (A380 FCOM DSC-31-15-20), and a PFD that moves to the ND DU later (its PFD DU fails afterwards) wins over a selection there.
- The weather radar and terrain picture (ndwxr) is hidden while its ND DU shows another display (the PFD, the EWD or the SD).

What you should see

Situation	What you should see
CAPT PFD display unit failed (or switched off)	The CAPT PFD DU is blank and the PFD appears on the CAPT ND DU, without weather or terrain on it.
EWD display unit failed	The E/WD appears on the SD DU.
PFD/ND pb pressed (CAPT or F/O)	The PFD and the ND of that side swap DUs; press again to swap back.
CAPT ND and MFD DUs failed, DU RECONF pressed	The CAPT PFD DU steps from the PFD to the ND (with no weather or terrain image), then back to the PFD.

CAPT PFD and MFD DUs failed, DU RECONF pressed	The CAPT ND DU, which shows the PFD, steps to the E/WD, then to the SD, then back to the PFD. While the E/WD (SD) is shown there, the F/O pb skips it.
SD DU failed, F/O DU RECONF pressed	The F/O ND DU steps from the ND to the E/WD, then to the SD pages (without the ATC mailbox), then back to the ND.
Failure cleared	Each DU goes back to its normal display.

The mouse on a DU with hidden displays

With the reconfiguration, every DU also holds the gauges of the displays it can show (the ND DUs hold a PFD, an E/WD and an SD as well). A hidden gauge stayed on top and took the mouse, so the OANS clicks on the F/O ND did not work (and the ATC mailbox on an ND DU of `core/a380x-mailbox-du-reconf`). Now the hosted E/WD and PFD never take the mouse, the hosted SD takes it only while it is shown, and the ND keeps it (OANS). The build adds one CSS rule per hosted gauge, and the display gate of each DU marks the hidden ones, found by their panel.cfg URL.

Limitations and why

Limitation	Why	What to do
The MFD is never drawn on another DU, and nothing is drawn on the MFD DUs; the DU RECONF pb skips them, so it does nothing when it acts on an MFD DU.	Every display is a gauge of its own on its DU texture: the MFD is one gauge over both MFD screens (one texture) that also runs the FMS. Design choice: the PFD and the ND on both DUs of a side, the E/WD and the SD also on the ND DUs, and the E/WD on the SD DU.	Use the other side's MFD.
The SD on an ND DU has no ATC mailbox.	The mailbox stays with the SD DU's own SD (design choice of the hosted SD). `core/a380x-mailbox-du-reconf` brings it to the ND DU.	With the SD DU failed, the ATC mailbox is lost on this branch alone.
A selected E/WD or SD replaces the PFD on that side.	With the PFD and MFD DUs of a side failed, the ND DU is the only DU left for the PFD (FCOM: the pb acts on the ND DU).	Press DU RECONF again to get the PFD back.
The ND on the PFD DU has no weather radar or terrain image.	The native ndwxr gauges stay on the ND DUs.	Use the other side's ND.
The KCCU keys do not reconfigure the DUs.	Not modelled.	Use the DU RECONF pb.
A full MSFS restart is needed after an update.	panel.cfg (new hosted gauges, the E/WD and the SD on the ND DUs included) and the cockpit behaviour XML change.	Restart the sim after installing.

How to use

1. Nothing to do for the automatic reconfiguration: a lost PFD or EWD DU moves its display by itself.
2. Press the **PFD/ND** pushbutton of your side to exchange your PFD and ND.
3. With a failed on-side DU, press **DU RECONF** to step the display of the DU it acts on.
4. With your PFD and MFD DUs failed, press **DU RECONF** to bring the E/WD, then the SD, onto your ND DU; once more for the PFD.

How to test

#	Do this	Expected result

1	A380X powered, all DUs on. flyPad > Failures > ATA 31: activate **CAPT PFD display unit**.	The CAPT PFD DU goes blank; the PFD appears on the CAPT ND DU (no weather / terrain image on it); CDS CAPT PFD DU FAULT.
2	Deactivate it. Press the CAPT **PFD/ND** pushbutton.	The CAPT PFD and ND swap DUs. Press again: back to normal.
3	Turn the F/O PFD DU brightness knob OFF.	The PFD appears on the F/O ND DU. Turn it back on: normal again.
4	Activate **EWD display unit**.	The E/WD appears on the SD DU. Deactivate it: the SD comes back.
5	Activate **CAPT ND display unit** and **CAPT MFD display unit**, then press the CAPT **DU RECONF** pushbutton twice.	The CAPT PFD DU shows the ND, then the PFD again.
6	Deactivate them. Activate **CAPT PFD display unit** and **CAPT MFD display unit**.	The PFD appears on the CAPT ND DU.
7	Press the CAPT **DU RECONF** pushbutton three times.	The CAPT ND DU shows the E/WD, then the SD (the current SD page), then the PFD again. No weather or terrain image on it meanwhile.
8	Press the CAPT DU RECONF pb once (E/WD on the CAPT ND DU). Then activate **F/O PFD display unit** and **F/O MFD display unit** and press the F/O **DU RECONF** pb.	The F/O ND DU skips the E/WD (shown on the CAPT side) and shows the SD.
9	Deactivate all failures. Activate **SD display unit** and press the F/O **DU RECONF** pb twice.	The F/O ND DU shows the E/WD, then the SD pages (no ATC mailbox); once more: the ND again.
10	Deactivate all failures.	Every DU shows its normal display.
11	No failure, on the ground at an airport: show the OANS airport map on the F/O ND (F/O EFIS control panel, range ZOOM). Drag the map with the mouse and click a runway or taxiway label.	The map pans and the click is taken on the F/O ND (before the mouse fix nothing reacted there).

5. A380X hosted EWD and PFD: hidden when not shown

> **Aircraft:** A380X | **Where:** SD DU and both ND DUs (where the EWD and PFD can be shown after a DU RECONF) | **Status:** Not sim-tested (added; unit tests: EWD/instrument.spec.ts 3 and PFD/instrument.spec.ts 2, loading the real instruments in a sim-like gauge panel; 3 fail before the change).

What it is for

The EWD and the PFD also run as hidden copies on the display units that can take them over (EWD: SD DU and both ND DUs; PFD: both ND DUs). Each copy has a gate that hides it and stops it while its DU shows its own display. The gate was never made: the instruments looked up their display unit before the sim had connected the gauge, found the wrong one and made no gate. The hidden copies then ran and rendered all the time and were never hidden. The gate is now made once the gauge is connected.

Situation	What you should see
Normal configuration	The hidden EWD and PFD copies are hidden and stopped on the SD and ND DUs.
EWD DU failed (or DU RECONF), EWD on the SD DU or an ND DU	The EWD copy shows there, as before.
PFD shown on the ND DU	The PFD copy shows there, as before.

Limitations and why

Limitation	Why	What to do
No visible change in the normal configuration.	The fix stops hidden copies; the effect is less work for the sim and no hidden copy in front of the DU.	-

How to use

1. Nothing to set: use the DU RECONF and PFD/ND pushbuttons as in the reconfiguration feature above.

How to test

#	Do this	Expected result
1	A380X powered, no failure. Use the SD mailbox buttons and click on the ND / OANS.	The clicks are taken (no hidden copy in front).
2	flyPad > Failures > ATA 31: **EWD display unit**.	The EWD shows on the SD DU.
3	Remove the failure; press the CAPT PFD/ND pb (PFD on the ND DU), then again.	The PFD shows on the ND DU, then the ND comes back.
4	Press a DU RECONF pb through its cycle.	Each display shows on the ND DU and hides again.

What PR testers should check

- SD mailbox and ND / OANS clicks.
- The EWD on the SD DU after an EWD DU failure; the PFD on the ND DU; the DU RECONF cycle.
- Not sim-tested yet.

6. A380X CAPT PFD reads the CAPT side FMS data

> **Aircraft:** A380X | **Where:** CAPT PFD (also when drawn on the CAPT ND DU) | **Status:** Not sim-tested (added; unit tests: PFD/instrument.spec.ts, 4 side cases, the CAPT case failing before).

What it is for

The captain PFD picked its FMS side before the sim had connected the gauge, and the side number it used was never 1: the CAPT PFD read the F/O side FMS data (the FM message flags, the TO waypoint, the MRP and the RNP). The side is now set when the gauge is connected, from the PFD's own display index (1 = CAPT, 2 = F/O, also when the PFD is drawn on an ND DU). The same bug is in FlyByWire upstream. The A32NX PFD is not affected.

Limitations and why

Limitation	Why	What to do
No visible change while both FMS sides agree.	The two sides normally show the same data.	Make them differ to see it (test below).

How to use

1. Nothing to set.

How to test

#	Do this	Expected result

1	A380X in flight on a flight plan. Set a different required RNP or FM message on each side (for example an RNP entered on the CAPT side only), then compare both PFDs.	The CAPT PFD shows the CAPT side data, the F/O PFD the F/O side data (before: both the F/O side).
2	CAPT PFD/ND pb: PFD on the CAPT ND DU.	Still the CAPT side data.

What PR testers should check

- The CAPT PFD FMA messages, RNP and TO waypoint match the CAPT FMS side.
- Not sim-tested yet.

A380X weather radar GAIN, ELEVN and TILT

The manual GAIN, ELEVN and TILT settings of the MFD SURV CONTROLS page reach the ND messages, the VD white line and the radar picture itself (GAIN moves the colour levels, a TILT or ELEVN in the ground cuts the weather), and start from their FCOM default values

Aircraft	A380X
Where to find it	MFD SURV > CONTROLS (WXR block: GAIN, ELEVN/TILT); ND weather picture and message (right side above the TERR box); VD
Branch	`core/a380x-wxr-controls`, based on `core/nd-weather-terrain` with `core/a380x-mfd-fms` merged in (stacked: needs core/nd-weather-terrain, core/surveillance and core/a380x-mfd-fms merged first). The terrain view resend fix of comes with its base `core/nd-weather-terrain`.
Status	ND messages tested in the sim by the user: OK; FCOM defaults user-checked in the sim; VD white line tested in the sim by the user: OK. GAIN acting on the radar picture tested in the sim by the user: OK; TILT / ELEVN cutting the picture not sim-tested (left out of that test flight); native test wxr_controls_test.
Pictures	No pictures: the messages and the line are drawn by the stock ND / VD code (no screenshot of this state).

1. ND messages of the manual GAIN, ELEVN and TILT
2. GAIN, TILT and ELEVN take the FCOM defaults at the first selection
3. VD white line of the manual ELEVN and TILT
4. GAIN and ELEVN/TILT change the radar picture

1. ND messages of the manual GAIN, ELEVN and TILT

> **Aircraft:** A380X | **Where:** ND, right side above the TERR box | **Status:** Tested in the sim by the user: OK.

What it is for

The GAIN button and the ELEVN/TILT option list of the MFD SURV CONTROLS page kept their state but showed nothing. The ND now shows the manual radar settings as the A380 FCOM WXR MESSAGES (DSC-34-20-30-20 P 17).

What you see

SURV CONTROLS setting	ND message
GAIN AUTO, ELEVN/TILT AUTO	**WX** in green (as before)
GAIN MAN, value 53	**GAIN 53%** in cyan, above the mode line; with ELEVN/TILT at AUTO the mode line is empty and GAIN takes its place
ELEVN/TILT at TILT, value -12.0	**TILT -12.0°** in cyan instead of WX
ELEVN/TILT at ELEVN, value 12500, QNH reference	**ELEVN 12500FT** in cyan instead of WX
ELEVN/TILT at ELEVN, STD baro reference on that side	**ELEVN FL125** in cyan
Manual mode without a value	GAIN ---%, TILT ---.-°, ELEVN -----FT (or ELEVN FL---)
MODE MAP	MAP in green (GAIN nn% above it with GAIN MAN); no ELEVN / TILT message
WXR OFF	WXR OFF in white, no GAIN line

Limitations and why

Limitation	Why	What to do
GAIN, ELEVN and TILT act on the radar picture only in part.	The MSFS radar has no gain, no cell heights and no tilt; tilting its beam crashed the gauge.	GAIN moves the colour levels; ELEVN / TILT only hide the weather where they reach the ground (see the last feature).
Cyan is a choice.	The A380 FCOM gives no colour; cyan is the colour of the manual modes of the same radar family in the A320 FCOM (DSC-34-SURV-30-30 P 10).	-
The grey background the FCOM figure prints behind GAIN is not drawn.	Not reproduced.	-

How to use

1. MFD **SURV > CONTROLS**, WXR block (WXR AUTO, MODE WX).
2. GAIN: press the button to **MAN**, enter a value in the field below it (0 to 100 %).
3. ELEVN/TILT: select **ELEVN** (enter an altitude or FL) or **TILT** (enter -15.0 to +15.0 degrees) in the field below the list.
4. Read the message on the ND (WX overlay selected). Back to **AUTO** for WX.

How to test

#	Do this	Expected result
1	A380X, CPT EFIS **WX** overlay, ND ARC. MFD SURV > CONTROLS: WXR AUTO, MODE WX, GAIN AUTO, ELEVN/TILT AUTO.	ND: **WX** in green on the right.
2	GAIN to **MAN**, enter **53**.	ND: **GAIN 53%** in cyan, the WX message gone.
3	ELEVN/TILT: select **TILT**, enter **-12**.	ND: **TILT -12.0°** in cyan, with GAIN 53% above it.
4	Select **ELEVN**, enter **12500** (QNH set on the CPT side).	ND: **ELEVN 12500FT** in cyan.
5	Set the CPT baro reference to **STD**.	ND: **ELEVN FL125**.
6	GAIN and ELEVN/TILT back to **AUTO**.	ND: **WX** in green, no GAIN line.
7	**MODE MAP** with GAIN MAN.	ND: **MAP** in green with the GAIN line above it.
8	WXR **OFF** (confirm).	ND: **WXR OFF** in white, no GAIN line.

2. GAIN, TILT and ELEVN take the FCOM defaults at the first selection

> **Aircraft:** A380X | **Where:** MFD SURV > CONTROLS, WXR block | **Status:** User-checked in the sim after a flight reload: GAIN 50 %, TILT +3.0° on the ground, ELEVN 100 FT at CYUL.

What it is for

A380 FCOM DSC-34-20-30-20-10 (GAIN and ELEVN knobs): "The default GAIN value is 50 %", "The default ELEVN value is the current aircraft altitude", the default TILT value is +3.00° on ground and 0° in flight. The page applies a default only when nothing is entered, but a value nobody had written read 0, a valid entry: after a power-up the first GAIN MAN, TILT or ELEVN selection kept 0. The MFD now starts with the three settings at no entry.

What you should see

Situation	What you should see
First GAIN MAN after the MFD starts	GAIN field 50 %; ND GAIN 50%.
First TILT selection on the ground	TILT +3.0°; ND TILT +3.0°.
First TILT selection in flight	TILT +0.0°.
First ELEVN selection	The current altitude, rounded to 100 ft (FL with the STD reference).
Later selections	The value entered or set before is kept.

How to use

1. Select GAIN **MAN**, **ELEVN** or **TILT** on the SURV CONTROLS page: the default value is entered.
2. Overwrite it in the entry field if needed.

How to test

#	Do this	Expected result
1	A380X on the ground, after a flight load or reload. MFD SURV > CONTROLS: GAIN to **MAN**.	GAIN field **50 %**; ND (WX overlay) **GAIN 50%**.
2	ELEVN/TILT: select **TILT**.	TILT field **+3.0°**; ND **TILT +3.0°**.
3	Select **ELEVN**.	ELEVN field = the airport elevation rounded to 100 ft (for example 100 FT at CYUL).
4	Enter TILT -5, go to AUTO, then select TILT again.	TILT **-5.0°** is kept (no default over an entry).
5	Reload the flight, take off, and select TILT for the first time in flight.	TILT **+0.0°**.

3. VD white line of the manual ELEVN and TILT

> **Aircraft:** A380X | **Where:** VD under the ND | **Status:** Tested in the sim by the user: OK.

What it is for

A380 FCOM DSC-34-20-30-20: "The VD displays a white line indicating the elevation or tilt value". In manual ELEVN the VD shows a horizontal white line at the selected altitude; in manual TILT a straight white line from the aircraft symbol at the selected angle (zero tilt = the horizon as seen by the ADIRS), as in the FCOM VD figures.

What you should see

Situation	What you should see
ELEVN selected	Horizontal white line at the selected altitude, from the aircraft to the end of the VD range.
TILT selected	Straight white line from the nose of the aircraft symbol, up for a positive tilt, down for a negative one.
Selected altitude off the VD scale	No line.
ELEVN/TILT AUTO, MODE MAP, WXR OFF or no WX overlay	No line (it goes with the ELEVN / TILT message of the ND).

Limitations and why

Limitation	Why	What to do
The line shows only while that side's ND shows the WX function.	Design choice: the line goes with the ND message.	-
The tilt line is straight.	Design choice, as in the FCOM figure: no earth curvature or beam refraction.	-
The weather on the VD does not follow ELEVN or TILT.	The MSFS radar has no height data; the ground cut of a manual ELEVN / TILT applies to the ND only. The VD weather colours follow the GAIN.	-

How to use

1. Select the **WX** overlay on the EFIS panel and ND ARC or ROSE NAV (the pages with the VD).
2. On MFD SURV > CONTROLS select **ELEVN** or **TILT** and enter the value.
3. Read the white line on the VD.

How to test

#	Do this	Expected result
1	A380X in flight, for example at 10 000 ft. CPT EFIS WX overlay, ND ARC 40 NM. SURV CONTROLS: ELEVN/TILT **ELEVN**, enter **12000**.	VD: a horizontal white line at 12 000 ft, ahead of the aircraft symbol.
2	Select **TILT**, enter **+3**.	VD: a straight white line from the nose of the aircraft symbol, rising to the end of the range.
3	Enter **-3**.	The line goes down from the aircraft symbol.
4	Select **ELEVN** with a value far above the VD scale (for example 60000).	No line.
5	ELEVN/TILT back to **AUTO**.	The line goes; ND shows WX.
6	With TILT selected, deselect the WX overlay.	The line goes with the ND message.

4. GAIN and ELEVN/TILT change the radar picture

> **Aircraft:** A380X | **Where:** ND weather picture; MFD SURV > CONTROLS, WXR block | **Status:** GAIN tested in the sim by the user: OK (FL150, ARC 40 NM: AUTO calibrated, 0 % sparse green, 100 % red cores, AUTO back to the calibrated picture, the VD following). TILT and ELEVN cutting the picture: not sim-tested. Native test wxr_controls_test (ndwxr).

What it is for

The manual GAIN, TILT and ELEVN entries changed the ND messages and the VD line only. They now act on the radar picture, within what the MSFS radar allows: its views take a table of rain rate bands, but have no cell heights and no tilt.

- **GAIN** moves the rain rate each colour starts at, as the receiver gain moves the reflectivity of each colour (A380 FCOM DSC-34-20-30-10/-20; Honeywell RDR-4000 Pilot manual: MIN about -16 dBZ, MAX about +10 dBZ, calibrated in AUTO). 50 %, the FCOM default, is the calibrated gain (the same picture as AUTO); 0 % is the MIN, 100 % the MAX. A lower gain turns red to yellow, yellow to green and removes green (Airbus Safety First #22); a higher gain the opposite. Turbulence (magenta) does not change.

- A manual **TILT** down shows the weather only up to the range where the tilted beam reaches the ground (the radar removes the ground returns with the TAWS terrain, DSC-34-20-30-10). A level or up tilt changes nothing.
- A manual **ELEVN** at or below the ground under the aircraft shows no weather; above it, the full picture.

What you should see

Situation	What you should see
GAIN AUTO, or MAN at 50 %	The calibrated picture (as before).
GAIN MAN 0 %	Red cells shrink or turn yellow, yellow turns green, light green disappears.
GAIN MAN 100 %	More and larger green, yellow and red areas.
TILT -10.0 at about 10 000 ft above the ground	Weather only inside about 9 NM.
TILT -3.0 at about 10 000 ft above the ground	Weather out to about 34 NM.
ELEVN at or below the ground elevation	No weather on the ND.
VD	Its weather colours follow the GAIN; the TILT / ELEVN cut is on the ND only.

Limitations and why

Limitation	Why	What to do
ELEVN and TILT cannot show a slice above or below the cells.	MSFS gives no cell tops: only the part in the ground can be removed.	-
The ground ahead is taken level at the elevation under the aircraft.	The gauge cannot read the terrain ahead back (design choice). The beam uses the earth curvature with the 4/3 radius of the standard refraction.	-
The GAIN scale is a design choice.	The FCOM gives the percentage but not its scale: 50 % calibrated, 0 % and 100 % the RDR-4000 MIN and MAX, linear in dB on each side.	-
A GAIN change may repaint the picture over a few seconds.	Not measured yet: the engine may repaint it sector by sector with its sweep.	-

How to use

1. MFD **SURV > CONTROLS**, WXR block (WXR AUTO, MODE WX).
2. GAIN **MAN** and a value: below 50 % to look into heavy cells (less colour), above 50 % to see weak returns. Back to **AUTO** for the calibrated picture.
3. ELEVN/TILT: **TILT** with a down angle (or **ELEVN**) to keep the weather only where the beam is above the ground. Back to **AUTO** for the full picture.

How to test

#	Do this	Expected result
1	A380X in flight at about FL100 (about 10 000 ft above the ground) with rain cells ahead. CPT EFIS: WX on, ND ARC 40 NM. MFD SURV > CONTROLS: WXR AUTO, MODE WX, TURB AUTO. Screenshot of the ND.	The calibrated picture.
2	GAIN button to **MAN** (field 50 %); wait 10 s.	Same picture as before.

3	Enter **0** in the GAIN field; wait 10 s.	Red cells shrink or turn yellow, yellow turns green, light green disappears. Magenta unchanged.
4	Enter **100**; wait 10 s.	More and larger green, yellow and red areas than at the start.
5	GAIN back to **AUTO**.	The first picture again. The VD weather colours followed each step.
6	ELEVN/TILT: select **TILT** (field +0.0).	The picture does not change.
7	Enter **-10**; wait 5 s.	Weather only inside about 9 NM; ND message TILT -10.0.
8	Enter **-3**; wait 5 s.	Weather out to about 34 NM; ND message TILT -3.0.
9	Over land at least 500 ft high: select **ELEVN**, enter 0 (FL 000 with STD); then 12000 (FL 120).	At 0: no weather on the ND. At 12000: the full picture.
10	ELEVN/TILT back to **AUTO**.	The full picture again.

A380X VD airports as flat ground

The A380X VD terrain profile shows the airports around the aircraft as flat ground at their elevation instead of spikes, and the ND / VD map views survive flight reloads

Aircraft	A380X (the map view fix: A32NX and A380X)
Where to find it	VD under the ND (terrain profile); ND weather and terrain
Branch	`core/a380x-vd-runways`, based on `core/nd-weather-terrain` with `core/oans-btv-raas` merged in (stacked: needs both merged first; the airports come from the nearby runway search of `core/oans-btv-raas`, merged again for its airports-with-a-runway fix).
Status	Airport circles: tested in the sim by the user: OK (one circle at RJTT); the empty list after take-off seen there is fixed and passed (group 2). The first version (runway rectangles) was tested. Map views across flight reloads: VD OK after 4 flight reloads, but the leak did not occur in that test, so the fix itself is not proven in the sim.
Pictures	No pictures.

1. VD airports drawn as flat ground

> **Aircraft:** A380X | **Where:** VD terrain profile | **Status:** Tested in the sim by the user: OK (one circle of radius 1.8 NM at RJTT, flat ground, no spikes). The airport list emptied at about 2,900 ft after take-off in that test; fixed by the runway search of `core/oans-btv-raas`, then OK (group 2 : 10 airports on the ground, 11 to 12 in the climb). The earlier runway rectangles: tested (VHHH OK, one short spike at CYUL). Unit tests: VdAirports.spec, ndwxr vd_airports_test.

What it is for

The MSFS terrain map draws the airport diagram in light grey over the terrain, and no setting hides it. The VD read that grey as terrain far above its plot: spikes wherever the vertical cut crossed an airport, the destination on approach included.

- The systems host searches the sim airports within 25 NM that have a land runway, and publishes up to 24 airports, closest first, as one circle each: the smallest circle around the corners of all its runways, plus 0.3 NM for the taxiways and aprons beside them, at the elevation of its highest runway threshold.
- Where the cut crosses such a circle, the VD draws flat ground at the airport elevation instead of the terrain map.
- The first version used a rectangle around each runway (plus 0.15 NM): taxiways and aprons between and beside the runways still showed as spikes. One circle per airport covers them (the user chose it over a fixed radius).

What you should see

Situation	What you should see
VD cut across an airport	Flat brown ground at the airport elevation over the whole airport (runways, taxiways, aprons), no spike.
Past the edge of the airport circle	The real terrain again.
Approach to the destination	The airport ahead shows as flat ground at its elevation.
After take-off, climbing away	The departure airport stays in the list (and flat) while it is within 25 NM.

Limitations and why

Limitation	Why	What to do

Real terrain inside an airport circle is flattened (for example a hill beside the runways).	Design choice: an airport is drawn as flat ground; the circle is the smallest one around the runways plus 0.3 NM.	-
Airports beyond 25 NM, or more than 24 airports, are not flattened.	Design choice of the search radius (enough for the VD ranges of the approach) and of the gauge list size.	-
Airports without a land runway of 100 m or more are not flattened.	The runway search leaves out heliports and water-only bases (`core/oans-btv-raas`), so they no longer push the real airports out of the list.	-

How to use

1. Nothing to set: read the VD terrain profile as usual (TERR SYS of the SURV page not OFF).

How to test

#	Do this	Expected result
1	A380X at a gate of an airport with parallel runways (for example CYUL or RJTT), flight plan from there. Both NDs ARC, range 10.	On the VD, flat brown ground at the airport elevation over the whole airport (no grey spikes from taxiways or aprons); past the airport edge, real terrain again.
2	Taxi, line up and take off; climb to 3,000 ft and beyond.	Flat ground under and ahead of the aircraft along the runway; after take-off the departure airport stays flat on the VD (the list does not empty).
3	A380X on approach to any airport, the VD cut crossing it.	The airport shows as flat ground at its elevation; no tall spikes.

2. ND and VD map views across flight reloads

> **Aircraft:** A32NX and A380X | **Where:** ND weather radar / terrain and A380X VD (native ndwxr gauges) | **Status:** VD OK after 4 flight reloads (user). In that test the sim restarted the module at each reload, so the leak this fixes did not occur: the fix itself is not proven in the sim. Unit test: ndwxr gauge_slots_test.

What it is for

The ND weather and terrain and the A380X VD are drawn by native gauges (ndwxr) on sim map views. Over flight reloads the gauges could leak their map views: the module keeps its state while the next flight's gauges start before the old ones are closed. Once the sim's view budget was used up, the VD went blank.

A slot table now hands each gauge over: a new gauge of the same display or role takes over or replaces the previous flight's gauge, deletes its views (never a view another gauge of the same display still uses, as view numbers are per display), and keeps the module within its view budget.

Limitations and why

Limitation	Why	What to do
The original blank VD after a reload was not reproduced.	In the reload test the sim restarted the module each time, so no old view was left; the blank VD seen before may come from another reload path.	Report the reload path (flight reload, return to the main menu, new flight) if the VD goes blank again.
A full MSFS restart is needed after an update.	The native module (ndwxr.wasm) changes.	Restart the sim after installing.

How to use

1. Nothing to set: reload the flight as usual; the ND weather / terrain and the VD come back.

How to test

#	Do this	Expected result
1	A380X at a gate, aircraft powered, both NDs ARC, TERR ON ND on one side, the VD shown. Note the VD and ND pictures.	Terrain on the ND and the VD profile.
2	Reload the flight (the same airport) four times, checking the NDs and the VD after each reload.	After every reload the ND terrain / weather and the VD profile come back; the VD never stays blank.
3	Repeat once on the A32NX with TERR ON ND.	The ND terrain comes back after each reload.

FCOM audit fixes

Small upstream bugs found by the FCOM audit of every A32NX and A380X system: eleven A380X fixes, six A32NX fixes, the fast ADIRS alignment of both aircraft and the A32NX ND chronometer colour, each checked against the FCOM, an A32NX climb thrust regression test, and a further set of FCOM gap fixes for both aircraft (ECAM lines, SD, brakes, GPWS, pressurization), the A380X FWS engine 4 take-off power fix, and the later A380X fixes: door alerts and outflow valves for every door, L/G, NAV disagree and F/CTL law STATUS alerts, the exhaust heat haze of each engine and GPWS mode 4B, and one contrail per running A380X engine; then the OANS ARPT NOT IN ACTIVE F/PLN message in PLAN mode only and A380X flap regression tests

Aircraft	A32NX and A380X
Where to find it	A380X: GPWS, OANS, SD (CRZ, PRESS, DOOR/OXY, F/CTL, G LOAD), FADEC, FWS (BRAKES HOT, T.O CONFIG, F/CTL ALTN LAW, HYD, CAB PRESS, DOOR, L/G, NAV), outflow valves, engine heat haze, contrails, ELEC panel, EWD EGT, brakes. A32NX: FWS (AIR PACK 2 OFF, ALL ENGINES FAILURE, F/CTL laws, ELAC FAULT, APU BLEED memo), BLEED, TCAS, SD (CRUISE, WHEEL, HYD), brakes, GPWS lights, ND chronometer. Both: ADIRS
Branch	`core/fcom-audit-fixes`, based on `master` (FlyByWire master 76db0aef1); independent of the other core branches. Each fix is small and can be taken upstream on its own. `core/failures-a32nx` merges this branch (its A32NX alerts build on the F/CTL law lines and APU memos): merge it there again when this branch changes. Of the A380X, only GPWS mode 4B is here; its RCL / RCL LAST, T.O CONFIG NORMAL, automatic SD pages and BTV INOP parts are on `core/failures-a380x`. The twelfth A380X fix of the audit (SD FUEL jettison lines white during a normal jettison) needs the jettison of `core/fuel` and is in that branch. Two A32NX fixes of the fifth set (T.O CONFIG NORMAL with the wing tank pump alerts, FUEL L(R) WING TK LO LVL) need the fuel pump alerts and STATUS fields of `core/failures-a32nx` and are in that branch.
Status	First four sets of fixes: all tested in the sim by the user; not sim-tested on the master base, unit tests and Rust tests pass there. Fifth set: A32NX APU BLEED memo and SD HYD PTU colour, and A380X nose wheel steering with A-SKID OFF, tested in the sim by the user: OK; A380X SD F/CTL droop circles tested in the sim by the user: OK (G LOAD line hidden at normal g, its display outside 0.7-1.4 g not sim-tested); the other fixes of that set not sim-tested: unit tests and Rust tests. FWS engine 4 take-off power: a normal four-lever take-off kept its phases (2, 3, 4, 6, 7, 8); the lever 3 / lever 4 split itself not sim-tested, unit test. Later fixes, tested in the sim by the user on a combined test build before the commit: door alerts, outflow valves with a door open, heat haze, L/G and NAV alerts: OK; the F/CTL law STATUS lines and GPWS mode 4B not sim-tested (unit tests). Per-engine contrails: engine 1 flameout at FL350 stopped only its contrail, relit at FL290 282 kt. None sim-tested on this core branch itself. OANS message fix: not sim-tested, unit test; flap regression tests: tests only.
Pictures	No pictures: every fix changes a stock FBW display, alert or system value.

1. A380X FCOM fixes
2. A32NX FCOM fixes
3. A380X FCOM fixes, fifth set
4. A32NX FCOM fixes, fifth set
5. ADIRS fast alignment in 30 s
6. A32NX ND chronometer in white
7. A32NX climb thrust regression test
8. A380X FWS flight phases: engine 4 take-off power from lever 4
9. A380X door alerts and outflow valves for every door
10. A380X L/G, NAV disagree and F/CTL law STATUS alerts
11. A380X exhaust heat haze of each engine
12. A380X GPWS mode 4B
13. A380X contrails from each running engine
14. A380X OANS: ARPT NOT IN ACTIVE F/PLN in PLAN mode only
15. A380X flap automatic retraction and extension: regression tests

1. A380X FCOM fixes

> **Aircraft:** A380X | **Where:** GPWS, OANS, SD, FADEC, FWS, ELEC panel, EWD | **Status:** Tested in the sim by the user: OK. Unit tests: LegacyGpws, OancSelectedRunway, FwsSystemDisplayLogicCrzPage, FadecWingAntiIce, BrakesHot, ElecPanelPbs, EgtLimits, CabinAltitudeIndication, FwsAbnormalSensedItems, OxygenIndication specs; Rust test in hydraulic/mod.rs (each fails without its fix).

What it is for

The FCOM audit of compared every A380X system with the A380 FCOM. These eleven findings were one or two wrong lines each (a variable nothing writes, a copy-paste index, an inverted condition). Each is now fixed as the FCOM describes.

What	Before	Now	FCOM
GPWS TOO LOW FLAPS	The FLAP MODE OFF selection was ignored (the GPWS read L:A32NX_GPWS_FLAP_OFF, which nothing writes)	TAWS FLAP MODE OFF on the MFD SURV CONTROLS page inhibits TOO LOW FLAPS	DSC-34-SURV
OANS selected runway	The FMS landing runway was compared with the FMS origin twice	The landing runway is green (selected) at the destination airport	DSC-34-10-70-20
SD automatic CRZ page	Engine 4 was not checked (engine 3 read twice)	Engine 4 alone at takeoff power also delays the CRZ page after takeoff	-
FADEC thrust limits	The wing anti-ice input read an A32NX variable that the A380X never writes	WING ANTI ICE ON (STRUCTURAL DEICE SWITCH) enters the thrust limit computation	DSC-70
BRAKES HOT and T.O CONFIG	Never triggered (L:A32NX_BRAKES_HOT is not written on the A380X)	From the 16 reported brake temperatures: one brake at or above 300 °C	PRO-ABN BRAKES HOT
ELEC panel COMMERCIAL 2 and ELMU pbs	All three pbs toggled COMMERCIAL (ELMU shed the commercial loads)	Each pb has its own state	DSC-24-20
EWD EGT amber limit	From the thrust lever angle	From the active thrust limit: none at FLEX/TOGA or in reverse, amber above the limit at MCT/CLB	DSC-70-90
SD PRESS CAB ALT needle	A320 delta P limits applied to feet	Red above 9 550 ft, like the value	DSC-21-30-20
F/CTL ALTN LAW procedure	MAX SPEED 310 KT shown only in ALT 1A (inverted)	Shown except in ALT 1A	PRO-ABN F/CTL ALTN LAW
HYD G RSVR LEVEL LO / G SYS OVHT	The G ELEC PMP A and B OFF line ticked from the yellow pumps; engine 3 and 4 pump disconnection on swapped DC buses	Green pumps; ENG 3 pump disconnection on DC 2, ENG 4 on DC 1	PRO-ABN HYD, DSC-29-40
SD DOOR/OXY crew supply	CREW SUPPLY OFF could not show; the CABIN line followed the crew oxygen pb	CKPT legend amber and REGUL PR LO on the ground with CREW SUPPLY OFF; the CABIN line amber on a low pressure only	DSC-35-20-20, DSC-35-30-20

Limitations and why

Limitation	Why	What to do

Only the wrong lines found by the audit were changed.	Each fix is kept to its finding so it can go upstream on its own; the rest of each system is as before.	-
ELMU and COMMERCIAL 2 have a state but no consumer yet.	Nothing in the electrical model sheds loads by those pbs (a380-simvars: A380X_OVHD_ELEC_COMMERCIAL_2_PB_IS_OFF, A380X_OVHD_ELEC_ELMU_PB_IS_OFF).	-
The EDP disconnection bus change has no cockpit test.	It only matters with DC 1 or DC 2 lost while disconnecting an engine pump; a Rust test covers it.	-

How to use

1. Nothing to do: the systems behave as in the FCOM.
2. TAWS FLAP MODE: MFD > SURV > CONTROLS, FLAP MODE OFF to inhibit TOO LOW FLAPS (e.g. a landing in a non-standard flap configuration).

How to test

#	Do this	Expected result
1	A380X on approach (FMS approach phase), gear down, flaps not in landing configuration, below 180 kt. MFD SURV CONTROLS: FLAP MODE on. Descend below 245 ft RA.	TOO LOW FLAPS.
2	Repeat with FLAP MODE OFF on the MFD SURV CONTROLS page.	No TOO LOW FLAPS.
3	Flight plan with a destination and a landing runway. At the destination, open the OANS map of the destination airport.	The FMS landing runway is shown green (selected runway).
4	After takeoff (climb), flaps and spoilers retracted, SD page not selected: keep thrust lever 4 alone at FLX/MCT or TOGA, levers 1-3 in CL.	The SD does not switch to the CRZ page at once: it waits for its 60 s timer, as with any engine at takeoff power. With all levers in CL it switches at once.
5	In flight, note the thrust limit on the E/WD; set WING ANTI ICE ON.	The thrust limit changes (the FADEC now sees the wing anti-ice).
6	After a high-energy stop (rejected takeoff or heavy braking), watch the SD WHEEL brake temperatures.	As soon as one brake reaches 300 °C: amber BRAKES HOT; the T.O CONFIG test then reports the hot brakes.
7	Overhead ELEC panel: press ELMU, then COMMERCIAL 2.	The COMMERCIAL pb stays as it was; each pb changes only itself.
8	Takeoff at FLX or TOGA, then thrust levers to CL.	EGT gauge: no amber EGT limit at FLX/TOGA; at CL the amber limit of the CLB thrust applies (EGT amber above it).
9	SD PRESS page with the cabin altitude above 9 550 ft (e.g. manual cabin altitude).	CAB ALT value and needle red.
10	Reach F/CTL ALTN LAW outside ALT 1A (e.g. with flyPad flight control failures).	The procedure shows MAX SPEED ..... 310 KT; in ALT 1A it does not.
11	flyPad > Failures > ATA 29: activate **Green reservoir leak**; when HYD G RSVR LEVEL LO shows, set the green electric pumps A and B OFF.	The procedure line of the green electric pumps A and B OFF ticks (the yellow pumps no longer tick it).
12	On the ground, overhead OXYGEN CREW SUPPLY pb OFF; SD DOOR/OXY page.	CKPT legend amber with REGUL PR LO; the CABIN line unchanged.

2. A32NX FCOM fixes

> **Aircraft:** A32NX | **Where:** E/WD, SD BLEED, CRUISE and WHEEL pages, TCAS | **Status:** Tested in the sim by the user: OK. Unit tests: DeltaPIndication, AutoBrakeIndication, AllEnginesFailure, TcasGpwsInhibit specs; Rust test in pneumatic.rs (each fails without its fix). The OPT RELIGHT SPD clean-up: AllEnginesFailure spec; the E/WD line shown is unchanged.

What it is for

The same audit on the A32NX found six one-line bugs. Each now follows the A320 FCOM. The ALL ENGINES FAILURE procedure was also cleaned up, with the LEAP-1A relight envelope of core/failures-a32nx.

What	Before	Now	FCOM
AIR PACK 2 OFF	Checked the ENG 1 BLEED pb	Checks the ENG 2 BLEED pb (each pack against its own bleed)	PRO-ABN AIR PACK 1(2) OFF
BLEED LO TEMP flag	Set only with the bleed valve closed (inverted)	Set when the bleed supplies air (valve open) below 150 °C in flight with WING ANTI ICE on; the bleed temperature turns amber on the SD BLEED page	DSC-36
TCAS TA ONLY on a GPWS alert	Read L:A32NX_GPWS_Warning_Active (an A380X variable)	A GPWS warning or alert (L:A32NX_GPWS_WARNING_LIGHT_ON / ALERT_LIGHT_ON) inhibits the RAs (TA ONLY)	DSC-34-SURV-60-10-10
SD CRUISE delta P pulse	Pulsed during the whole cruise	Pulses only in flight phase 7 above 1.5 PSI, resets at 1 PSI	DSC-21-20-40
SD WHEEL AUTO BRK colour	Amber with an engine off (single-engine taxi, engine failure)	Green when armed; amber only on an AUTO BRK failure (not modelled)	DSC-32-30-20
ENG ALL ENGINES FAILURE: APU START line	Only below 2 500 ft radio altitude	Below FL 250 (pressure altitude), text unchanged	PRO-ABN ENG ALL ENGINES FAILURE
ENG ALL ENGINES FAILURE: OPT RELIGHT SPD	Three unused lines (280/.77, 300/.77, 260/.77) in the message table next to the one shown; the line list written inline in the FWC	One line, OPT RELIGHT SPD.270/.77 (the line already shown, the LEAP-1A value); the line list and its conditions in FWC/Logic/AllEnginesFailure.ts with specs	Air Arabia A320 QRH ABN-19.01A ALL ENG FAIL (LEAP-1A; CFM56: 300/.77)

Limitations and why

Limitation	Why	What to do
The APU START line does not say "(BELOW FL 250)".	The 24-character ECAM line has no room; the condition gates the line instead, as for the other conditional lines.	-
AUTO BRK never turns amber.	The AUTO BRK failure that the FCOM links to the amber colour is not modelled.	-

How to use

1. Nothing to do: the alerts and pages behave as in the FCOM.

How to test

#	Do this	Expected result
1	A32NX in cruise. ENG 1 BLEED OFF, ENG 2 BLEED ON, X BLEED closed. Set PACK 2 pb OFF.	After 5 s: AIR PACK 2 OFF.
2	In flight with WING ANTI ICE ON, at low engine power (cold bleed air); SD BLEED page.	When a bleed supplies air below 150 °C, its bleed temperature is amber; with that ENG BLEED pb OFF (valve closed) it is not.
3	In flight with a TCAS RA possible (traffic), trigger a GPWS alert (e.g. SINK RATE).	The TCAS goes TA ONLY (no RA) while the GPWS alert is on.
4	SD CRUISE page in cruise with delta P above 1.5 PSI.	Delta P does not pulse. In the approach (flight phase 7) above 1.5 PSI it pulses, until it is below 1 PSI.
5	On the ground, one engine running, AUTO BRK MAX armed; SD WHEEL page.	AUTO BRK green (it was amber).
6	In flight above 2 500 ft RA and below FL 250, APU off: all engines flame out (e.g. fuel cut-off).	ENG ALL ENGINES FAILURE shows the APU ..... START line, and one OPT RELIGHT SPD line: 270/.77.

3. A380X FCOM fixes, fifth set

> **Aircraft:** A380X | **Where:** SD F/CTL and G LOAD line, FWS (HYD, CAB PRESS), GPWS, pressurization, brakes | **Status:** Nose wheel steering with A-SKID OFF tested in the sim by the user: OK. SD F/CTL droop circles tested in the sim by the user: OK (no circles at flaps 0 and 1, circles at flaps 2, no G LOAD line at 0.9-1.1 g; the G LOAD line outside 0.7-1.4 g not sim-tested). The other lines: not sim-tested; unit tests AileronDroop, GLoadIndication, HydOverheatFlags, FwsAbnormalSensedItems and LegacyGpws specs, Rust tests in air_conditioning/mod.rs, hydraulic/mod.rs and autobrakes.rs.

What it is for

Further FCOM gap fixes on the A380X: each line below differed from the A380 FCOM and now follows it.

What	Before	Now	FCOM
SD F/CTL aileron droop	The droop circle on the aileron scales in every configuration	Only in CONF 1+F, 2, 3 and FULL	DSC-27-10-20
SD G LOAD line	Shown at once whenever the value was outside 0.7-1.4 g, also on the ground; never XX	Only in flight, after more than 2 s outside 0.7-1.4 g; kept 5 s after the G load is back to normal; XX when the value is not valid	DSC-31-40-10
HYD G(Y) SYS OVHT	The two ENG PMP A+B DISC lines always shown	Shown only when turning the pumps OFF did not depressurise them	PRO-ABN HYD G (Y) SYS OVHT, DSC-29-20
HYD G(Y) SYS PRESS LO STATUS	BTV and CAT 3 DUAL missing; AVOID MAX TILLER ANGLE twice (Y); FUEL CONSUMPT INCRSD lines never shown	INOP SYS BTV and CAT 3 DUAL, INFO CAT 3 SINGLE ONLY; the Y STATUS shows TAXI WITH CARE and NO BRAKED PIVOT TURN; FUEL CONSUMPT INCRSD / FMS PRED UNRELIABLE when the PRIM also failed; Y SYS PRESS LO shows L/G GRVTY EXTN ONLY with the limitation MAX SPEED 220 KT (gravity extension)	PRO-ABN HYD G (Y) SYS PRESS LO

CAB PRESS EXCESS DIFF PRESS	Could trigger in flight phase 8 only	Can trigger on the ground before take-off and below 800 ft before landing (flight phases 1, 8 and 9)	PRO-ABN CAB PRESS EXCESS DIFF PRESS
GPWS mode 4 speeds	TOO LOW TERRAIN above 190 kt (4A) / 159 kt (4B), the A320 values, with a ceiling rising with the speed	Above 200 kt gear up (4A) and 180 kt flaps not in landing configuration (4B); a flat 500 ft (4A) / 245 ft (4B) ceiling while the terrain function is available, rising to 1 000 ft only with TERR SYS OFF or a TERR failure	DSC-34-20-20-10
Cabin descent rate	Limited to 350 ft/min	Limited to 300 ft/min	DSC-21-30-30
A-SKID switch OFF	Also cut the nose wheel steering (A320 logic)	The nose wheel steering stays: "A-SKID sw OFF does not cut off NWS"	DSC-32-30-10
AUTO BRK after take-off	Any mode disarmed 10 s after lift-off	Only RTO disarms 10 s after lift-off; a landing mode armed on the ground stays armed	DSC-32-10-30

Limitations and why

Limitation	Why	What to do
"Turning the pumps OFF was not successful" is read from the pump section pressure switch.	It is the only pump output the FWS has (design choice).	-
The GPWS "terrain function available" is the TERR SYS pb and the TERR failure of the selected system.	The FCOM names the aircraft position and the terrain database; the sim has no failure of either (design choice). The 1 000 ft ceiling is reached at 250 kt, as on the A320 (the FCOM figure is not precise enough to read the speed).	-
CAB PRESS EXCESS DIFF PRESS has no practical sim trigger.	It needs a differential pressure above the limit; the phase change is covered by the FwsAbnormalSensedItems spec.	-

How to use

1. Nothing to set: the pages, alerts and systems behave as in the FCOM.
2. A landing AUTO BRK mode can now be armed on the ground before take-off: it stays armed for the landing.

How to test

#	Do this	Expected result
1	A380X on the ground, aircraft powered, flaps lever at 0. ECAM control panel: **F/CTL** key. FLAPS lever to 1, wait; then to 2, wait.	Flaps 0 and 1: no white circles on the six ailerons. Flaps 2 (and 1+F on the way): a white circle on each aileron scale. The G LOAD line stays hidden (SD bottom, above the clock).
2	In flight, pull or push for more than 2 s outside 0.7-1.4 g (e.g. a firm pull-up above 1.4 g).	The G LOAD line shows the value; it stays 5 s after the load is back within 0.7-1.4 g. A shorter peak shows nothing.

3	Engines running, parking brake ON, ground speed 0. A-SKID switch (centre instrument panel, next to the L/G lever) OFF. Release the parking brake, taxi slowly and turn the tiller fully left.	The nose wheel turns and the aircraft turns left (before: no steering). Set A-SKID ON and the parking brake ON afterwards.
4	On the ground, AUTO BRK selector to a landing mode (e.g. LO). Take off.	The mode stays armed after lift-off (more than 10 s); with RTO armed, RTO disarms 10 s after lift-off.
5	In flight, flyPad > Failures > ATA 29: **Green reservoir leak**; wait for HYD G SYS PRESS LO. Open the STATUS.	INFO CAT 3 SINGLE ONLY; INOP SYS (APPR & LDG) includes BTV and CAT 3 DUAL.
6	Remove it; activate **Yellow reservoir leak**; wait for HYD Y SYS PRESS LO.	Procedure line L/G GRVTY EXTN ONLY; limitation MAX SPEED 220 KT (gravity extension) on the PFD and the ECAM; STATUS INFO TAXI WITH CARE, NO BRAKED PIVOT TURN, AVOID MAX TILLER ANGLE once, CAT 3 SINGLE ONLY.
7	A hydraulic system overheat (HYD G or Y SYS OVHT; set by the hydraulic model, this branch adds no failure for it): set the ENG PMP A and B pb-sws of the affected engines OFF.	The ENG PMP A+B DISC lines show only if a pump section keeps its pressure; with the pumps depressurised they do not show. Covered offline by HydOverheatFlags.spec.ts.
8	Approach over flat ground, gear up, flaps not in landing configuration, 220 kt, descend through 500 ft RA (MFD SURV CONTROLS: TERR SYS on).	TOO LOW TERRAIN from 500 ft RA, not higher. At 190 kt gear up: TOO LOW GEAR instead. (Gear up with the flaps in landing configuration: see "A380X GPWS mode 4B" below.)
9	Gear down, flaps not in landing configuration, 190 kt, descend through 245 ft RA.	TOO LOW TERRAIN from 245 ft RA (above 180 kt). Below 180 kt: TOO LOW FLAPS.
10	Repeat the gear-up pass at 250 kt with TERR SYS OFF.	TOO LOW TERRAIN from about 1 000 ft RA.
11	Descent at a high vertical speed; SD CAB PRESS page.	The cabin vertical speed goes down at no more than about 300 ft/min (was 350).

4. A32NX FCOM fixes, fifth set

> **Aircraft:** A32NX | **Where:** E/WD, SD HYD, brakes, GPWS/PULL UP pbs | **Status:** APU BLEED memo and SD HYD PTU colour tested in the sim by the user: OK. The other lines: not sim-tested; unit tests FlightControlLawAlerts, ApuMemos and PtuIndication specs, Rust tests in a320_systems (hydraulic brakes, lib.rs) and in the shared egpws.

What it is for

Further FCOM gap fixes on the A32NX. The T.O CONFIG NORMAL fix (not given with a FUEL L(R) TK PUMP 1+2 LO PR alert) and the FUEL L(R) WING TK LO LVL fix (FUEL MODE SEL MAN only with fuel in the centre tank, and its STATUS) of the same set are on `core/failures-a32nx`: they need its fuel pump alerts.

What	Before	Now	FCOM
F/CTL ALTN LAW	MAX SPEED and MANEUVER WITH CARE / MAX FL lines not as in the procedure; SPD BRK DO NOT USE not tied to an elevator fault	MAX SPEED 320 KT (320/.77 only with a dual hydraulic low pressure); SPD BRK DO NOT USE only with an L or R elevator fault; no MANEUVER WITH CARE, no MAX FL line	PRO-ABN F/CTL ALTN LAW
F/CTL DIRECT LAW	SPD BRK DO NOT USE next to USE SPD BRK WITH CARE	Only USE SPD BRK WITH CARE	PRO-ABN F/CTL DIRECT LAW

F/CTL ELAC 1(2) FAULT	FUEL CONSUMPT INCRSD and FMS PRED UNRELIABLE never shown	Shown after the procedure lines (hidden with them in the sidestick transducer case)	PRO-ABN F/CTL ELAC 1(2) FAULT
APU BLEED memo	Followed the APU bleed valve	Follows the APU BLEED pb-sw: APU available and pb-sw ON, at once	DSC-36-20
SD HYD PTU symbol	Amber also during the automatic PTU inhibitions (e.g. the second engine start with the parking brake set)	Amber only with the PTU pb-sw OFF	DSC-29-20
Alternate braking without anti-skid	Limited to 1 160 PSI	Limited to 1 000 PSI	DSC-32-30-10
Brake fans	Ran on the ground only	Run when the left main gear is down and locked, also in flight before landing	DSC-32-30-20
GPWS lights	The red PULL UP light for every GPWS alert	Red PULL UP only for the mode 1 PULL UP and mode 2 warnings; the other alerts (SINK RATE, DON'T SINK, TOO LOW ...) light the amber GPWS light of the GPWS/PULL UP pb	DSC-34-SURV-40

Limitations and why

Limitation	Why	What to do
An unused EGPWS test helper was removed with these fixes.	The Rust lint (clippy, all targets) failed on it; no behaviour change.	-

How to use

1. Nothing to set: the alerts, the SD and the systems behave as in the FCOM.

How to test

#	Do this	Expected result
1	A32NX on the ground, parking brake ON, APU running and AVAIL, APU BLEED pb OFF, engines off, PTU pb AUTO. Press APU BLEED ON, then OFF.	ON: the E/WD memo APU BLEED (green) replaces APU AVAIL at once. OFF: APU AVAIL comes back and APU BLEED goes at once.
2	APU BLEED ON. ECAM control panel: HYD page. Start engine 2 with the parking brake still ON (ENG MODE IGN START, ENG MASTER 2 ON), then press the PTU pb OFF and back to AUTO.	The PTU symbol stays green during the start (automatic inhibition); amber with the pb OFF, green again at AUTO.
3	In flight, flyPad > Failures > ATA 27: **ELAC 1**.	F/CTL ELAC 1 FAULT with its procedure lines and, at the end, FUEL CONSUMPT INCRSD and FMS PRED UNRELIABLE.
4	Remove it. ATA 29: **Green reservoir leak** and **Yellow reservoir leak**; wait for the G and Y pressures to be low.	F/CTL ALTN LAW (PROT LOST) with MAX SPEED 320/.77; no MANEUVER WITH CARE, no MAX FL 350, no SPD BRK DO NOT USE (no elevator fault).
5	When F/CTL DIRECT LAW shows.	USE SPD BRK WITH CARE, without a SPD BRK DO NOT USE line.
6	On the ground, A/SKID & N/W STRG switch OFF, press both brake pedals fully; triple brake indicator.	Brake pressure at most 1 000 PSI (was 1 160).

7	In flight on approach, BRAKE FAN pb ON, landing gear down and locked.	The brake fans run before touchdown (before: only on the ground).
8	In flight, trigger a GPWS alert other than PULL UP (e.g. SINK RATE, or TOO LOW GEAR).	The amber GPWS light of the GPWS/PULL UP pb on each pilot's panel; no red PULL UP. A mode 1 PULL UP or a mode 2 warning: red PULL UP.

5. ADIRS fast alignment in 30 s

> **Aircraft:** A32NX and A380X | **Where:** Overhead ADIRS panel | **Status:** Tested in the sim by the user: OK. Rust tests in adirs.rs (the existing fast alignment test tightened to 30 s).

What it is for

A fast (re)alignment, the IR mode selector set to OFF and back to NAV within 5 s, took 180 s. Both FCOMs give about 30 s (A320 FCOM DSC-22_20-20-40, IRS alignment page: "a fast alignment takes 30 s"; A380 FCOM DSC-34-10-30-30). The shared systems code now takes 30 s.

Limitations and why

Limitation	Why	What to do
Only the fast alignment changed.	A normal alignment from an unaligned IR is unchanged.	-

How to use

1. With the IRs aligned on the ground, set an IR mode selector to OFF and back to NAV within 5 s.

How to test

#	Do this	Expected result
1	Aircraft on the ground, IRs aligned (NAV). Set IR 1 to OFF and back to NAV within 5 s; start a stopwatch.	IR 1 is aligned again after about 30 s (not 3 min).
2	Repeat on the A380X.	Same: about 30 s.

6. A32NX ND chronometer in white

> **Aircraft:** A32NX | **Where:** ND, CHRONO pb on the EFIS control panel | **Status:** Tested in the sim by the user: OK. Spec ChronoColours.spec.ts (compiled styles of both aircraft).

What it is for

The ND chronometer was drawn in green digits on a light grey box, hard to read. The A320 FCOM gives "Chronometer Indication (white)" (DSC-31-45, INDICATIONS ON ND, item 9). The A32NX now shows white digits on a box in the display background colour.

Limitations and why

Limitation	Why	What to do
The dark box is a design choice.	The FCOM shows no box; a dark box keeps the digits readable over the map and terrain.	-
The A380X is unchanged.	Its FCOM gives no colour for the chronometer.	-

How to use

1. Press CHRONO on the EFIS control panel: the elapsed time appears at the bottom left of the ND. Press again to freeze it, a third time to reset and hide it.

How to test

#	Do this	Expected result
1	A32NX, ND powered: press CHRONO.	White digits on a black box at the bottom left of the ND, counting.
2	A380X: press CHRONO.	Unchanged: green digits on the grey box.

7. A32NX climb thrust regression test

> **Aircraft:** A32NX | **Where:** Unit tests (FMS and FADEC climb thrust tables) | **Status:** Test only, no code change: A320ClimbThrustLimit.spec.ts passes.

What it is for

A check of the A32NX climb against the A320neo FCOM climb tables (IndiGo FCOM PER-CLB-CLT, PDF pages 7248-7250 at ISA+10 and 7258-7260 at ISA+20, PW1127G-JM engines; no LEAP-1A table is available) found that the A32NX already climbs within 1-3 % of them, so no CLB thrust value was changed. The test keeps it so: the FADEC CLB N1 table (ThrustLimits_A32NX.hpp) and its FMS copy (cn1ClimbLimit) must stay identical row by row, and a climb computed with the FMS engine and flight models from FL 100 to FL 300 at 300 kt / M.78 must stay within 5 % of the FCOM times (8 % for each half) at 52 to 76 t.

Limitations and why

Limitation	Why	What to do
Nothing changes in the aircraft.	The finding was that the climb already matches the FCOM.	-
The reference is the PW1127G-JM table.	FBW models the LEAP-1A26; no LEAP climb table is available.	-

How to use

1. Nothing to do in the sim. When changing the CLB N1 table, change both copies and run the test.

How to test

#	Do this	Expected result
1	Run `npx vitest run fbw-a32nx/src/systems/fmgc/src/flightplanning/A320ClimbThrustLimit.spec.ts`.	All tests pass.
2	Optional in the sim: climbs at 52, 60 and 64 t, ISA+10, no wind, managed CLB speed, cruise FL 350; time from FL 100 to FL 300.	Within 5 % of 7.67 min (52 t), 9.08 min (60 t), 9.97 min (64 t); at ISA 2-5 % quicker.

8. A380X FWS flight phases: engine 4 take-off power from lever 4

> **Aircraft:** A380X | **Where:** FWS flight phases (take-off phase 3) | **Status:** No regression in the sim: a normal four-lever take-off went through phases 2, 3, 4, 6, 7, 8 in order. The lever 3 / lever 4 split that shows the fix: not sim-tested. Unit test FwsTakeoffPower.spec.ts (fails against the previous FwsFlightPhases.ts).

What it is for

The FWS flight phases need two engines at take-off power for phase 3. The take-off power of engine 4 was read from thrust lever 3 (`eng4TLAFullPwr = eng3TLA > 43.3`), so lever 3 alone at TOGA counted as two engines, and lever 4 was never read. Each engine now reads its own lever; the rule (TOGA, above FLX/MCT, or FLX/MCT with a FLEX TEMP) sits in `FwsTakeoffPower.ts`.

Limitations and why

Limitation	Why	What to do
Nothing changes with all four levers moved together.	Only a lever 3 / lever 4 split shows the fix.	-

How to use

1. Nothing to set: set the thrust levers for take-off as usual.

How to test

#	Do this	Expected result
1	A380X on the ground, PARKING BRK ON, engines running at idle, all levers at IDLE, no FLEX TEMP. Thrust lever 3 only to TOGA, hold 5 s.	The FWS stays in phase 2: no red CONFIG PARK BRK ON (before the fix, phase 3 and that warning).
2	Lever 3 back to IDLE; levers 3 and 4 to TOGA.	Phase 3: red CONFIG PARK BRK ON.
3	All levers to IDLE.	The warning goes.

9. A380X door alerts and outflow valves for every door

> **Aircraft:** A380X | **Where:** E/WD, T.O CONFIG pb, SD DOOR and CAB PRESS pages, outflow valves | **Status:** Tested in the sim by the user on a combined test build before the commit: DOOR UPPER 1L NOT CLOSED, DOOR FWD and AFT CARGO NOT CLOSED, T.O CONFIG with an upper deck door open and the outflow valves held open: OK. Unit tests: DoorAlerts.spec.ts; Rust tests in air_conditioning (cpiom_b.rs, mod.rs).

What it is for

The FWS and the pressurization read only some of the 18 A380X doors (10 main deck, 6 upper deck and 2 cargo doors). An open upper deck door or cargo door gave no alert, did not stop T.O CONFIG NORMAL and did not hold the outflow valves open. Every door now counts, as in the A380 FCOM (PRO-ABN-ECAM-10-52 DOOR alerts, PRO-ABN-ECAM-10-21 CAB PRESS INHIBITED BY DOORS, DSC-31-40-10 Takeoff Configuration Test, DSC-21-30-30).

What	Before	Now
DOOR ... NOT CLOSED cautions	Main deck doors only	Also **DOOR UPPER 1L** to **3R NOT CLOSED** and **DOOR FWD (AFT) CARGO NOT CLOSED**
IF ABNORM CAB V/S line of a DOOR alert	MAIN 5L/5R and UPPER 2L/2R showed the lines below it without the condition and never set the MAX FL 100/MEA limitation	A condition the crew confirms on every door; the MAX FL : 100/MEA-MORA limitation follows it
CAB PRESS INHIBITED BY DOORS	Not with MAIN 1R, the upper deck doors or the cargo doors	Any door not closed and locked at take-off power
T.O CONFIG test	Checked MAIN 1L, MAIN 2R and the cargo doors	Every main deck, upper deck and cargo door and the two cockpit sliding windows must be closed for T.O CONFIG NORMAL

Outflow valves with a door open at take-off power (AUTO)	The aft valves closed	The four outflow valves stay open (FCOM DSC-21-30-30: "maintaining the four outflow valves open")
Door inputs of the pressure controllers, cabin air and packs	Only MAIN 1L and 2R	All 16 passenger doors and both cargo doors

Limitations and why

Limitation	Why	What to do
No DOOR BULK CARGO NOT CLOSED.	The model has no bulk cargo door (no interactive point, no door in the systems).	-
A passenger door counts as not closed as soon as it moves.	The sim doors have no separate latch or lock.	-
The door alerts need an engine running.	They are inhibited in flight phase 1 (FCOM inhibitions).	-
A take-off with a door open keeps the outflow valves open until the next touchdown.	Design choice: the inhibition keeps its value from the lift-off. The manual pressurization mode is not affected (FCOM DSC-21-30-40).	-

How to use

1. Close every door and both cockpit sliding windows before the T.O CONFIG test.
2. With a DOOR ... NOT CLOSED caution, close that door (flyPad > Ground > Services, or the sim door controls) and follow the ECAM lines.

How to test

#	Do this	Expected result
1	A380X on the ground, PARKING BRK ON, engine 1 running at idle (engines 2-4 off), all doors closed. Open the upper deck door 1L (flyPad > Ground > Services: the second cabin door button at the bottom right; "Upper deck 1 L" on `core/efb-redesign`).	Amber **DOOR UPPER 1L NOT CLOSED**, MASTER CAUTION. Close the door: the alert goes.
2	All engines off, APU running. Open the forward cargo door (flyPad Services cargo door button), wait 30 s, then start engine 1.	With engine 1 running: amber **DOOR FWD CARGO NOT CLOSED** and **DOOR AFT CARGO NOT CLOSED**; SD DOOR page: FWD CARGO and AFT CARGO amber. (The cargo doors move only with all engines stopped: stop engine 1 to close them.)
3	All four engines running, flaps 1, FLEX entered (MFD PERF T.O), all doors closed. Open the upper deck door 1L; press the T.O CONFIG key on the ECAM control panel.	DOOR UPPER 1L NOT CLOSED; no T.O CONFIG NORMAL.
4	Door still open, PARKING BRK ON: ECAM control panel CAB PRESS key; all four thrust levers to FLX/MCT for 5 s, then IDLE.	SD CAB PRESS page: the four outflow valves stay fully open (before: the aft valves closed). Close the door afterwards.

10. A380X L/G, NAV disagree and F/CTL law STATUS alerts

> **Aircraft:** A380X | **Where:** E/WD, SD WHEEL and STATUS pages, PFD limitations | **Status:** Tested in the sim by the user on a combined test build before the commit: L/G CTL faults on the ground and BARO REF DISAGREE: OK; L/G GEAR NOT LOCKED UP after take-off: OK. L/G GEAR NOT LOCKED DOWN, L/G DOORS NOT CLOSED, L/G SYSTEM DISAGREE, ALT / ATT / HDG DISAGREE and the F/CTL law STATUS lines: not sim-tested. Unit tests: FwsGearAlerts, FwsNavAlerts and FwsFctlAlerts specs.

What it is for

FCOM gap audit: alerts of the A380 FCOM that the FWS did not have. The landing gear alerts come from the two landing gear control and indicating systems (LGCIS 1 and 2), the NAV alerts compare what the CAPT and F/O PFDs display.

Alert	When	What the crew sees
L/G CTL 1 (2) FAULT	One LGCIS failed	Crew awareness, SD WHEEL page, no chime
L/G CTL 1+2 FAULT	Both LGCIS failed, or invalid data from both	Single chime, MASTER CAUT; GPWS OFF line; NO L/G RETRACTION with the gear down, MAX SPEED 220 KT and gravity extension only with the gear up; STATUS INOP SYS (APPR & LDG) L/G CTL 1+2, GPWS 1+2
L/G GEAR NOT LOCKED DOWN	Lever DOWN, a gear not downlocked after about 60 s	Red warning with the recycle and gravity extension lines (gravity lines after 120 s); MAX SPEED 220 KT for the gravity extension
L/G GEAR NOT LOCKED UP	Lever UP, a gear not locked up after about 40 s	Single chime; recycle line (MAX SPEED 250 KT and AVOID EXCESS G LOAD with all gears down and a door open); STATUS INOP SYS L/G RETRACTION (L/G DOORS if a door is open), in flight INFO FUEL CONSUMPT INCRSD
L/G DOORS NOT CLOSED	A gear door not closed 30 s after the gear stopped moving (design choice: the FCOM gives no time)	MASTER CAUT, SD WHEEL page; STATUS INOP SYS L/G DOORS
L/G SYSTEM DISAGREE	The two LGCIS disagree for 5 s (design choice)	Crew awareness, SD WHEEL page
NAV CAPT AND F/O ALT DISAGREE	Displayed altitudes differ by more than 500 ft (STD) or 250 ft (QNH), same reference both sides	Single chime, MASTER CAUT; ALT ..... X CHECK, AIR DATA SWTG ..... AS RQRD
NAV CAPT AND F/O ATT DISAGREE	Pitch or roll differ by more than 5°	ATT ..... X CHECK, ATT HDG SWTG ..... AS RQRD
NAV CAPT AND F/O HDG DISAGREE	Headings differ by more than 5° (TRUE) or 7° (MAG)	HDG ..... X CHECK, ATT HDG SWTG ..... AS RQRD
NAV CAPT AND F/O BARO REF DISAGREE	One side in STD, the other on QNH	BARO REF ..... XCHECK
F/CTL ALTN LAW (PROT LOST), F/CTL DIRECT LAW	Existing alerts	New STATUS lines: INOP SYS (APPR & LDG) CAT 2 and GLS AUTOLAND; DIRECT LAW also LOAD ALLEVIATION, AP 1+2 and FD 1+2 (all phases) and INFO F/CTL BKUP CTL ACTIVE when all the PRIMs and SECs are failed

Limitations and why

Limitation	Why	What to do

The flight phase inhibitions are taken from the matching A320 alerts.	The FCOM gives them as figures only (design choice; the NAV alerts use the phases of the NAV ADR/IR FAULT alerts).	-
The NAV alerts wait 10 s.	The FCOM gives no confirm time; 10 s lets the two pilots set STD or a new QNH one after the other (design choice).	-
The model has three gears (nose, left, right).	The wing gears follow the body gears.	-
L/G GEAR NOT LOCKED UP needs a jammed gear.	The gear actuator failures (e.g. **Nose gear jammed actuator**) are on `core/failures-a380x`, not on this branch.	Merge it to test.

How to use

1. Nothing to set: follow the ECAM procedure when an alert shows, then read the STATUS page.

How to test

#	Do this	Expected result
1	A380X at the gate, engines running. flyPad > Failures > ATA 32: activate **LGCIU 1 Internal error**.	Amber L/G CTL 1 FAULT, no chime; SD WHEEL page.
2	Also activate **LGCIU 2 Power supply**.	L/G CTL 1+2 FAULT replaces it, single chime, lines GPWS OFF and NO L/G RETRACTION.
3	ECAM control panel: STS key.	INOP SYS APPR & LDG: L/G CTL 1+2, GPWS 1+2. Deactivate both failures.
4	On the ground, engines running. CAPT EFIS control panel: pull the BARO knob (STD); leave the F/O on QNH; wait 12 s.	NAV CAPT AND F/O BARO REF DISAGREE with BARO REF ..... XCHECK.
5	CAPT BARO knob back to QNH; wait 12 s.	The alert goes.
6	With `core/failures-a380x` merged: climbing above 1 500 ft, gear down, below 250 kt. flyPad > Failures > ATA 32: **Nose gear jammed actuator**; gear lever UP; wait 45 s.	About 40 s after the lever: amber L/G GEAR NOT LOCKED UP, single chime; STATUS L/G RETRACTION. Lever DOWN then UP: the recycle line ticks. Remove the failure, lever DOWN then UP.
7	In flight, reach F/CTL DIRECT LAW (flight control failures, flyPad > Failures > ATA 27); STATUS page.	INOP SYS: LOAD ALLEVIATION, AP 1+2, FD 1+2; APPR & LDG: CAT 2, GLS AUTOLAND.

11. A380X exhaust heat haze of each engine

> **Aircraft:** A380X | **Where:** Outside view behind the engines | **Status:** Tested in the sim by the user on a combined test build before the commit: OK. Unit test: EngineEffectIndexes.spec.ts.

What it is for

The exhaust heat haze effects of aircraft.cfg read the wrong engines: the haze of engine 2 followed engine 1, and engines 3 and 4 followed engine 2 (an upstream bug). Each exhaust now reads its own engine.

How to use

1. Nothing to set: look at the engines from the outside view.

How to test

#	Do this	Expected result
1	A380X on the ground, engine 1 running, engines 2-4 off. Outside view of both wings from behind, 10 s.	Heat haze behind engine 1 (left outer) only; none at the right wing.
2	Start engine 2.	Heat haze behind engines 1 and 2 (left wing) only.

12. A380X GPWS mode 4B

> **Aircraft:** A380X | **Where:** GPWS aural alerts (mode 4) | **Status:** Not sim-tested. Unit tests: LegacyGpws.spec.ts.

What it is for

The A380 FCOM splits the GPWS mode 4 by the gear and flaps (DSC-34-20-20-10): 4A with the gear up and the flaps not in landing configuration, 4B with the gear up and the flaps in landing configuration, or the gear down and the flaps not in landing configuration. Gear up with the landing flaps gave TOO LOW TERRAIN at high speed, and a mode 4 alert stayed on after the aircraft climbed back above its ceiling.

Gear / flaps	Now
Gear up, flaps not in landing configuration (4A)	Unchanged: TOO LOW GEAR below 500 ft under 200 kt, TOO LOW TERRAIN above 200 kt
Gear up, flaps in landing configuration (4B)	**TOO LOW GEAR** below 245 ft at any speed, never TOO LOW TERRAIN (the FCOM figure gives a flat 245 ft ceiling)
Gear down, flaps not in landing configuration (4B)	Unchanged: TOO LOW FLAPS below 245 ft under 180 kt, TOO LOW TERRAIN above 180 kt
Climbing back above the ceiling	The mode 4 alert stops

Limitations and why

Limitation	Why	What to do
Landing configuration = FLAPS FULL, or CONF 3 when CONF 3 is the landing configuration selected in the FMS.	As the existing FBW code (the flaps lever position, not the flap angle).	-
FLAP MODE OFF does not inhibit the gear-up 4B alert.	Design choice: the FLAP MODE pb is for a landing with an abnormal flaps configuration (FCOM); only the flaps alert changes with it.	-

How to use

1. Nothing to set: the GPWS calls out as in the FCOM.

How to test

#	Do this	Expected result
1	A380X on approach (FMS approach phase), MFD SURV CONTROLS: GPWS and TERR SYS on. Flaps FULL, gear UP, 220 kt. Descend through 245 ft RA.	TOO LOW GEAR below 245 ft; no TOO LOW TERRAIN at any height.
2	Climb back above 245 ft RA (go-around).	The alert stops.

13. A380X contrails from each running engine

> **Aircraft:** A380X | **Where:** Outside view behind the engines, in cruise | **Status:** Tested in the sim by the user: an engine 1 flameout at FL350 stopped only its contrail, the contrail came back after the relight at FL290, 282 kt. Unit tests: JettisonDripContrailEffects.spec.ts (7). Not sim-tested on this core branch itself.

What it is for

The fuselage part drew the contrails from the fuselage centre, whatever the engines did, and the wings part had a second copy on the exhaust nodes. Contrails come from the water vapour of the burnt fuel, so an engine that burns no fuel leaves none. The contrails are now declared once, on the exhaust of each of the four engines, and each shows only while the sim's contrail conditions are met and that engine burns fuel (MSFS engine combustion and a FADEC fuel flow above 0). A flamed out, shut down or seized engine leaves no contrail; the other engines keep theirs.

- One contrail per engine (engines 1 to 4), plus the ribbon contrail on engines 2 and 4 as before.
- Engine 2 now uses its own exhaust point (FX_EXHAUST_RIGH, the node of the left inner engine in the wing model despite its name); the old copy used the FX_EXHAUST_LEF point.
- No contrail is declared in the fuselage part any more.

Limitations and why

Limitation	Why	What to do
Whether contrails form is decided by MSFS.	The effect uses the sim's CONTRAILS CONDITIONS MET (altitude and temperature), as the stock contrail templates do.	-
No N3 threshold for "running".	No source gives an A380 core speed limit for it; the fuel flow and the MSFS combustion flag are used instead (design choice).	-

How to use

1. Nothing to set: fly in contrail conditions (cold air at cruise level) and look at the engines from the outside view.

How to test

#	Do this	Expected result
1	A380X in cruise at FL350, all four engines running, contrails showing behind the engines. Outside view from behind.	One contrail behind each of the four engines, none from the fuselage centre.
2	Shut down engine 1: ENG 1 MASTER OFF.	The contrail of engine 1 (left outer) stops; engines 2, 3 and 4 keep theirs.
3	Descend to FL290 at about 280 kt and relight engine 1 (ENG START selector IGN/START, ENG 1 MASTER ON).	Once engine 1 runs again and contrails form at that level, its contrail comes back.

14. A380X OANS: ARPT NOT IN ACTIVE F/PLN in PLAN mode only

> **Aircraft:** A380X | **Where:** ND with the OANS airport map | **Status:** Not sim-tested (added; unit test: OancAirportNotInActiveFpln.spec, failing first).

What it is for

The A380 FCOM OANS message ARPT NOT IN ACTIVE F/PLN means: "In PLAN mode, the flight crew has displayed an airport that is not the origin, destination, or alternate airport". Upstream showed it in every ND mode but ARC (ROSE NAV too). It now shows in PLAN mode only.

Situation	What you should see
ND in PLAN, OANS showing an airport that is not ORIGIN, DEST or ALTN	ARPT NOT IN ACTIVE F/PLN.
ND in ROSE NAV or ARC with the same airport	No message.
ND in PLAN with the origin, destination or alternate	No message.

Limitations and why

Limitation	Why	What to do
The other OANS messages are unchanged.	Only this message differed from the FCOM.	-

How to use

1. Nothing to set: the message follows the ND mode and the airport shown.

How to test

#	Do this	Expected result
1	A380X on the ground with a flight plan. Show the OANS map of an airport that is not ORIGIN, DEST or ALTN (OANS airport selection), ND in ROSE NAV, range ZOOM.	No ARPT NOT IN ACTIVE F/PLN.
2	Switch the ND to PLAN.	ARPT NOT IN ACTIVE F/PLN shows.
3	Select the origin airport on the OANS, ND in PLAN.	No message.

What PR testers should check

- The message only in PLAN mode, with an airport outside the flight plan.
- Not sim-tested yet.

15. A380X flap automatic retraction and extension: regression tests

> **Aircraft:** A380X | **Where:** Flaps lever at 1 after takeoff; SD F/CTL, PFD flap indication | **Status:** No change of behaviour (tests only, added). The behaviour itself was checked end to end after a user report: no bug.

What it is for

User report: the flaps seemed not to retract after a CONF 1+F takeoff. Checked end to end, no bug: with the flaps lever at 1 the SFCC gives CONF 1 above 212 kt and CONF 1+F again below 205 kt (A380 FCOM flaps and slats, automatic retraction system ARS). CONF 1 keeps the slats at 20 degrees: only the 8 degrees of flap retract, a small visible change. Two full-aircraft tests now guard this on the A380 hydraulics test bed (real SFCC, flap and slat assembly, four engines): a lever-1 takeoff through 212 kt and back below 205 kt, and an in-flight spawn with the lever at 1. The takeoff test gives each flap movement 60 s (30 s failed about once in ten runs with the random frame times of the test bed).

Situation	What you should see
Takeoff with CONF 1+F, lever left at 1, accelerating through 212 kt	CONF 1: the flaps go from 8 degrees to 0, the slats stay at 20 degrees; PFD and SD show 1.
Lever at 1, decelerating below 205 kt	CONF 1+F again: the flaps extend to 8 degrees.

Limitations and why

Limitation	Why	What to do
The retraction is hard to see from the cabin or the outside view.	Only 8 degrees of flap move; the slats stay out.	Watch the SD F/CTL page.

How to use

1. Take off with lever 1 (CONF 1+F); the flaps go to CONF 1 by themselves above 212 kt. Retract the lever to 0 as usual.

How to test

#	Do this	Expected result
1	A380X takeoff with the flaps lever at 1 (CONF 1+F). SD F/CTL page. Accelerate through 212 kt with the lever left at 1.	The flaps retract to 0 (CONF 1), the slats stay at 20 degrees; PFD flap indication 1.
2	Slow below 205 kt with the lever still at 1.	The flaps extend again (CONF 1+F).

What PR testers should check

- Nothing new to check in the sim: tests only. Report any CONF 1 / 1+F change at other speeds.

RAT ground stow and A32NX RAT performance

The ram air turbine (RAT) of both aircraft can be stowed again after the landing, on the ground only, as the FCOMs describe, and the A32NX RAT has the real propeller size and holds the FCOM pressure

Aircraft	A32NX and A380X
Where to find it	The aircraft systems (A32NX hydraulic RAT, A380X electrical RAT); requested from the flyPad Ground > Services page (row on `core/efb-redesign`)
Branch	`core/rat-ground-stow`, based on `master` (FlyByWire master 76db0aef1 merged in); independent of the other core branches. It holds the systems part (shared Rust, tests, simvar docs, CHANGELOG line). The flyPad row that sends the request is on `core/efb-redesign`, because it is a row of the Material 3 Services page: the feature works for the user only when both branches are merged. Without the flyPad row the request can be written as `L:A32NX_RAT_STOW_REQUEST` with any L:var tool; without this branch the flyPad row shows while the RAT is out but pressing it has no effect.
Status	RAT stow: tested in the sim by the user on both aircraft: OK. A32NX RAT propeller and pressure: tested in the sim by the user: OK (blue 2,558-2,567 psi). Not sim-tested on the master base; Rust tests pass there.
Pictures	No pictures: the flyPad row is one line of the Services equipment list; the A380X RAT rig and the A32NX HYD page are stock.

1. RAT stow on the ground

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, row "RAT stow (maintenance)" | **Status:** Tested in the sim by the user: OK on both aircraft. Rust tests in systems (ram_air_turbine_deployment.rs), a320_systems (hydraulic) and a380_systems (electrical); vitest ratStow.spec on `core/efb-redesign`.

What it is for

Once the RAT had extended (both AC buses lost, RAT MAN ON, A32NX EMER ELEC PWR MAN ON), nothing stowed it again: it stayed out for the rest of the session, and on the A32NX HYD RAT FAULT stayed on. On the real aircraft the ground crew stows it after the flight.

> **NOTE**
> A320 FCOM DSC-29-10-20 : "The RAT deploys automatically if AC BUS 1 and AC BUS 2 are both lost. It can be deployed manually from the overhead panel. It can be stowed only when the aircraft is on the ground."

> **NOTE**
> A380 FCOM DSC-29-10 : "The green hydraulic system powers RAT retraction, on ground."

The systems now accept a maintenance stow request (`L:A32NX_RAT_STOW_REQUEST`) on the ground, while no extension is commanded. One shared Rust model (`ram_air_turbine_deployment.rs`) moves the strut of both aircraft, so the A320 and the A380X behave the same.

What you should see

Situation	What you should see
RAT stowed	No RAT row in the Services list.
RAT out, aircraft on the ground and stopped	Row "RAT stow (maintenance)", status "RAT extended", button **Stow**.
Stow pressed and accepted	Status "Stowing" for about 3 s; the strut moves in 1 s; the row disappears. A380X: the RAT rig retracts. A32NX: the HYD page RAT indication goes back to stowed.

RAT out in flight, or on the ground but moving	Row greyed, status "RAT extended · stow when stopped on the ground".
Stow pressed while an extension is still commanded (emergency electrical configuration, RAT MAN ON or A32NX EMER ELEC PWR MAN ON held in)	Refused: the RAT stays out.
RAT MAN ON (or the automatic extension) after a stow	The RAT extends again.
A32NX on the ground with the RAT out	HYD RAT FAULT, as the FCOM says (PRO-ABN-HYD: "The RAT is not fully stowed"); it clears after the stow.

Limitations and why

Limitation	Why	What to do
The stow needs no hydraulic pressure.	Design choice: the FCOMs say only "on the ground" (A320) and that green hydraulics power the retraction on ground (A380); the stow is a ground crew action, so it is not tied to the aircraft hydraulics being pressurised.	-
The strut travels in 1 s.	Design choice: the same speed as the stock extension; no FCOM value.	-
No RAT on the A32NX model on this branch.	The visible A32NX RAT (doors, strut, propeller) is on `core/a32nx-exterior`; with this branch alone only the HYD page, HYD RAT FAULT and the RAT pump show the stow.	Merge `core/a32nx-exterior` to see the RAT go in.
A request in flight is dropped.	Each request is evaluated once (written back to 0), so a request in flight does not stow the RAT after the landing.	Request again on the ground.
The flyPad row needs the stationary aircraft.	Design choice of the flyPad (like the other ground services); the systems themselves only need "on the ground".	Stop the aircraft.

How to use

1. After a flight with the RAT out, stop on the ground.
2. Restore normal electrical power (engines or APU generators, or external power) so that no extension is commanded; on the A32NX the RAT MAN ON and EMER ELEC PWR MAN ON pbs are not pressed.
3. flyPad > Ground > Services: press **Stow** on the row "RAT stow (maintenance)".

How to test

The flyPad row is on `core/efb-redesign`: test with both branches merged. With this branch alone, replace each "press Stow" step by setting `L:A32NX_RAT_STOW_REQUEST` to 1 with an L:var tool (SimConnect); the "row" lines then do not apply.

#	Do this	Expected result
1	A380X on the ground, engines running, parking brake set. Overhead EMER ELEC PWR panel: open the guard and press RAT MAN ON.	The RAT extends (rig out); the flyPad Services list shows "RAT stow (maintenance)", active.

2	With RAT MAN ON released, press Stow on the flyPad.	"Stowing", the rig retracts in about 1 s and the row disappears. (While RAT MAN ON is held in, the extension is still commanded and the stow is refused.)
3	A380X in flight: press RAT MAN ON, then look at the flyPad.	Row greyed ("stow when stopped on the ground"); a request written in flight is dropped.
4	Land with the RAT out, stop, press Stow.	The RAT stows.
5	A32NX on the ground, engines running. HYD panel: RAT MAN ON (or EMER ELEC PWR MAN ON).	The RAT extends: HYD page RAT out, HYD RAT FAULT (FCOM: the RAT is not fully stowed); flyPad row "RAT stow (maintenance)".
6	Press Stow on the flyPad.	The HYD page RAT goes back to stowed, HYD RAT FAULT clears, the row disappears.
7	A32NX: press RAT MAN ON again.	The RAT extends again.

2. A32NX RAT propeller size and pressure

> **Aircraft:** A32NX | **Where:** Overhead HYD panel RAT MAN ON; ECAM HYD page | **Status:** Tested in the sim by the user: OK (group 2 : blue 2,558-2,567 psi on the RAT). Before this fix the same test gave 2,680 psi. Rust tests in a320_systems (hydraulic), systems (pumps, wind_turbine).

What it is for

The A32NX RAT propeller was 1 m wide and the RAT held the blue system at about 2,680 psi. The real A320 RAT has a 29.5 in (749 mm) propeller, and the FCOM gives 2,500 psi on the RAT.

> **NOTE**
> A320 FCOM DSC-29-10-10 : "3 000 PSI (2 500 PSI when powered by the RAT)". A320 FCOM PRO-ABN-ELEC ELEC EMER CONFIG: MIN RAT SPEED 140 kt; "The RAT is capable of supplying the EMER GEN down to 125 kt, except during flare".

The propeller now has the real diameter, and its performance is retuned to the real RAT (Hamilton Sundstrand data quoted in the code): rated output from 135.5 kt, governed speed 4,800 to 6,600 rpm. The blue pump settles at about 2,500 psi on the RAT, with or without the emergency generator. The A380X RAT is not changed.

What you should see

Situation	What you should see
RAT out, 140 kt or more	Blue system about 2,500 psi (2,400 to 2,600 psi) on the ECAM HYD page.
RAT out with normal electrics (RAT MAN ON, BLUE ELEC PUMP off)	The RAT alone holds the blue system at about 2,500 psi, at 200 kt and at 140 kt.
Emergency electrical configuration, slowing down	The emergency generator keeps running down to 125 kt. Below the governed band (about 117 kt) the blue pressure fades, and the emergency generator is lost at about 110 kt.

Limitations and why

Limitation	Why	What to do
The speeds of the fade (about 117 kt) and of the generator loss (about 110 kt) are model results.	The real values are not published: the model is tuned to give the rated output from 135.5 kt and to keep the generator down to 125 kt (FCOM); below the minimum governed speed (4,800 rpm) the pump unloads, a design choice.	-

How to use

1. Nothing to set: the RAT extends as before (both AC buses lost, RAT MAN ON, or EMER ELEC PWR MAN ON).
2. Keep 140 kt or more with the RAT as the only blue source (FCOM MIN RAT SPEED).

How to test

#	Do this	Expected result
1	A32NX in flight, clean, about 5,000 ft, 200 kt, AP and A/THR ON. Overhead HYD panel: BLUE ELEC PUMP pb OFF.	The blue system has lost its electric pump.
2	Overhead HYD panel: open the guard and press RAT MAN ON. ECAM control panel: HYD key. Wait 15 s.	The RAT is out on the HYD page; blue about 2,450-2,600 psi (not 2,680).
3	FCU SPD 160 kt for 30 s; flaps 1, SPD 140 kt for 30 s; flaps 2, SPD 135 kt for 30 s.	Blue stays about 2,450-2,600 psi at every step.
4	BLUE ELEC PUMP pb back to AUTO, accelerate and retract the flaps.	Blue back to about 3,000 psi. The RAT stays out until it is stowed on the ground (see the feature above).

A380X cargo doors follow the hydraulics

The A380X forward and aft cargo doors stay locked when the ground crew is asked to open them while an engine runs (their electric pumps are inhibited), the request is carried out once the engines are stopped, and a moving door holds its position

Aircraft	A380X
Where to find it	The aircraft systems (A380X hydraulics); the doors are requested from the flyPad Ground > Services page (row on `core/efb-redesign`) or from the sim door interactive point; SD DOOR page, E/WD
Branch	`core/a380x-cargo-doors`, based on `master` (FlyByWire master 76db0aef1); independent of the other core branches. It holds the systems part (Rust door controller and A380X hydraulics, tests, simvar documentation, CHANGELOG line). The flyPad part (the cargo door row of Ground > Services shows the real doors and says when the request waits for hydraulic power) is on `core/efb-redesign`, because it is a row of the Material 3 Services page: for the flyPad to show the new state, merge both branches (like the RAT stow row of `core/rat-ground-stow`). Without this branch the flyPad reads `L:A32NX_HYD_CARGO_DOOR_OPERATION_INHIBITED` as 0 (not inhibited); without the flyPad part the doors behave as described, but the row still shows the request.
Status	Sim-tested by the user on a combined test build: PASS, recorder-checked (engine running: both doors stayed locked; opened after the engine stopped; door held at 87 % while inhibited, then closed and locked 35 s after the inhibition ended). The close after a normal open (test 2, step 3) was skipped in that run. Rust tests: 4 (a380_systems hydraulic).
Pictures	No pictures: the behaviour is in the systems; the flyPad row is a single line of the Services doors list.

1. Cargo doors locked while their pumps are inhibited

> **Aircraft:** A380X | **Where:** Systems; flyPad > Ground > Services > Doors, row "Cargo fwd + aft" (`core/efb-redesign`) | **Status:** Sim-tested by the user, recorder-checked: PASS (the close after a normal open was not run). Rust tests: 4 in a380_systems (inhibited with an engine running; stays locked when opened with an engine running; opens once the engines are stopped; holds its position when an engine starts while it moves).

What it is for

On the A380X the cargo doors are moved by the green electric pumps. When the ground crew was asked to open a cargo door while an engine ran, the door unlocked anyway (0.24 % open in a sim run) although the pumps that move it are inhibited then. The cargo doors now follow the hydraulics.

> **NOTE**
> A380 FCOM DSC-29-10 P 3: the green electric pumps are the only power of the cargo door actuation, and they are inhibited as soon as an engine runs. DSC-52-30-40-10: an inhibited door is isolated from the hydraulics, there is no power to open it.

The systems now compute whether the cargo door operation is inhibited. It is inhibited as soon as any of the four engines runs; this is the same rule as the one that stops the green electric pumps in the A380X hydraulics (a placeholder based on the engine oil pressure, which covers the FCOM cases: engines 1 and 2 running, engines 3 and 4 running, taxi and flight). The state is published as `L:A32NX_HYD_CARGO_DOOR_OPERATION_INHIBITED` (1 = inhibited).

What you should see

Situation	What you should see
Engine running, a closed cargo door is asked to open	The door stays closed and locked. The request is kept.

Engines stopped, the request still pending (or asked again)	The door unlocks and opens (about 40 s); both doors, forward and aft.
Door moving or open, an engine starts	The door holds its position where it is (no further motion). It is not closed, so the usual open cargo door alerts apply (DOOR FWD CARGO / AFT CARGO NOT CLOSED, see the test).
Door held open, the request is switched off, then the engines are stopped	The inhibition ends, and the door closes and locks about 35 s later.
Engines stopped, normal open and close	As before: the doors open, then close and lock.
flyPad, engine running, door closed	Row "Cargo fwd + aft": "Closed · no hydraulic power, engine running"; the switch is greyed out (needs `core/efb-redesign`, see the next feature).

Limitations and why

Limitation	Why	What to do
The inhibition is "any engine running".	The A380X electric pump controllers use a placeholder (the oil pressure of the engines) for the same condition. The FCOM cases (engines 1 and 2 running, taxi, flight, engines 3 and 4 running) are all covered by that one rule, so the same rule is used here.	-
The A32NX is not changed.	Design choice: the change is in the A380X hydraulics (the green electric pumps); the A32NX cargo door model is unchanged.	-
A door that is not closed at engine start stays open until the engines are stopped.	Without the green electric pumps there is no power to move it (FCOM DSC-52-30-40-10).	Stop the engines.
The flyPad row shows the new state only with `core/efb-redesign`.	The row is a row of the Material 3 Services page, which is on that branch.	Merge both branches.

How to use

1. Stop on the ground and stop the engines (the green electric pumps are then allowed).
2. flyPad > Ground > Services > Doors: switch "Cargo fwd + aft" on to open both cargo doors, off to close them.
3. With an engine running, expect the doors to stay closed (the row says so); stop the engines first.

How to test

(it also lists the SimConnect recorder command). The flyPad lines need `core/efb-redesign` merged too; with this branch alone request the doors from the exterior door interactive point of the sim and read the E/WD, the SD DOOR page and the exterior model. After a change of the Rust code: one MSFS restart (WASM).

#	Do this	Expected result
1	A380X on the ground, PARKING BRK ON, APU running (AC power), engine 1 running at idle (engines 2-4 off), all doors closed, no amber alert on the E/WD. ECAM control panel: press the DOOR key (SD DOOR/OXY page). flyPad > Ground > Services > Doors: try the switch of the row "Cargo fwd + aft".	The row reads "Closed · no hydraulic power, engine running", its switch is greyed out and does nothing. No DOOR FWD CARGO / AFT CARGO alert on the E/WD; SD DOOR page: both cargo doors green; exterior: both doors shut. `L:A32NX_HYD_CARGO_DOOR_OPERATION_INHIBITED` = 1, `L:A32NX_FWD_DOOR_CARGO_LOCKED` = 1.

2	ENG MASTER 1 OFF; wait until engine 1 has stopped (N2 0 on the E/WD). flyPad: switch "Cargo fwd + aft" on; wait 40 s.	The row reads "Moving" with a progress bar, then "Open"; the FWD and AFT holds light up on the planform; exterior: both cargo doors open.
3	Switch "Cargo fwd + aft" off; wait 45 s.	The row reads "Closed"; SD DOOR page: both cargo doors green; exterior: both doors shut. (Not run.)
4	Switch "Cargo fwd + aft" on; wait 40 s (doors open). Engine start selector to IGN START, ENG MASTER 1 ON; wait until engine 1 is stable at idle.	E/WD amber DOOR FWD CARGO NOT CLOSED and DOOR AFT CARGO NOT CLOSED. The inhibition is 1.
5	Switch "Cargo fwd + aft" off; wait 30 s.	The doors stay open (exterior view); the row is amber "Not locked · no hydraulic power, engine running", its switch still usable. The recorder shows both positions constant (87 %).
6	ENG MASTER 1 OFF; wait until engine 1 has stopped, then 45 s more; engine start selector back to NORM.	The inhibition returns to 0, both doors close and lock about 35 s later; the row reads "Closed"; the door alerts leave the E/WD. (Any MASTER CAUTION after the shutdown came from other systems: FUEL NO ZFW OR ZFWCG DATA and SURV XPDR STBY, not from the doors.)

A380X ATC mailbox with the SD on an ND display unit

When the SD is shown on an ND display unit (DU RECONF pushbutton, e.g. with the SD DU off), the ATC mailbox comes with it and works there, as the A380 FCOM asks; the ATC MSG lights flash, the RING waits for the reading time and the KCCU MAILBOX key brings the cursor to the mailbox

Aircraft	A380X
Where to find it	F/O (or CAPT) DU RECONF pushbutton on the glareshield; the ATC mailbox at the bottom of the SD, on the SD DU or on an ND DU; ECAM CDS SD DU FAULT procedure
Branch	`core/a380x-mailbox-du-reconf`, based on `core/display-unit-failures` with `core/a380x-atccom-datalink` merged in (stacked: needs both, and their bases, merged first). `core/display-unit-failures` brings the DU reconfiguration (the SD on an ND DU) and the mouse fix for DUs holding hidden displays; `core/a380x-atccom-datalink` brings the ATC mailbox. Merge both into this branch again when they change. There is no cockpit printer here (`core/printer`): PRINT in the mailbox says PRINT NOT AVAIL.
Status	Tested in the sim by the user on a combined test build (with the mouse fix of `core/display-unit-failures`): mailbox on the F/O ND with the SD off: OK. Not sim-tested on this core branch itself. Unit tests: AtcMailbox.spec.tsx, MailboxLogic.spec.ts, MailboxBus.spec.ts, the CdsReconfiguration specs and HostedDisplay.spec.ts. ATC MSG flashing, RING delay and MAILBOX key: not sim-tested, unit tests only.
Pictures	No pictures: the mailbox on the ND DU looks like the mailbox at the bottom of the SD.

1. ATC mailbox follows the SD to the ND display unit

> **Aircraft:** A380X | **Where:** F/O DU RECONF pb; ATC mailbox on the F/O (or CAPT) ND DU | **Status:** Tested in the sim by the user on a combined test build: OK (SD off, F/O DU RECONF twice, mailbox on the F/O ND with its buttons, back on the SD with the same queue). Not sim-tested on this core branch. Unit tests: AtcMailbox.spec.tsx, MailboxLogic.spec.ts, MailboxBus.spec.ts, CdsReconfiguration specs, HostedDisplay.spec.ts.

What it is for

The ATC mailbox is the bottom section of the SD. The CDS SD DU FAULT procedure says "If the SD DU is turned OFF, use the F/O DU RECONF P/B for ATC mailbox" (A380 FCOM PRO-ABN-ECAM-10-31). On `core/display-unit-failures` the F/O DU RECONF pb brings the SD to the F/O ND DU, but without the mailbox, so the ATC messages could not be read or answered with the SD DU off. Now the mailbox comes with the SD.

What you should see

Situation	What you should see
Normal operation (SD DU on)	The mailbox at the bottom of the SD DU, as before.
SD DU off or failed, F/O DU RECONF pressed until the SD shows on the F/O ND DU	The SD with the mailbox on the F/O ND DU: the same queue and the same message; its buttons and the page arrows work with the F/O cursor.
SD DU on again	The mailbox is active on the SD DU again, with the same queue.
The SD shown both on the SD DU and on an ND DU (DU RECONF used because of another failed DU)	The SD DU's mailbox is the active one; the mailbox on the ND DU shows the same message and takes no input (design choice).

The SD shown nowhere	No mailbox marks a new message as displayed or read: the ATC MSG lights stay on until a DU shows the mailbox again (design choice).
PRINT in the mailbox	PRINT NOT AVAIL (no cockpit printer on this branch; printing is on `core/printer`).

How it works

- The systems host decides which DU has the active mailbox (`L:A380X_CDS_MAILBOX_DU`): the SD DU while it is working and shows the SD, else the ND DU that shows the SD (the SD is never on both sides at once, FCOM DSC-31-15-20), else none.
- Only the active mailbox answers the crew and sends to the ATC function; another mailbox on screen mirrors the displayed message.
- The A380X README and the simvar documentation describe the variable and the rule.

Limitations and why

Limitation	Why	What to do
Needs an ATC connection with a message to see anything in the mailbox.	The mailbox shows the messages of the FBW ATSU (`core/a380x-atccom-datalink`).	Log on to an ATC center (Hoppie or BeyondATC) or prepare a request with XFR TO MAILBOX.
No printing.	The cockpit printer is `core/printer`, not merged here.	Merge it for PRINT.
A full MSFS restart is needed after an update.	It comes with the panel.cfg and cockpit changes of `core/display-unit-failures`.	Restart the sim after installing.

How to use

1. With the SD DU off or failed (CDS SD DU FAULT), press the **F/O DU RECONF** pushbutton (glareshield, F/O side) until the F/O ND DU shows the SD: first the E/WD, then the SD with the mailbox.
2. Read and answer the ATC messages in the mailbox on the F/O ND with the F/O cursor, as on the SD.
3. Press F/O DU RECONF once more to get the ND back, or switch the SD DU back on.

How to test

#	Do this	Expected result
1	A380X powered, logged on to an ATC center (MFD ATC COM > CONNECT) with at least one message in the mailbox (e.g. a request prepared with REQUEST, XFR TO MAILBOX).	The message in the mailbox at the bottom of the SD.
2	Turn the SD DU off (ECAM control panel on the pedestal: SD knob fully left, OFF), or flyPad > Failures > ATA 31: **SD display unit**.	The SD DU goes blank (with the failure: CDS SD DU FAULT with IF SD DU OFF : F/O DU RECONF P/B FOR ATC MAILBOX).
3	Press the F/O **DU RECONF** pushbutton twice.	The F/O ND DU shows the E/WD, then the SD with the mailbox: the same queue and the same message.
4	Move the F/O cursor to the mailbox and click the previous-message arrow (or a message button).	The mailbox reacts on the F/O ND.
5	Press **PRINT** in the mailbox.	PRINT NOT AVAIL.
6	Turn the SD DU back on (or remove the failure) and wait for its self-test.	The SD mailbox shows the same queue and its buttons work.

2. ATC MSG lights flashing, RING delay and the KCCU MAILBOX key

> **Aircraft:** A380X | **Where:** Glareshield ATC MSG pushbuttons; ATC RING aural; KCCU MAILBOX key; SD mailbox | **Status:** Not sim-tested (added; unit tests: MailboxBus, AtcMsgLights and AtcMailbox specs, failing first).

What it is for

Three items of the A380 FCOM ATC datalink description (DSC-46-10-20):

- **ATC MSG lights flash at 2 Hz** (250 ms on, 250 ms off) while an ATC message waits in the mailbox, until either ATC MSG pushbutton is pressed; then both go off (FCOM: "The pushbutton flashes at a frequency of 2 Hz"). A new variable `L:A380X_GLARESHIELD_ATC_MSG_LIGHT` drives the two pushbuttons.
- **RING delay:** the ATC RING sounds only if the message is not read within 15 s (5 s for an urgent message), then repeats. The A32NX DCDU is unchanged.
- **KCCU MAILBOX key:** puts the KCCU cursor of that side on the SD mailbox (FCOM: "When pressed, the cursor moves to the mailbox section of the SD"); the MFD of that side hides its cursor. The key works with the KCCU keyboard ON. Design choice: the cursor goes to the middle of the mailbox (the FCOM gives no position).

Situation	What you should see
An ATC uplink arrives	Both ATC MSG pushbuttons flash at 2 Hz.
Message read within 15 s (5 s if urgent)	No RING.
Message not read after 15 s (5 s if urgent)	RING, then repeated until the message is read.
Either ATC MSG pushbutton pressed	Both lights go off.
KCCU MAILBOX key pressed (keyboard ON)	The cursor of that side jumps to the SD mailbox.

Limitations and why

Limitation	Why	What to do
The light changes need a flight reload.	The glareshield behaviour XML changes.	Reload the flight after installing.
The MAILBOX key does nothing with the KCCU keyboard OFF.	FCOM: the KCCU keys work with the keyboard switch ON.	Set the KBD switch ON.

How to use

1. When the ATC MSG lights flash, press one of them (or read the message) to stop them.
2. Press the KCCU **MAILBOX** key to bring the cursor to the SD mailbox, then answer the message.

How to test

#	Do this	Expected result
1	A380X, datalink logged on (BeyondATC or Hoppie). Send yourself a normal CPDLC uplink.	Both ATC MSG pushbuttons flash about twice a second; no RING in the first 15 s.
2	Do not read it for 15 s.	RING; it repeats until the message is read.
3	Send an urgent uplink.	RING after 5 s if not read.
4	Press one ATC MSG pushbutton.	Both lights go off.
5	KCCU keyboard ON, cursor on the MFD. Press the KCCU **MAILBOX** key.	The cursor moves to the middle of the SD mailbox; the MFD cursor goes.

What PR testers should check

- ATC MSG flashing and the RING delay with BeyondATC.
- The MAILBOX key cursor (both sides, also with the SD on an ND DU).
- Not sim-tested yet.

A380X ELEC, fuel and FCOM gap alerts

The A380X ECAM raises the FCOM combined busbar alerts, ELEC STATIC INV FAULT and the FUEL wing feed pump, wings balance and inner tank alerts, on top of the ELEC alerts and the A380X failures merged together; since it also holds the A380X part (new failures, ECAM alerts, memos, voices and system models: air data, cabin pressure, drive oil, APU, smoke, bleed, gear and brakes, engines, fuel, anti-ice, oxygen, avionics network, GPWS and windshear), the XFR pushbuttons and other fixes

Aircraft	A380X (A32NX: ADIRU power supplies, engine anti-ice valves and anti-ice key events, through shared code)
Where to find it	flyPad > Failures (ATA 21, 24, 26, 27, 28, 29, 30, 31, 32, 34, 35, 36, 42, 49, 73, 79); overhead ELEC, FUEL (XFR, EMER OUTR TK XFR), AIR (ENG BLEED), ANTI ICE, CARGO SMOKE, APU, EVAC and ADIRS panels; E/WD, STATUS (and STATUS MORE), PFD, SD ELEC, FUEL, ENGINE, PRESS, WHEEL and DOOR/OXY pages; ECP; T.O CONFIG key; flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs (with core/efb-redesign)
Branch	`core/a380x-elec-fuel-alerts`, based on `core/a380x-elec` with `core/failures-a380x` merged in (stacked: needs both merged first; core/a380x-elec itself needs core/surveillance, core/a380x-mfd-fms and core/gps). Why a separate branch: the combined busbar and wing feed alerts need the ELEC alerts of core/a380x-elec and the FWS of core/failures-a380x, and merging either of those two into the other would put each scope into the other. The merge keeps both failure lists, both flyPad realism settings and the ECAM texts of both once, and applies the merge notes of core/failures-a380x: T.O CONFIG NORMAL reads the ELEC GEN / DRIVE alerts, ELEC EMER CONFIG reads the FWS signal inhibited by ALL ENG FLAME OUT; the FUEL JETTISON pbs are still read directly (no core/fuel here). Since `core/fcom-audit-fixes` is merged in too (merge f0aaa86dc), a third parent: the A380X door, A-ICE and NAV alert files of that branch are the ones its later alerts extend. This branch is the A380X FCOM gap alert branch: it is the only one holding all the A380X alert code the later alerts build on (ELEC, XPDR and SURV of core/a380x-elec, engine, brake, air data and fire of core/failures-a380x, ECAM of core/ecam-fws). Merge core/a380x-elec, core/failures-a380x and core/fcom-audit-fixes into it again when they change. Dependencies on other core branches (merge both): the A380X smoke detection model and the CARGO SMOKE panel lights are on `core/failures-a32nx` (the SMOKE alerts here read its L:vars); the flyPad pin option of the smart FIVE HUNDRED callout is on `core/efb-redesign`. Merge note with `core/fuel`: the fuel transfer memos and the manual transfers of this branch read L:A380X_FUEL_JETTISON_IN_PROGRESS directly; with core/fuel they take its jettison signals. The features of the parents are described in their own manuals; the A32NX parts of these alerts are on core/failures-a32nx.
Status	Tested in the sim by the user on a combined test build: ELEC AC BUS 2+4 FAULT OK (flight) and FUEL L WING FEED PMPs FAULT OK. The other combined busbar alerts, STATIC INV FAULT, WINGS NOT BALANCED / BALANCED and INR TKs QTY LO: not sim-tested (unit tests FwsElecCombinedBusAlerts.spec.ts, FwsFuelFeedBalanceAlerts.spec.ts). On a combined test build: T.O CONFIG NORMAL with GEN 3 OFF recorded as passed; no ELEC EMER CONFIG during ALL ENG FLAME OUT tested. The core branch itself was not sim-tested. A full MSFS restart is needed after installing. The FCOM alert additions: not sim-tested; unit/Rust tests only.
Pictures	No pictures: the alerts use the stock E/WD, STATUS and SD pages.

1. Combined ELEC busbar alerts and ELEC STATIC INV FAULT
2. FUEL wing feed pumps, wings balance and inner tank alerts
3. ELEC alerts with the A380X failures: T.O CONFIG and ELEC EMER CONFIG
4. ADR and IR failures, ADIRU power supplies
5. CAB PRESS, RAM AIR, APU FAULT, HYD, A-ICE and SURV alerts
6. PFD FMA messages, AVAIL, EGT start limits and SD items
7. smart FIVE HUNDRED callout
8. ECAM advisories and STATUS MORE
9. EVAC ACTIVE memo
10. cabin pressure, generator drive oil, ELEC SD failures, e-pump inhibition
11. APU automatic shutdowns, smoke detection and cargo extinguishing alerts
12. bleed monitoring, leak detection, BLEED FAULT and LEAK alerts

13. landing gear and brakes
14. HP fuel valve, FADEC faults, motoring and auto-idle
15. LP valve, emergency outer tank transfer and fuel leak detection
16. anti-ice valves, wing anti-ice, oxygen, anti-ice key events
17. avionics network failures, SFCC channels, TAIL STRIKE
18. reactive windshear, GPWS mode 2, alpha floor Mach limit
19. NAV FM / GPS POS DISAGREE and NAV LS TUNING DISAGREE
20. A380X FWS voices: sidestick priority, intermediate RA callouts, WINDSHEAR, TERRAIN, GLIDE SLOPE, PITCH PITCH, COMPANY MESSAGE
21. No BRAKES BTV FAULT while an inner engine is not running
22. Engine exceedance marks only on displayed values
23. XFR pushbuttons start in AUTO and drive the manual fuel transfers

1. Combined ELEC busbar alerts and ELEC STATIC INV FAULT

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 24; E/WD, STATUS, SD ELEC AC / DC | **Status:** ELEC AC BUS 2+4 FAULT tested in the sim by the user on a combined test build: OK (flight). The other combined alerts and STATIC INV FAULT: not sim-tested (unit tests FwsElecCombinedBusAlerts.spec.ts).

What it is for

A380 FCOM PRO-ABN-ECAM-10-24 has combined alerts for the loss of several busbars together. The A380X showed the single busbar alerts side by side instead. Six combined alerts now trigger when all their busbars are lost and replace the single alerts they cover, with their procedure lines, flight phase inhibitions (phases 4, 5 and 10) and STATUS:

Alert	Busbars lost
ELEC AC BUS 1+2 & DC BUS 1 FAULT	AC 1, AC 2, DC 1
ELEC AC BUS 2+3 & DC BUS 1+2 FAULT	AC 2, AC 3, DC 1, DC 2
ELEC AC BUS 2+4 FAULT	AC 2, AC 4
ELEC AC BUS 3+4 & DC BUS 2 FAULT	AC 3, AC 4, DC 2
ELEC DC BUS 1+2 FAULT	DC 1, DC 2
ELEC DC BUS 1+ESS FAULT	DC 1, DC ESS

- A combined alert whose busbars are all part of another active one is replaced by it (DC BUS 1+2 FAULT inside AC BUS 2+3 & DC BUS 1+2 FAULT); combined alerts that only overlap are both shown.
- The fuel lines of these procedures depend on the flight time above FL 300 since departure (more or less than 30 min, counted from lift-off) and lead to gravity feeding (descent, CROSSFEED ... OFF, EXPECT NORM COLLECTOR CELL DEPLETION, AVOID NEG G LOAD), shown in flight only.
- The STATUS lists the FCOM lines that have an ECAM text, for example AC BUS 2+4 FAULT: MAX SPEED 310 KT, FUEL PARTLY UNUSABLE; INOP SYS F/CTL PROT, AP 1+2, A/THR, FD 1+2, ADR 2+3, TCAS 1+2, SATCOM, JETTISON, FEED TK 1 and 4 PMPs; INFO ALTN LAW : PROT LOST, FEED TK 1 (4) BY GRVTY ONLY. The JETTISON INOP SYS text (also in AC BUS 2+4, DC BUS 1+2 and DC BUS 1+ESS FAULT) is added here, identical to the one of `core/fuel`.
- **ELEC STATIC INV FAULT** (crew awareness, phases 3 to 10 inhibited): the static inverter runs only on batteries (no AC ESS power, both battery contactors closed); it counts as failed when it runs but gives no normal output.

Limitations and why

Limitation	Why	What to do
Some FCOM STATUS lines are missing (for example ATC COM VOICE ONLY, VHF 3, MOST FUEL XFR PMPs).	They have no ECAM text in FBW yet.	-

REDUND LOSS lines are not shown for these alerts.	They never reach the STATUS page in the FBW FWS.	-
The combined alerts do not trigger in ELEC EMER CONFIG.	Design choice: the bus fault alerts need one AC busbar powered; ELEC EMER CONFIG covers that case.	-
VENT AVNCS EXTRACT OVRD and OUTR TK XFR MAN lines are never ticked.	No avionics ventilation pb-sw; the manual outer tank transfer pb-sw is inoperative in the sim.	-

How to use

1. When a combined ELEC alert shows, follow its E/WD procedure (the sensed lines tick as you act).
2. Read the STATUS page; in flight follow the gravity feeding lines if they show.

How to test

Setup: cruise above FL300, AP ON.

#	Do this	Expected result
1	flyPad > Failures > ATA 24 Electrical Power > **AC 2**, then **AC 4**.	Master caution and single chime; E/WD ELEC AC BUS 2+4 FAULT, not the single AC BUS 2 / AC BUS 4 alerts; SD ELEC AC page.
2	ECAM control panel: STS.	STATUS: ALTN LAW : PROT LOST, MAX SPEED 310 KT, INOP SYS among others JETTISON.
3	Deactivate both. Fail **DC 1**, then **DC 2**.	ELEC DC BUS 1+2 FAULT instead of DC BUS 1 / DC BUS 2 FAULT. Deactivate both.
4	On the ground, aircraft on batteries only (BAT pbs AUTO, no EXT PWR, no APU). Fail **Static Inverter**.	ELEC STATIC INV FAULT. Deactivate it.

2. FUEL wing feed pumps, wings balance and inner tank alerts

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28; overhead FUEL panel; E/WD, SD FUEL | **Status:** FUEL L WING FEED PMPs FAULT tested in the sim by the user on a combined test build: OK. WINGS NOT BALANCED / BALANCED and INR TKs QTY LO: not sim-tested (unit tests FwsFuelFeedBalanceAlerts.spec.ts).

What it is for

Four FUEL alerts of A380 FCOM PRO-ABN-ECAM-10-28 with their crossfeed procedure lines:

- **FUEL L (R) WING FEED PMPs FAULT**: all pumps of feed tanks 1 and 2 (left) or 3 and 4 (right) at low pressure, abnormally not running or switched off. It replaces the two FEED TK n MAIN + STBY PMPs FAULT alerts of its tanks. Inhibited in phases 4 to 7, 9 and 10. Pumps switched off on request of FUEL WINGS NOT BALANCED do not count.
- **FUEL WINGS NOT BALANCED**: on the ground, an imbalance above 6 600 lb between the symmetric feed tanks or the inner tanks, 5 500 lb between the mid tanks or 2 600 lb between the outer tanks (not while refuelling); in flight, more than 6 600 lb between feed tanks 1 and 4 or 2 and 3. In flight it stays until that pair is balanced (the lighter tank is no longer lighter), then **FUEL WINGS BALANCED** shows until its pumps are back ON and its crossfeeds OFF. Inhibited in phases 3 to 7, 9 and 10.
- **FUEL INR TKs QTY LO**: fuel on board above 110 200 lb with an inner tank below 11 200 lb (ZFW and ZFCG entered, both outer tanks full). Inhibited in phases 3 to 11.

Limitations and why

Limitation	Why	What to do
The "balanced" point of WINGS NOT BALANCED is a design choice.	The FCOM gives no balanced threshold.	-
Landing ends the in-flight balance procedures.	Design choice.	-

How to use

1. When a FUEL alert shows, follow its E/WD lines with the CROSSFEED and feed pump pbs of the overhead FUEL panel.
2. Watch the feed tank quantities on the SD FUEL page until the alert clears.

How to test

Setup: cruise.

#	Do this	Expected result
1	flyPad > Failures > ATA 28 Fuel > **Feed tank 1 main pump**, **Feed tank 1 standby pump**, **Feed tank 2 main pump** and **Feed tank 2 standby pump**.	FUEL L WING FEED PMPs FAULT replaces the two FEED TK MAIN + STBY PMPs alerts.
2	Follow the E/WD lines (CROSSFEED pbs on the overhead FUEL panel).	The crossfeed lines tick.
3	Deactivate the four failures; CROSSFEED pbs back to normal.	The alert goes.
4	On the ground, engines running, set the left outer tank about 1.5 t (3 300 lb) above the right one (fuel set per tank, for example with the MSFS fuel menu).	FUEL WINGS NOT BALANCED. Set the tanks equal: it goes.

3. ELEC alerts with the A380X failures: T.O CONFIG and ELEC EMER CONFIG

> **Aircraft:** A380X | **Where:** T.O CONFIG key; E/WD | **Status:** On a combined test build: T.O CONFIG NORMAL blocked by GEN 3 OFF recorded as passed in the commit message; no ELEC EMER CONFIG during ALL ENG FLAME OUT tested OK. The core branch itself was not sim-tested.

What it is for

What this branch adds by having both parents together:

- **T.O CONFIG NORMAL** (the A380 rule of `core/failures-a380x`) also checks the ELEC GEN 1 (2)(3)(4) FAULT, GEN OFF and DRIVE DISCONNECTED alerts of `core/a380x-elec`, as in the FCOM list (DSC-31-40-10).
- **ELEC EMER CONFIG** (of `core/a380x-elec`) is not shown while ENG ALL ENG FLAME OUT is (FCOM: "This alert inhibits the ELEC EMER CONFIG alert").

How to use

1. Before take-off press T.O CONFIG with all four generators ON.
2. In an all engines flame out, follow ALL ENG FLAME OUT; ELEC EMER CONFIG does not come on top of it.

How to test

#	Do this	Expected result

1	On the ground, four engines running, ready for take-off. Overhead ELEC panel: GEN 3 pb OFF. ECAM control panel: T.O CONFIG key.	ELEC GEN 3 OFF; no T.O CONFIG NORMAL.
2	GEN 3 pb ON, wait 5 s, T.O CONFIG key again.	T.O CONFIG NORMAL.
3	In flight at FL200, 300 kt, APU OFF: activate the four "Engine n flameout (crew relight possible)" failures (flyPad > Failures > ATA 72). Wait 20 s.	Red ENG ALL ENG FLAME OUT; the RAT deploys; no ELEC EMER CONFIG. Relight as in the core/failures-a380x manual.

4. ADR and IR failures, ADIRU power supplies

> **Aircraft:** A380X and A32NX (ADIRU power supplies) | **Where:** flyPad > Failures > ATA 34; overhead ADIRS panel; E/WD | **Status:** Not sim-tested (unit tests: adirs.rs 17 Rust tests, FwsNavAlerts.spec.ts, failure id specs).

What it is for

Air data and inertial data could not be lost on their own, and the ADIRUs ran without their busbars. Now:

- Six new flyPad failures: **ADR 1 (2)(3) (air data lost, FAULT light)** and **IR 1 (2)(3) (inertial data lost, FAULT light)**. The outputs of that ADIRU part go to failure warning and its FAULT light comes on (new L:A32NX_ADIRS_ADR_n_FAULT_WARN_DISCRETE, shared ADIRS model).
- Each ADIRU is supplied by its busbars, from the FCOM ELEC INOP SYS lists: the ADR needs its AC and its DC busbar, the IR runs on either. Power cuts up to 200 ms, and the first 5 s after a flight load, are ignored, so a flight loaded on the runway or in the air keeps its alignment.
- An ADR that lost its DC busbar no longer gives a NAV ADR FAULT on top of the ELEC busbar alert.
- Shared code: the A32NX ADIRU supplies come with it (the ADIRS model now needs each aircraft's supplies). The A32NX ADR / IR failures, ON BAT light and alerts are on `core/failures-a32nx`.

What you should see

Situation	What you should see
ADR 1 failed	ADR 1 FAULT light on the ADIRS panel; E/WD NAV ADR 1 FAULT; the air data of ADR 1 are lost.
IR 1 failed	IR 1 FAULT light; E/WD NAV IR 1 FAULT; the inertial data of IR 1 are lost.
An ADR busbar lost (ELEC alert)	That ADR is lost, but only the ELEC alert shows, no NAV ADR FAULT.

Limitations and why

Limitation	Why	What to do
No high angle of attack / sideslip thresholds for the A380X ADRs.	The FCOM gives no boundary.	-
The A32NX IR 10 min alignment is not on this branch.	It is on core/failures-a32nx only (merge note: keep it when both are merged).	-

How to use

1. Open flyPad > Failures > ATA 34 Navigation > and activate an ADR or IR failure.
2. Follow the NAV procedure on the E/WD (ADIRS panel and source switching).

How to test

#	Do this	Expected result
1	A380X in cruise, AP ON. Activate **ADR 1 (air data lost, FAULT light)**.	ADR 1 FAULT light; NAV ADR 1 FAULT on the E/WD. Deactivate it.
2	Activate **IR 2 (inertial data lost, FAULT light)**.	IR 2 FAULT light; NAV IR 2 FAULT. Deactivate it.
3	Load a flight on the runway, then another one in cruise.	The ADIRUs stay aligned (no ALIGN, attitude and navigation available at once).

What PR testers should check

- Flights loaded on the runway and in cruise keep the IR alignment.
- The ADR and IR failures from the flyPad (lights, alerts, lost data).

5. CAB PRESS, RAM AIR, APU FAULT, HYD, A-ICE and SURV alerts

> **Aircraft:** A380X | **Where:** E/WD, STATUS, memos; SD PRESS; overhead RAM AIR, APU, ANTI ICE and OXYGEN panels | **Status:** Not sim-tested (unit tests: 8 new spec files CabPressAlerts, AutoFlightAlerts, ApuFaultAlerts, FuelTransferMemos, HydPumpStatus, AntiIceAlerts, CabinOxygenMasks, AbnormBleedConfig, and 5 tests in FwsSurvAlerts.spec.ts, all failing first).

What it is for

A380X FCOM alerts and memos that were missing or never triggered:

- **CAB PRESS** (FCOM PRO-ABN-ECAM-10-21-30): EXCESS NEGATIVE DIFF PRESS (red warning, phases 8 and 9) is held until the differential pressure is back above -0.32 PSI; DIFF PRESS HI (phase 8) only in automatic mode; DIFF PRESS LO in phases 8 and 9; new **CAB PRESS LDG ELEVN FAULT** (phase 8) when neither FMS sends a landing elevation (no destination), with the LDG ELEVN deferred procedure.
- **RAM AIR** memo with the RAM AIR pb ON (DSC-21-10-20): green on the ground with no engine running, amber otherwise.
- **AUTO FLT APPROACH CAPABILITY DOWNGRADED** (PRO-ABN-ECAM-10-22) when RA SYS B alone is failed: STATUS CAT 3 DUAL inoperative, CAT 3 SINGLE ONLY.
- **APU FAULT** (PRO-ABN-ECAM-10-49) after an APU emergency or automatic shutdown by its ECB: EMER SHUTDOWN or AUTO SHUTDOWN, -APU MASTER SW ... OFF; STATUS INOP SYS APU. Not shown while APU FIRE is.
- **Fuel memos** (DSC-28-20): FWD XFR IN PROGRESS (fuel leaving the trim tank), OUTR TK XFR IN PROG (fuel leaving the outer tanks), OUTR TKS XFRD and TRIM TK XFRD (tanks emptied). The fuel model tells no transfer state, so the FWS watches the tank weights: more than 5 kg out of a tank in 10 s is a transfer.
- **HYD:** ENG PMP PRESS LO lists the pumps of its system under STATUS INOP SYS (two or more failed) or REDUND LOSS (one failed); G (Y) RSVR AIR PRESS LO and G (Y) SYS OVHT get their FOR APPROACH deferred procedures (engine pumps back ON); G+Y SYS PRESS LO shows LDG PERF AFFECTED and the FCOM INOP SYS list (BTV, CAT 2, GLS AUTOLAND), without STABILIZER.
- **A-ICE ICE DETECTED** and **A-ICE SEVERE ICE DETECTED** (PRO-ABN-ECAM-10-30), cruise only (phase 8): ENG 1-4 A-ICE ... ON, and WING A-ICE ... ON for severe icing.
- **Cabin oxygen masks** drop by themselves above 13 800 ft cabin altitude (DSC-35-30-10, supplied by DC ESS).
- **SURV GPWS 1 (2)(1+2) FAULT** phases and lines; **SURV TAWS 1 (2)(1+2) FAULT** now triggers when the GPWS and the TERR SYS of a system are both failed, instead of the two separate alerts.
- **AIR ABNORM BLEED CONFIG** after ENG SHUT DOWN or ENG FIRE with an engine bleed lost: all the FCOM lines (XBLEED, AIR FLOW, PACK, WING A-ICE). The title is shown (the FWS has no procedure without a title).

Limitations and why

Limitation	Why	What to do
The fuel transfer memos come from the tank weights (5 kg in 10 s).	The fuel model gives no transfer state; the FCOM gives no detection rule (design choice).	-
No COLDFUEL OUTR TK XFR or GND XFR IN PROGRESS memo.	No fuel temperature model, no ground transfer function.	-
APU FAULT: the "failure but APU still available" and "failed start above FL 150" lines stay hidden.	The ECB tells neither case apart.	-
APPROACH CAPABILITY DOWNGRADED triggers only on RA SYS B.	The other FCOM triggers are PRIM monitorings the sim does not have.	-
The fuel transfer memos read L:A380X_FUEL_JETTISON_IN_PROGRESS directly.	The jettison signals of core/fuel are not on this branch.	With core/fuel merged they use its signals.

How to use

1. Nothing to set: the alerts and memos come by themselves; follow the E/WD lines and read the STATUS page.
2. For a FOR APPROACH deferred procedure, apply it in approach as the ECAM shows.

How to test

#	Do this	Expected result
1	A380X at the gate, engines off. Overhead AIR panel: RAM AIR pb ON.	Green RAM AIR memo. RAM AIR OFF.
2	Cruise, AP ON. flyPad > Failures > ATA 34 Navigation > **RA SYS B**.	AUTO FLT APPROACH CAPABILITY DOWNGRADED; STATUS: CAT 3 DUAL inoperative, CAT 3 SINGLE ONLY. Deactivate it.
3	Cruise. flyPad > Failures > ATA 34 Navigation > **GPWS 1**.	SURV GPWS 1 FAULT.
4	Also activate **TERR SYS 1**.	SURV TAWS 1 FAULT instead of SURV GPWS 1 FAULT and SURV TERR SYS 1 FAULT. Deactivate both.
5	Cruise. flyPad > Failures > ATA 29 Hydraulic Power > **Green reservoir air leak**; wait for HYD G RSVR AIR PRESS LO and follow it.	The FOR APPROACH deferred procedure G RSVR AIR PRESS LO (G ENG 1 PMP A AND B ... ON, G ENG 2 PMP A AND B ... ON) is listed.
6	Cruise above FL 300. flyPad > Failures > ATA 21 Air Conditioning > **Rapid Decompression**.	When the cabin altitude passes 13 800 ft the cabin oxygen masks drop by themselves; CAB PRESS alerts as before. Deactivate it, descend.
7	Long cruise with fuel in the outer tanks.	OUTR TK XFR IN PROG while the outer tanks empty, then OUTR TKS XFRD.
8	Cruise, no destination in the FMS (for example a flight plan without destination).	CAB PRESS LDG ELEVN FAULT (phase 8 only).

What PR testers should check

- The fuel transfer detection (5 kg in 10 s): no false memo in normal burn, memos during real transfers.
- The masks drop above 13 800 ft cabin altitude, not below.
- GPWS / TERR SYS / TAWS combinations of the SURV alerts.

6. PFD FMA messages, AVAIL, EGT start limits and SD items

> **Aircraft:** A380X | **Where:** PFD FMA; E/WD engine gauges; SD FUEL and WHEEL pages; E/WD | **Status:** Not sim-tested (unit tests: 11 spec files, 66 tests, all failing first, among them AvailIndication, EgtLimits, FMADefinitions, FuelImbalanceIndication, LgersIndications, ata70).

What it is for

- **PFD FMA** line 3: CHECK SPD MODE (its L:var was already set by the FMC) and CHECK EO SPD SETTING now show.
- **E/WD AVAIL**: stays 10 s after an engine start on the ground, pulses 1 min after a relight in flight, goes when the thrust lever is set above idle.
- **E/WD EGT**: the amber limit is 745 C during a ground start and 865 C during a relight in flight; no EEC trim during a start.
- **SD FUEL**: the tank quantities of an unbalanced pair are underlined amber, with the trigger of FUEL WINGS NOT BALANCED.
- **SD WHEEL**: L/G CTL and L/G GRVTY EXTN IN PROGRESS now show.
- **ENG 2 (3) SHUT DOWN**: the landing distance line reads LDG DIST IMPACT ON WET/CONTAM RWY ONLY.

Limitations and why

Limitation	Why	What to do
No EXTEND / RETRACT SPD BRK or CHECK APPR SEL on the A380X.	They need VNAV outputs the FMS does not give.	-
No WHEEL UP LOCK indication.	The gear model has no uplock state.	-
The ENG SHUT DOWN bleed lines are not added.	Not done yet.	-

How to use

1. Nothing to set: the indications follow the FCOM by themselves.

How to test

#	Do this	Expected result
1	A380X on the ground, engine start (ENG START selector, ENG MASTER 1 ON).	EGT amber limit at 745 C during the start; AVAIL shows when the engine is started and goes 10 s later.
2	On the ground, make one feed tank more than 6 600 lb heavier than its symmetric tank (MSFS fuel menu). SD FUEL page.	The quantities of that pair are underlined amber, with FUEL WINGS NOT BALANCED.
3	In flight, gear up: L/G GRVTY switch DOWN. SD WHEEL page.	L/G GRVTY EXTN IN PROGRESS until the gear is down.
4	Cruise: ENG MASTER 2 OFF (shut down engine 2).	ENG 2 SHUT DOWN; its landing distance line reads LDG DIST IMPACT ON WET/CONTAM RWY ONLY. Relight.

What PR testers should check

- The width of the fuel underline on the SD FUEL page.
- The EGT colour during a hot start.
- The AVAIL and fuel pulse animations.

7. smart FIVE HUNDRED callout

> **Aircraft:** A380X | **Where:** Aural callouts; flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs | **Status:** Not sim-tested (unit tests: LegacyGpws.spec.ts smart 500 block, failing without the change).

What it is for

The A380 FCOM (DSC-34-10-50 AUTOMATIC CALLOUTS) has a pin programmed "smart" FIVE HUNDRED callout: it is called only when the aircraft is more than half a dot off the glide slope, or without a valid glide slope. The callout logic (AutoCallOuts isFiveHundredCalloutDue, read by LegacyGpws) is on this branch.

Limitations and why

Limitation	Why	What to do
The flyPad option that sets the pin is not on this branch.	The flyPad settings pages are on core/efb-redesign.	Merge both: core/efb-redesign has the "Inhibited When Glideslope Captured" option.

How to use

1. flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs: under FIVE HUNDRED, set "Inhibited When Glideslope Captured" (needs core/efb-redesign).
2. Fly the approach: FIVE HUNDRED is only called when off the glide slope.

How to test

#	Do this	Expected result
1	With core/efb-redesign merged: set the option. ILS approach on the glide slope.	No FIVE HUNDRED callout at 500 ft RA.
2	Second approach, more than half a dot above or below the glide slope at 500 ft (or a non-ILS approach).	FIVE HUNDRED is called.
3	Option off, FIVE HUNDRED callout on.	FIVE HUNDRED is called on every approach.

What PR testers should check

- The callout on and off the glide slope with the pin option set and not set.

8. ECAM advisories and STATUS MORE

> **Aircraft:** A380X | **Where:** SD ENGINE and PRESS pages; ECP ENG and PRESS pbs, MORE pb; E/WD ADV; STATUS MORE | **Status:** Not sim-tested (unit tests: FwsSystemDisplayLogic.spec.ts and StatusMore.spec.ts, failing first, and FwsAdvisories.spec.ts).

What it is for

- **ECAM advisories** (FCOM DSC-31-40-10 ADVISORY): when an ENGINE value (oil quantity, oil temperature above 163 C, a rotor vibration above 5 units) or the CAB PRESS cabin V/S (above 1 800 ft/min) pulses on its SD page, the SD calls that page if it shows a flight phase page and its ECP pb comes on; if the crew had selected another page, the ECP pb flashes. The E/WD shows a pulsing ADV with the memos. ADV goes when the ECP pb is pressed or the value is back in range. New L:A380X_ECAM_SD_ADVISORY_PB.
- **STATUS MORE** (ECP MORE pb, DSC-31-40-10): new ALERTS IMPACTING LDG PERF section (the active alerts with a landing performance line), and the MORE INFO line of MISC OIS FAULT.

Limitations and why

Limitation	Why	What to do
No advisory for the nacelle temperature, delta P, cabin altitude or APU OIL LVL LO.	The SD pages do not pulse these values yet.	-

How to use

1. When ADV pulses on the E/WD, look at the called SD page; press its ECP pb to acknowledge.
2. When MORE shows at the bottom of the STATUS page, press the ECP MORE pb.

How to test

#	Do this	Expected result
1	A380X on the ground, engines at idle, SD on the flight phase page. flyPad > Failures > ATA 79 Oil > **Engine 1 oil overheat**; thrust lever 1 forward.	When the oil temperature passes 163 C and pulses, the SD shows the ENGINE page, the ECP ENG pb comes on, ADV pulses on the E/WD (with memos).
2	Press the ECP ENG pb.	ADV goes. Lever idle, deactivate the failure.
3	Flight with an alert that has a landing performance line (for example ENG 2 SHUT DOWN). STATUS, then ECP MORE pb.	STATUS MORE lists the alert under ALERTS IMPACTING LDG PERF.

What PR testers should check

- No ENG advisory at flight load while the oil quantity still reads 0.

9. EVAC ACTIVE memo

> **Aircraft:** A380X | **Where:** Overhead EVAC panel; E/WD memo | **Status:** Not sim-tested (unit tests: FwsMemosEvac.spec.ts).

What it is for

The A380 FCOM lists an EVAC ACTIVE memo while the evacuation command is on. The E/WD now shows EVAC ACTIVE while the EVAC COMMAND pb-sw is ON.

Limitations and why

Limitation	Why	What to do
The memo colour is green.	Assumed: the FCOM does not give it.	-

How to use

1. Set the EVAC COMMAND pb-sw ON (overhead EVAC panel): the memo shows until it is set OFF.

How to test

#	Do this	Expected result
1	A380X at the gate, powered. EVAC COMMAND pb-sw ON.	EVAC ACTIVE memo on the E/WD.
2	EVAC COMMAND pb-sw OFF.	The memo goes.

What PR testers should check

- The EVAC ACTIVE memo with the EVAC COMMAND pb-sw.

10. cabin pressure, generator drive oil, ELEC SD failures, e-pump inhibition

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 24; overhead PRESS, ELEC (DRIVE, APU GEN) and HYD panels; SD ELEC AC / DC; E/WD | **Status:** Not sim-tested (unit tests: CPC, DirectDrive, A380 pressurisation, fault light and e-pump Rust tests, failing first; FwsElecDriveOilAlerts.spec.ts and ElecFailureStates.spec.ts).

What it is for

- **Manual cabin pressure**: in MAN, the cabin altitude target holds the current cabin altitude and the V/S target 0 ft/min until the CABIN ALT TRGT or V/S TRGT selector is turned (it jumped to the selector position). Without the FMS landing elevation the controllers regulate to 7000 ft in descent (7500 ft also without ADIRU pressure) instead of 0 ft, and tell the FWS (new CPCS word bit 19).
- **Generator drive oil** (FCOM PRO-ABN-ECAM-10-24): new flyPad failures **DRIVE 1 (2)(3)(4) Oil Leak** and **DRIVE 1 (2)(3)(4) Oil Cooling Lost**; ELEC DRIVE n OIL LEVEL LO, OIL OVHT (above 200 C) and OIL PRESS LO (below 35 PSI); the DRIVE pb FAULT light (OVHT and PRESS LO); after 5 min of low level the generator switches off, after 5 min of low pressure or above 250 C the drive disconnects (then ELEC DRIVE n DISCONNECTED). Normal drive oil temperatures no longer reach 200 C at high N2.
- The **APU GEN A (B) pb-sw FAULT** light comes on when the GGPCU disconnects the APU generator.
- **SD ELEC AC and DC pages:** GEN, TR, STAT INV and RAT turn amber when they are failed.
- **Electric hydraulic pumps**: on the ground the green ones are inhibited only by engine 1 or 2 running, the yellow ones by engine 3 or 4; all are inhibited in flight (FCOM).
- Shared code: the engine generator drive oil model of the A32NX IDG (drive trait and oil failure types) is on this branch as the base of the A380X drive model; the A32NX IDG failures are not offered here (they are on core/failures-a32nx). The A32NX cabin pressure part is on core/failures-a32nx.

What you should see

Situation	What you should see
DRIVE 1 Oil Leak, engine 1 running	About 25 s after the leak starts the oil level is low (ELEC DRIVE 1 OIL LEVEL LO, shown only on the ground before the first engine start and after the last shutdown); about 50 s after, the pressure is low: ELEC DRIVE 1 OIL PRESS LO with the DRIVE 1 FAULT light and the SD ELEC AC page; 5 min later the drive disconnects (ELEC DRIVE 1 DISCONNECTED).
DRIVE 1 Oil Cooling Lost, engine 1 running	About 4 min later ELEC DRIVE 1 OIL OVHT with the DRIVE 1 FAULT light; about 100 s later (250 C) the drive disconnects.

Limitations and why

Limitation	Why	What to do
The leak and heating rates are design choices.	The FCOM gives no normal oil values.	-
A disconnected drive cannot be reconnected in flight.	FCOM DRIVE pb DISC.	Reconnect on the ground.

How to use

1. Drive oil alerts: follow the E/WD (GEN ... OFF or DRIVE ... DISC).
2. Manual cabin pressure: set MAN and turn the CABIN ALT TRGT or V/S TRGT selector to take control.

How to test

#	Do this	Expected result

1	A380X in cruise. flyPad > Failures > ATA 24 Electrical Power > **DRIVE 1 Oil Leak**. Wait 1 min.	ELEC DRIVE 1 OIL PRESS LO, DRIVE 1 FAULT light, SD ELEC AC page.
2	Wait 5 min more without action.	The drive disconnects: ELEC DRIVE 1 DISCONNECTED. Deactivate the failure.
3	Cruise. flyPad > Failures > ATA 24 Electrical Power > **DRIVE 2 Oil Cooling Lost**. Wait about 4 min.	ELEC DRIVE 2 OIL OVHT, DRIVE 2 FAULT light; about 100 s later the drive disconnects.
4	Cruise. flyPad > Failures > ATA 24 Electrical Power > **Static Inverter** or **TR 1**; SD ELEC DC page.	The failed item is amber. Deactivate it.
5	Cruise, pressurization: CABIN ALT MODE MAN.	The cabin altitude stays where it is until the CABIN ALT TRGT selector is turned; then it follows the selector.
6	On the ground, engines 3 and 4 running, 1 and 2 off. HYD panel: green electric pumps.	The green electric pumps can run; the yellow ones are inhibited.

What PR testers should check

- In MAN ALT / MAN V/S, the first selector turn jumps to the selector position.
- The drive oil failure timeline (level, pressure, disconnection).
- The APU GEN FAULT light during an APU start.

11. APU automatic shutdowns, smoke detection and cargo extinguishing alerts

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 26 and ATA 49; APU panel; CARGO SMOKE panel; E/WD, STATUS | **Status:** Not sim-tested (unit tests: Rust ECB, smoke detection, cargo extinguishing and fire tests, failing with the logic disabled; SmokeAlerts.spec.ts (FWS side only), failure list specs).

What it is for

- **APU ECB automatic shutdowns** (FCOM PRO-ABN-ECAM-10-49): the APU shuts down by itself, with APU FAULT, on an ECB failure, an overspeed, a low oil pressure, a high oil temperature (A380X: on the ground only) and when its DC supply is lost above N 70 % (1 s confirmation). Four new flyPad failures: **APU electronic control box (ECB)**, **APU overspeed (automatic shutdown)**, **APU low oil pressure (automatic shutdown)**, **APU high oil temperature (automatic shutdown on ground)**.
- **Smoke detection** (FCOM DSC-26): the A380X FWS raises the SMOKE warnings and SMOKE ... DET FAULT cautions of 9 zones: FWD, AFT and BULK CARGO, L / R MAIN AVNCS, L / R UPPER AVNCS, AFT AVNCS and IFE BAY (for example SMOKE FWD CARGO SMOKE, SMOKE FWD CARGO DET FAULT), with their procedure lines.
- **Cargo fire bottles**: SMOKE FWD+AFT CARGO BTL 1 (2) FAULT when a bottle has a low pressure; STATUS INOP SYS FWD+AFT CRGs BTL 1 (2).
- New flyPad failures (ATA 26): **Forward / Aft / Bulk cargo smoke**, **Left / Right main avionics bay smoke**, **Left / Right upper avionics bay smoke**, **Aft avionics bay smoke**, **IFE bay smoke**, the smoke detectors 1 and 2 of each zone (for example **Forward cargo smoke detector 1**), **Cargo fire bottle 1 (2) low pressure**.

Limitations and why

Limitation	Why	What to do
The SMOKE alerts need `core/failures-a32nx`.	The A380X smoke detection model (shared fire model) and the CARGO SMOKE panel lights (SMOKE, DISCH, BTL, AGENT pbs) are on that branch; the alerts here read its L:vars.	Merge both; without it the smoke failures give nothing.

No smoke detection in the crew rests, lavatories, CWS, showers and facilities.	Those zones are not modelled.	-
No bottle diverter / control alerts (FWD (AFT) CARGO BOTTLES FAULT).	Not modelled.	-
Some APU shutdown causes are missing.	No model for them.	-

How to use

1. APU FAULT: set the APU MASTER SW OFF as the E/WD asks.
2. SMOKE warnings: follow the E/WD (ISOL VALVES, AGENT, CAB FANS ...), with core/failures-a32nx merged.

How to test

#	Do this	Expected result
1	A380X at the gate, APU running (APU MASTER SW ON, START). flyPad > Failures > ATA 49 Airborne Auxiliary Power > **APU overspeed (automatic shutdown)**.	The APU shuts down; APU FAULT with AUTO SHUTDOWN and -APU MASTER SW ... OFF; STATUS INOP SYS APU.
2	APU MASTER SW OFF.	The line is done; APU FAULT goes when the ECB is off. Deactivate the failure.
3	With core/failures-a32nx merged, at the gate: flyPad > Failures > ATA 26 Fire Protection > **Forward cargo smoke**.	Red SMOKE FWD CARGO SMOKE with its lines (ISOL VALVES, AGENT ... DISCH). Deactivate it.
4	At the gate: activate **Forward cargo smoke detector 1** and **Forward cargo smoke detector 2**.	SMOKE FWD CARGO DET FAULT (crew awareness). Deactivate both.
5	At the gate: activate **Cargo fire bottle 1 low pressure**.	SMOKE FWD+AFT CARGO BTL 1 FAULT; STATUS INOP SYS FWD+AFT CRGs BTL 1.

What PR testers should check

- No APU shutdown during the normal power transfers (EXT PWR, APU GEN, engine generators).
- The DISCH / AGENT / BTL light emissives (with core/failures-a32nx).
- The smoke failures from the flyPad.

12. bleed monitoring, leak detection, BLEED FAULT and LEAK alerts

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 36; overhead AIR panel (ENG BLEED pbs); SD BLEED; E/WD | **Status:** Not sim-tested (unit tests: A380 bleed_monitoring 12 Rust tests, failing with the logic disabled; BleedMonitoringAlerts, EngineBleedValveIndication and AbnormBleedConfig specs).

What it is for

- **Bleed monitoring** (FCOM DSC-36-10-20): the CPIOM-A closes an engine bleed valve on an overtemperature (260 C, or 245 C above 45 PSI, for 55 s; the HP valve too) or an overpressure (60 PSI, or 45 PSI above 245 C, for 15 s), held until the ENG BLEED pb is set OFF. A failed bleed opens the crossbleed valves at AUTO.
- **Leak detection** (DSC-36-10-60): a leak of an engine pylon duct (above 180 C) or of a wing duct (above 124 C) closes that bleed valve and the crossbleed valves of its side.
- The **ENG 1 (2)(3)(4) BLEED FAULT** light comes on for these cases and when a bleed valve is not in its commanded position (DSC-36-20).

- New alerts (PRO-ABN-ECAM-10-36): **AIR ENG 1 (2)(3)(4) BLEED FAULT**; **AIR ENG 1 (2)(3)(4) BLEED LEAK** and **AIR L (R) OUTR (INR) WING LEAK** with the AIR LEAK ISOLATED line. A BLEED FAULT also triggers AIR ABNORM BLEED CONFIG.
- New flyPad failures (ATA 36): **Engine 1 (2)(3)(4) bleed valve stuck closed**, **... stuck open**, **Engine 1 (2)(3)(4) pylon bleed duct leak**, **Left / Right outer / inner wing bleed duct leak**.
- Shared code: the bleed air failure types of the A32NX pneumatic failures come with it (prerequisite commit, no behaviour of its own). The A32NX BLEED page part is on core/failures-a32nx.

Limitations and why

Limitation	Why	What to do
No A380X pack alerts.	No pack valve or regulation failure model.	-
One leak loop per zone; no APU, pack bay or hot-air duct leak; no HP stage pressure low.	Not modelled.	-
A pylon leak is only detected at high power.	The duct must be pressurised enough to heat the loop.	Test it with high thrust.

How to use

1. Follow the E/WD: ENG BLEED pb OFF for a leak or a bleed fault; AIR LEAK ISOLATED tells when it is done.
2. Use the AIR ABNORM BLEED CONFIG lines (crossbleed, air flow, packs, wing anti-ice) after a bleed is lost.

How to test

#	Do this	Expected result
1	A380X in climb or cruise. flyPad > Failures > ATA 36 Pneumatic > **Left outer wing bleed duct leak**.	ENG 1 BLEED FAULT light; AIR L OUTR WING LEAK; the engine 1 bleed valve and the left and center crossbleed valves close (SD BLEED).
2	ENG 1 BLEED pb OFF, deactivate the failure.	The FAULT light goes off once the overheat is gone. ENG 1 BLEED pb back to AUTO.
3	Climb thrust. Activate **Engine 2 pylon bleed duct leak**.	ENG 2 BLEED FAULT light; AIR ENG 2 BLEED LEAK; bleed valve 2 and the left crossbleed valve close. Deactivate it, ENG 2 BLEED OFF then AUTO.
4	Activate **Engine 3 bleed valve stuck open**, then ENG 3 BLEED pb OFF.	ENG 3 BLEED FAULT light (the valve is not where it is commanded). Deactivate it, pb AUTO.

What PR testers should check

- No false AIR ENG BLEED FAULT on a fast thrust increase at take-off.
- The leak failures from the flyPad (the pylon leak needs high power to be detected).

13. landing gear and brakes

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 32; L/G lever, L/G GRVTY switch, A-SKID switch, AUTO BRK; SD WHEEL; E/WD | **Status:** Not sim-tested (unit tests: 8 Rust gear / brake tests, failing first, 4 EmergencyBrakeLaw tests, FwsBrakeAlerts.spec.ts cases).

What it is for

- The gear extension is inhibited above **280 kt** (FCOM DSC-32-10-20; was 260 kt, the A320 value).

- Hydraulic supply as in the FCOM: the nose gear on GREEN, each main gear assembly (body and wing) on GREEN and YELLOW: with GREEN or YELLOW lost the main gears no longer retract, and the gear is extended by gravity.
- The **gravity extension** takes about 70 s from the L/G GRVTY switch to the last gear downlocked (was about 50 s).
- **Emergency braking** (A-SKID off, or normal and alternate braking lost) follows the EBCU laws: 1 700 psi for taxi-out and take-off, 1 000 psi from lift-off until the PARK BRK handle is set ON (was 1 160 psi).
- The **RTO autobrake** no longer activates below 40 kt, but an active one continues to the stop.
- New alerts: **BRAKES A-SKID FAULT ON ALL L/G** (emergency braking with the A-SKID switch ON) and **BRAKES AUTO BRK FAULT** with the new flyPad failure **Brakes autobrake (AUTO BRK)**.

Limitations and why

Limitation	Why	What to do
The wing and body gears are still one gear per side.	The gear model has no separate wing gears.	-
With GREEN lost the body gear, with YELLOW lost the nose gear do not retract.	Known deviation of the shared LGCIU: the gears move only with all doors open.	-
No braking LEHGS, separate accumulators or BCS split; 16 BRAKES alerts still missing.	Not modelled.	-

How to use

1. Extend the gear below 280 kt (and below VLO).
2. For a gravity extension, set the L/G GRVTY switch DOWN and wait about 70 s.

How to test

#	Do this	Expected result
1	A380X at the gate, engines running, A-SKID ON. flyPad > Failures > ATA 32 Landing Gear > **Brakes normal braking (BCS NORM BRK)** and **Brakes alternate braking (BCS ALTN BRK)**.	Emergency braking: BRAKES A-SKID FAULT ON ALL L/G (with the alerts of the two failures). Deactivate both.
2	At the gate: activate **Brakes autobrake (AUTO BRK)**.	BRAKES AUTO BRK FAULT; SD WHEEL page. Deactivate it.
3	In flight, gear up, below 250 kt: L/G GRVTY switch DOWN, start a timer.	The last gear is downlocked after about 70 s; SD WHEEL L/G GRVTY EXTN IN PROGRESS meanwhile.
4	Take-off roll with RTO armed, reject below 40 kt.	The RTO autobrake does not activate.
5	After landing with A-SKID OFF, brake with the pedals before setting the parking brake.	Emergency braking limited to 1 000 psi.

What PR testers should check

- The gravity extension timing in flight.
- The L/G alerts and indications with GREEN or YELLOW lost.
- The landing law after touchdown with A-SKID OFF.

14. HP fuel valve, FADEC faults, motoring and auto-idle

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 73 and ATA 79; ENG MASTER, ENG START; thrust reversers; E/WD, SD ENGINE | **Status:** Not sim-tested (unit tests: Rust systems / a380 tests, 10 failing first, and module tests; EngineMonitoringAlerts.spec.ts; failure list spec).

What it is for

- New flyPad failures **Engine 1 (2)(3)(4) HP fuel valve stuck** (ATA 73): the valve stays where it is; stuck closed, the engine gets no fuel and does not light up. When the valve position disagrees with the FADEC command for 2 s: **ENG 1 (2)(3)(4) HP FUEL VLV FAULT** (FCOM PRO-ABN-ECAM-10-70; master caution, ENGINE page) and the ENG FAULT light.
- New FADEC fault failures with their crew awareness alerts (no aural, no master light): **Engine n fuel strainer clog** - ENG n FUEL STRAINER CLOGGED; **Engine n fuel filter monitoring** - ENG n FUEL FILTER MONITORING FAULT; **Engine n oil chip (metal particles)** (ATA 79) - ENG n OIL CHIP DETECTED; **Engine n FADEC minor fault (internal sensor)** - ENG n MINOR FAULT; **Engine n overthrust protection** - ENG n OVTHR PROT LOST; **Engine n pressure and temperature probes** - ENG n SENSOR FAULT. 28 A380X failures in all.
- **Auto-idle**: engines 1 and 4 are held at idle while a thrust reverser operates (FCOM).
- Pre-start motoring of a hot engine (shared engine start code, as on core/failures-a32nx).

What you should see

Situation	What you should see
FUEL STRAINER CLOGGED	Shown in flight phases 1, 2, 11, 12 (inhibited 3 to 10).
FUEL FILTER MONITORING FAULT, OIL CHIP DETECTED	Shown only in phases 11 and 12 (after landing).
MINOR FAULT, OVTHR PROT LOST, SENSOR FAULT	Shown in phases 1, 11 and 12 (inhibited 2 to 10).
HP FUEL VLV FAULT	Inhibited in phases 3 to 7, 9 and 10.

Limitations and why

Limitation	Why	What to do
Other ENG alerts are still missing.	They need engine models the sim does not have.	-
No approach idle; no A380X derates.	No FCOM value; no A/THR model for the derates.	-

How to use

1. HP FUEL VLV FAULT during a start: ENG MASTER OFF (and ENG MAN START OFF for a manual start), as the E/WD asks.
2. The crew awareness FADEC alerts need no action in flight; report them after the flight.

How to test

#	Do this	Expected result
1	A380X at the gate, engines off. flyPad > Failures > ATA 73 Engine - Fuel And Control > **Engine 1 HP fuel valve stuck**. Start engine 1.	No fuel flow, no light-up; after 2 s ENG 1 HP FUEL VLV FAULT and the ENG 1 FAULT light. ENG MASTER 1 OFF, deactivate the failure.

2	At the gate, aircraft powered, engines off (phase 1). Activate **Engine 2 fuel strainer clog**, **Engine 3 FADEC minor fault (internal sensor)** and **Engine 4 pressure and temperature probes**.	ENG 2 FUEL STRAINER CLOGGED, ENG 3 MINOR FAULT, ENG 4 SENSOR FAULT (crew awareness, no chime). Deactivate them.
3	After landing, reversers deployed.	Engines 1 and 4 stay at idle while engines 2 and 3 give reverse thrust.
4	Engine running: activate **Engine 1 HP fuel valve stuck**, then ENG MASTER 1 OFF.	The engine still stops (the LP valve closes); ENG 1 HP FUEL VLV FAULT.

What PR testers should check

- Hot restart motoring.
- An A380X HP valve stuck open still stops the engine at ENG MASTER OFF.
- The outboard auto-idle in reverse.

15. LP valve, emergency outer tank transfer and fuel leak detection

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28; overhead FUEL panel (EMER OUTR TK XFR); SD FUEL; E/WD | **Status:** Not sim-tested (unit tests: Rust gravity_feed 16 and engine_lp_valve 5, 8 failing first; FwsFuelLeakLpValveAlerts, FuelLeak, FuelEmerOuterTransfer and FuelPumpsAndValves specs).

What it is for

- New flyPad failures **ENG 1 (2)(3)(4) LP fuel valve jammed in place**: **FUEL ENG n LP VLV FAULT** (FCOM PRO-ABN-ECAM-10-28) when the valve stays closed while commanded open (engine start) or stays open while commanded closed (engine shutdown), on the ground; before take-off -ENG n MASTER ... OFF. Jammed closed, the engine cannot start. STATUS INOP SYS ENG n LP VLV.
- The **EMER OUTR TK XFR** pb-sw opens the emergency outer tank transfer valves: the outer tank fuel flows by gravity into feed tanks 1 and 4 (flight_model.cfg gets the missing Feed 4 input line).
- **FQMS fuel leak detection** (DSC-28-40): slow leak (block fuel minus (FOB + fuel used) above 15 450 lb) and fast leak (FOB decrease minus fuel used above 4 400 lb in 30 min); **FUEL LEAK DETECTED** (phases 2 and 8) with the full FUEL LEAK procedure; **FUEL LEAK DET FAULT** (phase 8) with STATUS INOP SYS FUEL LEAK DET. New flyPad failures **Feed tank 1 (2)(3)(4) leak**.
- Shared code: the LP valve jam (engine_lp_valve.rs) is the same as on core/failures-a32nx, where the A32NX part is.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed.	flight_model.cfg changes.	Restart the sim after installing.
Other fuel alerts are still missing.	They need temperature, overflow, inerting or FQMS transfer models.	-

How to use

1. FUEL LEAK DETECTED: follow the FUEL LEAK procedure on the E/WD.
2. EMER OUTR TK XFR pb ON to move the outer tank fuel into feed tanks 1 and 4.

How to test

#	Do this	Expected result

1	A380X at the gate, engines off. flyPad > Failures > ATA 28 Fuel > **ENG 1 LP fuel valve jammed in place**. ENG MASTER 1 ON (start).	The engine does not start; FUEL ENG 1 LP VLV FAULT with -ENG 1 MASTER ... OFF. MASTER OFF, deactivate the failure.
2	Cruise with fuel in the outer tanks. FUEL panel: EMER OUTR TK XFR pb ON. SD FUEL page.	The outer tanks empty into feed tanks 1 and 4. pb OFF.
3	Cruise. Activate **Feed tank 1 leak**; wait up to 30 min.	FUEL LEAK DETECTED once the lost fuel is 4 400 lb more than the fuel used; follow the FUEL LEAK procedure. Deactivate the failure.
4	Normal flight without failure.	No FUEL LEAK DETECTED.

What PR testers should check

- EMER OUTR TK XFR into feed tanks 1 and 4.
- No false FUEL LEAK DETECTED.

16. anti-ice valves, wing anti-ice, oxygen, anti-ice key events

> **Aircraft:** A380X and A32NX (engine anti-ice valves, key events) | **Where:** flyPad > Failures > ATA 30 and ATA 35; overhead ANTI ICE panel; SD DOOR/OXY; E/WD | **Status:** Not sim-tested (unit tests: Rust engine_anti_ice, A380 ice_rain_protection and oxygen tests, failing first; AntiIceValveAlerts, OxygenAlerts and FadecWingAntiIce specs; the wasm key event mapping tests compile but cannot run on the host).

What it is for

- **A380X wing anti-ice system** (FCOM DSC-30): the WING ANTI ICE pb commands two valves per wing on bleed air: 30 s ground test, then 10 min inhibition; inhibited during the take-off until the thrust reduction; FAULT light. The flight controls and the FADECs now get the real wing anti-ice state.
- **Engine anti-ice valves** (both aircraft, shared model): open with the pb ON, or unpowered once the HP air is above 10 psig; 2 s travel; FAULT light. The MSFS engine anti-ice follows the valves.
- New A380X alerts (PRO-ABN-ECAM-10-30): **A-ICE ENG 1 (2)(3)(4) VLV CLOSED / VLV OPEN**, **A-ICE L (R) WING VLV CLOSED / VLV OPEN / VLV PRESS HI / VLV PRESS LO**.
- **A380X oxygen** (PRO-ABN-ECAM-10-35): cockpit and cabin bottle pressures on the DOOR/OXY page; **OXY CKPT BOTTLE PRESS LO**, **OXY CABIN BOTTLE PRESS LO**, **OXY CABIN VLV CLOSED**.
- New flyPad failures: ATA 30 **Left (Right) wing anti-ice valves stuck closed**, **Left (Right) wing outer anti-ice valve stuck open**, **Engine 1 (2)(3)(4) anti-ice valve stuck closed / stuck open**; ATA 35 **Cockpit oxygen bottle leak**, **Cabin oxygen bottles leak**, **Cabin oxygen regulator supply valve stuck closed**.
- **Key events** (both aircraft): MSFS anti-ice events (ANTI_ICE_TOGGLE_ENGn, ANTI_ICE_SET_ENGn, ANTI_ICE_ON / OFF / SET / TOGGLE, TOGGLE_STRUCTURAL_DEICE) now move the ENG and WING ANTI ICE pbs, so key bindings and hardware panels keep working.
- The A32NX engine anti-ice valves (with their ENG ANTI ICE pb FAULT light) come with the shared key event code: without them the A32NX would not build. The A32NX probe and window heat, alerts and failures are on core/failures-a32nx.

Limitations and why

Limitation	Why	What to do
No A380X probe / window heat model and its 15 alerts; no wing anti-ice on the SD BLEED page.	Not done yet.	-
No ADR probe heat bits.	Not done yet.	-

> **NOTE**
> Fix-up commit de9a236e6 removes a native stub of the systems WASM that these changes left unused (the workspace clippy check failed on it). No effect for the user.

How to use

1. Use the ENG and WING ANTI ICE pbs (or your key bindings / hardware panel) as usual.
2. On an A-ICE or OXY alert, follow the E/WD.

How to test

#	Do this	Expected result
1	A380X on the ground, engines running. WING ANTI ICE pb ON.	The wing valves open for the 30 s ground test, then close; ON again within 10 min: no new test.
2	Cruise. flyPad > Failures > ATA 30 Ice And Rain Protection > **Engine 1 anti-ice valve stuck closed**; ENG 1 ANTI ICE pb ON.	ENG 1 ANTI ICE FAULT light; A-ICE ENG 1 VLV CLOSED. pb OFF, deactivate the failure.
3	Cruise. Activate **Left wing anti-ice valves stuck closed**; WING ANTI ICE pb ON.	A-ICE L WING VLV CLOSED. pb OFF, deactivate it.
4	flyPad > Failures > ATA 35 Oxygen > **Cockpit oxygen bottle leak**; SD DOOR/OXY page.	The cockpit bottle pressure drops; then OXY CKPT BOTTLE PRESS LO. Deactivate it.
5	Bind a key to the MSFS "toggle engine 1 anti-ice" event and press it (either aircraft).	The ENG 1 ANTI ICE pb moves; no toggle loop.

What PR testers should check

- Key binding / hardware panel toggles of ENG and WING anti-ice: no loop, no toggle storm at flight load.
- A32NX ENG A.ICE FAULT light before about 25 % N2.
- A380X wing pb ground test and take-off inhibition.
- The oxygen leak failure on the DOOR/OXY page.

17. avionics network failures, SFCC channels, TAIL STRIKE

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 27, ATA 31 and ATA 42; E/WD, SD, STATUS | **Status:** Not sim-tested (unit tests: Rust avionics network and SFCC channel tests, 8 failing first; FwsTailStrikeAlerts.spec.ts and failure list specs, 14 failing first).

What it is for

- New ATA 42 failures (FCOM DSC-42-10/20) of the 16 **AFDX switches** (for example **AFDX switch 1 (network A)**), the 8 **IOMs** (**IOM A1** to **IOM A8**) and the 22 **CPIOMs** (for example **CPIOM C1 (FWS, FCDC)**, **CPIOM F1 (fuel)**). A failed switch is rerouted around; a failed module is no longer available, and the EWD, SD, PFD and FWS parts that already watch the CPIOM availability react to it.
- New ATA 27 failures **SFCC 1 (2) flap channel** and **SFCC 1 (2) slat channel**: the failed channel stops its motor (flaps or slats at half speed); F/CTL FLAP (SLAT) SYS 1 (2)(1+2) FAULT now have a trigger.
- **TAIL STRIKE** caution (PRO-ABN-ECAM-10-27-10, LAND ANSA) after a detected contact, and **TAIL STRIKE DET FAULT** (STATUS INOP SYS TAIL STRIKE DET) with the new failure **Tail strike detector** (ATA 31). Contact = pitch 11.5 deg or more with a main gear compressed (the FCOM LIM-10 geometry limit, design choice); it is remembered until the next flight (phase 1).

Limitations and why

Limitation	Why	What to do

Network losses have no effect on FMS, datalink, braking, steering, transponder or PFD yet.	Not modelled.	-
No actuator, sidestick, THS or WTB failures; no backup control (BCM / BPS); no ALT 1C reconfiguration.	They need the flight control model (Simulink).	-
The tail strike contact is a pitch / gear rule.	The sim has no fuselage contact sensor.	-

How to use

1. Activate the failures in the flyPad ATA 27, 31 and 42 lists.
2. After a TAIL STRIKE, land as soon as possible (LAND ANSA).

How to test

#	Do this	Expected result
1	A380X in flight below VFE. flyPad > Failures > ATA 27 Flight Controls > **SFCC 1 flap channel**; select the next flap position.	F/CTL FLAP SYS 1 FAULT; the flaps move at half speed. Deactivate it.
2	At the gate. flyPad > Failures > ATA 31 Indicating / Recording System > **Tail strike detector**.	TAIL STRIKE DET FAULT; STATUS INOP SYS TAIL STRIKE DET. Deactivate it.
3	Normal take-off and landing.	No TAIL STRIKE.
4	Cruise. flyPad > Failures > ATA 42 Integrated Modular Avionics > **AFDX switch 1 (network A)**.	Nothing lost (rerouted).
5	Activate **CPIOM C1 (FWS, FCDC)** and **CPIOM C2 (FWS, FCDC)**.	Note what the FWS, E/WD and SD lose and report it. Deactivate both.

What PR testers should check

- No false TAIL STRIKE on normal rotations and flares.
- Half-speed flaps / slats with a channel failure.
- CPIOM C1 / C2 or AFDX switch failures and their FWS / EWD / SD loss.

18. reactive windshear, GPWS mode 2, alpha floor Mach limit

> **Aircraft:** A380X | **Where:** PFD; GPWS aurals | **Status:** Not sim-tested (unit tests: Rust egpws 3, C++ alpha_floor_mach_inhibit_test, vitest windshear 13 (ReactiveWindshearDetection.spec.ts), GpwsMode2 8; all failing without the change).

What it is for

- **Reactive windshear** (FCOM DSC-22): the PRIMs detect a decreasing performance windshear at take-off and landing below 1 300 ft, CONF 1 or more: red WINDSHEAR on both PFDs (L:A32NX_REACTIVE_WINDSHEAR_WARNING). The FCOM gives no formula: F-factor above 0.13, filtered (design choice). The voice comes with the voices feature below.
- **Alpha floor** is inhibited above M 0.5 (was M 0.6), in the generated PRIM code.
- **GPWS mode 2** (excessive terrain closure rate) works again with a filtered radio altitude rate: TERRAIN then PULL UP on the PFD with the PULL UP aural, TERRAIN only in landing configuration. The PFD now shows the PULL UP of

mode 1 (excessive rate of descent): the mode 2 bit overwrote it.

- Shared code: the windshear detector, the PFD warning and the egpws Rust are the same as on core/failures-a32nx, where the A32NX part (FAC windshear, GPWS test, TCAS failure) is.

Limitations and why

Limitation	Why	What to do
No REAC W/S DET FAULT alert, SRS escape guidance, FLS modes, THR DES / DCLB, TAD / TCF or predictive windshear.	They need models, terrain or wind data.	-
The Simulink PRIM model still has M 0.6.	Only the generated code is changed.	Change the model before a regeneration.

How to use

1. On WINDSHEAR or PULL UP, apply the FCOM escape manoeuvre.

How to test

#	Do this	Expected result
1	A380X approach in gusty weather, CONF 1 or more, below 1 300 ft; normal turns.	No WINDSHEAR without a real shear.
2	Low flight over rising terrain, gear up, high terrain closure rate.	TERRAIN then PULL UP on the PFD with the PULL UP aural.
3	High rate of descent close to the ground (mode 1).	PULL UP on the PFD.
4	Cruise above M 0.5 at a high angle of attack (for example a tight high-altitude turn).	No alpha floor activation.

What PR testers should check

- No false WINDSHEAR in gusts or turns.
- A380X mode 2 over rough terrain.

19. NAV FM / GPS POS DISAGREE and NAV LS TUNING DISAGREE

> **Aircraft:** A380X | **Where:** E/WD, STATUS | **Status:** Not sim-tested (unit tests: FwsNavPositionAlerts.spec.ts, failing first).

What it is for

- **NAV FM / GPS POS DISAGREE** (FCOM PRO-ABN-ECAM-10-34-10) now triggers when the FMS finds its position too far from the GPS position (latitude-dependent FCOM thresholds; L:A32NX_FM_GPS_POS_DISAGREE of the FM position model of `core/gps`, merged here through core/a380x-elec): master caution, -ACFT POS ... CHECK, STATUS INFO FLS : RAW ONLY. Inhibited in phases 1, 3 to 6 and 12.
- **NAV LS TUNING DISAGREE** when the two MMRs (sim NAV 3 and NAV 4) disagree on frequency or course: master caution, phases 8 and 9 only; STATUS INOP SYS CAT 2, GLS AUTOLAND.

Limitations and why

Limitation	Why	What to do

The approach mode is not compared for LS TUNING DISAGREE.	The sim receivers have no approach mode.	-

How to use

1. On NAV FM / GPS POS DISAGREE, check the aircraft position (POSITION MONITOR page) as the E/WD asks.

How to test

#	Do this	Expected result
1	A380X normal flight with GPS.	No NAV FM / GPS POS DISAGREE.
2	ILS approach tuned by the FMS, below 1 500 ft.	No NAV LS TUNING DISAGREE.
3	If NAV FM / GPS POS DISAGREE shows, open STATUS.	INFO FLS : RAW ONLY.

What PR testers should check

- No false NAV FM / GPS POS DISAGREE or LS TUNING DISAGREE in normal flights.

20. A380X FWS voices: sidestick priority, intermediate RA callouts, WINDSHEAR, TERRAIN, GLIDE SLOPE, PITCH PITCH, COMPANY MESSAGE

> **Aircraft:** A380X | **Where:** Cockpit aurals; sidestick pbs | **Status:** Not sim-tested (unit tests: SidestickPriorityAurals, IntermediateHeightCallout, FwsVoiceCallouts, AutoCallOuts, FwsSidestickPriority, FwsWindshearAural, FwsVoiceSounds, LegacyGpws specs (27 failing first); FwsPitchAural 7, FwsCompanyMessageAural 5 and more AutoCallOuts / LegacyGpws / FwsVoiceSounds cases (24 failing first)).

What it is for

- **PRIORITY LEFT / PRIORITY RIGHT** (FCOM DSC-27-10-20) when a pilot holds the sidestick pb with the priority for 1 s (a short press to disconnect the autopilot does not call it; it waits during a GPWS alert).
- **Intermediate RA callout**: below 400 ft, the height rounded to 10 ft (for example TWO HUNDRED SEVENTY) 11 s after the previous callout; below 100 ft, the height to the foot (for example FORTY SEVEN) 4 s after it.
- **WINDSHEAR** said three times with the reactive windshear warning; it also silences the GPWS alerts.
- **TERRAIN** for GPWS mode 2 and **GLIDE SLOPE** (louder when hard) for mode 5.
- **PITCH PITCH** (DSC-22-27-20) in a manual landing below 400 ft RA with the pitch above 12 deg, once per event (the FWS computes it: the PRIM output is always false).
- **COMPANY MESSAGE** (DSC-46-40-20) when a company (AOC) message arrives, except in take-off and landing and below 10 000 ft in phase 8.
- The clips are made with tools/fws-voice (Piper TTS, public-domain LJ Speech voice) and play from the Wave Data slots of the A380X sound.xml.

Limitations and why

Limitation	Why	What to do
TERRAIN and GLIDE SLOPE may stay silent.	The A380X Wwise sound bank lacks 13 A32NX events (upstream large-files branch align-a380x-sounds).	Wait for that upstream merge.
No DUAL INPUT voice.	The F/O sidestick sends no input in the sim.	-

No SPEED SPEED, SELCAL or time marker voices.	No FCOM threshold / PRIM output, no SELCAL source, the FCOM contradicts itself on the time marker.	-
No COMPANY ALERT or CALL COMPANY.	The AOC has no urgent or call-company messages.	-
A full MSFS restart is needed.	sound.xml changes.	Restart the sim after installing.

How to use

1. Nothing to set: the voices come by themselves.

How to test

#	Do this	Expected result
1	A380X in flight, AP ON. Hold the captain sidestick pb 2 s.	AP disconnects; PRIORITY LEFT is said once. A short press only disconnects (no voice).
2	Manual approach: from 400 ft RA down to the touchdown.	The usual callouts plus the intermediate heights (rounded to 10 ft above 100 ft, to the foot below).
3	Manual landing: pitch above 12 deg below 400 ft RA (for example a hard flare).	PITCH PITCH, once.
4	In cruise, with Hoppie ACARS set up: receive a company telex.	COMPANY MESSAGE (with the COMPANY MSG memo).
5	GPWS mode 2 or mode 5 situation.	TERRAIN or GLIDE SLOPE (if the A380X sound bank has the events).

What PR testers should check

- The volume of the WAV clips against the Wwise voices, and their retrigger.
- The A380X Wave Data slots play at all (warning, misc and instrument slots).
- The bank + Piper voice mix below 100 ft (gap, volume); the intermediate callouts are not too chatty.
- PITCH PITCH once per manual flare; COMPANY MESSAGE per Hoppie telex.
- The TERRAIN / GLIDE SLOPE events are present in the A380X bank.

21. No BRAKES BTV FAULT while an inner engine is not running

> **Aircraft:** A380X | **Where:** E/WD; STATUS | **Status:** Not sim-tested (unit tests: FwsBrakeAlerts.spec.ts, 6 new, failing first).

What it is for

The FCDC sets BTV LOST when engine 2 or 3 is not running, and the FWS raised BRAKES BTV FAULT (single chime, MASTER CAUT) straight from it, for example during the engine start on the ground. The FCOM trigger is "The BTV auto-brake mode is failed" (PRO-ABN-ECAM-10-32); an inner engine out makes BTV inoperative through the ENG SHUT DOWN STATUS. The caution now needs both inner engines running; STATUS INOP SYS BTV and LDG DIST AFFECTED still follow the FCDC BTV LOST.

How to use

1. Nothing to set.

How to test

#	Do this	Expected result

1	A380X cold and dark at the gate: start the engines one by one.	No BRAKES BTV FAULT and no single chime from it at any time.
2	In flight: shut down engine 2.	ENG 2 SHUT DOWN; STATUS INOP SYS BTV; no BRAKES BTV FAULT. Relight.

22. Engine exceedance marks only on displayed values

> **Aircraft:** A380X | **Where:** E/WD EGT and N1 gauges; SD ENGINE vibrations | **Status:** Not sim-tested (unit tests: EgtLimits, N1 and EngineVibration specs, failing first).

What it is for

The red N1 / EGT exceedance marks and the N3 red cross latched any value, also while the gauge showed XX (FADEC not sending) or on an infinite value, and a NaN after an exceedance put the mark at NaN. The FCOM shows XX for a value that is not available. The exceedance memories now take only displayed, finite values (E/WD gauge active flag, SD FADEC powered) and keep the highest value through a NaN. The A32NX part is on core/failures-a32nx.

How to use

1. Nothing to set.

How to test

#	Do this	Expected result
1	A380X cold and dark, then power up and start the engines.	No red exceedance mark or red cross on the E/WD N1 / EGT gauges or the SD ENGINE page.
2	Load a flight in cruise.	No red mark.

23. XFR pushbuttons start in AUTO and drive the manual fuel transfers

> **Aircraft:** A380X | **Where:** Overhead FUEL panel (OUTR, MID, INR TK XFR, TRIM TK XFR pbs); SD FUEL | **Status:** Not sim-tested (unit tests: FuelManualTransfer.spec.ts 27, with all AUTO commands and lineups checked against flight_model.cfg / systems.cfg, and FuelEmerOuterTransfer gravity cases, failing first).

What it is for

The OUTR, MID and INR TK XFR and TRIM TK XFR pb L:vars were never set at load (read 0 = MAN) and MAN did nothing. Now (FCOM DSC-28-40, "fuel transfers are fully automatic"):

- They start in AUTO at every flight load (released, no light), also with the hangar, taxi and climb flights.
- MAN (FWD for the trim tank) suspends the automatic transfers and lines up the FCOM manual transfer: OUTR outer tanks into the feed tanks (with an outer tank pump OFF, the emergency valves feed tanks 1 and 4 by gravity); INR and MID with their aft pumps into the feed tanks; TRIM FWD trim tank into the inner tanks.
- Back to AUTO restarts the automatic logic. Jettison and refuel suspend the manual transfers; no lineup takes fuel out of a feed tank.

Limitations and why

Limitation	Why	What to do
No AUTO GND XFR.	The FCOM rule is incomplete and flight_model.cfg has no wing-to-trim line.	-

No outer gravity transfer to feed tanks 2 and 3, no XFR FAULT lights, no MAN XFR ECAM procedures.	Not done.	-
The jettison suspension reads L:A380X_FUEL_JETTISON_IN_PROGRESS.	core/fuel (its jettison signals) is not on this branch.	With core/fuel merged it uses them.

How to use

1. Leave the XFR pbs in AUTO for normal operation.
2. Press one to MAN (FWD) for a manual transfer, as the FCOM asks; press it again for AUTO.

How to test

Test this first, before the other fuel tests.

#	Do this	Expected result
1	Load the A380X cold and dark, then at the apron, then in flight. Overhead FUEL panel.	All four XFR pbs released, no light; on the ground no tank level changes by itself.
2	In flight: OUTR TK XFR pb MAN; SD FUEL page.	The outer tanks flow into the feed tanks.
3	Back to AUTO; then INR TK XFR MAN.	The automatic logic restarts; with INR MAN the inner tanks flow into the feed tanks (the right inner pump runs for the right side).
4	MID TK XFR MAN, then TRIM TK XFR FWD.	Mid tanks into the feed tanks; trim tank into the inner tanks. All back to AUTO.

What PR testers should check

- First: at load (cold and dark, apron, in flight) all four XFR pbs released / no light and no tank level change on the ground.
- The flow of each pb on the SD FUEL page, and back to AUTO.
- INR MAN runs the right inner pump (pump index swap).